Riemannian Geometry

— *EYNTKA* Series —

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Contents

Introduction

Riemannian geometry is the study of smooth manifolds carrying a metric: a smoothly varying inner product on the tangent spaces that lets one measure the lengths of curves, the angles between directions, and the volumes of regions. The metric forces a unique way to differentiate vector fields, the Levi-Civita connection, and the connection defines straight lines (geodesics) The extent to which those geodesics fail to behave as they would in Euclidean space is measured by the curvature tensor, and the organizing claim of this book is that curvature governs almost everything a Riemannian manifold does. Every chapter can be read as a chapter in the story of what curvature controls.

That story has two faces. The first face is metric and topological: curvature bends geodesics, refocuses or spreads neighboring ones, and thereby determines the completeness, the diameter, the fundamental group, and the shape of the manifold. This is the subject of Part I, which runs from the definition of a metric through the comparison theorems, submanifold geometry, the Gauss–Bonnet and sphere theorems, and the homogeneous and symmetric spaces where the geometry is as uniform as it can be. The second face is analytic: the same curvature, fed through the machinery of connections on vector bundles, manufactures topological invariants, and, turned into a differential operator, controls the harmonic analysis of the manifold. This is the subject of Part II, which develops Chern–Weil theory and the characteristic classes, proves the Hodge theorem identifying cohomology with harmonic forms, and closes with the Bochner technique, where a Weitzenböck formula couples curvature directly back to the harmonic forms of Part II's analysis.

The metric is taken to be of arbitrary signature wherever the theory permits it. The connection and the curvature tensor are introduced without any assumption of positive-definiteness, so that the Lorentzian case of general relativity travels alongside the Riemannian one. Where positive-definiteness is actually needed, in Hopf–Rinow, in the Hodge theorem, in Bochner's argument, the reliance is flagged and the Lorentzian divergence noted. The reader is assumed to be fluent with smooth manifolds, vector and tensor bundles, differential forms, and de Rham cohomology at the level of the companion volume *Differential Topology* ([6]); the analytic Part II draws in addition on measure theory and the elliptic theory of partial differential equations, imported at the points where they are used.

What the reader should carry away is a working command of both faces of the subject: the ability to compute with metrics, connections, geodesics, and curvature, to prove the comparison theorems and read their topological consequences, and to run the analytic machinery of Chern–Weil, Hodge, and Bochner. That command is the foundation for geometric analysis, general relativity, spin geometry and index theory, and the complex and Kähler refinement of Hodge theory. The closing chapter, chapter 14, maps each of these onward directions to where it is developed.

Part I

Metric Riemannian Geometry

The first Part develops the theme that curvature governs the geometry and topology of a manifold directly. Its starting datum is a metric, and from the metric everything is forced in turn. Chapters 1 through 3 build the local apparatus: the metric of arbitrary signature and its model spaces, the Levi-Civita connection that the metric determines uniquely, and the geodesics and exponential map that the connection defines. Chapter 4 introduces the curvature tensor as the obstruction to flatness, and chapter 5 measures its first effect infinitesimally, through the Jacobi fields that record how neighboring geodesics spread or refocus. With that infinitesimal measurement in hand, chapter 6 converts one-sided curvature bounds into hard global conclusions, completeness, diameter estimates, and the structure of the universal cover, the comparison theorems that are the technical heart of the metric theory.

The remaining chapters close the loop from local curvature to global shape. Chapter 7 studies how a manifold curves inside another through the second fundamental form, culminating in Gauss's Theorem Egregious that the intrinsic curvature is an isometry invariant; chapter 8 turns curvature integrals and curvature signs into topological constraints, from Gauss–Bonnet to Synge's theorem and the sphere theorem. Chapter 9 ends the Part with the homogeneous and symmetric spaces, the manifolds whose geometry looks the same at every point and whose curvature is as constrained as it can be. Throughout, the round sphere, hyperbolic space, the flat torus, and the surfaces of revolution serve as the recurring examples against which every general theorem is calibrated, and the pseudo-Riemannian case is carried along so that the reader emerges equipped for the Lorentzian geometry of general relativity as well as the Riemannian theory itself.

1

Riemannian and Pseudo-Riemannian Metrics

A smooth manifold, on its own, carries no geometry[1]. The constructions of *Differential Topology* [6] produce tangent spaces, vector fields, tensor bundles, and differential forms, but nothing among them measures the length of a tangent vector, the angle between two of them, or the volume of a region, in particular as none of these quantities are diffeomorphism-invariant. These are the first things one wants of a [traditional][2] space called geometric, and they are supplied by a single additional datum: a smoothly varying inner product on the tangent spaces, the *Riemannian metric.*

This chapter installs that datum and develops its immediate consequences. The metric is defined in the generality of arbitrary signature, so that the Riemannian and the pseudo-Riemannian (in particular the Lorentzian) cases are treated on the same footing from the outset; positive definiteness is invoked only where a result genuinely needs it. After establishing that Riemannian metrics always exist and that pseudo-Riemannian ones need not, the chapter fixes the model spaces that recur throughout the book, the round sphere, hyperbolic space, and the flat torus, and the isometries relating them. The algebraic apparatus the metric provides, the musical isomorphisms raising and lowering indices, the gradient and divergence, and the Riemannian volume form, is assembled next, and the chapter closes with the metric induced on a submanifold, the setting of classical surface theory. Everything built in later chapters, the connection, the geodesics, the curvature, is forced by the metric introduced here.

[1]Whitney's Approximations theorem from *Differential Topology* [6] shows that smoothness approximates continuity, hence is more a topological than geometric property

[2]this may be the wrong place for a footnote specifying how geometry have generizations that may not include what we "traditionally" think as a geometric thing, I also avoid using the word classical since that connotes classical geometry

1.1 Metrics of Arbitrary Signature

A smooth manifold, on its own, remembers almost nothing of geometry. It knows which functions are smooth, which curves are smooth, and how tangent spaces are glued together as one moves from chart to chart; it does not know how long a curve is, what angle two curves make where they cross, or whether one region is larger than another. All of that geometric information is the content of a single extra piece of data laid over the manifold: an inner product on each tangent space, varying smoothly from point to point. This section introduces that data in the generality that the rest of the theory requires, allowing the inner product to be indefinite, and isolates the one invariant that separates the cases.

The reason to allow indefiniteness from the outset is not abstraction for its own sake. The geometry of spacetime in general relativity is governed by a symmetric bilinear form that is not positive definite: the squared length of a tangent vector can be positive, negative, or zero, and those three signs encode the physical distinction between space, time, and light. The positive-definite (Riemannian) case and the indefinite (pseudo-Riemannian) case share a large body of formalism, and much of the connection and curvature theory in later chapters is written once and applies to both. Positive-definiteness is a genuine hypothesis in exactly a few places, most notably where lengths of curves must be nonnegative so that a distance function can be defined, and those places are flagged as they arise. Here the two notions are defined side by side, and the invariant that tells them apart, the signature, is shown to be locally constant, hence constant on each connected piece of the manifold.

Throughout, M denotes a smooth manifold of dimension n, and all objects are C^∞. The Einstein summation convention is in force: an index appearing once as a subscript and once as a superscript in a single term is summed from 1 to n. In a chart with coordinates $(x^1, \ldots, x^n)$ the coordinate vector fields are written $\partial_i = \partial/\partial x^i$ and the coordinate covectors dx^i.

Definition 1.1.1: Riemannian Metric

Let M be a smooth manifold. A *Riemannian metric* on M is a smooth symmetric covariant 2-tensor field g on M that is positive definite at every point: for each $p \in M$ the bilinear form

$$g_p \colon T_pM \times T_pM \to \mathbb{R}$$

is symmetric, $g_p(v, w) = g_p(w, v)$ for all $v, w \in T_pM$, and positive definite, $g_p(v, v) > 0$ for every nonzero $v \in T_pM$. A *Riemannian manifold* is a pair (M, g) consisting of a smooth manifold together with a Riemannian metric. The value $g_p(v, w)$ is written $\langle v, w\rangle$ or $\langle v, w\rangle_g$, and the induced *norm* of a tangent vector is $|v|_g = \sqrt{\langle v, v\rangle_g} \geq 0$.

Unwinding the definition of a covariant 2-tensor field, a Riemannian metric assigns to each point p an inner product on the tangent space T_pM, and it does so smoothly. Smoothness is the requirement that in every chart the component functions be smooth. Writing $g_{ij} = g(\partial_i, \partial_j)$ for the components in a chart with coordinates $(x^1, \ldots, x^n)$, one has

$$g = g_{ij}\, dx^i\, dx^j,$$

where the product of covectors is the symmetric product, and g is smooth precisely when each $g_{ij} \colon U \to \mathbb{R}$ is a smooth function on the chart domain U. Symmetry says the matrix (g_{ij}) is symmetric at every point; positive-definiteness says that this symmetric matrix is positive definite at every point. The tensor field machinery, the meaning of a smooth symmetric covariant 2-tensor field and its transformation law under a change of chart, is developed in [6]; the content added here is the pointwise positivity condition. Geometrically, once such a g is fixed, the length of a tangent vector, the angle $\arccos\big(\langle v, w\rangle / (|v|\,|w|)\big)$ between two nonzero vectors, and eventually the length of a curve, all acquire meaning as g must be preserved under Riemann-Manifold isomorphisms (isometries).

The first example is the flat model against which everything else is measured.

Example 1.1.2: The Euclidean and Minkowski Metrics

On $M = \mathbb{R}^n$, using the standard global coordinates $(x^1, \ldots, x^n)$, the *Euclidean metric* is

$$\bar{g} = \delta_{ij}\, dx^i\, dx^j = (dx^1)^2 + \cdots + (dx^n)^2,$$

so that $\bar{g}_{ij} = \delta_{ij}$ is the identity matrix at every point. Under the canonical identification of $T_p\mathbb{R}^n$ with $\mathbb{R}^n$, the value $\bar{g}_p(v, w) = \delta_{ij} v^i w^j$ is the usual dot product $v \cdot w$, which is symmetric and positive definite, and the component functions are constants, hence smooth. Thus $(\mathbb{R}^n, \bar{g})$ is a Riemannian manifold, the *Euclidean space* of dimension n; its norm $|v|_{\bar{g}} = \sqrt{(v^1)^2 + \cdots + (v^n)^2}$ is the Euclidean length.

Now alter a single sign. On $M = \mathbb{R}^{1,n}$, meaning $\mathbb{R}^{n+1}$ with coordinates written $(x^0, x^1, \ldots, x^n)$, the *Minkowski metric* is

$$\eta = -(dx^0)^2 + (dx^1)^2 + \cdots + (dx^n)^2,$$

with constant component matrix $\eta = \mathrm{diag}(-1, 1, \ldots, 1)$. The form η_p is symmetric, and its matrix is invertible at every point, so it is nondegenerate; but it is not positive definite, since $\eta_p(\partial_0, \partial_0) = -1 < 0$. Hence η is not a Riemannian metric. It is instead the basic pseudo-Riemannian example, defined precisely below, and it separates tangent vectors into three classes according to the sign of $\eta_p(v, v)$: a nonzero v is *timelike* if $\eta_p(v, v) < 0$, *spacelike* if $\eta_p(v, v) > 0$, and *null* (or *lightlike*) if $\eta_p(v, v) = 0$. For $v = (v^0, v^1, \ldots, v^n)$ this reads $-(v^0)^2 + \sum_{i\geq 1}(v^i)^2$, so v is timelike when its x^0-component dominates the spatial part, spacelike when the spatial part dominates, and null on the cone $-(v^0)^2 + \sum_{i\geq 1}(v^i)^2 = 0$. Unlike the Riemannian case, nonzero null vectors exist and have vanishing η-norm, so $\sqrt{\eta_p(v, v)}$ is not defined for all v; the classification into timelike, spacelike, and null takes the place of a norm.

The Euclidean and Minkowski forms differ in exactly one entry of their diagonal, and that single sign is what places one inside the Riemannian world and the other outside it. To capture both at once, the positive-definiteness requirement is relaxed to nondegeneracy, and the number of negative signs is recorded as a separate invariant.

Recall that a symmetric bilinear form b on a finite-dimensional real vector space V is *nondegenerate* if the only $v \in V$ with $b(v, w) = 0$ for all $w \in V$ is $v = 0$; equivalently, in any basis, the matrix of b is invertible. By Sylvester's law of inertia, for a nondegenerate symmetric b there is a basis in which the matrix of b is diagonal with entries ± 1, and the number ν of entries equal to -1 is independent of the diagonalizing basis. This number ν is the *index* of b, and the pair (number of -1's, number of $+1$'s) is its *signature*. The index is exactly the maximal dimension of a subspace on which b is negative definite, a description that will be used in the proof that the index cannot jump.

Definition 1.1.3: Pseudo-Riemannian Metric

Let M be a smooth manifold. A *pseudo-Riemannian metric* on M is a smooth symmetric covariant 2-tensor field g that is *nondegenerate* at every point and whose *index* ν, the number of -1's in a Sylvester diagonalization of g_p, is the same integer at every point of M. A pair (M, g) with g a pseudo-Riemannian metric is a *pseudo-Riemannian manifold*, and its *signature* is the pair $(\nu, n-\nu)$. The metric is called

1. *Riemannian* when $\nu = 0$, that is, when g_p is positive definite for every p; and
2. *Lorentzian* when $\nu = 1$, in which case, following the mostly-plus convention, the signature is $(-, +, \ldots, +)$.

The adjective *(pseudo-)Riemannian* refers to a pseudo-Riemannian metric of unspecified index, covering both the definite and indefinite cases.

Two conditions in this definition do work that the Riemannian definition handled automatically. First, nondegeneracy replaces positive-definiteness as the pointwise linear-algebraic requirement; it is exactly what is needed for the metric to identify tangent vectors with covectors (the musical isomorphisms of section 1.4) and thereby to raise and lower indices, and a positive-definite form is in particular nondegenerate, so definition 1.1.1 is the case $\nu = 0$ of definition 1.1.3. Second, the requirement that the index be a single fixed integer over all of M is a genuine hypothesis for a possibly disconnected manifold: on a manifold with two components one could in principle place a Riemannian metric on one piece and a Lorentzian metric on the other, and the resulting field would be smooth, symmetric, and nondegenerate but of no single signature, so it is excluded by fiat. On a *connected* manifold this constancy is automatic, and proposition 1.1.6 below proves exactly that. The index is a discrete invariant, an integer between 0 and n, so once it is known to vary continuously it can only be constant on a connected space; the content of the proposition is that it does vary continuously.

The running indefinite example is $(\mathbb{R}^{1,n}, \eta)$ of example 1.1.2: the matrix $\operatorname{diag}(-1, 1, \ldots, 1)$ is already in Sylvester form with a single -1, so η has index $\nu = 1$, signature $(-, +, \ldots, +)$, and $(\mathbb{R}^{1,n}, \eta)$ is a Lorentzian manifold, called *Minkowski space*. The classification of its tangent vectors into timelike, spacelike, and null carried out in that example is the model for the causal structure of any Lorentzian manifold.

A warning about terminology is worth stating before the word "metric" accumulates a second, incompatible meaning.

1.1.4 Note: A Riemannian Metric Is Not a Distance Function The word *metric* is overloaded. A Riemannian metric g is a smoothly varying inner product on tangent spaces; it is a tensor field, evaluated on pairs of tangent vectors at a single point, and $\langle v, w\rangle_g$ is a real number attached to $v, w \in T_pM$. It is not a metric in the sense of metric-space theory, that is, it is not a function $d\colon M \times M \to [0, \infty)$ measuring the separation of pairs of points. A Riemannian metric does *induce* such a distance function, by taking $d_g(p, q)$ to be the infimum of the lengths of curves from p to q, and (M, d_g) is then a metric space whose topology is the manifold topology; but that construction requires the length functional and a minimization argument, and it is carried out only in section 3.4, where the phrase *distance function* is reserved for d_g. Until then, "metric" means the tensor g, and no notion of point-to-point distance is in play. The distinction matters in the indefinite case for a structural reason: a Lorentzian metric assigns negative and zero squared lengths to vectors, so the naive infimum-of-lengths construction does not produce a metric space at all, and the passage from g to a distance function is special to the Riemannian setting.

The second family of examples, the round spheres, will be a recurring test case; here it appears only in its first form, as a metric induced by an ambient Euclidean metric, with the fuller treatment postponed.

Example 1.1.5: The Round Metric on the Sphere, First Look

Let $S^n = \left\{x \in \mathbb{R}^{n+1} \;\middle|\; |x|_{\bar g} = 1\right\}$ be the unit sphere, with inclusion $\iota\colon S^n \hookrightarrow \mathbb{R}^{n+1}$, and let $\bar g$ be the Euclidean metric on $\mathbb{R}^{n+1}$ from example 1.1.2. The inclusion is a smooth embedding, so at each $p \in S^n$ the differential $d\iota_p\colon T_pS^n \to T_p\mathbb{R}^{n+1}$ is an injective linear map, identifying T_pS^n with the hyperplane

$$\left\{w \in \mathbb{R}^{n+1} \;\middle|\; w \cdot p = 0\right\} \subset \mathbb{R}^{n+1}$$

tangent to the sphere. The *round metric* $\mathring{g}$ on S^n is the pullback

$$\mathring{g} = \iota^*\bar g, \qquad \mathring{g}_p(v, w) = \bar g_p\big(d\iota_p(v),\, d\iota_p(w)\big),$$

that is, the restriction of the ambient dot product to the tangent hyperplane. This is symmetric because $\bar g$ is, and it is smooth because ι is smooth and pullback of a smooth tensor field along a smooth map is smooth (see [6]). It is positive definite: for nonzero $v \in T_pS^n$ the vector $d\iota_p(v)$ is nonzero, since $d\iota_p$ is injective, and $\bar g$ is positive definite, so $\mathring{g}_p(v, v) = \bar g_p(d\iota_p v, d\iota_p v) > 0$. Hence $\mathring{g}$ is a Riemannian metric and $(S^n, \mathring{g})$ is a Riemannian manifold. The sphere of radius r, $S^n_r = \left\{x \;\middle|\; |x|_{\bar g} = r\right\}$, carries the analogous induced metric. The general statement that an immersion into a Riemannian manifold pulls back to a Riemannian metric, together with an explicit coordinate form of $\mathring{g}$, belongs to section 1.5, and the description of $(S^n_r, \mathring{g})$ as a homogeneous model space to section 1.3; here it suffices that the sphere carries a natural Riemannian metric, obtained by restricting the flat metric of the surrounding space.

It remains to justify the assertion, built into definition 1.1.3 as a hypothesis and claimed there to be automatic on connected manifolds, that the index of a nondegenerate metric cannot vary from point to point on a connected manifold. The precise statement is that the index is a locally constant function; connectedness then forces it to be globally constant.

Proposition 1.1.6: Constancy of the Signature

Let M be a connected smooth manifold and let g be a smooth symmetric covariant 2-tensor field on M that is nondegenerate at every point. Then the index $\nu(p)$ of g_p is the same for all $p \in M$; consequently the signature of a nondegenerate g is well defined, and the constancy required in definition 1.1.3 is automatic on a connected manifold.

***Proof*:**
The strategy is to show that the index function $\nu\colon M \to \{0, 1, \ldots, n\}$ is locally constant. A locally constant function on a connected space is constant, since for any value k the sets $\nu^{-1}(k)$ and $\nu^{-1}(\{0, \ldots, n\} \setminus \{k\})$ are both open, partition M, and connectedness forbids a nontrivial partition into two open sets. It therefore suffices to prove that every point of M has a neighborhood on which ν is constant.

Step 1: reduction to a symmetric matrix of functions. Fix $p_0 \in M$ and choose a chart (U, φ) with coordinates $(x^1, \ldots, x^n)$ about p_0, with U connected (shrink it to a coordinate ball if necessary). On U the metric has smooth components $g_{ij} = g(\partial_i, \partial_j)\colon U \to \mathbb{R}$, and at each $p \in U$ the index $\nu(p)$ is the index of the symmetric matrix

$$G(p) = \big(g_{ij}(p)\big)_{1 \le i,j \le n},$$

because $(\partial_1|_p, \ldots, \partial_n|_p)$ is a basis of T_pM and $G(p)$ is the matrix of the bilinear form g_p in that basis; the index of a symmetric form is a property of the form, computed in any basis. The entries $p \mapsto g_{ij}(p)$ are smooth, in particular continuous, so $p \mapsto G(p)$ is a continuous map from U into the space of symmetric $n \times n$ real matrices. By hypothesis $G(p)$ is invertible for every $p \in U$, so no eigenvalue of $G(p)$ is zero at any point. It suffices to show ν is constant on U; since every point of M lies in such a connected coordinate neighborhood, this establishes local constancy on M.

Step 2: the index counts negative eigenvalues, and none of them can cross zero. The index of a symmetric matrix equals the number of its negative eigenvalues, counted with multiplicity; this is one form of Sylvester's law of inertia, obtained by orthogonally diagonalizing the symmetric matrix and normalizing the basis. Fix a point $q \in U$ and let $\lambda_1(q) \le \lambda_2(q) \le \cdots \le \lambda_n(q)$ be the eigenvalues of $G(q)$, listed in increasing order with multiplicity; all are real because $G(q)$ is symmetric. The number of these that are negative is $\nu(q)$. Because $G(q)$ is invertible, 0 is never an eigenvalue, so each eigenvalue is either strictly positive or strictly negative at every point of U.

It remains to show that the count of negative eigenvalues cannot change as q moves within U. This follows from continuity of the eigenvalues together with the fact that none may pass through 0. Concretely, for a fixed $q \in U$ let

$$m = \min_{1 \le k \le n} |\lambda_k(q)| > 0,$$

the smallest absolute value of an eigenvalue of $G(q)$, which is positive since $G(q)$ is invertible. The characteristic polynomial $\det\big(G(p) - tI\big)$ has coefficients that are polynomials in the entries $g_{ij}(p)$, hence continuous functions of p; and its roots depend continuously on those coefficients, so the ordered eigenvalue functions $p \mapsto \lambda_k(p)$ are continuous on U (this continuity of ordered eigenvalues is a standard consequence of continuous dependence of roots on the coefficients of a real-rooted polynomial). Choose an open neighborhood $V \subseteq U$ of q small enough that

$$|\lambda_k(p) - \lambda_k(q)| < m \qquad \text{for all } p \in V \text{ and all } k = 1, \dots, n,$$

which is possible by continuity of each λ_k at q (take the intersection of the finitely many neighborhoods produced for the individual k). For $p \in V$ and each k: if $\lambda_k(q) > 0$ then $\lambda_k(q) \geq m$, so $\lambda_k(p) > \lambda_k(q) - m \geq 0$, whence $\lambda_k(p) > 0$; and if $\lambda_k(q) < 0$ then $\lambda_k(q) \leq -m$, so $\lambda_k(p) < \lambda_k(q) + m \leq 0$, whence $\lambda_k(p) < 0$. Thus each eigenvalue keeps its sign throughout V, and the number of negative eigenvalues is the same at p as at q. Therefore $\nu(p) = \nu(q)$ for all $p \in V$, so ν is constant on a neighborhood of q.

Step 3: conclusion. Since $q \in U$ was arbitrary, ν is locally constant on U, and U is connected, so ν is constant on U. As $p_0 \in M$ was arbitrary and each point admits such a connected coordinate neighborhood, ν is locally constant on all of M. By the opening remark, a locally constant integer-valued function on the connected manifold M is constant. Hence the index of g is a single integer, the signature is well defined, and no separate constancy hypothesis is needed on a connected manifold, as we sought to show.

The proof isolates why the signature is a robust invariant rather than an accident of a chosen basis or a chosen point: the index is an integer, and integers cannot vary continuously except by staying fixed, so the only way for the signature to change would be for an eigenvalue of the component matrix to pass through zero. Nondegeneracy forbids exactly that, at every point simultaneously, and connectedness ties the local values together. This is also the precise sense in which a Lorentzian metric can never degenerate into a Riemannian one across a connected region: the transition would require a null eigenvalue somewhere, contradicting nondegeneracy. The signature, once fixed, is a permanent feature of a connected (pseudo-)Riemannian manifold, and it is the single discrete datum distinguishing the Riemannian theory of the following sections from its Lorentzian counterpart.

1.2 Existence of Metrics

A Riemannian metric is a piece of geometric data laid on top of a smooth manifold, and nothing in the definition of a smooth manifold forces such data to exist. It is reasonable to ask whether the definition in definition 1.1.1 is ever vacuous: is there a smooth manifold so twisted that no smooth, symmetric, positive-definite 2-tensor field can be placed on it? The answer is no, and the reason is the mildest possible one. The obstruction that might have appeared, a failure to fit local choices of inner product together into one globally smooth field, is dissolved by the partition of unity, which is available on every smooth manifold that is Hausdorff and second countable. Every smooth manifold carries a Riemannian metric.

This should be compared against the situation for pseudo-Riemannian metrics, where existence can fail. The difference is the arithmetic of positive-definiteness: a positive combination of positive-definite forms is again positive-definite, so the gluing that averages local metrics never destroys the property that makes the result a metric. Nondegeneracy of a fixed index enjoys no such stability, and the failure is a topological obstruction. This section proves the existence theorem in the Riemannian case, isolates the algebraic fact that powers it, and then records, with its topological reason, why the Lorentzian case is different.

Throughout, M is a smooth manifold of dimension n, meaning Hausdorff, second countable, and C^∞. The Einstein summation convention (sum over a repeated upper and lower index) is in force.

The existence proof rests on a single algebraic observation, promoted to a statement about tensor fields. Two symmetric 2-tensors at a point can be added, and a symmetric 2-tensor can be scaled by a nonnegative number; the question is whether positive-definiteness survives these operations. It does, and with room to spare: as long as at least one summand contributes at a given point, the sum is already positive-definite there.

Proposition 1.2.1: Convex Combinations of Metrics

Let M be a smooth manifold and let $\{\rho_\alpha\}_{\alpha\in A}$ be a smooth partition of unity on M, so that each $\rho_\alpha\colon M\to\mathbb{R}$ is smooth with $\rho_\alpha\geq 0$, the supports $\{\operatorname{supp}\rho_\alpha\}$ form a locally finite family, and $\sum_{\alpha\in A}\rho_\alpha\equiv 1$. For each α let $V_\alpha\subseteq M$ be an open set with $\operatorname{supp}\rho_\alpha\subseteq V_\alpha$, and let g_α be a Riemannian metric on V_α. Suppose that for each α the product $\rho_\alpha g_\alpha$, extended by 0 outside $\operatorname{supp}\rho_\alpha$, is a smooth symmetric 2-tensor field on M. Then

$$g \;=\; \sum_{\alpha\in A}\rho_\alpha\, g_\alpha$$

is a Riemannian metric on M.

***Proof*:**
Step 1: the sum is a well-defined smooth symmetric 2*-tensor field.* Fix $p\in M$. Local finiteness of $\{\operatorname{supp}\rho_\alpha\}$ gives an open neighborhood U of p meeting only finitely many of the supports, say those indexed by $\alpha_1,\dots,\alpha_m$. On U every other term $\rho_\alpha g_\alpha$ vanishes identically, so on U the sum defining g is the finite sum $\rho_{\alpha_1}g_{\alpha_1}+\cdots+\rho_{\alpha_m}g_{\alpha_m}$ of smooth symmetric 2-tensor fields. A finite sum of smooth symmetric 2-tensor fields is a smooth symmetric 2-tensor field, so g restricts to one on U. Since such neighborhoods U cover M and the local expressions agree on overlaps (they are all the same locally finite sum), g is a globally defined smooth symmetric 2-tensor field. By construction g is symmetric at each point, because each g_α is.

Step 2: positive-definiteness. Fix $p\in M$ and a nonzero vector $v\in T_pM$. Here the value of the term $\rho_\alpha g_\alpha$ at p means its value as the globally defined tensor field of the hypothesis, namely $\rho_\alpha(p)\,g_{\alpha,p}$ when $p\in V_\alpha$ and 0 otherwise. Evaluate:

$$g_p(v,v) \;=\; \sum_{\alpha\in A}(\rho_\alpha g_\alpha)_p(v,v),$$

a finite sum by local finiteness. Consider a single index α. If $\rho_\alpha(p)=0$ the term $(\rho_\alpha g_\alpha)_p(v,v)$ vanishes. If $\rho_\alpha(p)>0$ then $p\in\operatorname{supp}\rho_\alpha\subseteq V_\alpha$, so $g_{\alpha,p}$ is defined; it is positive-definite, whence $g_{\alpha,p}(v,v)>0$, and the term equals $\rho_\alpha(p)\,g_{\alpha,p}(v,v)>0$. In either case the term is nonnegative, so $g_p(v,v)\geq 0$. It remains to exclude equality. Because $\sum_\alpha\rho_\alpha(p)=1$, at least one index β has $\rho_\beta(p)>0$. For that index $p\in V_\beta$ and $\rho_\beta(p)\,g_{\beta,p}(v,v)>0$, being a product of two strictly positive numbers, so the whole sum is strictly positive:

$$g_p(v,v) \;\geq\; \rho_\beta(p)\,g_{\beta,p}(v,v) \;>\; 0$$

Thus g_p is positive-definite for every p, and g is a Riemannian metric, as we sought to show.

The mechanism is worth stating in words, since it is exactly what fails for indefinite metrics. Positive-definiteness is a *closed cone* condition that is preserved under nonnegative combination:

the positive-definite symmetric bilinear forms on a fixed vector space form a convex cone, and the weights $\rho_\alpha(p)$ pick out a convex combination of points of that cone (after normalizing by the value $\sum \rho_\alpha(p) = 1$). A convex combination of points of a convex set lies in that set, and the strict conclusion above records that the combination cannot slip to the boundary as long as some weight is positive. No hypothesis links the g_α to one another; they may be wildly different metrics on the overlaps of their supports, and the sum is still a metric. This freedom is what lets the metrics on separate charts be chosen with no coordination at all.

A concrete instance makes the averaging visible. Take $M = \mathbb{R}$, covered by two overlapping intervals, and let $g_1 = dx^2$ and $g_2 = 4\,dx^2$ be two metrics on $\mathbb{R}$ (each a positive multiple of the standard one). Choose a partition of unity ρ_1, ρ_2 with $\rho_1 + \rho_2 \equiv 1$ and $0 \le \rho_1 \le 1$; for instance ρ_1 a smooth function increasing from 0 to 1 and $\rho_2 = 1 - \rho_1$. Then

$$g \;=\; \rho_1\, dx^2 + \rho_2\,(4\,dx^2) \;=\; \big(\rho_1 + 4\rho_2\big)\, dx^2 \;=\; \big(4 - 3\rho_1\big)\, dx^2$$

The coefficient $4 - 3\rho_1$ ranges over $[1, 4]$, so it is bounded below by 1 and never vanishes; g is a metric interpolating smoothly between the two scales, agreeing with dx^2 where $\rho_1 = 1$ and with $4\,dx^2$ where $\rho_1 = 0$. At every point at least one weight is positive, and the coefficient stays positive, which is proposition 1.2.1 in miniature.

With the algebra in hand, the existence theorem is a matter of choosing local metrics on chart domains and letting the partition of unity assemble them. The only inputs are the flat metric read off in each chart and the gluing guaranteed on any smooth manifold.

Theorem 1.2.2: Existence of Riemannian Metrics

Every smooth manifold M admits a Riemannian metric.

***Proof*:**
Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ be a smooth atlas for M, with each $\varphi_\alpha \colon U_\alpha \to \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$ a diffeomorphism onto an open subset of $\mathbb{R}^n$. The family $\{U_\alpha\}$ is an open cover of M.

Step 1: a metric on each chart domain. Let $\bar g = \delta_{ij}\, dx^i\, dx^j$ denote the Euclidean metric on $\mathbb{R}^n$, the standard inner product $\bar g_x(v, w) = \sum_{i=1}^n v^i w^i$ at each point x. On the chart domain U_α define

$$g_\alpha \;=\; \varphi_\alpha^*\, \bar g,$$

the pullback of $\bar g$ under φ_α (pullback of a covariant tensor field along a smooth map is treated in [6]). This g_α is a smooth symmetric 2-tensor field on U_α, being a pullback of one. It is positive-definite: for $p \in U_\alpha$ and nonzero $v \in T_pM$,

$$g_{\alpha,p}(v, v) \;=\; \bar g_{\varphi_\alpha(p)}\big(d(\varphi_\alpha)_p v,\, d(\varphi_\alpha)_p v\big) \;=\; \big|d(\varphi_\alpha)_p v\big|^2,$$

the squared Euclidean length of $d(\varphi_\alpha)_p v$. Since φ_α is a diffeomorphism its differential $d(\varphi_\alpha)_p$ is a linear isomorphism, so $d(\varphi_\alpha)_p v \ne 0$ whenever $v \ne 0$, and the squared length is strictly positive. Thus g_α is a Riemannian metric on the open submanifold U_α.

Step 2: gluing by a partition of unity. Since M is a smooth manifold (Hausdorff and second countable), there exists a smooth partition of unity $\{\rho_\alpha\}_{\alpha \in A}$ subordinate to the cover $\{U_\alpha\}$: each $\rho_\alpha \colon M \to \mathbb{R}$ is smooth with $\rho_\alpha \ge 0$ and $\operatorname{supp} \rho_\alpha \subseteq U_\alpha$, the family of supports is locally finite, and $\sum_{\alpha \in A} \rho_\alpha \equiv 1$ (see [6] for the construction of a partition of unity subordinate to an open cover). For each α the product $\rho_\alpha g_\alpha$ is defined and smooth on U_α, and it extends by 0 to a smooth symmetric 2-tensor field on all of M: on the open set $M \setminus \operatorname{supp} \rho_\alpha$ the factor ρ_α vanishes together with all its derivatives, so the extension is smooth across the boundary of $\operatorname{supp} \rho_\alpha$, and the two definitions agree on the overlap $U_\alpha \setminus \operatorname{supp} \rho_\alpha$, where both are 0.

Step 3: assembly. Take $V_\alpha = U_\alpha$ in proposition 1.2.1: each g_α is a Riemannian metric on the open set $U_\alpha \supseteq \operatorname{supp} \rho_\alpha$, the family $\{\rho_\alpha\}$ is a partition of unity, and the tensor fields $\rho_\alpha g_\alpha$, extended by 0 from Step 2, are globally defined, smooth, and symmetric. The hypotheses of that proposition are met, and it gives that

$$g \;=\; \sum_{\alpha \in A} \rho_\alpha \, g_\alpha$$

is a Riemannian metric on M. Hence M admits a Riemannian metric, completing the proof.

The proof extracts geometry from the flat model and spreads it over M with no compatibility conditions to check, because the convex-cone argument absorbs whatever discrepancies the charts introduce. What it does *not* produce is any particular metric: the output depends on the atlas and on the choice of partition of unity, and two such choices generally give metrics that are not isometric. The theorem asserts the existence of geometric structure, not its uniqueness or canonicity. A single manifold typically carries an infinite- dimensional space of Riemannian metrics, and much of the subject is the study of how the geometry, curvature, geodesics, distance, varies as the metric varies within that space.

When a manifold arises as a quotient of one that already carries a metric, the existence theorem can be bypassed by a more economical construction that descends the metric through the quotient, and the result is canonical rather than arbitrary. The next example carries this out for real projective space, which inherits a metric from the round sphere and thereby acquires a specific Riemannian structure without any recourse to a partition of unity.

Example 1.2.3: The Round Metric Descends to $\mathbb{RP}^n$

Real projective space $\mathbb{RP}^n$ is the quotient of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ by the antipodal identification $x \sim -x$; write $\pi \colon S^n \to \mathbb{RP}^n$ for the quotient map, which sends x to the line $[x] = \{x, -x\}$. The existence theorem theorem 1.2.2 already guarantees that $\mathbb{RP}^n$, being a smooth manifold, admits some Riemannian metric. A specific and canonical one is obtained instead by pushing the round metric g_{S^n} of example 1.3.4 down through π, using that the antipodal map is a symmetry of the round sphere.

Let $a \colon S^n \to S^n$, $a(x) = -x$, be the antipodal map, the restriction to S^n of $-\mathrm{id}$ on $\mathbb{R}^{n+1}$. Its differential at x is again $-\mathrm{id}$, now as a map $T_x S^n \to T_{-x} S^n$ (both tangent spaces are the same hyperplane $x^\perp = (-x)^\perp \subseteq \mathbb{R}^{n+1}$). The round metric is the restriction of the Euclidean inner product to each tangent space, so for $u, v \in T_x S^n$,

$$(a^* g_{S^n})_x(u,v) \;=\; (g_{S^n})_{-x}\big(da_x u,\, da_x v\big) \;=\; \langle -u, -v \rangle \;=\; \langle u, v \rangle \;=\; (g_{S^n})_x(u,v)$$

Thus $a^* g_{S^n} = g_{S^n}$, so a is an isometry of (S^n, g_{S^n}) in the sense of definition 1.3.1.

This is exactly what is needed for the metric to descend. The quotient map π is a smooth covering with $\pi \circ a = \pi$, and $da_x \colon T_x S^n \to T_{-x} S^n$ is precisely the identification of the two tangent spaces lying over the point $[x] \in \mathbb{RP}^n$ under $d\pi$. To define a metric $\bar{g}$ on $\mathbb{RP}^n$, set, for $\bar{u}, \bar{v} \in T_{[x]} \mathbb{RP}^n$,

$$\bar{g}_{[x]}(\bar{u}, \bar{v}) \;=\; (g_{S^n})_x(u,v), \qquad u = (d\pi_x)^{-1} \bar{u}, \quad v = (d\pi_x)^{-1} \bar{v},$$

where $d\pi_x \colon T_x S^n \to T_{[x]} \mathbb{RP}^n$ is a linear isomorphism because π is a local diffeomorphism. The formula uses a choice of preimage x over $[x]$; the two preimages are x and $-x$, and independence of the choice is exactly the isometry property just verified. Working over $-x$ replaces u, v by $da_x u = -u$ and $da_x v = -v$ and evaluates g_{S^n} at $-x$, giving $(g_{S^n})_{-x}(-u, -v) = (g_{S^n})_x(u, v)$, the same number. Hence $\bar{g}_{[x]}$ is well defined independently of the representative.

It remains to see that $\bar{g}$ is a smooth Riemannian metric. Smoothness is local, and near any $[x]$ the covering π restricts to a diffeomorphism from a neighborhood $\widetilde{U}$ of x onto a neighborhood of $[x]$, with local inverse $s = \pi|_{\widetilde{U}}^{-1}$. On that neighborhood the definition reads $\bar{g} = s^* g_{S^n}$, a pullback of a smooth 2-tensor field along a smooth map, hence smooth. Symmetry of $\bar{g}_{[x]}$ is inherited from symmetry of $(g_{S^n})_x$. For positive definiteness, if $\bar{u} \neq 0$ then $u = (d\pi_x)^{-1}\bar{u} \neq 0$ since $d\pi_x$ is an isomorphism, so $\bar{g}_{[x]}(\bar{u}, \bar{u}) = (g_{S^n})_x(u, u) > 0$. Therefore $\bar{g}$ is a Riemannian metric on $\mathbb{RP}^n$, and $\pi\colon (S^n, g_{S^n}) \to (\mathbb{RP}^n, \bar{g})$ is by construction a local isometry.

The output is not the arbitrary metric the partition-of-unity proof would have produced. It is the unique metric making π a local isometry, and it records all the local geometry of the round sphere while forgetting only the global antipodal identification. The same argument descends a metric through any quotient by a group of isometries acting freely and properly, a theme developed when quotient and covering metrics are treated in their own right.

The role of second countability deserves a word, since it is the one global hypothesis doing work. It is what guarantees a partition of unity, and hence what makes the gluing legal. On a manifold that is smooth and Hausdorff but not second countable, such as the long line, the argument breaks at Step 2, and the existence of a Riemannian metric is no longer automatic. For the manifolds of this book, all second countable by convention, no such pathology arises.

Example 1.2.4: A Metric on the Sphere by Charts

The construction can be run explicitly on S^2, where its output can be compared against a metric obtained by other means. Cover S^2 by the two stereographic charts, φ_N from the north pole and φ_S from the south pole, each a diffeomorphism onto $\mathbb{R}^2$. Pulling back the Euclidean metric $\bar{g} = du^2 + dv^2$ on $\mathbb{R}^2$ gives two metrics $g_N = \varphi_N^* \bar{g}$ and $g_S = \varphi_S^* \bar{g}$ on the respective chart domains $S^2 \setminus \{\text{north pole}\}$ and $S^2 \setminus \{\text{south pole}\}$. Choosing a subordinate partition of unity ρ_N, ρ_S (with ρ_N supported away from the north pole and ρ_S away from the south pole), the theorem produces

$$g \;=\; \rho_N\, g_N + \rho_S\, g_S,$$

a Riemannian metric on all of S^2. This metric is positive-definite at every point by the argument above: near the north pole only g_S contributes, near the south pole only g_N, and in the equatorial band both are present with positive weight. There is no claim that g is the round metric of example 1.1.5; it is one metric among many, built to demonstrate existence, and its curvature and symmetry depend entirely on the arbitrary weights ρ_N, ρ_S. The round metric is singled out from this crowd only later, by its symmetry properties.

The existence theorem should not be misread as saying that *any* symmetric 2-tensor structure one might want is available. The positive- definite case is the fortunate one. Allowing the signature to be indefinite changes the answer, and the change is topological rather than a defect of the gluing method.

1.2.5 Remark: Pseudo-Riemannian metrics need not exist The existence theorem is special to the Riemannian case. A smooth manifold need not admit a pseudo-Riemannian metric of a prescribed index (definition 1.1.3); for Lorentzian metrics, signature $(-,+,\ldots,+)$, the obstruction is topological.

The partition-of-unity argument fails at exactly the step where proposition 1.2.1 was used. Pulling back a fixed Minkowski form (as in example 1.1.2) in each chart produces local Lorentzian metrics, but a convex combination $\sum \rho_\alpha g_\alpha$ of Lorentzian forms need not be Lorentzian: the nondegenerate symmetric forms of index 1 do not form a convex set. Averaging two Lorentzian inner products whose negative directions disagree can yield a degenerate or positive-definite form, so the sum can fail to be a metric of any signature at some points. There is nothing to glue the local timelike directions into a coherent global choice.

A coherent global choice of timelike direction is precisely a nowhere- vanishing line field on M, a smooth assignment $p \mapsto L_p$ of a 1-dimensional subspace $L_p \subseteq T_pM$, and this is where topology enters. A Lorentzian metric g on M determines such a line field, but not by the naive route of looking at unit timelike vectors: the set $\{v \in T_pM : g_p(v,v) = -1\}$ is a two-sheeted hyperboloid whose vectors point in a continuum of directions, so it singles out no line. The correct construction compares g against an auxiliary Riemannian metric h, which exists by theorem 1.2.2. At each p define the endomorphism $A_p : T_pM \to T_pM$ by

$$g_p(u,v) \;=\; h_p(A_pu,\, v) \qquad \text{for all } u, v \in T_pM$$

Since g_p and h_p are symmetric and h_p is positive-definite, A_p is self-adjoint with respect to h_p, hence diagonalizable with real eigenvalues in some h_p-orthonormal eigenbasis $e_1, \ldots, e_n$. In that basis $g_p(e_i,e_j) = h_p(A_pe_i,e_j) = \lambda_i\delta_{ij}$, where λ_i is the eigenvalue of e_i, so the signs of the eigenvalues of A_p are exactly the diagonal signs of g_p, and hence give its signature. As g has signature $(-,+,\ldots,+)$, the operator A_p has exactly one negative eigenvalue, of multiplicity 1, and $n-1$ positive ones. Let $L_p \subseteq T_pM$ be the eigenline for the negative eigenvalue. This line is well defined and varies smoothly with p (the negative eigenspace is cut out by the continuously varying spectral projection of the smoothly varying operator A), and every nonzero $v \in L_p$ satisfies $g_p(v,v) < 0$, so L is a smooth timelike line field. Conversely a smooth line field $L \subseteq TM$ together with any Riemannian metric h produces a Lorentzian metric by reversing the sign of h along L, that is, $g(v,v) = h(v,v)$ for $v \perp_h L$ and $g(v,v) = -h(v,v)$ for $v \in L$. So a Lorentzian metric exists on M if and only if M carries a smooth line field. On a closed manifold the existence of a nowhere-vanishing line field is governed by the Euler characteristic: a closed surface admits a line field, and therefore a Lorentzian metric, if and only if its Euler characteristic vanishes.

The sphere S^2 is the clean counterexample. Its Euler characteristic is $\chi(S^2) = 2 \neq 0$, so by the Poincaré–Hopf theorem every vector field on S^2 has a zero, and in fact S^2 carries no nowhere-vanishing line field either. Hence S^2 admits no Lorentzian metric, even though, by theorem 1.2.2, it admits a Riemannian one (for instance the round metric, or the chart-glued metric of example 1.2.4). The torus T^2, with $\chi(T^2) = 0$, does admit a Lorentzian metric. The vanishing of the Euler characteristic and the Poincaré–Hopf count of zeros of a vector field belong to [6]; the point here is only that Riemannian existence is universal while Lorentzian existence is obstructed, and the obstruction has a name and a computation.

1.3 Isometries and Model Spaces

A metric turns each tangent space into an inner-product space, and a smooth map between manifolds carries tangent vectors forward. The question that organizes this section is when such a map respects the two metrics: when the length of a vector, measured before applying the map, equals the length of its image, measured afterward. A map with this property is an isometry, and it is the correct notion of sameness for Riemannian manifolds. Two manifolds that are isometric are, from the standpoint of every metric quantity built in the rest of this book (length, angle, geodesic, curvature), indistinguishable.

Isometries also let us single out the manifolds that will serve as reference points for everything that follows. Just as linear algebra is anchored by $\mathbb{R}^n$ with its standard inner product, Riemannian geometry is anchored by a short list of highly symmetric spaces, the *model spaces*: Euclidean space, the round sphere, hyperbolic space, and the flat quotients of Euclidean space. What distinguishes these is the size of their isometry groups. Each is homogeneous (its isometry group acts transitively, so the geometry looks the same at every point) and isotropic (the isometry group can rotate any direction at a point into any other, so the geometry looks the same in every direction). The bulk of this section constructs these spaces and records the symmetries that make them models.

Definition 1.3.1: Isometry

Let (M, g) and $(\tilde{M}, \tilde{g})$ be (pseudo-)Riemannian manifolds. A diffeomorphism $\varphi\colon M \to \tilde{M}$ is an *isometry* if $\varphi^*\tilde{g} = g$; that is, for every $p \in M$ and all $v, w \in T_pM$,

$$\tilde{g}_{\varphi(p)}\big(d\varphi_p(v),\, d\varphi_p(w)\big) = g_p(v, w)$$

When such a φ exists, (M, g) and $(\tilde{M}, \tilde{g})$ are *isometric*. A smooth map $\varphi\colon M \to \tilde{M}$ is a *local isometry* if every point of M has a neighborhood U on which φ restricts to a diffeomorphism onto its image with $(\varphi|_U)^*\tilde{g} = g|_U$; equivalently, each $d\varphi_p$ is a linear isometry of the inner-product spaces (T_pM, g_p) and $(T_{\varphi(p)}\tilde{M}, \tilde{g}_{\varphi(p)})$.

The condition $\varphi^*\tilde{g} = g$ is pointwise and linear-algebraic: at each p it asks that the differential $d\varphi_p$, a linear map between two inner-product spaces, preserve inner products. A linear map that preserves inner products is injective, so a local isometry is automatically an immersion; when M and $\tilde{M}$ have the same dimension it is a local diffeomorphism by the inverse function theorem. The distinction between a local isometry and a (global) isometry is therefore exactly the distinction between a local diffeomorphism and a diffeomorphism: a local isometry matches the metrics on small pieces but need not be injective or surjective. The flat torus below is the standard illustration that the two notions differ. The pullback $\varphi^*\tilde{g}$ is the pullback of a covariant 2-tensor field in the sense of [6]; that $\varphi^*\tilde{g}$ is again a smooth symmetric covariant 2-tensor field is part of that pullback machinery, and $\varphi^*\tilde{g}$ is positive definite whenever $\tilde{g}$ is and φ is an immersion.

Being isometric is an equivalence relation on (pseudo-)Riemannian manifolds. The identity map is an isometry; if $\varphi\colon M \to \tilde{M}$ is an isometry then so is φ^{-1}, because $(\varphi^{-1})^*g = (\varphi^{-1})^*\varphi^*\tilde{g} = (\varphi \circ \varphi^{-1})^*\tilde{g} = \tilde{g}$; and a composite of isometries is an isometry, since $(\psi \circ \varphi)^*\hat{g} = \varphi^*\psi^*\hat{g} = \varphi^*\tilde{g} = g$. The same computation with $\tilde{M} = M$ and $\tilde{g} = g$ shows that the isometries of a single manifold to itself are closed under composition and inversion.

Example 1.3.2: The Euclidean group as isometries

Let $(\mathbb{R}^n, \bar{g})$ be Euclidean space with $\bar{g} = \delta_{ij}\, dx^i\, dx^j$, the flat model of example 1.1.2 (the Einstein summation convention, summing over each repeated upper/lower index pair, is in force throughout this section). Under the canonical identification $T_x\mathbb{R}^n \cong \mathbb{R}^n$ the metric $\bar{g}_x$ is the standard dot product at every point. A linear map $A \in \mathrm{GL}(n, \mathbb{R})$, viewed as the diffeomorphism $x \mapsto Ax$, has constant differential $dA_x = A$, so $A^*\bar{g} = \bar{g}$ holds iff A preserves the dot product, that is iff $A \in O(n)$. More generally, for a fixed $b \in \mathbb{R}^n$ the rigid motion $x \mapsto Ax + b$ has differential A at every point, so it is an isometry of $(\mathbb{R}^n, \bar{g})$ precisely when $A \in O(n)$. Thus every element of the Euclidean group $\mathbb{R}^n \rtimes O(n)$ is an isometry of Euclidean space. That these are *all* of the isometries requires knowing that an isometry is determined by its value and differential at one point, which follows from the geodesic machinery of Chapter 3; the containment $\mathbb{R}^n \rtimes O(n) \subseteq \mathrm{Isom}(\mathbb{R}^n, \bar{g})$ is what the direct computation gives here.

The isometries of a fixed manifold form a group, and this group is the primary invariant that separates the model spaces from generic Riemannian manifolds.

Definition 1.3.3: Isometry group

Let (M, g) be a (pseudo-)Riemannian manifold. The set of all isometries $\varphi \colon M \to M$, under composition, is a group, the *isometry group* $\mathrm{Isom}(M, g)$. Its identity is id_M and the inverse of an isometry is its inverse map, both of which are isometries by the computation above.

The group axioms hold because composition of maps is associative and, as recorded above, isometries are closed under composition and inversion and include the identity. The isometry group is a subgroup of the diffeomorphism group $\mathrm{Diff}(M)$. For a Riemannian manifold it carries more structure than that of an abstract group: a theorem of Myers and Steenrod, taken up in Chapter 9, gives $\mathrm{Isom}(M, g)$ the structure of a Lie group acting smoothly on M, with dimension bounded by $\frac{1}{2}n(n+1)$. That bound is attained by the constant-curvature model spaces built next, Euclidean space, the round sphere, and hyperbolic space, which is one precise sense in which they are maximally symmetric; the flat torus (example 1.3.6), homogeneous but not isotropic, does not attain it. Until Chapter 9, $\mathrm{Isom}(M, g)$ is used only as a group; the size of its orbits already records homogeneity and isotropy.

A manifold (M, g) is *homogeneous* if $\mathrm{Isom}(M, g)$ acts transitively on M, and *isotropic at p* if the isotropy subgroup $\{\varphi \in \mathrm{Isom}(M, g) : \varphi(p) = p\}$ acts transitively on the set of unit vectors in T_pM; it is *isotropic* if it is isotropic at every point. Euclidean space is homogeneous by the translations $x \mapsto x + b$ and isotropic at the origin by the rotations $A \in O(n)$, which act transitively on unit vectors; together with homogeneity this makes it isotropic at every point. The next three examples build the remaining model spaces and verify the same two properties for each.

Example 1.3.4: The round sphere as a model space

Fix $r > 0$ and let

$$S^n_r = \left\{ x \in \mathbb{R}^{n+1} \;\middle|\; |x| = r \right\}$$

be the sphere of radius r, carrying the metric g_r induced by the inclusion $\iota \colon S^n_r \hookrightarrow (\mathbb{R}^{n+1}, \bar{g})$; that is, $g_r = \iota^*\bar{g}$, the first look at which was example 1.1.5. Concretely, for $p \in S^n_r$ the tangent space is $T_pS^n_r = \left\{ v \in \mathbb{R}^{n+1} \;\middle|\; \langle v, p \rangle = 0 \right\}$, the orthogonal complement of the position vector, and g_r is the restriction of the Euclidean dot product to this hyperplane. This metric is positive definite because it is the restriction of a positive-definite form to a subspace, so (S^n_r, g_r) is a Riemannian manifold in the sense of definition 1.1.1.

Symmetries. The orthogonal group $O(n+1)$ acts on $\mathbb{R}^{n+1}$ by linear isometries of $\bar{g}$ and preserves $|x|$, hence maps S^n_r to itself. For $A \in O(n+1)$ the restriction $A|_{S^n_r} : S^n_r \to S^n_r$ is a diffeomorphism, and since $A^*\bar{g} = \bar{g}$ its restriction pulls $g_r = \iota^*\bar{g}$ back to itself: writing $\iota \circ (A|_{S^n_r}) = A \circ \iota$ gives $(A|_{S^n_r})^* g_r = (A|_{S^n_r})^*\iota^*\bar{g} = (A \circ \iota)^*\bar{g} = \iota^* A^*\bar{g} = \iota^*\bar{g} = g_r$. Thus $O(n+1) \subseteq \operatorname{Isom}(S^n_r, g_r)$.

Homogeneity. Given $p, q \in S^n_r$, both have norm r, so p/r and q/r are unit vectors; extend each to an orthonormal basis and let $A \in O(n+1)$ be the orthogonal map sending the first basis to the second. Then $A(p) = q$, so $O(n+1)$ acts transitively and S^n_r is homogeneous.

Isotropy. Fix $p \in S^n_r$ and let $v, w \in T_pS^n_r$ be unit vectors, so $\{p/r, v\}$ and $\{p/r, w\}$ are each orthonormal. Extend to orthonormal bases of $\mathbb{R}^{n+1}$ agreeing in the first slot p/r; the orthogonal map A carrying one basis to the other fixes p and sends v to w. Its differential at p is A itself (a linear map is its own differential), so A lies in the isotropy subgroup at p and carries v to w. Hence S^n_r is isotropic.

Homogeneity and isotropy together force the sectional curvature to be constant, a statement made precise in Chapter 4; there the value for (S^n_r, g_r) is computed to be $\sec \equiv 1/r^2$, positive and shrinking as the radius grows. The round metric, its geodesics (great circles), and this curvature value recur as the constant-positive-curvature benchmark throughout the book.

The sphere is the model of constant positive curvature. Its constant-negative-curvature counterpart is hyperbolic space, which does not sit inside Euclidean space as a metric submanifold but does sit inside Minkowski space, where the indefinite ambient form restricts to a positive-definite metric on the appropriate hypersurface.

Example 1.3.5: Hyperbolic space and its models

Work in Minkowski space $\mathbb{R}^{1,n}$ of example 1.1.2, with coordinates $x = (x^0, x^1, \dots, x^n)$ and the Lorentzian form

$$m(x,y) = -x^0y^0 + \sum_{i=1}^{n} x^iy^i,$$

a pseudo-Riemannian metric of signature $(-,+,\dots,+)$ in the sense of definition 1.1.3. Define the *hyperboloid model*

$$H^n = \left\{x \in \mathbb{R}^{1,n} \;\middle|\; m(x,x) = -1,\ x^0 > 0\right\},$$

the upper sheet of the two-sheeted hyperboloid $-(x^0)^2 + \sum(x^i)^2 = -1$. This is a smooth hypersurface: -1 is a regular value of $f(x) = m(x,x)$ since $df_x(v) = 2m(x,v)$ is nonzero for $x \neq 0$. Its tangent space at p is $T_pH^n = \left\{v \in \mathbb{R}^{1,n} \;\middle|\; m(p,v) = 0\right\} = p^\perp$, the m-orthogonal complement of p.

The induced metric is Riemannian. Fix $p \in H^n$. Because $m(p,p) = -1 < 0$, the vector p spans a timelike line, and Minkowski space splits m-orthogonally as $\mathbb{R}p \oplus p^\perp$. On the timelike line $\mathbb{R}p$ the form m is negative definite; since m has index 1, its restriction to the complement $p^\perp$ must be positive definite. Thus $g_H := m|_{p^\perp}$ is a positive-definite inner product on each T_pH^n, and (H^n, g_H) is a Riemannian manifold. This is the mechanism by which an indefinite ambient form produces a genuine Riemannian metric on a spacelike hypersurface.

Symmetries. Let $O(1,n)$ be the group of linear maps preserving m, and let $O^+(1,n)$ be the subgroup preserving the condition $x^0 > 0$ on the set $\{m(x,x) = -1\}$ (the time-orientation-preserving Lorentz transformations). Each $A \in O^+(1,n)$ maps H^n to itself, and exactly as in the spherical case $(A|_{H^n})^*g_H = g_H$ because $A^*m = m$. Transitivity of $O^+(1,n)$ on H^n and on unit tangent vectors is proved by the same orthonormalization argument as for the sphere, now using

m-orthonormal frames (a timelike unit vector along p together with a spacelike orthonormal basis of $p^\perp$); the analog of extending to matching frames goes through because any two such frames are related by an element of $O^+(1,n)$. Hence H^n is homogeneous and isotropic. As with the sphere, this forces constant sectional curvature, computed in Chapter 4 to be $\sec \equiv -1$, negative.

The ball model. Stereographic projection of the hyperboloid from the point $(-1, 0, \ldots, 0)$ onto the hyperplane $\{x^0 = 0\}$ carries H^n diffeomorphically onto the open unit ball $B^n = \{u \in \mathbb{R}^n \mid |u| < 1\}$. Its inverse is parametrized by

$$x^0 = \frac{1+|u|^2}{1-|u|^2}, \qquad x^i = \frac{2u^i}{1-|u|^2} \quad (1 \le i \le n),$$

which one checks satisfies $m(x,x) = -1$ and $x^0 > 0$. Pulling $g_H = m$ back along this parametrization gives the *Poincaré ball metric*

$$g_B = \frac{4}{(1-|u|^2)^2}\,\bar{g}, \tag{1.1}$$

where $\bar{g} = \delta_{ij}\,du^i\,du^j$ is the Euclidean metric on the ball. Direct computation confirms (1.1): differentiating the parametrization and forming $m\big(\partial_a x, \partial_b x\big)$ yields $\frac{4}{(1-|u|^2)^2}\delta_{ab}$, with the cross terms vanishing. Thus (B^n, g_B), a conformal rescaling of the Euclidean metric that blows up at the boundary sphere, is isometric to the hyperboloid model.

The half-space model. The *upper half-space* is $U^n = \{y \in \mathbb{R}^n \mid y^n > 0\}$ with the metric

$$g_U = \frac{1}{(y^n)^2}\,\bar{g} = \frac{1}{(y^n)^2}\big((dy^1)^2 + \cdots + (dy^n)^2\big)$$

For $n = 2$ the map $z \mapsto i\dfrac{1+z}{1-z}$ carries the unit disc B^2 (coordinate $z = u^1 + iu^2$) biholomorphically onto the upper half-plane $U^2 = \{w \in \mathbb{C} \mid \mathrm{Im} w > 0\}$; a conformal map $w = F(z)$ pulls $g_U = (\mathrm{Im} w)^{-2}\,|dw|^2$ back to $|F'(z)|^2\,(\mathrm{Im} F(z))^{-2}\,|dz|^2$, and this equals the ball metric $\frac{4}{(1-|z|^2)^2}\,|dz|^2$ of (1.1) by direct substitution. So in dimension 2 the ball and half-plane models are isometric; the upper half-plane as a complex-analytic object is treated in [5]. In every dimension the analogous inversion in a boundary sphere provides an isometry $B^n \to U^n$, so all three models, hyperboloid, ball, and half-space, present the same Riemannian manifold. This is the origin of the phrase *hyperbolic space*: it names one manifold, presented by whichever model is convenient.

Homogeneity is compatible with a small isometry group in one important way: a manifold can be homogeneous under a discrete group of translations rather than a continuous one, and then it can carry curvature-zero geometry without being Euclidean space. The flat torus is the basic example, and it is also the example that separates local isometry from global isometry.

Example 1.3.6: The flat torus

Let $\mathbb{Z}^n \subset \mathbb{R}^n$ be the integer lattice, acting on $\mathbb{R}^n$ by translations $x \mapsto x + k$ for $k \in \mathbb{Z}^n$. The action is free and properly discontinuous, so the quotient $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ is a smooth compact n-manifold and the projection $\pi\colon \mathbb{R}^n \to \mathbb{T}^n$ is a smooth covering map. Each translation $\tau_k(x) = x + k$ is an isometry of $(\mathbb{R}^n, \bar{g})$ (it is the rigid motion with $A = \mathrm{id}$ in example 1.3.2), so the Euclidean metric is invariant under the group action and therefore descends to a metric on the quotient. Precisely: near any point of $\mathbb{T}^n$ the covering π restricts to a diffeomorphism from an open set

$\tilde{U} \subset \mathbb{R}^n$, and the local formula $g_{\text{flat}} := (\pi|_{\tilde{U}})_*(\bar{g}|_{\tilde{U}})$ is independent of the chosen sheet $\tilde{U}$ because any two sheets differ by a translation τ_k, which preserves $\bar{g}$. This defines the *flat torus* $(\mathbb{T}^n, g_{\text{flat}})$, and by construction $\pi\colon (\mathbb{R}^n, \bar{g}) \to (\mathbb{T}^n, g_{\text{flat}})$ satisfies $\pi^* g_{\text{flat}} = \bar{g}$.

π is a local isometry. The identity $\pi^* g_{\text{flat}} = \bar{g}$ says exactly that each $d\pi_x$ is a linear isometry $(T_x\mathbb{R}^n, \bar{g}_x) \to (T_{\pi(x)}\mathbb{T}^n, g_{\text{flat}})$, and π is a local diffeomorphism because it is a covering map. So π is a local isometry in the sense of definition 1.3.1, and the flat torus is *locally isometric* to Euclidean space: every point of $\mathbb{T}^n$ has a neighborhood carrying the same metric as an open set of $\mathbb{R}^n$. In particular the flat torus is flat, with the same constant zero curvature as $\mathbb{R}^n$ once curvature is defined in Chapter 4.

$\mathbb{T}^n$ is not globally isometric to $\mathbb{R}^n$. There is no isometry $\mathbb{T}^n \to \mathbb{R}^n$, and indeed no diffeomorphism, because $\mathbb{T}^n$ is compact while $\mathbb{R}^n$ is not. A local isometry between manifolds of equal dimension therefore need not be a global isometry: π matches the metrics on every small piece yet is far from injective, since $\pi(x) = \pi(x')$ whenever $x - x' \in \mathbb{Z}^n$. This is the promised separation of the two notions from definition 1.3.1.

Homogeneity, and non-uniqueness of the flat structure. The torus is homogeneous: the translations of $\mathbb{R}^n$ descend to a transitive action of $\mathbb{T}^n$ on itself by isometries (the group $\mathbb{T}^n$ acting on the manifold $\mathbb{T}^n$), since τ_b commutes with the lattice action and so passes to the quotient. It is not isotropic, however: the isometries just described are the identity on each tangent space up to lattice symmetries, and the geometry does record a preferred set of directions, those of the lattice. Replacing $\mathbb{Z}^n$ by a different lattice $\Lambda \subset \mathbb{R}^n$ produces another flat torus $\mathbb{R}^n/\Lambda$, and two such tori are isometric iff the lattices are related by an element of $O(n)$; distinct lattice shapes give non-isometric flat tori, all of them flat and all locally isometric to $\mathbb{R}^n$. The flat model space is thus not a single manifold but a family, distinguished by the metric data of the lattice rather than by curvature.

1.4 Musical Isomorphisms, Gradient, Divergence, and Volume

A metric does more than measure lengths and angles of tangent vectors. Because g_p is a nondegenerate bilinear form on each tangent space T_pM, it identifies that space with its dual T_p^*M, and this identification is the mechanism by which the metric reaches into every tensor bundle over M. On a bare smooth manifold there is no canonical way to turn a covector into a vector: the pairing $T_pM \times T_p^*M \to \mathbb{R}$ is available, but converting a functional ω into a vector requires a choice, and the tangent and cotangent bundles, though isomorphic, are not canonically so. A metric supplies exactly that choice, uniformly and smoothly across the manifold. Once vectors and covectors are interchangeable, quantities defined most directly in one bundle can be transported to the other: the differential df of a function is a covector field, but the metric lets one read it as a vector field, the gradient, whose direction is that of steepest ascent measured by g.

The same nondegeneracy that identifies TM with T^*M also singles out, on an oriented manifold, a preferred top-degree form. In Euclidean space one integrates functions against $dx^1 \wedge \cdots \wedge dx^n$ without comment; on a Riemannian manifold the analogous object is the volume form dV_g, and its coordinate expression $\sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$ carries the metric explicitly. With a volume form in hand one can integrate functions, and the failure of a flow to preserve dV_g becomes a scalar, the divergence of the generating vector field. This section assembles these four constructions, all of them consequences of nondegeneracy, and records the one computation, invariance of dV_g under change of chart, that makes the volume form well defined. Throughout, (M, g) is a (pseudo-)Riemannian manifold of dimension n, the Einstein summation convention is in force (a repeated index appearing once up and once down is summed from 1 to n), and the tensor-bundle and differential-form machinery is taken from [6].

Definition 1.4.1: Musical isomorphisms

Let (M, g) be a (pseudo-)Riemannian manifold. The *flat* operator is the bundle map

$$\flat\colon TM \to T^*M, \qquad v^\flat := g(v, -) = \langle v, - \rangle\,,$$

sending a tangent vector $v \in T_pM$ to the covector $w \mapsto \langle v, w\rangle$ on T_pM. Because g_p is nondegenerate, $\flat$ is a fiberwise linear isomorphism; its inverse is the *sharp* operator

$$\sharp\colon T^*M \to TM, \qquad \omega^\sharp := (\flat)^{-1}(\omega),$$

characterized by $\langle \omega^\sharp, w\rangle = \omega(w)$ for all $w \in T_pM$. In a chart $(U, (x^i))$, writing $g = g_{ij}\, dx^i\, dx^j$ and (g^{ij}) for the inverse of the matrix (g_{ij}), a vector $v = v^i\, \partial_i$ has

$$v^\flat = v_j\, dx^j, \qquad v_j = g_{ij} v^i,$$

and a covector $\omega = \omega_j\, dx^j$ has

$$\omega^\sharp = \omega^i\, \partial_i, \qquad \omega^i = g^{ij}\omega_j$$

The passages $v^i \mapsto v_j = g_{ij}v^i$ and $\omega_j \mapsto \omega^i = g^{ij}\omega_j$ are *lowering* and *raising* an index. More generally, for a tensor of type (k, l) the same contractions with g_{ij} or g^{ij} lower or raise any chosen slot, and (g^{ij}) is the *inverse metric*, the induced (pseudo-)Riemannian inner product g^{-1} on T^*M determined by $\langle \alpha, \beta\rangle_{g^{-1}} = g^{ij}\alpha_i\beta_j$.

The names are a mnemonic borrowed from musical notation: lowering an index (flat) pushes a symbol down, raising it (sharp) pushes it up. The content behind the notation is a single fact, nondegeneracy of g_p, which is exactly the statement that the linear map $v \mapsto g_p(v, -)$ from T_pM to T_p^*M has trivial kernel and hence, by equality of dimensions, is an isomorphism. In the Riemannian case positive-definiteness gives nondegeneracy for free; in the pseudo-Riemannian case nondegeneracy is imposed by definition precisely so that the musical isomorphisms exist. The matrix identity behind the coordinate formulas is $g^{ik}g_{kj} = \delta^i_j$, so raising an index one has just lowered returns the original object: $\sharp$ and $\flat$ are mutually inverse, as the notation $\sharp = \flat^{-1}$ demands. That g^{ij} defines an inner product on covectors, and not just a matrix of functions, is the assertion that these operations are geometric rather than coordinate artifacts; the pairing $\langle \alpha, \beta\rangle_{g^{-1}}$ is exactly the one for which $\flat$ and $\sharp$ are linear isometries between (T_pM, g_p) and (T_p^*M, g_p^{-1}).

Example 1.4.2: Musical isomorphisms in Euclidean and Minkowski space

On Euclidean space $(\mathbb{R}^n, \bar{g})$ with $\bar{g} = \delta_{ij}\, dx^i\, dx^j$ (see example 1.1.2), the matrix (g_{ij}) is the identity, so $g_{ij} = g^{ij} = \delta_{ij}$ and raising or lowering an index changes nothing numerically: for $v = v^i\, \partial_i$ one has $v^\flat = v^i\, dx^i$ and, for $\omega = \omega_i\, dx^i$, $\omega^\sharp = \omega_i\, \partial_i$. This is why in elementary vector calculus the distinction between a vector field and a 1-form is invisible; the Euclidean metric silently identifies them.

On Minkowski space $\mathbb{R}^{1,n}$ with $g = -(dx^0)^2 + \delta_{ab}\, dx^a\, dx^b$ (indices a, b running over $1, \ldots, n$) the identification is no longer inert. Here $(g_{ij}) = \operatorname{diag}(-1, 1, \ldots, 1)$ equals its own inverse, so for $v = v^0\, \partial_0 + v^a\, \partial_a$ one computes

$$v^\flat = -\, v^0\, dx^0 + v^a\, dx^a,$$

the timelike component acquiring a sign under lowering. Thus a timelike vector and the covector it corresponds to point in opposite temporal directions, a feature with no Euclidean analogue and the source of many sign conventions in relativity.

Among all covector fields on M, one arises on every smooth manifold without reference to a metric: the differential df of a smooth function f, defined by $df(X) = Xf$. It is precisely the object waiting to be sharpened.

Definition 1.4.3: Gradient

Let (M, g) be a (pseudo-)Riemannian manifold and $f \in C^\infty(M)$. The *gradient* of f is the vector field

$$\operatorname{grad} f := (df)^\sharp,$$

the sharp of the differential df. Equivalently, $\operatorname{grad} f$ is the unique vector field characterized by

$$\langle \operatorname{grad} f, X\rangle = df(X) = Xf \qquad \text{for every vector field } X$$

In a chart $(U, (x^i))$,

$$\operatorname{grad} f = g^{ij}\, \partial_i f\, \partial_j$$

The characterizing identity $\langle \operatorname{grad} f, X\rangle = Xf$ says that pairing the gradient against any direction X recovers the rate of change of f in that direction. The direction of $\operatorname{grad} f$ is therefore the one in which this pairing is largest for unit vectors, that is, the direction of steepest ascent as measured by the metric, and its magnitude $|\operatorname{grad} f|_g$ is that maximal rate. Because the identification uses g, the gradient is a metric-dependent object even though df is not: changing the metric rotates and rescales $\operatorname{grad} f$ while leaving df untouched. The coordinate formula makes this dependence explicit through the factor g^{ij}, and reduces, on Euclidean space where $g^{ij} = \delta^{ij}$, to the familiar $\operatorname{grad} f = \sum_i (\partial_i f)\, \partial_i$ of vector calculus.

Example 1.4.4: Gradient on the round sphere in stereographic coordinates

Take the round sphere S^2 and a stereographic chart in which the induced metric (whose full derivation belongs to example 1.1.5) has the conformal form $g = \lambda(x, y)^2\, (dx^2 + dy^2)$ for a smooth positive function λ; concretely, stereographic projection of the unit sphere from the north pole gives $\lambda(x, y) = 2/(1 + x^2 + y^2)$. Then $(g_{ij}) = \lambda^2 I$, so $(g^{ij}) = \lambda^{-2} I$, and for $f \in C^\infty(S^2)$

$$\operatorname{grad} f = \lambda^{-2}\big(\partial_x f\, \partial_x + \partial_y f\, \partial_y\big)$$

The gradient points in the same coordinate direction as the Euclidean gradient of f, but its length is scaled by λ^{-1}: near the north pole, where λ is small, a fixed coordinate slope of f corresponds to a large metric gradient, reflecting that stereographic coordinates compress large regions of the sphere into a small coordinate patch. On the flat model $(\mathbb{R}^2, \bar{g})$, where $\lambda \equiv 1$, the formula collapses to the ordinary gradient.

Turning from tensors to integration, the metric selects a preferred top-degree form. The construction requires an orientation to produce a genuine n-form; on a nonorientable manifold the same local data assembles instead into a density, against which functions can still be integrated. The distinction is recorded below and the orientable case is treated in full.

Definition 1.4.5: Riemannian volume form

Let (M, g) be an oriented (pseudo-)Riemannian manifold of dimension n. The *Riemannian volume form* dV_g is the unique smooth n-form on M that, in every oriented local chart $(U, (x^i))$ (one for which $dx^1 \wedge \cdots \wedge dx^n$ is positively oriented), is given by

$$dV_g = \sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n$$

In the Riemannian case $\det(g_{ij}) > 0$ and the absolute value is unnecessary; in the pseudo-Riemannian case the determinant may be negative and $|\det(g_{ij})|$ is intended. Equivalently, dV_g is the unique positively oriented n-form assigning the value 1 to every positively oriented g-orthonormal frame. If M is not orientable, the same local expression $\sqrt{|\det(g_{ij})|}\, \left|dx^1 \wedge \cdots \wedge dx^n\right|$ defines instead the *Riemannian density*, against which functions are integrated in the sense of [6]; the local computations below apply verbatim.

The characterization by orthonormal frames explains why the awkward-looking factor $\sqrt{|\det(g_{ij})|}$ is the right one. An oriented g-orthonormal coframe $(e^1, \ldots, e^n)$ has $e^1 \wedge \cdots \wedge e^n$ deserving the name "unit volume", so dV_g is decreed to send the dual frame to 1; expressing this dual frame in terms of $\partial_1, \ldots, \partial_n$ produces exactly the Jacobian-type factor $\sqrt{|\det(g_{ij})|}$, since the change-of-basis matrix from an orthonormal frame to the coordinate frame has determinant of absolute value $\sqrt{|\det(g_{ij})|}$. The form is what one integrates against: for $f \in C^\infty(M)$ with suitable support, $\int_M f\, dV_g$ generalizes the Riemann integral, and $\int_M dV_g$ is the total volume of M when M is compact. That the local formula does not depend on the chart, which is what entitles these local expressions to define a single global form, is the content of proposition 1.4.9 below.

Example 1.4.6: Volume forms of Euclidean space and the round sphere

On $(\mathbb{R}^n, \bar{g})$ the matrix (g_{ij}) is the identity, so $\det(g_{ij}) = 1$ and

$$dV_{\bar{g}} = dx^1 \wedge \cdots \wedge dx^n,$$

the standard volume form, against which integration reproduces ordinary Lebesgue integration on $\mathbb{R}^n$ (see [9]).

For the round sphere S^2 in a conformal chart $g = \lambda^2\,(dx^2 + dy^2)$ as in example 1.4.4, one has $(g_{ij}) = \lambda^2 I$, hence $\det(g_{ij}) = \lambda^4$ and

$$dV_g = \sqrt{\lambda^4}\, dx \wedge dy = \lambda^2\, dx \wedge dy$$

With $\lambda = 2/(1+x^2+y^2)$ this is $dV_g = \frac{4}{(1+x^2+y^2)^2}\, dx \wedge dy$, and integrating over the whole plane (the image of stereographic projection) recovers the area 4π of the unit sphere, as $\int_{\mathbb{R}^2} \frac{4}{(1+x^2+y^2)^2}\, dx\, dy = 4\pi$.

The volume form also converts a first-order differential property of a vector field into a scalar. A vector field X generates a flow, and one asks whether that flow expands or contracts volume. The infinitesimal answer is the Lie derivative of dV_g along X, which, being an n-form on an n-manifold, is a function multiple of dV_g.

Definition 1.4.7: Divergence

Let (M, g) be an oriented (pseudo-)Riemannian manifold and X a smooth vector field on M. The *divergence* $\mathrm{div}X \in C^\infty(M)$ is the unique smooth function defined by

$$\mathcal{L}_X(dV_g) = (\mathrm{div}X)\, dV_g,$$

where $\mathcal{L}_X$ denotes the Lie derivative along X (see [6]). Equivalently, by Cartan's formula $\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$ and $d(dV_g) = 0$ (as dV_g is a top form),

$$d(\iota_X\, dV_g) = (\mathrm{div}X)\, dV_g$$

On a nonorientable manifold the same identity, read against the Riemannian density, defines $\mathrm{div}X$; the local formula below is unaffected.

Because $\mathcal{L}_X(dV_g)$ is an n-form on an n-dimensional manifold, and dV_g is nowhere zero, the ratio $\mathcal{L}_X(dV_g)/dV_g$ is a well-defined smooth function, and this ratio is by definition $\mathrm{div}X$. The geometric reading is exact rather than metaphorical: if Φ_t is the flow of X, then $\frac{d}{dt}\big|_{t=0}\Phi_t^*(dV_g) = \mathcal{L}_X(dV_g) = (\mathrm{div}X)\, dV_g$, so $\mathrm{div}X$ measures the instantaneous logarithmic rate at which the flow of X changes volume, positive where the flow spreads volume out and negative where it concentrates volume. The definition uses only the volume form, hence only the metric through dV_g, and makes no reference to a connection; the coordinate expression below and the divergence theorem both follow from this alone. The connection-based formula $\mathrm{div}X = \mathrm{tr}(\nabla X)$ and its agreement with the present definition belong to the development of the Levi-Civita connection and are not needed here.

Example 1.4.8: Divergence in coordinates and on Euclidean space

Let $X = X^i\,\partial_i$ in an oriented chart, write $|g| := |\det(g_{ij})|$, and abbreviate $dx := dx^1 \wedge \cdots \wedge dx^n$, so that $dV_g = \sqrt{|g|}\, dx$. Contracting,

$$\iota_X\, dV_g = \sqrt{|g|}\ \iota_X\, dx = \sqrt{|g|}\ \sum_{i=1}^{n}(-1)^{i-1}X^i\, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n,$$

where the hat denotes omission. Applying the exterior derivative, only the ∂_i term of each summand survives the wedge, and with the sign $(-1)^{i-1}$ it lands back on dx:

$$d(\iota_X\, dV_g) = \sum_{i=1}^{n} \partial_i\big(\sqrt{|g|}\, X^i\big)\, dx = \frac{1}{\sqrt{|g|}}\, \partial_i\big(\sqrt{|g|}\, X^i\big)\ dV_g$$

Comparing with $d(\iota_X\, dV_g) = (\mathrm{div}X)\, dV_g$ gives the coordinate formula

$$\mathrm{div}X = \frac{1}{\sqrt{|g|}}\, \partial_i\big(\sqrt{|g|}\, X^i\big)$$

On $(\mathbb{R}^n, \bar{g})$, where $|g| \equiv 1$, this reduces to $\mathrm{div}X = \partial_i X^i = \sum_i \partial X^i/\partial x^i$, the classical divergence. Combining the gradient formula of definition 1.4.3 with this expression yields the *Laplace–Beltrami operator* $\Delta f := \mathrm{div}(\mathrm{grad}\, f) = \frac{1}{\sqrt{|g|}}\, \partial_i\big(\sqrt{|g|}\, g^{ij}\, \partial_j f\big)$, the metric generalization of the Laplacian.

It remains to justify the definition of dV_g: the local formula must be independent of the oriented chart used to write it, so that the local n-forms agree on overlaps and glue to a single global form. This is the one computation the section owes.

Proposition 1.4.9: Coordinate-independence of the volume form

Let (M, g) be an oriented (pseudo-)Riemannian manifold of dimension n. On the overlap of two oriented charts $(U, (x^i))$ and $(V, (y^\alpha))$, the local n-forms

$$\sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n \qquad \text{and} \qquad \sqrt{|\det(g_{\alpha\beta})|}\, dy^1 \wedge \cdots \wedge dy^n$$

coincide, where (g_{ij}) and $(g_{\alpha\beta})$ are the components of g in the two charts. Consequently the local expression of definition 1.4.5 defines a single smooth n-form dV_g on M.

***Proof*:**
Let $J = (\partial y^\alpha / \partial x^i)$ denote the Jacobian matrix of the transition map $y = y(x)$ on $U \cap V$, with entries $J^\alpha{}_i := \partial y^\alpha / \partial x^i$. Since both charts are positively oriented for the given orientation of M, the transition map is orientation-preserving, so

$$\det J > 0$$

Step 1: transformation of the metric components. The metric is a covariant 2-tensor, so its components transform by

$$g_{ij} = \frac{\partial y^\alpha}{\partial x^i} \frac{\partial y^\beta}{\partial x^j}\, g_{\alpha\beta} = J^\alpha{}_i\, J^\beta{}_j\, g_{\alpha\beta}$$

In matrix form, writing $G_x := (g_{ij})$ for the component matrix in the x-chart and $G_y := (g_{\alpha\beta})$ for that in the y-chart,

$$G_x = J^\mathsf{T}\, G_y\, J$$

Taking determinants and using multiplicativity together with $\det(J^\mathsf{T}) = \det J$,

$$\det(g_{ij}) = (\det J)^2\, \det(g_{\alpha\beta})$$

Because $(\det J)^2 > 0$, the sign of the determinant is the same in both charts, consistent with the constancy of the signature of g on the connected overlap, and taking absolute values and square roots,

$$\sqrt{|\det(g_{ij})|} = |\det J|\, \sqrt{|\det(g_{\alpha\beta})|} = (\det J)\, \sqrt{|\det(g_{\alpha\beta})|},$$

the last equality using $\det J > 0$.

Step 2: transformation of the coordinate n-form. The top-degree coordinate forms are related by the same Jacobian determinant. From $dy^\alpha = J^\alpha{}_i\, dx^i$ and the antisymmetry of the wedge product,

$$dy^1 \wedge \cdots \wedge dy^n = \det\!\left(\tfrac{\partial y^\alpha}{\partial x^i}\right) dx^1 \wedge \cdots \wedge dx^n = (\det J)\, dx^1 \wedge \cdots \wedge dx^n$$

This is the standard identity that the pullback of the volume element under a coordinate change multiplies by the Jacobian determinant; it holds for any smooth change of coordinates.

Step 3: assembling the two transformations. Multiply the two displayed relations. Using Step 2 followed by Step 1,

$$\sqrt{|\det(g_{\alpha\beta})|}\, dy^1 \wedge \cdots \wedge dy^n = \sqrt{|\det(g_{\alpha\beta})|}\, (\det J)\, dx^1 \wedge \cdots \wedge dx^n$$

By Step 1 the coefficient $(\det J)\sqrt{|\det(g_{\alpha\beta})|}$ equals $\sqrt{|\det(g_{ij})|}$, so the right-hand side is

$$\sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n$$

Thus the two local n-forms agree on $U \cap V$.

Since the local expressions coincide on every overlap of oriented charts, they patch together to a single well-defined n-form on M; it is smooth because on each chart it equals $\sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n$ with $\sqrt{|\det(g_{ij})|}$ a smooth nowhere-vanishing function (the determinant of the invertible matrix (g_{ij}) is nowhere zero and depends smoothly on the point). This is the asserted dV_g, completing the proof.

The orientation hypothesis entered at exactly one place: the sign $\det J > 0$, which let $|\det J|$ be dropped in favor of $\det J$ so that the two positively signed local forms could be matched. Without an orientation the coordinate n-forms $dx^1 \wedge \cdots \wedge dx^n$ transform with $\det J$, which can be negative, and no consistent choice of sign is available globally; this is precisely why the nonorientable case yields a density rather than a form, the density absorbing the sign into $\left|dx^1 \wedge \cdots \wedge dx^n\right|$ and transforming by $|\det J|$ instead. In either case the coefficient $\sqrt{|\det(g_{ij})|}$ is intrinsic, and the object it defines, form or density, is the geometric measure attached to g against which all integration over (M, g) is carried out.

1.5 Induced Metrics and Submanifolds

So far a Riemannian metric has been a structure placed on a manifold by hand: chosen, or shown to exist, or transported by an isometry. But the oldest source of metrics is more direct. A surface sitting in $\mathbb{R}^3$ inherits a way of measuring the length of tangent vectors from the ambient dot product, and this inherited measurement is what one actually computes with when doing geometry on the surface. The length of a curve on the sphere, the angle at which two great circles cross, the area of a spherical triangle: all of these are read off from the restriction of the Euclidean inner product to the tangent planes of the sphere. This is the mechanism by which classical differential geometry, the geometry of curves and surfaces in space, produced its metrics, and it is the mechanism by which most concrete Riemannian manifolds are still built today.

The construction generalizes with no change to any immersion into any (pseudo-)Riemannian manifold. Given a map $\iota\colon M \to \tilde{M}$ whose differential is injective at every point, and a metric $\tilde{g}$ on the target, one restricts $\tilde{g}$ to the image of each tangent space $d\iota_p(T_pM)$ and pulls the result back to T_pM. The outcome is a covariant 2-tensor on M, the pullback $\iota^*\tilde{g}$, and the point of this section is that when $\tilde{g}$ is Riemannian the injectivity of $d\iota$ is exactly what forces $\iota^*\tilde{g}$ to be positive definite, hence a Riemannian metric in its own right. This turns every immersed submanifold of a Riemannian manifold into a Riemannian manifold. The two running examples, the round sphere and the surfaces of revolution, are computed from this definition here; the sphere computation confirms that the metric obtained by restriction agrees with the round metric of example 1.3.4, and the surfaces of revolution supply the warped form $dr^2 + f(r)^2\,d\theta^2$ that later chapters use as a laboratory for geodesics and curvature. The Einstein summation convention, summation over each index that appears once raised and once lowered, is in force throughout.

Recall from [6] the pullback of a covariant tensor field. If $F\colon M \to \tilde{M}$ is smooth and A is a covariant k-tensor field on $\tilde{M}$, the pullback F^*A is the covariant k-tensor field on M defined pointwise by

$$(F^*A)_p(v_1,\dots,v_k) = A_{F(p)}\big(dF_p(v_1),\dots,dF_p(v_k)\big), \qquad v_1,\dots,v_k \in T_pM$$

For a metric $\tilde{g}$, which is a covariant 2-tensor, this reads $(F^*\tilde{g})_p(v,w) = \tilde{g}_{F(p)}(dF_pv,\, dF_pw)$. Pullback is smooth (it takes smooth tensor fields to smooth tensor fields), is functorial ($(F\circ G)^* = G^*\circ F^*$), and commutes with the symmetrization that makes a 2-tensor symmetric. These facts are established in [6] and are used freely below.

Definition 1.5.1: Induced metric and Riemannian submanifold

Let $(\tilde{M},\tilde{g})$ be a (pseudo-)Riemannian manifold and let $\iota\colon M \to \tilde{M}$ be an immersion, that is, a smooth map whose differential $d\iota_p\colon T_pM \to T_{\iota(p)}\tilde{M}$ is injective for every $p \in M$. The *induced metric* (or *pullback metric*) is the covariant symmetric 2-tensor field

$$\iota^*\tilde{g}$$

on M, given at each point by $(\iota^*\tilde{g})_p(v,w) = \tilde{g}_{\iota(p)}(d\iota_pv,\, d\iota_pw)$ for $v,w \in T_pM$. When $\iota^*\tilde{g}$ is a Riemannian metric, the pair $(M,\iota^*\tilde{g})$ is a *Riemannian submanifold* of $(\tilde{M},\tilde{g})$, and ι is an *isometric immersion*. If M is an embedded submanifold of $\tilde{M}$ and ι is the inclusion, $\iota^*\tilde{g}$ is the restriction of $\tilde{g}$ to tangent vectors of M, and one writes $\tilde{g}|_M$.
In the classical setting of a surface in $\mathbb{R}^3$, the induced metric is called the *first fundamental form* of the surface.

The name "first fundamental form" predates the language of tensors, and it fixes the perspective from which the object is read: it is the datum that records the intrinsic geometry of the surface, everything about the surface that a two-dimensional inhabitant could measure without reference to the surrounding space. Two surfaces with the same first fundamental form (in suitable coordinates) are indistinguishable to such an inhabitant, even if they sit in space quite differently; the plane and the cylinder are the standard illustration, and the isometry between them is worked out in example 1.5.4 below. The qualifier "first" anticipates a second fundamental form, which records instead how the surface curves inside the ambient space and is therefore extrinsic; that object belongs to the theory of submanifold curvature and is not needed here.

Two points of the definition deserve a word before any example. First, the immersion hypothesis is on the differential, not on the map ι itself: ι need not be injective, so its image may cross itself, yet the induced metric is still perfectly well defined because it is computed tangent space by tangent space. A figure-eight traced out by an immersion of the circle carries an induced metric on the circle even though the image has a self-crossing. Second, positive-definiteness of the induced metric is not automatic; it is a consequence of the immersion condition together with positive-definiteness of $\tilde{g}$, and this is precisely what the next proposition proves. Dropping the immersion hypothesis breaks it: the constant map has $d\iota_p = 0$, so $\iota^*\tilde{g} = 0$, which is not a metric.

Proposition 1.5.2: Immersions induce metrics

Let $(\tilde{M}, \tilde{g})$ be a Riemannian manifold and $\iota\colon M \to \tilde{M}$ an immersion. Then the induced metric $\iota^*\tilde{g}$ is a Riemannian metric on M. More generally, if $\tilde{g}$ is pseudo-Riemannian then $\iota^*\tilde{g}$ is a smooth symmetric 2-tensor field on M, and it is nondegenerate at p if and only if the subspace $d\iota_p(T_pM)$ is nondegenerate for $\tilde{g}$; in the Riemannian case this last condition holds at every point.

***Proof*:**
Step 1: smoothness and symmetry. The pullback of a smooth covariant 2-tensor field under a smooth map is again a smooth covariant 2-tensor field, by [6], so $\iota^*\tilde{g}$ is a smooth 2-tensor field on M. Symmetry is inherited pointwise: for $v, w \in T_pM$,

$$(\iota^*\tilde{g})_p(v, w) = \tilde{g}_{\iota(p)}(d\iota_p v,\, d\iota_p w) = \tilde{g}_{\iota(p)}(d\iota_p w,\, d\iota_p v) = (\iota^*\tilde{g})_p(w, v),$$

the middle equality being the symmetry of $\tilde{g}$. Thus $\iota^*\tilde{g}$ is a smooth symmetric covariant 2-tensor field, which is the type required of a (pseudo-)Riemannian metric by definition 1.1.1.

Step 2: the Riemannian case is positive definite. Fix $p \in M$ and a nonzero $v \in T_pM$. Because ι is an immersion, $d\iota_p$ is injective, so $d\iota_p v \neq 0$. Since $\tilde{g}$ is a Riemannian metric, $\tilde{g}_{\iota(p)}$ is positive definite on $T_{\iota(p)}\tilde{M}$, whence

$$(\iota^*\tilde{g})_p(v, v) = \tilde{g}_{\iota(p)}(d\iota_p v,\, d\iota_p v) > 0$$

For $v = 0$ both sides vanish. Therefore $(\iota^*\tilde{g})_p$ is positive definite on T_pM for every p, and $\iota^*\tilde{g}$ is a Riemannian metric.

Step 3: the pseudo-Riemannian case. Now let $\tilde{g}$ be pseudo-Riemannian. Step 1 shows $\iota^*\tilde{g}$ is a smooth symmetric 2-tensor field. Fix p and set $W = d\iota_p(T_pM)$, a linear subspace of $T_{\iota(p)}\tilde{M}$ of dimension $\dim M$, the equality of dimensions holding because $d\iota_p$ is injective. The map $d\iota_p\colon T_pM \to W$ is a linear isomorphism, and for $v, w \in T_pM$

$$(\iota^*\tilde{g})_p(v, w) = \tilde{g}_{\iota(p)}\big(d\iota_p v,\, d\iota_p w\big),$$

so $(\iota^*\tilde{g})_p$ is the pullback under this isomorphism of the bilinear form $\tilde{g}_{\iota(p)}|_{W\times W}$. A pullback of a bilinear form under a linear isomorphism is nondegenerate if and only if the form itself is nondegenerate: if $\varphi\colon V \to W$ is a linear isomorphism and b is a bilinear form on W, then $\varphi^*b(v, \cdot)$ is the zero functional on V exactly when $b(\varphi v, \cdot)$ is the zero functional on W, since φ is surjective, so the radical of φ^*b is φ^{-1} of the radical of b. Hence $(\iota^*\tilde{g})_p$ is nondegenerate if and only if $\tilde{g}_{\iota(p)}|_{W\times W}$ is, that is, if and only if W is a nondegenerate subspace of $(T_{\iota(p)}\tilde{M}, \tilde{g}_{\iota(p)})$. In the Riemannian case every subspace is nondegenerate, because the restriction of a positive-definite form to any subspace is again positive definite, hence nondegenerate; this recovers Step 2 and completes the proof.

The pseudo-Riemannian caveat in Step 3 is not a technicality to be skipped, and one concrete case makes the danger visible. In Minkowski space $\mathbb{R}^{1,3}$ with $\tilde{g} = -dt^2 + dx^2 + dy^2 + dz^2$, the plane spanned by ∂_t and ∂_x at a point is nondegenerate and inherits a Lorentzian metric, whereas the line spanned by the null vector $\partial_t + \partial_x$ inherits the zero form, since $\tilde{g}(\partial_t + \partial_x, \partial_t + \partial_x) = -1 + 1 = 0$. An immersion whose image is tangent to such a null direction fails to induce a pseudo-Riemannian metric there, even though it is a perfectly good immersion. Positive-definiteness of $\tilde{g}$ removes this possibility outright, which is why the Riemannian statement is clean and the pseudo-Riemannian statement carries a hypothesis. For the remainder of the section $\tilde{g}$ is Riemannian.

The computational content of an induced metric lives in coordinates. Suppose $\iota\colon M \to \tilde{M}$ is an immersion, $(x^1, \dots, x^n)$ are local coordinates on M, and $(y^1, \dots, y^{\tilde{n}})$ are local coordinates on $\tilde{M}$ near $\iota(p)$, in which $\tilde{g} = \tilde{g}_{ab}\, dy^a\, dy^b$. Writing the map in coordinates as $y^a = \iota^a(x^1, \dots, x^n)$, the differential sends $\partial/\partial x^i$ to $(\partial\iota^a/\partial x^i)\, \partial/\partial y^a$, so the components of the induced metric are

$$(\iota^*\tilde{g})_{ij} = \tilde{g}_{ab}\, \frac{\partial\iota^a}{\partial x^i}\, \frac{\partial\iota^b}{\partial x^j} \tag{1.2}$$

When $\tilde{M} = \mathbb{R}^{\tilde{n}}$ with the Euclidean metric, $\tilde{g}_{ab} = \delta_{ab}$, and (1.2) collapses to the dot product of the coordinate partials of ι, viewed as vectors in $\mathbb{R}^{\tilde{n}}$. This is the formula behind every classical first fundamental form, and it is the one applied in the two examples that follow.

Example 1.5.3: The induced metric on the sphere

Let $S^n = \{\, x \in \mathbb{R}^{n+1} : |x| = 1 \,\}$ with the inclusion $\iota\colon S^n \hookrightarrow \mathbb{R}^{n+1}$, and give $\mathbb{R}^{n+1}$ the Euclidean metric $\bar{g} = \delta_{ab}\, dy^a\, dy^b$. The inclusion is an embedding, in particular an immersion, so proposition 1.5.2 applies and $\iota^*\bar{g}$ is a Riemannian metric on S^n. The claim is that it equals the round metric described in example 1.3.4.

Work on the upper hemisphere $\{y^{n+1} > 0\}$ with the graph coordinates $(x^1, \dots, x^n)$, $|x| < 1$, under which

$$\iota(x) = \Big(x^1, \dots, x^n,\ \sqrt{1 - |x|^2}\Big), \qquad |x|^2 = (x^1)^2 + \dots + (x^n)^2$$

The other coordinate patches are treated identically. Then $\iota^a = x^a$ for $a \le n$ and $\iota^{n+1} = \sqrt{1 - |x|^2}$, so

$$\frac{\partial\iota^a}{\partial x^i} = \delta^a_i \quad (a \le n), \qquad \frac{\partial\iota^{n+1}}{\partial x^i} = \frac{-x^i}{\sqrt{1 - |x|^2}}$$

Substituting into (1.2) with $\tilde{g}_{ab} = \delta_{ab}$ gives

$$(\iota^*\bar{g})_{ij} = \sum_{a=1}^{n} \delta^a_i\, \delta^a_j + \frac{\partial\iota^{n+1}}{\partial x^i}\frac{\partial\iota^{n+1}}{\partial x^j} = \delta_{ij} + \frac{x^i x^j}{1 - |x|^2},$$

so the induced metric in these coordinates is

$$\iota^*\bar{g} = \delta_{ij}\, dx^i\, dx^j + \frac{(x^i\, dx^i)(x^j\, dx^j)}{1 - |x|^2} = |dx|^2 + \frac{(x \cdot dx)^2}{1 - |x|^2}, \tag{1.3}$$

where $|dx|^2 = \sum_i (dx^i)^2$ and $x \cdot dx = \sum_i x^i\, dx^i$.

To see that (1.3) is the round metric of example 1.3.4, evaluate the length it assigns to a curve and match it against the length measured in the ambient space. Let $c(t)$ be a curve on S^n and write it in the ambient coordinates as $\gamma(t) = \iota(c(t)) \in \mathbb{R}^{n+1}$. By the definition of the pullback, the $\iota^*\bar{g}$-length of the velocity of c equals the Euclidean length of the velocity of γ:

$$|\dot{c}(t)|^2_{\iota^*\bar{g}} = (\iota^*\bar{g})_{c(t)}\big(\dot{c}(t), \dot{c}(t)\big) = \bar{g}_{\gamma(t)}\big(d\iota\, \dot{c}(t),\, d\iota\, \dot{c}(t)\big) = |\dot{\gamma}(t)|^2$$

Thus the intrinsic length of any curve on S^n, measured with $\iota^*\bar{g}$, coincides with its length as a curve in $\mathbb{R}^{n+1}$, which is exactly the prescription defining the round metric as the restriction of $\bar{g}$ to tangent vectors of the sphere in example 1.3.4. A symmetric 2-tensor is determined by the quadratic form $v \mapsto g(v,v)$ it induces on tangent vectors, by polarization, and the two tensors induce the same quadratic form on every T_cS^n; hence $\iota^*\bar{g}$ equals the round metric, and the inclusion of the round sphere into Euclidean space is an isometric immersion.

The sphere computation is the prototype: a submanifold is described by a parametrization or a graph, the ambient inner product is restricted, and the resulting form is the metric one uses for all intrinsic measurements. The identity $|\dot{c}|_{\iota^*\bar{g}} = |\dot{\gamma}|$ derived above holds for every isometric immersion, not only for the sphere, and it is the reason the induced metric is the correct object: it is the unique metric on M for which ι preserves the lengths of tangent vectors, so that arc length computed upstairs on M agrees with arc length of the image curve downstairs. In the language of definition 1.3.1, an isometric immersion is a map that is a local isometry onto its image equipped with the induced metric, and the sphere inclusion realizes the abstract round metric of example 1.3.4 concretely as such an image.

The second running example produces the warped normal form that later chapters return to, and it introduces the class of surfaces of revolution.

Example 1.5.4: Surfaces of revolution

Let a *profile curve* in the xz-plane be given by $r \mapsto (\varphi(r), 0, \psi(r))$ for r in an interval I, with $\varphi(r) > 0$ and $(\varphi'(r), \psi'(r)) \neq (0,0)$ for all r; the condition $\varphi > 0$ keeps the curve off the axis of rotation. The associated *surface of revolution* $\Sigma \subset \mathbb{R}^3$ is swept out by rotating this curve about the z-axis, and is parametrized by

$$\iota(r,\theta) = \big(\varphi(r)\cos\theta,\ \varphi(r)\sin\theta,\ \psi(r)\big), \qquad r \in I,\ \theta \in \mathbb{R},$$

where θ is taken modulo 2π. Here r measures position along the profile and θ measures the angle of rotation.

This parametrization is an immersion. Its coordinate partials are

$$\frac{\partial \iota}{\partial r} = (\varphi'\cos\theta,\ \varphi'\sin\theta,\ \psi'), \qquad \frac{\partial \iota}{\partial \theta} = (-\varphi\sin\theta,\ \varphi\cos\theta,\ 0),$$

which are linearly independent at every point. The second is nonzero because $\varphi > 0$. The first is nonzero because $(\varphi', \psi') \neq (0,0)$, and it is not a scalar multiple of the second: the second vector is purely azimuthal, orthogonal to the radial direction $(\cos\theta, \sin\theta, 0)$ and to the z-axis, whereas the first has radial component φ' and z-component ψ', at least one of which is nonzero. Hence $d\iota$ has rank 2 everywhere and proposition 1.5.2 applies.

Compute the induced metric from (1.2) with $\bar{g}_{ab} = \delta_{ab}$, so that each component is the Euclidean dot product of two coordinate partials. Direct calculation gives

$$\begin{aligned}
(\iota^*\bar{g})_{rr} &= \frac{\partial \iota}{\partial r} \cdot \frac{\partial \iota}{\partial r} = \varphi'^2\cos^2\theta + \varphi'^2\sin^2\theta + \psi'^2 = \varphi'^2 + \psi'^2, \\
(\iota^*\bar{g})_{r\theta} &= \frac{\partial \iota}{\partial r} \cdot \frac{\partial \iota}{\partial \theta} = -\varphi'\varphi\cos\theta\sin\theta + \varphi'\varphi\sin\theta\cos\theta + 0 = 0, \\
(\iota^*\bar{g})_{\theta\theta} &= \frac{\partial \iota}{\partial \theta} \cdot \frac{\partial \iota}{\partial \theta} = \varphi^2\sin^2\theta + \varphi^2\cos^2\theta = \varphi^2
\end{aligned}$$

Therefore

$$\iota^*\bar{g} = (\varphi'^2 + \psi'^2)\, dr^2 + \varphi^2\, d\theta^2 \tag{1.4}$$

The vanishing of the cross term $(\iota^*\bar{g})_{r\theta}$ records that the meridians (θ constant) meet the parallels (r constant) orthogonally, a reflection of the rotational symmetry of the surface.

Two normalizations simplify (1.4). If the profile curve is parametrized by arc length, so that $\varphi'^2 + \psi'^2 \equiv 1$, the metric takes the *warped form*

$$\iota^*\bar{g} = dr^2 + f(r)^2\, d\theta^2, \qquad f(r) := \varphi(r),$$

in which the coefficient of dr^2 is constant and all the geometry is carried by the single warping function $f = \varphi$, the distance from a point of the surface to the axis. This is the form that reappears in the study of geodesics and curvature on rotationally symmetric surfaces.

The warped form specializes to two familiar surfaces at once. Taking $\varphi(r) = 1$ and $\psi(r) = r$ gives the right circular cylinder of radius 1, with induced metric $dr^2 + d\theta^2$; this is the flat Euclidean metric on the strip $\mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z})$, so the cylinder is locally isometric to the plane, illustrating the remark after definition 1.5.1 that the first fundamental form is an intrinsic datum blind to how the surface bends in space. Taking instead $\varphi(r) = \sin r$ and $\psi(r) = \cos r$ for $r \in (0, \pi)$, so that the profile is a unit semicircle parametrized by arc length, gives the unit sphere minus its poles, with induced metric

$$\iota^*\bar{g} = dr^2 + \sin^2 r\, d\theta^2,$$

the round metric in geodesic polar form, where r is the angle from the north pole. This is (1.3) rewritten in latitude and longitude, and it agrees with the round metric of example 1.3.4.

1.6 Warped Products

The metrics built so far have been either primitive, chosen or shown to exist on a single manifold, or inherited, pulled back along an immersion. Warped products are the first constructive recipe: they take two Riemannian manifolds and a positive function on the first, and assemble from these data a metric on the product of the two. The construction is simple to state and yet it accounts for a large fraction of the concrete metrics in the subject. Euclidean space written in polar coordinates, the round sphere in latitude and longitude, hyperbolic space in its several models, every surface of revolution, and the spatial slices of the standard cosmological spacetimes are all warped products, and recognizing a metric as a warped product is often the first step toward computing anything about it.

The idea begins with the ordinary product. If (M, g_M) and (N, g_N) are Riemannian manifolds, their product $M \times N$ carries a metric that measures the base direction with g_M and the fiber direction with g_N, with the two directions declared orthogonal. This product metric is the geometry of a manifold that is, at every point, a rigid orthogonal join of its two factors. A warped product loosens this join in one controlled way: the fiber metric g_N is rescaled by the square of a positive function f on the base, so that the size of the fiber over a point of M depends on where that point sits. Nothing else changes, the base metric and the orthogonality are kept, but allowing the fiber to shrink and grow is already enough to bend a flat product into a sphere or a hyperbolic space. The present section constructs both metrics at the level of the metric tensor and verifies that the warped-product tensor is a Riemannian metric; the Levi-Civita connection and the curvature of a warped product, which are where the bending becomes quantitative, are developed once the connection (Chapter 2) and the curvature tensor (Chapter 4) are available.

Throughout, $\pi_M \colon M \times N \to M$ and $\pi_N \colon M \times N \to N$ denote the two projections of the product manifold, both smooth submersions. The Einstein summation convention, summation over each index appearing once raised and once lowered, is in force.

Definition 1.6.1: Product metric

Let (M, g_M) and (N, g_N) be (pseudo-)Riemannian manifolds with projections $\pi_M \colon M \times N \to M$ and $\pi_N \colon M \times N \to N$. The *product metric* on $M \times N$ is the covariant 2-tensor field

$$g = \pi_M^* g_M + \pi_N^* g_N$$

Explicitly, at a point $(p, q) \in M \times N$ and for tangent vectors $u, v \in T_{(p,q)}(M \times N)$,

$$g_{(p,q)}(u, v) = (g_M)_p\big(d\pi_M\, u,\, d\pi_M\, v\big) + (g_N)_q\big(d\pi_N\, u,\, d\pi_N\, v\big)$$

The (pseudo-)Riemannian manifold $(M \times N,\, g)$ is the *Riemannian product* of (M, g_M) and (N, g_N), written $M \times N$ or $(M \times N,\, g_M \oplus g_N)$.

The canonical isomorphism $T_{(p,q)}(M \times N) \cong T_pM \oplus T_qN$ turns the two projection differentials into the two coordinate projections of this direct sum, and under this identification the product metric is the orthogonal direct sum of the two inner products. A vector tangent to the base slice $M \times \{q\}$ has $d\pi_N\, u = 0$ and is measured purely by g_M; a vector tangent to the fiber $\{p\} \times N$ has $d\pi_M\, u = 0$ and is measured purely by g_N; and the mixed term $g(u, v)$ between a base vector and a fiber vector vanishes, because one of the two differentials annihilates each of them. Thus the tangent space splits g-orthogonally into a horizontal part tangent to M and a vertical part tangent to N, and the product metric is exactly the metric for which this splitting is orthogonal and each summand keeps its own geometry. That $\pi_M^* g_M + \pi_N^* g_N$ is a Riemannian metric when both factors are is the special case $f \equiv 1$ of proposition 1.6.3 below, so its verification is folded into that proof.

The product metric records a manifold whose two factors do not interact: the geometry along M is the same over every point of N and conversely. A cylinder $S^1 \times \mathbb{R}$ with the product of the standard circle and line metrics is the picture to keep in mind, a fiber circle of the same radius sitting over every point of the line. The warped product breaks this uniformity in precisely one place, by letting the radius of the fiber vary over the base.

Definition 1.6.2: Warped product

Let (M, g_M) and (N, g_N) be (pseudo-)Riemannian manifolds and let $f \colon M \to (0, \infty)$ be a smooth positive function, the *warping function*. The *warped product* $M \times_f N$ is the product manifold $M \times N$ equipped with the covariant 2-tensor field

$$g = \pi_M^* g_M + (f \circ \pi_M)^2\, \pi_N^* g_N$$

Explicitly, at $(p, q) \in M \times N$ and for $u, v \in T_{(p,q)}(M \times N)$,

$$g_{(p,q)}(u, v) = (g_M)_p\big(d\pi_M\, u,\, d\pi_M\, v\big) + f(p)^2\, (g_N)_q\big(d\pi_N\, u,\, d\pi_N\, v\big)$$

The manifold (M, g_M) is the *base*, (N, g_N) the *fiber*, and g is written $g_M \oplus f^2 g_N$. The product metric of definition 1.6.1 is the case $f \equiv 1$.

Under the splitting $T_{(p,q)}(M \times N) \cong T_pM \oplus T_qN$ the warped metric is $(g_M)_p \oplus f(p)^2 (g_N)_q$: the base part is untouched, the base and fiber remain orthogonal, and the fiber inner product is scaled by the positive factor $f(p)^2$. Scaling an inner product by a positive constant scales every length it measures by the square root of that constant, so over the point $p \in M$ the fiber $\{p\} \times N$ carries the metric $f(p)^2 g_N$, a copy of (N, g_N) magnified by the factor $f(p)$. The warping function therefore controls the size of the fiber as it slides over the base, and the whole content of the construction is that this one positive function is allowed to vary. Where f is small the fiber is a small copy of N; where f is large the fiber is a large copy; and where f is constant the warped product is locally a Riemannian product. The square on f is a convention that makes the length scale, rather than the squared length, equal to f: a fiber curve of g_N-length ℓ over p has warped length $f(p)\,\ell$.

The metric g is manifestly a smooth symmetric covariant 2-tensor field, being a sum of pullbacks of such fields scaled by a smooth function; the point requiring proof is positive-definiteness, and this is where the two hypotheses, positivity of f and positive-definiteness of g_M and g_N, enter.

Proposition 1.6.3: The warped product is a Riemannian metric

Let (M, g_M) and (N, g_N) be Riemannian manifolds and $f\colon M \to (0, \infty)$ smooth. Then the warped-product tensor $g = \pi_M^* g_M + (f \circ \pi_M)^2\, \pi_N^* g_N$ of definition 1.6.2 is a Riemannian metric on $M \times N$. In particular, taking $f \equiv 1$, the product metric of definition 1.6.1 is a Riemannian metric. If instead g_M and g_N are pseudo-Riemannian of indices μ and ν, then g is pseudo-Riemannian of index $\mu + \nu$; positivity of f is all that is used of the warping function in either case.

***Proof*:**
Step 1: smoothness and symmetry. The projections π_M and π_N are smooth, and the pullback of a smooth symmetric covariant 2-tensor field under a smooth map is again a smooth symmetric covariant 2-tensor field, by [6]. Hence $\pi_M^* g_M$ and $\pi_N^* g_N$ are smooth symmetric covariant 2-tensor fields on $M \times N$. The function $f \circ \pi_M$ is smooth and strictly positive, so $(f \circ \pi_M)^2$ is a smooth positive function, and scaling a smooth symmetric 2-tensor field by a smooth function preserves smoothness and symmetry. Therefore $g = \pi_M^* g_M + (f \circ \pi_M)^2\, \pi_N^* g_N$ is a smooth symmetric covariant 2-tensor field on $M \times N$, which is the type required of a (pseudo-)Riemannian metric by definition 1.1.1.

Step 2: the orthogonal splitting of the tangent space. Fix a point $(p, q) \in M \times N$. The differentials of the projections furnish the canonical linear isomorphism

$$\Phi\colon T_{(p,q)}(M \times N) \longrightarrow T_pM \oplus T_qN, \qquad \Phi(u) = \big(d\pi_M\, u,\ d\pi_N\, u\big)$$

Its inverse identifies T_pM with the subspace tangent to the base slice $M \times \{q\}$ and T_qN with the subspace tangent to the fiber $\{p\} \times N$. Write a tangent vector as $u = (a, b)$ under Φ, with $a = d\pi_M\, u \in T_pM$ the *horizontal part* and $b = d\pi_N\, u \in T_qN$ the *vertical part*. In these terms the warped metric reads

$$g_{(p,q)}\big((a, b), (a', b')\big) = (g_M)_p(a, a') + f(p)^2\, (g_N)_q(b, b') \tag{1.5}$$

directly from the definition, since $d\pi_M\, u = a$ and $d\pi_N\, u = b$ and likewise for $u' = (a', b')$. Taking one vector horizontal, $b = 0$, and the other vertical, $a' = 0$, makes both terms of (1.5) vanish, so the horizontal subspace $T_pM \oplus 0$ and the vertical subspace $0 \oplus T_qN$ are g-orthogonal.

Step 3: positive-definiteness in the Riemannian case. Let $u = (a, b)$ be a nonzero tangent vector at (p, q). Then

$$g_{(p,q)}(u, u) = (g_M)_p(a, a) + f(p)^2\, (g_N)_q(b, b)$$

by (1.5). Both summands are nonnegative: $(g_M)_p$ and $(g_N)_q$ are positive definite because g_M and g_N are Riemannian, and the coefficient $f(p)^2$ is strictly positive because f takes values in $(0, \infty)$. Since $u \neq 0$, at least one of a, b is nonzero. If $a \neq 0$ then $(g_M)_p(a,a) > 0$; if $b \neq 0$ then $f(p)^2 > 0$ and $(g_N)_q(b,b) > 0$, so $f(p)^2(g_N)_q(b,b) > 0$. In either case the sum of the two nonnegative terms is strictly positive, so $g_{(p,q)}(u,u) > 0$. Thus $g_{(p,q)}$ is positive definite at every point, and g is a Riemannian metric. Setting $f \equiv 1$ throughout gives the same conclusion for the product metric.

Step 4: the pseudo-Riemannian case. Now suppose g_M and g_N are pseudo-Riemannian of indices μ and ν. Step 1 already shows g is a smooth symmetric 2-tensor field, and Step 2 gives the g-orthogonal splitting $T_{(p,q)}(M \times N) = (T_pM \oplus 0) \oplus_\perp (0 \oplus T_qN)$ with g restricting on the first summand to $(g_M)_p$ and on the second to $f(p)^2(g_N)_q$, by (1.5). Because $f(p)^2 > 0$, the form $f(p)^2(g_N)_q$ has the same signature as $(g_N)_q$: scaling a symmetric bilinear form by a positive constant preserves the sign of $b \mapsto (g_N)_q(b,b)$ on every vector, hence preserves the number of positive, negative, and zero eigenvalues. The index (number of negative eigenvalues) of a g-orthogonal direct sum of two nondegenerate forms is the sum of their indices, and the form is nondegenerate exactly when each summand is. Here $(g_M)_p$ has index μ and $f(p)^2(g_N)_q$ has index ν, both nondegenerate, so $g_{(p,q)}$ is nondegenerate of index $\mu + \nu$. As $\mu + \nu$ is independent of the point, g is a pseudo-Riemannian metric of constant index $\mu + \nu$ in the sense of definition 1.1.3, completing the proof.

The verification isolates exactly why the two hypotheses cannot be dropped. Positive-definiteness of the fiber metric g_N is needed so that the vertical term $f(p)^2(g_N)_q(b,b)$ is nonnegative; positivity of f is needed so that this term does not collapse. Were f allowed to vanish at a point p_0, the metric would degenerate along the entire fiber $\{p_0\} \times N$, where every vertical vector would have zero length, and g would fail to be a metric there. This is not an abstract worry: the polar and spherical presentations below have warping functions ($f(r) = r$, $f(r) = \sin r$) that do vanish at the endpoints of their domains, and it is precisely at those parameter values that the warped product breaks down as a manifold and a single point (the origin, or a pole) must be glued in by hand. The warped-product formalism describes the smooth geometry away from such collapse loci; the points where $f = 0$ are the seams where a warped product is capped off.

Two structural features of a warped product are worth naming before the examples, both concerning how the two factors sit inside $M \times_f N$. First, the base copy $M \times \{q\}$, for any fixed $q \in N$, is a Riemannian submanifold isometric to (M, g_M): the inclusion $p \mapsto (p,q)$ has image the base slice, and the induced metric of definition 1.5.1 is (g_M) unchanged, since the warping factor multiplies only the vertical term, which is absent for vectors tangent to the base. The base is embedded without distortion. Second, the fiber copy $\{p\} \times N$, for fixed $p \in M$, is a Riemannian submanifold isometric to $(N, f(p)^2 g_N)$, that is, to (N, g_N) rescaled by the constant $f(p)$; different fibers are rescaled copies of the same manifold, with scale governed by f. These two observations already fix the qualitative shape of a warped product, and the examples that follow make them quantitative.

The asymmetry between base and fiber, that only g_N is warped, and only by a function on M, is the source of the construction's utility. It is also what makes the Levi-Civita connection and curvature of a warped product computable in closed form in terms of the two factor geometries and the derivatives of f: the mixed connection terms involve $\operatorname{grad} f$ on the base, and the curvature of $M \times_f N$ splits into a base contribution, a fiber contribution, and a mixed contribution controlled by the Hessian of f. Those formulas, due in this generality to O'Neill, require the connection of Chapter 2 and the curvature tensor of Chapter 4 and are taken up there; the metric-level construction is complete as it stands, and the examples below are the payoff.

The first example is the one that motivates the whole formalism: Euclidean space, away from the origin, is a warped product over a half-line with spherical fibers, and the warping function is simply the radial distance.

Example 1.6.4: Euclidean space as a warped product
Let $(S^{n-1}, g_{S^{n-1}})$ be the unit sphere with its round metric, the case $r = 1$ of example 1.3.4, and take the base to be the half-line $(0, \infty)$ with the standard metric dr^2. Form the warped product

$$(0, \infty) \times_r S^{n-1}, \qquad g = dr^2 + r^2\, g_{S^{n-1}},$$

with warping function $f(r) = r$; here r denotes both the coordinate on the base and, as a function on the product, the warping function $f \circ \pi_M$. The claim is that this warped product is isometric to Euclidean space $\mathbb{R}^n$ with the origin removed, carrying the flat metric $\bar{g}$ of example 1.1.2.

Consider the map

$$\Psi\colon (0, \infty) \times S^{n-1} \longrightarrow \mathbb{R}^n \setminus \{0\}, \qquad \Psi(r, \omega) = r\,\omega,$$

which sends a radius and a unit direction to the point at that distance in that direction. This is a diffeomorphism, with inverse $x \mapsto (|x|\,,\, x/\,|x|)$, the passage to polar coordinates. To compute the pullback $\Psi^*\bar{g}$, write a tangent vector to the product at (r, ω) as (a, b) with $a \in T_r(0, \infty) = \mathbb{R}$ the radial part and $b \in T_\omega S^{n-1}$ the spherical part, and note $\langle \omega, \omega\rangle = 1$ and $\langle \omega, b\rangle = 0$, the latter because b is tangent to the unit sphere at ω, hence orthogonal to the position vector ω. The differential of Ψ carries (a, b) to

$$d\Psi_{(r,\omega)}(a, b) = a\,\omega + r\,b \;\in\; \mathbb{R}^n,$$

the first term from varying the radius, the second from varying the direction. Since $\bar{g}$ is the Euclidean dot product on $\mathbb{R}^n$ under the identification $T_x\mathbb{R}^n \cong \mathbb{R}^n$,

$$(\Psi^*\bar{g})\big((a, b), (a, b)\big) = \langle a\,\omega + r\,b, a\,\omega + r\,b\rangle = a^2\,\langle \omega, \omega\rangle + 2ar\,\langle \omega, b\rangle + r^2\,\langle b, b\rangle = a^2 + r^2\,|b|^2\,,$$

using $\langle \omega, \omega\rangle = 1$ and $\langle \omega, b\rangle = 0$. The cross term drops out precisely because the spherical directions are orthogonal to the radial direction. Now $|b|^2 = \langle b, b\rangle$ is the Euclidean square-length of b as a vector tangent to the unit sphere in $\mathbb{R}^n$, which by example 1.3.4 equals $g_{S^{n-1}}(b, b)$, the round metric being the restriction of the ambient dot product. Hence

$$(\Psi^*\bar{g})\big((a, b), (a, b)\big) = a^2 + r^2\, g_{S^{n-1}}(b, b) = \big(dr^2 + r^2\, g_{S^{n-1}}\big)\big((a, b), (a, b)\big),$$

and by polarization $\Psi^*\bar{g} = dr^2 + r^2 g_{S^{n-1}}$. Thus Ψ is an isometry from the warped product $(0, \infty) \times_r S^{n-1}$ onto $\mathbb{R}^n \setminus \{0\}$, so the flat metric in polar form is

$$\bar{g} = dr^2 + r^2\, g_{S^{n-1}}$$

This is the source of the warping function $f(r) = r$: the fiber over radius r is a sphere of that radius, a copy of the unit sphere magnified by r, exactly the geometry of the sphere $\{|x| = r\}$ inside $\mathbb{R}^n$. The warping degenerates as $r \to 0$, where $f(r) = r \to 0$ and the spheres collapse to the single missing point, the origin, which the warped product does not see. For $n = 2$ this reads $\bar{g} = dr^2 + r^2\, d\theta^2$, the elementary polar-coordinate expression for the plane, with $g_{S^1} = d\theta^2$.

The polar decomposition of $\mathbb{R}^n$ is the template, and the same shape, a half-line base with spherical fibers, produces the round sphere and hyperbolic space by changing only the warping function. Replacing r by $\sin r$ gives the sphere, as already seen in the geodesic polar form of example 1.5.4; replacing it by $\sinh r$ gives hyperbolic space, which the next example verifies.

Example 1.6.5: Hyperbolic space as a warped product

Hyperbolic space H^n of example 1.3.5 admits two natural warped-product presentations, one over a half-line with spherical fibers and one over a line with Euclidean fibers. The second is the quicker to verify from the models already in hand, so it is treated in full; the first is then stated with its warping function.

Half-space form: a line warped over Euclidean fibers. Recall the upper half-space model $U^n = \{y \in \mathbb{R}^n \mid y^n > 0\}$ from example 1.3.5, with metric

$$g_U = \frac{1}{(y^n)^2}\big((dy^1)^2 + \cdots + (dy^{n-1})^2 + (dy^n)^2\big)$$

Take the base to be $\mathbb{R}$ with coordinate t and metric dt^2, the fiber to be $\mathbb{R}^{n-1}$ with the Euclidean metric $\bar{g}_{n-1} = (dz^1)^2 + \cdots + (dz^{n-1})^2$, and the warping function $f(t) = e^t$. The resulting warped product is

$$\mathbb{R} \times_{e^t} \mathbb{R}^{n-1}, \qquad g = dt^2 + e^{2t}\,\bar{g}_{n-1}$$

Introduce the change of coordinates

$$\Xi\colon \mathbb{R} \times \mathbb{R}^{n-1} \longrightarrow U^n, \qquad \Xi(t,z) = \big(z^1, \ldots, z^{n-1},\ e^{-t}\big),$$

a diffeomorphism onto U^n with inverse $y \mapsto (-\log y^n,\ y^1, \ldots, y^{n-1})$. Along Ξ the horizontal coordinate is $y^i = z^i$ for $i < n$ and the vertical coordinate is $y^n = e^{-t}$, so $dy^i = dz^i$ for $i < n$ and $dy^n = -e^{-t}\,dt$. Substituting into g_U,

$$\Xi^* g_U = \frac{1}{(e^{-t})^2}\Big(\textstyle\sum_{i=1}^{n-1}(dz^i)^2 + e^{-2t}\,dt^2\Big) = e^{2t}\textstyle\sum_{i=1}^{n-1}(dz^i)^2 + e^{2t}\,e^{-2t}\,dt^2 = dt^2 + e^{2t}\,\bar{g}_{n-1}$$

Hence $\Xi^* g_U = g$, and the warped product $\mathbb{R} \times_{e^t} \mathbb{R}^{n-1}$ is isometric to the upper half-space model, so it presents hyperbolic space. In this form the fibers are flat Euclidean spaces $\mathbb{R}^{n-1}$ whose scale grows exponentially in the base coordinate t, matching the intuition, made precise once the curvature is available, that hyperbolic space spreads apart exponentially: the horospheres $t = \text{const}$ are flat, and moving in the t direction rescales them by e^t.

Geodesic polar form: a half-line warped over spherical fibers. The presentation parallel to the Euclidean polar form of example 1.6.4 is

$$(0,\infty) \times_{\sinh r} S^{n-1}, \qquad g = dr^2 + \sinh^2 r\; g_{S^{n-1}},$$

with warping function $f(r) = \sinh r$, where r is hyperbolic distance from a fixed center point. The metric $dr^2 + \sinh^2 r\, g_{S^{n-1}}$ is the geodesic polar form of H^n about that center: the fiber over radius r is a sphere whose size is governed by $\sinh r$ in place of the Euclidean r, and since $\sinh r > r$ for $r > 0$ the hyperbolic spheres are larger than Euclidean spheres of the same radius, the metric statement of negative curvature. The three warping functions $f(r) = r$ (Euclidean, example 1.6.4), $f(r) = \sin r$ (spherical, example 1.5.4), and $f(r) = \sinh r$ (hyperbolic) are exactly the solutions of $f'' + \kappa f = 0$ with $f(0) = 0$, $f'(0) = 1$, for $\kappa = 0$, $\kappa = 1$, and $\kappa = -1$ respectively; this differential equation is the one that later emerges, from the curvature computation of Chapter 4, as the condition for the warped product to have constant sectional curvature κ. Its identification is deferred to that chapter, but the three model spaces already exhibit the three cases side by side. As $r \to 0$ the warping $\sinh r \to 0$, and as in the Euclidean case the collapsing fibers are capped by the single center point.

The three constant-curvature model spaces are thus all warped products of the same base-fiber shape, distinguished by their warping functions. The final example turns from these to the class that gave the construction its geometric name, and reconciles the notation of this section with the surface computation of Chapter 1.

Example 1.6.6: Surfaces of revolution as warped products

The surfaces of revolution of example 1.5.4, with the profile curve parametrized by arc length, carry the induced metric

$$dr^2 + f(r)^2\, d\theta^2, \qquad f(r) = \varphi(r) > 0,$$

computed there, where r runs over an interval I, θ over $\mathbb{R}/2\pi\mathbb{Z}$, and $f = \varphi$ is the distance from a point of the surface to the axis of rotation. This is exactly a warped product. Take the base to be the interval $(I,\, dr^2)$, the fiber to be the circle $(S^1,\, d\theta^2) = (\mathbb{R}/2\pi\mathbb{Z},\, d\theta^2)$, and the warping function to be f; then

$$I \times_f S^1, \qquad g = dr^2 + f(r)^2\, d\theta^2,$$

and the induced metric of example 1.5.4 is by inspection this warped-product metric. The identification is immediate because the cross term $(\iota^*\bar{g})_{r\theta}$ was shown to vanish in example 1.5.4, which is the orthogonality of base and fiber demanded by definition 1.6.2, and the two diagonal terms are dr^2 and $f(r)^2\, d\theta^2$, the base and warped-fiber terms.

The symbol f carries the same meaning in both places, the warping function, so no reconciliation of notation is needed beyond noting that the fiber circle here is the unit circle $S^1 = (\mathbb{R}/2\pi\mathbb{Z}, d\theta^2)$, whose round metric is $d\theta^2$, the case $n = 1$ of the round sphere $g_{S^1} = d\theta^2$ used in example 1.6.4. Concretely, the fiber over the parameter value r is the circle of radius $f(r)$ traced out by rotating the corresponding point of the profile curve about the axis, a copy of the unit circle magnified by the distance $f(r)$ to the axis, precisely the geometry of a parallel of the surface. This is the origin of the term *warped product*: the fiber, here a circle, is warped, that is rescaled, by a function of position along the base.

The two specializations recorded in example 1.5.4 now read off as warped products of familiar type. The cylinder, $f(r) = 1$, is the unwarped Riemannian product $I \times S^1$ with metric $dr^2 + d\theta^2$, flat because it is a product of flat factors with constant warping. The unit sphere minus its poles, $f(r) = \sin r$ for $r \in (0, \pi)$, is the warped product $(0, \pi) \times_{\sin r} S^1$ with metric $dr^2 + \sin^2 r\, d\theta^2$, the two-dimensional case of the spherical polar form and the counterpart to the Euclidean $dr^2 + r^2\, d\theta^2$ and hyperbolic $dr^2 + \sinh^2 r\, d\theta^2$ forms of the preceding examples. The warping function $\sin r$ vanishes at $r = 0$ and $r = \pi$, where the parallels shrink to the north and south poles, the two points at which the warped product is capped off to recover the full sphere.

2

The Levi-Civita Connection

The metric of chapter 1 measures vectors at a single point, but it does not by itself say how to compare vectors at different points, nor how to differentiate a vector field along a curve. On a general manifold there is no canonical way to do either: the partial derivatives of the components of a vector field do not transform tensorially, so differentiation of tangent vectors requires an extra structure, a *connection*. This chapter shows that a metric selects one connection above all others.

The development proceeds from the general notion of an affine connection and its Christoffel symbols to the two properties that single out the geometrically correct one: freedom from torsion, which makes second covariant derivatives symmetric, and compatibility with the metric, which makes parallel transport preserve lengths and angles. The fundamental theorem of Riemannian geometry asserts that these two conditions are met by exactly one connection, the Levi-Civita connection, and the Koszul formula produces it explicitly. The chapter ends with parallel transport, the concrete transport of tangent vectors along curves that the connection defines, and the induced differentiation of tensor fields of every type. With the Levi-Civita connection in hand, the notion of a straight line on a manifold becomes available, and that is the subject of the next chapter.

2.1 Affine Connections and Christoffel Symbols

On a general smooth manifold there is no canonical way to compare tangent vectors at different points. A tangent vector at p lives in T_pM and a tangent vector at a nearby point q lives in T_qM, and these are different vector spaces with no preferred identification between them. This is not a defect of the setup; it reflects the fact that a bare smooth manifold carries no notion of "the same direction" from one point to the next. The consequence is that the derivative of a vector field is not well defined. If Y is a vector field and X_p a direction at p, one would like to differentiate Y along X_p by forming the difference quotient $\frac{1}{t}\left(Y_{\gamma(t)} - Y_p\right)$ along a curve γ with $\dot{\gamma}(0) = X_p$. The obstruction is immediate: $Y_{\gamma(t)}$ and Y_p sit in different tangent spaces, so their difference has no meaning.

An affine connection is precisely the extra structure that repairs this. It prescribes, as a primitive datum, how to differentiate one vector field along another, subject to the algebraic rules that a directional derivative ought to satisfy. On $\mathbb{R}^n$ there is an obvious choice: differentiate each component function separately, which works because the coordinate frame $\partial_1, \ldots, \partial_n$ furnishes a global identification of every tangent space with $\mathbb{R}^n$. On a manifold no such global frame need exist, and in general there is no distinguished connection at all; a connection is a choice. The purpose of this section is to axiomatize that choice, to record its coordinate description through the Christoffel symbols, to show that connections always exist and to measure exactly how much freedom the choice involves, and finally to transport the operation from vector fields on M to vector fields along a curve, where it becomes the covariant derivative D_t that governs geodesics and parallel transport in the sections to come. Throughout, M is a smooth n-manifold and $C^\infty(M)$ is its algebra of smooth real functions; $\mathfrak{X}(M)$ denotes the $C^\infty(M)$-module of smooth vector fields, following [6].

Definition 2.1.1: Affine Connection

Let M be a smooth manifold. An *affine connection* on TM is a map

$$\nabla\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M), \qquad (X, Y) \mapsto \nabla_X Y,$$

called *covariant differentiation*, satisfying the following three properties for all $X, X_1, X_2, Y, Y_1, Y_2 \in \mathfrak{X}(M)$, all $a, b \in \mathbb{R}$, and all $f \in C^\infty(M)$.

1. *$C^\infty(M)$-linearity in the differentiation slot:*

$$\nabla_{fX_1+X_2} Y = f\,\nabla_{X_1} Y + \nabla_{X_2} Y$$

2. *$\mathbb{R}$-linearity in the differentiated slot:*

$$\nabla_X (aY_1 + bY_2) = a\,\nabla_X Y_1 + b\,\nabla_X Y_2$$

3. *Leibniz rule in the differentiated slot:*

$$\nabla_X (fY) = (Xf)\,Y + f\,\nabla_X Y$$

The vector field $\nabla_X Y$ is the *covariant derivative* of Y in the direction X.

The asymmetry between the two slots is the heart of the definition and deserves comment. In the differentiation slot X, the connection is $C^\infty(M)$-linear: scaling the direction by a function scales the answer by the same function, pointwise. This is what makes $\nabla_X Y$ tensorial in X, and it has a concrete consequence recorded below in proposition 2.1.5, namely that $(\nabla_X Y)_p$ depends on X only through its value $X_p \in T_pM$, not on how X extends off p. In the differentiated slot Y, by contrast, the connection is only $\mathbb{R}$-linear and obeys the Leibniz rule; the extra term $(Xf)Y$ is exactly the derivative of the coefficient function, and it is what forces $\nabla_X Y$ to depend on how Y varies rather than only on its value at a point. A directional derivative must behave this way: differentiating f times a fixed field Y should produce a rate of change of f, which is the content of $(Xf)Y$. The failure of $C^\infty(M)$-linearity in Y is not a blemish but the defining feature that distinguishes a derivative from a tensor.

The first worked example is the one that motivates the whole apparatus.

Example 2.1.2: The Euclidean Connection

On $\mathbb{R}^n$ with global coordinates $(x^1, \ldots, x^n)$, write a vector field as $Y = Y^i\,\partial_i$ with $Y^i \in C^\infty(\mathbb{R}^n)$; the Einstein summation convention, summing over any index appearing once up and once down, is in force here and throughout the section. Define

$$\overline{\nabla}_X Y := (XY^i)\,\partial_i,$$

that is, differentiate each component function of Y in the direction X and reassemble. This is the *Euclidean* (or *flat*, or *standard*) connection.

Each axiom of definition 2.1.1 is immediate from the corresponding property of the directional derivative of functions. For property (1), $C^\infty(\mathbb{R}^n)$-linearity of $X \mapsto XY^i$ in X gives $\overline{\nabla}_{fX_1+X_2}Y = ((fX_1 + X_2)Y^i)\partial_i = f(X_1Y^i)\partial_i + (X_2Y^i)\partial_i = f\,\overline{\nabla}_{X_1}Y + \overline{\nabla}_{X_2}Y$. Property (2) follows because $X(aY_1^i + bY_2^i) = a\,XY_1^i + b\,XY_2^i$. For property (3), writing $fY = (fY^i)\partial_i$ and using the ordinary product rule $X(fY^i) = (Xf)Y^i + f(XY^i)$ gives

$$\overline{\nabla}_X(fY) = \big(X(fY^i)\big)\partial_i = (Xf)\,Y^i\partial_i + f\,(XY^i)\partial_i = (Xf)\,Y + f\,\overline{\nabla}_X Y,$$

which is the Leibniz rule. Thus $\overline{\nabla}$ is an affine connection on $\mathbb{R}^n$. The computation goes through because the coordinate frame (∂_i) is globally defined and its members are, so to speak, held fixed while the coefficients are differentiated. On a general manifold no global coordinate frame exists, so there is no canonical connection to write down; the definition above is the abstraction of what remains once the crutch of global coordinates is removed.

Away from $\mathbb{R}^n$ a connection cannot be described by "differentiate the components," because the coordinate frames on overlapping charts differ and the frame vectors themselves must be differentiated. The bookkeeping is carried by a family of functions on each chart.

Definition 2.1.3: Christoffel Symbols

Let ∇ be an affine connection on TM and let $(U, (x^1, \ldots, x^n))$ be a smooth chart with coordinate frame $(\partial_1, \ldots, \partial_n)$. Each $\nabla_{\partial_i}\partial_j$ is a smooth vector field on U, hence expands in the coordinate frame. The *Christoffel symbols* of ∇ in the chart are the n^3 functions $\Gamma^k_{ij} \in C^\infty(U)$ defined by

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\,\partial_k$$

They record the action of ∇ on the frame; the lower indices (i, j) label the direction and the field, the upper index k the output component.

Once the Christoffel symbols are known, the connection is completely determined on the chart. Writing $X = X^i\partial_i$ and $Y = Y^j\partial_j$ and applying the three axioms in turn, first C^∞-linearity in X, then $\mathbb{R}$-linearity in Y, then the Leibniz rule on each $Y^j\partial_j$, gives

$$\nabla_X Y = X^i\,\nabla_{\partial_i}\big(Y^j\partial_j\big) = X^i\big((\partial_i Y^j)\,\partial_j + Y^j\,\nabla_{\partial_i}\partial_j\big) = X^i(\partial_i Y^j)\,\partial_j + X^iY^j\,\Gamma^k_{ij}\,\partial_k$$

Relabelling the dummy index $j \to k$ in the first term collects the ∂_k-component:

$$\nabla_X Y = \Big(X^i\,\partial_i Y^k + \Gamma^k_{ij}\,X^iY^j\Big)\,\partial_k \tag{2.1}$$

The first summand is the Euclidean answer of example 2.1.2: the plain directional derivative of the components. The second summand, quadratic in the frame data through Γ^k_{ij}, is the correction that accounts for the variation of the frame itself. On $\mathbb{R}^n$ in Cartesian coordinates every Γ_{ij}^k vanishes, since $\overline{\nabla}_{\partial_i}\partial_j = (\partial_i\delta_j^k)\partial_k = 0$, and (2.1) reduces to $\overline{\nabla}_X Y = X^i(\partial_i Y^k)\partial_k = (XY^k)\partial_k$, recovering the flat connection. It should be stressed that the Γ_{ij}^k are not the components of a tensor: under a change of coordinates they acquire an inhomogeneous term (computed in proposition 2.1.5 below), so "the Christoffel symbols vanish" is a statement about a chart, not about a point.

Example 2.1.4: Christoffel Symbols in Polar Coordinates

Take $\mathbb{R}^2$ minus the origin with polar coordinates (r,θ), related to Cartesian coordinates by $x = r\cos\theta$, $y = r\sin\theta$, and equip it with the Euclidean connection $\overline{\nabla}$ of example 2.1.2. The coordinate frames are related by

$$\partial_r = \cos\theta\,\partial_x + \sin\theta\,\partial_y, \qquad \partial_\theta = -r\sin\theta\,\partial_x + r\cos\theta\,\partial_y$$

Since $\overline{\nabla}_{\partial_x}\partial_x = \overline{\nabla}_{\partial_x}\partial_y = \overline{\nabla}_{\partial_y}\partial_y = 0$, the connection acts on the polar frame by differentiating the coefficients above. Direct computation gives

$$\overline{\nabla}_{\partial_r}\partial_r = \overline{\nabla}_{\partial_r}\big(\cos\theta\,\partial_x + \sin\theta\,\partial_y\big) = (\partial_r\cos\theta)\,\partial_x + (\partial_r\sin\theta)\,\partial_y = 0,$$

because $\cos\theta$ and $\sin\theta$ are independent of r; hence $\Gamma_{rr}^r = \Gamma_{rr}^\theta = 0$. Next,

$$\overline{\nabla}_{\partial_r}\partial_\theta = \overline{\nabla}_{\partial_r}\big(-r\sin\theta\,\partial_x + r\cos\theta\,\partial_y\big) = -\sin\theta\,\partial_x + \cos\theta\,\partial_y = \tfrac{1}{r}\,\partial_\theta,$$

so $\Gamma_{r\theta}^\theta = \frac{1}{r}$ and $\Gamma_{r\theta}^r = 0$. The same value arises in the direction ∂_θ on ∂_r:

$$\overline{\nabla}_{\partial_\theta}\partial_r = \overline{\nabla}_{\partial_\theta}\big(\cos\theta\,\partial_x + \sin\theta\,\partial_y\big) = -\sin\theta\,\partial_x + \cos\theta\,\partial_y = \tfrac{1}{r}\,\partial_\theta,$$

giving $\Gamma_{\theta r}^\theta = \frac{1}{r}$, $\Gamma_{\theta r}^r = 0$. Finally,

$$\overline{\nabla}_{\partial_\theta}\partial_\theta = \overline{\nabla}_{\partial_\theta}\big(-r\sin\theta\,\partial_x + r\cos\theta\,\partial_y\big) = -r\cos\theta\,\partial_x - r\sin\theta\,\partial_y = -r\,\partial_r,$$

so $\Gamma_{\theta\theta}^r = -r$ and $\Gamma_{\theta\theta}^\theta = 0$. The single flat connection $\overline{\nabla}$ thus has all Christoffel symbols zero in Cartesian coordinates and the nonzero symbols $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$, $\Gamma_{\theta\theta}^r = -r$ in polar coordinates. This concretely exhibits the coordinate-dependence flagged above: the Christoffel symbols are not tensorial, and their vanishing in one chart says nothing about their vanishing in another.

Two questions are now pressing. Do connections exist on every manifold, given that no global-frame recipe is available in general? And how much does a connection depend on choices, once one exists? The next result answers both. Its proof uses a partition of unity, exactly as in the construction of Riemannian metrics; the analytic input is the standard fact that a smooth manifold admits a partition of unity subordinate to any open cover [6].

Proposition 2.1.5: Existence and Local Character of Connections

Let M be a smooth manifold. Then:

1. *(Locality.)* If ∇ is an affine connection and $X, Y \in \mathfrak{X}(M)$ agree with X', Y' on an open set U, then $\nabla_X Y = \nabla_{X'} Y'$ on U. Moreover $(\nabla_X Y)_p$ depends on X only through the value $X_p \in T_pM$. Consequently ∇ restricts to a well-defined connection on every open subset, and is determined on each chart by its Christoffel symbols through (2.1).

2. *(Difference tensor.)* If ∇ and $\widetilde{\nabla}$ are two affine connections on TM, then

$$A(X,Y) := \widetilde{\nabla}_X Y - \nabla_X Y$$

is $C^\infty(M)$-bilinear, hence defines a $(1,2)$-tensor field on M. Conversely, for any $(1,2)$-tensor field A and any affine connection ∇, the map $\widetilde{\nabla}_X Y := \nabla_X Y + A(X,Y)$ is an affine connection. The Christoffel symbols transform under a change of chart $(x^i) \to (\tilde{x}^a)$ by

$$\widetilde{\Gamma}^c_{ab} = \frac{\partial \tilde{x}^c}{\partial x^k} \frac{\partial x^i}{\partial \tilde{x}^a} \frac{\partial x^j}{\partial \tilde{x}^b} \Gamma^k_{ij} + \frac{\partial \tilde{x}^c}{\partial x^k} \frac{\partial^2 x^k}{\partial \tilde{x}^a \, \partial \tilde{x}^b},$$

the second, inhomogeneous term being what disqualifies (Γ^k_{ij}) from being a tensor.

3. *(Existence.)* M admits an affine connection on TM.

***Proof*:**
Step 1: locality of the differentiation slot. Fix an affine connection ∇. First $C^\infty(M)$-linearity in X (axiom (1)) is exactly the statement that $X \mapsto (\nabla_X Y)_p$ is $C^\infty(M)$-linear; by the tensor characterization lemma for $C^\infty(M)$-multilinear maps [6], such a map is pointwise, so $(\nabla_X Y)_p$ depends on X only through X_p. Explicitly: if $X_p = 0$ choose smooth functions f^i and vector fields E_i with $X = f^i E_i$ near p and $f^i(p) = 0$; then $(\nabla_X Y)_p = f^i(p)(\nabla_{E_i} Y)_p = 0$. Applied to $X - X'$ this shows $(\nabla_X Y)_p = (\nabla_{X'} Y)_p$ whenever $X_p = X'_p$.

Step 2: locality of the differentiated slot. Suppose $Y = Y'$ on an open set U and fix $p \in U$. Choose a bump function $\varphi \in C^\infty(M)$ with $\varphi \equiv 1$ on a neighborhood of p and $\operatorname{supp} \varphi \subset U$, which exists by [6]. Then $\varphi Y = \varphi Y'$ on all of M, so $\nabla_X(\varphi Y) = \nabla_X(\varphi Y')$. Expanding the left side with the Leibniz rule (axiom (3)) and evaluating at p, where $\varphi(p) = 1$ and $(X\varphi)(p) = 0$ because φ is constant near p, gives

$$(\nabla_X(\varphi Y))_p = (X\varphi)(p)\, Y_p + \varphi(p)\, (\nabla_X Y)_p = (\nabla_X Y)_p,$$

and likewise $(\nabla_X(\varphi Y'))_p = (\nabla_X Y')_p$; hence $(\nabla_X Y)_p = (\nabla_X Y')_p$. Combining Steps 1 and 2, if $(X, Y) = (X', Y')$ on U then $\nabla_X Y = \nabla_{X'} Y'$ on U. In particular ∇ descends to a connection on the open submanifold U, and on a chart the value $\nabla_X Y$ at any point is computed from the chart data alone; formula (2.1), derived from the axioms, then shows the Christoffel symbols determine ∇ on the chart. This proves (1).

Step 3: the difference is a tensor. Let $\nabla, \widetilde{\nabla}$ be connections and $A(X,Y) = \widetilde{\nabla}_X Y - \nabla_X Y$. Additivity of A in each slot is clear. For $f \in C^\infty(M)$, the differentiation slot gives $A(fX, Y) = \widetilde{\nabla}_{fX} Y - \nabla_{fX} Y = f \widetilde{\nabla}_X Y - f \nabla_X Y = f\, A(X,Y)$ by axiom (1) for both connections. For the differentiated slot, the Leibniz terms cancel:

$$A(X, fY) = \big((Xf)Y + f\widetilde{\nabla}_X Y\big) - \big((Xf)Y + f\nabla_X Y\big) = f\,A(X,Y)$$

Thus A is $C^\infty(M)$-bilinear, so by the tensor characterization [6] it is a $(1,2)$-tensor field: a section of $\mathrm{Hom}(TM \otimes TM, TM)$. Conversely, if A is any such tensor and ∇ any connection, then $\widetilde{\nabla}_X Y := \nabla_X Y + A(X,Y)$ is $C^\infty(M)$-linear in X (sum of two such), $\mathbb{R}$-linear in Y, and satisfies the Leibniz rule because A contributes no derivative term:

$$\widetilde{\nabla}_X(fY) = \nabla_X(fY) + A(X, fY) = (Xf)Y + f\nabla_X Y + f\,A(X,Y) = (Xf)Y + f\,\widetilde{\nabla}_X Y$$

Hence $\widetilde{\nabla}$ is an affine connection. For the transformation law one fixes a single connection ∇ and compares its Christoffel symbols in the two charts, applying (2.1) to $\nabla_{\partial/\partial\tilde{x}^a}(\partial/\partial\tilde{x}^b)$. Using $\partial/\partial\tilde{x}^a = (\partial x^i/\partial\tilde{x}^a)\partial_i$ together with C^∞-linearity in the direction and the Leibniz rule,

$$\nabla_{\partial/\partial\tilde{x}^a}\frac{\partial}{\partial\tilde{x}^b} = \frac{\partial x^i}{\partial\tilde{x}^a}\,\nabla_{\partial_i}\Big(\frac{\partial x^j}{\partial\tilde{x}^b}\,\partial_j\Big) = \frac{\partial x^i}{\partial\tilde{x}^a}\Big(\partial_i\frac{\partial x^j}{\partial\tilde{x}^b}\Big)\partial_j + \frac{\partial x^i}{\partial\tilde{x}^a}\frac{\partial x^j}{\partial\tilde{x}^b}\,\Gamma^k_{ij}\,\partial_k$$

The first term simplifies because $\frac{\partial x^i}{\partial\tilde{x}^a}\partial_i = \partial/\partial\tilde{x}^a$, so $\frac{\partial x^i}{\partial\tilde{x}^a}\big(\partial_i\frac{\partial x^j}{\partial\tilde{x}^b}\big) = \frac{\partial^2 x^j}{\partial\tilde{x}^a\partial\tilde{x}^b}$. Expanding $\partial_j = (\partial\tilde{x}^c/\partial x^j)\,\partial/\partial\tilde{x}^c$ and $\partial_k = (\partial\tilde{x}^c/\partial x^k)\,\partial/\partial\tilde{x}^c$ and reading off the $\partial/\partial\tilde{x}^c$-coefficient (renaming the summed index $j \to k$ in the first term) yields

$$\widetilde{\Gamma}^c_{ab} = \frac{\partial\tilde{x}^c}{\partial x^k}\,\frac{\partial x^i}{\partial\tilde{x}^a}\,\frac{\partial x^j}{\partial\tilde{x}^b}\,\Gamma^k_{ij} + \frac{\partial\tilde{x}^c}{\partial x^k}\,\frac{\partial^2 x^k}{\partial\tilde{x}^a\,\partial\tilde{x}^b},$$

the second term arising from differentiating the transition Jacobian. This establishes (2).

Step 4: existence. Cover M by charts $\{U_\alpha\}$ and let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to this cover [6], so $\operatorname{supp}\rho_\alpha \subset U_\alpha$, the family $\{\operatorname{supp}\rho_\alpha\}$ is locally finite, and $\sum_\alpha \rho_\alpha \equiv 1$. On each U_α the coordinate frame is a global frame, so the "differentiate the components" recipe of example 2.1.2 defines an affine connection ∇^α on U_α (all its Christoffel symbols vanish in that chart). Define

$$\nabla_X Y := \sum_\alpha \rho_\alpha\,\nabla^\alpha_X Y,$$

where $\nabla^\alpha_X Y$ is interpreted as extended by zero outside $\operatorname{supp}\rho_\alpha$; local finiteness makes the sum a finite sum near each point, hence a smooth vector field. Properties (1) and (2) of definition 2.1.1 are inherited termwise from the ∇^α, since $C^\infty(M)$-linearity in X and $\mathbb{R}$-linearity in Y are preserved by multiplication by the fixed functions ρ_α and by summation. For the Leibniz rule, compute using $\nabla^\alpha_X(fY) = (Xf)Y + f\nabla^\alpha_X Y$ and $\sum_\alpha \rho_\alpha = 1$:

$$\nabla_X(fY) = \sum_\alpha \rho_\alpha\big((Xf)Y + f\nabla^\alpha_X Y\big) = (Xf)Y\sum_\alpha \rho_\alpha + f\sum_\alpha \rho_\alpha\nabla^\alpha_X Y = (Xf)Y + f\,\nabla_X Y$$

The crucial point is that the convex-combination coefficients ρ_α sum to 1, so the derivative term $(Xf)Y$ appears with total weight 1 rather than being scaled; had the weights summed to anything else, the Leibniz rule would fail. Thus ∇ is an affine connection on TM, proving (3), completing the proof.

The proposition draws a clean picture of the space of connections. Part (2) says that connections form an affine space modeled on the vector space of $(1,2)$-tensor fields: fix one connection ∇, and every other is $\nabla + A$ for a unique such tensor A, while $\nabla - \nabla'$ is always a tensor even though neither connection is. This is the precise sense in which a connection is "a choice of order A": the ambiguity is a tensor field and can therefore be prescribed pointwise and patched. Part (1) is what makes the Christoffel symbols a legitimate local description rather than an artifact, and it licenses all coordinate computations to follow, including the polar-coordinate calculation of example 2.1.4. Existence in part (3) removes the last worry: although no connection is canonical on a bare manifold, at least one always exists, and once a Riemannian metric is present the Fundamental Theorem of the next chapter will single out a distinguished one.

With the connection understood as an operation on vector fields defined on all of M, the remaining task in this section is to differentiate a vector field that is defined only along a curve. A geodesic is a curve whose velocity does not turn, and the velocity of a curve is a vector field defined along that curve, not on any open set of M; the flat connection cannot be applied to it directly. The locality established above is exactly what allows the connection to be pulled back to curves.

Definition 2.1.6: Covariant Derivative Along a Curve

Let ∇ be an affine connection on TM and let $\gamma\colon I \to M$ be a smooth curve on an interval $I \subseteq \mathbb{R}$. A *vector field along* γ is a smooth map $V\colon I \to TM$ with $V(t) \in T_{\gamma(t)}M$ for every $t \in I$; the set of such fields is denoted $\mathfrak{X}(\gamma)$. The *covariant derivative along* γ is the unique map

$$D_t\colon \mathfrak{X}(\gamma) \to \mathfrak{X}(\gamma), \qquad V \mapsto D_t V,$$

written also $\frac{D}{dt}$, satisfying:

1. $\mathbb{R}$*-linearity:* $D_t(aV_1 + bV_2) = a\,D_tV_1 + b\,D_tV_2$ for $a, b \in \mathbb{R}$;
2. *Leibniz rule:* $D_t(fV) = \dot{f}\,V + f\,D_tV$ for $f \in C^\infty(I)$;
3. *compatibility with* ∇*:* if V is the restriction of a vector field $\widetilde{V} \in \mathfrak{X}(M)$, meaning $V(t) = \widetilde{V}_{\gamma(t)}$, then $D_tV(t) = \big(\nabla_{\dot\gamma(t)}\widetilde{V}\big)$.

In a chart $(U,(x^i))$ with $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$ and $V(t) = V^k(t)\,\partial_k|_{\gamma(t)}$, the covariant derivative is

$$D_tV = \Big(\dot V^k + \Gamma^k_{ij}\big(\gamma(t)\big)\,\dot\gamma^i\,V^j\Big)\,\partial_k\big|_{\gamma(t)} \tag{2.2}$$

***Proof*:**
Existence, uniqueness, and the formula are established together. On the chart domain, if D_t exists then property (2) applied to $V = V^k\partial_k|_\gamma$, viewing each $\partial_k|_\gamma$ as the restriction of the coordinate field $\partial_k \in \mathfrak{X}(U)$, gives by $\mathbb{R}$-linearity and the Leibniz rule

$$D_tV = \dot V^k\,\partial_k|_\gamma + V^k\,D_t\big(\partial_k|_\gamma\big)$$

By property (3), $D_t(\partial_k|_\gamma) = \nabla_{\dot\gamma}\partial_k = \dot\gamma^i\,\nabla_{\partial_i}\partial_k = \dot\gamma^i\Gamma^m_{ik}\partial_m|_\gamma$, using C^∞-linearity of ∇ in the direction slot (proposition 2.1.5(1)) to pass $\dot\gamma^i$ outside. Substituting and relabelling the summed indices $(m,k) \to (k,j)$ yields (2.2). This proves that any D_t satisfying (1)–(3) is given by the stated formula on each chart, hence is unique. Conversely, formula (2.2) defines a smooth vector field along γ on each chart; on an overlap the two coordinate expressions agree, because both compute the same operation applied to the same V, so they patch to a globally defined

$D_tV \in \mathfrak{X}(\gamma)$. Properties (1) and (2) are read off directly from (2.2), since $V \mapsto \dot{V}^k$ is $\mathbb{R}$-linear and satisfies the product rule $\frac{d}{dt}(fV^k) = \dot{f}\,V^k + f\dot{V}^k$ while the Γ-term is $\mathbb{R}$-linear and $C^\infty(I)$-linear in V. Property (3) holds because if $V(t) = \widetilde{V}_{\gamma(t)}$ with $\widetilde{V} = \widetilde{V}^k\partial_k$, then $V^k(t) = \widetilde{V}^k(\gamma(t))$ and the chain rule gives $\dot{V}^k = \dot{\gamma}^i(\partial_i\widetilde{V}^k)$, so (2.2) becomes $\big(\dot{\gamma}^i\partial_i\widetilde{V}^k + \Gamma^k_{ij}\dot{\gamma}^i\widetilde{V}^j\big)\partial_k = \nabla_{\dot{\gamma}}\widetilde{V}$ by (2.1), as required.

The content of this definition is that the connection on M induces, on each curve, an intrinsic notion of the rate of change of a vector field along the curve, and that this induced operation is pinned down by three requirements: it is linear, it differentiates coefficients by the product rule, and it agrees with ∇ whenever the field along the curve happens to extend to a field on M. The subtlety it resolves is real. A vector field along γ need not extend to any vector field on M, for instance when γ has a self-intersection and V takes different values at the two times the curve passes through the same point; there is then no $\widetilde{V}$ to which property (3) could be applied. Formula (2.2) shows the operation is nonetheless well defined, because it refers only to the values of V and of the Christoffel symbols along γ, never to an extension. The two summands mirror those in (2.1): $\dot{V}^k$ is the componentwise rate of change of V, and $\Gamma^k_{ij}\dot{\gamma}^iV^j$ corrects for the turning of the coordinate frame as the base point $\gamma(t)$ moves.

Example 2.1.7: Covariant Derivative of the Velocity on a Circle

Return to $\mathbb{R}^2$ minus the origin with the Euclidean connection and the polar Christoffel symbols of example 2.1.4. Let $\gamma(t) = (r(t), \theta(t)) = (1, t)$ describe the unit circle traversed once as t runs over $[0, 2\pi]$, so $\dot{\gamma} = (\dot{r}, \dot{\theta}) = (0, 1)$, that is, $\dot{\gamma} = \partial_\theta$ along γ. Its velocity is the vector field $V = \dot{\gamma}$ along γ, with components $V^r = 0$, $V^\theta = 1$ (constant). Formula (2.2) gives, for the r-component,

$$(D_t\dot{\gamma})^r = \dot{V}^r + \Gamma^r_{ij}\dot{\gamma}^iV^j = 0 + \Gamma^r_{\theta\theta}\,\dot{\theta}\,V^\theta = (-r)(1)(1) = -1,$$

using $r = 1$ along γ and that the only Γ^r with both lower indices θ is $\Gamma^r_{\theta\theta} = -r$; for the θ-component,

$$(D_t\dot{\gamma})^\theta = \dot{V}^\theta + \Gamma^\theta_{ij}\dot{\gamma}^iV^j = 0 + \Gamma^\theta_{\theta\theta}\,\dot{\theta}\,V^\theta = 0,$$

since $\Gamma^\theta_{\theta\theta} = 0$. Hence $D_t\dot{\gamma} = -\partial_r$ along the circle. The velocity has constant components in polar coordinates, yet its covariant derivative is nonzero: it points radially inward, reflecting that the unit circle is not a straight line and its velocity is continually turning toward the center. As a check, in Cartesian coordinates $\gamma(t) = (\cos t, \sin t)$ has velocity $(-\sin t, \cos t)$ and $D_t\dot{\gamma} = \overline{\nabla}_{\dot{\gamma}}\dot{\gamma} = (-\cos t, -\sin t)$, the inward unit radial vector, in agreement with $-\partial_r$. A curve whose velocity has vanishing covariant derivative, $D_t\dot{\gamma} = 0$, is a geodesic; the circle is not one, and its failure is measured precisely by this radial term.

2.2 Torsion and Metric Compatibility

An affine connection, as introduced in definition 2.1.1, is an extra piece of data laid on top of a smooth manifold: a rule for differentiating one vector field along another. On a bare manifold there is no preferred such rule, and proposition 2.1.5 shows that connections come in abundance, the difference of any two being a tensor. The geometry of a Riemannian manifold, by contrast, singles out one connection above all others. The point of this section and the next is to isolate the two properties that pin it down. Neither property refers to the connection's abundance; each is a clean structural constraint, and each has a transparent geometric reading.

The first property, *torsion-freeness*, measures the failure of the second-order behavior of ∇ to be symmetric in its two directional arguments. Torsion is the antisymmetric part of the connection that is not already accounted for by the Lie bracket of the two fields, and a connection is torsion-free precisely when its Christoffel symbols are symmetric in their lower indices. The second property, *metric compatibility*, ties the connection to the metric g of definition 1.1.1: it asks that differentiating an inner product of two fields obey the Leibniz rule with respect to ∇, so that the metric is constant from the connection's point of view. Torsion-freeness is a property a connection has entirely on its own, before any metric is chosen; metric compatibility is a property a connection has relative to a fixed g. Chapter 2's central theorem, proved in section 2.3, is that on a (pseudo-)Riemannian manifold exactly one connection has both. This section develops the two properties separately, so that the uniqueness argument there can proceed unencumbered.

Throughout, M is a smooth n-manifold and ∇ an affine connection on TM; when a metric is present it is denoted g, with the associated inner product $\langle v, w \rangle$ on each tangent space. The Einstein summation convention (sum over any index appearing once up and once down) is in force. Christoffel symbols Γ^k_{ij} are defined relative to a coordinate frame by $\nabla_{\partial_i} \partial_j = \Gamma^k_{ij}\, \partial_k$, as in definition 2.1.3.

Definition 2.2.1: Torsion Tensor

Let ∇ be an affine connection on a smooth manifold M. The *torsion* of ∇ is the map $T\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ defined by

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

The connection ∇ is called *torsion-free* (or *symmetric*) if $T \equiv 0$, that is, if

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

for all $X, Y \in \mathfrak{X}(M)$.

Two things must be checked before the definition earns its name. First, T is antisymmetric, $T(X, Y) = -T(Y, X)$, which is immediate from the antisymmetry of the Lie bracket. Second, and less transparently, T is a *tensor*: although each of the three terms $\nabla_X Y$, $\nabla_Y X$, and $[X, Y]$ fails on its own to be $C^\infty(M)$-linear in X or Y, the non-tensorial pieces cancel in the combination. The following lemma records this, and shows at the same time that torsion-freeness is equivalent to the coordinate condition $\Gamma^k_{ij} = \Gamma^k_{ji}$ promised in the contract for this section.

Lemma 2.2.2: Torsion is a tensor; symmetry of the Christoffel symbols

Let ∇ be an affine connection on M with torsion T. Then:

1. T is $C^\infty(M)$-bilinear, hence defines a $(1,2)$-tensor field on M; its value $T(X,Y)_p$ depends only on X_p and Y_p.
2. In any coordinate chart, with Γ^k_{ij} the Christoffel symbols of ∇,

$$T(\partial_i, \partial_j) = (\Gamma^k_{ij} - \Gamma^k_{ji})\, \partial_k$$

   Consequently ∇ is torsion-free if and only if $\Gamma^k_{ij} = \Gamma^k_{ji}$ for all i, j, k in every chart.

***Proof*:**
Step 1: bilinearity. By antisymmetry $T(X,Y) = -T(Y,X)$, so it suffices to establish $C^\infty(M)$-linearity in the second argument; linearity in the first then follows. Additivity in Y is clear from the additivity of $\nabla_X(\cdot)$, of $\nabla_{(\cdot)}X$ read off from the fixed first slot, and of the bracket, together with $\mathbb{R}$-linearity. For the function-linearity, fix $f \in C^\infty(M)$ and compute each term of $T(X, fY)$ using the Leibniz rule for ∇ (definition 2.1.1) and the corresponding identity for the Lie bracket. The Leibniz rule gives

$$\nabla_X(fY) = (Xf)\,Y + f\,\nabla_X Y,$$

while the definition of the bracket on a product gives $[X, fY] = (Xf)\,Y + f\,[X,Y]$. The term $\nabla_{fY} X = f\,\nabla_Y X$ is function-linear directly, since ∇ is $C^\infty(M)$-linear in its subscript. Subtracting,

$$T(X, fY) = (Xf)\,Y + f\,\nabla_X Y - f\,\nabla_Y X - (Xf)\,Y - f\,[X,Y] = f\,T(X,Y)$$

The two occurrences of $(Xf)\,Y$ cancel, leaving $T(X, fY) = f\,T(X,Y)$. Thus T is $C^\infty(M)$-bilinear. A $C^\infty(M)$-bilinear map $\mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ is a $(1,2)$-tensor field, and the value at p of such a map depends only on the values X_p, Y_p of the arguments at p.

Step 2: coordinate expression. In a chart, coordinate vector fields commute, $[\partial_i, \partial_j] = 0$. Hence

$$T(\partial_i, \partial_j) = \nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i - [\partial_i, \partial_j] = \Gamma^k_{ij}\,\partial_k - \Gamma^k_{ji}\,\partial_k = (\Gamma^k_{ij} - \Gamma^k_{ji})\,\partial_k,$$

using $\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\,\partial_k$ from definition 2.1.3. Since T is a tensor by Step 1, it vanishes identically if and only if it vanishes on every pair of coordinate frame fields in every chart, and by the display just above this occurs if and only if $\Gamma^k_{ij} = \Gamma^k_{ji}$ for all indices. This is the asserted equivalence, completing the proof.

The name *symmetric* is now transparent: a connection is torsion-free exactly when its Christoffel symbols are symmetric in the two lower indices. Geometrically, torsion detects a certain twisting. Given two commuting directions ∂_i and ∂_j, one may transport ∂_j infinitesimally in the ∂_i direction and, separately, ∂_i in the ∂_j direction; torsion is the leading-order discrepancy between the two resulting infinitesimal displacements, over and above the discrepancy already carried by the Lie bracket. When it vanishes, the connection introduces no twist of its own beyond what the bracket of the fields already prescribes. The bracket term is exactly the correction that makes T tensorial, and hence a pointwise object, rather than a differential operator on fields.

Example 2.2.3: The Euclidean connection is torsion-free

On $\mathbb{R}^n$ with the standard flat connection $\overline{\nabla}$ of example 2.1.2, all Christoffel symbols vanish in the standard coordinates: $\overline{\nabla}_{\partial_i}\partial_j = 0$, so $\Gamma^k_{ij} = 0$. Trivially $\Gamma^k_{ij} = \Gamma^k_{ji}$, and by lemma 2.2.2 the Euclidean connection is torsion-free. This can also be seen without coordinates: for $X = X^i\partial_i$ and $Y = Y^j\partial_j$ the flat connection reads off $\overline{\nabla}_X Y = X(Y^j)\,\partial_j$, so

$$\overline{\nabla}_X Y - \overline{\nabla}_Y X = X(Y^j)\,\partial_j - Y(X^j)\,\partial_j = [X, Y],$$

the last equality being the standard coordinate formula for the bracket on $\mathbb{R}^n$. Thus $T \equiv 0$.

For a connection with nonzero torsion, keep the same manifold $\mathbb{R}^2$ but define ∇ by prescribing $\Gamma^1_{12} = 1$ and all other symbols zero. Then $\Gamma^1_{12} = 1 \neq 0 = \Gamma^1_{21}$, so $T(\partial_1, \partial_2) = (\Gamma^1_{12} - \Gamma^1_{21})\,\partial_1 = \partial_1 \neq 0$. This ∇ is a perfectly good affine connection, differing from $\overline{\nabla}$ by a $(1, 2)$-tensor as in proposition 2.1.5; it simply carries torsion. The example shows that torsion-freeness is a genuine restriction, not an automatic feature.

Torsion-freeness constrains a connection using no metric at all. The second property of interest is of the opposite character: it makes sense only once a metric g has been fixed, and it asks the connection to respect that metric. The requirement is that the product rule hold when a directional derivative meets an inner product.

Definition 2.2.4: Metric-Compatible Connection

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM. Then ∇ is *compatible with* g (or *metric-compatible*) if

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

for all vector fields $X, Y, Z \in \mathfrak{X}(M)$, where $\langle -, -\rangle = g(-, -)$ and $X\langle Y, Z\rangle$ denotes the derivative of the function $p \mapsto \langle Y_p, Z_p\rangle$ along X.

The left-hand side, $X\langle Y, Z\rangle$, is the ordinary directional derivative of a real-valued function; the right-hand side distributes that derivative across the two slots of the inner product by way of ∇. The identity therefore says that ∇ obeys the Leibniz rule for the bilinear form g exactly as it does for the product of a function with a field. A connection satisfying it treats g as an invariant: lengths and angles measured by g are compatible with the notion of differentiation supplied by ∇. The compact restatement is that the covariant derivative of the metric tensor vanishes, $\nabla g = 0$, and the equivalence of the two formulations is the content of proposition 2.2.6 below. Before that, the direct verification on the flat model.

Example 2.2.5: The Euclidean connection is metric-compatible

Let $\bar{g} = \delta_{ij}\,dx^i\,dx^j$ be the Euclidean metric on $\mathbb{R}^n$ (definition 1.1.1) and $\overline{\nabla}$ the flat connection of example 2.1.2. For $Y = Y^i\partial_i$ and $Z = Z^j\partial_j$ the inner product is the function $\langle Y, Z\rangle = \delta_{ij}Y^iZ^j = \sum_i Y^iZ^i$, whose derivative along X is, by the ordinary product rule,

$$X\langle Y, Z\rangle = \sum_i \big(X(Y^i)\,Z^i + Y^i\,X(Z^i)\big)$$

On the other hand $\overline{\nabla}_X Y = X(Y^i)\,\partial_i$ and $\overline{\nabla}_X Z = X(Z^i)\,\partial_i$, so

$$\langle \overline{\nabla}_X Y, Z\rangle + \langle Y, \overline{\nabla}_X Z\rangle = \sum_i X(Y^i)\, Z^i + \sum_i Y^i\, X(Z^i),$$

which is the same expression. Hence $\overline{\nabla}$ is compatible with $\bar{g}$. The computation used only that the components $\bar{g}_{ij} = \delta_{ij}$ are constant, so that no derivative of the metric appears; the coordinate criterion of proposition 2.2.6 makes precise the sense in which constancy of the g_{ij} relative to a ∇-parallel frame is what compatibility demands.

Metric compatibility admits three faces, and it is convenient to have all of them available before the fundamental theorem. The first is the vanishing of ∇g; the second is the Leibniz identity of definition 2.2.4; the third is a statement about parallel transport, the operation of section 2.4, namely that transport along any curve is a linear isometry. The equivalence of the first two is a self-contained tensor computation and is proved in full here. The third is stated as part of the same package and proved in section 2.4, once parallel transport is available; the forward reference is precise, and no circularity results, since the argument there uses only the Leibniz identity established here.

To state the first formulation, recall from definition 1.4.1 that g extends canonically to all tensor bundles, so that ∇g is the covariant derivative of g regarded as a covariant 2-tensor field: for vector fields X, Y, Z,

$$(\nabla_X g)(Y, Z) = X\big(g(Y, Z)\big) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z)$$

This is the standard Leibniz definition of the induced connection on the tensor bundle acting on a $(0, 2)$-tensor, and the reader may take it as the working definition of ∇g for the purposes of this section; it is developed for general tensor bundles in section 2.4.

Proposition 2.2.6: Characterizations of metric compatibility

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM. The following are equivalent.

1. $\nabla g = 0$, i.e. $\nabla_X g = 0$ for every $X \in \mathfrak{X}(M)$.
2. For all $X, Y, Z \in \mathfrak{X}(M)$,
$$X\,\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$
3. Parallel transport along every smooth curve in M is a linear isometry between the tangent spaces at its endpoints.

In any coordinate chart, conditions (i) and (ii) are further equivalent to

$$\partial_k g_{ij} = \Gamma^l_{ki}\, g_{lj} + \Gamma^l_{kj}\, g_{il} \qquad \text{for all } i, j, k,$$

where $g_{ij} = g(\partial_i, \partial_j)$.

***Proof*:**
Equivalence of (i) *and* (ii). Unwinding the definition of ∇g recorded above, for any X, Y, Z,

$$(\nabla_X g)(Y, Z) = X \langle Y, Z \rangle - \langle \nabla_X Y, Z \rangle - \langle Y, \nabla_X Z \rangle$$

The right-hand side is exactly the difference of the two sides of the identity in (ii). Hence $(\nabla_X g)(Y, Z) = 0$ for all Y, Z if and only if the identity of (ii) holds for all Y, Z (with this same X); requiring this for all X gives $\nabla g = 0$ if and only if (ii) holds for all X, Y, Z. Thus (i) $\iff$ (ii).

It remains to justify that the pointwise vanishing of the tensor $\nabla_X g$ is detected by the identity in (ii), i.e. that $\nabla_X g$ really is a tensor so that "$(\nabla_X g)(Y, Z) = 0$ for all Y, Z" is equivalent to $\nabla_X g = 0$. This is standard for the induced connection on a tensor bundle, but the tensoriality can be seen directly from the display: the map $(Y, Z) \mapsto X \langle Y, Z \rangle - \langle \nabla_X Y, Z \rangle - \langle Y, \nabla_X Z \rangle$ is $C^\infty(M)$-bilinear. Indeed, for $f \in C^\infty(M)$,

$$X \langle fY, Z \rangle = X\big(f \langle Y, Z \rangle\big) = (Xf) \langle Y, Z \rangle + f\, X \langle Y, Z \rangle ,$$

while $\langle \nabla_X (fY), Z \rangle = \langle (Xf)Y + f \nabla_X Y, Z \rangle = (Xf) \langle Y, Z \rangle + f \langle \nabla_X Y, Z \rangle$ and $\langle fY, \nabla_X Z \rangle = f \langle Y, \nabla_X Z \rangle$. Subtracting, the $(Xf) \langle Y, Z \rangle$ terms cancel and the expression scales by f; linearity in Z is identical. So $\nabla_X g$ is a well-defined $(0, 2)$-tensor and vanishes iff it vanishes on all coordinate pairs.

Coordinate form. Apply (ii) to $X = \partial_k$, $Y = \partial_i$, $Z = \partial_j$. Then $X \langle Y, Z \rangle = \partial_k g_{ij}$, while $\nabla_{\partial_k} \partial_i = \Gamma^l_{ki} \partial_l$ and $\nabla_{\partial_k} \partial_j = \Gamma^l_{kj} \partial_l$ give

$$\langle \nabla_{\partial_k} \partial_i, \partial_j \rangle + \langle \partial_i, \nabla_{\partial_k} \partial_j \rangle = \Gamma^l_{ki}\, g_{lj} + \Gamma^l_{kj}\, g_{il}$$

Equating the two, (ii) restricted to coordinate frames is precisely the displayed identity $\partial_k g_{ij} = \Gamma^l_{ki} g_{lj} + \Gamma^l_{kj} g_{il}$. Conversely, since both sides of the identity in (ii) are $C^\infty(M)$-linear in Y and Z (shown above) and $\mathbb{R}$-linear in X together with the Leibniz rule in X, an identity that holds on coordinate frames holds for all fields: write $X = X^k \partial_k$, $Y = Y^i \partial_i$, $Z = Z^j \partial_j$ and expand. Hence the coordinate identity is equivalent to (ii), and therefore to (i).

Equivalence with (iii). Parallel transport along curves is constructed in section 2.4; there proposition 2.4.4 proves that when ∇ satisfies the Leibniz identity of (ii), the transport $P^\gamma_{t_0, t_1}$ along any curve γ is a linear isometry, and, conversely, that if every parallel transport is a linear isometry then (ii) holds. That argument uses only (ii) and the defining equation $D_t V = 0$ of a parallel field, so it may be invoked here without circularity. Combining it with the equivalence (i) $\iff$ (ii) above yields the equivalence of all three, as we sought to show.

The coordinate identity in proposition 2.2.6 is the working form of metric compatibility. Read as a formula for the derivatives $\partial_k g_{ij}$ in terms of the Christoffel symbols, it says that the rate of change of the metric coefficients is entirely dictated by the connection. This is precisely the constraint that will, in the next section, be solved together with the symmetry $\Gamma^k_{ij} = \Gamma^k_{ji}$ of lemma 2.2.2 to recover the Christoffel symbols uniquely from g: two linear conditions on the Γ^k_{ij}, symmetry from torsion-freeness and the derivative identity from compatibility, together determine them. The example below illustrates the derivative identity in the one setting where it is nontrivial yet fully explicit, and previews how it forces the values of the Christoffel symbols.

Example 2.2.7: Metric compatibility on a warped metric
Consider the upper half-plane coordinates (x, y) with the metric

$$g = y^2\,dx^2 + y^2\,dy^2, \qquad y > 0,$$

so that $g_{11} = g_{22} = y^2$ and $g_{12} = g_{21} = 0$; write $\partial_1 = \partial_x$, $\partial_2 = \partial_y$. The only nonzero coordinate derivatives are $\partial_2 g_{11} = \partial_2 g_{22} = 2y$. Metric compatibility, in the form of proposition 2.2.6, is the system

$$\partial_k g_{ij} = \Gamma^l_{ki}\,g_{lj} + \Gamma^l_{kj}\,g_{il}$$

Take, for instance, $k = 2$, $i = j = 1$: the left side is $\partial_2 g_{11} = 2y$, and since g is diagonal the right side is $\Gamma^1_{21} g_{11} + \Gamma^1_{21} g_{11} = 2y^2\,\Gamma^1_{21}$. Compatibility therefore forces $\Gamma^1_{21} = 1/y$. Taking $k = 1$, $i = j = 1$ gives $\partial_1 g_{11} = 0 = 2y^2\,\Gamma^1_{11}$, so $\Gamma^1_{11} = 0$; taking $k = 2$, $i = j = 2$ gives $\partial_2 g_{22} = 2y = 2y^2\,\Gamma^2_{22}$, so $\Gamma^2_{22} = 1/y$. Each equation of the compatibility system pins down a combination of Christoffel symbols against a known derivative of g. On its own this system does not determine every Γ^k_{ij}, since compatibility alone leaves the antisymmetric-in-(i, j) part free; adjoining the symmetry $\Gamma^k_{ij} = \Gamma^k_{ji}$ from lemma 2.2.2 removes exactly that freedom. The next section carries out this solution in general and in closed form.

The two properties are logically independent. The flat connection on $\mathbb{R}^n$ has both, as example 2.2.3 and example 2.2.5 record. A connection can be torsion-free without being compatible with a given g: on $\mathbb{R}^n$ scale the metric to $\bar{g}$ but keep some non-flat torsion-free ∇ whose symbols are symmetric yet do not satisfy $\partial_k \bar{g}_{ij} = \Gamma^l_{ki}\bar{g}_{lj} + \Gamma^l_{kj}\bar{g}_{il}$, which for the constant Euclidean metric reads $\Gamma^l_{ki}\delta_{lj} + \Gamma^l_{kj}\delta_{il} = 0$, i.e. $\Gamma^j_{ki} + \Gamma^i_{kj} = 0$; a nonzero symmetric Γ violating this fails compatibility while retaining torsion-freeness. Conversely a connection can be compatible with g while carrying torsion, since the derivative identity of proposition 2.2.6 constrains only the symmetric-in-(i, j) combination $\Gamma^l_{ki} g_{lj} + \Gamma^l_{kj} g_{il}$ and leaves the torsion $\Gamma^k_{ij} - \Gamma^k_{ji}$ unconstrained. That the two conditions are independent, yet jointly determine a connection, is what makes the theorem of the next section possible: the symmetric part of Γ is fixed by compatibility and the antisymmetric part by torsion-freeness.

2.3 The Fundamental Theorem of Riemannian Geometry

Two structures now sit on a (pseudo-)Riemannian manifold, and so far they have led separate lives. The metric g measures lengths and angles; a connection ∇ differentiates one vector field along another. The metric knows nothing about differentiation, and a general affine connection knows nothing about the metric: a smooth manifold carries a vast supply of connections (their difference is any $(1, 2)$-tensor, by proposition 2.1.5), and none of them is singled out by the smooth structure alone. On a Riemannian manifold, though, we have grown accustomed to differentiating vectors in a way that respects lengths, as one does for the position and velocity of a curve in $\mathbb{R}^n$. The question is whether the metric picks out a connection of its own, and if so, how many.

Two conditions on a connection have already earned their names. Metric compatibility (definition 2.2.4) is the demand that ∇ and g agree, so that differentiating an inner product obeys a product rule and, equivalently, parallel transport preserves lengths; without it, a vector could grow or shrink under transport and the geometry of ∇ would drift away from the geometry of g. Torsion-freeness (definition 2.2.1) is the demand that the connection add no antisymmetric part beyond the Lie bracket, so that second covariant derivatives are as symmetric as possible and, in coordinates, $\Gamma^k_{ij} = \Gamma^k_{ji}$. Each condition is natural on its own; neither, on its own, determines ∇. The content of this section is that the two together determine ∇ completely and are simultaneously satisfiable. The argument is short and self-contained, and it is worth watching how the two conditions interlock: the same three-term cyclic manipulation that proves uniqueness also produces the formula that proves existence.

Theorem 2.3.1: Fundamental Theorem of Riemannian Geometry

Let (M, g) be a (pseudo-)Riemannian manifold. There exists a unique affine connection ∇ on TM that is both torsion-free and compatible with g. It is characterized by the *Koszul formula*: for all vector fields X, Y, Z on M,

$$\begin{aligned} 2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle \\ - \langle X, [Y, Z]\rangle + \langle Y, [Z, X]\rangle + \langle Z, [X, Y]\rangle . \end{aligned} \tag{2.3}$$

***Proof*:**
The proof has two halves. First uniqueness: any torsion-free, metric-compatible connection is forced to satisfy (2.3), and the right-hand side determines $\nabla_X Y$ because g is nondegenerate. Then existence: the right-hand side of (2.3) defines a $\nabla_X Y$, and this ∇ is verified to be a connection that is torsion-free and metric-compatible.

Step 1: derivation of the Koszul formula (uniqueness). Suppose ∇ is torsion-free and compatible with g. Fix vector fields X, Y, Z. Compatibility (definition 2.2.4) applied to the three pairs, cyclically, gives

$$\begin{aligned} X\langle Y, Z\rangle &= \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle , \\ Y\langle Z, X\rangle &= \langle \nabla_Y Z, X\rangle + \langle Z, \nabla_Y X\rangle , \\ Z\langle X, Y\rangle &= \langle \nabla_Z X, Y\rangle + \langle X, \nabla_Z Y\rangle \end{aligned}$$

Add the first two and subtract the third:

$$X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle + \langle \nabla_Y Z, X\rangle + \langle Z, \nabla_Y X\rangle - \langle \nabla_Z X, Y\rangle - \langle X, \nabla_Z Y\rangle$$

Now eliminate every covariant derivative whose subscript is not X by torsion-freeness (definition 2.2.1), which supplies $\nabla_X Z = \nabla_Z X + [X, Z]$, $\nabla_Y X = \nabla_X Y + [Y, X]$, and $\nabla_Y Z = \nabla_Z Y + [Y, Z]$. Substituting these into the second, third, and fourth terms on the right,

$$\begin{aligned} X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle = {} & \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_Z X\rangle + \langle Y, [X, Z]\rangle + \langle \nabla_Z Y, X\rangle + \langle [Y, Z], X\rangle \\ & + \langle Z, \nabla_X Y\rangle + \langle Z, [Y, X]\rangle - \langle \nabla_Z X, Y\rangle - \langle X, \nabla_Z Y\rangle . \end{aligned}$$

The pair $\langle Y, \nabla_Z X\rangle - \langle \nabla_Z X, Y\rangle$ vanishes and the pair $\langle \nabla_Z Y, X\rangle - \langle X, \nabla_Z Y\rangle$ vanishes, by symmetry of g; the two remaining $\nabla_X Y$ terms combine as $\langle \nabla_X Y, Z\rangle + \langle Z, \nabla_X Y\rangle = 2\langle \nabla_X Y, Z\rangle$. What survives is

$$X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle = 2\langle \nabla_X Y, Z\rangle + \langle Y, [X, Z]\rangle + \langle X, [Y, Z]\rangle + \langle Z, [Y, X]\rangle$$

Solve for $2\langle \nabla_X Y, Z\rangle$, then normalize the brackets using $[X, Z] = -[Z, X]$ and $[Y, X] = -[X, Y]$:

$$2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle X, [Y, Z]\rangle + \langle Y, [Z, X]\rangle + \langle Z, [X, Y]\rangle ,$$

which is (2.3). Every quantity on the right depends only on g, on the bracket, and on X, Y, Z; no reference to ∇ survives. Hence a torsion-free metric-compatible connection must satisfy (2.3).

This already gives uniqueness. Fix X, Y and a point p. As Z ranges over all vector fields, the right-hand side of (2.3) assigns to each $Z_p \in T_pM$ a number, and by inspection this assignment is $\mathbb{R}$-linear in Z_p once X, Y are fixed (each term is $\mathbb{R}$-linear in Z, and the value at p depends only on Z_p, a point verified in Step 2 below). Thus the right-hand side is the pairing of a well-defined covector with Z_p. Because g_p is nondegenerate (definition 1.1.3), the equation $2\langle W, Z_p\rangle = (\text{that covector})(Z_p)$ for all Z_p has at most one solution $W = (\nabla_X Y)_p$. So any two torsion-free metric-compatible connections agree.

Step 2: the Koszul formula defines a connection (existence). Define $\nabla_X Y$ by requiring that (2.3) hold for all Z; concretely, for fixed X, Y let $\omega(Z)$ denote the right-hand side of (2.3), and set $\nabla_X Y$ to be the unique vector field with $2\langle \nabla_X Y, Z\rangle = \omega(Z)$ for all Z. For this to make sense two things must be checked: that ω is a 1-form (that is, $C^\infty(M)$-linear and pointwise in Z, so that the musical isomorphism $\sharp$ of definition 1.4.1 produces a vector field $\nabla_X Y := \frac{1}{2}\,\omega^\sharp$), and that the resulting operator is a connection.

First, ω is additive in Z termwise. For $C^\infty(M)$-linearity, replace Z by fZ with $f \in C^\infty(M)$. Each of the six terms of ω either scales by f or produces one additional term through a Leibniz rule, using $[Y, fZ] = f[Y, Z] + (Yf)Z$ and $[fZ, X] = f[Z, X] - (Xf)Z$:

$$\begin{aligned}
X\langle Y, fZ\rangle &= fX\langle Y, Z\rangle + (Xf)\langle Y, Z\rangle, & Y\langle fZ, X\rangle &= fY\langle Z, X\rangle + (Yf)\langle Z, X\rangle,\\
-fZ\langle X, Y\rangle &= f\big(-Z\langle X, Y\rangle\big), & -\langle X, [Y, fZ]\rangle &= -f\langle X, [Y, Z]\rangle - (Yf)\langle X, Z\rangle,\\
\langle Y, [fZ, X]\rangle &= f\langle Y, [Z, X]\rangle - (Xf)\langle Y, Z\rangle, & \langle fZ, [X, Y]\rangle &= f\langle Z, [X, Y]\rangle
\end{aligned}$$

The four terms beyond $f\,\omega(Z)$ are $(Xf)\langle Y, Z\rangle$, $(Yf)\langle Z, X\rangle$, $-(Yf)\langle X, Z\rangle$, and $-(Xf)\langle Y, Z\rangle$. By symmetry of g, $\langle Y, Z\rangle = \langle Z, Y\rangle$ and $\langle Z, X\rangle = \langle X, Z\rangle$, so $(Xf)\langle Y, Z\rangle - (Xf)\langle Y, Z\rangle = 0$ and $(Yf)\langle Z, X\rangle - (Yf)\langle X, Z\rangle = 0$. All four cancel, so $\omega(fZ) = f\,\omega(Z)$. Hence ω is $C^\infty(M)$-linear in Z, and by the tensor characterization of 1-forms ([6]) it is a 1-form, its value at p depends only on Z_p, and $\nabla_X Y := \frac{1}{2}\,\omega^\sharp$ is a well-defined smooth vector field.

Now verify the connection axioms (definition 2.1.1). Write $\Phi(X, Y, Z)$ for the right-hand side of (2.3), so $2\langle \nabla_X Y, Z\rangle = \Phi(X, Y, Z)$ for all Z.

$\mathbb{R}$-bilinearity. Φ is $\mathbb{R}$-linear in X and in Y termwise (each metric-derivative and bracket term is $\mathbb{R}$-linear in each slot), so $\nabla_X Y$ is $\mathbb{R}$-linear in X and in Y.

$C^\infty(M)$-linearity in X. Replace X by fX. The terms of $\Phi(fX, Y, Z)$ that differ from $f\,\Phi(X, Y, Z)$ arise where X acts as a derivation or enters a bracket, using $[Z, fX] = f[Z, X] + (Zf)X$ and $[fX, Y] = f[X, Y] - (Yf)X$:

$$\begin{aligned}
Y\langle Z, fX\rangle &= fY\langle Z, X\rangle + (Yf)\langle Z, X\rangle, & -Z\langle fX, Y\rangle &= -fZ\langle X, Y\rangle - (Zf)\langle X, Y\rangle,\\
\langle Y, [Z, fX]\rangle &= f\langle Y, [Z, X]\rangle + (Zf)\langle Y, X\rangle, & \langle Z, [fX, Y]\rangle &= f\langle Z, [X, Y]\rangle - (Yf)\langle Z, X\rangle,
\end{aligned}$$

while $X\langle Y, Z\rangle \mapsto fX\langle Y, Z\rangle$ and $-\langle fX, [Y, Z]\rangle = -f\langle X, [Y, Z]\rangle$ produce no extra terms. The four extra terms are $(Yf)\langle Z, X\rangle$, $-(Zf)\langle X, Y\rangle$, $+(Zf)\langle Y, X\rangle$, and $-(Yf)\langle Z, X\rangle$. The two (Yf) terms cancel outright, and by symmetry of g the two (Zf) terms give $-(Zf)\langle X, Y\rangle + (Zf)\langle Y, X\rangle = 0$. Hence $\Phi(fX, Y, Z) = f\,\Phi(X, Y, Z)$ for all Z, so $\nabla_{fX} Y = f\nabla_X Y$.

Leibniz in Y. Replace Y by fY. The terms that differ from $f\,\Phi(X, Y, Z)$ arise where Y acts as a derivation or enters a bracket, using $[fY, Z] = f[Y, Z] - (Zf)Y$ and $[X, fY] = f[X, Y] + (Xf)Y$:

$$\begin{aligned}
X\langle fY, Z\rangle &= fX\langle Y, Z\rangle + (Xf)\langle Y, Z\rangle, & -Z\langle X, fY\rangle &= -fZ\langle X, Y\rangle - (Zf)\langle X, Y\rangle,\\
-\langle X, [fY, Z]\rangle &= -f\langle X, [Y, Z]\rangle + (Zf)\langle X, Y\rangle, & \langle Z, [X, fY]\rangle &= f\langle Z, [X, Y]\rangle + (Xf)\langle Z, Y\rangle,
\end{aligned}$$

while $Y\langle Z,X\rangle \mapsto fY\langle Z,X\rangle$ and $\langle fY,[Z,X]\rangle = f\langle Y,[Z,X]\rangle$ produce no extra terms. The four extra terms are $(Xf)\langle Y,Z\rangle$, $-(Zf)\langle X,Y\rangle$, $+(Zf)\langle X,Y\rangle$, and $(Xf)\langle Z,Y\rangle$. The two (Zf) terms cancel, and by symmetry of g the two (Xf) terms combine to $2(Xf)\langle Y,Z\rangle$. Therefore

$$\Phi(X,fY,Z) = f\,\Phi(X,Y,Z) + 2(Xf)\langle Y,Z\rangle = 2f\langle \nabla_X Y,Z\rangle + 2(Xf)\langle Y,Z\rangle\,,$$

so $2\langle \nabla_X(fY),Z\rangle = 2\langle f\nabla_X Y + (Xf)Y,Z\rangle$ for all Z, and nondegeneracy gives $\nabla_X(fY) = f\nabla_X Y + (Xf)Y$. Thus ∇ is a connection.

Step 3: torsion-freeness. By the defining relation,

$$2\langle \nabla_X Y - \nabla_Y X, Z\rangle = \Phi(X,Y,Z) - \Phi(Y,X,Z)$$

Interchanging X and Y in (2.3), the first three (symmetric) metric-derivative terms $X\langle Y,Z\rangle + Y\langle Z,X\rangle - Z\langle X,Y\rangle$ are invariant under $X \leftrightarrow Y$ (using symmetry of g in the last), so they cancel in the difference. Among the bracket terms, $-\langle X,[Y,Z]\rangle \to -\langle Y,[X,Z]\rangle$ and $\langle Y,[Z,X]\rangle \to \langle X,[Z,Y]\rangle$ and $\langle Z,[X,Y]\rangle \to \langle Z,[Y,X]\rangle$. Hence

$$\Phi(X,Y,Z)-\Phi(Y,X,Z) = -\langle X,[Y,Z]\rangle+\langle Y,[Z,X]\rangle+\langle Z,[X,Y]\rangle+\langle Y,[X,Z]\rangle-\langle X,[Z,Y]\rangle-\langle Z,[Y,X]\rangle$$

Now $[Z,Y] = -[Y,Z]$ and $[X,Z] = -[Z,X]$ and $[Y,X] = -[X,Y]$, so $-\langle X,[Y,Z]\rangle - \langle X,[Z,Y]\rangle = 0$, $\langle Y,[Z,X]\rangle + \langle Y,[X,Z]\rangle = 0$, and $\langle Z,[X,Y]\rangle - \langle Z,[Y,X]\rangle = 2\langle Z,[X,Y]\rangle$. Therefore

$$2\langle \nabla_X Y - \nabla_Y X, Z\rangle = 2\langle [X,Y],Z\rangle \quad \text{for all } Z,$$

and nondegeneracy gives $\nabla_X Y - \nabla_Y X = [X,Y]$, that is, the torsion vanishes (definition 2.2.1).

Step 4: metric compatibility. Fix X,Y,Z. From the defining relation, $2\langle \nabla_X Y,Z\rangle = \Phi(X,Y,Z)$ and $2\langle Y,\nabla_X Z\rangle = 2\langle \nabla_X Z,Y\rangle = \Phi(X,Z,Y)$. Add:

$$2\big(\langle \nabla_X Y,Z\rangle + \langle Y,\nabla_X Z\rangle\big) = \Phi(X,Y,Z) + \Phi(X,Z,Y)$$

Writing out both and using symmetry of g, the term $X\langle Y,Z\rangle$ appears in $\Phi(X,Y,Z)$ and $X\langle Z,Y\rangle$ appears in $\Phi(X,Z,Y)$, together giving $2X\langle Y,Z\rangle$. The remaining metric-derivative terms are $Y\langle Z,X\rangle - Z\langle X,Y\rangle$ from the first and $Z\langle Y,X\rangle - Y\langle X,Z\rangle$ from the second; since $\langle Z,X\rangle = \langle X,Z\rangle$ and $\langle Y,X\rangle = \langle X,Y\rangle$, these cancel in pairs. For the bracket terms, $\Phi(X,Y,Z)$ contributes $-\langle X,[Y,Z]\rangle + \langle Y,[Z,X]\rangle + \langle Z,[X,Y]\rangle$ and $\Phi(X,Z,Y)$ contributes $-\langle X,[Z,Y]\rangle + \langle Z,[Y,X]\rangle + \langle Y,[X,Z]\rangle$; using $[Z,Y] = -[Y,Z]$, $[Y,X] = -[X,Y]$, $[X,Z] = -[Z,X]$, each pair cancels. Hence

$$\langle \nabla_X Y,Z\rangle + \langle Y,\nabla_X Z\rangle = X\langle Y,Z\rangle\,,$$

which is exactly the compatibility identity of definition 2.2.4. So ∇ is metric-compatible.

Existence and uniqueness are both established, completing the proof.

The proof deserves a second look for what makes it work. Uniqueness is the sturdier half: it uses nothing about M beyond nondegeneracy of g, and the same cyclic combination of the three compatibility equations that eliminates the unknown ∇ from the right-hand side is the standard move whenever a symmetric bilinear form is to be recovered from its derivatives, as in the polarization identity for an inner product. Existence is the more delicate half only because one must confirm that the formula, read as a definition, respects the tensorial and Leibniz bookkeeping of a connection; the cancellations in Steps 2 through 4 are all the same cancellation, driven by the symmetry of

g and the antisymmetry of the bracket. Notice that positive-definiteness was never invoked, only nondegeneracy: the theorem holds verbatim for Lorentzian and general pseudo-Riemannian metrics, which is why every spacetime carries a canonical connection.

Definition 2.3.2: Levi-Civita Connection

Let (M, g) be a (pseudo-)Riemannian manifold. The unique torsion-free, metric-compatible affine connection on TM provided by theorem 2.3.1 is called the *Levi-Civita connection* of (M, g), also called the *Riemannian connection*. It is denoted ∇ (or ∇^g when the metric must be displayed).

From now on, the symbol ∇ on a (pseudo-)Riemannian manifold means the Levi-Civita connection unless another connection is named. Two consequences of definition 2.3.2 organize the rest of the theory. First, every metric invariant of ∇, the geodesics, parallel transport, and the curvature tensor built later, is an invariant of g alone, because ∇ is manufactured from g; an isometry φ (definition 1.3.1) therefore carries the Levi-Civita connection of the source to that of the target, since $\varphi^*\tilde{g} = g$ forces φ to intertwine the two connections that g and $\tilde{g}$ uniquely determine. Second, the connection is a local object: the Koszul formula reads off $\nabla_X Y$ from g and derivatives of g on any neighborhood, so ∇ on an open set U is the Levi-Civita connection of $(U, g|_U)$. Both statements will be used repeatedly and without further comment.

To make the connection computable one converts the Koszul formula into coordinates. In a chart the coordinate frame (∂_i) has vanishing brackets, $[\partial_i, \partial_j] = 0$, so (2.3) collapses to a formula for the Christoffel symbols in terms of the metric components and their first derivatives. Throughout what follows the Einstein summation convention is in force: an index appearing once as an upper and once as a lower index is summed over $1, \dots, n$.

Proposition 2.3.3: Christoffel Symbols of the Levi-Civita Connection

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇. In any smooth chart $(U, (x^i))$, with metric components $g_{ij} = \langle \partial_i, \partial_j \rangle$ and inverse matrix (g^{ij}), the Christoffel symbols of ∇ (definition 2.1.3) are

$$\Gamma^k_{ij} = \tfrac{1}{2}\, g^{kl} \big(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij} \big)$$

In particular $\Gamma^k_{ij} = \Gamma^k_{ji}$, and the connection is symmetric in the lower indices.

***Proof*:**
Apply the Koszul formula (2.3) to the coordinate frame, taking $X = \partial_i$, $Y = \partial_j$, $Z = \partial_l$. Every Lie bracket of coordinate fields vanishes, $[\partial_i, \partial_j] = 0$, so the three bracket terms drop out and the three metric-derivative terms survive:

$$2 \langle \nabla_{\partial_i} \partial_j, \partial_l \rangle = \partial_i \langle \partial_j, \partial_l \rangle + \partial_j \langle \partial_l, \partial_i \rangle - \partial_l \langle \partial_i, \partial_j \rangle = \partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}$$

By definition 2.1.3, $\nabla_{\partial_i} \partial_j = \Gamma^m_{ij} \partial_m$, so the left-hand side is

$$2 \left\langle \Gamma^m_{ij} \partial_m, \partial_l \right\rangle = 2\Gamma^m_{ij} \langle \partial_m, \partial_l \rangle = 2\Gamma^m_{ij} g_{ml}$$

Equating the two expressions,

$$2\Gamma^m_{ij} g_{ml} = \partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}$$

To solve for Γ^k_{ij}, contract both sides with the inverse metric g^{kl}, summing over l. Since $g^{kl}g_{ml} = \delta^k_m$,

$$2\Gamma^m_{ij}\delta^k_m = 2\Gamma^k_{ij} = g^{kl}\big(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\big),$$

which is the stated formula after dividing by 2. Symmetry in i, j is visible from the right-hand side: interchanging i and j swaps the first two terms $\partial_i g_{jl} \leftrightarrow \partial_j g_{il}$ and fixes the third, using $g_{ij} = g_{ji}$; the sum is unchanged. Equivalently, symmetry follows from torsion-freeness of ∇ together with $[\partial_i, \partial_j] = 0$, as required.

The formula for Γ^k_{ij} is the working definition of the Levi-Civita connection in practice: to differentiate covariantly, one reads off g_{ij} from the chart, inverts the matrix to get g^{kl}, and differentiates. Everything downstream, the geodesic equation, parallel transport, and curvature, is assembled from these symbols. Because the Γ^k_{ij} are built from first derivatives of the metric, they encode no invariant information at a single point (they can be made to vanish at any chosen point by passing to normal coordinates, developed in section 3.2); the invariant content of the metric first appears in the second derivatives of g, which is where curvature will come from.

Example 2.3.4: Christoffel Symbols of a Surface of Revolution

Consider a surface of revolution with the warped metric

$$g = dr^2 + f(r)^2\, d\theta^2, \qquad f(r) > 0,$$

in coordinates $(x^1, x^2) = (r, \theta)$; this is the form computed in example 1.5.4. The metric matrix and its inverse are

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & f(r)^2 \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & f(r)^{-2} \end{pmatrix}$$

The only nonzero partial derivatives of the components are $\partial_r g_{\theta\theta} = 2ff'$, where $f' = df/dr$. Since (g^{ij}) is diagonal, the sum over l in proposition 2.3.3 collapses to the single term $l = k$:

$$\Gamma^k_{ij} = \tfrac{1}{2}\, g^{kk}\big(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}\big) \qquad \text{(no sum on } k\text{)}$$

Run through the components. For $k = r$ ($g^{rr} = 1$): the only surviving term needs a nonzero ∂g, so $\Gamma^r_{\theta\theta} = \frac{1}{2} \cdot 1 \cdot (-\partial_r g_{\theta\theta}) = -ff'$, and $\Gamma^r_{rr} = \Gamma^r_{r\theta} = 0$. For $k = \theta$ ($g^{\theta\theta} = f^{-2}$): $\Gamma^\theta_{r\theta} = \frac{1}{2} f^{-2}\big(\partial_r g_{\theta\theta} + \partial_\theta g_{r\theta} - \partial_\theta g_{r\theta}\big) = \frac{1}{2} f^{-2}(2ff') = f'/f$, and by symmetry $\Gamma^\theta_{\theta r} = f'/f$, while $\Gamma^\theta_{rr} = \Gamma^\theta_{\theta\theta} = 0$. The nonzero symbols are therefore

$$\Gamma^r_{\theta\theta} = -f(r)f'(r), \qquad \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{f'(r)}{f(r)}$$

Two checks confirm the answer. Symmetry in the lower indices holds, as proposition 2.3.3 guarantees. And in the flat case $f(r) = r$ (polar coordinates on $\mathbb{R}^2$, whose induced metric is exactly $dr^2 + r^2 d\theta^2$) these become $\Gamma^r_{\theta\theta} = -r$ and $\Gamma^\theta_{r\theta} = 1/r$, the familiar Christoffel symbols of the plane in polar coordinates; the metric $\bar{g} = \delta_{ij} dx^i dx^j$ of example 1.1.2 carries no curvature, yet its Christoffel symbols in polar coordinates are nonzero, a reminder that the Γ^k_{ij} measure the chart, not the geometry.

The round 2-sphere is a special case of the previous computation, and its Christoffel symbols are worth having explicitly, since they drive the geodesic and curvature calculations of later chapters.

Example 2.3.5: Christoffel Symbols of the Round Sphere

On the round unit 2-sphere S^2, away from the poles, use the standard spherical coordinates $(x^1, x^2) = (\theta, \varphi)$ in which the induced metric is

$$g = d\theta^2 + \sin^2\theta\, d\varphi^2, \qquad 0 < \theta < \pi,$$

with θ the polar angle measured from the north pole and φ the azimuthal angle. The metric matrix and its inverse are

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^{-2}\theta \end{pmatrix}$$

Only one component depends on the coordinates, and its single nonzero partial derivative is

$$\partial_\theta g_{\varphi\varphi} = 2\sin\theta\cos\theta,$$

while every other $\partial_k g_{ij}$ vanishes. Since (g^{ij}) is diagonal, the contraction over l in proposition 2.3.3 keeps only the term $l = k$, giving

$$\Gamma^k_{ij} = \tfrac{1}{2}\, g^{kk}\big(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}\big) \qquad \text{(no sum on } k\text{)}$$

For $k = \theta$ ($g^{\theta\theta} = 1$) the only combination that meets a nonzero derivative is $i = j = \varphi$, where the two symmetric terms $\partial_\varphi g_{\varphi\theta}$ vanish and the subtracted term survives:

$$\Gamma^\theta_{\varphi\varphi} = \tfrac{1}{2}\cdot 1 \cdot \big(-\partial_\theta g_{\varphi\varphi}\big) = -\tfrac{1}{2}(2\sin\theta\cos\theta) = -\sin\theta\cos\theta,$$

and $\Gamma^\theta_{\theta\theta} = \Gamma^\theta_{\theta\varphi} = 0$. For $k = \varphi$ ($g^{\varphi\varphi} = \sin^{-2}\theta$) the surviving case is $\{i, j\} = \{\theta, \varphi\}$:

$$\Gamma^\varphi_{\theta\varphi} = \tfrac{1}{2}\,\sin^{-2}\theta\big(\partial_\theta g_{\varphi\varphi} + \partial_\varphi g_{\theta\varphi} - \partial_\varphi g_{\theta\varphi}\big) = \tfrac{1}{2}\,\sin^{-2}\theta\,(2\sin\theta\cos\theta) = \cot\theta,$$

with $\Gamma^\varphi_{\varphi\theta} = \cot\theta$ by symmetry, and $\Gamma^\varphi_{\theta\theta} = \Gamma^\varphi_{\varphi\varphi} = 0$. The nonzero Christoffel symbols of the round sphere are therefore

$$\Gamma^\theta_{\varphi\varphi} = -\sin\theta\cos\theta, \qquad \Gamma^\varphi_{\theta\varphi} = \Gamma^\varphi_{\varphi\theta} = \cot\theta,$$

and all others vanish. This is the case $f(\theta) = \sin\theta$ of example 2.3.4, with the radial coordinate r there playing the role of θ here: substituting $f = \sin\theta$ gives $-ff' = -\sin\theta\cos\theta$ and $f'/f = \cot\theta$, matching term by term, since the round sphere is the surface of revolution generated by a unit-speed meridian. The blow-up of $\cot\theta$ as $\theta \to 0, \pi$ reflects the breakdown of the coordinates at the poles, not of the geometry, exactly as $1/r$ does at the origin in polar coordinates.

2.4 Parallel Transport

The obstruction that motivated the connection in the first place was the absence of any canonical way to compare tangent vectors at different points of a manifold: T_pM and T_qM are distinct vector spaces with no preferred identification. A connection repairs this infinitesimally, by prescribing how to differentiate a vector field along a curve. Parallel transport is the finite, integrated form of that repair. Given a curve γ from p to q and a tangent vector at p, one drags the vector along γ so that it never changes from the connection's point of view, that is, so that its covariant derivative along γ

stays zero. The vector that arrives at q is the parallel transport of the starting vector. This furnishes exactly the missing identification $T_pM \to T_qM$, with one caveat that will run through the rest of the theory: the identification depends on the curve. Transporting the same vector along two different paths from p to q can produce two different vectors at q, and the discrepancy is the first visible symptom of curvature.

The construction is a matter of solving a linear ordinary differential equation. In a chart the condition $D_tV = 0$ becomes a first-order linear system for the component functions of V, and the existence and uniqueness theory for such systems hands over, at once, that a parallel field is determined by its value at any single time and that parallel transport is a linear isomorphism between tangent spaces. When the connection is the Levi-Civita connection of a metric, transport preserves lengths and angles: it is a linear isometry, the promised third face of metric compatibility. This section carries out the construction, records these two properties in full, and then extends the covariant derivative from the tangent bundle to all tensor bundles, so that ∇ can differentiate metrics, forms, and endomorphisms, and does so compatibly with contraction and with the raising and lowering of indices. Throughout, M is a smooth n-manifold, ∇ an affine connection on TM (definition 2.1.1), and $\gamma\colon I \to M$ a smooth curve on an interval $I \subseteq \mathbb{R}$; the covariant derivative D_t along γ is the operator of definition 2.1.6. The Einstein summation convention, summing over any index appearing once up and once down, is in force.

Definition 2.4.1: Parallel Vector Field and Parallel Transport

Let ∇ be an affine connection on TM and $\gamma\colon I \to M$ a smooth curve. A vector field $V \in \mathfrak{X}(\gamma)$ along γ is *parallel* (along γ, with respect to ∇) if

$$D_tV \equiv 0 \quad \text{on } I$$

For $t_0, t_1 \in I$, the *parallel transport* from $\gamma(t_0)$ to $\gamma(t_1)$ along γ is the map

$$P^{\gamma}_{t_0,t_1}\colon T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M, \qquad P^{\gamma}_{t_0,t_1}(v) = V(t_1),$$

where V is the parallel field along γ with $V(t_0) = v$. This is well defined once every such v extends to a unique parallel field, which is established in theorem 2.4.2 below.

A parallel field is one that the connection reads as constant along the curve. On $\mathbb{R}^n$ with the Euclidean connection $\overline{\nabla}$ the condition $D_tV = 0$ reads $\dot{V}^k = 0$ in Cartesian coordinates (all Γ^k_{ij} vanish), so parallel fields are exactly those with constant Cartesian components, and parallel transport is ordinary translation of a vector to a new base point, keeping its Euclidean components fixed. This is the picture the word *parallel* is meant to evoke: transporting a vector while keeping it, from the flat connection's vantage, pointed the same way. On a curved manifold no global Cartesian frame is available, and the condition $D_tV = 0$ couples the components of V to the Christoffel symbols along γ; the resulting fields are no longer given by any fixed-components recipe, and, as the sphere example below shows, transporting a vector around a closed loop can return it rotated. The map $P^{\gamma}_{t_0,t_1}$ is the whole point of the construction: it converts the infinitesimal comparison supplied by ∇ into a bona fide linear map between the two tangent spaces $T_{\gamma(t_0)}M$ and $T_{\gamma(t_1)}M$, an identification that the bare manifold could not provide.

The existence and uniqueness of parallel fields, and the linearity and invertibility of $P^{\gamma}_{t_0,t_1}$, all follow from reading $D_tV = 0$ as a linear system and invoking the standard theory of linear ordinary differential equations.

Theorem 2.4.2: Existence and Uniqueness of Parallel Transport

Let ∇ be an affine connection on TM and $\gamma\colon I \to M$ a smooth curve. For every $t_0 \in I$ and every $v \in T_{\gamma(t_0)}M$ there is a unique parallel vector field $V \in \mathfrak{X}(\gamma)$ along γ with $V(t_0) = v$. Consequently, for all $t_0, t_1 \in I$ the parallel transport

$$P^{\gamma}_{t_0,t_1} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$$

is a well-defined linear isomorphism, with inverse $P^{\gamma}_{t_1,t_0}$ and with the composition law $P^{\gamma}_{t_1,t_2} \circ P^{\gamma}_{t_0,t_1} = P^{\gamma}_{t_0,t_2}$ for all $t_0, t_1, t_2 \in I$.

***Proof*:**
Step 1: reduction to a linear system on a chart. Suppose first that the image $\gamma(I)$ lies in the domain of a single chart $(U, (x^1, \dots, x^n))$. Writing $V(t) = V^k(t)\,\partial_k|_{\gamma(t)}$, the local formula (2.2) for the covariant derivative gives

$$D_t V = \Big(\dot V^k + \Gamma^k_{ij}\big(\gamma(t)\big)\,\dot\gamma^i(t)\,V^j(t)\Big)\,\partial_k\big|_{\gamma(t)}$$

Hence V is parallel if and only if its components satisfy the system

$$\dot V^k(t) = -\,\Gamma^k_{ij}\big(\gamma(t)\big)\,\dot\gamma^i(t)\,V^j(t), \qquad k = 1, \dots, n \tag{2.4}$$

Set $A^k{}_j(t) := -\,\Gamma^k_{ij}(\gamma(t))\,\dot\gamma^i(t)$. Each $A^k{}_j$ is a smooth (in particular continuous) real function on I, since γ, $\dot\gamma$, and the Christoffel symbols are smooth. In the vector notation $V(t) = (V^1(t), \dots, V^n(t)) \in \mathbb{R}^n$, the system (2.4) is the linear ordinary differential equation

$$\dot V(t) = A(t)\,V(t), \tag{2.5}$$

with $A(t) = (A^k{}_j(t))$ a matrix of continuous functions on I.

Step 2: existence and uniqueness on a chart. For a linear system (2.5) with continuous coefficient matrix $A(\cdot)$ on an interval I and any prescribed $t_0 \in I$, $v \in \mathbb{R}^n$, the existence-and-uniqueness theorem for linear ordinary differential equations provides a unique solution $V\colon I \to \mathbb{R}^n$ with $V(t_0) = v$, and this solution is defined on all of I. Linearity of the equation is what guarantees the solution does not escape to infinity in finite time, so no maximal-interval shrinking occurs: the solution exists on the full parameter interval, not on a proper subinterval only. The solution is smooth because $A(\cdot)$ is smooth. Translating back, $V(t) = V^k(t)\partial_k|_{\gamma(t)}$ is the unique parallel field along γ with $V(t_0) = v$, establishing the theorem when $\gamma(I)$ lies in one chart.

Step 3: patching over the general curve. For a general smooth curve, cover the compact image of each closed subinterval by finitely many charts and propagate the solution across the overlaps. Precisely, fix $t_0 \in I$ and $v \in T_{\gamma(t_0)}M$. Consider the set

$$J = \{\, t \in I : \text{a unique parallel field along } \gamma|_{[t_0,t]} \text{ (or } \gamma|_{[t,t_0]}) \text{ with value } v \text{ at } t_0 \text{ exists} \,\}$$

The point t_0 lies in J. For any $s \in I$, the point $\gamma(s)$ lies in some chart domain U_s, and $\gamma^{-1}(U_s)$ is an open neighborhood of s in I containing an interval $(s-\varepsilon, s+\varepsilon) \cap I$ whose image lies in U_s; by Step 2 a unique parallel field exists over that interval with any prescribed value at any one of its points. Thus if $s \in J$ then, using the value already determined at s as the new initial condition, the solution extends uniquely to $(s-\varepsilon, s+\varepsilon) \cap I$, so J is open in I. It is also closed:

if s is a limit point of J lying in I, choose a chart interval about s as above meeting J at some point s'; the unique solution on the chart interval agrees with the already-constructed field at s' by uniqueness (Step 2), hence extends the field to s, so $s \in J$. Since I is an interval, hence connected, and J is nonempty, open, and closed in I, one concludes $J = I$. The resulting field V is parallel on all of I and satisfies $V(t_0) = v$, and it is unique because any two such fields agree on the connected set where both the argument applies, namely all of I.

Step 4: linearity and invertibility of $P^{\gamma}_{t_0,t_1}$. Fix t_0, t_1. For $v \in T_{\gamma(t_0)}M$ let V_v denote the unique parallel field with $V_v(t_0) = v$; then $P^{\gamma}_{t_0,t_1}(v) = V_v(t_1)$. If $v, w \in T_{\gamma(t_0)}M$ and $a, b \in \mathbb{R}$, then $aV_v + bV_w$ is parallel by $\mathbb{R}$-linearity of D_t (definition 2.1.6) and has value $av + bw$ at t_0; by uniqueness it equals V_{av+bw}. Evaluating at t_1,

$$P^{\gamma}_{t_0,t_1}(av + bw) = (aV_v + bV_w)(t_1) = a\,V_v(t_1) + b\,V_w(t_1) = a\,P^{\gamma}_{t_0,t_1}(v) + b\,P^{\gamma}_{t_0,t_1}(w),$$

so $P^{\gamma}_{t_0,t_1}$ is linear. For the composition law, let V be parallel with $V(t_0) = v$; then $V(t_1) = P^{\gamma}_{t_0,t_1}(v)$, and since the *same* field V is the unique parallel field with value $V(t_1)$ at t_1, one has $P^{\gamma}_{t_1,t_2}(V(t_1)) = V(t_2) = P^{\gamma}_{t_0,t_2}(v)$. As v was arbitrary, $P^{\gamma}_{t_1,t_2} \circ P^{\gamma}_{t_0,t_1} = P^{\gamma}_{t_0,t_2}$. Taking $t_2 = t_0$ gives $P^{\gamma}_{t_1,t_0} \circ P^{\gamma}_{t_0,t_1} = P^{\gamma}_{t_0,t_0} = \mathrm{id}_{T_{\gamma(t_0)}M}$, and symmetrically the other composite is the identity, so $P^{\gamma}_{t_0,t_1}$ is a linear isomorphism with inverse $P^{\gamma}_{t_1,t_0}$, completing the proof.

The theorem does the work the connection was designed for: it delivers, for each curve γ and each pair of parameter values, a well-defined linear identification of the two tangent spaces along the curve. The isomorphisms $P^{\gamma}_{t_0,t_1}$ depend smoothly on the endpoints and compose as a path is traversed, so a single curve organizes its tangent spaces into a coherent one-parameter family of identifications. What the theorem does *not* say is that this identification is independent of the curve. Because (2.5) depends on γ through both the Christoffel symbols evaluated along γ and the velocity $\dot{\gamma}$, two curves with the same endpoints generally produce different transports; the failure of $P^{\gamma}_{t_0,t_1}$ to depend only on the endpoints, quantified along an infinitesimal loop, is the curvature of ∇, to be constructed later. The next example exhibits this path-dependence in a form one can compute by hand.

Example 2.4.3: Parallel Transport on the Round Sphere

Take the unit sphere S^2 with the round metric and its Levi-Civita connection (definition 2.3.2), and transport a tangent vector around a geodesic triangle to see the path-dependence directly. Fix the north pole N and consider a spherical triangle with vertices at N and two points A, B on the equator, whose two meridian sides from N to A and from N to B subtend an angle α at N, and whose third side is the equatorial arc from A to B. All three sides are geodesics (great-circle arcs).

Transport is governed by two features of the Levi-Civita connection established below and in section 3.1. First, along a geodesic σ the velocity $\dot{\sigma}$ is itself parallel, since a geodesic satisfies $D_t\dot{\sigma} = 0$; so a vector making a fixed angle with the geodesic and of fixed length, transported along that geodesic, keeps that angle and length, because transport preserves both length and the inner product with the parallel field $\dot{\sigma}$ (the isometry property proved in proposition 2.4.4). Begin at A with the vector v tangent to the meridian AN, pointing from A toward N. Carry v up the meridian to N: it stays tangent to the meridian, arriving at N pointing back down along NA. Now carry it down the second meridian from N to B: the two meridians meet at N at angle α, and the transported vector, which points along NA at N, therefore makes angle α with the meridian NB; transported down NB it keeps that angle α, arriving at B as a vector making angle α with the meridian and hence with the southward direction. Finally carry it along the equator from B back to A: the equator is a geodesic and the vector keeps its angle to it, so it returns to A

still rotated by the same angle relative to where it would sit had it been carried only along the meridian.

The upshot is that transport around the closed triangular loop returns v rotated, and the rotation angle equals the angular excess of the triangle. For this triangle the three interior angles are α at N and $\pi/2$ at each of A, B, summing to $\alpha + \pi$, so the excess over π is exactly α, which equals the spherical area enclosed. Transport of v around the loop is thus rotation by $\alpha \neq 0$: the identification $T_A S^2 \to T_A S^2$ produced by going around the triangle is not the identity. On the flat plane the same loop returns every vector unchanged, since Euclidean transport keeps Cartesian components fixed. The nonzero rotation on S^2 is the concrete signature of curvature, and it shows in a single computable instance that parallel transport depends on the path and not only on the endpoints.

Path-dependence is the qualitative feature of transport. Its quantitative counterpart on a (pseudo-)Riemannian manifold is that, whatever the path, transport by the Levi-Civita connection is rigid: it preserves the metric. This is the third characterization of metric compatibility promised in proposition 2.2.6, and its proof completes the equivalence stated there.

Proposition 2.4.4: Parallel Transport is a Linear Isometry

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM. Then ∇ is compatible with g (definition 2.2.4) if and only if, for every smooth curve γ and all t_0, t_1 in its parameter interval, the parallel transport $P^{\gamma}_{t_0,t_1}$ is a linear isometry:

$$\left\langle P^{\gamma}_{t_0,t_1}(v), P^{\gamma}_{t_0,t_1}(w)\right\rangle_g = \langle v, w\rangle_g \qquad \text{for all } v, w \in T_{\gamma(t_0)}M$$

In particular the Levi-Civita connection of (M, g) (definition 2.3.2), being metric-compatible, transports vectors isometrically along every curve, preserving lengths and angles.

***Proof*:**
Compatibility implies isometry. Assume ∇ is compatible with g. Let $\gamma\colon I \to M$ be a smooth curve and $V, W \in \mathfrak{X}(\gamma)$ two parallel fields along it, so $D_t V = 0$ and $D_t W = 0$. Consider the real function

$$h(t) := \langle V(t), W(t)\rangle_g = g_{\gamma(t)}\big(V(t), W(t)\big)$$

The compatibility identity of definition 2.2.4 has a version for fields along a curve: for any $V, W \in \mathfrak{X}(\gamma)$,

$$\frac{d}{dt}\langle V, W\rangle_g = \langle D_t V, W\rangle_g + \langle V, D_t W\rangle_g \tag{2.6}$$

To verify (2.6), work in a chart $(U, (x^i))$ where $\gamma(t) = (\gamma^i(t))$, $V = V^k \partial_k|_\gamma$, $W = W^l \partial_l|_\gamma$, and $g_{kl} = g(\partial_k, \partial_l)$. Then $\langle V, W\rangle_g = g_{kl}(\gamma(t))\, V^k W^l$, and the ordinary product rule gives

$$\frac{d}{dt}\langle V, W\rangle_g = \big(\dot\gamma^i\, \partial_i g_{kl}\big) V^k W^l + g_{kl}\, \dot V^k W^l + g_{kl}\, V^k \dot W^l$$

The coordinate form of metric compatibility, $\partial_i g_{kl} = \Gamma^m_{ik} g_{ml} + \Gamma^m_{il} g_{km}$ from proposition 2.2.6, converts the first term:

$$\big(\dot\gamma^i\, \partial_i g_{kl}\big) V^k W^l = \dot\gamma^i \big(\Gamma^m_{ik} g_{ml} + \Gamma^m_{il} g_{km}\big) V^k W^l = g_{ml}\big(\Gamma^m_{ik}\dot\gamma^i V^k\big) W^l + g_{km} V^k \big(\Gamma^m_{il}\dot\gamma^i W^l\big)$$

Adding $g_{kl}\dot V^k W^l + g_{kl} V^k \dot W^l$ and grouping by W^l and by V^k respectively, the sum becomes

$$g_{ml}\big(\dot V^m + \Gamma^m_{ik}\dot\gamma^i V^k\big) W^l + g_{km} V^k \big(\dot W^m + \Gamma^m_{il}\dot\gamma^i W^l\big) = \langle D_t V, W\rangle_g + \langle V, D_t W\rangle_g\,,$$

by the local formula (2.2) for $D_t V$ and $D_t W$; this is (2.6). Now for parallel V, W both covariant derivatives vanish, so $h'(t) = \langle D_t V, W\rangle + \langle V, D_t W\rangle = 0$, and h is constant on I. Given $v, w \in T_{\gamma(t_0)}M$, let V, W be the parallel fields with $V(t_0) = v$, $W(t_0) = w$ (theorem 2.4.2); then $V(t_1) = P^\gamma_{t_0,t_1}(v)$ and $W(t_1) = P^\gamma_{t_0,t_1}(w)$, and constancy of h gives

$$\left\langle P^\gamma_{t_0,t_1}(v), P^\gamma_{t_0,t_1}(w)\right\rangle_g = h(t_1) = h(t_0) = \langle v, w\rangle_g$$

Thus $P^\gamma_{t_0,t_1}$ preserves g, and being also a linear isomorphism (theorem 2.4.2) it is a linear isometry.

Isometry implies compatibility. Conversely, assume that $P^\gamma_{t_0,t_1}$ is a linear isometry for every curve γ and all parameters. To recover the compatibility identity of definition 2.2.4, fix $p \in M$, a tangent vector $X_p \in T_pM$, and vector fields $Y, Z \in \mathfrak{X}(M)$; it suffices to show $X_p \langle Y, Z\rangle = \left\langle \nabla_{X_p} Y, Z\right\rangle + \left\langle Y, \nabla_{X_p} Z\right\rangle$ at p, since p and X_p are arbitrary and both sides are pointwise in the direction (proposition 2.1.5). Choose a curve γ with $\gamma(0) = p$ and $\dot\gamma(0) = X_p$. Let $(e_1, \ldots, e_n)$ be a basis of T_pM and let $E_a(t) := P^\gamma_{0,t}(e_a)$ be the parallel fields extending them, so each E_a is parallel, $D_t E_a = 0$, and $(E_1(t), \ldots, E_n(t))$ is a basis of $T_{\gamma(t)}M$ for every t (each $P^\gamma_{0,t}$ is an isomorphism). By hypothesis $P^\gamma_{0,t}$ is an isometry, so the functions

$$\langle E_a(t), E_b(t)\rangle_g = \left\langle P^\gamma_{0,t}(e_a), P^\gamma_{0,t}(e_b)\right\rangle_g = \langle e_a, e_b\rangle_g$$

are constant in t. Expand Y and Z along γ in this parallel frame: $Y(\gamma(t)) = Y^a(t)E_a(t)$ and $Z(\gamma(t)) = Z^a(t)E_a(t)$ with smooth coefficient functions Y^a, Z^a. Because each E_a is parallel, the covariant derivative along γ of Y restricted to γ is

$$D_t\big(Y^a E_a\big) = \dot Y^a\, E_a + Y^a\, D_t E_a = \dot Y^a\, E_a,$$

and by definition 2.1.6(3) this equals $\nabla_{\dot\gamma} Y$; at $t = 0$, $\nabla_{X_p} Y = \dot Y^a(0)\, E_a(0) = \dot Y^a(0)\, e_a$, and similarly $\nabla_{X_p} Z = \dot Z^a(0)\, e_a$. On the other hand, writing $c_{ab} := \langle e_a, e_b\rangle_g = \langle E_a(t), E_b(t)\rangle_g$ (constant in t),

$$\langle Y, Z\rangle_g\big(\gamma(t)\big) = \left\langle Y^a E_a, Z^b E_b\right\rangle_g = Y^a(t) Z^b(t)\, c_{ab},$$

so differentiating this function of t and evaluating at 0, where the c_{ab} carry no t-dependence,

$$X_p \langle Y, Z\rangle = \frac{d}{dt}\bigg|_0 \big(Y^a Z^b c_{ab}\big) = \dot Y^a(0) Z^b(0) c_{ab} + Y^a(0)\dot Z^b(0) c_{ab}$$

The first summand is $\left\langle \dot Y^a(0) e_a, Z^b(0) e_b\right\rangle_g = \left\langle \nabla_{X_p} Y, Z\right\rangle$ and the second is $\left\langle Y, \nabla_{X_p} Z\right\rangle$, both at p. Hence $X_p \langle Y, Z\rangle = \left\langle \nabla_{X_p} Y, Z\right\rangle + \left\langle Y, \nabla_{X_p} Z\right\rangle$, which is the compatibility identity. As p, X_p, Y, Z were arbitrary, ∇ is compatible with g.

The Levi-Civita connection is metric-compatible by definition 2.3.2, so the forward direction applies to it, completing the proof.

The proposition supplies the geometric reading of metric compatibility that its algebraic definition only hinted at. A metric-compatible connection is one whose parallel transport is a rigid motion between tangent spaces: it carries orthonormal bases to orthonormal bases, keeps the length of a transported vector fixed, and keeps the angle between two transported vectors fixed. For the Levi-Civita connection this holds along every curve, so a vector's length is a transport invariant even though the vector's direction, in any external sense, is not. With the equivalence proposition 2.2.6(iii) now proved, all three formulations of compatibility, $\nabla g = 0$, the Leibniz identity for inner products, and isometric transport, stand on equal footing, and one may invoke whichever is most convenient. The isometry property is what makes the sphere computation of example 2.4.3 legitimate: the angle and length arguments used there rest precisely on transport preserving g.

Example 2.4.5: Isometric Transport and Constant Speed of Geodesics

One immediate consequence of proposition 2.4.4 is worth isolating, as it recurs throughout the geodesic theory. Let γ be a geodesic of a (pseudo-)Riemannian manifold, meaning $D_t\dot{\gamma} = 0$, so that the velocity $\dot{\gamma}$ is a parallel field along γ (definition 2.4.1). Then for any t,

$$\dot{\gamma}(t) = P^{\gamma}_{t_0,t}\big(\dot{\gamma}(t_0)\big),$$

since both sides are the value at t of the parallel field with initial value $\dot{\gamma}(t_0)$, and these agree by uniqueness (theorem 2.4.2). Applying the isometry property to $v = w = \dot{\gamma}(t_0)$ gives

$$|\dot{\gamma}(t)|_g^2 = \langle \dot{\gamma}(t), \dot{\gamma}(t)\rangle_g = \left\langle P^{\gamma}_{t_0,t}\dot{\gamma}(t_0), P^{\gamma}_{t_0,t}\dot{\gamma}(t_0)\right\rangle_g = \langle \dot{\gamma}(t_0), \dot{\gamma}(t_0)\rangle_g = |\dot{\gamma}(t_0)|_g^2,$$

so $|\dot{\gamma}|_g$ is constant along γ: geodesics have constant speed. The same conclusion is reached by direct differentiation, since $\frac{d}{dt}\langle \dot{\gamma}, \dot{\gamma}\rangle = 2\langle D_t\dot{\gamma}, \dot{\gamma}\rangle = 0$ by (2.6) and $D_t\dot{\gamma} = 0$; the transport formulation makes the invariance conceptual rather than computational. This constant-speed property is taken up in its own right in section 3.1, where geodesics are the subject; here it serves to show the isometry of transport at work on the one parallel field every geodesic carries, its own velocity.

The covariant derivative has so far acted only on vector fields and on vector fields along curves. Curvature, the Hessian of a function, the second fundamental form, and the very tensor g that has been differentiated informally above all live in tensor bundles built from TM and T^*M. The connection extends to all of these in exactly one way consistent with the operations that relate them, namely the Leibniz rule, contraction, and the pairing of a vector with a covector.

Definition 2.4.6: Induced Connection on Tensor Bundles

Let ∇ be an affine connection on TM of a smooth manifold M. The *induced connection*, also denoted ∇, on each tensor bundle $T^{(r,s)}M$ of (r,s)-tensors is the unique family of operators ∇_X on tensor fields, for $X \in \mathfrak{X}(M)$, determined by the following requirements.

1. On functions ($(0,0)$-tensors), $\nabla_X f = Xf$; on vector fields ($(1,0)$-tensors), ∇_X is the given connection on TM.
2. $\mathbb{R}$*-linearity and the Leibniz rule over tensor products:* for tensor fields S, T,
$$\nabla_X(S \otimes T) = (\nabla_X S) \otimes T + S \otimes (\nabla_X T)$$
3. *Commutation with contractions:* for every contraction C of an upper against a lower index, $\nabla_X(\mathrm{C}\,T) = \mathrm{C}(\nabla_X T)$.

Equivalently, on a covector field ω the induced connection is fixed by requiring the pairing $\langle \omega, Y\rangle = \omega(Y)$ to obey the product rule,

$$(\nabla_X \omega)(Y) = X\big(\omega(Y)\big) - \omega(\nabla_X Y), \tag{2.7}$$

and on a general (r,s)-tensor T by the Leibniz-type formula

$$(\nabla_X T)(\omega^1, \dots, \omega^r, Y_1, \dots, Y_s) = X\big(T(\omega^1, \dots, \omega^r, Y_1, \dots, Y_s)\big) - \sum_a T(\dots, \nabla_X \omega^a, \dots) - \sum_b T(\dots, \nabla_X Y_b, \dots)$$

***Proof*:**
The requirements determine ∇_X on all tensor bundles and are consistent. On a 1-form ω, apply requirement (3) to the contraction $\mathrm{C}(\omega \otimes Y) = \omega(Y)$ and requirement (2) to the product $\omega \otimes Y$:

$$X\big(\omega(Y)\big) = \nabla_X\big(\omega(Y)\big) = \nabla_X\, \mathrm{C}(\omega \otimes Y) = \mathrm{C}\big(\nabla_X(\omega \otimes Y)\big) = (\nabla_X \omega)(Y) + \omega(\nabla_X Y),$$

using requirement (1) for the first equality (the contraction $\omega(Y)$ is a function). Solving for $(\nabla_X \omega)(Y)$ gives (2.7), so the action on 1-forms is forced. That (2.7) defines a genuine 1-form, that is, that $Y \mapsto (\nabla_X \omega)(Y)$ is $C^\infty(M)$-linear, follows from a cancellation of the derivative term: for $f \in C^\infty(M)$,

$$(\nabla_X \omega)(fY) = X\big(f\,\omega(Y)\big) - \omega\big((Xf)Y + f\nabla_X Y\big) = (Xf)\omega(Y) + fX\big(\omega(Y)\big) - (Xf)\omega(Y) - f\,\omega(\nabla_X Y) = f\,(\nabla_X \omega)(Y),$$

the $(Xf)\omega(Y)$ terms cancelling. On a general (r,s)-tensor T, iterating the Leibniz rule (2) across the tensor product and contracting against the arguments $\omega^1, \dots, \omega^r, Y_1, \dots, Y_s$ yields the displayed Leibniz-type formula; the same cancellation as for 1-forms shows the right-hand side is $C^\infty(M)$-linear in each argument, so $\nabla_X T$ is again an (r,s)-tensor field. Conversely, the operators so defined satisfy (1), (2), and (3) by construction: (2) because the formula was built from the product rule, and (3) because contracting the Leibniz-type formula in a pair of slots reproduces the covariant derivative of the contracted tensor, the extra terms matching. Hence the family is the unique extension of ∇ compatible with (1)–(3), completing the proof.

The induced connection is what lets ∇ differentiate any tensorial object of the theory, and the requirements defining it are exactly the compatibilities one needs downstream. Requirement (2), the Leibniz rule over tensor products, means ∇ never needs a separate definition for each new bundle; every tensor is assembled from vectors and covectors, and its covariant derivative is assembled the same way. Requirement (3), commutation with contractions, is the statement that differentiating and then contracting gives the same result as contracting and then differentiating, so that traces and pairings are transparent to ∇. Two consequences are used constantly. First, the covariant derivative commutes with the musical isomorphisms of definition 1.4.1 when ∇ is metric-compatible: since $\nabla g = 0$ (proposition 2.2.6), lowering an index $Y \mapsto Y^\flat = g(Y, -)$ commutes with ∇_X, because $\nabla_X(Y^\flat) = \nabla_X\big(\mathrm{C}(g \otimes Y)\big) = \mathrm{C}\big((\nabla_X g) \otimes Y + g \otimes \nabla_X Y\big) = (\nabla_X Y)^\flat$, the first term vanishing; raising is the inverse operation and commutes likewise. So for a metric-compatible connection one may freely raise and lower indices before or after differentiating. Second, applied to g itself, requirement (2) recovers the informal expansion of ∇g used already in proposition 2.2.6: contraction of $\nabla_X(g \otimes Y \otimes Z)$ produces $(\nabla_X g)(Y, Z) = X\langle Y, Z\rangle - \langle \nabla_X Y, Z\rangle - \langle Y, \nabla_X Z\rangle$, the definition on which the whole compatibility discussion rested.

Example 2.4.7: Covariant Derivative of a One-Form on the Plane

Take $\mathbb{R}^2$ minus the origin in polar coordinates (r, θ) with the Euclidean connection $\overline{\nabla}$, whose nonzero Christoffel symbols are $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = 1/r$ and $\Gamma^r_{\theta\theta} = -r$ (example 2.1.4). Differentiate the coordinate 1-form $d\theta$ in the radial direction $X = \partial_r$. Formula (2.7) gives, on each coordinate field ∂_j,

$$(\overline{\nabla}_{\partial_r}\, d\theta)(\partial_j) = \partial_r\big(d\theta(\partial_j)\big) - d\theta(\overline{\nabla}_{\partial_r}\partial_j)$$

The pairing $d\theta(\partial_j) = \delta^\theta_j$ is constant, so its ∂_r-derivative is zero, and $d\theta$ selects the θ-component of $\overline{\nabla}_{\partial_r}\partial_j$. For $j = \theta$: $\overline{\nabla}_{\partial_r}\partial_\theta = \Gamma^k_{r\theta}\partial_k = \frac{1}{r}\partial_\theta$, whose θ-component is $1/r$, so $(\overline{\nabla}_{\partial_r} d\theta)(\partial_\theta) = -1/r$. For $j = r$: $\overline{\nabla}_{\partial_r}\partial_r = 0$, so $(\overline{\nabla}_{\partial_r} d\theta)(\partial_r) = 0$. Therefore

$$\overline{\nabla}_{\partial_r}\, d\theta = -\tfrac{1}{r}\, d\theta$$

The sign is opposite to that seen when the same connection acts on the vector field ∂_θ, where $\overline{\nabla}_{\partial_r}\partial_\theta = +\frac{1}{r}\partial_\theta$; the covector transforms contravariantly to its dual vector, exactly as (2.7) dictates through the minus sign. As a consistency check with commutation over contraction, the pairing $\langle d\theta, \partial_\theta\rangle = d\theta(\partial_\theta) = 1$ is constant, so its covariant derivative must vanish; indeed $\overline{\nabla}_{\partial_r}(d\theta(\partial_\theta)) = (\overline{\nabla}_{\partial_r} d\theta)(\partial_\theta) + d\theta(\overline{\nabla}_{\partial_r}\partial_\theta) = (-\frac{1}{r})(1) + (1)(\frac{1}{r}) = 0$, confirming requirement (3) of definition 2.4.6 in this instance.

3

Geodesics and the Exponential Map

The Levi-Civita connection of chapter 2 makes it possible to ask which curves are straight. A straight line in Euclidean space is one whose velocity does not change, and the covariant derivative supplies the correct notion of an unchanging velocity on a manifold: a curve is a *geodesic* when its acceleration, the covariant derivative of its own velocity, vanishes. This chapter develops geodesics and the map that organizes them, the exponential map, and establishes the sense in which geodesics are the shortest paths.

The geodesic equation is a second-order ordinary differential equation, so a geodesic is determined by an initial point and an initial velocity; the exponential map packages these solutions by sending a tangent vector to the point reached by travelling along the corresponding geodesic for unit time. Near a point it is a diffeomorphism, and it produces the normal coordinates in which the metric is standard to first order. The Gauss lemma, which records that the exponential map sends radii to orthogonal geodesics, then yields the local minimizing property: within a small ball the radial geodesic is the unique shortest path, and the metric induces a genuine distance function on the manifold, resolving the terminological caution raised in chapter 1. The chapter closes with the variational picture, in which geodesics reappear as the critical points of the length and energy functionals, the point of view that governs the comparison theory of later chapters.

3.1 The Geodesic Equation

A straight line in Euclidean space is the trace of a particle that feels no force: its velocity is constant, so its acceleration vanishes. On a curved manifold there is no ambient vector space in which to demand that the velocity vector literally not change, but the Levi-Civita connection of definition 2.3.2 supplies the intrinsic replacement. The velocity $\dot{\gamma}$ of a curve γ is a vector field along γ, and its intrinsic rate of change is the covariant derivative $D_t\dot{\gamma}$ of definition 2.1.6. Requiring this to vanish is the manifold analogue of "the velocity does not change," and the curves that satisfy it are the geodesics. They are the straightest curves the geometry allows, the paths of free particles, and, as the next chapter shows, the locally shortest ones.

This section extracts the differential equation that governs geodesics and settles the two questions any second-order equation raises: given a starting point and a starting velocity, does a solution exist, and is it unique. The answers come from the theory of ordinary differential equations, applied to the coordinate form of the equation $D_t\dot\gamma = 0$. A geodesic turns out to be determined by its initial position and initial velocity exactly as a Newtonian trajectory is determined by initial data, and it depends smoothly on both. Throughout, (M, g) is a (pseudo-)Riemannian manifold with its Levi-Civita connection ∇, and the Einstein summation convention, summing over any index appearing once as an upper and once as a lower index, is in force.

Definition 3.1.1: Geodesic

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇, and let $\gamma\colon I \to M$ be a smooth curve on an interval $I \subseteq \mathbb{R}$. Then γ is a *geodesic* if its velocity field is parallel along γ, that is,

$$D_t\dot\gamma = 0 \qquad \text{on } I,$$

where D_t is the covariant derivative along γ of definition 2.1.6. The vector field $D_t\dot\gamma \in \mathfrak{X}(\gamma)$ is the *acceleration* of γ; a geodesic is thus a curve of zero acceleration, an *autoparallel* curve.

The word autoparallel names the geometric content: a geodesic transports its own velocity parallel to itself. The velocity vector at one instant is carried, by the connection, exactly onto the velocity vector at the next, so the curve never turns relative to the geometry it lives in. On the flat model $(\mathbb{R}^n, \bar g)$ the Levi-Civita connection is the Euclidean connection, whose covariant derivative along a curve is componentwise ordinary differentiation, so $D_t\dot\gamma = \ddot\gamma$ and the geodesic condition reads $\ddot\gamma = 0$: the geodesics of Euclidean space are the affinely parametrized straight lines $t \mapsto p + tv$, recovering the classical picture. One caution is built into the definition. The condition $D_t\dot\gamma = 0$ constrains not only the trace of the curve but its parametrization: a straight line traversed with varying speed, such as $t \mapsto p + t^2 v$, has $\ddot\gamma = 2v \neq 0$ and is not a geodesic in this sense, even though its image is a geodesic image. Geodesics are curves, parametrization included, not point sets.

To compute with geodesics one passes to coordinates. In a chart $(U, (x^1, \dots, x^n))$ write $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$; the velocity has components $\dot\gamma^k$ and, by the coordinate formula (2.2) for the covariant derivative along a curve applied to $V = \dot\gamma$ (so $V^k = \dot\gamma^k$ and $\dot V^k = \ddot\gamma^k$),

$$D_t\dot\gamma = \Big(\ddot\gamma^k + \Gamma^k_{ij}\big(\gamma(t)\big)\,\dot\gamma^i\,\dot\gamma^j\Big)\,\partial_k\big|_{\gamma(t)}$$

Since the coordinate frame (∂_k) is a basis of each tangent space, the vanishing of $D_t\dot\gamma$ is the vanishing of every component. This yields the local form of the geodesic condition.

$$\ddot\gamma^k + \Gamma^k_{ij}\big(\gamma(t)\big)\,\dot\gamma^i\,\dot\gamma^j = 0, \qquad k = 1, \dots, n \tag{3.1}$$

Equation (3.1) is the *geodesic equation.* It is a system of n coupled second-order ordinary differential equations for the coordinate functions $\gamma^k(t)$, one for each k. The leading term $\ddot\gamma^k$ is the naive coordinate acceleration; the quadratic term $\Gamma^k_{ij}\dot\gamma^i\dot\gamma^j$ is the correction that accounts for the curving of the coordinate frame, exactly as in the covariant derivative it descends from. The whole geometric content of the connection enters through the Christoffel symbols Γ^k_{ij}, which by proposition 2.3.3 are built from the metric g_{ij} and its first derivatives. Two features of the equation should be registered before it is solved. The coefficients $\Gamma^k_{ij}(\gamma(t))$ depend on the unknown γ itself, so the system is nonlinear; and the nonlinearity is quadratic in the velocities $\dot\gamma^i$, never higher, which is what will make the reduction to a first-order system in the next theorem clean.

Example 3.1.2: Meridians on a Surface of Revolution

Take a surface of revolution with the warped metric $g = dr^2 + f(r)^2\, d\theta^2$, $f(r) > 0$, in coordinates $(x^1, x^2) = (r, \theta)$, whose Christoffel symbols were computed in example 2.3.4: the only nonzero symbols are

$$\Gamma^r_{\theta\theta} = -f(r)f'(r), \qquad \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{f'(r)}{f(r)}$$

The geodesic equation (3.1) has one component for each of $k = r$ and $k = \theta$. Reading off the nonzero symbols, the r-component receives a contribution only from the pair $(i, j) = (\theta, \theta)$, and the θ-component only from the pairs (r, θ) and (θ, r), which are equal and so combine with a factor 2:

$$\begin{aligned} \ddot r - f(r)f'(r)\,\dot\theta^2 &= 0, \\ \ddot\theta + \frac{2f'(r)}{f(r)}\,\dot r\,\dot\theta &= 0 \end{aligned}$$

Consider a *meridian*, a curve of constant longitude $\theta(t) \equiv \theta_0$ traced out in the r-direction; parametrize it by $\gamma(t) = (r(t), \theta_0)$. Then $\dot\theta \equiv 0$ and $\ddot\theta \equiv 0$, so the second equation is satisfied identically, and the first equation reduces to $\ddot r = 0$, whose solutions are the affine functions $r(t) = r_0 + ct$. Thus every meridian, traversed at constant r-speed, is a geodesic. The geometric reading is that a meridian is the profile curve of the surface, and moving along it the velocity never turns sideways, because the rotational symmetry of the surface leaves no preferred direction for it to turn toward. A parallel of constant latitude $r(t) \equiv r_0$, by contrast, has $\dot r \equiv 0$ and $\dot\theta = \text{const}$, and the first equation demands $-f(r_0)f'(r_0)\dot\theta^2 = 0$; since $f > 0$ this forces $f'(r_0) = 0$. A circle of latitude is therefore a geodesic exactly when it sits at a critical radius of the profile function f, for instance the equator of a sphere of revolution, where f is maximal. Specializing to the flat plane $f(r) = r$ recovers the earlier finding: the equations become $\ddot r - r\dot\theta^2 = 0$ and $\ddot\theta + \frac{2}{r}\dot r\dot\theta = 0$, the polar-coordinate form of $\ddot\gamma = 0$, whose radial-line solutions $\theta \equiv \theta_0$, $r = r_0 + ct$ are precisely the straight lines through the origin.

The geodesic equation is nonlinear and second order, so its solvability is not automatic; it is delivered by the existence and uniqueness theory for ordinary differential equations, once the second-order system is recast as a first-order one on a suitable space. The natural space is the tangent bundle TM, whose points record a position together with a velocity, precisely the data that a geodesic evolves.

Theorem 3.1.3: Existence and Uniqueness of Geodesics

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection. For each $p \in M$ and each $v \in T_pM$ there is a unique *maximal geodesic*

$$\gamma_v \colon I_v \to M, \qquad \gamma_v(0) = p, \quad \dot{\gamma}_v(0) = v,$$

defined on an open interval $I_v \subseteq \mathbb{R}$ containing 0, in the sense that any geodesic with these initial conditions is the restriction of γ_v to a subinterval. Moreover γ_v depends smoothly on the initial data: the map $(t, p, v) \mapsto \gamma_v(t)$ is smooth on its (open) domain of definition inside $\mathbb{R} \times TM$.

***Proof*:**
The argument reduces the second-order geodesic equation to a first-order system on the tangent bundle, invokes the standard theorem on ordinary differential equations, and then globalizes the local solutions into a single maximal one.

Step 1: reduction to a first-order system in a chart. Fix a chart $(U, (x^i))$ and identify a curve γ in U with its coordinate functions $(\gamma^1, \ldots, \gamma^n)$. Introduce the velocity coordinates $w^k := \dot{\gamma}^k$, doubling the number of unknowns to $2n$, namely (γ^k, w^k). The geodesic equation (3.1) then becomes the first-order system

$$\dot{\gamma}^k = w^k, \qquad \dot{w}^k = -\Gamma^k_{ij}(\gamma)\, w^i\, w^j, \qquad k = 1, \ldots, n \tag{3.2}$$

A curve γ in U solves (3.1) if and only if the pair $(\gamma, \dot{\gamma}) = (\gamma^k, w^k)$ solves (3.2): the first block of equations is the definition $w^k = \dot{\gamma}^k$, and substituting it into the second reproduces $\ddot{\gamma}^k = \dot{w}^k = -\Gamma^k_{ij}(\gamma)\dot{\gamma}^i\dot{\gamma}^j$. Write $\xi = (\gamma^1, \ldots, \gamma^n, w^1, \ldots, w^n)$ for the unknown taking values in the open set $\pi^{-1}(U) \cong U \times \mathbb{R}^n \subseteq TM$, where $\pi \colon TM \to M$ is the bundle projection; the system (3.2) has the form $\dot{\xi} = F(\xi)$ with

$$F(\gamma, w) = \big(w^1, \ldots, w^n,\ -\Gamma^k_{ij}(\gamma) w^i w^j\big)$$

The map F is smooth: its first n components are the linear projections $\xi \mapsto w^k$, and its last n components are polynomials of degree two in the w's whose coefficients Γ^k_{ij} are smooth functions of γ on U, by proposition 2.3.3. Thus F is a smooth vector field on the open subset $\pi^{-1}(U)$ of TM.

Step 2: local existence, uniqueness, and smooth dependence. The system $\dot{\xi} = F(\xi)$ with F smooth is the setting of the fundamental theorem on ordinary differential equations, equivalently the theorem on the local flow of a smooth vector field [6]. It provides, for each initial value $\xi_0 = (p, v) \in \pi^{-1}(U)$, a number $\varepsilon > 0$ and a unique solution $\xi \colon (-\varepsilon, \varepsilon) \to \pi^{-1}(U)$ with $\xi(0) = \xi_0$; uniqueness means any two solutions with the same initial value agree on the intersection of their domains. The theorem further furnishes smooth dependence on the initial value: there is a neighborhood of ξ_0 and a single $\varepsilon > 0$ such that the map $(t, \xi_0) \mapsto \xi(t; \xi_0)$ is smooth in t and in ξ_0 jointly. Translating back, the first block of ξ is a curve γ in U solving (3.1) with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$, and the smooth dependence of ξ on $\xi_0 = (p, v)$ is exactly the asserted smooth dependence of $\gamma_v(t)$ on (t, p, v) within the chart. Uniqueness of the geodesic follows: two geodesics through p with velocity v lift to two solutions of (3.2) with the same initial value, hence coincide where both are defined.

Step 3: the solution is independent of the chart. The first-order system (3.2) was written in a particular chart, but the condition it encodes, $D_t\dot\gamma = 0$, is coordinate-free: the covariant derivative along a curve is chart-independent by definition 2.1.6. Consequently, if a geodesic passes from the domain of one chart into that of another, the two coordinate representations describe the same curve on the overlap, and by the uniqueness of Step 2 they patch. This makes the local geodesics globally well defined as curves in M and lets the maximality argument proceed on M rather than in a fixed chart.

Step 4: the maximal geodesic. Fix p and v. Consider the collection of all geodesics $\gamma\colon I \to M$ with $\gamma(0) = p$ and $\dot\gamma(0) = v$, where I ranges over open intervals containing 0; this collection is nonempty by Step 2. Let I_v be the union of all such intervals I, again an open interval containing 0. Define $\gamma_v\colon I_v \to M$ by setting $\gamma_v(t) = \gamma(t)$ for any geodesic γ in the collection whose domain contains t. This is well defined: if $\gamma_1\colon I_1 \to M$ and $\gamma_2\colon I_2 \to M$ are two geodesics with the same initial data, then the set $\{t \in I_1 \cap I_2 : \gamma_1(t) = \gamma_2(t) \text{ and } \dot\gamma_1(t) = \dot\gamma_2(t)\}$ is nonempty (it contains 0), closed in the connected set $I_1 \cap I_2$ by continuity, and open by the local uniqueness of Step 2 applied at each of its points (a geodesic through $\gamma_1(t_0)$ with velocity $\dot\gamma_1(t_0)$ is unique near t_0, forcing $\gamma_1 = \gamma_2$ on a neighborhood of t_0); hence it is all of $I_1 \cap I_2$, so γ_1 and γ_2 agree on the overlap. Therefore the assignment $\gamma_v(t) = \gamma(t)$ does not depend on which γ is used, and γ_v is a well-defined curve on I_v. It is a geodesic, because the geodesic equation is local in t and near every $t_0 \in I_v$ the curve γ_v coincides with some member γ of the collection, which satisfies the equation there; and it is smooth for the same reason. By construction $\gamma_v(0) = p$, $\dot\gamma_v(0) = v$, and every geodesic with these initial conditions has domain contained in I_v and agrees with γ_v there, so it is a restriction of γ_v. This is the asserted maximal geodesic, and it is unique because any curve with the same defining property has the same domain I_v and the same values.

Finally, the smooth dependence on (t, p, v) is inherited from Step 2. The domain

$$\mathcal{D} = \{(t, p, v) \in \mathbb{R} \times TM : t \in I_v\}$$

is open: around any $(t_0, p_0, v_0) \in \mathcal{D}$, the local solution of Step 2 exists for initial data near (p_0, v_0) and for times in a neighborhood of t_0, obtained by chaining finitely many local flow boxes along the compact segment $\gamma_{v_0}([0, t_0])$; on that neighborhood $(t, p, v) \mapsto \gamma_v(t)$ agrees with a composition of smooth local flows, hence is smooth. As this holds near every point of $\mathcal{D}$, the map $(t, p, v) \mapsto \gamma_v(t)$ is smooth on $\mathcal{D}$, completing the proof.

The theorem is the geometric counterpart of the determinism of Newtonian mechanics: a geodesic is fixed by where it starts and how fast, in what direction, it leaves, and nothing more. The passage to TM is the reason the second-order equation on M becomes a first-order flow: a point of TM carries both the position and the velocity, so evolving it forward advances both at once, and the geodesics are exactly the projections to M of the integral curves of the vector field F built in the proof. That field, defined here only in charts, is a genuine global object on TM, the *geodesic spray*, and its flow is the *geodesic flow*; the exponential map of section 3.2 is nothing but this flow read off at time one, and the smooth dependence just proved is what makes the exponential map smooth. One point deserves emphasis. The interval I_v need not be all of $\mathbb{R}$: a geodesic can run off the manifold in finite time, as it does on $\mathbb{R}^n \setminus \{0\}$ with the Euclidean metric, where the geodesic aimed at the deleted origin reaches it at a finite parameter and cannot continue. When every maximal geodesic is defined on all of $\mathbb{R}$ the manifold is called geodesically complete, a condition analyzed in the Hopf–Rinow theorem of a later chapter.

Uniqueness of the geodesic through given initial data is a computational tool as much as a structural fact: to identify all geodesics of a model space it suffices to exhibit, through each point and in each direction, one curve that solves (3.1), and theorem 3.1.3 then forces it to be the geodesic. The two constant-curvature models make this concrete.

Example 3.1.4: Great circles are the geodesics of the sphere

On the round 2-sphere in coordinates $(x^1, x^2) = (\theta, \varphi)$ with $g = d\theta^2 + \sin^2\theta\, d\varphi^2$, the Christoffel symbols were computed in example 2.3.5: the only nonzero ones are

$$\Gamma^{\theta}_{\varphi\varphi} = -\sin\theta\cos\theta, \qquad \Gamma^{\varphi}_{\theta\varphi} = \Gamma^{\varphi}_{\varphi\theta} = \cot\theta$$

The geodesic equation (3.1) therefore has the two components

$$\begin{aligned} \ddot\theta - \sin\theta\cos\theta\,\dot\varphi^2 &= 0, \\ \ddot\varphi + 2\cot\theta\,\dot\theta\,\dot\varphi &= 0, \end{aligned}$$

the θ-component receiving its contribution from the pair $(i, j) = (\varphi, \varphi)$ and the φ-component from the pairs (θ, φ) and (φ, θ), equal and combining with a factor 2.

The equator. Set $\theta(t) \equiv \pi/2$, so $\dot\theta \equiv 0$, $\ddot\theta \equiv 0$, and $\cos\theta \equiv 0$. The first equation reads $0 - 0 = 0$, satisfied identically, and the second reduces to $\ddot\varphi = 0$, whose solutions are $\varphi(t) = \varphi_0 + ct$. Thus the equator, traversed at constant φ-speed, is a geodesic. It is a great circle, the intersection of S^2 with the plane $\{x^3 = 0\}$ through the center.

The meridians. Set $\varphi(t) \equiv \varphi_0$, so $\dot\varphi \equiv 0$ and $\ddot\varphi \equiv 0$. The second equation holds identically, and the first reduces to $\ddot\theta = 0$, giving $\theta(t) = \theta_0 + ct$. Each meridian, a half great circle of constant longitude, is a geodesic; this is the specialization of example 3.1.2 to the sphere of revolution with profile $f(\theta) = \sin\theta$.

General great circles. Every remaining geodesic is obtained from these by symmetry, without solving the equations again. Fix $p \in S^2$ and a unit $v \in T_pS^2$, and let $\Pi \subset \mathbb{R}^3$ be the plane through the origin spanned by the position vector p and the ambient velocity v; the great circle $C = S^2 \cap \Pi$ passes through p in the direction v. Let $\rho\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the reflection fixing Π pointwise and negating the orthogonal direction. Then ρ is a linear isometry of $\mathbb{R}^3$ preserving S^2, hence an isometry of the round sphere, and its fixed-point set on S^2 is exactly C. An isometry carries geodesics to geodesics, because $\varphi^* g = g$ preserves the Levi-Civita connection and so takes solutions of (3.1) to solutions; therefore $\rho \circ \gamma_v$ is a geodesic. It has the same initial data as γ_v, namely $\rho(p) = p$ and $d\rho_p(v) = v$ (both p and v lie in the fixed plane Π), so by the uniqueness in theorem 3.1.3, $\rho \circ \gamma_v = \gamma_v$. Thus γ_v is fixed pointwise by ρ, which means its image lies in the fixed set C. Since some suitably rotated equator realizes any prescribed (p, v) (rotations act transitively on unit tangent vectors), every geodesic of S^2 is a great circle traversed at constant speed, and conversely each great circle is a geodesic. The same reflection argument identifies the geodesics of S^n for every n as its great circles.

The hyperbolic plane furnishes the negatively curved counterpart, where the geodesics are visibly not the Euclidean straight lines of the coordinate model.

Example 3.1.5: Geodesics of the hyperbolic plane

Take the upper half-plane model of example 1.3.5 in dimension two, with coordinates $(x^1, x^2) =$

(x, y) on $U^2 = \{(x, y) \mid y > 0\}$ and metric

$$g = \frac{1}{y^2}\left(dx^2 + dy^2\right)$$

Here $g_{11} = g_{22} = y^{-2}$, $g_{12} = 0$, so the inverse metric is $g^{11} = g^{22} = y^2$, and the only nonzero partial derivatives of the components are $\partial_y g_{11} = \partial_y g_{22} = -2y^{-3}$. Feeding these into proposition 2.3.3 gives the nonzero Christoffel symbols

$$\Gamma^x_{xy} = \Gamma^x_{yx} = -\frac{1}{y}, \qquad \Gamma^y_{xx} = \frac{1}{y}, \qquad \Gamma^y_{yy} = -\frac{1}{y}$$

For instance $\Gamma^y_{xx} = \frac{1}{2} g^{yy}\left(2\partial_x g_{xy} - \partial_y g_{xx}\right) = \frac{1}{2} y^2\left(0 - (-2y^{-3})\right) = y^{-1}$, and the others follow by the same computation. The geodesic equation (3.1) for $\gamma(t) = (x(t), y(t))$ becomes

$$\begin{aligned} \ddot{x} - \frac{2}{y}\,\dot{x}\,\dot{y} &= 0, \\ \ddot{y} + \frac{1}{y}\,\dot{x}^2 - \frac{1}{y}\,\dot{y}^2 &= 0 \end{aligned}$$

The vertical rays. Set $x(t) \equiv x_0$, so $\dot{x} \equiv 0$ and $\ddot{x} \equiv 0$; the first equation holds identically. The second reduces to $\ddot{y} = \dot{y}^2/y$, solved by $y(t) = y_0\, e^{ct}$, for which $\dot{y} = c y_0 e^{ct}$, $\ddot{y} = c^2 y_0 e^{ct}$, and $\dot{y}^2/y = c^2 y_0 e^{ct}$ agree. So each vertical line $x = x_0$, traversed as $y = y_0 e^{ct}$, is a geodesic. Its speed $g(\dot{\gamma}, \dot{\gamma})^{1/2} = |\dot{y}|/y = |c|$ is constant, consistent with the parametrization being affine; the exponential dependence, rather than a linear one, is what constant speed demands in this metric, since intervals near the boundary $y = 0$ are metrically long.

The semicircles on the real axis. For the non-vertical geodesics, consider the curve

$$x(t) = a + R\tanh t, \qquad y(t) = \frac{R}{\cosh t} \qquad (R > 0),$$

whose image is the upper semicircle $(x-a)^2 + y^2 = R^2$, $y > 0$, centered on the real axis. Direct differentiation gives $\dot{x} = R\operatorname{sech}^2 t$, $\ddot{x} = -2R\operatorname{sech}^2 t \tanh t$, $\dot{y} = -R\operatorname{sech} t \tanh t$, and $\ddot{y} = -R\operatorname{sech} t\left(\operatorname{sech}^2 t - \tanh^2 t\right)$. For the first equation,

$$\frac{2}{y}\,\dot{x}\,\dot{y} = \frac{2}{R\operatorname{sech} t}\left(R\operatorname{sech}^2 t\right)\left(-R\operatorname{sech} t \tanh t\right) = -2R\operatorname{sech}^2 t \tanh t = \ddot{x},$$

so $\ddot{x} - \frac{2}{y}\dot{x}\dot{y} = 0$. For the second, using $\operatorname{sech}^2 t + \tanh^2 t = 1$,

$$\frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 = R\operatorname{sech}^3 t - R\operatorname{sech} t\tanh^2 t = R\operatorname{sech} t\left(\operatorname{sech}^2 t - \tanh^2 t\right) = -\ddot{y},$$

so $\ddot{y} + \frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 = 0$. Both equations hold, and the speed is computed from $y^{-2} = R^{-2}\operatorname{sech}^{-2} t$ and $\dot{x}^2 + \dot{y}^2 = R^2\operatorname{sech}^2 t\,(\operatorname{sech}^2 t + \tanh^2 t)$: the R^2 and the $\operatorname{sech}^{\pm 2} t$ factors cancel, leaving $g(\dot{\gamma}, \dot{\gamma}) = y^{-2}(\dot{x}^2 + \dot{y}^2) = \operatorname{sech}^2 t + \tanh^2 t = 1$ by the identity $\operatorname{sech}^2 t + \tanh^2 t = 1$, so the parametrization is by arc length.

These exhaust the geodesics. Through a point $p = (x_0, y_0)$ with $y_0 > 0$, a unit direction v is either vertical, matched by the vertical ray through p, or non-vertical, in which case exactly one circle centered on the real axis passes through p tangent to v (its center is the intersection of the axis with the Euclidean normal line to v at p), realized by a suitable (a, R) above. In every case a solution of the geodesic equation with initial data (p, v) has been produced, so by the uniqueness in theorem 3.1.3 it is *the* geodesic through (p, v). The geodesics of the hyperbolic plane are therefore precisely the vertical rays and the semicircles meeting the real axis at right angles, a family that looks nothing like the straight lines of the coordinate plane it is drawn in.

The remaining structural fact about geodesics is that they move at constant speed. This is not built into the definition; it is a consequence of metric compatibility, the property of the Levi-Civita connection that its covariant derivative respects the inner product.

Proposition 3.1.6: Geodesics Have Constant Speed

Let (M, g) be a (pseudo-)Riemannian manifold and let γ be a geodesic. Then the quantity $\langle\dot\gamma, \dot\gamma\rangle$ is constant along γ. In the Riemannian case the speed $|\dot\gamma|_g = \langle\dot\gamma, \dot\gamma\rangle^{1/2}$ is constant, and γ is either constant or an immersion with nowhere-vanishing velocity.

***Proof*:**
Consider the smooth real function $s(t) = \langle\dot\gamma(t), \dot\gamma(t)\rangle$ along γ, where $\langle -, -\rangle = g(-, -)$. Its derivative is computed by the along-a-curve Leibniz rule for the covariant derivative, the compatibility identity (2.6) established in section 2.4: for any two vector fields $V, W \in \mathfrak{X}(\gamma)$,

$$\frac{d}{dt}\langle V, W\rangle = \langle D_t V, W\rangle + \langle V, D_t W\rangle$$

This identity is the along-a-curve counterpart of the compatibility of definition 2.2.4, and, unlike the on-M statement, it holds for every field along γ whether or not it extends to a field on M; that is why it applies to $\dot\gamma$ directly. Taking $V = W = \dot\gamma$ gives

$$\dot s(t) = \frac{d}{dt}\langle\dot\gamma, \dot\gamma\rangle = 2\,\langle D_t\dot\gamma, \dot\gamma\rangle$$

Since γ is a geodesic, $D_t\dot\gamma = 0$ by definition 3.1.1, so $\dot s \equiv 0$ and s is constant on the interval of definition. Hence $\langle\dot\gamma, \dot\gamma\rangle$ is constant, as claimed.

In the Riemannian case g is positive definite, so $s(t) = |\dot\gamma|_g^2 \geq 0$ and $|\dot\gamma|_g = s^{1/2}$ is a constant $c \geq 0$. If $c = 0$ then $\dot\gamma \equiv 0$, so γ is constant. If $c > 0$ then $\dot\gamma(t) \neq 0$ for all t, so γ is an immersion, completing the proof.

Constant speed has two consequences that shape everything downstream. First, a geodesic comes with a preferred class of parametrizations. If γ is a geodesic and $t \mapsto at + b$ is an affine change of parameter, then $\sigma(t) = \gamma(at + b)$ satisfies $\dot\sigma = a\,\dot\gamma$ and $D_t\dot\sigma = a^2\,(D_t\dot\gamma) \circ (at + b) = 0$, so σ is again a geodesic; but a nonaffine reparametrization changes the speed and destroys the geodesic property, precisely because constant speed is forced. Geodesics are therefore *affinely parametrized* curves, and among all parametrizations of a given geodesic image only the affine ones, running at constant speed, are geodesics. Second, for a nonconstant Riemannian geodesic the speed is a positive constant, so one may rescale time to arrange unit speed $|\dot\gamma|_g \equiv 1$; this arc-length parametrization is the standing normalization when geodesics are compared with arbitrary curves as candidate shortest paths, the theme taken up through the Gauss lemma and the distance function in the sections that follow. In the pseudo-Riemannian case the proof shows only that $\langle\dot\gamma, \dot\gamma\rangle$ is constant, not that a speed can be normalized to one: the constant may be negative or zero, and a geodesic is called timelike, spacelike, or null according to the sign of this conserved quantity, its causal character preserved for all time.

3.2 The Exponential Map and Normal Coordinates

The geodesic equation of section 3.1 produces, for each starting point p and each initial velocity $v \in T_pM$, a unique maximal geodesic γ_v (theorem 3.1.3). The data of the problem live in the tangent space: a point p is fixed, and the free parameter is the vector v. It is natural to assemble these solutions into a single map that reads a tangent vector and returns the point the geodesic reaches. Fixing the travel time at one unit, the map sends $v \in T_pM$ to $\gamma_v(1)$, the endpoint of the geodesic that leaves p with velocity v after unit time. This is the exponential map at p, and it is the device that transports the linear structure of the tangent space onto the manifold near p.

The payoff is a coordinate system tailored to the geometry rather than to an arbitrary chart. The Christoffel symbols Γ^k_{ij} measure the chart, not the geometry (this was the closing lesson of section 2.3, where polar coordinates on the flat plane had nonzero symbols), and one would like coordinates in which the metric looks as Euclidean as a chart can be forced to look. The exponential map delivers exactly this: its inverse is a chart, the *normal coordinates*, in which the metric agrees with the Euclidean metric to first order at p, the Christoffel symbols vanish at p, and every geodesic through p is a straight ray. These coordinates are the standard tool for computing pointwise metric invariants and for reducing tensor identities at a point to their flat-space form.

Definition 3.2.1: Exponential Map

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and fix $p \in M$. For $v \in T_pM$ let γ_v denote the maximal geodesic with $\gamma_v(0) = p$ and $\dot\gamma_v(0) = v$ (theorem 3.1.3). Let

$$\mathcal{E}_p = \{v \in T_pM \mid \gamma_v \text{ is defined on an interval containing } [0, 1]\}$$

be the set of vectors for which the geodesic exists up to time 1. The *exponential map* at p is

$$\exp_p : \mathcal{E}_p \to M, \qquad \exp_p(v) = \gamma_v(1)$$

Two features of this definition ask to be recorded at once. The domain $\mathcal{E}_p$ is a subset of the vector space T_pM, and it contains the origin, since γ_0 is the constant geodesic at p, defined for all time, so $\exp_p(0) = p$. It need not be all of T_pM: a geodesic can run off the manifold in finite time, as on an open ball in $\mathbb{R}^n$, where a geodesic aimed at the boundary ceases to exist once it reaches the edge, so long vectors v fall outside $\mathcal{E}_p$. A manifold on which every γ_v extends to all of $\mathbb{R}$, so that $\mathcal{E}_p = T_pM$ for every p, is called geodesically complete; the equivalence of this condition with metric completeness is the content of the Hopf–Rinow theorem of a later chapter, and is not needed here. What is needed is that $\mathcal{E}_p$ is an open neighborhood of 0 containing a ball, which the smooth dependence of geodesics on their initial data guarantees and which the differential computation below makes precise.

The defining feature that makes $\exp_p$ tractable is the homogeneity of the geodesic equation under rescaling of the velocity: travelling twice as fast for half as long lands at the same point. This is the rescaling identity, and it is what lets the single travel time $t = 1$ in the definition capture geodesics of every length.

Proposition 3.2.2: Rescaling Identity for Geodesics

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, $v \in T_pM$, and $\lambda \in \mathbb{R}$. Then for every t for which either side is defined,

$$\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$$

Consequently, if $v \in \mathcal{E}_p$ then $tv \in \mathcal{E}_p$ for all $t \in [0, 1]$, and

$$\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t);$$

that is, $t \mapsto \exp_p(tv)$ is the geodesic γ_v itself, reparametrized on $[0, 1]$.

***Proof*:**
Fix v and λ and set $\sigma(t) := \gamma_v(\lambda t)$, defined for all t with λt in the domain of γ_v. This σ is a smooth curve with $\sigma(0) = \gamma_v(0) = p$, and by the chain rule its velocity is $\dot\sigma(t) = \lambda\,\dot\gamma_v(\lambda t)$, so $\dot\sigma(0) = \lambda\,\dot\gamma_v(0) = \lambda v$. It remains to check that σ is a geodesic. The geodesic condition is $D_t\dot\sigma = 0$ (definition 3.1.1), where D_t is the covariant derivative along the curve. In a chart, $\sigma^k(t) = \gamma_v^k(\lambda t)$, so $\dot\sigma^k(t) = \lambda\,\dot\gamma_v^k(\lambda t)$ and $\ddot\sigma^k(t) = \lambda^2\,\ddot\gamma_v^k(\lambda t)$, and the local form of the geodesic equation (3.1) gives

$$\ddot\sigma^k + \Gamma^k_{ij}(\sigma)\,\dot\sigma^i\dot\sigma^j = \lambda^2\,\ddot\gamma_v^k(\lambda t) + \Gamma^k_{ij}(\gamma_v(\lambda t))\,\lambda\dot\gamma_v^i(\lambda t)\,\lambda\dot\gamma_v^j(\lambda t) = \lambda^2\Big(\ddot\gamma_v^k + \Gamma^k_{ij}(\gamma_v)\,\dot\gamma_v^i\dot\gamma_v^j\Big)\Big|_{\lambda t}$$

The bracketed quantity vanishes for all arguments because γ_v is a geodesic, so $D_t\dot\sigma = 0$. Thus σ is a geodesic with $\sigma(0) = p$ and $\dot\sigma(0) = \lambda v$. By uniqueness of geodesics (theorem 3.1.3), $\sigma = \gamma_{\lambda v}$ on their common domain, which is the claimed identity $\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$.

For the consequence, suppose $v \in \mathcal{E}_p$, so γ_v is defined on an interval containing $[0, 1]$. Fix $t \in [0, 1]$. Applying the identity with $\lambda = t$ shows $\gamma_{tv}(s) = \gamma_v(ts)$ is defined for all s with $ts \in [0, 1]$, in particular for $s \in [0, 1]$ since $t \in [0, 1]$; hence $tv \in \mathcal{E}_p$. Evaluating at $s = 1$, $\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t)$, as we sought to show.

The rescaling identity has a clean picture behind it. The image $\exp_p(\{tv : t \in [0, 1]\})$ of a radial segment in T_pM is precisely the geodesic γ_v run from p to $\exp_p(v)$; the exponential map straightens each geodesic through p into a ray emanating from the origin of T_pM. This is why $\exp_p$ deserves to be thought of as spreading the tangent space out over the manifold along geodesics, and it is the geometric fact that the normal coordinates below make into an algebraic one.

Example 3.2.3: The exponential map of Euclidean space

On $(\mathbb{R}^n, \bar g)$ with $\bar g = \delta_{ij}\,dx^i\,dx^j$ (example 1.1.2) the Christoffel symbols vanish in the standard chart, so the geodesic equation (3.1) reads $\ddot\gamma^k = 0$, whose solutions are the affine lines $\gamma_v(t) = p + tv$. These are defined for all $t \in \mathbb{R}$, so $\mathcal{E}_p = T_p\mathbb{R}^n$ for every p, and under the canonical identification $T_p\mathbb{R}^n \cong \mathbb{R}^n$,

$$\exp_p(v) = \gamma_v(1) = p + v$$

The exponential map at p is translation by p, a diffeomorphism of $\mathbb{R}^n$ onto itself. The rescaling identity $\gamma_{\lambda v}(t) = p + \lambda tv = \gamma_v(\lambda t)$ is visible directly, and the radial rays of $T_p\mathbb{R}^n$ map to the straight lines through p. Euclidean space is the case in which $\exp_p$ is a global diffeomorphism, not only a local one, and the normal coordinates it produces are the standard Cartesian coordinates translated to place p at the origin.

On the round sphere the picture is already less trivial, and it exhibits the finite-time behaviour that Euclidean space suppresses.

Example 3.2.4: The exponential map of the round sphere

On the unit sphere (S^n, g) of example 1.3.4 the geodesics through a point p are the great circles traced at constant speed; a unit vector $v \in T_pS^n$ generates the great circle

$$\gamma_v(t) = (\cos t)\, p + (\sin t)\, v,$$

a curve on S^n (since $|\gamma_v(t)|^2 = \cos^2 t + \sin^2 t = 1$ using $\langle p, v\rangle = 0$) with $\gamma_v(0) = p$, $\dot\gamma_v(0) = v$, and unit speed. For a general $v \in T_pS^n$ with $|v| = \rho > 0$, the rescaling identity gives $\gamma_v(t) = (\cos \rho t)\, p + (\sin \rho t)\, v/\rho$, so

$$\exp_p(v) = \gamma_v(1) = (\cos |v|)\, p + (\sin |v|)\, \frac{v}{|v|}, \qquad \exp_p(0) = p$$

Every geodesic extends to all of $\mathbb{R}$, so $\mathcal{E}_p = T_pS^n$. Here $\exp_p$ is far from injective: the vector v with $|v| = \pi$ sends every direction to the antipode $-p$, and adding $2\pi\, v/|v|$ returns to the same image. The map is a diffeomorphism only on the open ball $\{v \mid |v| < \pi\}$, whose image is S^n minus the antipode. This is the generic situation the next proposition addresses: $\exp_p$ is a local diffeomorphism near 0, and how far the local statement extends is a global question governed by the injectivity radius $\mathrm{inj}(p)$, which for the unit sphere equals π.

The differential of $\exp_p$ at the origin is the computation that turns the exponential map into a coordinate chart. The key is again the rescaling identity: it forces the derivative of $\exp_p$ along a radial direction to reproduce that same direction, so that $\exp_p$ moves the origin at unit rate in every direction at once.

Proposition 3.2.5: The Differential of the Exponential Map at the Origin

Let (M, g) be a (pseudo-)Riemannian manifold and $p \in M$. Under the canonical identification $T_0(T_pM) \cong T_pM$ of the tangent space to the vector space T_pM at its origin with T_pM itself, the differential of $\exp_p$ at 0 is the identity:

$$d(\exp_p)_0 = \mathrm{id}_{T_pM}$$

Consequently there is an open neighborhood of 0 in T_pM on which $\exp_p$ is a diffeomorphism onto an open neighborhood of p in M.

***Proof*:**

The domain $\mathcal{E}_p$ is a neighborhood of 0 in T_pM: by smooth dependence of geodesics on their initial data (theorem 3.1.3), the set of (q, w) for which the geodesic exists up to time 1 is open in the tangent bundle, and it contains $(p, 0)$, so its slice $\mathcal{E}_p$ over p is an open neighborhood of 0. Hence $\exp_p$ is a smooth map between open subsets of manifolds of the same dimension n, and its differential $d(\exp_p)_0 \colon T_0(T_pM) \to T_pM$ is defined.

Because T_pM is a vector space, the canonical identification $T_0(T_pM) \cong T_pM$ sends a vector $v \in T_pM$ to the velocity at $s = 0$ of the straight line $s \mapsto sv$ in T_pM. Compute $d(\exp_p)_0$ on this vector by differentiating $\exp_p$ along that line. For $|s|$ small enough that $sv \in \mathcal{E}_p$, the rescaling identity (proposition 3.2.2) gives

$$\exp_p(sv) = \gamma_v(s),$$

so the curve $s \mapsto \exp_p(sv)$ in M is exactly the geodesic γ_v. By the chain rule, applied to the composition of $s \mapsto sv$ with $\exp_p$,

$$d(\exp_p)_0(v) = \frac{d}{ds}\Big|_{s=0} \exp_p(sv) = \frac{d}{ds}\Big|_{s=0} \gamma_v(s) = \dot{\gamma}_v(0) = v$$

Thus $d(\exp_p)_0(v) = v$ for every $v \in T_pM$, which is the identity map under the stated identification.

The identity map is invertible, so $d(\exp_p)_0$ is a linear isomorphism. By the inverse function theorem, $\exp_p$ restricts to a diffeomorphism from some open neighborhood of 0 in T_pM onto an open neighborhood of $\exp_p(0) = p$ in M, completing the proof.

The statement that $d(\exp_p)_0$ is the identity is the precise form of the intuition that $\exp_p$ agrees with the linear structure of T_pM to first order at the base point. It is a first-order statement only: the second derivatives of $\exp_p$ at 0 encode the metric's departure from flatness, and this is where the curvature will eventually enter (through the Jacobi equation of Chapter 5). For the present purposes, first order is enough. The inverse function theorem now upgrades the local isomorphism of differentials to a local diffeomorphism of maps, and it is this diffeomorphism whose inverse serves as a chart.

Definition 3.2.6: Normal Neighborhoods and Normal Coordinates

Let (M, g) be a (pseudo-)Riemannian manifold and $p \in M$. An open neighborhood U of p is a *normal neighborhood* of p if there is a star-shaped open neighborhood V of 0 in T_pM (meaning $v \in V$ implies $tv \in V$ for all $t \in [0, 1]$) such that $\exp_p$ restricts to a diffeomorphism $\exp_p : V \to U$. When V is an open ball $\{v \mid |v| < \varepsilon\}$, its image U is a *geodesic ball* (or *normal ball*) $B(p, \varepsilon)$ of radius ε.
Given a normal neighborhood U of p and an orthonormal basis $(e_1, \ldots, e_n)$ of (T_pM, g_p), let $E : \mathbb{R}^n \to T_pM$ be the linear isomorphism $E(x^1, \ldots, x^n) = x^i e_i$ (Einstein summation over the repeated index is in force). The map

$$\varphi := E^{-1} \circ (\exp_p|_V)^{-1} : U \to \mathbb{R}^n$$

is a smooth chart on U, called a system of *normal coordinates* (or *geodesic normal coordinates*) centered at p. In these coordinates p has coordinates 0, and the geodesic γ_v with $v = v^i e_i$ is the straight ray $t \mapsto (tv^1, \ldots, tv^n)$.

The three ingredients of the chart deserve to be separated. The exponential map converts a neighborhood of p into a star-shaped set in the vector space T_pM; the choice of an orthonormal basis identifies that vector space with $\mathbb{R}^n$ in a way that makes g_p the standard dot product; and the composite is the coordinate map. The requirement that V be star-shaped is exactly what makes the last sentence of the definition hold: for $v \in V$ the whole segment $\{tv : t \in [0, 1]\}$ lies in V, so by the rescaling identity (proposition 3.2.2) its image $t \mapsto \exp_p(tv) = \gamma_v(t)$ is a geodesic lying in U, and in coordinates this image is the straight line through the origin with direction $(v^1, \ldots, v^n)$. Normal coordinates are the coordinates in which radial geodesics from p are literally the straight lines through the origin, parametrized proportionally to arc length. The freedom in the construction is exactly the choice of orthonormal basis, so any two normal coordinate systems at p differ by an element of $O(n)$ (or of the appropriate pseudo-orthogonal group in the indefinite case) acting on $\mathbb{R}^n$.

Example 3.2.7: Normal coordinates on the sphere are geodesic polar coordinates

On the unit sphere (S^n, g) of example 1.3.4, fix a point p and an orthonormal basis $(e_1, \dots, e_n)$ of T_pS^n. From example 3.2.4, for $v = v^i e_i$ with $|v| = \rho < \pi$,

$$\exp_p(v) = (\cos\rho)\, p + (\sin\rho)\, \frac{v}{\rho},$$

and this restricts to a diffeomorphism on the ball $V = \{v \mid |v| < \pi\}$, which is star-shaped about 0. The image $U = \exp_p(V)$ is S^n minus the antipode $-p$, a normal neighborhood, and $\varphi = E^{-1} \circ (\exp_p|_V)^{-1}$ is a normal coordinate chart. A point at normal-coordinate distance $\rho = |v|$ from p lies on the great circle in direction v/ρ at arc length ρ from p, so the coordinate $|\varphi(q)|$ of a point q is exactly its distance from p along the connecting geodesic. Writing $\rho = |v|$ and letting $\theta = v/|v|$ range over the unit sphere of T_pS^n recovers the classical geodesic polar coordinates (ρ, θ) on S^n, in which the metric will take the warped form $d\rho^2 + \sin^2\rho\, d\theta^2$ once the Gauss lemma of section 3.3 is available. The failure of $\exp_p$ to be a diffeomorphism at $|v| = \pi$, where the entire coordinate sphere collapses to $-p$, is the reason the normal chart cannot be extended past radius π.

The value of normal coordinates is that the metric assumes its simplest possible pointwise form at the center, and its first derivatives there vanish. This is the sharp version of the claim that $\exp_p$ agrees with the flat model to first order, transferred from the map to the metric components.

Proposition 3.2.8: The Metric in Normal Coordinates

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, and let $(x^1, \dots, x^n)$ be normal coordinates centered at p arising from an orthonormal basis $(e_1, \dots, e_n)$ of (T_pM, g_p) (definition 3.2.6). Write g_{ij} for the components of g and Γ^k_{ij} for the Christoffel symbols of the Levi-Civita connection in these coordinates. Then, at the center p:

1. $g_{ij}(p) = \varepsilon_i \delta_{ij}$, where $\varepsilon_i = \langle e_i, e_i \rangle_g \in \{+1, -1\}$; in the Riemannian case $g_{ij}(p) = \delta_{ij}$;
2. $\Gamma^k_{ij}(p) = 0$ for all i, j, k;
3. $\partial_k g_{ij}(p) = 0$ for all i, j, k.

***Proof*:**

The three statements are proved in order; each rests on the fact, from definition 3.2.6 via the rescaling identity (proposition 3.2.2), that in normal coordinates every geodesic through p is a straight ray from the origin.

Step 1: the metric at p. The coordinate vector field ∂_i at p corresponds under the chart to the basis vector e_i. Indeed, the normal coordinate map is $\varphi = E^{-1} \circ (\exp_p|_V)^{-1}$, so its inverse is $\varphi^{-1} = \exp_p \circ E$; the coordinate vector $\partial_i|_p$ is the image under $d(\varphi^{-1})_0$ of the standard basis vector $\partial/\partial x^i$ of $\mathbb{R}^n$ at 0. Now $d(\varphi^{-1})_0 = d(\exp_p)_0 \circ dE_0$; the linear map E is its own differential and sends $\partial/\partial x^i$ to e_i, and $d(\exp_p)_0 = \mathrm{id}$ by proposition 3.2.5. Hence $\partial_i|_p = e_i$. Therefore

$$g_{ij}(p) = \langle \partial_i, \partial_j \rangle_p = \langle e_i, e_j \rangle_g = \varepsilon_i \delta_{ij},$$

since $(e_1, \dots, e_n)$ is g_p-orthonormal, so $\langle e_i, e_j \rangle_g = 0$ for $i \neq j$ and $\langle e_i, e_i \rangle_g = \pm 1$; in the Riemannian case all $\varepsilon_i = +1$ and $g_{ij}(p) = \delta_{ij}$.

Step 2: the Christoffel symbols at p. Fix any $v = v^i e_i \in T_pM$ small enough that the ray $t \mapsto tv$ lies in V. By definition 3.2.6, the geodesic γ_v has, in normal coordinates, the components

$$\gamma_v^k(t) = t\, v^k,$$

linear in t. Substituting into the local geodesic equation (3.1), and using $\dot{\gamma}_v^k(t) = v^k$ and $\ddot{\gamma}_v^k(t) = 0$, gives for every t (with the point $\gamma_v(t)$ in the chart)

$$0 = \ddot{\gamma}_v^k + \Gamma_{ij}^k(\gamma_v(t))\, \dot{\gamma}_v^i \dot{\gamma}_v^j = \Gamma_{ij}^k(\gamma_v(t))\, v^i v^j$$

Evaluate at $t = 0$, where $\gamma_v(0) = p$:

$$\Gamma_{ij}^k(p)\, v^i v^j = 0 \qquad \text{for every } v = (v^1, \dots, v^n) \in \mathbb{R}^n \text{ near } 0$$

Both sides are homogeneous of degree 2 in v, so the identity, holding on a neighborhood of 0, holds for all $v \in \mathbb{R}^n$. For fixed k, the map $v \mapsto \Gamma_{ij}^k(p)\, v^i v^j$ is a quadratic form whose associated symmetric matrix is $\frac{1}{2}\big(\Gamma_{ij}^k(p) + \Gamma_{ji}^k(p)\big) = \Gamma_{ij}^k(p)$, the last equality because the Levi-Civita connection is torsion-free, so $\Gamma_{ij}^k = \Gamma_{ji}^k$ (proposition 2.3.3). A symmetric bilinear form whose quadratic form vanishes identically is zero: taking $v = e_i$ gives $\Gamma_{ii}^k(p) = 0$, and taking $v = e_i + e_j$ gives $\Gamma_{ii}^k(p) + 2\Gamma_{ij}^k(p) + \Gamma_{jj}^k(p) = 0$, whence $\Gamma_{ij}^k(p) = 0$. Thus $\Gamma_{ij}^k(p) = 0$ for all i, j, k.

Step 3: the first derivatives of the metric at p. The Christoffel symbols and the first derivatives of the metric determine one another. By proposition 2.3.3,

$$\Gamma_{ij}^k = \tfrac{1}{2}\, g^{kl}\big(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\big)$$

Lowering the free index by contracting with g_{km} and summing over k, using $g_{km} g^{kl} = \delta_m^l$, yields the equivalent relation

$$g_{km}\Gamma_{ij}^k = \tfrac{1}{2}\big(\partial_i g_{jm} + \partial_j g_{im} - \partial_m g_{ij}\big)$$

Evaluate at p. The left-hand side vanishes by Step 2, so at p

$$\partial_i g_{jm}(p) + \partial_j g_{im}(p) - \partial_m g_{ij}(p) = 0 \qquad \text{for all } i, j, m \tag{3.3}$$

To solve this for the individual derivatives, abbreviate $a_{ijm} := \partial_i g_{jm}(p)$. This array is symmetric in its last two indices, $a_{ijm} = a_{imj}$, because $g_{jm} = g_{mj}$. In this notation, and rewriting the third term of (3.3) as $\partial_m g_{ij}(p) = a_{mij} = a_{mji}$, the relation reads

$$a_{ijm} + a_{jim} = a_{mij} \qquad \text{for all } i, j, m \tag{3.4}$$

Form the three relations obtained from (3.4) by cyclically rotating the ordered triple (i, j, m): the relation itself, its version under $(i, j, m) \mapsto (j, m, i)$, and its version under $(i, j, m) \mapsto (m, i, j)$,

$$a_{ijm} + a_{jim} = a_{mij}, \qquad a_{jmi} + a_{mji} = a_{ijm}, \qquad a_{mij} + a_{imj} = a_{jmi}$$

Add all three. Using the last-two-index symmetry to identify $a_{jmi} = a_{jim}$, $a_{mji} = a_{mij}$, and $a_{imj} = a_{ijm}$, every term on the right of one equation is matched by an identical term on the left of another, and the total collapses to

$$a_{ijm} + a_{jim} + a_{mij} = 0 \tag{3.5}$$

Substitute the original relation (3.4), in the form $a_{ijm} + a_{jim} = a_{mij}$, into (3.5):

$$a_{mij} + a_{mij} = 0, \qquad \text{hence} \qquad a_{mij} = 0$$

The indices are arbitrary, so $a_{ijm} = 0$ for every i, j, m; that is, $\partial_i g_{jm}(p) = 0$, completing the proof.

The three properties are the working content of normal coordinates. The first two say that at the center p the metric is the standard inner product and the connection has no first-order part: covariant differentiation at p agrees with ordinary partial differentiation, because the correction terms $\Gamma^k_{ij}(p)$ are gone. The third is the sharper statement that the metric is flat to first order at p, so the Taylor expansion of g_{ij} about p begins $g_{ij}(x) = \varepsilon_i \delta_{ij} + O(|x|^2)$ with no linear term. What survives is the quadratic term, and its coefficients are the second derivatives of the metric at p; these cannot in general be removed by any choice of coordinates, and their obstruction to being made to vanish is precisely the curvature of Chapter 4. Normal coordinates are thus the exact expression of the principle that a (pseudo-)Riemannian metric is Euclidean to first order but not to second: the first-order deviation is coordinate artifact, the second-order deviation is geometry.

The computational use of proposition 3.2.8 is constant in what follows. Any tensor identity that is $C^\infty(M)$-linear, hence pointwise, may be verified at an arbitrary point p by passing to normal coordinates there, where the metric is $\varepsilon_i \delta_{ij}$ and all Christoffel symbols vanish. Terms involving ∇ collapse to partial derivatives at p, and expressions that would be forbidding in an arbitrary chart become tractable. The device is the geometer's counterpart of choosing an orthonormal basis in linear algebra, and it will be used to establish the symmetries of the curvature tensor, the divergence and Laplacian formulas, and the second variation, each time by reducing an invariant statement at a point to its flat-space form.

Example 3.2.9: Normal coordinates confirm the flat model

On $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, the exponential map at p is $\exp_p(v) = p + v$ (example 3.2.3), so with the standard orthonormal basis the normal coordinate map $\varphi(q) = q - p$ is translation carrying p to the origin. In these coordinates $g_{ij} \equiv \delta_{ij}$ is constant, so every claim of proposition 3.2.8 holds not only at p but everywhere: $g_{ij}(p) = \delta_{ij}$, the Christoffel symbols $\Gamma^k_{ij} = \frac{1}{2}\delta^{kl}(\partial_i \delta_{jl} + \partial_j \delta_{il} - \partial_l \delta_{ij}) = 0$ vanish identically, and $\partial_k g_{ij} \equiv 0$. This is the degenerate case in which the second-order term also vanishes, consistent with the flatness of Euclidean space; on a curved manifold such as the sphere the analogous normal coordinates make $g_{ij}(p) = \delta_{ij}$, $\Gamma^k_{ij}(p) = 0$, $\partial_k g_{ij}(p) = 0$ hold at the center but leave a nonzero $\partial_k \partial_l g_{ij}(p)$, the seed of the curvature computation, as for the sphere of example 3.2.7.

3.3 The Gauss Lemma

The exponential map $\exp_p$ of definition 3.2.1 turns a neighborhood of the origin in the tangent space T_pM into a neighborhood of p in M, and by proposition 3.2.5 it does so as a diffeomorphism, its differential at the origin being the identity. This much is a statement about the point p alone. What it does not yet reveal is how $\exp_p$ distorts the linear geometry of T_pM as one moves away from the origin. In T_pM there is a clean picture: the rays through the origin are orthogonal to the spheres $|v| = r$ centered at the origin, and travelling out along a ray at unit speed increases r at unit rate. The exponential map carries the rays to geodesics through p and carries the spheres to certain hypersurfaces in M; whether it also carries the orthogonality of that flat picture to M is not obvious, since $\exp_p$ is only a diffeomorphism, not an isometry, and its differential away from the origin is unknown.

The Gauss lemma answers this. It states that $\exp_p$ preserves exactly the part of the flat orthogonality that involves the radial direction: radial geodesics from p remain orthogonal to the images of the spheres, and the differential of $\exp_p$ sends the radial direction of T_pM to the unit-speed velocity of the radial geodesic without any distortion in length. The tangential directions may be stretched or sheared, and how much they are is the business of curvature; but the radial–tangential angle is rigid. From this single orthogonality statement the metric near p acquires a normal form in polar coordinates, $g = dr^2 + h_r$, in which the radial part is exactly Euclidean and all the geometry is packed into the family h_r of metrics on the geodesic spheres. That normal form is what proves the local minimizing property proved in the next section, and it is the starting point for the comparison theory of later chapters. Throughout, the Einstein summation convention is in force: an index that appears once raised and once lowered is summed over $1, \ldots, n$.

Definition 3.3.1: Geodesic Balls and Geodesic Spheres

Let (M, g) be a Riemannian manifold, $p \in M$, and let $\mathcal{O} \subseteq T_pM$ be a star-shaped open neighborhood of the origin on which $\exp_p$ (definition 3.2.1) restricts to a diffeomorphism onto a normal neighborhood $U = \exp_p(\mathcal{O})$ (definition 3.2.6). For $r > 0$ small enough that the open ball $B_r(0) = \left\{ v \in T_pM \;\middle|\; |v|_g < r \right\}$ has closure contained in $\mathcal{O}$, the image

$$B_r(p) := \exp_p\bigl(B_r(0)\bigr)$$

is called the *geodesic ball* (or *normal ball*) of radius r about p, and the image of the sphere,

$$S_r(p) := \exp_p\Bigl(\left\{ v \in T_pM \;\middle|\; |v|_g = r \right\}\Bigr),$$

is called the *geodesic sphere* of radius r about p. The number r is the *radius* of the geodesic ball or sphere.

The set $\left\{ v \;\middle|\; |v|_g = r \right\}$ is a round sphere of radius r in the inner product space (T_pM, g_p), and since $\exp_p$ is a diffeomorphism on $\mathcal{O}$, its image $S_r(p)$ is an embedded hypersurface of M diffeomorphic to S^{n-1}, bounding the geodesic ball $B_r(p)$. The name is a warning as much as a description: a geodesic sphere is not in general a metric sphere for any obvious distance, nor need it look round from inside M; it is only the exponential image of a Euclidean sphere in the tangent space. What the Gauss lemma will show is that, despite this, the radial geodesics leaving p cross every geodesic sphere at a right angle, just as radii cross spheres in the flat model.

Example 3.3.2: Geodesic Spheres in Euclidean Space and on the Round Sphere

On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2 the geodesics through a point p are the straight lines $t \mapsto p + tv$, so $\exp_p(v) = p + v$ for every $v \in T_pM \cong \mathbb{R}^n$, and $\exp_p$ is a global diffeomorphism. A geodesic ball $B_r(p)$ is then the ordinary open Euclidean ball of radius r about p, and the geodesic sphere $S_r(p)$ is the ordinary Euclidean sphere of radius r. Here the geodesic sphere really is a round sphere, and the radial lines $t \mapsto p + t\,v/\,|v|$ meet it orthogonally by the elementary geometry of $\mathbb{R}^n$; the Gauss lemma below recovers this as a special case.

On the round sphere S^n_r of example 1.3.4 the picture is already less trivial. Fix a point p; the geodesics through p are the great circles, traversed at constant speed. A geodesic ball $B_\rho(p)$ of radius $\rho < \pi r$ (a scale on which $\exp_p$ is still a diffeomorphism) is a spherical cap, and the geodesic sphere $S_\rho(p)$ is the boundary circle of that cap, the set of points at great-circle distance ρ from p. This boundary circle is a Euclidean circle in the ambient $\mathbb{R}^{n+1}$, but its radius as measured inside S^n_r is not ρ: for $n = 2$ and unit sphere $r = 1$ the circumference of $S_\rho(p)$ is $2\pi \sin\rho$, not $2\pi\rho$. The tangential directions are compressed relative to the flat model, precisely the distortion the Gauss lemma does not control. What it does control is that the great circles through p still meet each cap boundary at a right angle, which one sees directly from the rotational symmetry of the sphere about the axis through p.

The proof of the Gauss lemma rests on a single computation with a two-parameter family of geodesics, so it is worth isolating the mechanism before stating the result. Let $v \in T_pM$ lie in the domain $\mathcal{O}$ of definition 3.3.1, and let $w \in T_pM$ be any tangent vector, which under the canonical identification $T_v(T_pM) \cong T_pM$ is regarded as a tangent vector to T_pM at the point v. The vector $d(\exp_p)_v(w) \in T_{\exp_p(v)}M$ is computed by running a curve in T_pM through v with velocity w, applying $\exp_p$, and differentiating. The next lemma records the one identity about such families that the argument needs: the symmetry of the mixed covariant derivatives of a variation, a consequence of the connection being torsion-free.

Lemma 3.3.3: Symmetry of the Mixed Covariant Derivative

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and let $\Gamma\colon (-\varepsilon, \varepsilon) \times [a, b] \to M$, $(s, t) \mapsto \Gamma(s, t)$, be a smooth map (a *variation*). Write $\partial_s\Gamma$ and $\partial_t\Gamma$ for the coordinate velocity fields of Γ, each a smooth vector field along Γ, and let D_s, D_t denote covariant differentiation along the s- and t-parameter curves (definition 2.1.6). Then

$$D_s\,\partial_t\Gamma = D_t\,\partial_s\Gamma$$

***Proof*:**
The identity is local, so it suffices to verify it in a chart $(U, (x^k))$ meeting the image of Γ, using the coordinate expression for D_t from definition 2.1.6. Write the components of Γ as $\Gamma^k(s, t)$. Then

$$\partial_t\Gamma = \frac{\partial\Gamma^k}{\partial t}\,\partial_k, \qquad \partial_s\Gamma = \frac{\partial\Gamma^k}{\partial s}\,\partial_k,$$

where (∂_k) is the coordinate frame. By the local formula for the covariant derivative of a vector field along a curve, applied to the s-curves,

$$D_s\,\partial_t\Gamma = \left(\frac{\partial^2\Gamma^k}{\partial s\,\partial t} + \Gamma^k_{ij}\,\frac{\partial\Gamma^i}{\partial s}\,\frac{\partial\Gamma^j}{\partial t}\right)\partial_k,$$

the second term arising because the frame vectors ∂_k themselves vary along the s-curve, contributing $\frac{\partial\Gamma^i}{\partial s}\,\nabla_{\partial_i}\partial_j = \frac{\partial\Gamma^i}{\partial s}\,\Gamma^k_{ij}\partial_k$ against the j-component $\frac{\partial\Gamma^j}{\partial t}$ of $\partial_t\Gamma$. Interchanging the roles of s and t,

$$D_t\,\partial_s\Gamma = \left(\frac{\partial^2\Gamma^k}{\partial t\,\partial s} + \Gamma^k_{ij}\,\frac{\partial\Gamma^i}{\partial t}\,\frac{\partial\Gamma^j}{\partial s}\right)\partial_k$$

The pure second-derivative terms agree because mixed partial derivatives of the smooth functions Γ^k commute. The connection terms agree because the Christoffel symbols are symmetric in their lower indices, $\Gamma^k_{ij} = \Gamma^k_{ji}$ (proposition 2.3.3), which lets the relabelling $i \leftrightarrow j$ turn $\Gamma^k_{ij}\,\frac{\partial\Gamma^i}{\partial t}\,\frac{\partial\Gamma^j}{\partial s}$ into $\Gamma^k_{ij}\,\frac{\partial\Gamma^i}{\partial s}\,\frac{\partial\Gamma^j}{\partial t}$. Both bracketed expressions are therefore equal, so $D_s\,\partial_t\Gamma = D_t\,\partial_s\Gamma$, as claimed.

Symmetry of the Christoffel symbols is the whole of the connection input to the lemma, and it is what makes the Gauss lemma a statement about the Levi-Civita connection in particular; a connection with torsion would introduce a discrepancy $\partial_s\Gamma$ and $\partial_t\Gamma$ equal to the torsion tensor evaluated on the two velocity fields, and the orthogonality below would fail. With this in hand the Gauss lemma is a short computation.

Theorem 3.3.4: The Gauss Lemma

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, and let $v \in T_pM$ be a nondegenerate vector, meaning $\langle v, v \rangle_g \neq 0$, lying in the domain on which $\exp_p$ is defined and a local diffeomorphism (definition 3.2.1). Identify $T_v(T_pM)$ with T_pM in the canonical way. Then for every $w \in T_pM$,

$$\left\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \right\rangle_g = \langle v, w \rangle_g$$

In particular:

1. $d(\exp_p)_v(v) = \dot{\gamma}_v(1)$, the velocity at $t = 1$ of the radial geodesic $\gamma_v(t) = \exp_p(tv)$, and $\left\langle d(\exp_p)_v(v), d(\exp_p)_v(v) \right\rangle_g = \langle v, v \rangle_g$: the radial direction is carried to the radial geodesic without change of the conserved quantity $\langle \dot{\gamma}_v, \dot{\gamma}_v \rangle_g$; in the Riemannian case this reads $\left| d(\exp_p)_v(v) \right|_g = |v|_g$, no distortion of length; and

2. if $\langle v, w \rangle_g = 0$, then $d(\exp_p)_v(v) \perp d(\exp_p)_v(w)$: in the Riemannian case, where $\langle v, w \rangle_g = 0$ means $w \perp v$ in the inner product space (T_pM, g_p), this says that radial geodesics from p meet the geodesic spheres $S_r(p)$ of definition 3.3.1 orthogonally.

***Proof*:**
Fix v in the domain with $\langle v, v \rangle_g \neq 0$, and let $w \in T_pM$ be arbitrary. Both sides of the asserted identity are $\mathbb{R}$-linear in w with v fixed, so it suffices to treat the two cases $w = v$ and $\langle v, w \rangle_g = 0$ separately; the general case follows in *Step 3* by writing w as the sum of its component along v and its g_p-orthogonal complement and using linearity.

Step 1: the radial direction. By the rescaling identity of definition 3.2.1, $\gamma_v(t) = \exp_p(tv)$ is the geodesic with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$, defined at least for $t \in [0, 1]$. Differentiating $t \mapsto \exp_p(tv)$ at $t = 1$ using the chain rule, and noting that the curve $t \mapsto tv$ in T_pM has velocity v at the point v,

$$d(\exp_p)_v(v) = \left.\frac{d}{dt}\right|_{t=1} \exp_p(tv) = \dot{\gamma}_v(1)$$

Along a geodesic the quantity $\langle \dot{\gamma}_v, \dot{\gamma}_v \rangle_g$ is constant (proposition 3.1.6), so $\langle \dot{\gamma}_v(1), \dot{\gamma}_v(1) \rangle_g = \langle \dot{\gamma}_v(0), \dot{\gamma}_v(0) \rangle_g = \langle v, v \rangle_g$, which gives both $d(\exp_p)_v(v) = \dot{\gamma}_v(1)$ and, taking $w = v$,

$$\left\langle d(\exp_p)_v(v), d(\exp_p)_v(v) \right\rangle_g = \langle \dot{\gamma}_v(1), \dot{\gamma}_v(1) \rangle_g = \langle v, v \rangle_g$$

This proves the identity when $w = v$ and establishes claim (1).

Step 2: the orthogonal directions. Suppose now $\langle v, w \rangle_g = 0$; the goal is $\left\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \right\rangle_g = 0$. Choose a curve $s \mapsto v(s)$ in T_pM with $v(0) = v$, with $\langle v(s), v(s) \rangle_g = \langle v, v \rangle_g$ constant in s, and with $v'(0) = w$. Such a curve exists: the level set $\Sigma = \left\{ u \in T_pM \;\middle|\; \langle u, u \rangle_g = \langle v, v \rangle_g \right\}$ is, near v, a smooth hypersurface of T_pM, because the function $u \mapsto \langle u, u \rangle_g$ has differential $2 \langle v, - \rangle_g$ at v, which is nonzero: the inner product g_p is nondegenerate, and $v \neq 0$ since $\langle v, v \rangle_g \neq 0$ by hypothesis. The tangent space to Σ at v is the kernel of this differential, namely $v^\perp = \left\{ u \;\middle|\; \langle v, u \rangle_g = 0 \right\}$. Since $w \in v^\perp$ by hypothesis, there is a smooth curve in Σ through v with velocity w at $s = 0$, and any such curve serves as $v(s)$. Along it $\langle v(s), v(s)\rangle_g$ is constant. Define the variation of geodesics

$$\Gamma(s,t) := \exp_p\bigl(t\,v(s)\bigr), \qquad (s,t) \in (-\varepsilon,\varepsilon)\times[0,1],$$

which is smooth by smoothness of $\exp_p$ and of $s \mapsto v(s)$, and is defined on this rectangle for ε small because v lies in the open domain of $\exp_p$. For each fixed s the curve $t \mapsto \Gamma(s,t) = \gamma_{v(s)}(t)$ is the geodesic with initial velocity $v(s)$, so

$$D_t\,\partial_t\Gamma = 0 \qquad \text{and} \qquad \langle \partial_t\Gamma(s,t), \partial_t\Gamma(s,t)\rangle_g = \langle v(s), v(s)\rangle_g = \langle v, v\rangle_g \tag{3.6}$$

for all (s,t), the first because each t-curve is a geodesic (definition 3.1.1) and the second because $\langle \partial_t\Gamma, \partial_t\Gamma\rangle_g$ is constant along each geodesic t-curve (proposition 3.1.6), together with the constancy of $\langle v(s), v(s)\rangle_g$ in s.

Now consider the function

$$f(s,t) := \langle \partial_t\Gamma, \partial_s\Gamma\rangle_g\,(s,t),$$

the g-inner product of the two velocity fields of the variation along Γ. The claim of Step 2 is exactly that $f(0,1) = 0$, since at $(s,t) = (0,1)$ one has $\partial_t\Gamma(0,1) = \dot\gamma_v(1) = d(\exp_p)_v(v)$ by Step 1, and

$$\partial_s\Gamma(0,1) = \frac{\partial}{\partial s}\bigg|_{s=0} \exp_p\bigl(v(s)\bigr) = d(\exp_p)_v\bigl(v'(0)\bigr) = d(\exp_p)_v(w),$$

using $t = 1$ so that $t\,v(s) = v(s)$ and $v'(0) = w$. The strategy is to differentiate f in t and integrate.

Differentiate f along the t-curve, using metric compatibility of ∇ (definition 2.2.4), which turns the derivative of an inner product into inner products of covariant derivatives:

$$\frac{\partial f}{\partial t} = \langle D_t\,\partial_t\Gamma, \partial_s\Gamma\rangle_g + \langle \partial_t\Gamma, D_t\,\partial_s\Gamma\rangle_g$$

The first term vanishes by the geodesic equation (3.6). For the second term, the symmetry lemma (lemma 3.3.3) replaces $D_t\,\partial_s\Gamma$ by $D_s\,\partial_t\Gamma$, so

$$\frac{\partial f}{\partial t} = \langle \partial_t\Gamma, D_s\,\partial_t\Gamma\rangle_g = \tfrac{1}{2}\,\frac{\partial}{\partial s}\,\langle \partial_t\Gamma, \partial_t\Gamma\rangle_g\,,$$

where the second equality is again metric compatibility, now applied to the s-derivative of $\langle \partial_t\Gamma, \partial_t\Gamma\rangle_g$. By (3.6) the quantity $\langle \partial_t\Gamma, \partial_t\Gamma\rangle_g$ equals $\langle v, v\rangle_g$, a constant independent of both s and t, so its s-derivative is zero. Therefore

$$\frac{\partial f}{\partial t}(s,t) = 0 \qquad \text{for all } (s,t),$$

and $f(s,t)$ does not depend on t: for each fixed s, $f(s,\cdot)$ is constant.

It remains to evaluate this constant. Consider the value at $t = 0$. For every s, $\Gamma(s,0) = \exp_p(0) = p$, so the s-curve $s \mapsto \Gamma(s,0)$ is constant at p, and hence its velocity vanishes: $\partial_s\Gamma(s,0) = 0$. Consequently

$$f(s,0) = \langle \partial_t\Gamma(s,0), \partial_s\Gamma(s,0)\rangle_g = \langle \partial_t\Gamma(s,0), 0\rangle_g = 0$$

Since $f(s,\cdot)$ is constant in t, it follows that $f(s,t) = 0$ for all t, in particular $f(0,1) = 0$. As computed above, $f(0,1) = \bigl\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\bigr\rangle_g$, so

$$\bigl\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\bigr\rangle_g = 0 = \langle v, w\rangle_g\,,$$

which is the asserted identity when $\langle v, w\rangle_g = 0$ and establishes claim (2).

Step 3: the general case. For arbitrary $w \in T_pM$, decompose $w = \lambda v + w^\perp$ with $\lambda \in \mathbb{R}$ and $\left\langle v, w^\perp\right\rangle_g = 0$; this is possible with $\lambda = \langle v, w\rangle_g / \langle v, v\rangle_g$ because $\langle v, v\rangle_g \neq 0$ by hypothesis, taking $w^\perp = w - \lambda v$, which satisfies $\left\langle v, w^\perp\right\rangle_g = \langle v, w\rangle_g - \lambda \langle v, v\rangle_g = 0$. By linearity of $d(\exp_p)_v$ and bilinearity of g, using Step 1 for the λv part and Step 2 for the $w^\perp$ part,

$$\left\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\right\rangle_g = \lambda \langle v, v\rangle_g + 0 = \langle v, w\rangle_g$$

This proves the identity for all w. The orthogonality assertion of claim (2) is the special case $\langle v, w\rangle_g = 0$, saying that the radial geodesic γ_v, whose velocity at $t = 1$ is $d(\exp_p)_v(v)$, is g-orthogonal to every vector $d(\exp_p)_v(w)$ with $\langle v, w\rangle_g = 0$. In the Riemannian case these vectors $d(\exp_p)_v(w)$, for $w \perp v$ in (T_pM, g_p), are exactly the vectors tangent to the geodesic sphere $S_{|v|_g}(p)$, since they are the images under $d(\exp_p)_v$ of the vectors w tangent to the Euclidean sphere through v; so the radial geodesic meets that geodesic sphere orthogonally. This completes the proof.

The lemma is named for Gauss because the surface case, that the radial geodesics from a point cut the geodesic circles at right angles, is one of the computations behind his *Theorema Egregium*; the argument above is the modern packaging, and it isolates precisely the two ingredients that are used, metric compatibility (to differentiate the inner product) and the torsion-free symmetry (through lemma 3.3.3). Both are the defining properties of the Levi-Civita connection, so the Gauss lemma is a property of the metric alone. Notice what is *not* asserted: nothing is claimed about $\left\langle d(\exp_p)_v(w), d(\exp_p)_v(w')\right\rangle_g$ for w, w' both tangential. Those inner products measure how the geodesic spheres are shaped, and they are governed by curvature, which is why the round sphere and Euclidean space, which share the flat radial structure, have geodesic spheres of different sizes as example 3.3.2 records. The rigidity the Gauss lemma provides is only in the radial direction, but that is enough to put the metric into a normal form.

Proposition 3.3.5: The Metric in Geodesic Polar Coordinates

Let (M, g) be a Riemannian manifold, $p \in M$, and let $B_R(p)$ be a geodesic ball about p (definition 3.3.1). On the punctured ball $B_R(p) \setminus \{p\}$ introduce *geodesic polar coordinates* (r, θ), where $r = \left|\exp_p^{-1}(\cdot)\right|_g \in (0, R)$ is the radial distance parameter and θ ranges over the unit sphere $\left\{u \in T_pM \;\middle|\; |u|_g = 1\right\} \cong S^{n-1}$, so that $q = \exp_p\bigl(r\,\theta\bigr)$. Then in these coordinates the metric takes the form

$$g = dr^2 + h_r,$$

where for each fixed $r \in (0, R)$ the expression h_r is a Riemannian metric on the geodesic sphere $S_r(p)$, and dr^2 and h_r contain no cross term: g has no $dr\,d\theta$ component. Equivalently, the radial vector field ∂_r is a g-unit vector field everywhere orthogonal to each geodesic sphere, and h_r is the metric induced on $S_r(p)$ by restriction of g.

***Proof*:**
The map $\Psi\colon (0, R) \times S^{n-1} \to B_R(p) \setminus \{p\}$, $\Psi(r, \theta) = \exp_p(r\theta)$, is a diffeomorphism: it is the composition of the diffeomorphism $\exp_p$ on the punctured ball $B_R(0) \setminus \{0\} \subset T_pM$ with the polar-coordinate diffeomorphism $(0, R) \times S^{n-1} \to B_R(0) \setminus \{0\}$, $(r, \theta) \mapsto r\theta$, of the punctured Euclidean

ball in (T_pM, g_p). The coordinate vector fields of Ψ are computed by pushing forward the corresponding vector fields on $(0, R) \times S^{n-1}$. At a point $q = \exp_p(r_0\theta_0)$, write $v = r_0\theta_0 \in T_pM$, so $|v|_g = r_0$.

The radial field. The r-coordinate curve through q is $r \mapsto \exp_p(r\theta_0) = \gamma_{\theta_0}(r)$, the unit-speed radial geodesic in direction θ_0. Hence

$$\partial_r\big|_q = \dot{\gamma}_{\theta_0}(r_0) = d(\exp_p)_v(\theta_0),$$

and by claim (1) of the Gauss lemma (theorem 3.3.4), applied with the radial vector $v = r_0\theta_0$ and using linearity of $d(\exp_p)_v$ so that $d(\exp_p)_v(\theta_0) = \frac{1}{r_0} d(\exp_p)_v(v)$,

$$g(\partial_r, \partial_r)\big|_q = \left|d(\exp_p)_v(\theta_0)\right|_g^2 = \frac{1}{r_0^2}\left|d(\exp_p)_v(v)\right|_g^2 = \frac{1}{r_0^2}\,|v|_g^2 = \frac{1}{r_0^2}\,r_0^2 = 1$$

So ∂_r is a g-unit vector field, which accounts for the coefficient 1 of dr^2 and shows the dr^2 term is exactly Euclidean.

No cross term. Let ∂_{θ^α} $(\alpha = 1, \ldots, n-1)$ be any coordinate vector field of the sphere factor S^{n-1}, tangent to the level set $\{r = r_0\}$. Its image under Ψ at q is

$$\partial_{\theta^\alpha}\big|_q = d(\exp_p)_v\big(r_0\,\partial_{\theta^\alpha}\theta_0\big),$$

where $\partial_{\theta^\alpha}\theta_0 \in T_{\theta_0}S^{n-1}$ is tangent to the unit sphere at θ_0 and hence g_p-orthogonal to θ_0, so that the vector $w := r_0\,\partial_{\theta^\alpha}\theta_0$ satisfies $\langle v, w\rangle_g = r_0^2\,\langle\theta_0, \partial_{\theta^\alpha}\theta_0\rangle_g = 0$. By claim (2) of the Gauss lemma (theorem 3.3.4),

$$g\big(\partial_r, \partial_{\theta^\alpha}\big)\big|_q = \frac{1}{r_0}\,\big\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\big\rangle_g = \frac{1}{r_0}\,\langle v, w\rangle_g = 0$$

So g has no $dr\,d\theta^\alpha$ cross terms, and in the coordinate frame $(\partial_r, \partial_{\theta^1}, \ldots, \partial_{\theta^{n-1}})$ the matrix of g is block diagonal, with the 1×1 radial block equal to 1 and the remaining $(n-1) \times (n-1)$ block equal to the restriction of g to the level set $\{r = r_0\} = S_{r_0}(p)$.

The tangential block is a metric on the sphere. Define h_r to be this tangential block, that is the 2-tensor on $S_r(p)$ obtained by restricting g to vectors tangent to the geodesic sphere; it is the first fundamental form of $S_r(p)$ as a submanifold of the Riemannian manifold (M, g) in the sense of definition 1.5.1. By proposition 1.5.2, the restriction of the Riemannian metric g to the tangent spaces of the embedded hypersurface $S_r(p)$ is positive definite, hence h_r is a Riemannian metric on $S_r(p)$ for each $r \in (0, R)$. Assembling the two blocks, in geodesic polar coordinates

$$g = dr^2 + h_r,$$

with h_r a Riemannian metric on the geodesic sphere of radius r and no cross term, which is the assertion, completing the proof.

The form $g = dr^2 + h_r$ separates the geometry near p into two pieces of unequal difficulty. The radial piece dr^2 is universal, the same on every Riemannian manifold, and it is the Gauss lemma that certifies this; travelling out along a radial geodesic, one gains arclength at exactly the rate dr, with no interference from the tangential directions. The tangential piece h_r carries everything that distinguishes one manifold from another near p. For Euclidean space, $\exp_p(r\theta) = p + r\theta$ gives

$h_r = r^2\, g_{S^{n-1}}$, the round metric on the sphere of radius r, so $g = dr^2 + r^2\, g_{S^{n-1}}$ recovers the polar form of the flat metric of example 1.1.2; the coefficient r^2 is what makes the geodesic sphere of radius r have the Euclidean size. On the round unit sphere S^n the same construction yields $h_r = \sin^2 r\; g_{S^{n-1}}$, and the deficit between $\sin^2 r$ and r^2 is the tangential compression seen in example 3.3.2, the first appearance of positive curvature. The rate at which h_r departs from the Euclidean $r^2\, g_{S^{n-1}}$ as r grows is measured by the second derivatives of the metric, which is where the curvature tensor of the next chapter enters; the Gauss lemma has already fixed the first-order radial behaviour, so the polar form is the correct coordinate system in which to read curvature off the metric.

Example 3.3.6: The Polar Form on a Surface of Revolution

The normal form $g = dr^2 + h_r$ is not only an abstract consequence of the Gauss lemma; on a surface it is a coordinate system one can write down. Take the warped metric

$$g = d\rho^2 + f(\rho)^2\, d\varphi^2, \qquad f(\rho) > 0,$$

of a surface of revolution as in example 1.5.4, in coordinates (ρ, φ), and suppose $f(0) = 0$ with $f'(0) = 1$, the condition that makes $\rho = 0$ a smooth pole rather than a cone point. The meridian curves $\varphi = \text{const}$ are the curves along which only ρ varies; their velocity has g-length $|\partial_\rho|_g = \sqrt{g_{\rho\rho}} = 1$, so each meridian is parametrized by arclength, and a direct check from the Christoffel symbols of this warped metric, whose only nonzero entries are $\Gamma^\rho_{\varphi\varphi} = -ff'$ and $\Gamma^\varphi_{\rho\varphi} = \Gamma^\varphi_{\varphi\rho} = f'/f$ by proposition 2.3.3, shows the meridians satisfy the geodesic equation (3.1): with $\dot\varphi = 0$ one has $\ddot\rho = -\Gamma^\rho_{\varphi\varphi}\dot\varphi^2 = 0$ and $\ddot\varphi + 2\Gamma^\varphi_{\rho\varphi}\dot\rho\dot\varphi = 0$, both satisfied identically. So the meridians are unit-speed radial geodesics emanating from the pole $\rho = 0$, and ρ is exactly the radial distance parameter r of proposition 3.3.5. Reading off the metric, $dr^2 = d\rho^2$ and

$$h_r = f(r)^2\, d\varphi^2,$$

a metric on the geodesic circle $S_r(p)$ (here a one-dimensional sphere, so h_r is a single positive coefficient $f(r)^2$). The Gauss lemma is visible in the absence of a $d\rho\, d\varphi$ cross term in the warped metric, which was built in from the start; the content of proposition 3.3.5 is that this absence is forced by the geometry, not a lucky feature of the coordinates. The circumference of the geodesic circle of radius r is $\int_0^{2\pi} \sqrt{h_r}\, d\varphi = 2\pi f(r)$, and the pole condition $f(0) = 0,\ f'(0) = 1$ makes this $2\pi r + O(r^3)$ for small r, matching the Euclidean circumference to leading order exactly as the polar form $g = dr^2 + h_r$ with $h_r \to r^2 g_{S^{n-1}}$ predicts.

3.4 Geodesics Locally Minimize and the Distance Function

A geodesic was defined in section 3.1 as a curve of vanishing acceleration, a curve that is as straight as the connection allows. Straightness and shortness are separate ideas, and the passage from one to the other is the business of this section. On $\mathbb{R}^n$ a straight segment is the shortest path between its endpoints, and every student meets the two descriptions of a line as interchangeable; on a curved manifold they come apart. A great circle on the sphere is a geodesic, yet the long arc of a great circle between two nearby points is a geodesic that is far from shortest, and no path realizes the distance between antipodal points uniquely. The correct statement is local: near any point a geodesic really is the shortest path to nearby points, and it is the only shortest path up to how it is parametrized. The tool that makes this precise is the Gauss lemma of section 3.3, which says that inside a normal ball the geodesics emanating from the center are orthogonal to the spheres they cross. That single orthogonality forces any competing path to spend length crossing spheres it need not cross.

Once shortest paths exist locally, the length of the shortest path becomes a candidate for the distance between two points, and the infimum of lengths over all connecting curves defines a function $d_g\colon M \times M \to [0, \infty)$. This section constructs d_g, proves that it is a metric in the sense of metric-space theory, and proves that the topology it generates is the topology M already carried as a manifold. That last statement closes the loop opened in box 1.1.4: the Riemannian metric, a tensor, and the distance function, a metric-space datum, are different objects, but the second is built from the first, and the resulting metric space recovers the manifold exactly. Throughout, (M, g) is Riemannian; positive-definiteness is used everywhere here, since the length of a curve must be a nonnegative real number for the infimum to behave, and the indefinite case is excluded.

Recall that a piecewise-smooth curve on $[a, b]$ is a continuous map $\gamma\colon [a, b] \to M$ for which there is a partition $a = t_0 < t_1 < \cdots < t_k = b$ such that γ restricted to each $[t_{i-1}, t_i]$ is smooth; at the partition points γ has a right and a left velocity, possibly different. The velocity $\dot\gamma(t)$ is defined for all but finitely many t, and $|\dot\gamma(t)|_g$ is a bounded function with at most finitely many discontinuities, hence integrable.

Definition 3.4.1: Length and the Riemannian Distance

Let (M, g) be a Riemannian manifold. The *length* of a piecewise-smooth curve $\gamma\colon [a, b] \to M$ is

$$L(\gamma) = \int_a^b |\dot\gamma(t)|_g \, dt = \int_a^b \sqrt{\langle \dot\gamma(t), \dot\gamma(t) \rangle_g} \, dt$$

Two points $p, q \in M$ are joined by a piecewise-smooth curve when M is connected. The *Riemannian distance* between p and q is

$$d_g(p, q) = \inf \{ L(\gamma) \mid \gamma \text{ a piecewise-smooth curve from } p \text{ to } q \},$$

the infimum of the lengths of all piecewise-smooth curves γ with $\gamma(a) = p$ and $\gamma(b) = q$.

The length is defined for a piecewise-smooth curve, not only a smooth one, because the curves that compete to be shortest are built by concatenation: to go from p to q one might travel along one geodesic and then another, and the join is a corner. The integral does not see the corner, a finite set of points of measure zero, so length is additive over concatenation and unchanged by inserting extra partition points. Length is also invariant under reparametrization. If $\varphi\colon [c, d] \to [a, b]$ is a piecewise-smooth increasing bijection and $\tilde\gamma = \gamma \circ \varphi$, then $\dot{\tilde\gamma}(s) = \varphi'(s)\, \dot\gamma(\varphi(s))$ where $\varphi' > 0$, so $\left|\dot{\tilde\gamma}(s)\right|_g = \varphi'(s)\ |\dot\gamma(\varphi(s))|_g$, and the substitution $t = \varphi(s)$ gives $L(\tilde\gamma) = L(\gamma)$. Thus length is a property of the image curve together with its orientation, and the infimum defining d_g may as well range over curves parametrized on $[0, 1]$, or over unit-speed curves, without changing its value. The choice of parametrization that makes analysis cleanest is arc length: a curve γ on $[a, b]$ with $|\dot\gamma|_g \equiv 1$ has $L(\gamma) = b - a$, and every piecewise-smooth curve with nowhere-vanishing speed admits a unit-speed reparametrization.

That d_g is a nonnegative real number and not just a formal infimum requires that some curve join p to q and that lengths be bounded below, both of which hold: on a connected manifold any two points are joined by a piecewise-smooth curve, and lengths are nonnegative, so the infimum is a well-defined element of $[0, \infty)$.

Example 3.4.2: Length and Distance in Euclidean Space

On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, with $\bar{g} = \delta_{ij}\, dx^i\, dx^j$, the length of a piecewise-smooth curve $\gamma = (\gamma^1, \ldots, \gamma^n)$ is

$$L(\gamma) = \int_a^b \sqrt{(\dot{\gamma}^1)^2 + \cdots + (\dot{\gamma}^n)^2}\, dt,$$

the ordinary arc length of vector calculus. For the straight segment $\gamma(t) = p + t(q-p)$ on $[0,1]$ one has $\dot{\gamma} \equiv q - p$, so $L(\gamma) = |q-p|_{\bar{g}}$, the Euclidean norm of the displacement. For any other piecewise-smooth curve σ from p to q, the fundamental theorem of calculus applied to each coordinate gives $q - p = \int_a^b \dot{\sigma}(t)\, dt$, and the triangle inequality for the vector-valued integral yields

$$|q-p|_{\bar{g}} = \left| \int_a^b \dot{\sigma}(t)\, dt \right|_{\bar{g}} \leq \int_a^b |\dot{\sigma}(t)|_{\bar{g}}\, dt = L(\sigma)$$

Hence $L(\sigma) \geq |q-p|_{\bar{g}}$ for every competitor, with equality for the straight segment, so

$$d_{\bar{g}}(p,q) = |q-p|_{\bar{g}}$$

The Riemannian distance on Euclidean space is exactly the Euclidean distance, and the infimum is attained precisely by the straight segments, that is, by the geodesics of $\bar{g}$ (whose geodesic equation $\ddot{\gamma}^k = 0$, from eq. (3.1) with vanishing Christoffel symbols, has the affine lines as solutions). This is the model that the local minimizing theorem generalizes: a geodesic is locally the unique shortest path, and the equality case of a triangle inequality is what pins down uniqueness.

The Euclidean computation is a template. The general local statement replaces the straight segment by a radial geodesic inside a normal ball and replaces the coordinatewise triangle inequality by the Gauss lemma. Fix $p \in M$ and a normal ball $B = \exp_p(\left\{ v \in T_pM \;\middle|\; |v|_g < \varepsilon \right\})$ on which $\exp_p$ is a diffeomorphism onto its image, as produced in section 3.2. For $q \in B$ with $q = \exp_p(v)$, the *radial geodesic* to q is $\gamma_v(t) = \exp_p(tv)$ on $[0,1]$; it has constant speed $|v|_g$ by proposition 3.1.6, so its length is $L(\gamma_v) = |v|_g$, the radius at which q sits. The claim is that no piecewise-smooth path from p to q is shorter, and that the radial geodesic is the only one that ties.

Theorem 3.4.3: Radial Geodesics Locally Minimize

Let (M,g) be a Riemannian manifold, $p \in M$, and let $B = \exp_p(B_\varepsilon(0))$ be a geodesic ball about p, where $B_\varepsilon(0) = \left\{ v \in T_pM \;\middle|\; |v|_g < \varepsilon \right\}$ and $\exp_p$ is a diffeomorphism from $B_\varepsilon(0)$ onto B. Let $q \in B$, write $q = \exp_p(v)$ with $r = |v|_g < \varepsilon$, and let $\gamma_v(t) = \exp_p(tv)$, $t \in [0,1]$, be the radial geodesic from p to q. Then for every piecewise-smooth curve $\sigma\colon [a,b] \to M$ with $\sigma(a) = p$ and $\sigma(b) = q$,

$$L(\sigma) \geq r = L(\gamma_v),$$

and equality holds if and only if σ is a monotone reparametrization of γ_v. Consequently the radial geodesic is the unique shortest piecewise-smooth path from p to q up to reparametrization, and $d_g(p,q) = r$.

***Proof*:**
Write $q = \exp_p(v)$, $r = |v|_g$. Introduce geodesic polar coordinates on the punctured ball $B \setminus \{p\}$: by theorem 3.3.4 and proposition 3.3.5, on $B \setminus \{p\}$ the metric takes the form

$$g = dr^2 + h_r, \tag{3.7}$$

where r is the radial coordinate and h_r is for each fixed r a Riemannian metric on the geodesic sphere $\{x \in B \mid r(x) = r\}$; in particular h_r is positive semidefinite on the whole tangent space and vanishes exactly on the radial direction ∂_r. The radial coordinate function $r\colon B \setminus \{p\} \to (0, \varepsilon)$ is smooth, being $\left|\exp_p^{-1}(\cdot)\right|_g$, and its differential dr is dual to ∂_r; the decomposition (3.7) is the content of the Gauss lemma, namely that ∂_r has unit length and is g-orthogonal to every vector tangent to a geodesic sphere.

Step 1: the length of any path dominates the change in radius. Let $\sigma\colon [a,b] \to M$ be piecewise smooth with $\sigma(a) = p$ and $\sigma(b) = q$. Reduce to the case that σ maps into the closed ball $\overline{B_r} = \exp_p(\left\{w \;\middle|\; |w|_g \le r\right\})$. If $\sigma([a,b]) \subseteq \overline{B_r}$ this holds already. Otherwise σ leaves $\overline{B_r}$, so there is a time t^* with $\sigma(t^*) \notin \overline{B_r}$. Choose a radius ρ_0 with $r < \rho_0 < \varepsilon$ small enough that $\sigma(t^*)$ lies outside the open geodesic ball $B^\circ_{\rho_0} := \exp_p(\left\{w \;\middle|\; |w|_g < \rho_0\right\})$: if $\sigma(t^*) \in B$ it has some radius in (r, ε) and any ρ_0 below that radius serves, while if $\sigma(t^*) \notin B$ then $\sigma(t^*)$ lies outside every such ball with $\rho_0 < \varepsilon$. Let $c = \inf\left\{t \in [a,b] \;\middle|\; \sigma(t) \notin B^\circ_{\rho_0}\right\}$ be the first time σ exits $B^\circ_{\rho_0}$; the defining set is nonempty (it contains t^*) and $\sigma(a) = p \in B^\circ_{\rho_0}$, so $a < c$. On $[a, c)$ the curve stays inside $B^\circ_{\rho_0}$, hence inside B, and by continuity $\sigma(c)$ lies in the closed ball $\overline{B_{\rho_0}}$ but not in $B^\circ_{\rho_0}$, so $\sigma(c)$ has radius exactly ρ_0, that is, $\sigma(c)$ lies on the geodesic sphere $S_{\rho_0} = \exp_p(\left\{w \;\middle|\; |w|_g = \rho_0\right\})$. Thus $\sigma|_{[a,c]}$ is a path from p to a point of radius ρ_0 contained in $\overline{B_{\rho_0}}$. The estimate proved below, applied with ρ_0 in place of r to this portion, gives $L(\sigma) \ge L(\sigma|_{[a,c]}) \ge \rho_0 > r$, so the claimed inequality $L(\sigma) \ge r$ holds outright. It therefore suffices to prove the bound, and to analyze the equality case, for a path σ with $\sigma([a,b]) \subseteq \overline{B_r}$; assume this from now on.

Let $a_0 = \sup\{t \in [a,b] \mid \sigma(t) = p\}$ be the last time σ sits at p. On $(a_0, b]$ the curve avoids p, so $\rho(t) := r(\sigma(t))$ is defined and piecewise smooth there, with $\rho(t) \to 0$ as $t \downarrow a_0$ and $\rho(b) = r$. Where σ is smooth and away from p, decompose the velocity using (3.7):

$$|\dot\sigma|_g^2 = \langle \dot\sigma, \dot\sigma \rangle_g = dr(\dot\sigma)^2 + h_r(\dot\sigma, \dot\sigma) = \dot\rho^2 + h_r(\dot\sigma, \dot\sigma),$$

using $dr(\dot\sigma) = \frac{d}{dt} r(\sigma(t)) = \dot\rho$. Since h_r is positive semidefinite, $h_r(\dot\sigma, \dot\sigma) \ge 0$, so

$$|\dot\sigma(t)|_g = \sqrt{\dot\rho(t)^2 + h_r(\dot\sigma, \dot\sigma)} \ge \sqrt{\dot\rho(t)^2} = |\dot\rho(t)| \ge \dot\rho(t) \tag{3.8}$$

at every t where $\dot\sigma$ is defined and $\sigma(t) \ne p$.

Step 2: integrate. For $a_0 < c' < b$, integrating (3.8) over $[c', b]$ and using the fundamental theorem of calculus for the piecewise-smooth function ρ gives

$$\int_{c'}^b |\dot\sigma(t)|_g \, dt \ge \int_{c'}^b \dot\rho(t)\, dt = \rho(b) - \rho(c') = r - \rho(c')$$

Letting $c' \downarrow a_0$, the right side tends to $r - 0 = r$, since $\rho(c') \to 0$, while the left side increases to $\int_{a_0}^b |\dot\sigma|_g \, dt \le L(\sigma)$. Hence

$$L(\sigma) \ge \int_{a_0}^b |\dot\sigma|_g \, dt \ge r = L(\gamma_v),$$

which is the asserted inequality for a path contained in $\overline{B_r}$; by the reduction of Step 1 the bound $L(\sigma) \geq r$ then holds for every competitor.

Step 3: the equality case. Suppose $L(\sigma) = r$. Then every inequality above is an equality. In particular $\int_{a_0}^{b} |\dot\sigma|_g \, dt = L(\sigma)$, forcing $|\dot\sigma(t)|_g = 0$ for almost every $t \in [a, a_0]$, so σ is constant at p on $[a, a_0]$ and may be discarded by reparametrization. On $(a_0, b]$ equality in (3.8) at almost every t requires, first, $h_r(\dot\sigma, \dot\sigma) = 0$, hence $\dot\sigma$ has no component tangent to the geodesic spheres, so $\dot\sigma$ is a radial vector, a multiple of ∂_r; and second, $|\dot\rho| = \dot\rho$, hence $\dot\rho \geq 0$, so ρ is nondecreasing. Thus wherever σ moves it moves radially and outward. A curve whose velocity is everywhere a nonnegative multiple of ∂_r stays on the radial line through q: in the coordinates $\exp_p^{-1}$, its direction $v/|v|_g$ is fixed and only its radius ρ varies, so $\sigma(t) = \exp_p\big(\rho(t)\, v/r\big)$ with ρ nondecreasing from 0 to r. This is precisely a monotone reparametrization of $\gamma_v(t) = \exp_p(tv)$, obtained by the increasing change of parameter $t \mapsto \rho(t)/r$. Conversely, any monotone reparametrization of γ_v has length r by reparametrization invariance. Therefore equality holds if and only if σ is a monotone reparametrization of γ_v.

Finally, the inequality $L(\sigma) \geq r$ over all competitors gives $d_g(p, q) \geq r$, while the radial geodesic itself has length r, so $d_g(p, q) = r$, completing the proof.

The theorem is the exact local analogue of the Euclidean computation of example 3.4.2, and the two proofs run in parallel. In both, one bounds the length of an arbitrary path by the length of the geodesic, using an inequality that becomes an equality only for the geodesic itself; in the Euclidean case the inequality is the triangle inequality for the integral of the velocity, and in the general case it is the pointwise inequality (3.8) coming from the Gauss-lemma decomposition $g = dr^2 + h_r$. The decomposition is what does the work: it says that the only length that counts toward reaching a farther sphere is the radial part $\dot\rho$, and any tangential motion is wasted length that a path could avoid by going straight out. A radial geodesic wastes nothing, which is why it, and it alone up to reparametrization, realizes the distance. The word *local* is not decorative. The conclusion holds only for q inside a normal ball on which $\exp_p$ is a diffeomorphism; beyond that ball the exponential map may fail to be injective, radial geodesics may cross, and a radial geodesic can stop being the shortest path, as the long great-circle arc on the sphere shows.

The round sphere makes both the local minimizing property and its limits concrete: on it the Riemannian distance is a closed-form function of the ambient inner product, the minimizer is a short great-circle arc, and the one place the minimizer fails to be unique is exactly the pair of antipodal points that no normal ball contains.

Example 3.4.4: The Distance Function on the Round Sphere

On the round sphere $S^n_r = \{x \in \mathbb{R}^{n+1} \mid |x| = r\}$ of example 1.3.4, with the metric g_r induced from $(\mathbb{R}^{n+1}, \bar g)$, the Riemannian distance between two points is

$$d_{g_r}(p, q) = r \arccos\Big(\frac{\langle p, q\rangle}{r^2}\Big),$$

where $\langle -, - \rangle$ is the Euclidean inner product of $\mathbb{R}^{n+1}$. This is the length of the shorter of the two great-circle arcs joining p to q, and the minimizing geodesic is unique unless p and q are antipodal.

To compute it, first record the competitor whose length the formula reports. The geodesics of S^n_r are the great circles, the intersections of S^n_r with 2-planes through the origin, traced at constant speed, as established in example 3.1.4. Fix $p, q \in S^n_r$ that are not antipodal, so $q \neq -p$, and set

$$\alpha = \arccos\Big(\frac{\langle p, q\rangle}{r^2}\Big) \in [0, \pi),$$

the angle at the origin subtended by p and q; the restriction $\alpha < \pi$ is the non-antipodal hypothesis, since $\langle p, q\rangle = -r^2$ is the antipodal case. If $q = p$ then $\alpha = 0$ and both sides of the asserted formula vanish, so assume $q \neq p$. Then $q \neq \pm p$, and the vectors p, q span a 2-plane; let $u = p/r$ and let w be the unit vector in that plane obtained by orthonormalizing q against u, so that

$$q = r\cos\alpha\, u + r\sin\alpha\, w, \qquad \langle u, u\rangle = \langle w, w\rangle = 1, \qquad \langle u, w\rangle = 0$$

Define the unit-speed great-circle arc $\gamma\colon [0, r\alpha] \to S^n_r$ by

$$\gamma(s) = r\cos\Big(\frac{s}{r}\Big)u + r\sin\Big(\frac{s}{r}\Big)w$$

Then $|\gamma(s)|^2 = r^2\cos^2(s/r) + r^2\sin^2(s/r) = r^2$, so γ lands in S^n_r; it satisfies $\gamma(0) = ru = p$ and $\gamma(r\alpha) = r\cos\alpha\, u + r\sin\alpha\, w = q$. Its velocity is $\dot\gamma(s) = -\sin(s/r)\, u + \cos(s/r)\, w$, a unit vector by orthonormality of u, w, so γ is a geodesic of unit speed. Its length is

$$L(\gamma) = \int_0^{r\alpha} |\dot\gamma(s)|_{g_r}\, ds = \int_0^{r\alpha} 1\, ds = r\alpha,$$

the length of the short great-circle arc from p to q. Hence $d_{g_r}(p, q) \le r\alpha$.

The reverse inequality is where theorem 3.4.3 enters. The exponential map $\exp_p$ of S^n_r is a diffeomorphism from the open ball $\Big\{v \in T_pS^n_r \;\Big|\; |v|_{g_r} < \pi r\Big\}$ onto $S^n_r \setminus \{-p\}$: a unit-speed geodesic from p returns to p after arc length $2\pi r$ and reaches the antipode $-p$ at arc length πr, so no two radial directions meet before that, and every point other than $-p$ is hit exactly once at a radius below πr. Thus for non-antipodal q the geodesic ball $B = \exp_p(\Big\{v \;\Big|\; |v|_{g_r} < \pi r\Big\})$ contains q, and $q = \exp_p(v)$ with $r\alpha = |v|_{g_r} < \pi r$. The theorem applies verbatim: $d_{g_r}(p, q) = |v|_{g_r} = r\alpha$, and the radial geodesic, which is the short great-circle arc γ above, is the unique shortest path up to reparametrization. This proves the formula for $q \neq -p$.

For antipodal $q = -p$ the formula still reads $d_{g_r}(p, q) = r\arccos(-1) = r\pi$, and this too is correct, by continuity or directly: every great-circle arc from p to $-p$ is a half great circle of length $r\pi$, and any path attaining a smaller length would, by the estimate (3.8) applied on a normal ball about p out to radius approaching πr, be forced to have length at least πr in the limit, so $d_{g_r}(p, -p) = r\pi$. Uniqueness fails here in an extreme way: through p and $-p$ there pass infinitely many great circles (one for every 2-plane containing the line $\mathbb{R}p$), each contributing a half-circle of length $r\pi$, so infinitely many distinct geodesics realize the distance. The antipode $-p$ is the first point at which radial geodesics from p cease to minimize uniquely, the simplest instance of the *cut locus*, taken up in a later chapter; here it appears as the exact boundary of the normal ball, the one place the local theorem cannot reach.

An immediate consequence is a lower bound for d_g in terms of the exponential radius, the tool that turns the local minimizing property into a statement about the topology.

Corollary 3.4.5: Distance to Points of a Normal Ball

With the notation of theorem 3.4.3, for every $q = \exp_p(v) \in B$ one has $d_g(p,q) = |v|_g$. Consequently, for every δ with $0 < \delta < \varepsilon$ small enough that the metric ball $\{x \in M \mid d_g(p,x) < \delta\}$ is contained in B, the metric ball and the geodesic ball of radius δ coincide:

$$\{x \in M \mid d_g(p,x) < \delta\} = \exp_p\left(\left\{v \;\middle|\; |v|_g < \delta\right\}\right)$$

***Proof*:**
The equality $d_g(p,q) = |v|_g$ is the last assertion of theorem 3.4.3, applied to q in place of the running point. Such a δ exists, and in fact any δ with $0 < \delta < \varepsilon$ works. To see this, it suffices to check that $d_g(p,x) \geq \varepsilon$ for every $x \in M \setminus B$, for then $d_g(p,x) < \delta < \varepsilon$ forces $x \in B$, so the metric ball of radius δ is contained in B. Fix $x \in M \setminus B$ and let γ be any piecewise-smooth curve from p to x. Since $x \notin B$, the curve γ leaves B, so for each $\rho < \varepsilon$ it reaches radius at least ρ; as in Step 1 of theorem 3.4.3, the intermediate value theorem gives a first time at which γ meets the geodesic sphere $S_\rho = \exp_p(\left\{v \;\middle|\; |v|_g = \rho\right\})$, and the estimate (3.8) integrated out to radius ρ shows that the portion of γ up to that meeting has length at least ρ, whence $L(\gamma) \geq \rho$. As this holds for every $\rho < \varepsilon$, $L(\gamma) \geq \varepsilon$; taking the infimum over γ gives $d_g(p,x) \geq \varepsilon$. Thus $d_g(p, M \setminus B) \geq \varepsilon > 0$ (the same argument used later in the diagonal-vanishing part of theorem 3.4.6), so any $\delta < \varepsilon$ works.

Fix such a δ, so that $\{x \mid d_g(p,x) < \delta\} \subseteq B$, and verify the two inclusions. If $d_g(p,x) < \delta$, then $x \in B$ by the choice of δ, so $x = \exp_p(w)$ for a unique $w \in B_\varepsilon(0)$, and the first part gives $|w|_g = d_g(p,x) < \delta$; hence $x \in \exp_p(\left\{v \;\middle|\; |v|_g < \delta\right\})$. Conversely, if $x = \exp_p(w)$ with $|w|_g < \delta$, then $w \in B_\varepsilon(0)$ (as $\delta < \varepsilon$) and the first part gives $d_g(p,x) = |w|_g < \delta$; hence $x \in \{x \mid d_g(p,x) < \delta\}$. The two sets therefore coincide, as required.

With distances to nearby points controlled, the global object d_g can be shown to satisfy the axioms of a metric and to reproduce the manifold topology.

Theorem 3.4.6: The Riemannian Distance Is a Metric

Let (M,g) be a connected Riemannian manifold. Then d_g is a metric on M: for all $p,q,x \in M$,

1. $d_g(p,q) \geq 0$, with $d_g(p,q) = 0$ if and only if $p = q$;
2. $d_g(p,q) = d_g(q,p)$;
3. $d_g(p,x) \leq d_g(p,q) + d_g(q,x)$.

Moreover the metric-space topology induced by d_g coincides with the manifold topology of M. In particular (M, d_g) is the metric space anticipated in box 1.1.4, built from the tensor g yet recovering the original topology.

***Proof*:**
Nonnegativity and symmetry. Lengths are integrals of the nonnegative function $|\dot{\gamma}|_g$, so $L(\gamma) \geq 0$ and hence $d_g \geq 0$. Symmetry follows from reparametrization: if $\gamma\colon [a,b] \to M$ runs from p to q, then $\gamma^-(t) = \gamma(a+b-t)$ runs from q to p with $|\dot{\gamma}^-(t)|_g = |\dot{\gamma}(a+b-t)|_g$, so $L(\gamma^-) = L(\gamma)$; the set of curves from p to q and the set from q to p are thus in length-preserving bijection, and their infima agree, giving $d_g(p,q) = d_g(q,p)$.

Triangle inequality. Fix p, q, x and $\eta > 0$. Choose a piecewise-smooth curve α from p to q with $L(\alpha) < d_g(p,q) + \eta/2$ and a piecewise-smooth curve β from q to x with $L(\beta) < d_g(q,x) + \eta/2$; these exist by definition of the infimum. The concatenation $\alpha * \beta$, which follows α and then β, is a piecewise-smooth curve from p to x (the join at q is at worst a corner, and length is additive over concatenation), so

$$d_g(p,x) \leq L(\alpha * \beta) = L(\alpha) + L(\beta) < d_g(p,q) + d_g(q,x) + \eta$$

As $\eta > 0$ was arbitrary, $d_g(p,x) \leq d_g(p,q) + d_g(q,x)$.

Vanishing exactly on the diagonal. If $p = q$ the constant curve has length 0, so $d_g(p,p) = 0$. The content is the converse: $d_g(p,q) = 0$ forces $p = q$. Suppose $p \neq q$. Choose a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p small enough that $q \notin B$; this is possible because B is an open neighborhood of p and M is Hausdorff, so shrinking ε excludes the fixed point $q \neq p$. Any piecewise-smooth curve γ from p to q must leave B, hence must cross the geodesic sphere $S_\delta = \exp_p\left(\left\{v \,\middle|\, |v|_g = \delta\right\}\right)$ for every $0 < \delta < \varepsilon$; in particular it crosses $S_{\varepsilon/2}$. By the argument of theorem 3.4.3 (the estimate (3.8) integrated out to radius $\varepsilon/2$), the portion of γ from p to its first meeting with $S_{\varepsilon/2}$ has length at least $\varepsilon/2$, so $L(\gamma) \geq \varepsilon/2$. Taking the infimum over γ gives $d_g(p,q) \geq \varepsilon/2 > 0$. Hence $d_g(p,q) = 0$ implies $p = q$, and d_g is positive off the diagonal. This is the one place positive-definiteness of g is used decisively: it makes $|\dot{\gamma}|_g$ a genuine norm, so that a nonconstant radial motion contributes strictly positive length and distinct points cannot be at distance zero.

The topologies agree. It must be shown that a set is open in the manifold topology if and only if it is open in the d_g-metric topology. It suffices to compare bases of neighborhoods at an arbitrary point p, that is, to show that every manifold-open neighborhood of p contains a metric ball about p, and every metric ball about p contains a manifold-open neighborhood of p.

Fix p and a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p on which corollary 3.4.5 applies. For $0 < \delta < \varepsilon$ small enough that the metric ball $\{x \mid d_g(p,x) < \delta\}$ is contained in B, that corollary gives

$$\{x \in M \mid d_g(p,x) < \delta\} = \exp_p\left(\left\{v \in T_pM \,\middle|\, |v|_g < \delta\right\}\right) \tag{3.9}$$

The right-hand side is the image under the diffeomorphism $\exp_p$ of a Euclidean-open ball in T_pM, hence a manifold-open set; so (3.9) exhibits every sufficiently small metric ball about p as a manifold-open neighborhood of p. This shows every metric ball about p contains a manifold-open neighborhood of p (indeed small ones are manifold-open), giving one inclusion of topologies: every d_g-open set is manifold-open.

Conversely, let W be a manifold-open neighborhood of p. Since $\exp_p$ is a diffeomorphism on $B_\varepsilon(0)$ and continuous, $\exp_p^{-1}(W \cap B)$ is an open neighborhood of 0 in T_pM, so it contains a Euclidean ball $\left\{v \,\middle|\, |v|_g < \delta\right\}$ for some $0 < \delta < \varepsilon$, which may be taken small enough that (3.9) holds. Then, applying $\exp_p$,

$$\{x \mid d_g(p,x) < \delta\} = \exp_p\Big(\Big\{v \;\Big|\; |v|_g < \delta\Big\}\Big) \subseteq W \cap B \subseteq W$$

Thus W contains the metric ball of radius δ about p, so W is a d_g-neighborhood of p; every manifold-open set is therefore d_g-open. The two topologies have the same neighborhoods at every point and hence coincide, as we sought to show.

The theorem resolves the caution recorded at the very start of the book. In box 1.1.4 the word "metric" was flagged as carrying two incompatible meanings, the tensor g and a metric-space distance, and the passage from the first to the second was promised here. The promise is now kept: the tensor g produces, through the length functional and an infimum, a function d_g that satisfies the three axioms of a metric, and the metric space (M, d_g) carries exactly the topology M already had. Nothing about the point-set topology of the manifold is lost or gained by remembering the metric; what is gained is quantitative, a numerical distance between points where before there was only nearness. That the metric topology reproduces the manifold topology is what licenses the free interchange, in all later chapters, between analytic statements phrased with d_g (Cauchy sequences, completeness, bounded sets) and topological statements phrased with charts. Completeness of (M, d_g) and the equivalence of metric completeness with geodesic completeness are the subject of the Hopf–Rinow theorem later; here the point is only that d_g is a metric at all, and the correct one for the manifold.

One direction of the correspondence between geodesics and shortest paths has been established: radial geodesics minimize locally. The reverse direction asks whether a curve that minimizes length must be a geodesic. It is, once parametrized correctly, and the cleanest proof is variational.

Proposition 3.4.7: Minimizers Are Geodesics

Let (M, g) be a Riemannian manifold and let $\gamma\colon [a,b] \to M$ be a piecewise-smooth, unit-speed curve that minimizes length among piecewise-smooth curves with the same endpoints:

$$L(\gamma) = d_g\big(\gamma(a), \gamma(b)\big)$$

Then γ is a geodesic; in particular γ is smooth (with no corners) and satisfies $D_t\dot\gamma = 0$.

***Proof*:**
The argument has two parts: first, a length minimizer is locally a shortest path between nearby points of its image, hence locally a radial geodesic and so smooth with $D_t\dot\gamma = 0$; second, a global verification that the pieces fit with no corner. Both follow from theorem 3.4.3 and its uniqueness clause.

Step 1: local coincidence with a radial geodesic. Fix $t_0 \in [a,b]$ and put $p = \gamma(t_0)$. Choose a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p on which theorem 3.4.3 applies. By continuity of γ, there is $\tau > 0$ with $\gamma([t_0 - \tau, t_0 + \tau] \cap [a,b]) \subseteq B$. Take any $s_1 < s_2$ in this parameter interval and set $q_i = \gamma(s_i)$. The restriction $\gamma|_{[s_1,s_2]}$ is a piecewise-smooth curve from q_1 to q_2, and it is length-minimizing among such curves: if some curve σ from q_1 to q_2 were shorter, replacing $\gamma|_{[s_1,s_2]}$ by σ inside γ would produce a piecewise-smooth curve from $\gamma(a)$ to $\gamma(b)$ of length

$$L(\gamma|_{[a,s_1]}) + L(\sigma) + L(\gamma|_{[s_2,b]}) < L(\gamma|_{[a,s_1]}) + L(\gamma|_{[s_1,s_2]}) + L(\gamma|_{[s_2,b]}) = L(\gamma),$$

contradicting minimality of γ. Hence $L(\gamma|_{[s_1,s_2]}) = d_g(q_1, q_2)$. Now $q_1, q_2 \in B$, and q_2 lies in a geodesic ball about q_1 once $s_2 - s_1$ is small (shrink τ so that the image lies in a ball on which theorem 3.4.3 applies with center q_1); by the uniqueness clause of that theorem, the unique shortest path from q_1 to q_2 up to reparametrization is the radial geodesic. Therefore $\gamma|_{[s_1,s_2]}$ is a monotone reparametrization of a geodesic. Since γ is unit-speed by hypothesis and geodesics have constant speed by proposition 3.1.6, the reparametrization is affine with unit derivative, so $\gamma|_{[s_1,s_2]}$ is itself a geodesic. As $s_1 < s_2$ near t_0 were arbitrary, γ agrees near t_0 with a geodesic and hence is smooth there and satisfies $D_t\dot{\gamma} = 0$ on a neighborhood of t_0.

Step 2: no corners, and global smoothness. Step 1 applies at every $t_0 \in [a,b]$, including the finitely many partition points where γ was only assumed piecewise smooth. At such a point t_0 the curve coincides on a two-sided neighborhood with a single geodesic $\tilde{\gamma}$: the local minimizer through t_0 is a radial geodesic from $\gamma(t_0)$, and by the argument of Step 1 applied on $[s_1, t_0]$ and on $[t_0, s_2]$, both the incoming and outgoing pieces are unit-speed geodesics through $\gamma(t_0)$ realizing the distance to the same nearby points; the uniqueness clause of theorem 3.4.3 forces them to be the two halves of one radial geodesic, so their velocities match at t_0, $\dot{\gamma}(t_0^-) = \dot{\gamma}(t_0^+)$. Thus γ has no corner at t_0, and being a solution of the geodesic equation on each side with matching position and velocity, it is by the uniqueness in theorem 3.1.3 the restriction of a single geodesic across t_0. Hence γ is smooth on all of $[a,b]$ and satisfies $D_t\dot{\gamma} = 0$ throughout, so γ is a geodesic, as required.

Together with theorem 3.4.3 this closes the local circle between the two descriptions of a straight line. Straightness, the vanishing of covariant acceleration, and shortness, the realization of Riemannian distance, are locally the same condition, tied by the exponential map and the Gauss lemma: a radial geodesic is the unique local minimizer, and a unit-speed local minimizer is a geodesic. The proposition is stated for a curve minimizing over its full parameter interval, but its proof is entirely local, using only that short subarcs of a minimizer minimize; this is the property that makes the variational characterization of section 3.5 the natural next tool, where the same conclusion is obtained by differentiating the length and energy functionals and reading off that their critical points are exactly the geodesics. The two viewpoints, the direct distance estimate of this section and the variational calculus of the next, are complementary: the estimate gives uniqueness of the local minimizer outright, while the variation gives the geodesic equation as the vanishing of a derivative and extends without change to the indefinite setting where lengths alone no longer suffice.

3.5 First Variation of Energy and Length

The geodesic equation was introduced in section 3.1 as an equation of motion: a geodesic is a curve whose velocity does not turn, $D_t\dot{\gamma} = 0$. In section 3.4 geodesics acquired a second character, that of locally shortest paths, established inside a normal ball by the Gauss lemma. These two descriptions, autoparallel and length-minimizing, are logically distinct, and the bridge between them is variational. A shortest path is one that cannot be made shorter by any small deformation, and to make that idea precise means differentiating the length of a curve with respect to a parameter that deforms it. This section carries out that differentiation and reads off its consequence: the curves for which the derivative vanishes, the *critical points* of the length, are exactly the geodesics.

Length is not the most convenient functional to differentiate. Its integrand $|\dot{\gamma}|_g$ carries a square root, which is singular wherever the velocity vanishes and which obscures the algebra of the variation. The energy functional, whose integrand is $\frac{1}{2}|\dot{\gamma}|_g^2$, has no square root and no such singularity, and its critical points turn out to be the geodesics on the nose, with no reparametrization ambiguity.

The two functionals are tied together by a Cauchy–Schwarz inequality, $L(\gamma)^2 \leq 2(b-a)E(\gamma)$, with equality precisely for curves of constant speed. The strategy of the section is therefore to compute the first variation of energy first, extract the geodesic equation from it, and then transfer the conclusion to length by way of the inequality. Throughout, (M, g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ of definition 2.3.2, and D_t is the induced covariant derivative along a curve from definition 2.1.6; the Einstein summation convention is in force.

Definition 3.5.1: Energy, Length, and Admissible Variations

Let $\gamma\colon [a, b] \to M$ be a piecewise-smooth curve. The *length* of γ is

$$L(\gamma) = \int_a^b |\dot\gamma(t)|_g \, dt,$$

as in definition 3.4.1, and the *energy* of γ is

$$E(\gamma) = \tfrac{1}{2}\int_a^b |\dot\gamma(t)|_g^2 \, dt = \tfrac{1}{2}\int_a^b \langle\dot\gamma(t), \dot\gamma(t)\rangle \, dt$$

A *variation* of γ is a continuous map $\Gamma\colon (-\varepsilon, \varepsilon) \times [a, b] \to M$, for some $\varepsilon > 0$, that is piecewise smooth in the sense that there is a partition $a = t_0 < t_1 < \cdots < t_m = b$ with Γ smooth on each rectangle $(-\varepsilon, \varepsilon) \times [t_{i-1}, t_i]$, and such that $\Gamma(0, t) = \gamma(t)$ for all t. Each curve $\gamma_s := \Gamma(s, \cdot)$ is a *longitudinal* curve of the variation. The *variation field* of Γ is the piecewise-smooth vector field along γ

$$V(t) = \frac{\partial}{\partial s}\bigg|_{s=0} \Gamma(s, t) = \frac{\partial \Gamma}{\partial s}(0, t) \in T_{\gamma(t)}M$$

The variation is *proper*, or has *fixed endpoints*, if $\Gamma(s, a) = \gamma(a)$ and $\Gamma(s, b) = \gamma(b)$ for all s, equivalently if $V(a) = V(b) = 0$. A curve γ is a *critical point* of E (respectively of L) for proper variations if $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = 0$ (respectively $\frac{d}{ds}\big|_{s=0} L(\gamma_s) = 0$) for every proper variation Γ of γ.

A variation is a one-parameter family of curves through γ, and its variation field V records the initial velocity of that family: at each time t, the vector $V(t)$ is the direction in which the point $\gamma(t)$ begins to move as the parameter s is switched on. Deforming γ while holding its endpoints fixed is exactly the operation whose infinitesimal effect on length one wishes to measure, and the properness condition $V(a) = V(b) = 0$ encodes "holding the endpoints fixed" at the level of the variation field. The energy functional weights the curve by the square of its speed, so it penalizes both length and uneven parametrization; a curve that covers a fixed image but speeds up and slows down carries more energy than one that traverses the same image at constant speed. This double sensitivity is what makes energy the sharper of the two functionals, and it is the reason its critical points are geodesics without the reparametrization slack that length carries.

The variation field need not come from a prescribed variation to be useful; conversely, any vector field along γ is the variation field of some variation, as the following construction shows and as the criticality arguments below will use.

Example 3.5.2: Variations built from a vector field

Let $\gamma\colon [a,b] \to M$ be a smooth curve and let V be any piecewise-smooth vector field along γ. A variation of γ with variation field V can be produced from the exponential map of definition 3.2.1. For each t, the geodesic $s \mapsto \exp_{\gamma(t)}(sV(t))$ starts at $\gamma(t)$ with initial velocity $V(t)$; setting

$$\Gamma(s,t) = \exp_{\gamma(t)}\big(s\,V(t)\big)$$

defines, for $|s|$ small enough that $sV(t)$ lies in the domain of $\exp_{\gamma(t)}$ for every t in the compact interval $[a,b]$, a piecewise-smooth variation with $\Gamma(0,t) = \exp_{\gamma(t)}(0) = \gamma(t)$. Its variation field is

$$\frac{\partial}{\partial s}\bigg|_{s=0} \exp_{\gamma(t)}\big(sV(t)\big) = d(\exp_{\gamma(t)})_0\big(V(t)\big) = V(t),$$

the last equality by proposition 3.2.5. If moreover $V(a) = V(b) = 0$, then $\Gamma(s,a) = \exp_{\gamma(a)}(0) = \gamma(a)$ and likewise at b, so the variation is proper. Thus every (piecewise-smooth) vector field along γ vanishing at the endpoints arises as the variation field of a proper variation, a fact that will let the first-variation formula be tested against an arbitrary such field.

As a concrete instance, take $\gamma(t) = (t,0)$ in the plane $\mathbb{R}^2$ with its Euclidean metric, $t \in [0,1]$, and $V(t) = \big(0, \sin(\pi t)\big)$, which vanishes at both endpoints. Here $\exp_p(w) = p + w$, so $\Gamma(s,t) = \big(t,\, s\sin(\pi t)\big)$: the segment is bowed into a family of arches, pinned at $(0,0)$ and $(1,0)$, and $\frac{\partial\Gamma}{\partial s}(0,t) = (0, \sin\pi t) = V(t)$ as required.

The computation of the first variation rests on one identity, the symmetry of the two covariant derivatives that a variation makes available. A smooth variation Γ has two coordinate directions, s and t, and hence two families of vector fields along it: $\partial_t\Gamma = \frac{\partial\Gamma}{\partial t}$, the velocities of the longitudinal curves, and $\partial_s\Gamma = \frac{\partial\Gamma}{\partial s}$, the variation fields of the transverse curves. Each can be differentiated covariantly in either direction. Because ∇ is torsion-free and the coordinate fields ∂_s, ∂_t commute, the two mixed covariant derivatives agree.

Lemma 3.5.3: Symmetry Lemma

Let $\Gamma\colon (-\varepsilon,\varepsilon)\times[a,b] \to M$ be a smooth map, and write D_t and D_s for the covariant derivatives along the t- and s-parameter curves, respectively. Then

$$D_s\,\partial_t\Gamma = D_t\,\partial_s\Gamma$$

***Proof*:**

The statement is local in Γ, so it suffices to verify it in a coordinate chart $(U,(x^k))$ about a point $\Gamma(s_0,t_0)$. Write the coordinate expression $\Gamma^k(s,t) = x^k \circ \Gamma$. The two velocity fields have components

$$\partial_t\Gamma = \frac{\partial\Gamma^k}{\partial t}\,\partial_k, \qquad \partial_s\Gamma = \frac{\partial\Gamma^k}{\partial s}\,\partial_k$$

Applying the coordinate formula for the covariant derivative along a curve from definition 2.1.6 to the field $\partial_t\Gamma$, differentiated in the s-direction, gives

$$D_s\,\partial_t\Gamma = \left(\frac{\partial^2\Gamma^k}{\partial s\,\partial t} + \Gamma^k_{ij}(\Gamma)\,\frac{\partial\Gamma^i}{\partial s}\,\frac{\partial\Gamma^j}{\partial t}\right)\partial_k,$$

where Γ^k_{ij} are the Christoffel symbols and the argument Γ indicates evaluation at $\Gamma(s,t)$. Interchanging the roles of s and t,

$$D_t\,\partial_s\Gamma = \left(\frac{\partial^2\Gamma^k}{\partial t\,\partial s} + \Gamma^k_{ji}(\Gamma)\,\frac{\partial\Gamma^j}{\partial t}\,\frac{\partial\Gamma^i}{\partial s}\right)\partial_k$$

The mixed second partial derivatives of the smooth functions Γ^k are equal, so the first terms agree. In the second terms, relabelling the summed indices $i \leftrightarrow j$ in the display for $D_t\,\partial_s\Gamma$ turns its Christoffel factor into $\Gamma^k_{ij}\,\frac{\partial\Gamma^i}{\partial s}\,\frac{\partial\Gamma^j}{\partial t}$, which equals the second term of $D_s\,\partial_t\Gamma$ because the Levi-Civita connection is torsion-free, $\Gamma^k_{ij} = \Gamma^k_{ji}$, by definition 2.3.2 together with lemma 2.2.2. Hence the two expressions coincide, as we sought to show.

With the symmetry lemma in hand the first variation of energy is a direct calculation, differentiating under the integral sign and integrating by parts. The metric compatibility of ∇ enters at the step where the derivative of an inner product is distributed across its two slots; the symmetry lemma enters at the step where a derivative in the s-direction of the velocity is converted into a derivative in the t-direction of the variation field, so that an integration by parts in t becomes available.

Theorem 3.5.4: First Variation of Energy

Let $\gamma\colon [a,b] \to M$ be a piecewise-smooth curve, smooth on each subinterval $[t_{i-1}, t_i]$ of a partition $a = t_0 < t_1 < \cdots < t_m = b$, and let Γ be a piecewise-smooth variation of γ with variation field V. Then the energy of the longitudinal curves is differentiable in s at $s = 0$, with

$$\left.\frac{d}{ds}\right|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\,dt - \sum_{i=1}^{m-1}\langle V(t_i), \Delta_i\dot\gamma\rangle + \Big[\,\langle V,\dot\gamma\rangle\,\Big]_a^b,$$

where $\Delta_i\dot\gamma := \dot\gamma(t_i^+) - \dot\gamma(t_i^-)$ is the jump in the velocity at the interior breakpoint t_i, and $\big[\,\langle V,\dot\gamma\rangle\,\big]_a^b = \langle V(b),\dot\gamma(b)\rangle - \langle V(a),\dot\gamma(a)\rangle$. In particular, if γ is smooth and Γ is proper, then

$$\left.\frac{d}{ds}\right|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\,dt$$

Consequently a smooth curve γ is a critical point of E for proper variations if and only if $D_t\dot\gamma \equiv 0$, that is, if and only if γ is a geodesic.

***Proof*:**
Step 1: differentiation under the integral. Fix first a subinterval $[t_{i-1}, t_i]$ on which Γ is smooth. On $(-\varepsilon,\varepsilon) \times [t_{i-1}, t_i]$ the function $(s,t) \mapsto \langle\partial_t\Gamma, \partial_t\Gamma\rangle$ is smooth, so the energy contribution

$$E_i(s) = \tfrac{1}{2}\int_{t_{i-1}}^{t_i}\langle\partial_t\Gamma(s,t), \partial_t\Gamma(s,t)\rangle\,dt$$

may be differentiated under the integral sign. Using the metric compatibility of the Levi-Civita connection (definition 2.2.4) to differentiate the inner product along the s-curve,

$$\frac{d}{ds}E_i(s) = \tfrac{1}{2}\int_{t_{i-1}}^{t_i}\frac{\partial}{\partial s}\langle\partial_t\Gamma, \partial_t\Gamma\rangle\,dt = \int_{t_{i-1}}^{t_i}\langle D_s\,\partial_t\Gamma, \partial_t\Gamma\rangle\,dt,$$

the factor $\frac{1}{2}$ cancelling against the two equal terms produced by the Leibniz rule in the symmetric inner product.

Step 2: the symmetry lemma and integration by parts. By lemma 3.5.3, $D_s\,\partial_t\Gamma = D_t\,\partial_s\Gamma$, so the integrand becomes $\langle D_t\,\partial_s\Gamma, \partial_t\Gamma\rangle$. Apply metric compatibility again, now to the t-derivative of the inner product $\langle \partial_s\Gamma, \partial_t\Gamma\rangle$:

$$\frac{d}{dt}\langle \partial_s\Gamma, \partial_t\Gamma\rangle = \langle D_t\,\partial_s\Gamma, \partial_t\Gamma\rangle + \langle \partial_s\Gamma, D_t\,\partial_t\Gamma\rangle$$

Solving for the first term on the right and substituting,

$$\frac{d}{ds}E_i(s) = \int_{t_{i-1}}^{t_i} \frac{d}{dt}\langle \partial_s\Gamma, \partial_t\Gamma\rangle\, dt - \int_{t_{i-1}}^{t_i} \langle \partial_s\Gamma, D_t\,\partial_t\Gamma\rangle\, dt$$

The first integral is elementary by the fundamental theorem of calculus, and the second is already in the desired form.

Step 3: evaluation at $s=0$. At $s=0$ the transverse field is the variation field, $\partial_s\Gamma(0,t) = V(t)$, and the longitudinal velocity is $\partial_t\Gamma(0,t) = \dot\gamma(t)$, whence $D_t\,\partial_t\Gamma(0,t) = D_t\dot\gamma(t)$. Setting $s=0$ in the last display,

$$\left.\frac{d}{ds}\right|_{s=0} E_i = \Big[\,\langle V, \dot\gamma\rangle\,\Big]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle V, D_t\dot\gamma\rangle\, dt$$

Since $E(\gamma_s) = \sum_{i=1}^m E_i(s)$, summing over the subintervals gives

$$\left.\frac{d}{ds}\right|_{s=0} E(\gamma_s) = \sum_{i=1}^m \Big[\,\langle V, \dot\gamma\rangle\,\Big]_{t_{i-1}}^{t_i} - \int_a^b \langle V, D_t\dot\gamma\rangle\, dt$$

The variation field V is continuous across the breakpoints, so $V(t_i^-) = V(t_i^+) = V(t_i)$, whereas $\dot\gamma$ may jump. Collecting the boundary contributions, the term at t_i appears once with a plus sign from the subinterval $[t_{i-1}, t_i]$ (evaluated at its right end, using $\dot\gamma(t_i^-)$) and once with a minus sign from $[t_i, t_{i+1}]$ (at its left end, using $\dot\gamma(t_i^+)$); their sum is $\big\langle V(t_i), \dot\gamma(t_i^-)\big\rangle - \big\langle V(t_i), \dot\gamma(t_i^+)\big\rangle = -\langle V(t_i), \Delta_i\dot\gamma\rangle$. The two extreme endpoints $t_0 = a$ and $t_m = b$ are not cancelled and contribute $\langle V(b), \dot\gamma(b)\rangle - \langle V(a), \dot\gamma(a)\rangle = \big[\,\langle V, \dot\gamma\rangle\,\big]_a^b$. Assembling,

$$\left.\frac{d}{ds}\right|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\, dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i\dot\gamma\rangle + \Big[\,\langle V, \dot\gamma\rangle\,\Big]_a^b,$$

which is the asserted formula.

Step 4: criticality. Suppose now γ is smooth, so that there are no interior breakpoints and the jump sum is absent, and suppose Γ is proper, so that $V(a) = V(b) = 0$ and the boundary term $\big[\,\langle V, \dot\gamma\rangle\,\big]_a^b$ vanishes. The formula reduces to

$$\left.\frac{d}{ds}\right|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\, dt,$$

as claimed. If γ is a geodesic then $D_t\dot\gamma \equiv 0$ and the integral vanishes for every V, so γ is critical. Conversely, suppose γ is critical for all proper variations. By example 3.5.2, every smooth vector field V along γ with $V(a) = V(b) = 0$ is the variation field of a proper variation, so

$\int_a^b \langle V, D_t\dot\gamma\rangle\, dt = 0$ for all such V. Take in particular $V = \varphi\, D_t\dot\gamma$, where $\varphi\colon [a,b] \to [0,\infty)$ is smooth, positive on (a,b), and vanishing at a and b; this V is admissible. In the Riemannian case $\langle -,-\rangle$ is positive definite, so

$$0 = \int_a^b \langle V, D_t\dot\gamma\rangle\, dt = \int_a^b \varphi\, |D_t\dot\gamma|_g^2\, dt,$$

an integral of a nonnegative continuous function that is strictly positive wherever $D_t\dot\gamma \neq 0$ on (a,b). Its vanishing forces $D_t\dot\gamma \equiv 0$ on (a,b), hence on $[a,b]$ by continuity, so γ is a geodesic. In the pseudo-Riemannian case, where the inner product is indefinite and the choice $V = \varphi\, D_t\dot\gamma$ need not give a positive integrand, argue componentwise instead: fix a point $t_* \in (a,b)$ and, working in a coordinate chart around $\gamma(t_*)$, apply the criticality condition to the fields $V = \varphi\, \partial_k|_\gamma$ for each k and each bump function φ supported near t_*. This yields $\int \varphi\, g_{kl}(\gamma)(D_t\dot\gamma)^l\, dt = 0$ for all such φ, so by the fundamental lemma of the calculus of variations $g_{kl}(\gamma)(D_t\dot\gamma)^l = 0$ near t_* for every k; nondegeneracy of (g_{kl}) then gives $(D_t\dot\gamma)^l = 0$ near t_*. As t_* was arbitrary, $D_t\dot\gamma \equiv 0$ on (a,b), and again γ is a geodesic, completing the proof.

The formula isolates three ways in which a curve can fail to be a critical point of energy, and reading them off is the substance of the theorem. The interior integral $-\int \langle V, D_t\dot\gamma\rangle\, dt$ measures the curve's own acceleration $D_t\dot\gamma$, tested against the direction V in which it is deformed; it vanishes for every deformation exactly when the acceleration is zero, that is, when the curve is a geodesic. The jump sum penalizes corners: at a breakpoint where the velocity jumps by $\Delta_i\dot\gamma$, deforming in the direction of that jump decreases the energy, so a piecewise-smooth minimizer can have no genuine corners and must be smooth. The boundary term records what happens when the endpoints are allowed to move; for proper variations it is zero, but its form $\big[\langle V, \dot\gamma\rangle\big]_a^b$ already signals that a curve meeting a boundary condition freely will do so orthogonally to $\dot\gamma$, the transversality that governs free-endpoint problems. The clean equivalence "critical for energy $\iff$ geodesic" is the payoff: unlike length, energy has no reparametrization freedom, and its critical points are geodesics parametrized exactly as the equation $D_t\dot\gamma = 0$ dictates, at constant speed.

Example 3.5.5: First variation on the flat plane

Return to the segment $\gamma(t) = (t,0)$ in $\mathbb{R}^2$, $t \in [0,1]$, with the variation $\Gamma(s,t) = (t, s\sin\pi t)$ and variation field $V(t) = (0, \sin\pi t)$ from example 3.5.2. Since the metric is Euclidean and γ is a straight segment, $\dot\gamma = (1,0)$ is constant and $D_t\dot\gamma = \bar\nabla_{\dot\gamma}\dot\gamma = 0$. The theorem predicts $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = 0$, and this is confirmed directly: the longitudinal velocity is $\partial_t\Gamma = (1, \pi s\cos\pi t)$, so

$$E(\gamma_s) = \tfrac12 \int_0^1 \big(1 + \pi^2 s^2 \cos^2 \pi t\big)\, dt = \tfrac12 + \tfrac{\pi^2}{4}\, s^2,$$

whose derivative at $s = 0$ vanishes. The energy is minimized at $s = 0$: the straight segment is critical, consistent with its being a geodesic. The bowed curves γ_s all cost strictly more energy, the excess being quadratic in s, which foreshadows the second-variation analysis of later chapters. Had γ instead been a curved arc with $D_t\dot\gamma \neq 0$, the same variation field would generally produce a nonzero first variation, exhibiting the acceleration through the pairing $-\int \langle V, D_t\dot\gamma\rangle\, dt$.

Energy having been dispatched, the first variation of length is obtained by the same calculation with the integrand $|\dot\gamma|_g$ in place of $\frac12 |\dot\gamma|_g^2$. The square root introduces a factor $1/|\dot\gamma|_g$ in the integrand, which is harmless as long as the curve is regular, meaning $\dot\gamma$ never vanishes; the cleanest statement assumes unit speed, $|\dot\gamma|_g \equiv 1$, which can always be arranged for a regular curve by reparametrizing by arc length, as recorded in section 3.4.

Theorem 3.5.6: First Variation of Length

Let $\gamma\colon [a,b] \to M$ be a piecewise-smooth, regular, unit-speed curve, $|\dot\gamma|_g \equiv 1$, in a Riemannian manifold, smooth on each subinterval of a partition $a = t_0 < \cdots < t_m = b$, and let Γ be a piecewise-smooth variation of γ with variation field V. Then

$$\left.\frac{d}{ds}\right|_{s=0} L(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\, dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i\dot\gamma\rangle + \Big[\,\langle V,\dot\gamma\rangle\,\Big]_a^b$$

In particular, a smooth regular curve is a critical point of L for proper variations if and only if $D_t\dot\gamma \equiv 0$, that is, if and only if it is a geodesic after reparametrization by arc length. Moreover, for every piecewise-smooth curve $\gamma\colon [a,b] \to M$,

$$L(\gamma)^2 \le 2(b-a)\,E(\gamma),$$

with equality if and only if γ has constant speed, $|\dot\gamma|_g$ constant on $[a,b]$.

***Proof*:**
Step 1: the length variation. As in the proof of theorem 3.5.4, work first on a subinterval $[t_{i-1}, t_i]$ on which Γ is smooth, and write $|\partial_t\Gamma| = \langle \partial_t\Gamma, \partial_t\Gamma\rangle^{1/2}$, which is smooth and positive near $s = 0$ because γ is regular with $|\dot\gamma|_g \equiv 1 > 0$, so $\partial_t\Gamma$ stays nonzero for small $|s|$. Then

$$L_i(s) = \int_{t_{i-1}}^{t_i} |\partial_t\Gamma|\, dt, \qquad \frac{d}{ds}L_i(s) = \int_{t_{i-1}}^{t_i} \frac{\partial}{\partial s}\langle \partial_t\Gamma, \partial_t\Gamma\rangle^{1/2}\, dt$$

By the chain rule and metric compatibility (definition 2.2.4),

$$\frac{\partial}{\partial s}\langle \partial_t\Gamma, \partial_t\Gamma\rangle^{1/2} = \frac{\langle D_s\,\partial_t\Gamma, \partial_t\Gamma\rangle}{|\partial_t\Gamma|}$$

Apply the symmetry lemma (lemma 3.5.3) to replace $D_s\,\partial_t\Gamma$ by $D_t\,\partial_s\Gamma$, and evaluate at $s = 0$, where $|\partial_t\Gamma| = |\dot\gamma|_g = 1$, $\partial_s\Gamma = V$, and $\partial_t\Gamma = \dot\gamma$:

$$\left.\frac{d}{ds}\right|_{s=0} L_i = \int_{t_{i-1}}^{t_i} \langle D_tV, \dot\gamma\rangle\, dt$$

The unit-speed hypothesis is what removes the denominator here; without it the factor $1/|\dot\gamma|_g$ would persist inside the integral.

Step 2: integration by parts. By metric compatibility again, $\frac{d}{dt}\langle V,\dot\gamma\rangle = \langle D_tV,\dot\gamma\rangle + \langle V, D_t\dot\gamma\rangle$, so $\langle D_tV,\dot\gamma\rangle = \frac{d}{dt}\langle V,\dot\gamma\rangle - \langle V, D_t\dot\gamma\rangle$ and

$$\left.\frac{d}{ds}\right|_{s=0} L_i = \Big[\,\langle V,\dot\gamma\rangle\,\Big]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle V, D_t\dot\gamma\rangle\, dt$$

Summing over the subintervals and collecting the boundary contributions exactly as in Step 3 of the proof of theorem 3.5.4, with the continuity of V across breakpoints and the possible jumps of $\dot\gamma$, yields

$$\frac{d}{ds}\bigg|_{s=0} L(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\, dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i\dot\gamma\rangle + \Big[\langle V, \dot\gamma\rangle\Big]_a^b,$$

the asserted formula. For a smooth curve and a proper variation the jump sum and boundary term drop, leaving $\frac{d}{ds}\big|_{s=0} L(\gamma_s) = -\int_a^b \langle V, D_t\dot\gamma\rangle\, dt$. The criticality equivalence follows exactly as in Step 4 of the proof of theorem 3.5.4: $D_t\dot\gamma \equiv 0$ makes the integral vanish for all V, and conversely testing against $V = \varphi\, D_t\dot\gamma$ with φ a positive interior bump function forces $D_t\dot\gamma \equiv 0$, using positive-definiteness of the Riemannian inner product. A unit-speed curve with $D_t\dot\gamma \equiv 0$ is a geodesic by definition 3.1.1; a general regular critical curve becomes such after arc-length reparametrization.

Step 3: the Cauchy–Schwarz inequality. For any piecewise-smooth $\gamma\colon [a,b] \to M$, apply the Cauchy–Schwarz inequality in $L^2([a,b])$ to the pair of functions $|\dot\gamma|_g$ and the constant function 1:

$$L(\gamma)^2 = \left(\int_a^b |\dot\gamma|_g \cdot 1\, dt\right)^2 \le \left(\int_a^b |\dot\gamma|_g^2\, dt\right)\left(\int_a^b 1^2\, dt\right) = (b-a)\int_a^b |\dot\gamma|_g^2\, dt = 2(b-a)\,E(\gamma),$$

which is the claimed inequality. Equality in Cauchy–Schwarz holds if and only if the two functions are linearly dependent in L^2, that is, if and only if $|\dot\gamma|_g$ is a constant multiple of 1, hence constant on $[a,b]$. Thus equality holds precisely for constant-speed curves, completing the proof.

The length variation carries the same interior term $-\int \langle V, D_t\dot\gamma\rangle\, dt$ as the energy variation, and for that reason the two functionals have the same critical points among regular curves, namely the geodesics; the difference is only that length, being insensitive to reparametrization, detects geodesics only up to how they are parametrized, whereas energy fixes the parametrization to be by constant speed. The inequality $L(\gamma)^2 \le 2(b-a)E(\gamma)$ makes the relationship quantitative. Among all curves from p to q on a fixed parameter interval, the energy is at least $L^2/2(b-a)$, with equality exactly for the constant-speed ones; so minimizing energy over a fixed interval is minimizing length among constant-speed reparametrizations, and a minimizer of energy is automatically a length-minimizer traversed at constant speed. This is the technical reason energy is the functional of choice in the existence theory for geodesics: its critical points are exactly geodesics, and its minimizers are the shortest constant-speed paths.

Example 3.5.7: The inequality is strict for uneven parametrization

On the Euclidean interval, take the two curves in $\mathbb{R}$ that trace the segment from 0 to 1 over $t \in [0,1]$: first $\alpha(t) = t$, at unit speed, and second $\beta(t) = t^2$, which speeds up. Both have length $L = 1$, since $\int_0^1 |\dot\beta|\, dt = \int_0^1 2t\, dt = 1$. Their energies differ:

$$E(\alpha) = \tfrac{1}{2}\int_0^1 1\, dt = \tfrac{1}{2}, \qquad E(\beta) = \tfrac{1}{2}\int_0^1 (2t)^2\, dt = \tfrac{1}{2}\int_0^1 4t^2\, dt = \tfrac{2}{3}$$

The inequality reads $L^2 = 1 \le 2(b-a)E = 2E$ in both cases: for α it is $1 \le 1$, equality, matching the constant speed $|\dot\alpha| \equiv 1$; for β it is $1 \le \frac{4}{3}$, strict, matching the non-constant speed $|\dot\beta| = 2t$. The extra energy $\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ is the penalty β pays for uneven parametrization while covering the same image. This is the sense in which energy sees more than length: the two curves are indistinguishable to L but separated by E.

The two theorems together settle the relationship between the differential-equation description of geodesics and the variational one. Restated for emphasis and cross-reference, the conclusion for energy is the following.

Corollary 3.5.8: Geodesics Are the Critical Points of Energy

Let (M, g) be a (pseudo-)Riemannian manifold and $\gamma\colon [a, b] \to M$ a smooth curve. Then γ is a critical point of the energy functional E for proper variations if and only if γ is a geodesic, that is, $D_t\dot{\gamma} \equiv 0$.

***Proof*:**
This is the final assertion of theorem 3.5.4, recorded here for reference. For a smooth curve and proper variations the first-variation formula reduces to $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t\dot{\gamma}\rangle \, dt$; this vanishes for every proper variation exactly when $D_t\dot{\gamma} \equiv 0$, by the argument given there, which is the geodesic equation of definition 3.1.1. Hence the critical points of E are precisely the geodesics, as claimed.

This variational characterization completes the promise made in section 3.4: a curve that minimizes length, once parametrized by arc length, is a critical point of length and hence, by theorem 3.5.6, satisfies $D_t\dot{\gamma} \equiv 0$, so it is a smooth geodesic. The jump term in the first-variation formula supplies the smoothness: a length-minimizing piecewise-smooth curve can have no corner, since a variation supported near a corner and pointing along the velocity jump would decrease the length. A minimizer is therefore a single smooth geodesic, not a broken one, which is the regularity statement that closes the loop between the metric and the differential-geometric pictures of shortest paths.

4

The Curvature Tensor

The geodesics and the exponential map of chapter 3 give a manifold its straight lines, but they do not yet measure how the manifold bends. On Euclidean space the second covariant derivatives of a vector field commute and parallel transport is path-independent; on a general Riemannian manifold neither holds, and the precise failure of both is a single tensor, the Riemann curvature tensor. This chapter defines that tensor, extracts the invariants built from it, and identifies the spaces in which it is as simple as possible.

The development moves from the general to the specific. The curvature endomorphism and its algebraic symmetries come first, together with the two Bianchi identities that every curvature tensor obeys. Contracting and restricting the tensor then produces the hierarchy of curvature invariants: the sectional curvature, which records the curvature of each tangent plane and in fact determines the whole tensor; the Ricci and scalar curvatures, the averaged quantities that govern volume and enter the Einstein condition; and the Weyl tensor, the trace-free remainder. The chapter closes by pinning down the extreme cases, the flat manifolds and the constant-curvature model spaces, and by computing curvature through the warped-product construction of chapter 1, which reproduces the sphere, hyperbolic space, and Euclidean space from a single formula. A fixed sign convention, stated at the outset, is in force throughout: with it, the sphere has positive curvature.

4.1 The Riemann Curvature Tensor

On Euclidean space the order of differentiation does not matter: the mixed second partial derivatives of a smooth function agree, and a vector field differentiated first along ∂_i and then along ∂_j gives the same result as the reverse. The Levi-Civita connection of definition 2.3.2 lets one differentiate vector fields on any manifold, and the question of whether the order matters becomes a question about the manifold rather than about calculus. On the round sphere it does matter, as the parallel-transport computation around a geodesic triangle in section 2.4 already showed: carrying a vector around a closed loop and back returns it rotated, so the infinitesimal version, comparing $\nabla_X \nabla_Y Z$ with $\nabla_Y \nabla_X Z$, cannot vanish. The exact discrepancy is the object this chapter is built around.

That discrepancy is a tensor. The expression $\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z$ looks like a second-order differential operator applied to Z, and a differential operator is not tensorial in Z; but subtracting the correction $\nabla_{[X,Y]} Z$ removes every derivative that falls on Z, and what remains depends only on the values of X, Y, and Z at a single point. The result is a $(1,3)$-tensor field, the Riemann curvature endomorphism, and its vanishing is exactly the condition that second covariant derivatives commute. This section defines the tensor, proves that it is a tensor, records its coordinate components in terms of the Christoffel symbols, and checks that Euclidean space, where nothing bends, has curvature zero. Before any formula is written, the sign convention is fixed, because the various conventions in the literature differ by signs that propagate into every subsequent invariant.

4.1.1 Convention: The curvature sign convention Throughout this book the *Riemann curvature endomorphism* of a (pseudo-)Riemannian manifold (M, g) with Levi-Civita connection ∇ is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

The associated $(0,4)$-tensor is $\mathrm{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle$. The sectional curvature of a 2-plane $\Pi = \mathrm{span}(X, Y)$ is

$$K(\Pi) = \frac{\mathrm{Riem}(X, Y, Y, X)}{|X|^2\,|Y|^2 - \langle X, Y\rangle^2}$$

the Ricci tensor is $\mathrm{Ric}(X, Y) = \sum_i \langle R(e_i, X)Y, e_i\rangle$ for an orthonormal frame (e_i), the scalar curvature is $\mathrm{scal} = \mathrm{tr}_g \mathrm{Ric}$, and a manifold has constant sectional curvature c exactly when $R(X, Y)Z = c\big(\langle Y, Z\rangle X - \langle X, Z\rangle Y\big)$. With these choices the round sphere S^n_r has $K \equiv 1/r^2 > 0$, hyperbolic space H^n has $K \equiv -1$, and Euclidean space has $K \equiv 0$.
Conventions in the literature diverge here. Some texts define the endomorphism with the opposite overall sign, and some place the arguments of the $(0,4)$-tensor in a different order, so that a quoted formula for Ric or for the constant-curvature form may carry the reverse sign. Every curvature formula in this book is written relative to the convention stated above; a sign copied from another source should be reconciled with it before use.

The definition of the endomorphism repeats the display just fixed, now as a formal object attached to the connection.

Definition 4.1.2: Riemann curvature endomorphism

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2). The *Riemann curvature endomorphism* is the map

$$R\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

defined for vector fields X, Y, Z by

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

For fixed X, Y the assignment $Z \mapsto R(X, Y)Z$ is written $R(X, Y)$ and called the curvature operator of the pair (X, Y).

The three terms are read as a corrected commutator. The bracket $[\nabla_X, \nabla_Y]Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z$ measures the failure of the two second covariant derivatives to agree, and the third term $\nabla_{[X,Y]}Z$ is the derivative one must subtract because X and Y need not commute as vector fields. When X and Y are coordinate frame fields, $[X, Y] = [\partial_i, \partial_j] = 0$ and the correction disappears, so on a chart the tensor is literally the commutator of the two coordinate second derivatives; the general definition is what makes this a well-defined object independent of any chart. The interpretation that drives everything else is the one from the opening: R is the infinitesimal holonomy of the connection, the amount by which parallel transport around an infinitesimal loop spanned by X and Y fails to be the identity, and it vanishes identically precisely when parallel transport is path-independent, that is, when second covariant derivatives commute.

The expression for $R(X, Y)Z$ involves derivatives of Z, so it is not evident that its value at a point depends only on the values of X, Y, and Z at that point. The next proposition establishes exactly that: R is $C^\infty(M)$-linear in each of its three arguments, and hence, by the tensor characterization lemma of [6], is a tensor field. This is the reason the corrected commutator, rather than the bare one, is the right object.

Proposition 4.1.3: Tensoriality of the curvature endomorphism

The Riemann curvature endomorphism R of definition 4.1.2 is $C^\infty(M)$-linear in each of the three arguments X, Y, Z. Consequently R is a smooth $(1, 3)$-tensor field: its value $R(X, Y)Z$ at a point p depends only on X_p, Y_p, Z_p, and it defines, for each p, a trilinear map $T_pM \times T_pM \times T_pM \to T_pM$.

***Proof*:**
Additivity in each argument is immediate from the additivity of ∇ and of the Lie bracket, so it remains to check homogeneity over $C^\infty(M)$: that $R(fX, Y)Z = R(X, fY)Z = R(X, Y)(fZ) = fR(X, Y)Z$ for every $f \in C^\infty(M)$. The antisymmetry $R(X, Y)Z = -R(Y, X)Z$ is visible from the defining formula, since interchanging X and Y negates each of the three terms; hence linearity in the first argument follows from linearity in the second, and only the second and third arguments need separate treatment.

Step 1: linearity in Y. Fix $f \in C^\infty(M)$. Two identities for the Levi-Civita connection and the bracket are used. First, the Leibniz rule of definition 2.1.1,

$$\nabla_X(fW) = (Xf)W + f\nabla_X W$$

for any vector field W. Second, the bracket identity $[X, fY] = f[X, Y] + (Xf)Y$, established for the Lie bracket of vector fields in [6]. Compute the three terms of $R(X, fY)Z$ in turn. For the first,

$$\nabla_X\nabla_{fY}Z = \nabla_X\big(f\nabla_Y Z\big) = (Xf)\nabla_Y Z + f\nabla_X\nabla_Y Z$$

using $\nabla_{fY}Z = f\nabla_Y Z$ (the connection is $C^\infty(M)$-linear in the differentiating slot, again by definition 2.1.1) and then the Leibniz rule. For the second term,

$$\nabla_{fY}\nabla_X Z = f\nabla_Y\nabla_X Z$$

directly. For the third term, using the bracket identity and then the linearity of ∇ in its lower slot,

$$\nabla_{[X,fY]}Z = \nabla_{f[X,Y]+(Xf)Y}Z = f\nabla_{[X,Y]}Z + (Xf)\nabla_Y Z$$

Assembling the three,

$$R(X, fY)Z = (Xf)\nabla_Y Z + f\nabla_X\nabla_Y Z - f\nabla_Y\nabla_X Z - f\nabla_{[X,Y]}Z - (Xf)\nabla_Y Z$$

The two $(Xf)\nabla_Y Z$ terms cancel, and the remaining three terms are f times the three terms of $R(X,Y)Z$, so $R(X, fY)Z = fR(X,Y)Z$. By the antisymmetry noted above, $R(fX,Y)Z = fR(X,Y)Z$ as well.

Step 2: linearity in Z. Fix $f \in C^\infty(M)$ and expand each term of $R(X,Y)(fZ)$. Applying the Leibniz rule twice to the first term,

$$\begin{aligned}\nabla_X\nabla_Y(fZ) &= \nabla_X\big((Yf)Z + f\nabla_Y Z\big)\\ &= \big(X(Yf)\big)Z + (Yf)\nabla_X Z + (Xf)\nabla_Y Z + f\nabla_X\nabla_Y Z\end{aligned}$$

Interchanging X and Y gives the second term,

$$\nabla_Y\nabla_X(fZ) = \big(Y(Xf)\big)Z + (Xf)\nabla_Y Z + (Yf)\nabla_X Z + f\nabla_Y\nabla_X Z$$

and the Leibniz rule once gives the third,

$$\nabla_{[X,Y]}(fZ) = \big([X,Y]f\big)Z + f\nabla_{[X,Y]}Z$$

Subtracting the second and third displays from the first, the terms $(Yf)\nabla_X Z$ cancel between the first and second lines, as do the terms $(Xf)\nabla_Y Z$. The coefficient of Z that survives is

$$X(Yf) - Y(Xf) - [X,Y]f = 0$$

which vanishes by the very definition of the Lie bracket, $[X,Y]f = X(Yf) - Y(Xf)$ from [6]. What remains is exactly f times the three terms of $R(X,Y)Z$, so $R(X,Y)(fZ) = fR(X,Y)Z$.

This is where the correction term earns its place: without $-\nabla_{[X,Y]}Z$ the coefficient of Z would be $X(Yf) - Y(Xf) \neq 0$ in general, and the bare commutator would fail to be tensorial in Z.

Step 3: from linearity to a tensor. A map that is $C^\infty(M)$-multilinear in its vector-field arguments and takes values in vector fields is induced by a smooth tensor field, by the tensor characterization of [6]; concretely, its value at p depends only on the values of the arguments at p. Smoothness of the tensor field is inherited from the smoothness of ∇ and the bracket. Hence R is a smooth $(1,3)$-tensor field, and at each point p it restricts to a trilinear map on T_pM with values in T_pM, as we sought to show.

Because R is a tensor, its value on vectors at a point may be computed by extending those vectors to any convenient vector fields, and the result does not depend on the extension. This is the license under which every later computation freely chooses coordinate or orthonormal frames. It is also what makes the pointwise trilinear form the object of study: the algebraic identities it satisfies, taken up in the next section, are constraints at each point, and the whole apparatus of sectional, Ricci, and scalar curvature is extracted from this single tensor.

Lowering the output index with the metric turns the endomorphism into a $(0,4)$-tensor, which is the form in which the symmetries are cleanest and in which sectional curvature is read off.

Definition 4.1.4: The (0,4) curvature tensor

Let R be the Riemann curvature endomorphism of definition 4.1.2. The $(0,4)$ *Riemann curvature tensor* is the covariant 4-tensor field Riem defined by

$$\mathrm{Riem}(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle$$

for vector fields X, Y, Z, W.

The passage from R to Riem is the metric contraction of the top index of the $(1,3)$-tensor against a fourth argument, an operation that is $C^\infty(M)$-linear in W because g is bilinear over $C^\infty(M)$ and $C^\infty(M)$-linear in X, Y, Z by proposition 4.1.3. Hence Riem is a genuine $(0,4)$-tensor field. It carries the same information as R: since g is nondegenerate, $\langle R(X,Y)Z, W\rangle = 0$ for all W forces $R(X,Y)Z = 0$, so Riem vanishes if and only if R does. The order of the four arguments is part of the convention; the sectional curvature in box 4.1.1 is $\mathrm{Riem}(X,Y,Y,X)$ over the squared area of the parallelogram spanned by X and Y, and this pairing is what the sign convention was arranged to make positive for the sphere.

To compute with R in charts, and to connect the abstract tensor to the Christoffel symbols of definition 2.1.3, one records its components. The Einstein summation convention, summing over each repeated upper-lower index pair, is in force for the remainder of the section.

Definition 4.1.5: Components of the curvature tensor

Let $(U,(x^i))$ be a smooth chart on (M,g), with coordinate frame (∂_i). The *components* of the Riemann curvature endomorphism in this chart are the functions $R^l{}_{ijk}$ defined by

$$R(\partial_i, \partial_j)\partial_k = R^l{}_{ijk}\,\partial_l$$

and the *components* of the $(0,4)$-tensor are

$$R_{lijk} = \mathrm{Riem}(\partial_l, \partial_i, \partial_j, \partial_k) = \langle R(\partial_l, \partial_i)\partial_j, \partial_k\rangle$$

which are related to the endomorphism components by lowering the top index, $R_{mijk} = g_{ml}\,R^l{}_{ijk}$.

The components are computable directly from the Christoffel symbols. Since $[\partial_i, \partial_j] = 0$ for coordinate frame fields, the correction term in definition 4.1.2 drops out, and

$$R(\partial_i, \partial_j)\partial_k = \nabla_{\partial_i}\nabla_{\partial_j}\partial_k - \nabla_{\partial_j}\nabla_{\partial_i}\partial_k$$

Writing $\nabla_{\partial_j}\partial_k = \Gamma^m_{jk}\partial_m$ and applying ∇_{∂_i} with the Leibniz rule,

$$\nabla_{\partial_i}\nabla_{\partial_j}\partial_k = \nabla_{\partial_i}\left(\Gamma^m_{jk}\partial_m\right) = (\partial_i\Gamma^m_{jk})\partial_m + \Gamma^m_{jk}\,\Gamma^l_{im}\partial_l = \left(\partial_i\Gamma^l_{jk} + \Gamma^l_{im}\Gamma^m_{jk}\right)\partial_l$$

after relabelling the summed index in the first term from m to l. Subtracting the same expression with i and j interchanged gives the component formula

$$R^l{}_{ijk} = \partial_i\Gamma^l_{jk} - \partial_j\Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik} \tag{4.1}$$

Every quantity on the right is built from the metric and its first and second derivatives, through the formula $\Gamma^k_{ij} = \frac{1}{2}g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$ of proposition 2.3.3. Thus the curvature is a second-order differential invariant of the metric: $R^l{}_{ijk}$ depends on g_{ij}, ∂g_{ij}, and $\partial^2 g_{ij}$. The antisymmetry

$R^l{}_{ijk} = -R^l{}_{jik}$ in the first two lower indices is visible in (4.1), matching $R(\partial_i, \partial_j) = -R(\partial_j, \partial_i)$; the deeper symmetries of R_{lijk} are proved in section 4.2.

A remark on index placement is in order, because it interacts with the sign convention. The two lower indices i, j that are antisymmetrized are the first two, the arguments of the operator $R(\partial_i, \partial_j)$; the third lower index k is the vector being acted on. Different books order these differently, and reordering them changes the sign of (4.1). The placement here matches definition 4.1.2 and box 4.1.1 exactly, and it is the placement used in every component computation that follows.

The first test of the definition is the flat model, where the answer must be zero.

Example 4.1.6: Curvature of Euclidean space

Consider Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, with $\bar{g} = \delta_{ij}\, dx^i\, dx^j$ in the standard chart. The metric components are the constants $g_{ij} = \delta_{ij}$, so all their partial derivatives vanish, and by the Christoffel formula of proposition 2.3.3,

$$\Gamma^k_{ij} = \tfrac{1}{2}\, \delta^{kl}\big(\partial_i \delta_{jl} + \partial_j \delta_{il} - \partial_l \delta_{ij}\big) = 0$$

identically. Substituting $\Gamma \equiv 0$ into the component formula (4.1), every term is either a derivative of a Christoffel symbol or a product of two, so

$$R^l{}_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik} = 0$$

for all indices. Hence $R \equiv 0$ on $\mathbb{R}^n$, and therefore Riem $\equiv 0$ as well. Every invariant built from R vanishes: the sectional curvature of every 2-plane is 0, and $\mathbb{R}^n$ is flat, in agreement with the value $K \equiv 0$ recorded in box 4.1.1.

The computation also exhibits, in the simplest case, why R measures the manifold and not the chart. In polar coordinates on $\mathbb{R}^2$ the induced metric is $dr^2 + r^2\, d\theta^2$, and its Christoffel symbols are not all zero: as computed in proposition 2.3.3 one has $\Gamma^r_{\theta\theta} = -r$ and $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = 1/r$, with the rest vanishing. Feeding these into (4.1) and using the one relevant nonzero derivative $\partial_r \Gamma^r_{\theta\theta} = -1$ gives, for the component with $(l, i, j, k) = (r, r, \theta, \theta)$,

$$R^r{}_{r\theta\theta} = \partial_r \Gamma^r_{\theta\theta} - \partial_\theta \Gamma^r_{r\theta} + \Gamma^r_{rm}\Gamma^m_{\theta\theta} - \Gamma^r_{\theta m}\Gamma^m_{r\theta} = (-1) - 0 + 0 - \Gamma^r_{\theta\theta}\, \Gamma^\theta_{r\theta} = -1 - (-r)\big(\tfrac{1}{r}\big) = 0$$

and every other independent component vanishes by the same cancellation. The nonzero Christoffel symbols conspire so that R still vanishes: the plane is flat regardless of the coordinates used to describe it, and it is the curvature tensor, not the Christoffel symbols, that detects this.

The Euclidean computation isolates the two features that the rest of the chapter builds on. First, curvature is a second-order quantity in the metric, so it cannot be removed by a first-order coordinate change; the normal coordinates of proposition 3.2.8 arrange $g_{ij}(p) = \delta_{ij}$, $\Gamma^k_{ij}(p) = 0$, and $\partial_k g_{ij}(p) = 0$ at a single point, yet on a curved manifold the second derivatives $\partial_k \partial_l g_{ij}(p)$ cannot all be made to vanish, and (4.1) extracts exactly the combination of them that is coordinate-independent. Second, the vanishing of R is the precise flatness criterion: a manifold with $R \equiv 0$ is locally isometric to Euclidean space, a statement proved in section 4.5.

4.2 Symmetries and the Bianchi Identities

The curvature endomorphism $R(X,Y)Z$ of definition 4.1.2 and the four-tensor $\mathrm{Riem}(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle$ of definition 4.1.4 are, as they stand, three vector inputs and one covector input away from being unconstrained. In dimension n a general $(0,4)$-tensor at a point has n^4 independent components; the Riemann tensor of a (pseudo-)Riemannian manifold has far fewer, because it inherits rigid algebraic relations from the way it was built. Two of those relations, antisymmetry in the first two slots and antisymmetry in the last two, express that $R(X,Y)$ is a bundle map that swaps sign under exchange of X, Y and takes values in the skew-symmetric endomorphisms of each tangent space. A third, the symmetry that exchanges the first pair of arguments with the second, is less transparent and turns out to be a consequence of the other relations together with the algebraic Bianchi identity. These are the symmetries that make sectional curvature well defined in section 4.3 and that cut the number of independent components down to $n^2(n^2-1)/12$.

Alongside the pointwise (algebraic) symmetries there is a single relation among the covariant derivatives of R, the differential Bianchi identity. Where the algebraic symmetries are statements about R at one point, the differential identity constrains how R changes from point to point; contracting it produces the divergence identity that forces Schur's lemma and, in general relativity, singles out the Einstein tensor as the divergence-free combination of curvature. This section proves all of these, first the four algebraic symmetries in one proposition, then the differential identity, and finally records the same relations in index notation for later coordinate computations. Throughout, the sign convention fixed in box 4.1.1 is in force, and Einstein's summation convention (sum over any index appearing once up and once down) is used in the component statements.

The starting point is a small observation that removes all the Lie-bracket clutter from the proofs. Every identity below is $C^\infty(M)$-linear in each argument, hence tensorial, hence pointwise: to check it at a point p it suffices to check it for *any* convenient choice of vector fields extending prescribed vectors at p. Choosing extensions whose pairwise Lie brackets vanish at p collapses the bracket term $\nabla_{[X,Y]}Z$ and reduces each proof to a computation with the two double-derivative terms. Such extensions exist: in the normal coordinates at p of proposition 3.2.8, the coordinate vector fields ∂_i have identically vanishing brackets $[\partial_i, \partial_j] = 0$, and they also satisfy $\nabla_{\partial_i}\partial_j(p) = \Gamma^k_{ij}(p)\partial_k = 0$ because the Christoffel symbols vanish at the center. Both features are exploited below.

Proposition 4.2.1: Algebraic symmetries of the curvature tensor

Let (M,g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ and Riemann tensor Riem. For all vector fields X, Y, Z, W on M:

1. (antisymmetry in the first pair) $\mathrm{Riem}(X,Y,Z,W) = -\mathrm{Riem}(Y,X,Z,W)$;
2. (antisymmetry in the second pair) $\mathrm{Riem}(X,Y,Z,W) = -\mathrm{Riem}(X,Y,W,Z)$;
3. (pair symmetry) $\mathrm{Riem}(X,Y,Z,W) = \mathrm{Riem}(Z,W,X,Y)$;
4. (first Bianchi identity) $\mathrm{Riem}(X,Y,Z,W) + \mathrm{Riem}(Y,Z,X,W) + \mathrm{Riem}(Z,X,Y,W) = 0$.

***Proof*:**
Each identity is $\mathbb{R}$-linear in all four arguments, and is $C^\infty(M)$-linear because R is a tensor (proposition 4.1.3) and the metric pairing $\langle -,-\rangle$ is $C^\infty(M)$-bilinear. Hence each is a pointwise statement about the values X_p, Y_p, Z_p, W_p, and it suffices to prove it at an arbitrary $p \in M$ using any vector fields realizing those values.

1. Exchanging X and Y in the defining formula
$$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$$
negates the first two terms and, since $[Y,X] = -[X,Y]$, the third as well, so $R(Y,X)Z = -R(X,Y)Z$. Pairing with W gives $\operatorname{Riem}(Y,X,Z,W) = -\operatorname{Riem}(X,Y,Z,W)$, which is (1).

2. It suffices to prove the equivalent statement $\operatorname{Riem}(X,Y,Z,Z) = 0$ for all X,Y,Z; the general form follows by polarizing $Z \mapsto Z + W$ and using the bilinearity of Riem in the last two slots. Indeed, expanding $0 = \operatorname{Riem}(X,Y,Z+W,Z+W) = \operatorname{Riem}(X,Y,Z,Z) + \operatorname{Riem}(X,Y,Z,W) + \operatorname{Riem}(X,Y,W,Z) + \operatorname{Riem}(X,Y,W,W)$ and cancelling the two vanishing diagonal terms leaves $\operatorname{Riem}(X,Y,Z,W) = -\operatorname{Riem}(X,Y,W,Z)$, which is (2).

   So fix p and, by the reduction above, choose vector fields X,Y,Z with $[X,Y]_p = 0$ (for instance coordinate fields of a chart at p, whose brackets vanish identically). At p then
$$\operatorname{Riem}(X,Y,Z,Z) = \langle \nabla_X\nabla_Y Z, Z\rangle - \langle \nabla_Y\nabla_X Z, Z\rangle$$
since the bracket term drops out. Metric compatibility (definition 2.2.4), $U\langle A,B\rangle = \langle \nabla_U A, B\rangle + \langle A, \nabla_U B\rangle$, applied twice gives
$$\begin{aligned} X\,Y\,\langle Z,Z\rangle &= X\big(2\,\langle \nabla_Y Z, Z\rangle\big) \\ &= 2\,\langle \nabla_X\nabla_Y Z, Z\rangle + 2\,\langle \nabla_Y Z, \nabla_X Z\rangle \end{aligned}$$
and, symmetrically, $Y\,X\,\langle Z,Z\rangle = 2\,\langle \nabla_Y\nabla_X Z, Z\rangle + 2\,\langle \nabla_X Z, \nabla_Y Z\rangle$. Subtracting, the two mixed terms $\langle \nabla_Y Z, \nabla_X Z\rangle$ and $\langle \nabla_X Z, \nabla_Y Z\rangle$ coincide and cancel, so
$$(X\,Y - Y\,X)\,\langle Z,Z\rangle = 2\,\langle \nabla_X\nabla_Y Z, Z\rangle - 2\,\langle \nabla_Y\nabla_X Z, Z\rangle = 2\,\operatorname{Riem}(X,Y,Z,Z)$$
using the previous display for the right-hand side. But $(X\,Y - Y\,X)f = [X,Y]f$ for any function f, and $[X,Y]_p = 0$, so the left-hand side vanishes at p. Hence $\operatorname{Riem}(X,Y,Z,Z)_p = 0$, and since p was arbitrary, $\operatorname{Riem}(X,Y,Z,Z) \equiv 0$, proving (2).

3. Pair symmetry follows formally from (1), (2), and the first Bianchi identity (4), which is proved in part (4) below independently of (3). Write $b(A,B,C,D)$ for the first-Bianchi cyclic sum in the first three arguments,
$$b(A,B,C,D) := \operatorname{Riem}(A,B,C,D) + \operatorname{Riem}(B,C,A,D) + \operatorname{Riem}(C,A,B,D)$$
so that $b \equiv 0$ by (4). Form the specific combination
$$\Sigma := b(X,Y,Z,W) + b(Y,Z,W,X) - b(Z,W,X,Y) - b(W,X,Y,Z)$$
which is 0 because each of the four summands is 0. Expanding the four b-terms gives twelve summands:
$$\begin{aligned} \Sigma = {} & \operatorname{Riem}(X,Y,Z,W) + \operatorname{Riem}(Y,Z,X,W) + \operatorname{Riem}(Z,X,Y,W) \\ & + \operatorname{Riem}(Y,Z,W,X) + \operatorname{Riem}(Z,W,Y,X) + \operatorname{Riem}(W,Y,Z,X) \\ & - \operatorname{Riem}(Z,W,X,Y) - \operatorname{Riem}(W,X,Z,Y) - \operatorname{Riem}(X,Z,W,Y) \\ & - \operatorname{Riem}(W,X,Y,Z) - \operatorname{Riem}(X,Y,W,Z) - \operatorname{Riem}(Y,W,X,Z) \end{aligned}$$

Reduce each of the twelve summands to a canonical form by ordering its first pair with (1) and its second pair with (2), each reordering contributing a sign, and collect the results. The twelve summands fall into six canonical classes, each fed by two of them. Four of the six classes cancel outright, by the following identities, in which (1) and (2) bring both members to a common canonical form:

$$\begin{aligned}
\mathrm{Riem}(Y,Z,X,W) + \mathrm{Riem}(Y,Z,W,X) &= \mathrm{Riem}(Y,Z,X,W) - \mathrm{Riem}(Y,Z,X,W) = 0,\\
\mathrm{Riem}(Z,X,Y,W) - \mathrm{Riem}(X,Z,W,Y) &= -\mathrm{Riem}(X,Z,Y,W) + \mathrm{Riem}(X,Z,Y,W) = 0,\\
\mathrm{Riem}(W,Y,Z,X) - \mathrm{Riem}(Y,W,X,Z) &= \mathrm{Riem}(Y,W,X,Z) - \mathrm{Riem}(Y,W,X,Z) = 0,\\
-\mathrm{Riem}(W,X,Z,Y) - \mathrm{Riem}(W,X,Y,Z) &= -\mathrm{Riem}(X,W,Y,Z) + \mathrm{Riem}(X,W,Y,Z) = 0
\end{aligned}$$

The two remaining classes are the diagonal ones. The class of $\mathrm{Riem}(X,Y,Z,W)$ is fed by $\mathrm{Riem}(X,Y,Z,W)$ and by $-\mathrm{Riem}(X,Y,W,Z) = +\mathrm{Riem}(X,Y,Z,W)$, contributing $+2\ \mathrm{Riem}(X,Y,Z,W)$; the class of $\mathrm{Riem}(Z,W,X,Y)$ is fed by $-\mathrm{Riem}(Z,W,X,Y)$ and by $\mathrm{Riem}(Z,W,Y,X) = -\mathrm{Riem}(Z,W,X,Y)$, contributing $-2\ \mathrm{Riem}(Z,W,X,Y)$. Summing the six classes,

$$0 = \Sigma = 2\ \mathrm{Riem}(X,Y,Z,W) - 2\ \mathrm{Riem}(Z,W,X,Y)$$

so $\mathrm{Riem}(X,Y,Z,W) = \mathrm{Riem}(Z,W,X,Y)$, which is (3). The reduction is purely algebraic, using only (1), (2), and $b \equiv 0$; no additional hypothesis on X,Y,Z,W is needed.

4. By tensoriality the identity is pointwise; fix p and choose vector fields X,Y,Z with vanishing pairwise brackets at p, so that every $\nabla_{[\,\cdot\,,\cdot\,]}$ term below vanishes at p. Then

$$\begin{aligned}
R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = {}& \big(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z\big)\\
&+ \big(\nabla_Y\nabla_Z X - \nabla_Z\nabla_Y X\big) + \big(\nabla_Z\nabla_X Y - \nabla_X\nabla_Z Y\big)
\end{aligned}$$

at p. Regroup the six terms so that each covariant derivative ∇_U is applied to a difference:

$$= \nabla_X(\nabla_Y Z - \nabla_Z Y) + \nabla_Y(\nabla_Z X - \nabla_X Z) + \nabla_Z(\nabla_X Y - \nabla_Y X)$$

Because the Levi-Civita connection is torsion-free (definition 2.2.1, theorem 2.3.1), $\nabla_U V - \nabla_V U = [U,V]$, so this equals

$$\nabla_X[Y,Z] + \nabla_Y[Z,X] + \nabla_Z[X,Y]$$

Now use torsion-freeness once more, in the form $\nabla_U[V,W] = [U,[V,W]] + \nabla_{[V,W]}U$, together with the vanishing of the bracket terms at p: at p the fields $[Y,Z],[Z,X],[X,Y]$ all vanish (the pairwise brackets were chosen to vanish at p), so $\nabla_{[V,W]}U(p) = 0$, and

$$\nabla_X[Y,Z] + \nabla_Y[Z,X] + \nabla_Z[X,Y] = [X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$$

by the Jacobi identity for the Lie bracket of vector fields. Hence $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$ at p, and pairing with any W gives (4). Since p is arbitrary, (4) holds everywhere.

This establishes all four symmetries, completing the proof.

The four relations are not independent. Antisymmetry in each pair, (1) and (2), and the first Bianchi identity (4) are the primitive facts; the pair symmetry (3) is their formal consequence, as the proof shows. Their provenance is worth separating: (1) is purely combinatorial, built into the antisymmetric definition of $R(X, Y)$; (4) is the signature of torsion-freeness, and would fail for a connection with torsion; (2) is the one place metric compatibility enters, which is why it is special to the Levi-Civita connection among all torsion-free connections. A general affine connection has a curvature satisfying (1) always, and (4) when it is torsion-free, but (2) and hence (3) require the metric. This is the algebraic form of the fact that the Riemann tensor of (M, g) takes values in a smaller space than the curvature of an arbitrary connection.

A restatement clarifies (1) and (2). For fixed X, Y the map $Z \mapsto R(X, Y)Z$ is an endomorphism of each tangent space T_pM, and (2) says $\langle R(X, Y)Z, W\rangle = -\langle R(X, Y)W, Z\rangle$, that is, this endomorphism is skew-adjoint with respect to g: $R(X, Y) \in \mathfrak{so}(T_pM, g)$, the Lie algebra of the orthogonal (or, in the pseudo-Riemannian case, the indefinite-orthogonal) group of (T_pM, g_p). Meanwhile (1) says the assignment $(X, Y) \mapsto R(X, Y)$ is itself alternating. So R is an alternating bilinear map into skew-adjoint endomorphisms, exactly the object that will reappear in section 4.3 as the curvature operator on two-forms.

The flat model exhibits all four symmetries in the empty way. On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, every component of R vanishes (example 4.1.6), so each identity of proposition 4.2.1 reads $0 = 0$. A first nontrivial check comes on the round sphere, computed in section 4.3, where $\mathrm{Riem}(X, Y, Z, W) = r^{-2}\big(\langle X, W\rangle \langle Y, Z\rangle - \langle X, Z\rangle \langle Y, W\rangle\big)$; the reader may verify directly from this closed form that (1)–(4) all hold, and that the pattern of signs is forced by them.

Example 4.2.2: The constant-curvature tensor obeys the symmetries

Any (pseudo-)Riemannian manifold whose curvature has the constant-curvature form

$$R(X, Y)Z = c\big(\langle Y, Z\rangle X - \langle X, Z\rangle Y\big)$$

of box 4.1.1, for a constant $c \in \mathbb{R}$, provides a fully explicit test of proposition 4.2.1. Lowering an index,

$$\mathrm{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle = c\big(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle\big)$$

Each symmetry is now a symmetry of this bracketed expression in the four inner products.

1. Swapping $X \leftrightarrow Y$ turns the bracket into $\langle X, Z\rangle \langle Y, W\rangle - \langle Y, Z\rangle \langle X, W\rangle$, the negative of the original; so $\mathrm{Riem}(X, Y, Z, W) = -\mathrm{Riem}(Y, X, Z, W)$.
2. Swapping $Z \leftrightarrow W$ turns it into $\langle Y, W\rangle \langle X, Z\rangle - \langle X, W\rangle \langle Y, Z\rangle$, again the negative; so $\mathrm{Riem}(X, Y, Z, W) = -\mathrm{Riem}(X, Y, W, Z)$.
3. Swapping the pairs $(X, Y) \leftrightarrow (Z, W)$ sends the bracket to $\langle W, X\rangle \langle Z, Y\rangle - \langle Z, X\rangle \langle W, Y\rangle$, which equals the original by symmetry of the inner product; so $\mathrm{Riem}(X, Y, Z, W) = \mathrm{Riem}(Z, W, X, Y)$.
4. The cyclic sum in the first three slots is

$$\begin{aligned} &c\big(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle\big) \\ &\quad + c\big(\langle Z, X\rangle \langle Y, W\rangle - \langle Y, X\rangle \langle Z, W\rangle\big) \\ &\quad + c\big(\langle X, Y\rangle \langle Z, W\rangle - \langle Z, Y\rangle \langle X, W\rangle\big) \end{aligned}$$

in which the six products cancel in pairs ($\langle Y, Z\rangle \langle X, W\rangle$ against $-\langle Z, Y\rangle \langle X, W\rangle$, and so on), giving 0; so the first Bianchi identity holds.

The round sphere S^n_r of example 1.3.4 realizes this with $c = 1/r^2$ and hyperbolic space H^n of example 1.3.5 with $c = -1$, so both model spaces carry curvature tensors of exactly this form; the computation above is then their curvature symmetries in one stroke. That $R(X,Y)Z = c(\langle Y,Z\rangle X - \langle X,Z\rangle Y)$ is in fact the *only* curvature-type tensor compatible with constant sectional curvature is proved in section 4.5; here it serves only as a worked instance of the algebraic symmetries.

The four algebraic identities constrain the value of R at each point. The next identity constrains its derivative and is the one relation among the components of ∇R. It is the geometric analogue of the fact that the exterior derivative of a curvature two-form is controlled by the connection, and its contraction is the source of the conservation law behind Einstein's equations. The covariant derivative ∇R appearing below is the $(1,4)$-tensor obtained by differentiating the $(1,3)$-tensor R; recall from definition 2.4.6 that ∇ extends to all tensor bundles and obeys the tensor Leibniz rule, so that

$$(\nabla_X R)(Y,Z)W = \nabla_X\big(R(Y,Z)W\big) - R(\nabla_X Y,Z)W - R(Y,\nabla_X Z)W - R(Y,Z)\nabla_X W$$

is the value of the differentiated tensor on the arguments Y, Z, W.

Theorem 4.2.3: Second (differential) Bianchi identity

Let (M,g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ and Riemann curvature R. For all vector fields X, Y, Z, W,

$$(\nabla_X R)(Y,Z)W + (\nabla_Y R)(Z,X)W + (\nabla_Z R)(X,Y)W = 0$$

that is, the cyclic sum over X, Y, Z of $(\nabla_X R)(Y,Z)$ vanishes.

***Proof*:**
Both sides are $C^\infty(M)$-linear in each of X, Y, Z, W: the tensor ∇R is $C^\infty(M)$-linear in all its arguments by construction (definition 2.4.6), and a cyclic sum of tensors is a tensor. Hence the identity is pointwise, and it suffices to verify it at an arbitrary point $p \in M$.

Fix p and work in normal coordinates (x^i) centered at p (definition 3.2.6), applied to the coordinate frame $X = \partial_a$, $Y = \partial_b$, $Z = \partial_c$, $W = \partial_d$. By $C^\infty(M)$-multilinearity it suffices to prove the identity for these frame fields, since general X, Y, Z, W are C^∞-combinations of them and both sides are multilinear. Two features of normal coordinates are used at the single point p, from proposition 3.2.8:

- $[\partial_i, \partial_j] = 0$ identically (coordinate fields commute); and
- $\Gamma^k_{ij}(p) = 0$, so $\nabla_{\partial_i}\partial_j(p) = 0$ for all i, j.

The first feature simplifies the curvature endomorphism on the frame to $R(\partial_i,\partial_j) = \nabla_{\partial_i}\nabla_{\partial_j} - \nabla_{\partial_j}\nabla_{\partial_i}$, with no bracket term, everywhere in the chart. The second collapses the covariant derivative of a tensor to the plain directional derivative of its frame components *at* p: for any tensor T built from the frame, since all $\nabla_{\partial_a}\partial_i(p) = 0$, the correction terms in the tensor Leibniz rule vanish at p, so

$$(\nabla_{\partial_a} R)(\partial_b,\partial_c)\partial_d\big|_p = \nabla_{\partial_a}\big(R(\partial_b,\partial_c)\partial_d\big)\big|_p$$

with all four frame-argument correction terms absent at p.

Compute the last quantity. Using $R(\partial_b, \partial_c) = \nabla_{\partial_b}\nabla_{\partial_c} - \nabla_{\partial_c}\nabla_{\partial_b}$,

$$(\nabla_{\partial_a} R)(\partial_b, \partial_c)\partial_d\big|_p = \nabla_{\partial_a}\left(\nabla_{\partial_b}\nabla_{\partial_c}\partial_d - \nabla_{\partial_c}\nabla_{\partial_b}\partial_d\right)\big|_p = \left(\nabla_{\partial_a}\nabla_{\partial_b}\nabla_{\partial_c} - \nabla_{\partial_a}\nabla_{\partial_c}\nabla_{\partial_b}\right)\partial_d\big|_p$$

Now form the cyclic sum over (a, b, c), that is, over $(\partial_a, \partial_b, \partial_c)$ in the pattern $(a, b, c) \to (b, c, a) \to (c, a, b)$. Writing ∇_{abc} for $\nabla_{\partial_a}\nabla_{\partial_b}\nabla_{\partial_c}\partial_d\big|_p$, the three terms of the cyclic sum are

$$(\nabla_{abc} - \nabla_{acb}) + (\nabla_{bca} - \nabla_{bac}) + (\nabla_{cab} - \nabla_{cba})$$

These six third-order derivatives cancel in pairs. To see this, note that at p the operators ∇_{∂_i} and ∇_{∂_j} applied in the first two slots may be interchanged with an error measured by curvature: for any vector field V,

$$\nabla_{\partial_i}\nabla_{\partial_j} V - \nabla_{\partial_j}\nabla_{\partial_i} V = R(\partial_i, \partial_j)V$$

since $[\partial_i, \partial_j] = 0$. Apply this to the outer two derivatives of each triple, taking $V = \nabla_{\partial_c}\partial_d$ and its permutations. Pairing ∇_{abc} with ∇_{bac}: their difference is

$$\nabla_{abc} - \nabla_{bac} = \left(\nabla_{\partial_a}\nabla_{\partial_b} - \nabla_{\partial_b}\nabla_{\partial_a}\right)\nabla_{\partial_c}\partial_d\big|_p = R(\partial_a, \partial_b)\left(\nabla_{\partial_c}\partial_d\right)\big|_p = 0$$

because $\nabla_{\partial_c}\partial_d(p) = 0$ and $R(\partial_a, \partial_b)$ is a pointwise-linear (tensorial) operator, so it annihilates a vector field vanishing at p. The same argument, using $\nabla_{\partial_i}\partial_j(p) = 0$ for the appropriate inner index, gives

$$\nabla_{bca} - \nabla_{cba} = R(\partial_b, \partial_c)\left(\nabla_{\partial_a}\partial_d\right)\big|_p = 0, \qquad \nabla_{cab} - \nabla_{acb} = R(\partial_c, \partial_a)\left(\nabla_{\partial_b}\partial_d\right)\big|_p = 0$$

each vanishing for the same reason: the innermost covariant derivative $\nabla_{\partial_c}\partial_d$, $\nabla_{\partial_a}\partial_d$, or $\nabla_{\partial_b}\partial_d$ is a vector field vanishing at p, and the tensorial curvature operator sends it to 0 at p. The six terms of the cyclic sum therefore partition into the three vanishing differences

$$(\nabla_{abc} - \nabla_{bac}) + (\nabla_{bca} - \nabla_{cba}) + (\nabla_{cab} - \nabla_{acb}) = 0$$

which is exactly the cyclic sum rearranged. Hence

$$(\nabla_{\partial_a} R)(\partial_b, \partial_c)\partial_d + (\nabla_{\partial_b} R)(\partial_c, \partial_a)\partial_d + (\nabla_{\partial_c} R)(\partial_a, \partial_b)\partial_d\big|_p = 0$$

Since the frame vectors $\partial_a, \partial_b, \partial_c, \partial_d$ span T_pM and both sides are multilinear over $\mathbb{R}$, the identity holds for arbitrary X, Y, Z, W at p. As $p \in M$ was arbitrary, the second Bianchi identity holds on all of M, as we sought to show.

The normal-coordinate device is what makes this proof short. In an arbitrary chart the third covariant derivatives of ∂_d would carry the derivatives of the Christoffel symbols, and the cancellations would be buried under those terms; centering normal coordinates at p kills the first-order data $\Gamma^k_{ij}(p)$ and leaves only the second-order data, in which the identity is manifest. The identity itself is the covariant form of a differential relation the reader may already know for connection and curvature forms: it is $dR = 0$ in the appropriate exterior-covariant sense. Its most consequential corollary is obtained by contracting twice, which produces the *contracted* second Bianchi identity governing the divergence of the Ricci tensor and of the Einstein tensor $\mathrm{Ric} - \frac{1}{2}\,\mathrm{scal}\, g$. That contraction, and the Schur lemma it yields, are carried out in section 4.4 once the Ricci tensor has been defined.

For Euclidean space the identity is again vacuous, since $R \equiv 0$ makes $\nabla R \equiv 0$ and every term is zero (example 4.1.6). More informatively, any manifold whose curvature is *parallel*, $\nabla R \equiv 0$ (a locally symmetric space), satisfies the second Bianchi identity trivially term by term; the constant-curvature models of example 4.2.2 are of this kind, because differentiating $\operatorname{Riem}(X, Y, Z, W) = c(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle)$ with c constant and $\nabla g = 0$ gives $\nabla \operatorname{Riem} = 0$. Thus for the sphere and for hyperbolic space the differential Bianchi identity reduces, correctly, to $0 = 0$, and the content of theorem 4.2.3 is that the same cyclic relation persists even where ∇R does not vanish.

For coordinate computations, and for the contractions of section 4.4 and the classification of section 4.5, it is convenient to have the symmetries recorded in components. Fix a chart with coordinate frame (∂_i). Following the component convention of definition 4.1.5, write the endomorphism components $R^l{}_{ijk}$ through $R(\partial_i, \partial_j)\partial_k = R^l{}_{ijk}\,\partial_l$, and lower the free upper index with the metric to obtain the fully covariant components

$$R_{ijkl} = \operatorname{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) = \langle R(\partial_i, \partial_j)\partial_k, \partial_l\rangle = g_{lm}\, R^m{}_{ijk}$$

so that R_{ijkl} carries the first pair (i, j) in the endomorphism slots and the second pair (k, l) in the two Riemann arguments, matching the four-argument order of Riem.

Proposition 4.2.4: Curvature symmetries in components

In any coordinate frame the fully covariant curvature components satisfy

$$R_{ijkl} = -R_{jikl}, \qquad R_{ijkl} = -R_{ijlk}, \qquad R_{ijkl} = R_{klij}$$

and the first Bianchi identity

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0$$

while the second Bianchi identity reads, in terms of the covariant-derivative components $\nabla_m R_{ijkl}$,

$$\nabla_m R_{ijkl} + \nabla_i R_{jmkl} + \nabla_j R_{mikl} = 0$$

the cyclic sum being over the three indices (m, i, j) in the first two curvature slots.

***Proof*:**
The three equalities and the first Bianchi identity are the componentwise transcriptions of parts (1), (2), (3), and (4) of proposition 4.2.1, obtained by evaluating each on the coordinate frame (∂_i). For the first equality, $R_{ijkl} = \operatorname{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) = -\operatorname{Riem}(\partial_j, \partial_i, \partial_k, \partial_l) = -R_{jikl}$ by part (1). For the second, part (2) gives $R_{ijkl} = -\operatorname{Riem}(\partial_i, \partial_j, \partial_l, \partial_k) = -R_{ijlk}$. For the third, part (3) gives $R_{ijkl} = \operatorname{Riem}(\partial_k, \partial_l, \partial_i, \partial_j) = R_{klij}$. For the Bianchi relation, part (4) applied to $(\partial_i, \partial_j, \partial_k)$ in the first three slots and paired with ∂_l gives

$$\operatorname{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) + \operatorname{Riem}(\partial_j, \partial_k, \partial_i, \partial_l) + \operatorname{Riem}(\partial_k, \partial_i, \partial_j, \partial_l) = 0$$

that is $R_{ijkl} + R_{jkil} + R_{kijl} = 0$.

For the differential identity, evaluate theorem 4.2.3 on the coordinate frame, taking $X = \partial_m$, $Y = \partial_i$, $Z = \partial_j$ in the derivative and first two curvature slots and pairing the endomorphism value with ∂_l while its argument is ∂_k:

$$(\nabla_{\partial_m} R)(\partial_i, \partial_j)\partial_k + (\nabla_{\partial_i} R)(\partial_j, \partial_m)\partial_k + (\nabla_{\partial_j} R)(\partial_m, \partial_i)\partial_k = 0$$

Pairing each term with ∂_l under $\langle -, -\rangle$ and writing $\nabla_m R_{ijkl}$ for the components of the tensor ∇R (so that $\langle(\nabla_{\partial_m} R)(\partial_i, \partial_j)\partial_k, \partial_l\rangle = \nabla_m R_{ijkl}$, the metric being parallel) turns the three terms into $\nabla_m R_{ijkl}$, $\nabla_i R_{jmkl}$, and $\nabla_j R_{mikl}$, giving the stated cyclic relation. This proves all the component identities, as required.

The component antisymmetries $R_{ijkl} = -R_{jikl} = -R_{ijlk}$ say that R_{ijkl} is skew in (i, j) and skew in (k, l), so it descends to a symmetric bilinear form on the space $\Lambda^2 T_p^* M$ of two-forms, the pair symmetry $R_{ijkl} = R_{klij}$ being exactly this symmetry; this is the curvature operator, and it is the natural home for the eigenvalue estimates of later chapters. Counting independent components with these relations, together with the first Bianchi identity, gives $\frac{1}{12}n^2(n^2 - 1)$, which is 0 for $n = 1$, 1 for $n = 2$, 6 for $n = 3$, and 20 for $n = 4$; the single component in dimension two is the Gauss curvature, and the twenty in dimension four are the arena of general relativity. The decomposition of this space into scalar, Ricci, and Weyl pieces is taken up in section 4.4, and the fact that the scalar quantity K on two-planes already recovers the whole of R_{ijkl} is proved in section 4.3.

4.3 Sectional Curvature

The Riemann tensor R is a formidable object: at each point it is a $(1, 3)$-tensor, and even after the algebraic symmetries of proposition 4.2.1 cut down its independent components, it records $\frac{1}{12}n^2(n^2 - 1)$ numbers at every point of an n-manifold. That is a large amount of data, and none of it is visibly geometric. The purpose of this section is to repackage the same information as a single real-valued function on a concrete geometric object, the set of tangent 2-planes, in a way that connects the curvature tensor to the shape of the manifold one can actually see.

The idea is old and comes from surfaces. On a surface the Gaussian curvature at a point is one number, and it measures how the surface bends: positive where the surface is dome-like, negative at a saddle, zero on a plane or cylinder. In higher dimensions there is no single bending number at a point, because the manifold can bend differently in different two-dimensional directions. The remedy is to attach a curvature number to each of those directions. A tangent 2-plane $\Pi \subseteq T_p M$ is spanned by two vectors, and the quantity $\mathrm{Riem}(X, Y, Y, X)$ built from a spanning pair, once normalized so that it depends only on the plane and not on the pair, is the *sectional curvature* of Π. It equals the Gaussian curvature of the little surface swept out by geodesics leaving p in the directions of Π, a statement that will be made precise once the second fundamental form is available. Two facts justify the reduction. First, the sectional curvature is well defined: it is a function of the plane alone, independent of the pair used to span it. Second, and less obviously, it loses nothing: the numbers $K(\Pi)$, as Π ranges over all tangent 2-planes, determine the full curvature tensor. Sectional curvature is therefore not a coarsening of R but a faithful re-encoding of it, and it is the encoding through which the model spaces S^n, H^n, and $\mathbb{R}^n$ reveal themselves as the spaces of constant curvature $+1/r^2$, -1, and 0.

Throughout, (M, g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇, R is the curvature endomorphism of definition 4.1.2, and $\mathrm{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle$ is the associated $(0, 4)$-tensor of definition 4.1.4. The algebraic symmetries recorded in proposition 4.2.1 are used constantly; they are (i) antisymmetry in the first pair, (ii) antisymmetry in the second pair, (iii) symmetry under exchange of the two pairs, and (iv) the first Bianchi identity. Sectional curvature in the stated normalization requires the metric to be nondegenerate on the plane in question, so the definition is given first in the Riemannian case, where every 2-plane is admissible.

Definition 4.3.1: Sectional curvature

Let (M, g) be a Riemannian manifold, $p \in M$, and let $\Pi \subseteq T_pM$ be a 2-dimensional linear subspace spanned by vectors $X, Y \in T_pM$. The *sectional curvature* of Π is

$$K(\Pi) = \frac{\operatorname{Riem}(X, Y, Y, X)}{|X|^2 |Y|^2 - \langle X, Y\rangle^2}$$

The denominator is the squared area of the parallelogram spanned by X and Y; by the Cauchy–Schwarz inequality it is positive precisely because X, Y are linearly independent, so the quotient is defined. That $K(\Pi)$ depends only on Π, and not on the chosen spanning pair, is the content of proposition 4.3.2. When emphasis on the point is needed the value is written $K_p(\Pi)$; the notations $\sec(\Pi)$ and $K(X, Y)$ are also used, the latter for the sectional curvature of the plane spanned by X and Y.

The denominator deserves a name. Write

$$G(X, Y) = |X|^2 |Y|^2 - \langle X, Y\rangle^2$$

for the *Gram determinant* of the pair (X, Y), the determinant of the matrix $\left(\begin{smallmatrix} \langle X,X\rangle & \langle X,Y\rangle \\ \langle Y,X\rangle & \langle Y,Y\rangle \end{smallmatrix}\right)$. It is the square of the area of the parallelogram on X and Y, vanishes exactly when the two vectors are dependent, and transforms under a change of spanning pair by the square of the determinant of the change-of-basis matrix. That last property is what makes the quotient in definition 4.3.1 basis-independent, as the next proposition shows. In the pseudo-Riemannian setting $G(X, Y)$ can vanish or be negative for independent vectors (a plane on which g degenerates, or a timelike plane), and the definition is restricted to *nondegenerate* 2-planes, those on which g restricts to a nondegenerate form so that $G(X, Y) \neq 0$; the well-definedness argument below is unchanged in that case.

A single worked instance fixes the meaning before the general theory. Take Euclidean space $(\mathbb{R}^n, \bar{g})$. Its curvature tensor vanishes identically, since in standard coordinates the metric components are constant, every Christoffel symbol is zero, and R is built from the Γ^k_{ij} and their derivatives (example 4.1.6). Hence $\operatorname{Riem}(X, Y, Y, X) = 0$ for every pair, and $K(\Pi) = 0/G(X, Y) = 0$ for every 2-plane Π at every point. Euclidean space has vanishing sectional curvature in all directions, the analytic statement of its flatness. The two model spaces with nonzero curvature, S^n_r and H^n, are computed at the end of the section.

Proposition 4.3.2: Sectional curvature is independent of the spanning pair

Let (M, g) be Riemannian, $p \in M$, and $\Pi \subseteq T_pM$ a 2-plane. If (X, Y) and (X', Y') are two bases of Π, then

$$\frac{\operatorname{Riem}(X', Y', Y', X')}{G(X', Y')} = \frac{\operatorname{Riem}(X, Y, Y, X)}{G(X, Y)}$$

so $K(\Pi)$ in definition 4.3.1 is well defined. The same holds for a nondegenerate 2-plane in the pseudo-Riemannian case.

***Proof*:**
Since (X, Y) and (X', Y') are bases of the same 2-dimensional space Π, there is an invertible

matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in GL(2, \mathbb{R})$ with

$$X' = aX + bY, \qquad Y' = cX + dY$$

It suffices to prove that both the numerator and the denominator of the quotient scale by the same factor $(\det A)^2 = (ad - bc)^2$ under this substitution; the ratio is then unchanged.

The numerator. Write $Q(U, V) = \operatorname{Riem}(U, V, V, U)$ for the numerator functional. Expanding by multilinearity of Riem in each slot,

$$Q(X', Y') = \operatorname{Riem}(aX + bY,\ cX + dY,\ cX + dY,\ aX + bY)$$

is a sum of terms $\operatorname{Riem}(\,\cdot\,, \cdot\,, \cdot\,, \cdot\,)$ in which each of the four arguments is one of X, Y with a scalar coefficient. By the antisymmetry $\operatorname{Riem}(U, V, \,\cdot\,, \cdot\,) = -\operatorname{Riem}(V, U, \,\cdot\,, \cdot\,)$ of proposition 4.2.1(i), any term whose first two arguments are equal (both X or both Y) vanishes; likewise by antisymmetry in the last pair, proposition 4.2.1(ii), any term whose last two arguments are equal vanishes. The surviving terms have first pair a permutation of $\{X, Y\}$ and last pair a permutation of $\{X, Y\}$. Reading off the scalar coefficient of each from the two factors $cX + dY$ in the middle and the outer factors $aX + bY$, and collecting,

$$Q(X', Y') = abcd \operatorname{Riem}(X, Y, X, Y) + a^2d^2 \operatorname{Riem}(X, Y, Y, X) + b^2c^2 \operatorname{Riem}(Y, X, X, Y) + abcd \operatorname{Riem}(Y, X, Y, X)$$

Using the two antisymmetries, $\operatorname{Riem}(X, Y, X, Y) = -\operatorname{Riem}(X, Y, Y, X)$, $\operatorname{Riem}(Y, X, X, Y) = -\operatorname{Riem}(X, Y, X, Y) = \operatorname{Riem}(X, Y, Y, X)$, and $\operatorname{Riem}(Y, X, Y, X) = -\operatorname{Riem}(X, Y, Y, X)$, so every surviving term is a multiple of $\operatorname{Riem}(X, Y, Y, X)$:

$$Q(X', Y') = \left[a^2d^2 - abcd - abcd + b^2c^2\right] \operatorname{Riem}(X, Y, Y, X) = (ad - bc)^2\, Q(X, Y)$$

The denominator. The Gram determinant $G(X, Y) = \det\bigl(\langle X_i, X_j\rangle\bigr)$ with $(X_1, X_2) = (X, Y)$ is the determinant of the matrix of inner products. Under the substitution $(X', Y') = (X, Y)A^{\mathsf{T}}$ in the sense above, the Gram matrix transforms as $\bigl(\langle X_i', X_j'\rangle\bigr) = A\bigl(\langle X_i, X_j\rangle\bigr)A^{\mathsf{T}}$, so

$$G(X', Y') = \det\bigl(A\bigl(\langle X_i, X_j\rangle\bigr)A^{\mathsf{T}}\bigr) = (\det A)^2\, G(X, Y) = (ad - bc)^2\, G(X, Y)$$

This uses only bilinearity and symmetry of $\langle -, - \rangle$, which hold for any (pseudo-)Riemannian g; for a nondegenerate plane $G(X, Y) \neq 0$, so the quotient is defined.

Dividing, the factor $(ad - bc)^2$ cancels:

$$\frac{Q(X', Y')}{G(X', Y')} = \frac{(ad - bc)^2\, Q(X, Y)}{(ad - bc)^2\, G(X, Y)} = \frac{Q(X, Y)}{G(X, Y)}$$

Hence the quotient is the same for the two bases, and $K(\Pi)$ depends only on Π, as we sought to show.

In practice the formula simplifies whenever a convenient spanning pair is available. If X, Y are orthonormal, then $|X|^2 = |Y|^2 = 1$ and $\langle X, Y\rangle = 0$, so $G(X, Y) = 1$ and $K(\Pi) = \operatorname{Riem}(X, Y, Y, X)$ with no denominator at all. Proposition 4.3.2 guarantees that this stripped-down value agrees with the general one, so orthonormal pairs may always be used to compute sectional curvature, and this is how the examples below proceed.

The heart of the section is that no information is lost in passing from R to K. The precise statement concerns algebraic curvature tensors: multilinear maps carrying the four symmetries of the Riemann tensor, of which the value Riem_p at a point is the geometric example.

Theorem 4.3.3: Sectional curvature determines the curvature tensor

Let V be a finite-dimensional real vector space with a nondegenerate symmetric bilinear form $\langle -,- \rangle$, and let $\mathcal{R}, \mathcal{R}' \colon V^4 \to \mathbb{R}$ be two *algebraic curvature tensors*, that is, quadrilinear maps each satisfying the symmetries of proposition 4.2.1:

1. $\mathcal{R}(x,y,z,w) = -\mathcal{R}(y,x,z,w)$;
2. $\mathcal{R}(x,y,z,w) = -\mathcal{R}(x,y,w,z)$;
3. $\mathcal{R}(x,y,z,w) = \mathcal{R}(z,w,x,y)$;
4. $\mathcal{R}(x,y,z,w) + \mathcal{R}(y,z,x,w) + \mathcal{R}(z,x,y,w) = 0$.

If $\mathcal{R}(x,y,y,x) = \mathcal{R}'(x,y,y,x)$ for all $x,y \in V$, then $\mathcal{R} = \mathcal{R}'$. Consequently, on a Riemannian manifold the sectional curvature function $\Pi \mapsto K(\Pi)$ on 2-planes determines the full curvature tensor Riem_p at each point, and two Riemannian manifolds whose curvature tensors have the same sectional curvatures at corresponding points have equal curvature tensors there.

***Proof*:**
Set $D = \mathcal{R} - \mathcal{R}'$. The symmetries (i)–(iv) are preserved under subtraction, so D is again an algebraic curvature tensor, and the hypothesis says $D(x,y,y,x) = 0$ for all $x,y \in V$. The claim $\mathcal{R} = \mathcal{R}'$ is equivalent to $D \equiv 0$, and this is what will be shown: an algebraic curvature tensor whose associated quadratic function $Q(x,y) := D(x,y,y,x)$ vanishes identically is itself zero. The argument is polarization in three steps, each extracting more of D from the vanishing of Q.

Step 1: linearize the first argument. Fix $y \in V$ and expand $Q(x+z,y)$ using multilinearity of D:

$$D(x+z,y,y,x+z) = D(x,y,y,x) + D(x,y,y,z) + D(z,y,y,x) + D(z,y,y,z)$$

The outer terms are $Q(x,y)$ and $Q(z,y)$. The two middle terms are equal: by the pair symmetry (iii), $D(z,y,y,x) = D(y,x,z,y)$; then the first-pair antisymmetry (i) gives $D(y,x,z,y) = -D(x,y,z,y)$, and the second-pair antisymmetry (ii) gives $-D(x,y,z,y) = D(x,y,y,z)$. Chaining these, $D(z,y,y,x) = D(y,x,z,y) = D(x,y,y,z)$. Therefore

$$Q(x+z,y) - Q(x,y) - Q(z,y) = 2\,D(x,y,y,z)$$

Since $Q \equiv 0$ by hypothesis, the left side vanishes for all x,y,z, and hence

$$D(x,y,y,z) = 0 \qquad \text{for all } x,y,z \in V \tag{4.2}$$

Step 2: linearize the repeated middle argument. In (4.2) the second and third arguments coincide. Replace y by $y+w$ and expand:

$$0 = D(x,y+w,y+w,z) = D(x,y,y,z) + D(x,y,w,z) + D(x,w,y,z) + D(x,w,w,z)$$

The first and last terms vanish by (4.2), leaving

$$D(x,y,w,z) + D(x,w,y,z) = 0 \qquad \text{for all } x,y,w,z \in V \tag{4.3}$$

so D is antisymmetric in its second and third arguments.

Step 3: apply the first Bianchi identity. Fix $x, y, z, w \in V$. The first Bianchi identity (iv), applied with the four arguments x, z, y in the cyclic slots and w in the last slot, reads

$$D(x, z, y, w) + D(z, y, x, w) + D(y, x, z, w) = 0$$

Rewrite two of these terms. By the first-pair antisymmetry (i), $D(y, x, z, w) = -D(x, y, z, w)$; and by the pair symmetry (iii), $D(z, y, x, w) = D(x, w, z, y)$, which by the second-pair antisymmetry (ii) equals $-D(x, w, y, z)$, and by (4.3) equals $+D(x, y, w, z)$, which by (ii) equals $-D(x, y, z, w)$. Substituting,

$$D(x, z, y, w) - D(x, y, z, w) - D(x, y, z, w) = 0$$

that is $D(x, z, y, w) = 2\,D(x, y, z, w)$. On the other hand, applying (4.3) directly with the roles of the second and third arguments gives $D(x, z, y, w) = -D(x, y, z, w)$. Comparing,

$$2\,D(x, y, z, w) = D(x, z, y, w) = -D(x, y, z, w)$$

so $3\,D(x, y, z, w) = 0$, and therefore $D(x, y, z, w) = 0$ for all $x, y, z, w \in V$.

Thus $D \equiv 0$, i.e. $\mathcal{R} = \mathcal{R}'$. For the geometric consequence, apply this to $V = T_pM$ with $\langle -, - \rangle = g_p$: the value $\mathcal{R} = \mathrm{Riem}_p$ is an algebraic curvature tensor by proposition 4.2.1, and for linearly independent x, y the number $\mathrm{Riem}_p(x, y, y, x) = G(x, y)\,K(\Pi)$ is recovered from the sectional curvature of $\Pi = \mathrm{span}(x, y)$, while for dependent x, y it vanishes by antisymmetry. Hence the function K on 2-planes determines $\mathrm{Riem}_p(x, y, y, x)$ for all x, y, and by the theorem determines Riem_p completely, as claimed.

The theorem is the reason sectional curvature is the primary invariant of Riemannian geometry. It says that the seemingly modest data of one number per tangent plane carry every part of the curvature tensor, so any curvature condition can in principle be phrased as a condition on sectional curvatures, and conversely a hypothesis such as "all sectional curvatures are positive" or "all sectional curvatures equal c" is a full constraint on the tensor R. The constant case is the sharpest illustration. Consider the $(1, 3)$-tensor $R_c(X, Y)Z = c\big(\langle Y, Z\rangle\, X - \langle X, Z\rangle\, Y\big)$ and its $(0, 4)$-form $\mathcal{R}_c(X, Y, Z, W) = \langle R_c(X, Y)Z, W\rangle = c\big(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle\big)$; this $\mathcal{R}_c$ is an algebraic curvature tensor, and $\mathcal{R}_c(x, y, y, x) = c\big(|x|^2\,|y|^2 - \langle x, y\rangle^2\big) = c\,G(x, y)$, so its sectional curvature is c on every plane. If a Riemannian manifold has $K(\Pi) = c$ for every 2-plane at every point, then Riem_p and $\mathcal{R}_c$ agree on the quantity $(x, y) \mapsto \mathrm{Riem}_p(x, y, y, x)$ at each point, and by theorem 4.3.3 the two tensors coincide: $\mathrm{Riem}_p = \mathcal{R}_c$. This is how the constant-curvature form of R is pinned down in the classification of space forms, taken up later in the chapter; here it already lets the model spaces be identified from a single sectional curvature computation, since each is isotropic and homogeneous.

The first computation is the round sphere, the constant-positive-curvature benchmark. Its Christoffel symbols were found in example 2.3.5; feeding them into the coordinate curvature formula of definition 4.1.5 gives the sectional curvature directly.

Example 4.3.4: The round sphere has constant sectional curvature $1/r^2$

Work first on the unit sphere S^2 with the coordinates (θ, φ) of example 2.3.5, in which

$$g = d\theta^2 + \sin^2\theta\, d\varphi^2, \qquad 0 < \theta < \pi$$

and the only nonzero Christoffel symbols are $\Gamma^{\theta}_{\varphi\varphi} = -\sin\theta\cos\theta$ and $\Gamma^{\varphi}_{\theta\varphi} = \Gamma^{\varphi}_{\varphi\theta} = \cot\theta$. The curvature endomorphism of definition 4.1.2 on the coordinate frame is $R(\partial_i, \partial_j)\partial_k = R^l{}_{ijk}\partial_l$, and by the coordinate formula of definition 4.1.5,

$$R^l{}_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik}$$

(the sign convention is that of definition 4.1.2; the coordinate vector fields commute, so the $[\partial_i, \partial_j]$ term is absent). The single sectional plane at each point is $\Pi = \mathrm{span}(\partial_\theta, \partial_\varphi)$, and the number that enters the sectional curvature is $\mathrm{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta) = \langle R(\partial_\theta, \partial_\varphi)\partial_\varphi, \partial_\theta\rangle$, the ∂_θ-coefficient of $R(\partial_\theta, \partial_\varphi)\partial_\varphi$ times $g_{\theta\theta}$. That coefficient is the component $R^\theta{}_{\theta\varphi\varphi}$ (upper index $l = \theta$; operator pair $i = \theta$, $j = \varphi$; acted-on index $k = \varphi$):

$$R^\theta{}_{\theta\varphi\varphi} = \partial_\theta \Gamma^\theta_{\varphi\varphi} - \partial_\varphi \Gamma^\theta_{\theta\varphi} + \Gamma^\theta_{\theta m}\Gamma^m_{\varphi\varphi} - \Gamma^\theta_{\varphi m}\Gamma^m_{\theta\varphi}$$

Here $\partial_\theta \Gamma^\theta_{\varphi\varphi} = \partial_\theta(-\sin\theta\cos\theta) = -\cos^2\theta + \sin^2\theta = \sin^2\theta - \cos^2\theta$; the term $\partial_\varphi \Gamma^\theta_{\theta\varphi} = 0$; the third sum vanishes because $\Gamma^\theta_{\theta m} = 0$ for every m; and the fourth sum keeps only $m = \varphi$, giving $-\Gamma^\theta_{\varphi\varphi}\Gamma^\varphi_{\theta\varphi} = -(-\sin\theta\cos\theta)(\cot\theta) = \sin\theta\cos\theta \cdot \frac{\cos\theta}{\sin\theta} = \cos^2\theta$. Adding,

$$R^\theta{}_{\theta\varphi\varphi} = (\sin^2\theta - \cos^2\theta) + \cos^2\theta = \sin^2\theta$$

The component $R^\theta{}_{\theta\varphi\varphi}$ is by definition the ∂_θ-coefficient of $R(\partial_\theta, \partial_\varphi)\partial_\varphi$. Lowering that index with $g_{\theta\theta} = 1$ gives the $(0,4)$-component in the order needed for the sectional curvature,

$$\mathrm{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta) = \langle R(\partial_\theta, \partial_\varphi)\partial_\varphi, \partial_\theta\rangle = g_{\theta\theta}\, R^\theta{}_{\theta\varphi\varphi} = \sin^2\theta$$

The Gram determinant of the coordinate frame is

$$G(\partial_\theta, \partial_\varphi) = g_{\theta\theta}g_{\varphi\varphi} - g_{\theta\varphi}^2 = 1 \cdot \sin^2\theta - 0 = \sin^2\theta$$

so the sectional curvature of the only 2-plane at each point is

$$K = \frac{\mathrm{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta)}{G(\partial_\theta, \partial_\varphi)} = \frac{\sin^2\theta}{\sin^2\theta} = 1$$

independent of the point. Thus the unit round sphere has $K \equiv 1$. The blow-up of $\cot\theta$ at the poles, noted in example 2.3.5, cancels out of K, as it must: the curvature is a geometric quantity and the poles are ordinary points of S^2.

For the sphere of radius r, the metric of S^n_r scales as $g_r = r^2 g_1$ relative to the unit metric on the same underlying manifold. Under a constant rescaling $g \mapsto \lambda^2 g$ the Christoffel symbols are unchanged (they depend only on ratios and derivatives of metric components, and the λ^2 cancels in $\frac{1}{2}g^{kl}(\partial_i g_{jl} + \cdots)$), so the $(1,3)$-curvature $R^l{}_{ijk}$ is unchanged, and the $(0,4)$-tensor Riem scales by λ^2 from the single lowered index. The Gram determinant scales by λ^4. Hence K scales by $\lambda^2/\lambda^4 = \lambda^{-2}$, and with $\lambda = r$ the round S^n_r has

$$K \equiv \frac{1}{r^2}$$

For $n \geq 3$ this value holds for *every* tangent 2-plane, not only the coordinate one: by example 1.3.4 the sphere S^n_r is homogeneous and isotropic, so its isometry group carries any 2-plane at any point to any other 2-plane at any other point, and sectional curvature is invariant under isometries (isometries commute with ∇ and preserve g, hence preserve Riem and G). The sectional curvature is therefore the same on all planes, equal to the value $1/r^2$ computed on one of them. The round sphere is the space of constant curvature $+1/r^2$.

The second computation is the constant-negative-curvature model. The upper half-plane realization of H^2 carries the Christoffel symbols already found in example 3.1.5, and the sectional curvature follows from the same coordinate formula.

Example 4.3.5: The hyperbolic plane has constant sectional curvature -1

Take the upper half-plane model $H^2 = \{(x,y) \mid y > 0\}$ of example 1.3.5 with the metric of example 3.1.5,

$$g = \frac{1}{y^2}\left(dx^2 + dy^2\right)$$

whose nonzero Christoffel symbols are $\Gamma^x_{xy} = \Gamma^x_{yx} = -\frac{1}{y}$, $\Gamma^y_{xx} = \frac{1}{y}$, and $\Gamma^y_{yy} = -\frac{1}{y}$. As in the spherical case the coordinate vector fields commute, so

$$R^l{}_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik}$$

by definition 4.1.5, with the sign convention of definition 4.1.2. The one 2-plane at each point is $\operatorname{span}(\partial_x, \partial_y)$; compute $R^x{}_{xyy}$, taking $(i,j) = (x,y)$ acting on ∂_y with upper index x:

$$R^x{}_{xyy} = \partial_x \Gamma^x_{yy} - \partial_y \Gamma^x_{xy} + \Gamma^x_{xm}\Gamma^m_{yy} - \Gamma^x_{ym}\Gamma^m_{xy}$$

Evaluate term by term. First, $\Gamma^x_{yy} = 0$, so $\partial_x \Gamma^x_{yy} = 0$. Second, $\partial_y \Gamma^x_{xy} = \partial_y(-y^{-1}) = y^{-2}$. Third, the sum $\Gamma^x_{xm}\Gamma^m_{yy}$ over m: the factor Γ^x_{xm} is nonzero only for $m = y$ (value $\Gamma^x_{xy} = -y^{-1}$), and then $\Gamma^y_{yy} = -y^{-1}$, contributing $(-y^{-1})(-y^{-1}) = y^{-2}$. Fourth, the sum $\Gamma^x_{ym}\Gamma^m_{xy}$ over m: the factor Γ^x_{ym} is nonzero only for $m = x$ (value $\Gamma^x_{yx} = -y^{-1}$), and then $\Gamma^x_{xy} = -y^{-1}$, contributing $(-y^{-1})(-y^{-1}) = y^{-2}$, entering with a minus sign. Collecting,

$$R^x{}_{xyy} = 0 - y^{-2} + y^{-2} - y^{-2} = -y^{-2}$$

Lower the index with $g_{xx} = y^{-2}$: the $(0,4)$-component is

$$\operatorname{Riem}(\partial_x, \partial_y, \partial_y, \partial_x) = \langle R(\partial_x, \partial_y)\partial_y, \partial_x \rangle = g_{xx} R^x{}_{xyy} = y^{-2} \cdot (-y^{-2}) = -y^{-4}$$

The Gram determinant of the coordinate frame is

$$G(\partial_x, \partial_y) = g_{xx}g_{yy} - g_{xy}^2 = y^{-2} \cdot y^{-2} - 0 = y^{-4}$$

so the sectional curvature is

$$K = \frac{\operatorname{Riem}(\partial_x, \partial_y, \partial_y, \partial_x)}{G(\partial_x, \partial_y)} = \frac{-y^{-4}}{y^{-4}} = -1$$

at every point of H^2, independent of (x,y). The hyperbolic plane has constant curvature -1.

The extension to H^n follows the same isotropy argument as for the sphere. By example 1.3.5, hyperbolic space H^n is homogeneous and isotropic, its isometry group $O^+(1,n)$ acting transitively on tangent 2-planes; and the several models (hyperboloid, upper half-space, Poincaré ball) of example 1.3.5 are mutually isometric, so the value of K is model-independent. Since sectional curvature is preserved by isometries and computed to be -1 on one 2-plane, it equals -1 on all of them:

$$K \equiv -1$$

Rescaling the metric by λ^2 multiplies K by λ^{-2}, exactly as for the sphere, so the rescaled hyperbolic space of "radius" ρ has constant curvature $-1/\rho^2$; the value -1 corresponds to the standard normalization. Hyperbolic space is the space of constant negative curvature, completing the trio of constant-curvature models alongside S^n_r with $K = +1/r^2$ and Euclidean space $\mathbb{R}^n$ with $K \equiv 0$.

4.4 Ricci and Scalar Curvature; Einstein Metrics

The Riemann curvature tensor of an n-dimensional manifold is an intricate object: at each point it is a section of a bundle whose fibers, once the algebraic symmetries of proposition 4.2.1 are imposed, have dimension $\frac{1}{12}n^2(n^2-1)$. For $n = 4$ that is already 20 independent components at every point. Much of Riemannian geometry proceeds not by confronting Riem in its entirety but by extracting from it coarser tensors that retain part of the geometric information while being far more tractable. The sectional curvature of the previous section is one such extract, and it loses nothing: by theorem 4.3.3 the sectional curvatures determine Riem completely. The tensors introduced here, the Ricci and scalar curvatures, are genuine simplifications. They are traces of Riem, and passing to a trace discards information. What one gains in exchange is a symmetric 2-tensor and a scalar function that couple directly to the geometry of volumes, to the Laplacian, and to the equations of general relativity.

The Ricci curvature measures how the volume of a small geodesic ball differs from its Euclidean counterpart in a given direction, and the scalar curvature aggregates this over all directions into a single number at each point. Both arise by contracting the Riemann tensor. The order and the slots in which one contracts are governed by the sign convention fixed for this chapter, and the payoff of that convention is that the round sphere and hyperbolic space acquire the expected signs: positive Ricci and scalar curvature for the sphere, negative for hyperbolic space. This section defines these traces, records their symmetry, decomposes the full curvature tensor into the pieces that Ricci and scalar curvature see together with the trace-free remainder (the Weyl tensor), and singles out the metrics whose Ricci tensor is a constant multiple of g, the Einstein metrics. The section closes with Schur's lemma, a rigidity statement showing that in dimension at least three a pointwise proportionality $\mathrm{Ric} = f\,g$ forces f to be constant.

Throughout, (M, g) is an n-dimensional (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and R, Riem denote the curvature endomorphism and its $(0,4)$-form as fixed in section 4.1. The Einstein summation convention is in force: an index repeated once up and once down is summed from 1 to n. Recall the sign convention:

$$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z, \qquad \mathrm{Riem}(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle$$

Definition 4.4.1: Ricci Curvature

Let (M, g) be a (pseudo-)Riemannian manifold. The *Ricci curvature* Ric is the covariant 2-tensor field obtained from the curvature endomorphism R of definition 4.1.2 by tracing over its first and last arguments: for $X, Y \in T_pM$,

$$\operatorname{Ric}(X, Y) = \operatorname{tr}\big(Z \mapsto R(Z, X)Y\big)$$

the trace of the linear endomorphism $Z \mapsto R(Z, X)Y$ of T_pM. Equivalently, if $(e_i)_{i=1}^n$ is a local orthonormal frame for a Riemannian metric, then

$$\operatorname{Ric}(X, Y) = \sum_{i=1}^{n} \langle R(e_i, X)Y, e_i\rangle = \sum_{i=1}^{n} \operatorname{Riem}(e_i, X, Y, e_i)$$

In local coordinates (x^i), writing the components of the curvature endomorphism as $R(\partial_i, \partial_j)\partial_k = R^l{}_{ijk}\,\partial_l$ as in definition 4.1.5, the components $R_{ij} := \operatorname{Ric}(\partial_i, \partial_j)$ are the contraction

$$R_{ij} = R^k{}_{kij}$$

the trace taken over the upper index and the first lower index.

Two remarks fix the bookkeeping, because tracing a $(1, 3)$-tensor over one of its four index slots is exactly where sign and normalization errors enter. First, the coordinate formula $R_{ij} = R^k{}_{kij}$ is the same contraction as the frame formula: with $Z = \partial_k$ playing the role summed against e_i, the endomorphism $Z \mapsto R(Z, \partial_i)\partial_j$ has ∂_l-component $R^l{}_{kij}$, and its trace is $\sum_k R^k{}_{kij}$. Second, the orthonormal-frame expression is the definition one uses in practice: it makes manifest that Ric is obtained from Riem by summing $\operatorname{Riem}(e_i, X, Y, e_i)$ over an orthonormal basis, so that Ric is a metric contraction of Riem over its first and fourth slots. In the pseudo-Riemannian setting the frame sum acquires the signs $\varepsilon_i = \langle e_i, e_i\rangle = \pm 1$ of the orthonormal frame, $\operatorname{Ric}(X, Y) = \sum_i \varepsilon_i \operatorname{Riem}(e_i, X, Y, e_i)$; the coordinate formula $R_{ij} = R^k{}_{kij}$ is unchanged and signature-agnostic, which is why it is the cleaner one to carry.

Interpreting the definition: $\operatorname{Ric}(X, X)$ is a sum of sectional-curvature-like quantities. If X is a unit vector and (e_i) is an orthonormal basis with $e_n = X$, then $\operatorname{Riem}(e_i, X, X, e_i)$ is, for $i \neq n$, exactly the numerator of the sectional curvature $K(\operatorname{span}(e_i, X))$ of the plane spanned by e_i and X (the denominator is 1 since the vectors are orthonormal), while the $i = n$ term vanishes by the antisymmetry of proposition 4.2.1. Hence

$$\operatorname{Ric}(X, X) = \sum_{i=1}^{n-1} K\big(\operatorname{span}(e_i, X)\big)$$

the sum of the sectional curvatures of the $n-1$ coordinate planes containing X. The Ricci curvature in the direction X is the total sectional curvature of the planes through X; up to the normalization $n-1$ it is their average. This is the precise sense in which $\operatorname{Ric}(X, X)$ controls the spreading of nearby geodesics leaving p in directions near X, and thereby the infinitesimal volume of a geodesic cone about X: positive Ricci curvature means geodesics converge on average, negative means they spread.

Before recording the first structural property, a computation on the model spaces shows the definition in action and pins down the signs.

Example 4.4.2: Ricci Curvature of the Space Forms

Suppose (M, g) has constant sectional curvature c, meaning $K(\Pi) = c$ for every 2-plane Π. By theorem 4.3.3 the curvature endomorphism is then forced to be

$$R(X,Y)Z = c\big(\langle Y, Z\rangle X - \langle X, Z\rangle Y\big)$$

the constant-curvature form of the sign convention (this identity is derived in full in section 4.5; here it is used only to read off the trace). Fix a point and an orthonormal basis (e_i). For unit $X = e_a$,

$$\begin{aligned}\operatorname{Ric}(X,X) &= \sum_{i=1}^{n} \langle R(e_i, X)X, e_i\rangle = \sum_{i=1}^{n} c\big(\langle X, X\rangle \langle e_i, e_i\rangle - \langle e_i, X\rangle \langle X, e_i\rangle\big) \\ &= c\sum_{i=1}^{n}\big(1 - \langle e_i, X\rangle^2\big) = c\big(n-1\big)\end{aligned}$$

since $\sum_i \langle e_i, X\rangle^2 = |X|^2 = 1$. By polarization $\operatorname{Ric}(X,Y) = c(n-1)\langle X, Y\rangle$ for all X, Y, that is,

$$\operatorname{Ric} = (n-1)c\, g$$

For the round sphere S^n_r of example 1.3.4 the sectional curvature is $c = 1/r^2$ (example 4.3.4), so $\operatorname{Ric} = \frac{n-1}{r^2}\, g$, a positive multiple of g. For Euclidean space $c = 0$ and $\operatorname{Ric} \equiv 0$, consistent with $R \equiv 0$ from example 4.1.6. For hyperbolic space H^n the sectional curvature is $c = -1$ (example 4.3.5), so $\operatorname{Ric} = -(n-1)\, g$, a negative multiple. The signs are exactly those the convention was chosen to produce.

The Ricci tensor is symmetric. This is not automatic from the definition, which distinguishes the slots X and Y; it is a consequence of the pair symmetry of Riem.

Proposition 4.4.3: Symmetry of the Ricci Tensor

The Ricci curvature of a (pseudo-)Riemannian manifold is a symmetric 2-tensor:

$$\operatorname{Ric}(X,Y) = \operatorname{Ric}(Y,X) \qquad \text{for all } X, Y \in T_pM$$

Equivalently, in coordinates $R_{ij} = R_{ji}$.

***Proof*:**

Fix $p \in M$ and an orthonormal basis $(e_i)_{i=1}^n$ of T_pM (Riemannian case; the pseudo-Riemannian case is identical after inserting the signs ε_i, which are common to the two expressions below and cancel in the comparison). By definition,

$$\operatorname{Ric}(X,Y) = \sum_{i=1}^{n} \operatorname{Riem}(e_i, X, Y, e_i)$$

Apply the pair symmetry $\operatorname{Riem}(A,B,C,D) = \operatorname{Riem}(C,D,A,B)$ from proposition 4.2.1(iii) to each summand, with $(A,B,C,D) = (e_i, X, Y, e_i)$:

$$\operatorname{Riem}(e_i, X, Y, e_i) = \operatorname{Riem}(Y, e_i, e_i, X)$$

Now apply the two antisymmetries of proposition 4.2.1(i)–(ii): swapping the first pair introduces a sign, $\operatorname{Riem}(Y, e_i, e_i, X) = -\operatorname{Riem}(e_i, Y, e_i, X)$, and swapping the last pair introduces a second sign, $-\operatorname{Riem}(e_i, Y, e_i, X) = \operatorname{Riem}(e_i, Y, X, e_i)$. The two signs cancel, giving

$$\operatorname{Riem}(e_i, X, Y, e_i) = \operatorname{Riem}(e_i, Y, X, e_i)$$

Summing over i,

$$\operatorname{Ric}(X, Y) = \sum_{i=1}^{n} \operatorname{Riem}(e_i, X, Y, e_i) = \sum_{i=1}^{n} \operatorname{Riem}(e_i, Y, X, e_i) = \operatorname{Ric}(Y, X)$$

as we sought to show.

Symmetry of Ric is what makes it a legitimate candidate to be proportional to g, and it is what allows one to take its metric trace unambiguously. That trace is the scalar curvature.

Definition 4.4.4: Scalar Curvature

Let (M, g) be a (pseudo-)Riemannian manifold with Ricci tensor Ric. The *scalar curvature* is the function $\operatorname{scal}\colon M \to \mathbb{R}$ obtained as the metric trace of Ric:

$$\operatorname{scal} = \operatorname{tr}_g \operatorname{Ric} = g^{ij} R_{ij}$$

Equivalently, for a local orthonormal frame (e_i) (Riemannian case),

$$\operatorname{scal} = \sum_{i=1}^{n} \operatorname{Ric}(e_i, e_i) = \sum_{i,j=1}^{n} \operatorname{Riem}(e_i, e_j, e_j, e_i)$$

the sum of the sectional curvatures over all ordered pairs of distinct frame directions.

Unwinding the frame expression makes the geometric content of scal transparent. Using $\operatorname{Ric}(e_i, e_i) = \sum_j \operatorname{Riem}(e_j, e_i, e_i, e_j)$ and the antisymmetry that kills the diagonal terms $i = j$, the double sum runs over ordered pairs $i \neq j$, and each unordered pair $\{i, j\}$ contributes twice the sectional curvature $K(\operatorname{span}(e_i, e_j))$ of the plane it spans. Thus

$$\operatorname{scal} = 2 \sum_{i<j} K\bigl(\operatorname{span}(e_i, e_j)\bigr)$$

The scalar curvature at p is twice the sum of the sectional curvatures of the $\binom{n}{2}$ coordinate planes at p; equivalently, up to the constant $n(n-1)$ it is the average sectional curvature over all 2-planes. It carries the least pointwise information of the three traces of Riem, one number per point, but it is the trace that appears in the Gauss–Bonnet integrand in dimension two, in the total-scalar-curvature (Einstein–Hilbert) functional, and as the coefficient governing the leading volume defect of small geodesic balls.

Example 4.4.5: Scalar Curvature of the Space Forms

Continuing example 4.4.2, suppose (M, g) has constant sectional curvature c, so that $\operatorname{Ric} = (n-1)c\, g$. Taking the metric trace and using $\operatorname{tr}_g g = g^{ij} g_{ij} = n$,

$$\operatorname{scal} = \operatorname{tr}_g \operatorname{Ric} = (n-1)c \operatorname{tr}_g g = n(n-1)c$$

For the round sphere S^n_r with $c = 1/r^2$ this gives $\mathrm{scal} = \frac{n(n-1)}{r^2}$, a positive constant. For Euclidean space $\mathrm{scal} \equiv 0$, and for H^n with $c = -1$ one gets $\mathrm{scal} = -n(n-1)$. In dimension $n = 2$ the formula reads $\mathrm{scal} = 2c$, so for the unit sphere S^2 the scalar curvature is 2 while the Gauss curvature (the single sectional curvature) is 1: the scalar curvature is twice the Gauss curvature in dimension two, matching the frame formula $\mathrm{scal} = 2\sum_{i<j} K = 2K(\mathrm{span}(e_1, e_2))$.

The passage from Riem to Ric to scal can be organized as an orthogonal decomposition of the full curvature tensor. The Ricci and scalar curvatures capture the "trace part" of Riem; the trace-free remainder is a tensor with the same algebraic symmetries as Riem but with all of its contractions vanishing, the Weyl tensor. To state this cleanly it is convenient to use the Kulkarni–Nomizu product, which builds a curvature-type tensor out of two symmetric 2-tensors.

Definition 4.4.6: Ricci Decomposition and the Weyl Tensor

Let (M, g) be a (pseudo-)Riemannian manifold of dimension $n \geq 3$. For symmetric 2-tensors h, k, define their *Kulkarni–Nomizu product* $h \owedge k$, a $(0, 4)$-tensor, by

$$(h \owedge k)(X, Y, Z, W) = h(X, W)k(Y, Z) + h(Y, Z)k(X, W) - h(X, Z)k(Y, W) - h(Y, W)k(X, Z)$$

It shares the algebraic symmetries of Riem in proposition 4.2.1: it is antisymmetric in (X, Y) and in (Z, W), symmetric under exchange of the pairs, and satisfies the first Bianchi identity. Write the *trace-free Ricci tensor* as

$$\mathring{\mathrm{Ric}} = \mathrm{Ric} - \frac{\mathrm{scal}}{n} g$$

The *Ricci decomposition* of the curvature tensor is

$$\mathrm{Riem} = \frac{\mathrm{scal}}{2n(n-1)} g \owedge g + \frac{1}{n-2} \mathring{\mathrm{Ric}} \owedge g + W$$

which *defines* the *Weyl tensor* W as the remainder. The three summands are the *scalar part*, the *trace-free Ricci part*, and the *Weyl (conformal) part*.

The defining property of the Weyl tensor, and the reason the coefficients above are what they are, is that W is *totally trace-free*: every contraction of W over a pair of its indices vanishes. This is proved directly.

Proposition 4.4.7: The Weyl Tensor Is Trace-Free

For $n \geq 3$ the Weyl tensor W defined in definition 4.4.6 satisfies $g^{il} W_{ijkl} = 0$; that is, the metric contraction of W over its first and last slots vanishes, and hence (by the symmetries) every contraction of W vanishes.

***Proof*:**
Fix $p \in M$ and an orthonormal basis (e_i), so that the metric contraction over the first and fourth slots of a $(0, 4)$-tensor T is the map $(Y, Z) \mapsto \sum_i T(e_i, Y, Z, e_i)$; for $T = \mathrm{Riem}$ this is Ric by definition. It suffices to compute this contraction on each of the three summands of definition 4.4.6 and check that the sum of the first two equals Ric, forcing the contraction of W

to vanish.

Step 1: contraction of a Kulkarni–Nomizu product. Let h, k be symmetric 2-tensors. Viewing h, k as symmetric endomorphisms of T_pM via the metric, write hk and kh for the two composite endomorphisms and $\mathrm{tr}_g h, \mathrm{tr}_g k$ for their traces. Contracting $h \owedge k$ over its first and fourth slots term by term,

$$\begin{aligned}\sum_i (h \owedge k)(e_i, Y, Z, e_i) &= \sum_i \Big[h(e_i, e_i)k(Y, Z) + h(Y, Z)k(e_i, e_i) \\ &\qquad - h(e_i, Z)k(Y, e_i) - h(Y, e_i)k(e_i, Z)\Big] \\ &= (\mathrm{tr}_g h)\, k(Y, Z) + (\mathrm{tr}_g k)\, h(Y, Z) - (kh)(Y, Z) - (hk)(Y, Z)\end{aligned}$$

the third and fourth terms being the two endomorphism composites $(kh)(Y, Z) = \sum_i k(Y, e_i)h(e_i, Z)$ and $(hk)(Y, Z) = \sum_i h(Y, e_i)k(e_i, Z)$. Two special cases are needed, and in each the two composites coincide because one factor is g.

Taking $h = k = g$: then $\mathrm{tr}_g g = n$ and $gg = g$ as an endomorphism, so

$$\sum_i (g \owedge g)(e_i, Y, Z, e_i) = n\, g(Y, Z) + n\, g(Y, Z) - g(Y, Z) - g(Y, Z) = 2(n - 1)\, g(Y, Z)$$

Taking $h = \mathring{\mathrm{Ric}}$ and $k = g$: then $\mathrm{tr}_g g = n$, $\mathrm{tr}_g \mathring{\mathrm{Ric}} = \mathrm{scal} - \frac{\mathrm{scal}}{n} \cdot n = 0$, and both composites $\mathring{\mathrm{Ric}}\, g$ and $g\, \mathring{\mathrm{Ric}}$ equal $\mathring{\mathrm{Ric}}$ as endomorphisms, so

$$\sum_i (\mathring{\mathrm{Ric}} \owedge g)(e_i, Y, Z, e_i) = 0 \cdot g(Y, Z) + n\, \mathring{\mathrm{Ric}}(Y, Z) - \mathring{\mathrm{Ric}}(Y, Z) - \mathring{\mathrm{Ric}}(Y, Z) = (n - 2)\, \mathring{\mathrm{Ric}}(Y, Z)$$

Step 2: assembling the trace of the decomposition. Contract the identity

$$\mathrm{Riem} = \frac{\mathrm{scal}}{2n(n - 1)}\, g \owedge g + \frac{1}{n - 2}\, \mathring{\mathrm{Ric}} \owedge g + W$$

over the first and fourth slots. The left side gives Ric. Using Step 1, the first summand contributes

$$\frac{\mathrm{scal}}{2n(n - 1)} \cdot 2(n - 1)\, g = \frac{\mathrm{scal}}{n}\, g$$

and the second contributes

$$\frac{1}{n - 2} \cdot (n - 2)\, \mathring{\mathrm{Ric}} = \mathring{\mathrm{Ric}} = \mathrm{Ric} - \frac{\mathrm{scal}}{n}\, g$$

Write $C(Y, Z) := \sum_i W(e_i, Y, Z, e_i)$ for the first-and-fourth contraction of the Weyl summand. Adding the three contributions gives

$$\mathrm{Ric} = \frac{\mathrm{scal}}{n}\, g + \Big(\mathrm{Ric} - \frac{\mathrm{scal}}{n}\, g\Big) + C = \mathrm{Ric} + C$$

Hence $C = 0$: the contraction of W over its first and fourth slots vanishes. Because W inherits the antisymmetries and pair symmetry of Riem (each summand of the decomposition satisfies them, and Riem does by proposition 4.2.1), every other pairwise contraction of W reduces to C up to sign: contracting the first two or the last two slots gives zero by antisymmetry, and contracting any mixed pair reduces to C by the pair symmetry together with the antisymmetries. Therefore W is totally trace-free, as we sought to show.

The decomposition is an orthogonal one with respect to the natural inner product on curvature-type tensors, and it is the pointwise decomposition of the space of algebraic curvature tensors into irreducible representations of the orthogonal group $O(n)$: the scalar part, the trace-free Ricci part, and the Weyl part are the three irreducible summands for $n \geq 4$. Two low-dimensional cases collapse this and are worth recording, since they explain why curvature is so much more rigid in low dimensions.

For $n = 2$ the space of algebraic curvature tensors is one-dimensional, and Riem is completely determined by the scalar curvature: explicitly $\mathrm{Riem} = \frac{\mathrm{scal}}{2}\frac{1}{2} g \owedge g$ reduces to $\mathrm{Riem}(X, Y, Z, W) = \frac{\mathrm{scal}}{2}\big(\langle X, W\rangle\langle Y, Z\rangle - \langle X, Z\rangle\langle Y, W\rangle\big)$, so a surface has a single curvature function, the Gauss curvature $K = \frac{\mathrm{scal}}{2}$. (The decomposition of definition 4.4.6 was stated for $n \geq 3$ because of the $\frac{1}{n-2}$; in dimension two the Weyl and trace-free-Ricci pieces are absent.) For $n = 3$ the Weyl tensor vanishes identically: the dimension of the space of algebraic curvature tensors is $\frac{1}{12} \cdot 9 \cdot 8 = 6$, which is exactly the dimension of the space of symmetric 2-tensors on $\mathbb{R}^3$, so the map $\mathrm{Riem} \mapsto \mathrm{Ric}$ is injective on curvature tensors and $W \equiv 0$. Consequently, in dimension three Riem is determined by Ric through

$$\mathrm{Riem} = \frac{\mathrm{scal}}{12}\, g \owedge g + \mathring{\mathrm{Ric}} \owedge g$$

For $n \geq 4$ the Weyl tensor is an independent piece of the curvature, invisible to Ric and scal; it is the obstruction to a metric being conformally flat.

Among all metrics, those for which the Ricci tensor is proportional to the metric occupy a distinguished place, both because they are the critical points of the total scalar curvature functional and because they are the natural higher-dimensional analogues of constant-curvature surfaces.

Definition 4.4.8: Einstein Metric

A (pseudo-)Riemannian metric g on M is an *Einstein metric*, and (M, g) is an *Einstein manifold*, if there is a constant $\lambda \in \mathbb{R}$ with

$$\mathrm{Ric} = \lambda\, g$$

The constant λ is the *Einstein constant* of g.

Taking the metric trace of $\mathrm{Ric} = \lambda g$ gives $\mathrm{scal} = \lambda\, \mathrm{tr}_g g = n\lambda$, so the Einstein constant is determined by the scalar curvature, $\lambda = \mathrm{scal}/n$, and the scalar curvature of an Einstein metric is constant. An Einstein metric is exactly one whose Ricci tensor is a constant multiple of g, equivalently one whose trace-free Ricci tensor $\mathring{\mathrm{Ric}}$ of definition 4.4.6 vanishes identically. Geometrically the condition says that the Ricci curvature is the same in every direction: $\mathrm{Ric}(X, X) = \lambda\, |X|^2$ for all X, so the average sectional curvature of the planes through any unit vector is the constant $\lambda/(n-1)$, independent of the vector. This is a far weaker requirement than constant sectional curvature, which asks that *every* sectional curvature be equal; an Einstein metric only asks that the directional averages agree. In dimensions two and three the two notions collapse (in dimension three by the vanishing of the Weyl tensor, $\mathring{\mathrm{Ric}} = 0$ together with $W = 0$ forces the constant-curvature form of example 4.4.2), but from dimension four onward Einstein metrics that are not of constant curvature exist in abundance.

The running example makes the definition concrete and reconnects with the space forms.

Example 4.4.9: The Sphere Is Einstein

Consider the round sphere S^n_r of example 1.3.4, of constant sectional curvature $c = 1/r^2$ (example 4.3.4). By the computation of example 4.4.2,

$$\mathrm{Ric} = (n-1)c\, g = \frac{n-1}{r^2}\, g$$

so S^n_r is an Einstein manifold with Einstein constant $\lambda = \frac{n-1}{r^2} > 0$, and by example 4.4.5 its scalar curvature is the constant

$$\mathrm{scal} = n(n-1)c = \frac{n(n-1)}{r^2}$$

Both are positive, as the sign convention requires for the sphere. The same computation shows the other two space forms are Einstein: Euclidean space is Einstein with $\lambda = 0$ (indeed $\mathrm{Ric} \equiv 0$), and H^n is Einstein with $\lambda = -(n-1) < 0$ and $\mathrm{scal} = -n(n-1)$. For $n = 2$ the sphere gives $\mathrm{Ric} = \frac{1}{r^2} g$ and $\mathrm{scal} = \frac{2}{r^2}$, so every surface is trivially Einstein: the Ricci tensor of a surface is always $\frac{\mathrm{scal}}{2} g$, whence the Einstein condition in dimension two carries no information and the interesting range is $n \geq 3$.

In defining an Einstein metric the constancy of λ was imposed by hand. It is worth asking whether it must be imposed at all: if the Ricci tensor is pointwise a multiple $f(p)\, g_p$ of the metric, with f a function that is a priori allowed to vary, is f automatically constant? For surfaces the answer is no, since $\mathrm{Ric} = \frac{\mathrm{scal}}{2} g$ always and scal can be any function. For $n = 3$ the same phenomenon is blocked, and in fact for every $n \geq 3$ pointwise proportionality forces constancy on a connected manifold. This is Schur's lemma, and its proof is the first substantial application of the second Bianchi identity.

Theorem 4.4.10: Schur's Lemma

Let (M, g) be a connected (pseudo-)Riemannian manifold of dimension $n \geq 3$. Suppose there is a smooth function $f : M \to \mathbb{R}$ with

$$\mathrm{Ric} = f\, g$$

at every point. Then f is constant; consequently g is an Einstein metric and $\mathrm{scal} = nf$ is constant.

***Proof*:**

The argument contracts the second Bianchi identity twice and reads off a first-order equation for scal that forces df to vanish. Throughout, fix $p \in M$ and work in a local orthonormal frame (e_i) (Riemannian case; the pseudo-Riemannian case is identical with the frame signs ε_i inserted, which cancel as before). All contractions below are metric contractions, written as sums over the frame.

Step 1: the once-contracted second Bianchi identity. Recall first that the covariant derivative $\nabla\,\mathrm{Riem}$ inherits the algebraic symmetries of Riem (proposition 4.2.1): since $\nabla g = 0$, differentiating an identity such as $\mathrm{Riem}(A, B, C, D) = -\,\mathrm{Riem}(B, A, C, D)$ gives $(\nabla_E\,\mathrm{Riem})(A, B, C, D) = -(\nabla_E\,\mathrm{Riem})(B, A, C, D)$, and likewise for the other symmetries. The differential Bianchi identity (theorem 4.2.3), written for the $(0, 4)$-tensor Riem and evaluated on frame vectors, reads

$$(\nabla_{e_a}\,\mathrm{Riem})(e_b, e_c, e_d, e_e) + (\nabla_{e_b}\,\mathrm{Riem})(e_c, e_a, e_d, e_e) + (\nabla_{e_c}\,\mathrm{Riem})(e_a, e_b, e_d, e_e) = 0$$

the sum being cyclic in the differentiation direction together with the first two arguments, (a, b, c). Contract this over the differentiation slot and the last argument, that is, set $e_a = e_e = e_i$ and sum over i. Each summand is evaluated by the first-and-fourth-slot contraction $\sum_i (\nabla_E \operatorname{Riem})(e_i, \cdot, \cdot, e_i)$, which by definition 4.4.1 is the covariant derivative of Ric. The first summand is

$$\sum_i (\nabla_{e_i} \operatorname{Riem})(e_b, e_c, e_d, e_i)$$

the metric contraction of $\nabla \operatorname{Riem}$ over its differentiation direction and its last argument. The second summand becomes

$$\sum_i (\nabla_{e_b} \operatorname{Riem})(e_c, e_i, e_d, e_i) = -\sum_i (\nabla_{e_b} \operatorname{Riem})(e_i, e_c, e_d, e_i) = -(\nabla_{e_b} \operatorname{Ric})(e_c, e_d)$$

using the antisymmetry proposition 4.2.1(i) in the first two arguments and then the first-and-fourth contraction. The third summand becomes

$$\sum_i (\nabla_{e_c} \operatorname{Riem})(e_i, e_b, e_d, e_i) = (\nabla_{e_c} \operatorname{Ric})(e_b, e_d)$$

directly by the first-and-fourth contraction. Adding the three and rearranging,

$$(\nabla_{e_c} \operatorname{Ric})(e_b, e_d) - (\nabla_{e_b} \operatorname{Ric})(e_c, e_d) = -\sum_i (\nabla_{e_i} \operatorname{Riem})(e_b, e_c, e_d, e_i)$$

Relabel $(e_c, e_b, e_d) = (X, Y, Z)$ and use the antisymmetry $\operatorname{Riem}(Y, X, Z, e_i) = -\operatorname{Riem}(X, Y, Z, e_i)$ on the right to obtain the once-contracted differential Bianchi identity

$$(\nabla_X \operatorname{Ric})(Y, Z) - (\nabla_Y \operatorname{Ric})(X, Z) = \sum_i (\nabla_{e_i} \operatorname{Riem})(X, Y, Z, e_i) \tag{4.4}$$

valid for all X, Y, Z since both sides are tensorial.

Step 2: the twice-contracted identity $\operatorname{div} \operatorname{Ric} = \frac{1}{2}\, d \operatorname{scal}$. Define the divergence of the symmetric 2-tensor Ric by $(\operatorname{div} \operatorname{Ric})(X) := \sum_i (\nabla_{e_i} \operatorname{Ric})(e_i, X)$. Contract (4.4) a second time, over the pair (Y, Z): put $Y = Z = e_j$ and sum over j. On the left,

$$\sum_j (\nabla_X \operatorname{Ric})(e_j, e_j) - \sum_j (\nabla_{e_j} \operatorname{Ric})(X, e_j) = \nabla_X \operatorname{scal} - (\operatorname{div} \operatorname{Ric})(X)$$

using $\sum_j (\nabla_X \operatorname{Ric})(e_j, e_j) = \nabla_X (\operatorname{tr}_g \operatorname{Ric}) = \nabla_X \operatorname{scal}$ (since ∇ commutes with the metric trace) and the symmetry of Ric in the second sum. On the right,

$$\sum_{i,j} (\nabla_{e_i} \operatorname{Riem})(X, e_j, e_j, e_i) = \sum_i (\nabla_{e_i} \operatorname{Ric})(X, e_i) = (\operatorname{div} \operatorname{Ric})(X)$$

where the inner sum over j is evaluated by the antisymmetries: $\operatorname{Riem}(X, e_j, e_j, e_i) = -\operatorname{Riem}(e_j, X, e_j, e_i) = \operatorname{Riem}(e_j, X, e_i, e_j)$ by proposition 4.2.1(i)–(ii), and summing over j this is the first-and-fourth-slot contraction, giving $\operatorname{Ric}(X, e_i)$ (the derivative $\nabla_{e_i} \operatorname{Riem}$ inherits the algebraic symmetries of Riem because $\nabla g = 0$). Equating the two sides,

$$\nabla_X \operatorname{scal} - (\operatorname{div} \operatorname{Ric})(X) = (\operatorname{div} \operatorname{Ric})(X)$$

that is $\nabla_X \operatorname{scal} = 2\,(\operatorname{div}\operatorname{Ric})(X)$ for every X, which is the twice-contracted differential Bianchi identity

$$\operatorname{div}\operatorname{Ric} = \frac{1}{2}\,d\operatorname{scal} \tag{4.5}$$

The factor $\frac{1}{2}$ is fixed by the sign and contraction conventions of this chapter, and (4.5) is the identity underlying the divergence-freeness of the Einstein tensor $\operatorname{Ric} - \frac{1}{2}\operatorname{scal} g$.

Step 3: imposing $\operatorname{Ric} = f\,g$. Now use the hypothesis. Since $\nabla g = 0$ (metric compatibility, definition 2.2.4), the covariant derivative of $\operatorname{Ric} = f\,g$ is

$$(\nabla_X \operatorname{Ric})(Y,Z) = (\nabla_X f)\,g(Y,Z) + f\,(\nabla_X g)(Y,Z) = (Xf)\,g(Y,Z)$$

Taking the divergence, for any Y,

$$(\operatorname{div}\operatorname{Ric})(Y) = \sum_i (\nabla_{e_i}\operatorname{Ric})(e_i,Y) = \sum_i (e_i f)\,g(e_i,Y) = Yf = df(Y)$$

since $\sum_i (e_i f)\,g(e_i,Y) = \sum_i (e_i f)\,\langle e_i, Y\rangle$ is the expansion of Y against the orthonormal frame applied to df, namely $df(Y)$. On the other hand the scalar curvature of $\operatorname{Ric} = f\,g$ is $\operatorname{scal} = \operatorname{tr}_g \operatorname{Ric} = f\operatorname{tr}_g g = nf$, so $d\operatorname{scal} = n\,df$. Substituting both into (4.5),

$$df = \operatorname{div}\operatorname{Ric} = \frac{1}{2}\,d\operatorname{scal} = \frac{n}{2}\,df$$

Hence $\left(1 - \frac{n}{2}\right) df = 0$, that is $\frac{2-n}{2}\,df = 0$. Because $n \geq 3$ the scalar factor $\frac{2-n}{2}$ is nonzero, so $df = 0$ identically. Therefore f is locally constant, and since M is connected, f is constant on M. With f constant, $\operatorname{Ric} = f\,g$ exhibits g as an Einstein metric with Einstein constant f, and $\operatorname{scal} = nf$ is constant, as we sought to show.

Schur's lemma is the reason the definition of an Einstein metric may safely be stated with a constant λ rather than a function: in dimension at least three, pointwise proportionality of Ric to g already delivers constancy. The mechanism is worth isolating. The single input beyond the algebraic symmetries is the differential Bianchi identity, contracted to (4.5); the hypothesis $\operatorname{Ric} = f\,g$ then turns that identity into $df = \frac{n}{2}df$, and the dimensional restriction $n \geq 3$ is exactly what makes the coefficient $1 - \frac{n}{2}$ invertible. The failure in dimension two is visible in the same equation: there $1 - \frac{n}{2} = 0$, the identity $df = \frac{n}{2}df$ is vacuous, and f is free. The contracted Bianchi identity (4.5) recorded along the way, $\operatorname{div}\operatorname{Ric} = \frac{1}{2}d\operatorname{scal}$, is itself one of the identities the subject depends on: rearranged as $\operatorname{div}\!\left(\operatorname{Ric} - \frac{1}{2}\operatorname{scal} g\right) = 0$, it says the Einstein tensor is divergence-free, the identity that makes the field equations of general relativity consistent with local conservation of energy and momentum.

4.5 Flat Manifolds and Space Forms

The curvature tensor was introduced as the obstruction to commuting second covariant derivatives. The extreme case in which the obstruction vanishes deserves separate treatment: a manifold on which $R \equiv 0$ is called *flat*, and the first result of this section shows that flatness is exactly local indistinguishability from Euclidean space. This is the sharpest possible converse to the computation of example 4.1.6, where the vanishing of the Euclidean curvature followed from the constancy of the metric coefficients. Here the implication runs the other way: if the curvature vanishes then coordinates can be manufactured in which the metric coefficients are constant, and the manifold looks, patch by patch, like a piece of $\mathbb{R}^n$.

Between the flat case and the general case sits a class of manifolds whose curvature is as symmetric as it can be without vanishing: those whose sectional curvature $K(\Pi)$ takes a single value c on every 2-plane Π at every point. These are the *space forms*, and they generalize the three model geometries that have run through the book. The plane $\mathbb{R}^n$ has $c = 0$, the round sphere S^n_r has $c = 1/r^2$ (example 4.3.4), and hyperbolic space H^n has $c = -1$ (example 4.3.5). The main structural fact is that these three, after rescaling, exhaust the local possibilities: any manifold of constant sectional curvature c is locally isometric to exactly one of them. Constant sectional curvature is a scalar condition on a 2-plane, yet it pins down the full $(0, 4)$ curvature tensor and, with more work, the local geometry.

Throughout, (M, g) is a (pseudo-)Riemannian manifold of dimension n with Levi-Civita connection ∇ (definition 2.3.2) and curvature endomorphism R (definition 4.1.2); the associated $(0, 4)$ tensor is $\operatorname{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle$ (definition 4.1.4). The sign convention of box 4.1.1 is in force. The Einstein summation convention is used where indices appear. The results on flatness and the local classification are stated for the Riemannian case; the frame constructions go through verbatim in the pseudo-Riemannian setting with orthonormal replaced by pseudo-orthonormal, and this is noted where it matters.

Theorem 4.5.1: Flat Manifolds are Locally Euclidean

Let (M, g) be a Riemannian manifold. Then the curvature endomorphism satisfies $R \equiv 0$ if and only if every point of M has a neighborhood isometric to an open subset of Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2. A manifold with $R \equiv 0$ is called *flat*.

***Proof*:**
One direction is immediate from example 4.1.6. If a neighborhood U of p is isometric to an open subset of $(\mathbb{R}^n, \bar{g})$, then the curvature endomorphism of U agrees under the isometry with that of Euclidean space, which vanishes; since R is a tensor determined by the metric (proposition 4.1.3, definition 2.3.2), it vanishes on U, and as p was arbitrary, $R \equiv 0$ on M. The content is the converse.

Assume $R \equiv 0$ and fix $p \in M$. The strategy is to build, on a neighborhood of p, a frame $(E_1, \ldots, E_n)$ that is simultaneously orthonormal and parallel, and then to show that this frame is the coordinate frame of a chart in which g is the Euclidean metric.

Step 1: a parallel orthonormal frame on a coordinate cube. Choose a chart $x = (x^1, \ldots, x^n)\colon W \to (-\varepsilon, \varepsilon)^n$ centered at p, so $x(p) = 0$, and let $(e_1, \ldots, e_n)$ be an orthonormal basis of (T_pM, g_p). Fix a and drop the subscript, writing $v = e_a$; the field $E = E_a$ is built on the cube by parallel transport along coordinate directions, and the vanishing of curvature is what makes it parallel in every direction. For $0 \le m \le n$ let

$$S_m = \{q \in W : x^{m+1}(q) = \cdots = x^n(q) = 0\}$$

be the m-dimensional coordinate slab through p, so $S_0 = \{p\}$ and $S_n = W$. The construction proceeds by induction on m, producing E on S_m with the property

$$P(m): \quad \nabla_{\partial_j} E = 0 \text{ on } S_m \text{ for all } j \le m$$

Set $E(p) = v$, which is $P(0)$ vacuously. Given E on S_m satisfying $P(m)$, extend it to S_{m+1} as follows: each point of S_{m+1} lies on a unique line segment in the x^{m+1}-direction issuing from a point of S_m, and E is defined on S_{m+1} by parallel transporting its value at that base point along the segment (theorem 2.4.2). This makes

$$\nabla_{\partial_{m+1}} E = 0 \text{ on } S_{m+1} \tag{4.6}$$

by construction, and it remains only to recover the earlier conditions $\nabla_{\partial_j} E = 0$ for $j \leq m$ on the enlarged slab S_{m+1}.

Fix $j \leq m$ and consider the field $F = \nabla_{\partial_j} E$ on S_{m+1}. Its covariant derivative in the ∂_{m+1}-direction is controlled by curvature: since $[\partial_{m+1}, \partial_j] = 0$, the definition of R (definition 4.1.2) gives

$$\nabla_{\partial_{m+1}} F = \nabla_{\partial_{m+1}} \nabla_{\partial_j} E = \nabla_{\partial_j} \nabla_{\partial_{m+1}} E + R(\partial_{m+1}, \partial_j) E$$

The first term on the right vanishes because $\nabla_{\partial_{m+1}} E = 0$ throughout S_{m+1} by (4.6), and the second vanishes because $R \equiv 0$. Hence $\nabla_{\partial_{m+1}} F = 0$: the field F is parallel along each x^{m+1}-segment. At the base point of the segment, on S_m, one has $F = \nabla_{\partial_j} E = 0$ by $P(m)$. By uniqueness of parallel transport (theorem 2.4.2), $F \equiv 0$ along the whole segment, so $\nabla_{\partial_j} E = 0$ on all of S_{m+1}. Together with (4.6) this is $P(m+1)$.

By induction $P(n)$ holds: $\nabla_{\partial_j} E = 0$ on $S_n = W$ for every j. Since $(\partial_1, \ldots, \partial_n)$ frames TW and $\nabla_Y E$ is $C^\infty(W)$-linear in Y, this gives $\nabla_Y E = 0$ for every vector field Y, so E is parallel on W. Carrying out the construction for each $a = 1, \ldots, n$ produces parallel fields $E_1, \ldots, E_n$ with $E_a(p) = e_a$.

Step 2: the frame is orthonormal. Metric compatibility (definition 2.2.4) gives, for any Y,

$$Y \langle E_a, E_b \rangle = \langle \nabla_Y E_a, E_b \rangle + \langle E_a, \nabla_Y E_b \rangle = 0$$

because both E_a and E_b are parallel. Hence each function $\langle E_a, E_b \rangle$ is constant on the (connected) cube, equal to its value $\langle e_a, e_b \rangle = \delta_{ab}$ at p. So $(E_1, \ldots, E_n)$ is orthonormal at every point.

Step 3: the frame commutes and gives Euclidean coordinates. Because ∇ is torsion-free (definition 2.2.1) and each E_a is parallel,

$$[E_a, E_b] = \nabla_{E_a} E_b - \nabla_{E_b} E_a = 0$$

A frame of pairwise-commuting vector fields is, on a possibly smaller neighborhood U of p, the coordinate frame of a chart: by the canonical form for commuting vector fields ([6], the simultaneous-flow-box theorem), there are coordinates $y = (y^1, \ldots, y^n)$ on U with $\partial/\partial y^a = E_a$ for each a. In these coordinates the metric components are

$$g_{ab} = \left\langle \frac{\partial}{\partial y^a}, \frac{\partial}{\partial y^b} \right\rangle = \langle E_a, E_b \rangle = \delta_{ab}$$

constant and equal to the Euclidean values by Step 2. Thus $y\colon U \to y(U) \subset \mathbb{R}^n$ pulls the Euclidean metric $\bar{g} = \delta_{ab}\, dy^a\, dy^b$ back to g, so y is an isometry of (U, g) onto the open subset $y(U)$ of $(\mathbb{R}^n, \bar{g})$ (definition 1.3.1). As p was arbitrary, every point of M has a Euclidean neighborhood, completing the proof.

The theorem isolates curvature as the sole local obstruction to being Euclidean. Two flat manifolds are locally identical no matter how different they look globally: the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ of example 1.3.6 is compact and has closed geodesics, yet each of its points has a neighborhood indistinguishable from a neighborhood in the plane, precisely because it is a quotient of $\mathbb{R}^n$ by isometries and so inherits $R \equiv 0$. The distinction between local and global is therefore genuine, and it is topology, not curvature, that separates the plane from the torus. The same phenomenon organizes the constant-curvature case, treated next: constant sectional curvature is a purely local condition, and it too leaves the global classification to topology.

The parallel-frame argument used only that the direction of R was zero. When the curvature is instead a fixed nonzero constant on every plane, the algebraic content of that constancy must first be extracted. The following definition names the class, and the proposition after it converts the scalar constancy of K into a rigid formula for the full curvature tensor.

Definition 4.5.2: Constant Sectional Curvature

A Riemannian manifold (M, g) has *constant sectional curvature* $c \in \mathbb{R}$ if $K(\Pi) = c$ for every point $p \in M$ and every 2-plane $\Pi \subset T_pM$, where K is the sectional curvature of definition 4.3.1. When M is understood, one says M is a *space form* of curvature c.

Proposition 4.5.3: The Curvature Tensor of a Space of Constant Curvature

Let (M, g) be a Riemannian manifold and $c \in \mathbb{R}$. Then (M, g) has constant sectional curvature c if and only if the curvature endomorphism has the form

$$R(X, Y)Z = c\big(\langle Y, Z\rangle X - \langle X, Z\rangle Y\big) \tag{4.7}$$

for all vector fields X, Y, Z, equivalently

$$\mathrm{Riem}(X, Y, Z, W) = c\big(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle\big) \tag{4.8}$$

***Proof*:**
The equivalence of (4.7) and (4.8) is a matter of pairing with W and using $\mathrm{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle$ (definition 4.1.4); the two are the same statement.

The formula implies constant curvature. Suppose (4.8) holds. For a 2-plane Π spanned by X, Y,

$$\mathrm{Riem}(X, Y, Y, X) = c\big(\langle Y, Y\rangle \langle X, X\rangle - \langle X, Y\rangle \langle Y, X\rangle\big) = c\big(|X|^2 |Y|^2 - \langle X, Y\rangle^2\big)$$

The denominator in the definition of sectional curvature (definition 4.3.1) is exactly $|X|^2 |Y|^2 - \langle X, Y\rangle^2$, which is positive because X, Y are linearly independent (the Cauchy–Schwarz inequality is strict for a nonproportional pair). Dividing gives $K(\Pi) = c$, for every plane at every point.

Constant curvature implies the formula. Suppose $K(\Pi) = c$ for all planes. Define a $(0, 4)$ tensor T by the right-hand side of (4.8):

$$T(X, Y, Z, W) = c\big(\langle Y, Z\rangle \langle X, W\rangle - \langle X, Z\rangle \langle Y, W\rangle\big)$$

A direct check shows T satisfies the algebraic symmetries of a curvature tensor. It is antisymmetric in the first pair (X, Y) and in the last pair (Z, W), symmetric under interchange of the two pairs, and satisfies the first Bianchi identity:

$$T(X, Y, Z, W) + T(Y, Z, X, W) + T(Z, X, Y, W)$$

expands to

$$c\big[(\langle Y,Z\rangle\langle X,W\rangle-\langle X,Z\rangle\langle Y,W\rangle)+(\langle Z,X\rangle\langle Y,W\rangle-\langle Y,X\rangle\langle Z,W\rangle)+(\langle X,Y\rangle\langle Z,W\rangle-\langle Z,Y\rangle\langle X,W\rangle)\big]=0$$

since the six terms cancel in pairs. Thus both Riem (by proposition 4.2.1) and T are algebraic curvature tensors on each T_pM. By the computation in the previous paragraph applied to T in place of Riem, the sectional curvatures of T are also constant equal to c; that is, T and Riem induce the same sectional-curvature function on every 2-plane. By theorem 4.3.3, an algebraic curvature tensor is determined by its sectional curvatures, so Riem $= T$, which is (4.8). Passing back to the endomorphism form gives (4.7), as required.

The proposition is the algebraic heart of the theory of space forms. The condition $K \equiv c$ names one number on each 2-plane, a function on the Grassmannian of planes, and a priori this is far less data than the whole tensor Riem; the polarization result theorem 4.3.3 is what closes the gap, forcing the entire curvature to be the universal expression (4.7). The right-hand side of (4.7) depends on the metric alone once c is fixed, so all local curvature information in a space form is carried by the single scalar c. The three model spaces realize the three signs of c, as the next example records.

Example 4.5.4: The Model Space Forms

The three running model geometries have constant sectional curvature and so satisfy (4.7); they are the canonical representatives of the three signs of c.

Euclidean space $(\mathbb{R}^n, \bar{g})$ (example 1.1.2) has $R \equiv 0$ by example 4.1.6, so it is a space form with $c = 0$, and (4.7) reads $R(X,Y)Z = 0$.

The round sphere S^n_r (example 1.3.4) has $K \equiv 1/r^2$: for S^2 this was computed from the Christoffel symbols in example 4.3.4, and the value extends to all S^n_r by the homogeneity and isotropy recorded in example 1.3.4. Thus for the sphere (4.7) becomes

$$R(X,Y)Z = \frac{1}{r^2}\big(\langle Y,Z\rangle X - \langle X,Z\rangle Y\big)$$

As a check, for orthonormal X, Y this gives $\langle R(X,Y)Y, X\rangle = \frac{1}{r^2}(\langle Y,Y\rangle\langle X,X\rangle - 0) = \frac{1}{r^2}$, matching $K = 1/r^2$.

Hyperbolic space H^n (example 1.3.5) has $K \equiv -1$: for the hyperbolic plane this was computed in example 4.3.5, and again it extends to all dimensions by the homogeneity and isotropy of the model. Here (4.7) reads

$$R(X,Y)Z = -\big(\langle Y,Z\rangle X - \langle X,Z\rangle Y\big)$$

so that $\langle R(X,Y)Y, X\rangle = -1$ on an orthonormal pair, matching $K = -1$. Rescaling the hyperbolic metric by the constant factor $1/(-c)$ for $c < 0$ produces the model of curvature c, in the same way that S^n_r realizes curvature $1/r^2$; the sign of c is the invariant that cannot be scaled away.

The models settle the three signs of c one manifold at a time. The classification theorem asserts that these are the only local possibilities: a space form of curvature c is, near each point, an isometric copy of the model with the same c (after normalizing the scale). The proof produces the local isometry explicitly through geodesic normal coordinates, in which the metric of a constant-curvature manifold is forced into a universal form depending only on c.

Theorem 4.5.5: Local Classification of Space Forms

Let (M, g) be a Riemannian n-manifold of constant sectional curvature c. Then every point of M has a neighborhood isometric to an open subset of the model space of curvature c, namely:

1. Euclidean space $(\mathbb{R}^n, \bar{g})$ if $c = 0$;
2. the round sphere $S^n_{1/\sqrt{c}}$ of radius $1/\sqrt{c}$ if $c > 0$;
3. hyperbolic space H^n rescaled to curvature c if $c < 0$.

In particular any two Riemannian manifolds of the same constant sectional curvature c are locally isometric to each other.

***Proof*:**
The case $c = 0$ is theorem 4.5.1: constant curvature 0 means $K \equiv 0$, hence $R \equiv 0$ by proposition 4.5.3, hence local isometry to $(\mathbb{R}^n, \bar{g})$. Assume $c \neq 0$.

Fix $p \in M$ and let $(x^1, \ldots, x^n)$ be geodesic normal coordinates centered at p on a normal ball B, arising from an orthonormal basis of (T_pM, g_p) (definition 3.2.6). Introduce geodesic polar coordinates (ρ, θ), where $\rho = |x|$ is the geodesic distance from p and θ ranges over the unit sphere $\Sigma = \{u \in T_pM : |u| = 1\}$; a point of the punctured ball is $\exp_p(\rho\,\theta)$. By the Gauss lemma (theorem 3.3.4) and its corollary on the polar form of the metric (proposition 3.3.5), the metric on the punctured normal ball is

$$g = d\rho^2 + h_\rho \tag{4.9}$$

where h_ρ is, for each fixed $\rho > 0$, a metric on the geodesic sphere $\{|x| = \rho\}$, degenerating to the point p as $\rho \to 0$. The strategy is to compute h_ρ from the constant-curvature hypothesis and to recognize the result as the model metric.

Step 1: the radial variation equation. The metric h_ρ is read off from the differential of the exponential map. Fix a unit vector $\theta \in \Sigma$ and let $\gamma(\rho) = \exp_p(\rho\theta)$ be the radial geodesic in direction θ, so $\dot\gamma(\rho) = (d\exp_p)_{\rho\theta}(\theta)$; it has constant speed $|\dot\gamma| \equiv |\theta| = 1$ (proposition 3.1.6). For a tangent vector $w \in T_\theta\Sigma \subset T_pM$ (so $w \perp \theta$, $|w| = 1$), consider the variation of geodesics

$$\alpha(\rho, s) = \exp_p\big(\rho\,\theta(s)\big), \qquad \theta(0) = \theta, \quad \tfrac{d}{ds}\big|_0 \theta(s) = w$$

where $\theta(s)$ is a curve in Σ with the stated initial data. For each fixed s the curve $\rho \mapsto \alpha(\rho, s)$ is a geodesic, and the variation field

$$J(\rho) = \frac{\partial \alpha}{\partial s}\Big|_{s=0}$$

is a vector field along γ measuring the infinitesimal spreading of the radial geodesics. Its value $J(\rho) = (d\exp_p)_{\rho\theta}(\rho w)$ is exactly the image under $d\exp_p$ of the vector ρw tangent to the sphere of radius ρ in T_pM, so the norm $|J(\rho)|$ is the factor by which h_ρ stretches the direction w: for $w_1, w_2 \in T_\theta\Sigma$,

$$h_\rho(w_1, w_2) = \big\langle (d\exp_p)_{\rho\theta}(\rho w_1), (d\exp_p)_{\rho\theta}(\rho w_2) \big\rangle = \langle J_1(\rho), J_2(\rho) \rangle \tag{4.10}$$

with J_i the variation field associated to w_i. By the Gauss lemma $J(\rho) \perp \dot\gamma(\rho)$ for all ρ.

The field J satisfies a second-order equation involving curvature. Let $T = \partial\alpha/\partial\rho$ and $S = \partial\alpha/\partial s$, so along γ (at $s = 0$) one has $T = \dot\gamma$ and $S = J$. Since ∂_ρ and ∂_s are coordinate fields on the parameter rectangle, the symmetry lemma (lemma 3.5.3) gives $D_s T = D_\rho S$, where D_ρ, D_s are the covariant derivatives along the ρ- and s-curves (definition 2.1.6). Each ρ-curve is a geodesic, so $D_\rho T = 0$. The curvature endomorphism governs the failure of D_ρ and D_s to commute on a field along α: with the sign convention of box 4.1.1, and because the parameter fields commute, the definition of R (definition 4.1.2) yields for any field V along α

$$D_\rho D_s V - D_s D_\rho V = R(T, S)V$$

Applying this to $V = T$ and using $D_s T = D_\rho S$ together with $D_\rho T = 0$,

$$D_\rho D_\rho S = D_\rho D_s T = D_s D_\rho T + R(T, S)T = R(T, S)T$$

Evaluated at $s = 0$, where $T = \dot\gamma$ and $S = J$, this is $D_\rho^2 J = R(\dot\gamma, J)\dot\gamma$, equivalently

$$D_\rho^2 J + R(J, \dot\gamma)\dot\gamma = 0 \tag{4.11}$$

by the antisymmetry $R(\dot\gamma, J) = -R(J, \dot\gamma)$ (proposition 4.2.1). This is the radial variation equation, derived here directly from torsion-freeness and the definition of R.

Now insert the constant-curvature form (4.7). Since $|\dot\gamma| = 1$ and $J \perp \dot\gamma$,

$$R(J, \dot\gamma)\dot\gamma = c\big(\langle\dot\gamma, \dot\gamma\rangle\, J - \langle J, \dot\gamma\rangle\, \dot\gamma\big) = c\, J$$

so (4.11) becomes the linear equation $D_\rho^2 J + cJ = 0$. Its initial data are fixed by the behavior of J at p. From $J(\rho) = (d\exp_p)_{\rho\theta}(\rho w)$ one has $J(0) = 0$; and since $J(\rho) = (d\exp_p)_0(\rho w) + O(\rho^2) = \rho w + O(\rho^2)$ by $d(\exp_p)_0 = \mathrm{id}$ (proposition 3.2.5), the covariant initial velocity is $D_\rho J|_0 = w$, a unit vector. Let $e(\rho)$ be the parallel field along γ with $e(0) = w$, obtained by parallel transport of w; it is a unit field ($|e| \equiv |w| = 1$) and stays orthogonal to $\dot\gamma$, because parallel transport preserves inner products (proposition 2.4.4), $\dot\gamma$ is itself parallel, and $\langle w, \dot\gamma(0)\rangle = \langle w, \theta\rangle = 0$. Let ψ solve the scalar initial value problem

$$\psi'' + c\,\psi = 0, \qquad \psi(0) = 0,\ \psi'(0) = 1 \tag{4.12}$$

and set $\tilde J(\rho) = \psi(\rho)\, e(\rho)$. Then $D_\rho^2 \tilde J = \psi'' e = -c\psi e = -c\tilde J$, so $\tilde J$ solves $D_\rho^2 \tilde J + c\tilde J = 0$ with $\tilde J(0) = 0$ and $D_\rho \tilde J|_0 = \psi'(0)e(0) = w$; the same initial value problem as J. By uniqueness of solutions of this linear second-order equation along γ (a linear ODE in components, theorem 2.4.2 providing the parallel frame in which it is a constant-coefficient system), $J = \tilde J = \psi\, e$. The equation (4.12) is independent of the direction θ and of the choice of unit $w \perp \theta$, which is the analytic form of the isotropy forced by constant sectional curvature.

Step 2: solving the radial equation. The solution of (4.12) is

$$\psi(\rho) = \begin{cases} \dfrac{1}{\sqrt{c}} \sin\!\big(\sqrt{c}\,\rho\big) & c > 0 \\[2ex] \dfrac{1}{\sqrt{-c}} \sinh\!\big(\sqrt{-c}\,\rho\big) & c < 0 \end{cases}$$

as is verified by substitution. For $c > 0$, $\psi'(\rho) = \cos(\sqrt{c}\,\rho)$ and $\psi''(\rho) = -\sqrt{c}\,\sin(\sqrt{c}\,\rho) = -c\,\psi$, so $\psi'' + c\psi = 0$, with $\psi(0) = 0$ and $\psi'(0) = 1$; for $c < 0$, $\psi''(\rho) = (-c)\cdot\frac{1}{\sqrt{-c}}\sinh(\sqrt{-c}\,\rho) = (-c)\psi$, that is $\psi'' + c\psi = 0$, again with $\psi(0) = 0$, $\psi'(0) = 1$. The metric h_ρ is now read off on an orthonormal basis. Fix an orthonormal basis $w_1, \dots, w_{n-1}$ of $T_\theta\Sigma$, and let J_i be the variation field with $D_\rho J_i|_0 = w_i$; by Step 1, $J_i(\rho) = \psi(\rho)\, e_i(\rho)$ where e_i is the parallel field with $e_i(0) = w_i$.

Since parallel transport is a linear isometry (proposition 2.4.4) and $e_i(0) = w_i$ are orthonormal, $e_1(\rho), \ldots, e_{n-1}(\rho)$ are orthonormal for every ρ. Hence by (4.10),

$$h_\rho(w_i, w_j) = \langle J_i(\rho), J_j(\rho)\rangle = \psi(\rho)^2 \langle e_i(\rho), e_j(\rho)\rangle = \psi(\rho)^2\, \delta_{ij} = \psi(\rho)^2 \langle w_i, w_j\rangle$$

As both h_ρ and $\psi(\rho)^2 g_\Sigma$ are bilinear forms on $T_\theta\Sigma$ agreeing on a basis, where g_Σ denotes the round metric of radius 1 on the unit sphere $\Sigma \subset T_pM$ (whose value on w_i, w_j is $\langle w_i, w_j\rangle$),

$$h_\rho = \psi(\rho)^2\, g_\Sigma \tag{4.13}$$

Combining (4.9) and (4.13),

$$g = d\rho^2 + \psi(\rho)^2\, g_\Sigma \tag{4.14}$$

Step 3: recognizing the model. The form (4.14) is exactly the geodesic polar form of the model space of curvature c. For $c > 0$, set $r = 1/\sqrt{c}$; then $\psi(\rho) = r\sin(\rho/r)$, and (4.14) becomes

$$g = d\rho^2 + r^2 \sin^2\!\big(\rho/r\big)\, g_\Sigma$$

which is the round metric of S^n_r written in geodesic polar coordinates about a point, as recorded for the sphere in example 1.3.4 and section 3.2. For $c < 0$, set $k = \sqrt{-c}$; then $\psi(\rho) = \frac{1}{k}\sinh(k\rho)$, and

$$g = d\rho^2 + \frac{1}{k^2}\sinh^2\!\big(k\rho\big)\, g_\Sigma$$

which is the metric of hyperbolic space of curvature $c = -k^2$ in geodesic polar coordinates, matching the model of example 1.3.5 after the rescaling by $1/k$. The point is that the model space M_c of curvature c also has constant sectional curvature c, so Steps 1 and 2 apply to it verbatim: about any of its points q_0 its metric takes, in geodesic polar coordinates on a normal ball, the same expression (4.14) with the same radial factor ψ, since ψ is determined by c alone through (4.12).

The isometry is now explicit. Fix a point q_0 of the model M_c and a linear isometry $\iota\colon (T_pM, g_p) \to (T_{q_0}M_c, g_{q_0})$; such an ι exists because both are inner-product spaces of dimension n. On the normal ball B about p, define

$$\Phi = \exp_{q_0} \circ\, \iota \circ (\exp_p)^{-1}$$

which is a diffeomorphism of B onto a normal ball about q_0 (composition of the diffeomorphism $(\exp_p)^{-1}$, the linear isomorphism ι, and the diffeomorphism $\exp_{q_0}$ on the appropriate domains, by proposition 3.2.5). Because ι is a linear isometry, it carries the unit sphere $\Sigma \subset T_pM$ isometrically to the unit sphere in $T_{q_0}M_c$ and preserves the radial coordinate ρ; hence in polar coordinates Φ is $(\rho, \theta) \mapsto (\rho, \iota\theta)$. Both metrics are $d\rho^2 + \psi(\rho)^2 g_\Sigma$ in these coordinates with the identical ψ, and ι pulls back g_Σ on the model unit sphere to g_Σ on Σ, so $\Phi^* g_{M_c} = g$. Thus Φ is an isometry (definition 1.3.1) of (B, g) onto an open subset of the model. As p was arbitrary this gives the local isometry to the model space of curvature c; for two manifolds of the same c the corresponding maps to the common model compose to a local isometry between them, completing the proof.

The classification reduces the local geometry of a constant-curvature manifold to a single scalar and one of three universal metrics. Every such manifold, viewed through a small enough window, is a piece of a sphere, a plane, or a hyperbolic space; there is no further local invariant. The role of the Gauss lemma and the radial variation equation (4.11) is to convert the algebraic rigidity of proposition 4.5.3 into an explicit metric normal form (4.14), whose radial factor ψ satisfies the elementary equation (4.12) and so is completely pinned down by c. The variation field J of Step 1 and its equation (4.11) are the first appearance of a construction taken up systematically under the name of Jacobi fields in a later chapter; here it is derived from scratch, using only the definition of curvature and torsion-freeness, so the argument rests on the tools of the present chapter.

What the theorem does not settle is the global shape. Local isometry to the round sphere does not make a manifold a sphere: the real projective space $\mathbb{RP}^n$ inherits constant curvature 1 from S^n through the antipodal quotient (example 1.2.3), yet it is not homeomorphic to S^n, and the flat torus is locally Euclidean without being $\mathbb{R}^n$. The passage from local to global requires completeness and simple connectivity, and it is the content of the Killing–Hopf theorem, stated next as the natural companion to the local classification.

4.5.6 Foreshadowing: The Killing–Hopf Theorem The local classification theorem 4.5.5 has a global counterpart. A *complete*, *simply connected* Riemannian n-manifold of constant sectional curvature c is (globally) isometric to the model space of curvature c: to the sphere $S^n_{1/\sqrt{c}}$ for $c > 0$, to Euclidean space $\mathbb{R}^n$ for $c = 0$, and to hyperbolic space H^n (rescaled) for $c < 0$. This is the *Killing–Hopf theorem.* A general complete space form of curvature c is then a quotient $\widetilde{M}/\Gamma$ of the corresponding model $\widetilde{M}$ by a discrete group Γ of isometries acting freely and properly discontinuously, so that the classification of complete space forms becomes a problem about such groups; the flat tori $\mathbb{R}^n/\mathbb{Z}^n$ and $\mathbb{RP}^n = S^n/\{\pm 1\}$ are the first instances.
The proof requires two tools not yet available. Completeness, in the sense that $\exp_p$ is defined on all of T_pM, is developed through the Hopf–Rinow theorem, and the globalization of a local isometry into a covering map uses the completeness of both source and model together with a monodromy argument along paths. Both belong to the completeness machinery of a later chapter, and the theorem is invoked there once that machinery is in place. A complete self-contained account may be found in do Carmo, *Riemannian Geometry*, and in John M. Lee, *Introduction to Riemannian Manifolds*, where the Killing–Hopf theorem is proved in full from the local normal form (4.14) established above.

4.6 Curvature of Warped Products

The warped products of section 1.6 were built at the level of the metric tensor, and the promise made there was that their curvature is computable in closed form from the geometry of the two factors and the derivatives of the single warping function. That promise is redeemed here. The point of the construction is not that it produces new manifolds, but that it produces manifolds whose curvature can be read off a one-dimensional differential expression: a warped product $I \times_f N$ over an interval has, in the radial direction, sectional curvature exactly $-f''/f$, so a metric of the form $dr^2 + f(r)^2 g_N$ has constant curvature κ in that direction precisely when f solves the linear equation $f'' + \kappa f = 0$. This is the mechanism, foreshadowed in example 1.6.5, by which the three warping functions r, $\sin r$, and $\sinh r$ deliver the three constant-curvature model spaces; the present section makes it a theorem and then reads the models off it.

The route to the curvature runs through the Levi-Civita connection of the warped product, which is itself determined by the two factor connections and the gradient of f. The connection of Chapter 2 is now available, so the derivation can proceed without deferral: a short lemma computes ∇ on horizontal and vertical vectors, and the curvature theorem then substitutes those formulas into the defining expression $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$ of definition 4.1.2. The sign convention of box 4.1.1 is in force throughout, so that the round sphere carries $K = +1/r^2$ and hyperbolic space $K = -1$; the model computations at the end confirm both signs. The Einstein summation convention is not needed here, since the entire computation is carried out invariantly in terms of vector fields.

Fix, once and for all, a warped product $M \times_f N$ with base (M, g_M), fiber (N, g_N), and smooth warping function $f\colon M \to (0,\infty)$, carrying the metric $g = g_M \oplus f^2 g_N$ of definition 1.6.2. Its Levi-Civita connection is written ∇, and the Levi-Civita connections of the two factors are ∇^M and ∇^N. Two classes of vector field on $M \times N$ organize every computation. A vector field is *horizontal* if it is everywhere tangent to the base slices $M \times \{q\}$, and *vertical* if it is everywhere tangent to the fibers $\{p\} \times N$; by the orthogonal splitting of proposition 1.6.3 these are the g-orthogonal complements of one another, and every vector field is uniquely the sum of a horizontal and a vertical one. A vector field X on M has a unique horizontal lift to $M \times N$, characterized by $d\pi_M\,\tilde{X} = X$ and $d\pi_N\,\tilde{X} = 0$; the lift is again written X where no confusion results, and such lifts are called *basic horizontal* fields. Likewise a vector field on N lifts to a *basic vertical* field. Basic lifts are π_M- and π_N-related to the originals, so their brackets and the metric pairings among them are controlled by the factors: for basic horizontal X, Y (lifts of base fields) and basic vertical V, W (lifts of fiber fields),

$$g(X,Y) = g_M(X,Y)\circ\pi_M, \qquad g(V,W) = (f\circ\pi_M)^2\, g_N(V,W)\circ\pi_N, \qquad g(X,V) = 0$$

and the bracket of a basic horizontal with a basic vertical field vanishes, $[X,V] = 0$, while $[X,Y]$ is the horizontal lift of the base bracket and $[V,W]$ the vertical lift of the fiber bracket. These facts are the standard behaviour of lifts under the projections π_M, π_N ([6]); they are used without further comment below.

One further object enters, the Hessian of the warping function on the base. For a smooth function f on (M, g_M) the gradient $\operatorname{grad} f = (df)^\sharp$ of definition 1.4.3 and the Levi-Civita connection ∇^M combine into the *Hessian*

$$\operatorname{Hess}^M f(X,Y) = \left\langle \nabla^M_X \operatorname{grad} f, Y\right\rangle_{g_M} = X(Yf) - (\nabla^M_X Y)f$$

a symmetric covariant 2-tensor field on M; its metric trace $\Delta_M f = \operatorname{tr}_{g_M}\operatorname{Hess}^M f = \operatorname{div}^M(\operatorname{grad} f)$ is the Laplacian met in definition 1.4.7. The gradient here is the base gradient, so $Xf = \langle \operatorname{grad} f, X\rangle_{g_M}$ for horizontal X, and $|\operatorname{grad} f|^2$ means the base norm $|\operatorname{grad} f|^2_{g_M} = g_M(\operatorname{grad} f, \operatorname{grad} f)$. These are the only invariants of f that appear in the curvature.

Lemma 4.6.1: The connection of a warped product

Let $M \times_f N$ be a warped product with Levi-Civita connection ∇, and let X, Y be basic horizontal fields and V, W basic vertical fields. Then

1. $\nabla_X Y = \nabla^M_X Y$, again basic horizontal;
2. $\nabla_X V = \nabla_V X = \dfrac{Xf}{f}\, V$, vertical;
3. $\nabla_V W = \nabla^N_V W - \dfrac{g(V,W)}{f}\, \operatorname{grad} f$, whose vertical part is the fiber connection $\nabla^N_V W$ and whose horizontal part is $-\, g_N(V,W)\, f\, \operatorname{grad} f$.

Here ∇^M and ∇^N are the Levi-Civita connections of the factors, $Xf = \langle \operatorname{grad} f, X\rangle_{g_M}$, and $\operatorname{grad} f$ is the base gradient, lifted horizontally.

***Proof*:**
The Levi-Civita connection is characterized by the Koszul formula (2.3) of theorem 2.3.1: for vector fields A, B, C,

$$2\langle \nabla_A B, C\rangle = A\langle B, C\rangle + B\langle A, C\rangle - C\langle A, B\rangle + \langle [A,B], C\rangle - \langle [A,C], B\rangle - \langle [B,C], A\rangle$$

where all inner products are those of g. Each claim is proved by evaluating $2\langle \nabla_A B, C\rangle$ from this formula as C ranges over horizontal and vertical fields separately, using the pairings and brackets recorded above; two vectors with the same inner product against every horizontal and every vertical field are equal, since these span the tangent space g-orthogonally.

1. Take $A = X$, $B = Y$ horizontal. Test first against a horizontal $C = Z$. Every term of the Koszul formula then involves only horizontal fields and the function $g(-,-) = g_M(-,-) \circ \pi_M$, and the brackets $[X,Y], [X,Z], [Y,Z]$ are horizontal lifts of the corresponding base brackets. The formula is therefore literally the base Koszul formula for $2\, g_M(\nabla^M_X Y, Z)$ pulled back to $M \times N$, so

   $$2\langle \nabla_X Y, Z\rangle = 2\, g_M(\nabla^M_X Y, Z) \circ \pi_M = 2\left\langle \nabla^M_X Y, Z\right\rangle$$

   Now test against a vertical $C = V$. Then $\langle X, V\rangle = \langle Y, V\rangle = 0$, so the first three terms vanish after noting that $X\langle Y, V\rangle$, $Y\langle X, V\rangle$, and $V\langle X, Y\rangle$ are derivatives of a horizontal pairing: the first two vanish because $\langle Y, V\rangle = \langle X, V\rangle = 0$ identically, and the third is $V(g_M(X,Y)\circ\pi_M) = 0$ because $g_M(X,Y)\circ\pi_M$ is constant along fibers, being a function of the base point alone, and V is vertical. Of the bracket terms, $\langle [X,Y], V\rangle = 0$ since $[X,Y]$ is horizontal, while $[X,V] = [Y,V] = 0$, so $\langle [X,V], Y\rangle = \langle [Y,V], X\rangle = 0$. Hence $2\langle \nabla_X Y, V\rangle = 0$ for every vertical V. Thus $\nabla_X Y$ has the horizontal value $\nabla^M_X Y$ and no vertical part, which is claim (1).

2. Take $A = X$ horizontal, $B = V$ vertical. First test against a horizontal $C = Z$. The Koszul terms are

   $$X\langle V, Z\rangle + V\langle X, Z\rangle - Z\langle X, V\rangle + \langle [X,V], Z\rangle - \langle [X,Z], V\rangle - \langle [V,Z], X\rangle$$

Here $\langle V, Z\rangle = \langle X, V\rangle = 0$, so the first and third terms vanish; $[X, V] = 0$; $[X, Z]$ is horizontal so $\langle [X, Z], V\rangle = 0$; and $[V, Z] = -[Z, V] = 0$ since Z is horizontal and V vertical, so the last term vanishes. The surviving term is $V\langle X, Z\rangle$, and $\langle X, Z\rangle = g_M(X, Z) \circ \pi_M$ is constant along fibers, so $V\langle X, Z\rangle = 0$. Therefore $2\langle \nabla_X V, Z\rangle = 0$ for all horizontal Z: the field $\nabla_X V$ is vertical. Now test against a vertical $C = W$. The Koszul terms are

$$X\langle V, W\rangle + V\langle X, W\rangle - W\langle X, V\rangle + \langle [X, V], W\rangle - \langle [X, W], V\rangle - \langle [V, W], X\rangle$$

The terms with $\langle X, W\rangle = \langle X, V\rangle = 0$ drop, as do $\langle [X, V], W\rangle = 0$ (since $[X, V] = 0$) and $\langle [X, W], V\rangle = 0$ (since $[X, W] = 0$); and $\langle [V, W], X\rangle = 0$ because $[V, W]$ is vertical. There remains $X\langle V, W\rangle$. By the fiber pairing $\langle V, W\rangle = f^2\, g_N(V, W) \circ \pi_N$, and $g_N(V, W) \circ \pi_N$ is constant along the base direction while f^2 is a function of the base, so

$$X\langle V, W\rangle = \left(Xf^2\right) g_N(V, W) \circ \pi_N = 2f\,(Xf)\, g_N(V, W) \circ \pi_N = \frac{2\,Xf}{f}\,\langle V, W\rangle$$

Comparing, $2\langle \nabla_X V, W\rangle = \dfrac{2\,Xf}{f}\,\langle V, W\rangle$ for every vertical W, and since $\nabla_X V$ is vertical this forces $\nabla_X V = \dfrac{Xf}{f}\,V$. Because X and V are basic and $[X, V] = 0$, the connection is symmetric there: $\nabla_V X - \nabla_X V = [V, X] = 0$, so $\nabla_V X = \nabla_X V$, which is claim (2).

3. Take $A = V$, $B = W$ vertical. Test first against a vertical $C = U$. The bracket $[V, U], [W, U]$ and $[V, W]$ are all vertical, and the pairings $\langle V, W\rangle, \langle V, U\rangle, \langle W, U\rangle$ are $f^2\, g_N(-,-) \circ \pi_N$. Because f^2 is constant along fibers, a vertical field differentiates it to zero, so for the three derivative terms

$$V\langle W, U\rangle + W\langle V, U\rangle - U\langle V, W\rangle = f^2\big(V\,g_N(W, U) + W\,g_N(V, U) - U\,g_N(V, W)\big) \circ \pi_N$$

and, combined with the three vertical bracket terms, the whole right side is f^2 times the fiber Koszul formula for $2\,g_N(\nabla^N_V W, U)$. Thus

$$2\langle \nabla_V W, U\rangle = f^2 \cdot 2\,g_N(\nabla^N_V W, U) \circ \pi_N = 2\left\langle \nabla^N_V W, U\right\rangle$$

the last equality by the vertical pairing again. So the vertical part of $\nabla_V W$ is $\nabla^N_V W$. Now test against a horizontal $C = X$. The Koszul terms are

$$V\langle W, X\rangle + W\langle V, X\rangle - X\langle V, W\rangle + \langle [V, W], X\rangle - \langle [V, X], W\rangle - \langle [W, X], V\rangle$$

The pairings $\langle W, X\rangle = \langle V, X\rangle = 0$ kill the first two terms; $[V, W]$ is vertical so $\langle [V, W], X\rangle = 0$; and $[V, X] = [W, X] = 0$ kill the last two. There remains $-X\langle V, W\rangle$, and by the computation of part (2),

$$-X\langle V, W\rangle = -\frac{2\,Xf}{f}\,\langle V, W\rangle = -2\,(Xf)\,f\,g_N(V, W) \circ \pi_N$$

Writing $Xf = \langle \operatorname{grad} f, X\rangle_{g_M} = \langle \operatorname{grad} f, X\rangle$ (the base and warped pairings agree on horizontal vectors), this is $-2\,g_N(V, W)\,f\,\langle \operatorname{grad} f, X\rangle$. Hence $2\langle \nabla_V W, X\rangle = -2\,g_N(V, W)\,f\,\langle \operatorname{grad} f, X\rangle$ for all horizontal X, so the horizontal part of $\nabla_V W$ is $-\,g_N(V, W)\,f\,\operatorname{grad} f$. Using $g(V, W) = f^2 g_N(V, W)$ this horizontal part is $-\dfrac{g(V, W)}{f}\,\operatorname{grad} f$, and combining with the vertical part gives

$$\nabla_V W = \nabla^N_V W - \frac{g(V, W)}{f}\,\operatorname{grad} f$$

which is claim (3), as we sought to show.

The three formulas say something geometric about how the fiber sits inside the warped product. The base slices are totally geodesic: by (1) the connection of $M \times_f N$ restricted to horizontal fields is exactly the base connection, so a geodesic of the base, lifted into a slice, remains a geodesic of the whole. The fibers, by contrast, are generally not totally geodesic; the horizontal correction $-\frac{g(V,W)}{f} \operatorname{grad} f$ in (3) is the second fundamental form of the fiber, and it vanishes exactly where $\operatorname{grad} f = 0$, that is where the warping is momentarily stationary. Formula (2) is the one that will drive the curvature: it records that transporting a vertical vector in a horizontal direction rescales it by the logarithmic derivative Xf/f of the warping, the infinitesimal version of the fact that fibers over nearby base points are rescaled copies of one another.

With the connection in hand the curvature is a substitution. The next theorem is the O'Neill formula for the sectional curvatures of a warped product, organized by the three types of tangent 2-plane: a plane spanned by two horizontal vectors, a mixed plane spanned by one horizontal and one vertical vector, and a plane spanned by two vertical vectors.

Theorem 4.6.2: Curvature of a warped product

Let $M \times_f N$ be a warped product with the sign convention of box 4.1.1, and write K, K^M, K^N for the sectional curvatures of $M \times_f N$, of the base, and of the fiber respectively. Let X, Y be g-orthonormal horizontal vectors and V, W be g-orthonormal vertical vectors at a point.

1. *(Base planes.)* The plane $\operatorname{span}(X, Y)$ has
$$K(X, Y) = K^M(X, Y)$$
the base sectional curvature of the corresponding base 2-plane.
2. *(Mixed planes.)* The plane $\operatorname{span}(X, V)$ has
$$K(X, V) = -\frac{\operatorname{Hess}^M f(X, X)}{f}$$
independent of V; here $\operatorname{Hess}^M f$ is the base Hessian and X is unit.
3. *(Fiber planes.)* The plane $\operatorname{span}(V, W)$ has
$$K(V, W) = \frac{K^N(\bar{V}, \bar{W}) - |\operatorname{grad} f|^2}{f^2}$$
where $\bar{V}, \bar{W}$ are the images of V, W in the fiber and $|\operatorname{grad} f|^2 = g_M(\operatorname{grad} f, \operatorname{grad} f)$ is the squared base gradient; $K^N(\bar{V}, \bar{W})$ is the sectional curvature of the unscaled fiber (N, g_N).

Consequently the Ricci curvature, with $d = \dim M$ and $m = \dim N$, satisfies, for X, Y horizontal and V, W vertical,

$$\operatorname{Ric}(X, Y) = \operatorname{Ric}^M(X, Y) - \frac{m}{f} \operatorname{Hess}^M f(X, Y), \qquad \operatorname{Ric}(X, V) = 0$$
$$\operatorname{Ric}(V, W) = \operatorname{Ric}^N(\bar{V}, \bar{W}) - g(V, W) \left(\frac{\Delta_M f}{f} + (m-1) \frac{|\operatorname{grad} f|^2}{f^2} \right)$$

where $\operatorname{Ric}^M, \operatorname{Ric}^N$ are the factor Ricci tensors and $\Delta_M f = \operatorname{tr}_{g_M} \operatorname{Hess}^M f$. In particular, for a one-dimensional base interval I with the metric $g = dr^2 + f(r)^2 g_N$, a plane containing the radial direction ∂_r has sectional curvature

$$K = -\frac{f''}{f}$$

and a plane tangent to the fiber has $K = \dfrac{K^N - (f')^2}{f^2}$.

***Proof*:**
It suffices to compute the $(0,4)$ curvature $\mathrm{Riem}(A,B,B,A) = \langle R(A,B)B, A\rangle$ of definition 4.1.4 on basic fields whose values at the point are the given orthonormal vectors, since for a g-orthonormal pair the sectional curvature of definition 4.3.1 is $K(A,B) = \mathrm{Riem}(A,B,B,A)$, the denominator $|A|^2\,|B|^2 - \langle A,B\rangle^2$ being 1. Extend X,Y to basic horizontal fields and V,W to basic vertical fields; because curvature is tensorial (proposition 4.1.3), the value at the point depends only on the values there. Throughout, R is the curvature endomorphism of definition 4.1.2, and the connection formulas are those of lemma 4.6.1. It is convenient to abbreviate the logarithmic derivative Xf/f and to record that $\operatorname{grad} f$ is horizontal, so $g(\operatorname{grad} f, V) = 0$ for vertical V.

Step 1: base planes. Let X,Y be basic horizontal. By lemma 4.6.1(1) each of $\nabla_X Y$, $\nabla_Y X$ is horizontal and equals the base covariant derivative, and $[X,Y]$ is a basic horizontal field, so $\nabla_{[X,Y]}Y$ is again the base covariant derivative by (1). Every term in $R(X,Y)Y = \nabla_X\nabla_Y Y - \nabla_Y\nabla_X Y - \nabla_{[X,Y]}Y$ is thus computed entirely within the horizontal fields by the base connection, and reproduces the base curvature endomorphism $R^M(X,Y)Y$ lifted horizontally. Pairing with the horizontal X and using $\langle -,-\rangle = g_M(-,-)\circ\pi_M$ on horizontal fields,

$$\mathrm{Riem}(X,Y,Y,X) = \langle R(X,Y)Y, X\rangle = g_M\big(R^M(X,Y)Y, X\big)\circ\pi_M = \mathrm{Riem}^M(X,Y,Y,X)$$

For g-orthonormal X,Y (which are then g_M-orthonormal, the two metrics agreeing on horizontal vectors) this is $K(X,Y) = K^M(X,Y)$, establishing (1).

Step 2: mixed planes. Let X be basic horizontal and V basic vertical, with $[X,V]=0$. Compute $R(X,V)V = \nabla_X\nabla_V V - \nabla_V\nabla_X V$ (the bracket term drops). For the first term, by lemma 4.6.1(3),

$$\nabla_V V = \nabla^N_V V - \frac{g(V,V)}{f}\operatorname{grad} f$$

Differentiate horizontally. The vertical part $\nabla^N_V V$ is a basic vertical field, so by (2) $\nabla_X(\nabla^N_V V) = \dfrac{Xf}{f}\nabla^N_V V$, which is vertical. The horizontal part is $-\dfrac{g(V,V)}{f}\operatorname{grad} f$; since V is basic, $g(V,V) = f^2 g_N(V,V)$ with $g_N(V,V)$ constant along the base, so $\dfrac{g(V,V)}{f} = f\,g_N(V,V)$ and

$$\nabla_X\left(-\frac{g(V,V)}{f}\operatorname{grad} f\right) = -g_N(V,V)\,\nabla_X\big(f\,\operatorname{grad} f\big) = -g_N(V,V)\big((Xf)\operatorname{grad} f + f\,\nabla^M_X \operatorname{grad} f\big)$$

using that on horizontal fields ∇ is the base connection (part (1)) and that $\operatorname{grad} f$ is a basic horizontal field. Only the horizontal component of $R(X,V)V$ contributes to $\langle R(X,V)V, X\rangle$, so the vertical pieces may be discarded from here on. The first term of $R(X,V)V$ contributes the horizontal part

$$\big(\nabla_X\nabla_V V\big)^{\mathrm{h}} = -g_N(V,V)\big((Xf)\operatorname{grad} f + f\,\nabla^M_X\operatorname{grad} f\big)$$

For the second term, $\nabla_X V = \dfrac{Xf}{f}V$ by lemma 4.6.1(2), a vertical field, so applying (3),

$$\nabla_V\left(\frac{Xf}{f}V\right) = \frac{Xf}{f}\,\nabla_V V = \frac{Xf}{f}\left(\nabla^N_V V - \frac{g(V,V)}{f}\operatorname{grad} f\right)$$

the coefficient Xf/f being a function of the base and hence constant along the fiber, so $V(Xf/f) = 0$. Its horizontal part is $-\dfrac{Xf}{f}\cdot\dfrac{g(V,V)}{f}\operatorname{grad} f = -\,(Xf)\,g_N(V,V)\operatorname{grad} f$. Subtracting,

$$\big(R(X,V)V\big)^{\mathrm{h}} = -g_N(V,V)\big((Xf)\operatorname{grad} f + f\,\nabla^M_X \operatorname{grad} f\big) + (Xf)\,g_N(V,V)\operatorname{grad} f = -g_N(V,V)\,f\,\nabla^M_X \operatorname{grad} f$$

the two $(Xf)\operatorname{grad} f$ terms cancelling. Pairing with the horizontal X, and using $\langle \nabla^M_X \operatorname{grad} f, X\rangle = \operatorname{Hess}^M f(X,X)$ and $g(V,V) = f^2 g_N(V,V)$,

$$\operatorname{Riem}(X,V,V,X) = \langle R(X,V)V, X\rangle = -g_N(V,V)\,f\ \operatorname{Hess}^M f(X,X) = -\frac{g(V,V)}{f}\ \operatorname{Hess}^M f(X,X)$$

For g-orthonormal X, V this is $g(V,V) = 1$ and $|X| = 1$, giving $K(X,V) = -\dfrac{\operatorname{Hess}^M f(X,X)}{f}$, independent of the choice of unit vertical V, which is (2).

Step 3: fiber planes. Let V, W be basic vertical with $[V,W]$ vertical (the vertical lift of the fiber bracket). Compute

$$R(V,W)W = \nabla_V\nabla_W W - \nabla_W\nabla_V W - \nabla_{[V,W]}W$$

and take the vertical part, which is what pairs with the vertical V. By lemma 4.6.1(3), $\nabla_W W = \nabla^N_W W - \dfrac{g(W,W)}{f}\operatorname{grad} f$. Apply ∇_V. The vertical field $\nabla^N_W W$ differentiates by (3) to $\nabla_V\nabla^N_W W = \nabla^N_V\nabla^N_W W - \dfrac{g(V,\nabla^N_W W)}{f}\operatorname{grad} f$, whose vertical part is $\nabla^N_V\nabla^N_W W$. The horizontal field $-\dfrac{g(W,W)}{f}\operatorname{grad} f = -g_N(W,W)\,f\,\operatorname{grad} f$ differentiates by (2), since $\operatorname{grad} f$ is basic horizontal and $g_N(W,W)$ is constant along the fiber:

$$\nabla_V(-g_N(W,W)\,f\,\operatorname{grad} f) = -g_N(W,W)\,f\,\nabla_V(\operatorname{grad} f) = -g_N(W,W)\,f\cdot\frac{(\operatorname{grad} f)f}{f}\,V = -g_N(W,W)\ |\operatorname{grad} f|^2\ V$$

where $(\operatorname{grad} f)f = \langle \operatorname{grad} f, \operatorname{grad} f\rangle = |\operatorname{grad} f|^2$ and the factor of f cancels the $1/f$ from (2). This term is already vertical. Hence

$$\big(\nabla_V\nabla_W W\big)^{\mathrm{v}} = \nabla^N_V\nabla^N_W W - g_N(W,W)\ |\operatorname{grad} f|^2\ V$$

By the same computation with V, W exchanged in the appropriate slots,

$$\big(\nabla_W\nabla_V W\big)^{\mathrm{v}} = \nabla^N_W\nabla^N_V W - g_N(V,W)\ |\operatorname{grad} f|^2\ W$$

and the bracket term, $[V,W]$ being vertical, gives by (3) the vertical part $\nabla_{[V,W]}W = \nabla^N_{[V,W]}W - \dfrac{g([V,W],W)}{f}\operatorname{grad} f$, whose vertical part is $\nabla^N_{[V,W]}W$. Assembling the vertical part of $R(V,W)W$,

$$\big(R(V,W)W\big)^{\mathrm{v}} = \underbrace{\nabla^N_V\nabla^N_W W - \nabla^N_W\nabla^N_V W - \nabla^N_{[V,W]}W}_{=R^N(V,W)W} - |\operatorname{grad} f|^2\,\big(g_N(W,W)V - g_N(V,W)W\big)$$

The underbraced part is the fiber curvature endomorphism $R^N(\bar V,\bar W)\bar W$, lifted vertically. Pair with V. The metric pairing of vertical fields is $\langle -,-\rangle = f^2 g_N(-,-)$, so

$$\big\langle R^N(V,W)W, V\big\rangle = f^2\,g_N\big(R^N(\bar V,\bar W)\bar W,\bar V\big) = f^2\ \operatorname{Riem}^N(\bar V,\bar W,\bar W,\bar V)$$

and

$$\Big\langle -|\text{grad}\, f|^2 \left(g_N(W,W)V - g_N(V,W)W\right), V\Big\rangle = -|\text{grad}\, f|^2\ f^2\big(g_N(W,W)g_N(V,V) - g_N(V,W)^2\big)$$

Therefore

$$\text{Riem}(V,W,W,V) = f^2\,\text{Riem}^N(\bar V,\bar W,\bar W,\bar V) - |\text{grad}\, f|^2\, f^2\big(g_N(V,V)g_N(W,W) - g_N(V,W)^2\big)$$

Now take V,W to be g-orthonormal. Then $f^2 g_N(V,V) = f^2 g_N(W,W) = 1$ and $g_N(V,W) = 0$, so $g_N(V,V) = g_N(W,W) = 1/f^2$ and the Gram determinant $g_N(V,V)g_N(W,W) - g_N(V,W)^2 = 1/f^4$. The fiber vectors $\bar V, \bar W$ have g_N-Gram determinant $1/f^4$ as well, so

$$\text{Riem}^N(\bar V,\bar W,\bar W,\bar V) = K^N(\bar V,\bar W)\cdot \frac{1}{f^4}$$

by the definition of fiber sectional curvature (definition 4.3.1). Substituting,

$$K(V,W) = \text{Riem}(V,W,W,V) = f^2\cdot\frac{K^N(\bar V,\bar W)}{f^4} - |\text{grad}\, f|^2\, f^2\cdot\frac{1}{f^4} = \frac{K^N(\bar V,\bar W) - |\text{grad}\, f|^2}{f^2}$$

which is (3).

Step 4: the Ricci curvature. By definition $\text{Ric}(A,A) = \sum_i \langle R(e_i,A)A, e_i\rangle$ over a g-orthonormal frame $\{e_i\}$ (definition 4.4.1), and it is convenient to take a frame adapted to the splitting: d horizontal unit fields $X_1,\dots,X_d$ forming a g_M-orthonormal base frame, and m vertical unit fields $E_1,\dots,E_m$ forming a g-orthonormal fiber frame, so that $g(E_\alpha,E_\beta) = \delta_{\alpha\beta}$ and the images $\bar E_\alpha$ are g_N-orthonormal up to the scaling $g_N(\bar E_\alpha,\bar E_\beta) = \delta_{\alpha\beta}/f^2$. For X horizontal unit,

$$\text{Ric}(X,X) = \sum_{i=1}^{d}\text{Riem}(X_i,X,X,X_i) + \sum_{\alpha=1}^{m}\text{Riem}(E_\alpha,X,X,E_\alpha)$$

The horizontal terms sum by Step 1 to the base Ricci curvature $\text{Ric}^M(X,X)$. Each mixed term is, by the $(0,4)$ identity of Step 2 (with the roles of the two arguments arranged so that the horizontal X is the differentiated direction), $\text{Riem}(E_\alpha,X,X,E_\alpha) = -\dfrac{g(E_\alpha,E_\alpha)}{f}\,\text{Hess}^M f(X,X) = -\dfrac{1}{f}\,\text{Hess}^M f(X,X)$, using the symmetry $\text{Riem}(E_\alpha,X,X,E_\alpha) = \text{Riem}(X,E_\alpha,E_\alpha,X)$ of proposition 4.2.1. There are m such terms, so

$$\text{Ric}(X,X) = \text{Ric}^M(X,X) - \frac{m}{f}\,\text{Hess}^M f(X,X)$$

and polarization in X gives the stated horizontal formula for $\text{Ric}(X,Y)$. The mixed component $\text{Ric}(X,V)$ vanishes, but this needs the curvature endomorphisms $R(e_i,X)V$ that Steps 1–3 did not compute, so it is verified directly. Take X horizontal and V vertical, both basic and unit, and evaluate

$$\text{Ric}(X,V) = \sum_{i=1}^{d}\text{Riem}(X_i,X,V,X_i) + \sum_{\alpha=1}^{m}\text{Riem}(E_\alpha,X,V,E_\alpha)$$

where $\text{Riem}(e_i,X,V,e_i) = \langle R(e_i,X)V, e_i\rangle$, over the same adapted frame. Consider first a horizontal frame vector X_j. Since X_j and X are basic horizontal and V basic vertical, and

$\nabla_A V = \dfrac{Af}{f} V$ for horizontal A by lemma 4.6.1(2), each covariant derivative appearing in $R(X_j, X)V = \nabla_{X_j} \nabla_X V - \nabla_X \nabla_{X_j} V - \nabla_{[X_j,X]} V$ is vertical: indeed $\nabla_X V = \dfrac{Xf}{f} V$ is vertical, and differentiating a vertical field in the horizontal direction X_j keeps it vertical by (2), as does the bracket term with the horizontal $[X_j, X]$. Hence $R(X_j, X)V$ is vertical and pairs to zero with the horizontal X_j:

$$\operatorname{Riem}(X_j, X, V, X_j) = \langle R(X_j, X)V, X_j \rangle = 0$$

for every j, so the horizontal sum vanishes. Now consider a vertical frame vector E_α. Here $[E_\alpha, X] = 0$, so $R(E_\alpha, X)V = \nabla_{E_\alpha} \nabla_X V - \nabla_X \nabla_{E_\alpha} V$, and only the vertical part of this contributes to the pairing with the vertical E_α. For the first term, $\nabla_X V = \dfrac{Xf}{f} V$ by (2), and Xf/f is a function of the base and hence constant along the fiber, so $E_\alpha(Xf/f) = 0$ and

$$\nabla_{E_\alpha} \nabla_X V = \frac{Xf}{f} \nabla_{E_\alpha} V = \frac{Xf}{f} \left(\nabla^N_{E_\alpha} V - \frac{g(E_\alpha, V)}{f} \operatorname{grad} f \right)$$

by lemma 4.6.1(3), whose vertical part is $\dfrac{Xf}{f} \nabla^N_{E_\alpha} V$. For the second term, applying (3) to $\nabla_{E_\alpha} V = \nabla^N_{E_\alpha} V - \dfrac{g(E_\alpha, V)}{f} \operatorname{grad} f$ and then ∇_X: the vertical field $\nabla^N_{E_\alpha} V$ is basic, so by (2) $\nabla_X (\nabla^N_{E_\alpha} V) = \dfrac{Xf}{f} \nabla^N_{E_\alpha} V$, which is vertical, while the horizontal field $-\dfrac{g(E_\alpha, V)}{f} \operatorname{grad} f = -g_N(E_\alpha, V)\, f \operatorname{grad} f$ differentiates by part (1) to $-g_N(E_\alpha, V)\big((Xf) \operatorname{grad} f + f\, \nabla^M_X \operatorname{grad} f\big)$, which is horizontal. So the vertical part of $\nabla_X \nabla_{E_\alpha} V$ is again $\dfrac{Xf}{f} \nabla^N_{E_\alpha} V$. Subtracting, the vertical part of $R(E_\alpha, X)V$ is

$$\frac{Xf}{f} \nabla^N_{E_\alpha} V - \frac{Xf}{f} \nabla^N_{E_\alpha} V = 0$$

so $R(E_\alpha, X)V$ has no vertical component and pairs to zero with E_α:

$$\operatorname{Riem}(E_\alpha, X, V, E_\alpha) = \langle R(E_\alpha, X)V, E_\alpha \rangle = 0$$

for every α. Both sums vanish, so $\operatorname{Ric}(X, V) = 0$, and by bilinearity the mixed Ricci component vanishes on all horizontal-vertical pairs. For V vertical unit,

$$\operatorname{Ric}(V, V) = \sum_{i=1}^{d} \operatorname{Riem}(X_i, V, V, X_i) + \sum_{\alpha=1}^{m} \operatorname{Riem}(E_\alpha, V, V, E_\alpha)$$

Each horizontal term is a mixed plane, equal by Step 2 to $-\dfrac{1}{f} \operatorname{Hess}^M f(X_i, X_i)$, and summing over the g_M-orthonormal base frame gives $-\dfrac{1}{f} \operatorname{tr}_{g_M} \operatorname{Hess}^M f = -\dfrac{\Delta_M f}{f}$. Each fiber term with α such that $E_\alpha \neq V$ is a fiber plane, equal by Step 3 to $\dfrac{K^N(\bar{E}_\alpha, \bar{V}) - |\operatorname{grad} f|^2}{f^2}$ (the term $E_\alpha = V$ contributes nothing, being $\operatorname{Riem}(V, V, V, V) = 0$). Summing over the $m-1$ fiber directions orthogonal to V, the sectional curvatures K^N assemble into the fiber Ricci curvature $\mathrm{Ric}^N(\bar{V},\bar{V})$ evaluated on the unit fiber vector $\bar{V}$ (which for the scaled fiber frame equals the intrinsic $\mathrm{Ric}^N(\bar{V},\bar{V})$ since Ricci is a sum of sectional curvatures over an orthonormal complement), and the $-\left|\mathrm{grad}\, f\right|^2/f^2$ contributes $m-1$ times. Collecting,

$$\mathrm{Ric}(V,V) = \mathrm{Ric}^N(\bar{V},\bar{V}) - \frac{\Delta_M f}{f} - (m-1)\frac{\left|\mathrm{grad}\, f\right|^2}{f^2}$$

and since $g(V,V)=1$ this is the stated formula, polarizing to general vertical V,W.

Step 5: the one-dimensional base. When $M=I$ is an interval with $g_M = dr^2$, the unit horizontal field is ∂_r, the base is flat so $K^M=0$, and $\mathrm{grad}\, f = f'\,\partial_r$ with $\left|\mathrm{grad}\, f\right|^2 = (f')^2$. The base Hessian of a function on a 1-manifold with the flat metric is $\mathrm{Hess}^M f(\partial_r,\partial_r) = \partial_r(\partial_r f) - (\nabla^M_{\partial_r}\partial_r)f = f''$, since $\nabla^M_{\partial_r}\partial_r = 0$ for the flat metric dr^2. A plane containing ∂_r is a mixed plane, so by (2) its sectional curvature is

$$K = -\frac{\mathrm{Hess}^M f(\partial_r,\partial_r)}{f} = -\frac{f''}{f}$$

and a plane tangent to the fiber is a fiber plane, so by (3) its sectional curvature is $\dfrac{K^N-(f')^2}{f^2}$, completing the proof.

The theorem separates the curvature of a warped product into three mechanisms. Base planes see only the base: warping the fiber does not change how the base is curved among itself, consistent with the base slices being totally geodesic. Mixed planes see only the second derivative of the warping, through the base Hessian, and are blind to the fiber: bending the fiber sizes creates curvature in the directions that couple base and fiber, and the sign is opposite to that of $\mathrm{Hess}^M f$, so a warping function that is concave along a base geodesic produces positive mixed curvature there. Fiber planes see a competition between the intrinsic fiber curvature K^N and the steepness $\left|\mathrm{grad}\, f\right|^2$ of the warping: even a flat fiber acquires nonzero curvature once f has nonzero gradient, and a fiber of positive curvature can be flattened, or driven negative, by a sufficiently steep warping. The one-dimensional specialization $K=-f''/f$ is the case that recovers the models, and it is the reason the constant curvature condition is the linear equation $f''+\kappa f=0$ announced in example 1.6.5.

Example 4.6.3: The model spaces from the warped-product formula

The three constant-curvature models were presented in example 1.6.4 and example 1.6.5 as warped products $(0,\cdot)\times_f S^{n-1}$ over a half-line, with unit-sphere fibers and metric $g = dr^2 + f(r)^2 g_{S^{n-1}}$, distinguished by their warping functions $f(r) = r$, $\sin r$, $\sinh r$. Their sectional curvatures follow from the one-dimensional case of theorem 4.6.2, and reproduce the values computed independently in example 4.3.4 and example 4.3.5. Here $\dim M = 1$, so the only plane types are the radial (mixed) planes containing ∂_r and the tangential (fiber) planes; there are no base planes.

Radial planes. By the one-dimensional formula, a plane containing ∂_r has $K=-f''/f$. For the three warpings,

$$f(r)=r\colon\quad -\frac{f''}{f} = -\frac{0}{r} = 0,\qquad f(r)=\sin r\colon\quad -\frac{f''}{f} = -\frac{-\sin r}{\sin r} = 1,$$

$$f(r)=\sinh r\colon\quad -\frac{f''}{f} = -\frac{\sinh r}{\sinh r} = -1$$

so the radial sectional curvatures are 0, $+1$, and -1 for the Euclidean, spherical, and hyperbolic models respectively.

Tangential planes. The fiber is the unit sphere S^{n-1}, which by example 4.3.4 has constant sectional curvature $K^N = 1$. The fiber formula gives $K = \dfrac{K^N - (f')^2}{f^2} = \dfrac{1-(f')^2}{f^2}$. For the three warpings, using $f' = 1$, $\cos r$, $\cosh r$ and the identities $1-\cos^2 r = \sin^2 r$, $1-\cosh^2 r = -\sinh^2 r$,

$$f(r) = r\colon \quad \frac{1-1}{r^2} = 0, \qquad f(r) = \sin r\colon \quad \frac{1-\cos^2 r}{\sin^2 r} = \frac{\sin^2 r}{\sin^2 r} = 1,$$

$$f(r) = \sinh r\colon \quad \frac{1-\cosh^2 r}{\sinh^2 r} = \frac{-\sinh^2 r}{\sinh^2 r} = -1$$

so the tangential sectional curvatures are again 0, $+1$, and -1. In each model the radial and tangential curvatures coincide, so the sectional curvature is constant over all 2-planes at every point: $K \equiv 0$ for $\mathbb{R}^n = (0,\infty) \times_r S^{n-1}$, $K \equiv +1$ for the unit sphere $S^n = (0,\pi) \times_{\sin r} S^{n-1}$, and $K \equiv -1$ for hyperbolic space $H^n = (0,\infty) \times_{\sinh r} S^{n-1}$. This matches the sign convention of box 4.1.1, under which S^n has $K = +1$ and H^n has $K = -1$, and reproduces example 4.3.4 and example 4.3.5 from a single formula.

The three warpings are exactly the solutions of $f'' + \kappa f = 0$ with $f(0) = 0$ and $f'(0) = 1$ for $\kappa = 0, 1, -1$, as anticipated in example 1.6.5. The radial curvature $-f''/f = \kappa$ is constant by construction; that the tangential curvature is also κ is the content of the identity $(1-(f')^2)/f^2 = \kappa$, which follows by differentiating the first integral $(f')^2 + \kappa f^2 = 1$ of the equation $f'' + \kappa f = 0$ subject to $f(0) = 0$, $f'(0) = 1$. Thus a warped product $(0, \cdot) \times_f S^{n-1}$ has constant sectional curvature κ precisely when f solves $f'' + \kappa f = 0$, and the sphere, Euclidean space, and hyperbolic space are the three solutions capped smoothly at $r = 0$, where the collapsing fibers close up the missing center point.

For a sphere of radius ρ the warping is $f(r) = \rho \sin(r/\rho)$ on $(0, \pi\rho)$, giving metric $dr^2 + \rho^2 \sin^2(r/\rho)\, g_{S^{n-1}}$; then $-f''/f = 1/\rho^2$ and $(1-(f')^2)/f^2 = (1-\cos^2(r/\rho))/(\rho^2 \sin^2(r/\rho)) = 1/\rho^2$, so $K \equiv 1/\rho^2$, recovering the value $K(S^n_\rho) = 1/\rho^2$ of example 4.3.4 for the sphere of radius ρ.

5

Jacobi Fields and Conjugate Points

The first variation of chapter 3 identified geodesics as the critical points of the energy, and the curvature tensor of chapter 4 measured how the manifold bends. This chapter joins the two. Differentiating the energy a second time along a geodesic produces a quadratic form in which the curvature appears explicitly, and the sign of that curvature term decides whether a geodesic is a genuine minimum or merely a critical point. Curvature, in other words, controls stability, and the vehicle for the analysis is the Jacobi field.

A Jacobi field is the infinitesimal variation of a geodesic through nearby geodesics; equivalently, it is a solution of a single linear second-order equation, the Jacobi equation, in which the curvature enters as the coefficient. When the curvature is positive the neighboring geodesics are pulled back together and may refocus at a conjugate point, where the exponential map degenerates; when it is nonpositive they spread and never refocus. The chapter develops the second variation, the Jacobi equation and its solutions, the conjugate points and their identification with the singularities of the exponential map, and the index form whose positivity governs minimization. The recurring computation is the constant-curvature model, where the Jacobi equation is $f'' + cf = 0$ and every statement becomes explicit: the round sphere refocuses geodesics at the antipode, and nonpositively curved spaces never do. These results are the analytic core on which the comparison theorems of the next chapter rest.

5.1 The Second Variation of Energy

The first variation of section 3.5 located the geodesics as the critical points of the energy: a smooth curve γ is a geodesic exactly when $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = 0$ for every proper variation. Criticality alone, however, does not distinguish a minimum from a saddle. A straight segment in the plane minimizes energy, but a great-circle arc on the sphere that runs past the antipode is still a geodesic and so still critical, yet it can be shortened, so it is not a minimum. To separate these cases one differentiates the energy a second time and asks whether the resulting quadratic form is positive.

That second derivative is the subject of this section. Differentiating the first-variation formula along a critical geodesic produces a quadratic form in the variation field V, and curvature enters it explicitly. The formula

$$E''(0) = \int_0^b \left(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle\right) dt$$

carries all the geometry of the chapter in its second term. The pairing $\langle R(V,\dot\gamma)\dot\gamma, V\rangle$ equals $\mathrm{Riem}(V,\dot\gamma,\dot\gamma,V)$, which for V orthogonal to $\dot\gamma$ is the sectional curvature of the plane spanned by V and $\dot\gamma$ times $|V|^2\,|\dot\gamma|^2$. It enters with a minus sign, so positive curvature drives $E''(0)$ down and can make it negative, destabilizing the geodesic, while nonpositive curvature keeps $E''(0)$ nonnegative and preserves minimization. The remainder of the chapter is a study of this competition between the bending term $|D_t V|^2$, which resists deformation, and the curvature term, which may abet it. Throughout, (M,g) is a Riemannian manifold with Levi-Civita connection ∇ of definition 2.3.2, curvature endomorphism R of definition 4.1.2 in the sign convention of box 4.1.1, and D_t the induced covariant derivative along a curve from definition 2.1.6; geodesics are parametrized by constant speed as in definition 3.1.1, and $[0,b]$ is the parameter interval.

Theorem 5.1.1: Second Variation of Energy

Let $\gamma\colon [0,b]\to M$ be a geodesic, and let $\Gamma\colon (-\varepsilon,\varepsilon)\times[0,b]\to M$ be a smooth proper variation of γ, with variation field $V(t)=\partial_s\Gamma(0,t)$ satisfying $V(0)=V(b)=0$. Then the energy $E(\gamma_s)$ of the longitudinal curves is twice differentiable in s at $s=0$, and

$$E''(0) := \left.\frac{d^2}{ds^2}\right|_{s=0} E(\gamma_s) = \int_0^b \left(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle\right) dt$$

Equivalently, integrating the first term by parts,

$$E''(0) = -\int_0^b \left\langle D_t^2 V + R(V,\dot\gamma)\dot\gamma, V\right\rangle\, dt$$

***Proof*:**
The variation Γ is smooth on the whole rectangle $(-\varepsilon,\varepsilon)\times[0,b]$, so all the derivatives below may be taken under the integral sign. Write $T=\partial_t\Gamma$ and $S=\partial_s\Gamma$ for the coordinate velocity fields along Γ; at $s=0$ these restrict to $T=\dot\gamma$ and $S=V$ along γ. The energy of the longitudinal curve $\gamma_s=\Gamma(s,\cdot)$ is

$$E(\gamma_s) = \tfrac12 \int_0^b \langle T,T\rangle\, dt$$

Step 1: the first s-derivative. Metric compatibility of ∇ (definition 2.2.4), applied to the s-derivative of the inner product $\langle T,T\rangle$, gives

$$\frac{\partial}{\partial s}\langle T,T\rangle = 2\langle D_s T, T\rangle, \qquad \text{so} \qquad E'(s) = \frac{d}{ds}E(\gamma_s) = \int_0^b \langle D_s T, T\rangle\, dt$$

By the symmetry lemma (lemma 3.5.3), $D_s T = D_s\,\partial_t\Gamma = D_t\,\partial_s\Gamma = D_t S$, so

$$E'(s) = \int_0^b \langle D_t S, T\rangle\, dt$$

Step 2: the second s-derivative. Differentiate $E'(s)$ again in s, using metric compatibility once more:

$$E''(s) = \int_0^b \frac{\partial}{\partial s} \langle D_t S, T \rangle \, dt = \int_0^b \Big(\langle D_s D_t S, T \rangle + \langle D_t S, D_s T \rangle \Big) \, dt$$

The second summand is already in usable form: $D_s T = D_t S$ by the symmetry lemma, so $\langle D_t S, D_s T \rangle = \langle D_t S, D_t S \rangle = |D_t S|^2$. The first summand requires commuting the two covariant derivatives D_s and D_t acting on the field S along Γ. This is exactly what the curvature endomorphism measures. Since ∂_s and ∂_t are coordinate fields on the parameter rectangle, their bracket vanishes, and the definition of R (definition 4.1.2) in the convention of box 4.1.1 gives, for any field W along Γ,

$$D_s D_t W - D_t D_s W = R\big(\partial_s \Gamma, \partial_t \Gamma\big) W = R(S, T) W$$

Applying this with $W = S$,

$$D_s D_t S = D_t D_s S + R(S, T) S$$

Therefore

$$E''(s) = \int_0^b \Big(\langle D_t D_s S, T \rangle + \langle R(S, T) S, T \rangle + |D_t S|^2 \Big) \, dt$$

Step 3: evaluation at $s = 0$ and integration by parts. At $s = 0$ one has $S = V$, $T = \dot\gamma$, and $D_t S = D_t V$. The first integrand is rewritten by metric compatibility, moving one D_t off of $D_s S$ and onto $T = \dot\gamma$:

$$\langle D_t D_s S, T \rangle = \frac{d}{dt} \langle D_s S, T \rangle - \langle D_s S, D_t T \rangle$$

Along the base curve, $D_t T = D_t \dot\gamma = 0$ because γ is a geodesic (definition 3.1.1), so the last term vanishes at $s = 0$ and the first integrand becomes the total derivative $\frac{d}{dt} \langle D_s S, \dot\gamma \rangle$. Its integral over $[0, b]$ is the boundary difference

$$\int_0^b \frac{d}{dt} \langle D_s S, \dot\gamma \rangle \, dt = \Big[\langle D_s S, \dot\gamma \rangle \Big]_0^b$$

This boundary term vanishes. Indeed, the variation is proper, so $\Gamma(s, 0) = \gamma(0)$ and $\Gamma(s, b) = \gamma(b)$ are constant in s; hence $\partial_s \Gamma(s, 0) = 0$ and $\partial_s \Gamma(s, b) = 0$ for all s, and the covariant s-derivative $D_s S = D_s \, \partial_s \Gamma$ is the covariant derivative along the constant s-curves $t = 0$ and $t = b$, which is 0 there. Thus $\langle D_s S, \dot\gamma \rangle = 0$ at both endpoints, and the boundary difference is 0. Collecting the surviving terms at $s = 0$,

$$E''(0) = \int_0^b \Big(\langle R(V, \dot\gamma) V, \dot\gamma \rangle + |D_t V|^2 \Big) \, dt$$

Finally, the symmetries of the curvature tensor (proposition 4.2.1) convert the curvature term into the stated form. By the definition of the $(0, 4)$-tensor (definition 4.1.4) and antisymmetry in the second pair,

$$\langle R(V, \dot\gamma) V, \dot\gamma \rangle = \mathrm{Riem}(V, \dot\gamma, V, \dot\gamma) = -\,\mathrm{Riem}(V, \dot\gamma, \dot\gamma, V) = -\,\langle R(V, \dot\gamma) \dot\gamma, V \rangle$$

Substituting,

$$E''(0) = \int_0^b \big(|D_t V|^2 - \langle R(V, \dot\gamma) \dot\gamma, V \rangle \big) \, dt,$$

which is the asserted formula.

Step 4: the integrated-by-parts form. The field V is smooth on $[0,b]$ with $V(0) = V(b) = 0$. Metric compatibility gives $\frac{d}{dt}\langle D_tV, V\rangle = \left\langle D_t^2V, V\right\rangle + |D_tV|^2$, so

$$\int_0^b |D_tV|^2\, dt = \Big[\langle D_tV, V\rangle\Big]_0^b - \int_0^b \left\langle D_t^2V, V\right\rangle dt = -\int_0^b \left\langle D_t^2V, V\right\rangle dt,$$

the boundary term vanishing because $V(0) = V(b) = 0$. Substituting into the first form,

$$E''(0) = -\int_0^b \left\langle D_t^2V, V\right\rangle dt - \int_0^b \langle R(V,\dot\gamma)\dot\gamma, V\rangle\, dt = -\int_0^b \left\langle D_t^2V + R(V,\dot\gamma)\dot\gamma, V\right\rangle dt,$$

completing the proof.

The two forms of the formula read off two different aspects of the same quadratic form. The first, $\int_0^b(|D_tV|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle)\,dt$, displays the tension that governs the whole chapter: the term $|D_tV|^2$ is a nonnegative bending cost, always working to raise $E''(0)$, while the curvature term enters with a minus sign, so a plane of positive sectional curvature lowers $E''(0)$ and a plane of negative sectional curvature raises it. Writing $V = V^\perp + V^\top$ into components normal and tangent to $\dot\gamma$, and using $\langle R(V,\dot\gamma)\dot\gamma, V\rangle = \mathrm{Riem}(V,\dot\gamma,\dot\gamma,V)$, the curvature term sees only the normal part: with $V^\perp$ orthogonal to $\dot\gamma$,

$$\langle R(V,\dot\gamma)\dot\gamma, V\rangle = \mathrm{Riem}(V^\perp,\dot\gamma,\dot\gamma,V^\perp) = K(V^\perp,\dot\gamma)\big(\left|V^\perp\right|^2|\dot\gamma|^2 - \left\langle V^\perp,\dot\gamma\right\rangle^2\big) = K(V^\perp,\dot\gamma)\,\left|V^\perp\right|^2|\dot\gamma|^2,$$

by definition 4.3.1 together with the tangential vanishing $\mathrm{Riem}(\dot\gamma,\dot\gamma,\dot\gamma,V) = 0$ from the antisymmetry of proposition 4.2.1. Positive sectional curvature in the plane of $V^\perp$ and $\dot\gamma$ therefore subtracts a positive quantity from $E''(0)$; enough of it, spread over a long enough geodesic, can outweigh the bending term and make $E''(0)$ negative. The second form, $-\int_0^b \left\langle D_t^2V + R(V,\dot\gamma)\dot\gamma, V\right\rangle dt$, exposes the operator $V \mapsto D_t^2V + R(V,\dot\gamma)\dot\gamma$ acting on the variation field; the fields on which this operator vanishes are the Jacobi fields of section 5.2, the directions in which γ can be deformed through geodesics at no second-order cost.

The bilinear form obtained by polarizing $E''(0)$ is called the *index form* of γ; on piecewise-smooth fields V, W along γ vanishing at the endpoints it is

$$I(V,W) = \int_0^b \big(\langle D_tV, D_tW\rangle - \langle R(V,\dot\gamma)\dot\gamma, W\rangle\big)\,dt,$$

so that $I(V,V) = E''(0)$. Its systematic study, including the index lemma and the connection between its signature and the conjugate points of γ, is carried out in section 5.4; the present section records only the formula and its first consequence for minimization.

The immediate consequence is a one-sided sign condition. If a geodesic actually minimizes energy, then perturbing it in any admissible direction cannot decrease the energy, and this constrains the sign of the second variation.

Corollary 5.1.2: Minimizers Have Nonnegative Second Variation

Let $\gamma\colon [0,b] \to M$ be a geodesic that locally minimizes energy among fixed-endpoint curves, in the sense that $E(\gamma) \le E(c)$ for every piecewise-smooth curve $c\colon [0,b] \to M$ with $c(0) = \gamma(0)$, $c(b) = \gamma(b)$ sufficiently close to γ. Then $E''(0) \ge 0$ for every smooth proper variation of γ; equivalently, $I(V,V) \ge 0$ for every smooth field V along γ with $V(0) = V(b) = 0$.

***Proof*:**
Let Γ be a smooth proper variation of γ with variation field V, and set $E(s) = E(\gamma_s)$. Each longitudinal curve γ_s has the same endpoints as γ, and $\gamma_s \to \gamma$ uniformly as $s \to 0$ (indeed in C^1, by smoothness of Γ on the compact rectangle), so for $|s|$ small γ_s lies in the range where the minimizing hypothesis applies. Hence $E(s) \ge E(0)$ for all $|s|$ small, so $s = 0$ is a local minimum of the smooth function $E(s)$. Since γ is a geodesic, $E'(0) = 0$ by corollary 3.5.8, and the second-derivative test for a smooth function with a local minimum at an interior critical point gives $E''(0) \ge 0$. By theorem 5.1.1 this is exactly $\int_0^b (|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle)\,dt = I(V,V) \ge 0$, as claimed.

The contrapositive is the tool for detecting non-minimizing geodesics: if some proper variation has $E''(0) < 0$, then γ does not minimize energy, because the energy strictly decreases in that direction. Producing such a variation is a matter of finding a field V vanishing at the endpoints for which the negative curvature contribution overwhelms the bending term. On a positively curved space a geodesic long enough for the curvature to accumulate will admit one, and this is what fails minimization past the antipode on the sphere.

Example 5.1.3: A long great-circle arc does not minimize

On the round sphere S^n_r of radius r, which has constant sectional curvature $c = 1/r^2$ by example 4.3.4, take a unit-speed great-circle geodesic $\gamma\colon [0,b] \to S^n_r$ of length b, one of the great circles identified in example 3.1.4. The claim is that if $b > \pi r$, so that γ runs strictly past the antipode of its starting point, then γ does not minimize energy, and a proper variation with $E''(0) < 0$ can be written down.

Choose a parallel unit vector field $E(t)$ along γ with $E(t) \perp \dot\gamma(t)$ for all t; such a field exists because parallel transport preserves both length and orthogonality to the parallel field $\dot\gamma$, so it suffices to parallel-transport one unit vector orthogonal to $\dot\gamma(0)$, which is available since $n \ge 2$. Define the variation field

$$V(t) = \sin\!\left(\frac{\pi t}{b}\right) E(t),$$

which vanishes at $t = 0$ and $t = b$, hence is the field of a proper variation, for instance $\Gamma(s,t) = \exp_{\gamma(t)}(sV(t))$ as in example 3.5.2. Since E is parallel, $D_t E = 0$, so

$$D_t V = \frac{\pi}{b}\cos\!\left(\frac{\pi t}{b}\right) E(t), \qquad |D_t V|^2 = \frac{\pi^2}{b^2}\cos^2\!\left(\frac{\pi t}{b}\right)$$

using $|E| \equiv 1$. For the curvature term, $V(t) = \sin(\pi t/b)\,E(t)$ is orthogonal to $\dot\gamma$ and γ has unit speed, so by the constant-curvature reduction above,

$$\langle R(V,\dot\gamma)\dot\gamma, V\rangle = c\,|V|^2 = \frac{1}{r^2}\sin^2\!\left(\frac{\pi t}{b}\right)$$

Feeding both into the second-variation formula of theorem 5.1.1 and using $\int_0^b \cos^2(\pi t/b)\,dt = \int_0^b \sin^2(\pi t/b)\,dt = b/2$,

$$E''(0) = \int_0^b \Big(\frac{\pi^2}{b^2}\cos^2\frac{\pi t}{b} - \frac{1}{r^2}\sin^2\frac{\pi t}{b}\Big)\,dt = \frac{\pi^2}{b^2}\cdot\frac{b}{2} - \frac{1}{r^2}\cdot\frac{b}{2} = \frac{b}{2}\Big(\frac{\pi^2}{b^2} - \frac{1}{r^2}\Big) = \frac{\pi^2 r^2 - b^2}{2b\,r^2}$$

This is negative precisely when $b^2 > \pi^2 r^2$, that is, when $b > \pi r$. Thus a great-circle arc longer than πr has $E''(0) < 0$ for this variation, so by the contrapositive of corollary 5.1.2 it does not minimize energy, and by the length-energy inequality of theorem 3.5.6 it does not minimize length either: the arc can be shortened by sliding it off toward the shorter way around. At the threshold $b = \pi r$, exactly the antipodal distance, this particular variation gives $E''(0) = 0$, the borderline where the great circle stops minimizing; the same value πr reappears in section 5.3 as the first conjugate distance on S^n_r. For $b < \pi r$ the variation gives $E''(0) > 0$, consistent with the fact that a great-circle arc shorter than the antipodal distance is the unique shortest path between its endpoints.

5.2 The Jacobi Equation

The second variation of energy, computed for a geodesic γ and a proper variation with field V, produced the quadratic form

$$E''(0) = \int_0^b \big(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle\big)\,dt$$

in which the curvature enters through the term $\langle R(V,\dot\gamma)\dot\gamma, V\rangle = \mathrm{Riem}(V,\dot\gamma,\dot\gamma,V)$. Whether a geodesic minimizes energy against a given deformation is decided by the sign of this integral, and the direction V that makes it smallest is the direction along which the geodesic is closest to failing to minimize. Setting the associated Euler–Lagrange equation to zero singles out those variation fields, and the equation they satisfy is the object of this section.

There is a second route to the same equation, and it is the one that gives the fields their geometric meaning. A geodesic can be deformed not through arbitrary curves but through other geodesics: a one-parameter family γ_s of geodesics, all issuing from a common construction, sweeps out a variation whose field $J(t) = \partial_s\gamma_s(t)|_{s=0}$ records how neighboring geodesics separate or converge. Differentiating the geodesic equation of the whole family in the transverse direction, and invoking the symmetry lemma and the definition of curvature, turns $D_t\dot\gamma_s = 0$ into a linear second-order equation for J. The two derivations meet: the fields that arise from variations through geodesics are exactly the critical fields of the second variation, and both are the solutions of the Jacobi equation. Throughout, (M,g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), curvature endomorphism R of definition 4.1.2 under the sign convention of box 4.1.1, D_t is the covariant derivative along a curve (definition 2.1.6), and $\gamma\colon [0,b] \to M$ is a geodesic (definition 3.1.1). The Einstein summation convention is in force where indices appear.

Definition 5.2.1: Jacobi field

Let $\gamma\colon [0,b] \to M$ be a geodesic. A *Jacobi field* along γ is a smooth vector field J along γ satisfying the *Jacobi equation*

$$D_t^2 J + R(J,\dot\gamma)\dot\gamma = 0 \tag{5.1}$$

where R is the curvature endomorphism of definition 4.1.2 and $D_t^2 J = D_t(D_t J)$ is the second covariant derivative of J along γ.

The equation (5.1) is a linear second-order ordinary differential equation for the field J, and the curvature term is what couples it to the geometry of the ambient manifold. Read as an equation of motion, it describes an infinitesimal separation vector J subject to an acceleration $-R(J,\dot\gamma)\dot\gamma$ produced entirely by curvature; the field $D_t^2 J$ is the relative acceleration of two neighboring geodesics, and the curvature dictates its sign. Pairing (5.1) with J and using metric compatibility to write $\left\langle D_t^2 J, J\right\rangle = \frac{d}{dt}\left\langle D_t J, J\right\rangle - |D_t J|^2$ ties the equation to the second variation of the opening: the curvature coefficient there and the curvature coefficient here are the same object, so a Jacobi field is precisely a critical field of the second-variation integral. The interpretation to keep in mind is the one the next theorem makes exact: J is the velocity, at the base geodesic, of a family of geodesics fanning out from γ, and the Jacobi equation is the linearization of the geodesic equation along that fan.

The variational meaning is established first, because it justifies calling (5.1) the equation of a variation through geodesics and supplies both the fields that appear in the second variation and the geometric picture of geodesic spreading.

Theorem 5.2.2: Jacobi fields are the variation fields of geodesic variations

Let $\gamma\colon [0,b] \to M$ be a geodesic.

1. If $\Gamma\colon (-\varepsilon,\varepsilon)\times[0,b] \to M$ is a smooth variation of γ (so $\Gamma(0,t) = \gamma(t)$) such that every longitudinal curve $\gamma_s = \Gamma(s,\cdot)$ is a geodesic, then the variation field $J(t) = \partial_s\Gamma(0,t)$ is a Jacobi field along γ.
2. Conversely, every Jacobi field J along γ arises as the variation field of some smooth variation of γ through geodesics.

***Proof*:**
Write $T = \partial_t\Gamma$ and $S = \partial_s\Gamma$ for the two coordinate velocity fields of the variation, and D_t, D_s for the covariant derivatives along the t- and s-parameter curves (definition 2.1.6). At $s = 0$ one has $T = \dot\gamma$ and $S = J$.

1. Since each longitudinal curve γ_s is a geodesic, its velocity T is autoparallel along it: $D_t T = 0$ for every s. Differentiating this identity covariantly in the s-direction gives $D_s D_t T = 0$ identically. The plan is to convert $D_s D_t T$ into an expression in $D_t^2 S$ and curvature, evaluated at $s = 0$.

   Step 1: commuting the derivatives. The parameter fields ∂_s, ∂_t on the rectangle $(-\varepsilon,\varepsilon)\times[0,b]$ commute, and Γ carries them to $S = \partial_s\Gamma$ and $T = \partial_t\Gamma$. For any smooth vector field W along Γ, the definition of the curvature endomorphism (definition 4.1.2), applied with the commuting parameter fields so that the bracket term drops, gives

$$D_s D_t W - D_t D_s W = R(S,T)W \tag{5.2}$$

This is the standard passage from the endomorphism $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$ to the covariant derivatives along the two parameter directions of a variation; it holds with the sign of box 4.1.1 because S and T are the images of commuting coordinate fields, so no $\nabla_{[S,T]}$ correction survives.

Step 2: applying the identity. Take $W = T$ in (5.2):

$$D_s D_t T = D_t D_s T + R(S,T)T$$

The left-hand side vanishes because $D_t T = 0$ for every s. By the symmetry lemma (lemma 3.5.3), $D_s T = D_s \partial_t \Gamma = D_t \partial_s \Gamma = D_t S$, so $D_t D_s T = D_t D_t S = D_t^2 S$. Hence

$$0 = D_t^2 S + R(S,T)T$$

identically on the rectangle.

Step 3: evaluation at $s = 0$. Set $s = 0$, where $S = J$ and $T = \dot\gamma$:

$$D_t^2 J + R(J,\dot\gamma)\dot\gamma = 0$$

This is the Jacobi equation (5.1), so J is a Jacobi field along γ.

2. Let J be a Jacobi field along γ; a variation of γ through geodesics with variation field J is constructed by exponentiating a suitable family of initial data. Set $p = \gamma(0)$ and $v = \dot\gamma(0)$, so $\gamma(t) = \exp_p(tv)$ (definition 3.2.1). Choose a smooth curve $c\colon (-\varepsilon,\varepsilon) \to M$ with $c(0) = p$ and $\dot c(0) = J(0)$, and let $W(s)$ be a smooth vector field along c with $W(0) = v$ and $D_s W(0) = D_t J(0)$; such a c and W exist because $J(0) \in T_pM$ and $D_t J(0) \in T_pM$ are prescribed vectors, realized for instance by taking c a short curve with the given initial velocity and W any field along it matching the two conditions at $s = 0$. Define

$$\Gamma(s,t) = \exp_{c(s)}\bigl(t\,W(s)\bigr)$$

for $|s| < \varepsilon$ small enough that $tW(s)$ lies in the domain of $\exp_{c(s)}$ for all $t \in [0,b]$, which is possible by smoothness of the exponential map and compactness of $[0,b]$. For each fixed s the curve $t \mapsto \Gamma(s,t)$ is the geodesic through $c(s)$ with initial velocity $W(s)$, so Γ is a variation of γ through geodesics with $\Gamma(0,t) = \exp_p(tv) = \gamma(t)$. By part (1), its variation field $\widetilde J(t) = \partial_s \Gamma(0,t)$ is a Jacobi field. Its initial data are

$$\widetilde J(0) = \partial_s \Gamma(s,0)\big|_{s=0} = \dot c(0) = J(0)$$

since $\Gamma(s,0) = \exp_{c(s)}(0) = c(s)$, and, using the symmetry lemma once more,

$$D_t \widetilde J(0) = D_t \partial_s \Gamma(0,0) = D_s \partial_t \Gamma(0,0) = D_s W(0) = D_t J(0)$$

because $\partial_t \Gamma(s,0) = \frac{d}{dt}\big|_{t=0} \exp_{c(s)}(tW(s)) = W(s)$. Thus $\widetilde J$ and J are Jacobi fields with the same value and the same covariant derivative at $t = 0$; by the uniqueness statement of theorem 5.2.3 below they coincide, $\widetilde J = J$. Hence Γ is a variation through geodesics whose variation field is J, as required.

Both directions are established, completing the proof.

The theorem gives the Jacobi field its name and its picture. A Jacobi field is the derivative of a family of geodesics with respect to the family parameter, so it measures the rate at which two infinitesimally close geodesics drift apart or draw together; the Jacobi equation is what remains of the geodesic equation after this differentiation, the curvature term being the trace left by the noncommutativity of covariant derivatives. The converse half is the more useful for computation: it says that no generality is lost in thinking of a Jacobi field as an actual spread of geodesics, since every solution of (5.1) is realized by one. The construction in part (2) also isolates the two pieces of initial data that determine a Jacobi field, the initial value $J(0)$ (how far the neighboring geodesic starts) and the initial covariant derivative $D_tJ(0)$ (how fast its initial velocity turns away), which is the content of the existence and uniqueness theorem taken up next.

Because the Jacobi equation is a linear second-order equation, its solution theory is the linear-ODE theory already used for geodesics and for parallel transport. Choosing a parallel orthonormal frame along γ converts (5.1) into a constant-form linear system, from which existence, uniqueness, and the dimension count follow.

Theorem 5.2.3: The Jacobi fields form a $2n$-dimensional space

Let $\gamma\colon [0,b] \to M$ be a geodesic in an n-dimensional manifold. For each pair of vectors $u, w \in T_{\gamma(0)}M$ there is a unique Jacobi field J along γ with $J(0) = u$ and $D_tJ(0) = w$. Consequently the set $\mathcal{J}(\gamma)$ of Jacobi fields along γ is a real vector space of dimension $2n$, and the map

$$\mathcal{J}(\gamma) \longrightarrow T_{\gamma(0)}M \oplus T_{\gamma(0)}M, \qquad J \longmapsto \big(J(0), D_tJ(0)\big)$$

is a linear isomorphism.

***Proof*:**
Let $(E_1, \dots, E_n)$ be a parallel orthonormal frame along γ, obtained by parallel transporting an orthonormal basis of $T_{\gamma(0)}M$ (theorem 2.4.2); it is orthonormal at every t because parallel transport is a linear isometry (proposition 2.4.4). Write a vector field J along γ in this frame as $J(t) = J^i(t)E_i(t)$ with scalar coefficients $J^i\colon [0,b] \to \mathbb{R}$. Since the E_i are parallel, $D_tE_i = 0$, so covariant differentiation of J reduces to ordinary differentiation of its coefficients:

$$D_tJ = \dot{J}^iE_i, \qquad D_t^2J = \ddot{J}^iE_i$$

The curvature term is linear in J, so its frame coefficients are linear in the J^i. Define the smooth functions

$$a^k{}_i(t) = \big\langle R\big(E_i(t), \dot\gamma(t)\big)\dot\gamma(t), E_k(t)\big\rangle$$

the components of the endomorphism $J \mapsto R(J,\dot\gamma)\dot\gamma$ in the frame; they are smooth because R, $\dot\gamma$, and the frame are smooth. Then $R(J,\dot\gamma)\dot\gamma = J^ia^k{}_iE_k$, and the Jacobi equation (5.1) becomes, coefficient by coefficient, the linear system

$$\ddot{J}^k(t) + a^k{}_i(t)\, J^i(t) = 0, \qquad k = 1, \dots, n \tag{5.3}$$

This is a homogeneous linear second-order system of n equations with smooth coefficients on the compact interval $[0,b]$. Rewriting it as a first-order system in the $2n$ unknowns $(J^1, \dots, J^n, \dot{J}^1, \dots, \dot{J}^n)$, the standard existence and uniqueness theorem for linear ordinary differential equations (as used for the geodesic equation in theorem 3.1.3 and for the parallel-transport equation in theorem 2.4.2) provides, for each initial datum $(J^i(0), \dot{J}^i(0))$, a unique solution on all of $[0, b]$. The initial datum corresponds exactly to $(J(0), D_t J(0))$ through the frame: $J(0) = J^i(0)E_i(0)$ and $D_t J(0) = \dot{J}^i(0)E_i(0)$. Hence for each $u, w \in T_{\gamma(0)}M$ there is a unique Jacobi field with $J(0) = u$, $D_t J(0) = w$.

Linearity of (5.1) in J makes $\mathcal{J}(\gamma)$ a real vector space, and the evaluation map $J \mapsto (J(0), D_t J(0))$ is linear. It is injective by the uniqueness just proved (a Jacobi field with zero initial data is the zero solution) and surjective by the existence, so it is a linear isomorphism onto $T_{\gamma(0)}M \oplus T_{\gamma(0)}M$, which has dimension $2n$. Therefore $\dim \mathcal{J}(\gamma) = 2n$, as claimed.

The theorem records that a Jacobi field is pinned down by exactly the same amount of data as a geodesic through a point, but now transverse to the base geodesic: its starting displacement and the rate at which that displacement is turning. The $2n$-dimensional count will be split in the next proposition into the part along the geodesic, which is geometrically inert, and the part transverse to it, which carries all the information about how geodesics focus. Two of the $2n$ dimensions are visibly present already, and they are the least interesting: multiples of $\dot{\gamma}$ and of $t\dot{\gamma}$ solve the Jacobi equation trivially, because the curvature term annihilates $\dot{\gamma}$. Separating these off is the content of the following splitting.

Proposition 5.2.4: Tangential and normal splitting of Jacobi fields

Let $\gamma \colon [0, b] \to M$ be a geodesic, and suppose γ has constant speed $|\dot{\gamma}| \equiv \ell$.

1. A vector field of the form $J(t) = (a + bt)\dot{\gamma}(t)$ with $a, b \in \mathbb{R}$ is a Jacobi field, and these are exactly the Jacobi fields everywhere tangent to γ.

2. Every Jacobi field J splits uniquely as $J = J^\top + J^\perp$, where $J^\top$ is tangent to $\dot{\gamma}$ and $J^\perp$ is everywhere orthogonal to $\dot{\gamma}$, and each of $J^\top$, $J^\perp$ is again a Jacobi field. If γ is nonconstant ($\ell \neq 0$), the tangential part is $J^\top(t) = (a+bt)\dot{\gamma}(t)$ with $a = \langle J(0), \dot{\gamma}(0)\rangle / \ell^2$ and $b = \langle D_t J(0), \dot{\gamma}(0)\rangle / \ell^2$.

In particular the space $\mathcal{J}(\gamma)$ decomposes as an internal direct sum $\mathcal{J}(\gamma) = \mathcal{J}^\top(\gamma) \oplus \mathcal{J}^\perp(\gamma)$ with $\dim \mathcal{J}^\top(\gamma) = 2$ and $\dim \mathcal{J}^\perp(\gamma) = 2n - 2$.

***Proof*:**

1. Let $J(t) = (a + bt)\dot{\gamma}(t)$. Since γ is a geodesic, $D_t\dot{\gamma} = 0$ (definition 3.1.1), so

$$D_t J = b\,\dot{\gamma} + (a + bt)D_t\dot{\gamma} = b\,\dot{\gamma}, \qquad D_t^2 J = b\,D_t\dot{\gamma} = 0$$

   The curvature term vanishes because R is antisymmetric in its first pair of arguments, $R(\dot{\gamma}, \dot{\gamma}) = 0$ (definition 4.1.2), so $R(J, \dot{\gamma})\dot{\gamma} = (a+bt)R(\dot{\gamma}, \dot{\gamma})\dot{\gamma} = 0$. Hence $D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$ and J is a Jacobi field. These form a two-dimensional space, spanned by $\dot{\gamma}$ and $t\dot{\gamma}$, corresponding to initial data $(J(0), D_t J(0)) \in \{(a\dot{\gamma}(0), b\dot{\gamma}(0)) : a, b \in \mathbb{R}\}$. Conversely a Jacobi field tangent to γ at every t may be written $J(t) = \varphi(t)\dot{\gamma}(t)$ for a smooth scalar φ; then $D_t^2 J = \ddot{\varphi}\,\dot{\gamma}$ as above and the curvature term vanishes, so (5.1) reduces to $\ddot{\varphi}\,\dot{\gamma} = 0$, forcing $\ddot{\varphi} \equiv 0$ (for nonconstant γ, where $\dot{\gamma} \neq 0$), hence $\varphi(t) = a + bt$. Thus the tangential Jacobi fields are exactly the $(a + bt)\dot{\gamma}$.

2. Consider the scalar function $f(t) = \langle J(t), \dot{\gamma}(t)\rangle$ for a Jacobi field J. Using metric compati-

bility (definition 2.2.4) and $D_t\dot\gamma = 0$,

$$\dot f = \langle D_t J, \dot\gamma\rangle, \qquad \ddot f = \langle D_t^2 J, \dot\gamma\rangle$$

By the Jacobi equation $D_t^2 J = -R(J,\dot\gamma)\dot\gamma$, so

$$\ddot f = -\langle R(J,\dot\gamma)\dot\gamma, \dot\gamma\rangle = -\operatorname{Riem}(J,\dot\gamma,\dot\gamma,\dot\gamma) = 0$$

the pairing vanishing by the antisymmetry of Riem in its last pair of arguments (definition 4.1.2 gives $R(J,\dot\gamma)\dot\gamma \perp \dot\gamma$ directly, since $\langle R(X,Y)Z, Z\rangle = 0$ from the antisymmetry $\operatorname{Riem}(X,Y,Z,W) = -\operatorname{Riem}(X,Y,W,Z)$). Hence f is affine, $f(t) = f(0) + \dot f(0)\,t = \langle J(0), \dot\gamma(0)\rangle + \langle D_t J(0), \dot\gamma(0)\rangle\, t$. Assume γ nonconstant, $\ell \neq 0$, and define the tangential part

$$J^\top(t) = \frac{f(t)}{\ell^2}\,\dot\gamma(t) = \Big(\tfrac{\langle J(0),\dot\gamma(0)\rangle}{\ell^2} + \tfrac{\langle D_t J(0),\dot\gamma(0)\rangle}{\ell^2}\,t\Big)\dot\gamma(t)$$

which is of the form $(a+bt)\dot\gamma$ with the stated a, b, and so is a tangential Jacobi field by part (1). Set $J^\perp = J - J^\top$. Since $|\dot\gamma|^2 = \ell^2$ is constant, $\langle J^\top(t), \dot\gamma(t)\rangle = \frac{f(t)}{\ell^2}\,\ell^2 = f(t) = \langle J(t), \dot\gamma(t)\rangle$, whence $\langle J^\perp(t), \dot\gamma(t)\rangle = 0$ for all t; thus $J^\perp$ is everywhere orthogonal to $\dot\gamma$. As a difference of two Jacobi fields, $J^\perp$ is a Jacobi field by linearity of (5.1). The decomposition $J = J^\top + J^\perp$ is unique, since a field simultaneously tangent to and orthogonal to the nonzero $\dot\gamma$ is zero. (When γ is constant the statement is vacuous, as there is no distinguished direction; the substantive case is $\ell \neq 0$, and one may take $\ell = 1$ by unit-speed reparametrization.)

For the dimension count, the tangential Jacobi fields $\mathcal{J}^\top(\gamma)$ form the two-dimensional space of part (1). The normal Jacobi fields $\mathcal{J}^\perp(\gamma)$ are those with $J \perp \dot\gamma$ throughout; by the splitting, every Jacobi field is uniquely a sum of one of each type, so $\mathcal{J}(\gamma) = \mathcal{J}^\top(\gamma) \oplus \mathcal{J}^\perp(\gamma)$, and $\dim \mathcal{J}^\perp(\gamma) = 2n - \dim \mathcal{J}^\top(\gamma) = 2n - 2$ by theorem 5.2.3, completing the proof.

The splitting quarantines the trivial part of the Jacobi equation. A tangential field $(a+bt)\dot\gamma$ is nothing more than a reparametrization of the geodesic itself: sliding γ along its own image, or speeding it up linearly, produces a variation through geodesics whose field points along $\dot\gamma$ and carries no transverse information. All the geometry of how nearby geodesics focus is in the normal part $J^\perp$, whose $2n-2$ dimensions match the $n-1$ transverse directions at each of the two ends of the interval. From this point on the object of interest is the normal Jacobi field, and the model computation that follows treats exactly that case, with $J \perp \dot\gamma$ and $J(0) = 0$, the initial condition that will detect conjugate points in the next section.

The Jacobi equation becomes fully explicit on a space of constant curvature, where the curvature endomorphism is the universal expression of proposition 4.5.3. There the equation collapses to a scalar equation $f'' + cf = 0$, and its solutions are the trigonometric or hyperbolic functions whose zeros locate the conjugate points.

Example 5.2.5: Normal Jacobi fields on a space of constant curvature

Let (M, g) have constant sectional curvature c (definition 4.5.2), and let $\gamma\colon [0,b] \to M$ be a unit-speed geodesic, $|\dot\gamma| \equiv 1$. The normal Jacobi fields with $J(0) = 0$ are computed here in closed form; they are the fields governing the focusing of geodesics issuing from $\gamma(0)$.

Let E be a parallel unit field along γ with $E \perp \dot\gamma$, obtained by parallel transporting a unit vector orthogonal to $\dot\gamma(0)$; it stays orthogonal to $\dot\gamma$ and of unit length because parallel transport is a linear isometry and $\dot\gamma$ is itself parallel (proposition 2.4.4). Seek a Jacobi field of the form $J(t) = f(t)E(t)$ with f scalar. Since E is parallel, $D_t J = \dot f E$ and $D_t^2 J = \ddot f E$. For the curvature term, the constant-curvature form (4.7) of proposition 4.5.3 gives, using $|\dot\gamma|^2 = 1$ and $\langle E, \dot\gamma\rangle = 0$,

$$R(J,\dot\gamma)\dot\gamma = f\,R(E,\dot\gamma)\dot\gamma = f\,c\big(\langle\dot\gamma,\dot\gamma\rangle\,E - \langle E,\dot\gamma\rangle\,\dot\gamma\big) = c\,f\,E$$

Substituting into the Jacobi equation (5.1),

$$D_t^2 J + R(J,\dot\gamma)\dot\gamma = \big(\ddot f + c\,f\big)E = 0$$

and since E is nowhere zero this is the scalar equation

$$f'' + c\,f = 0 \tag{5.4}$$

With the initial conditions $J(0) = 0$ and $D_t J(0) = E(0)$, that is $f(0) = 0$ and $f'(0) = 1$, the solution of (5.4) is

$$f_c(t) = \begin{cases} \dfrac{\sin(\sqrt{c}\,t)}{\sqrt{c}} & c > 0 \\ t & c = 0 \\ \dfrac{\sinh(\sqrt{-c}\,t)}{\sqrt{-c}} & c < 0 \end{cases}$$

as is checked by substitution: for $c > 0$, $f_c'' = -\sqrt{c}\,\sin(\sqrt{c}\,t) = -c\,f_c$; for $c = 0$, $f_c'' = 0$; for $c < 0$, $f_c'' = \sqrt{-c}\,\sinh(\sqrt{-c}\,t) = -c\,f_c$, and in every case $f_c(0) = 0$, $f_c'(0) = 1$. The Jacobi field is $J(t) = f_c(t)E(t)$, and the general normal Jacobi field with $J(0) = 0$ is a sum $\sum_{i=1}^{n-1} f_c(t)\,c_i E_i(t)$ over a parallel orthonormal frame $E_1, \ldots, E_{n-1}$ of the orthogonal complement of $\dot\gamma$, with constants c_i; all $n-1$ such fields share the single radial factor f_c, the analytic signature of isotropy.

The three cases carry the geometry directly. On the round sphere S^n_r, which has constant curvature $c = 1/r^2$ by example 4.3.4 and example 1.3.4, the field is

$$J(t) = r\sin(t/r)\,E(t)$$

whose norm $|J(t)| = r\sin(t/r)$ starts at 0, grows to its maximum r at $t = \pi r/2$, and returns to 0 at $t = \pi r$: geodesics leaving a point of the sphere spread apart and then refocus, meeting again at the antipodal point a distance πr away. On Euclidean space $\mathbb{R}^n$ ($c = 0$) the field is $J(t) = t\,E(t)$, with $|J(t)| = t$ growing without bound, so geodesics spread linearly and never refocus. On hyperbolic space H^n (curvature $c = -1$, example 4.3.5) the field is $J(t) = \sinh(t)\,E(t)$, with $|J(t)| = \sinh t$ growing exponentially, so geodesics spread even faster and again never meet again. The vanishing of f_c at a positive time, which happens only for $c > 0$, is precisely the event that the next section names a conjugate point.

5.3 Conjugate Points

The Jacobi equation of section 5.2 governs how a family of geodesics fans out from a common starting point. A Jacobi field J that vanishes at $t = 0$ is the infinitesimal record of a pencil of geodesics all leaving $\gamma(0)$, and its value $J(t)$ measures, to first order, how far the neighboring geodesics have drifted from γ by time t. On a nonpositively curved space these fields grow and the geodesics keep spreading. On a positively curved space the curvature term in $D_t^2 J + R(J,\dot\gamma)\dot\gamma = 0$ acts as a restoring force, pulling the fanned geodesics back toward one another, and a Jacobi field that started at zero can return to zero. The instant at which it does is a *conjugate point*: a point along γ at which the nearby geodesics from $\gamma(0)$ attempt to refocus.

The phenomenon is visible on the round sphere. Every great circle through the north pole passes through the south pole, so all the geodesics leaving the north pole in every direction reconverge at the antipode. The antipode is the point at which the exponential map, which spread the tangent space out over the sphere, folds it back to a single point. This is the second face of a conjugate point, and the central result of the section identifies the two: q is conjugate to p along a geodesic exactly when the exponential map degenerates in the direction of q, and the number of independent geodesic pencils that refocus there is the dimension of the kernel of its differential. Conjugate points are therefore where the linear model of the manifold provided by $\exp_p$ breaks down, and their absence, guaranteed by nonpositive curvature, is what makes that model global. Both directions of this dictionary are used repeatedly in the sequel: conjugate points obstruct minimization, studied through the index form in section 5.4, and their nonexistence under $K \le 0$ is the analytic input to the Cartan–Hadamard theorem of the next chapter.

Throughout, (M,g) is a Riemannian manifold with Levi-Civita connection ∇, curvature endomorphism R (definition 4.1.2), and sectional curvature K (definition 4.3.1); γ is a geodesic (definition 3.1.1) and D_t the covariant derivative along it. Jacobi fields and the Jacobi equation (5.1) are those of section 5.2. The sign conventions fixed there and in Chapter 4 are in force.

Definition 5.3.1: Conjugate Point

Let $\gamma\colon [0,b] \to M$ be a geodesic and set $p = \gamma(0)$. A point $\gamma(t_0)$ with $0 < t_0 \le b$ is *conjugate to p along γ* if there exists a Jacobi field J along γ, not identically zero, with

$$J(0) = 0 \qquad \text{and} \qquad J(t_0) = 0$$

The *multiplicity* of the conjugate point $\gamma(t_0)$ is the dimension of the vector space

$$\mathcal{J}_{0,t_0} = \{J \text{ a Jacobi field along } \gamma \mid J(0) = 0,\ J(t_0) = 0\}$$

of Jacobi fields vanishing at both endpoints. The relation is symmetric: if $\gamma(t_0)$ is conjugate to p along γ, then $p = \gamma(0)$ is conjugate to $\gamma(t_0)$ along the reversed geodesic $t \mapsto \gamma(t_0 - t)$, with the same multiplicity, since $J(t_0 - t)$ is again a Jacobi field vanishing at both ends. One therefore speaks of the two points as *conjugate along γ*.

The definition isolates the geometric event described in the opening: a nonzero Jacobi field vanishing at 0 and at t_0 is the infinitesimal generator of a family of geodesics that leave $\gamma(0)$ together and, to first order, meet again at $\gamma(t_0)$. That the multiplicity is a dimension, and not a bare count, reflects that several independent families may refocus at once, as on the sphere, where every direction transverse to a great circle produces a Jacobi field vanishing at the antipode. Two structural remarks bound the multiplicity. A Jacobi field vanishing at 0 and t_0 has no tangential part: the tangential Jacobi fields are $(a+bt)\dot\gamma$ (proposition 5.2.4), and such a field vanishes at two distinct times only if $a = b = 0$. Hence every field in $\mathcal{J}_{0,t_0}$ is normal to γ, and the space of normal Jacobi fields vanishing at 0 is already $(n-1)$-dimensional; imposing $J(t_0) = 0$ can only cut this down. The multiplicity of a conjugate point therefore lies between 1 and $n-1$.

A first concrete instance grounds the definition before the general theory. On the unit sphere S^n a geodesic γ is a unit-speed great circle, and by example 5.2.5, applied with curvature $c = 1$, a normal Jacobi field J with $J(0) = 0$ has the form $J(t) = \sin(t)\,E(t)$ with E a parallel unit normal field, so $|J(t)| = |\sin t|$. This vanishes at $t = 0$ and next at $t = \pi$, and at no interior time $0 < t < \pi$. Consequently no point $\gamma(t_0)$ with $0 < t_0 < \pi$ is conjugate to p: any Jacobi field vanishing at 0 is normal (its tangential part $(a + bt)\dot\gamma$ must vanish at $t = 0$, forcing $a = 0$, and it is a sum of terms $\sin(t)E_i(t)$, none of which vanishes on $(0, \pi)$), so it cannot vanish again before $t = \pi$. The first conjugate point is the antipode $\gamma(\pi)$, computed in full in example 5.3.4. Below the antipodal distance the geodesics from p have not yet had time to refocus, and $\exp_p$ remains a diffeomorphism on the corresponding ball, as example 3.2.4 recorded.

The sphere calculation already suggests that conjugate points are where the exponential map ceases to be a local diffeomorphism. The connection is exact, and it is the organizing theorem of the section. It rests on a single computation: the Jacobi fields along a radial geodesic that vanish at the base point are precisely the images, under the differential of $\exp_p$, of the radial rays in the tangent space. The idea goes back to the definition of the exponential map as the time-one flow of geodesics, and to the fact from section 5.2 that Jacobi fields are the variation fields of variations through geodesics.

Theorem 5.3.2: Conjugate Points are the Singularities of the Exponential Map

Let $p \in M$, let $v \in T_pM$ lie in the domain of $\exp_p$, and let $\gamma(t) = \exp_p(tv)$ be the geodesic with $\gamma(0) = p$ and $\dot\gamma(0) = v$, defined for t in an interval containing $[0, 1]$. Fix $t_0 > 0$ with $t_0 v$ in the domain of $\exp_p$, and set $q = \exp_p(t_0 v) = \gamma(t_0)$. Then:

1. For every $w \in T_pM$, the field
$$J_w(t) = d(\exp_p)_{tv}(t\,w) \tag{5.5}$$
is the Jacobi field along γ with $J_w(0) = 0$ and $D_t J_w(0) = w$, where on the right $tw \in T_{tv}(T_pM)$ is identified with an element of T_pM in the usual way. Every Jacobi field along γ vanishing at 0 arises this way.
2. The point $q = \gamma(t_0)$ is conjugate to p along γ if and only if the differential $d(\exp_p)_{t_0 v} \colon T_{t_0 v}(T_pM) \to T_qM$ is singular (not injective).
3. The multiplicity of the conjugate point q equals $\dim\ker d(\exp_p)_{t_0 v}$.

***Proof*:**
The tangent space $T_a(T_pM)$ to the vector space T_pM at a point a is canonically identified with T_pM itself, as in proposition 3.2.5; this identification is used without further comment, so $d(\exp_p)_a$ is read as a linear map $T_pM \to T_{\exp_p(a)}M$.

1. Fix $w \in T_pM$. Build a variation of γ through geodesics whose variation field is J_w. In the tangent space T_pM, consider the two-parameter family of vectors
$$(s, t) \longmapsto t\,(v + sw), \qquad s \in (-\varepsilon, \varepsilon),\ t \in [0, t_0 + \delta]$$
For ε and δ small the vectors $t(v + sw)$ all lie in the (open) domain of $\exp_p$, because $t_0 v$ lies in that open set and the map $(s, t) \mapsto t(v + sw)$ is continuous with $(0, t) \mapsto tv$ in the

domain for $t \in [0, t_0]$. Define

$$\Gamma(s,t) = \exp_p\bigl(t(v+sw)\bigr)$$

For each fixed s, the rescaling identity (proposition 3.2.2) shows that $t \mapsto \Gamma(s,t) = \exp_p(t(v+sw))$ is the geodesic with initial point p and initial velocity $v+sw$; thus Γ is a variation of $\gamma = \Gamma(0,\cdot)$ through geodesics. By theorem 5.2.2, its variation field

$$J(t) = \partial_s \Gamma(s,t)\big|_{s=0}$$

is a Jacobi field along γ. Compute it from the chain rule. The map $s \mapsto t(v+sw)$ is a straight line in T_pM through tv with velocity tw at $s=0$, so

$$J(t) = \frac{\partial}{\partial s}\Big|_{s=0} \exp_p\bigl(t(v+sw)\bigr) = d(\exp_p)_{tv}(t\,w) = J_w(t)$$

which is (5.5). It remains to identify the initial data of J_w. At $t=0$ the differential $d(\exp_p)_0(0\cdot w) = 0$, so $J_w(0) = 0$. For the initial derivative, all curves $s \mapsto \Gamma(s,0) = \exp_p(0) = p$ are constant, and each $t \mapsto \Gamma(s,t)$ is a geodesic starting at p with velocity $v+sw$, so $\partial_t\Gamma(s,0) = v+sw$. Using the symmetry lemma (lemma 3.5.3) and evaluating at $t=0$,

$$D_t J_w(0) = D_t\,\partial_s\Gamma\big|_{(0,0)} = D_s\,\partial_t\Gamma\big|_{(0,0)} = D_s\big|_{s=0}(v+sw) = w,$$

the last covariant derivative being computed along the constant curve $s \mapsto \Gamma(s,0) = p$, where covariant differentiation of the T_pM-valued curve $s \mapsto v+sw$ reduces to the ordinary derivative w. Thus J_w is the Jacobi field with $J_w(0) = 0$ and $D_tJ_w(0) = w$.

Conversely, let J be any Jacobi field along γ with $J(0) = 0$. Set $w = D_tJ(0)$. Then J_w is a Jacobi field with the same initial data $J_w(0) = 0 = J(0)$ and $D_tJ_w(0) = w = D_tJ(0)$, so $J = J_w$ by the uniqueness of Jacobi fields from their initial data (theorem 5.2.3). Hence every Jacobi field vanishing at 0 has the form (5.5).

2. By part (1), the Jacobi fields along γ vanishing at 0 are exactly the fields J_w, $w \in T_pM$, and the assignment $w \mapsto J_w$ is linear and injective (distinct w give distinct initial derivatives $D_tJ_w(0) = w$, hence distinct fields). Evaluating at $t = t_0$,

$$J_w(t_0) = d(\exp_p)_{t_0v}(t_0\,w) = t_0\,d(\exp_p)_{t_0v}(w),$$

using linearity of the differential. Since $t_0 > 0$, this vanishes if and only if $d(\exp_p)_{t_0v}(w) = 0$. Now $q = \gamma(t_0)$ is conjugate to p if and only if there is some $w \neq 0$ with J_w not identically zero and $J_w(t_0) = 0$. A field J_w with $w \neq 0$ is not identically zero, because $D_tJ_w(0) = w \neq 0$. Therefore q is conjugate to p if and only if there exists $w \neq 0$ with $d(\exp_p)_{t_0v}(w) = 0$, that is, if and only if $d(\exp_p)_{t_0v}$ has nontrivial kernel, which is exactly the statement that it is singular.

3. The multiplicity of q is $\dim \mathcal{J}_{0,t_0}$ (definition 5.3.1). Every field in $\mathcal{J}_{0,t_0}$ vanishes at 0, so by part (1) it equals J_w for a unique w, and the map $J \mapsto D_tJ(0)$ is a linear isomorphism from $\mathcal{J}_{0,t_0}$ onto

$$\{w \in T_pM \mid J_w(t_0) = 0\} = \bigl\{w \in T_pM \bigm| d(\exp_p)_{t_0v}(w) = 0\bigr\} = \ker d(\exp_p)_{t_0v},$$

where the first equality is the computation of part (2) and the injectivity of $w \mapsto J_w$ gives the isomorphism. Hence $\dim \mathcal{J}_{0,t_0} = \dim\ker d(\exp_p)_{t_0v}$, as claimed.

This establishes all three parts, completing the proof.

The theorem converts a statement about Jacobi fields, which are solutions of a differential equation, into a statement about the rank of a single linear map, the differential of the exponential map. Both viewpoints are needed. The Jacobi side is where the curvature enters and where the constant-curvature models are computed; the exponential side is where conjugate points connect to the failure of $\exp_p$ to be a chart, and hence to the geometry of geodesic balls. The formula (5.5) is the bridge, and it says something concrete: to differentiate the exponential map in a direction w at the point t_0v, follow the Jacobi field that starts at $\gamma(0)$ with zero value and initial velocity w, and read off its value at time t_0, up to the factor t_0. A conjugate point is where some such Jacobi field, having started at zero, has returned to zero, so that the exponential map has collapsed that infinitesimal direction. The absence of conjugate points on $[0, b]$ is thus equivalent to $\exp_p$ being an immersion, hence a local diffeomorphism, along the whole radial geodesic, a fact used constantly in the minimization theory of section 5.4.

One consequence is worth recording before the curvature hypotheses are imposed. A geodesic has only finitely many conjugate points on any compact parameter interval, and so, whenever conjugate points exist at all, there is a well-defined *first* one. The mechanism is the linear structure of the Jacobi equation, not any general property of smooth functions: a C^∞ function can vanish on a whole subinterval without being identically zero, so isolation of zeros must come from the differential equation, and it does.

Set up the count in a frame. Let $(E_1, \ldots, E_{n-1})$ be a parallel orthonormal frame of the normal bundle $\dot\gamma^\perp$ along γ, obtained by parallel transport of an orthonormal basis of $\dot\gamma(0)^\perp$, and let $J_1, \ldots, J_{n-1}$ be the normal Jacobi fields with $J_i(0) = 0$ and $D_tJ_i(0) = E_i(0)$. These are linearly independent (their initial derivatives $E_1(0), \ldots, E_{n-1}(0)$ are), and by the discussion following definition 5.3.1 every Jacobi field vanishing at 0 and at a second time is normal, so $J_1, \ldots, J_{n-1}$ span the space of normal Jacobi fields vanishing at 0, which is the only space that can produce conjugate points. Let $A(t)$ be the $(n-1) \times (n-1)$ matrix whose ith column records the components of $J_i(t)$ in the frame, $J_i(t) = \sum_k A_{ki}(t)E_k(t)$; then $A(0) = 0$ and, since the E_k are parallel, $A'(0) = I$. A point $\gamma(t_0)$ with $t_0 \in (0, b]$ is conjugate to p exactly when some nonzero combination $\sum_i c_iJ_i$ vanishes at t_0, that is, exactly when $A(t_0)c = 0$ for some $c \neq 0$, so the conjugate parameters are the zeros of $t \mapsto \det A(t)$ on $(0, b]$.

The kernel data at a conjugate parameter is constrained by a conserved quantity. Writing the curvature operator $S(t)\colon \dot\gamma^\perp \to \dot\gamma^\perp$, $S(t)X = R(X, \dot\gamma)\dot\gamma$, in the frame gives a symmetric matrix $S(t)$, symmetric because the pair symmetry of the curvature tensor (proposition 4.2.1) yields $\langle R(X, \dot\gamma)\dot\gamma, Y\rangle = \langle R(Y, \dot\gamma)\dot\gamma, X\rangle$; and since the E_k are parallel, the Jacobi equation for each column reads $A''(t) + S(t)A(t) = 0$ in matrix form. The Wronskian-type matrix $W(t) = A'(t)^\intercal A(t) - A(t)^\intercal A'(t)$ has derivative

$$W'(t) = A''^\intercal A - A^\intercal A'' = (-SA)^\intercal A - A^\intercal(-SA) = -A^\intercal S^\intercal A + A^\intercal SA = 0,$$

using $S^\intercal = S$, so W is constant; and $W(0) = 0$ because $A(0) = 0$. Hence $A'(t)^\intercal A(t) = A(t)^\intercal A'(t)$ for all t: the matrix $A^\intercal A'$ is symmetric throughout.

Each conjugate parameter is an isolated zero. Fix $t_0 \in (0, b]$ with $\det A(t_0) = 0$ and choose $0 \neq c \in \ker A(t_0)$. The field $J = \sum_i c_iJ_i$ is a nonzero normal Jacobi field with $J(0) = 0$ and $J(t_0) = 0$; because the frame (E_k) is parallel, its components in the frame are the entries of $A(t)c$ and those of D_tJ are the entries of $A'(t)c$. Were $A'(t_0)c = 0$ as well, the field J would satisfy $J(t_0) = 0$ and $D_tJ(t_0) = 0$, forcing $J \equiv 0$ by the uniqueness of Jacobi fields from their value and covariant derivative at t_0 (theorem 5.2.3, whose linear-ODE proof determines a solution equally from data at any initial time), contradicting $c \neq 0$; hence $A'(t_0)c \neq 0$, so the restriction of $A'(t_0)$ to $\ker A(t_0)$ is injective.

The conserved symmetry pins down its image: for $c \in \ker A(t_0)$ and any $d \in \mathbb{R}^{n-1}$,

$$\langle A'(t_0)c, A(t_0)d\rangle = c^\mathsf{T} A'(t_0)^\mathsf{T} A(t_0)d = c^\mathsf{T} A(t_0)^\mathsf{T} A'(t_0)d = (A(t_0)c)^\mathsf{T} A'(t_0)d = 0,$$

so $A'(t_0)\big(\ker A(t_0)\big)$ is orthogonal to $\operatorname{im} A(t_0)$. Let $m = \dim\ker A(t_0) \geq 1$, so $\dim\operatorname{im} A(t_0) = n-1-m$. Choose a basis $c^{(1)}, \dots, c^{(n-1)}$ of $\mathbb{R}^{n-1}$ whose first m vectors span $\ker A(t_0)$. For $j \leq m$ the column $A(t)c^{(j)}$ vanishes at t_0, so $A(t)c^{(j)} = (t-t_0)B_j(t)$ with B_j smooth and $B_j(t_0) = A'(t_0)c^{(j)}$. Expanding the determinant multilinearly in the columns,

$$\det\big(A(t)c^{(1)}, \dots, A(t)c^{(n-1)}\big) = (t-t_0)^m\,\varphi(t),$$

where

$$\varphi(t) = \det\big(B_1(t), \dots, B_m(t), A(t)c^{(m+1)}, \dots, A(t)c^{(n-1)}\big)$$

The left side is $\det A(t)$ times the fixed nonzero factor $\det(c^{(1)}, \dots, c^{(n-1)})$. At $t = t_0$ the first m columns of φ are $A'(t_0)c^{(1)}, \dots, A'(t_0)c^{(m)}$, a basis of $A'(t_0)(\ker A(t_0))$ by injectivity, and the last $n-1-m$ columns are $A(t_0)c^{(m+1)}, \dots, A(t_0)c^{(n-1)}$, a basis of $\operatorname{im} A(t_0)$; by the orthogonality just shown these two spaces meet only in 0, and their dimensions sum to $n-1$, so the $n-1$ columns are linearly independent and $\varphi(t_0) \neq 0$. Thus $\det A(t) = (t-t_0)^m\psi(t)$ with ψ smooth and $\psi(t_0) \neq 0$, so $\det A$ is nonzero on a punctured neighborhood of t_0 and t_0 is an isolated zero of finite order m, the order equal to the multiplicity of the conjugate point.

Isolated zeros in the compact interval $[0, b]$ are finite in number: an infinite set of zeros would accumulate at some $\bar{t} \in [0, b]$ by the Bolzano–Weierstrass theorem, and $\det A(\bar{t}) = 0$ by continuity, but then $\bar{t}$ would be a non-isolated zero, contradicting the previous paragraph (the argument there applies to every zero, including $\bar{t}$). Hence the conjugate points along γ are finite in number and do not accumulate, and there is a well-defined *first* conjugate point, the smallest zero of $\det A$ in $(0, b]$, whenever any exist at all. This recovers the exponential-map picture. By part (1) of theorem 5.3.2 the columns satisfy $J_i(t) = t\, d(\exp_p)_{tv}(E_i(0))$, so $A(t)$ is t times the matrix of $d(\exp_p)_{tv}$ restricted to $\dot{\gamma}^\perp$, and $\det A(t) = t^{n-1}\det\big(d(\exp_p)_{tv}|_{\dot{\gamma}^\perp}\big)$. For $t > 0$ this vanishes exactly when $d(\exp_p)_{tv}$ is singular, since $d(\exp_p)_{tv}$ carries the radial direction v to the nonzero radial geodesic velocity and preserves the orthogonal splitting into radial and normal parts (theorem 3.3.4), so its singularity is that of its normal block. Thus the conjugate parameters are precisely the zeros of $t \mapsto \det\big(d(\exp_p)_{tv}\big)$ on $(0, b]$, in agreement with theorem 5.3.2, and the finiteness just established is the finiteness of those zeros.

Whether any exist is decided by the sign of the curvature. The restoring term in the Jacobi equation has the sign of the sectional curvature, and where that sign is nonpositive the term cannot bend a nonzero Jacobi field back to the zero it started from. The next proposition makes this precise, and it is the mechanism behind the global rigidity of nonpositively curved spaces.

Proposition 5.3.3: Nonpositive Curvature Excludes Conjugate Points

Let (M, g) be a Riemannian manifold with sectional curvature $K \leq 0$, that is, $K(\Pi) \leq 0$ for every 2-plane Π at every point (definition 4.3.1). Then no geodesic in M has a pair of conjugate points. Equivalently, for every $p \in M$ the differential $d(\exp_p)_v$ is nonsingular at every v in the domain of $\exp_p$.

***Proof*:**
Let $\gamma\colon [0,b] \to M$ be a geodesic and J a Jacobi field along γ with $J(0) = 0$. It suffices to show that $J(t) \neq 0$ for all $t \in (0,b]$ unless $J \equiv 0$; then no nonzero Jacobi field vanishes at two points, so γ has no conjugate points, and by theorem 5.3.2 this is the asserted nonsingularity of $d(\exp_p)_v$.

Consider the smooth function $f(t) = |J(t)|^2 = \langle J, J\rangle$ on $[0,b]$. Differentiating with metric compatibility (definition 2.2.4), and writing D_t for the covariant derivative along γ,

$$f'(t) = \frac{d}{dt}\langle J, J\rangle = 2\langle D_t J, J\rangle$$

Differentiating once more,

$$f''(t) = 2\left\langle D_t^2 J, J\right\rangle + 2\langle D_t J, D_t J\rangle = 2\left\langle D_t^2 J, J\right\rangle + 2|D_t J|^2$$

The Jacobi equation (5.1) gives $D_t^2 J = -R(J,\dot\gamma)\dot\gamma$, so

$$\left\langle D_t^2 J, J\right\rangle = -\langle R(J,\dot\gamma)\dot\gamma, J\rangle = -\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J),$$

using the $(0,4)$ notation $\operatorname{Riem}(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle$. The quantity $\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J)$ is related to sectional curvature: for a geodesic $\dot\gamma$ is nowhere zero (a geodesic has constant speed, and $J(0) = 0$ with $J \not\equiv 0$ forces the speed to be positive on the interval of interest), and where J and $\dot\gamma$ are linearly independent, spanning a 2-plane Π,

$$\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J) = K(\Pi)\big(|J|^2|\dot\gamma|^2 - \langle J,\dot\gamma\rangle^2\big)$$

by the definition of sectional curvature (definition 4.3.1); where J and $\dot\gamma$ are dependent, the Gram determinant $|J|^2|\dot\gamma|^2 - \langle J,\dot\gamma\rangle^2$ vanishes and, by the same identity read as an equality of quadratic expressions in J, so does $\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J)$. In either case

$$\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J) = K(\Pi)\big(|J|^2|\dot\gamma|^2 - \langle J,\dot\gamma\rangle^2\big) \leq 0,$$

because $K \leq 0$ by hypothesis and the Gram determinant is nonnegative by the Cauchy–Schwarz inequality. Therefore $\left\langle D_t^2 J, J\right\rangle = -\operatorname{Riem}(J,\dot\gamma,\dot\gamma,J) \geq 0$, and

$$f''(t) = 2\left\langle D_t^2 J, J\right\rangle + 2|D_t J|^2 \geq 2|D_t J|^2 \geq 0 \tag{5.6}$$

So $f = |J|^2$ is a convex function on $[0,b]$.

Now suppose, for contradiction, that $J(t_0) = 0$ for some $t_0 \in (0,b]$, while $J \not\equiv 0$. Then $f(0) = 0$ and $f(t_0) = 0$, and $f \geq 0$; a convex nonnegative function vanishing at two points $0 < t_0$ vanishes on the whole interval between them, so $f \equiv 0$ on $[0,t_0]$, that is, $J \equiv 0$ on $[0,t_0]$. In particular $J(0) = 0$ and $D_t J(0) = 0$ (the derivative of the identically zero field), so by uniqueness of Jacobi fields from initial data (theorem 5.2.3) $J \equiv 0$ on all of $[0,b]$, contradicting $J \not\equiv 0$. Hence no nonzero Jacobi field with $J(0) = 0$ vanishes again on $(0,b]$, so γ has no conjugate points, completing the proof.

The proof extracts the whole force of the hypothesis from the single inequality (5.6): nonpositive curvature makes $|J|^2$ convex, and a nonnegative convex function that starts at zero and is not identically zero can never return to zero. Geometrically, the geodesics emanating from p spread at least as fast as in Euclidean space and never refocus. The equivalent statement, that $\exp_p$ is everywhere nonsingular, is the local half of the Cartan–Hadamard theorem, which upgrades it on a complete simply connected manifold of nonpositive curvature to the conclusion that $\exp_p$ is a diffeomorphism of all of T_pM onto M; the completeness argument belongs to the next chapter and is flagged here only as the destination of this proposition. The flat case $K \equiv 0$ is the boundary: there $f'' = 2\,|D_tJ|^2$, and a Jacobi field with $J(0) = 0$ has $J(t) = t\,D_tJ(0)$ growing linearly, again with no second zero, consistent with $\mathbb{R}^n$ having no conjugate points.

The complementary picture, positive curvature forcing refocusing, is exhibited in its sharpest form by the sphere, where the constant-curvature computation of section 5.2 makes every conjugate point explicit.

Example 5.3.4: Conjugate points on the round sphere

Let S^n_r be the round sphere of radius r, with constant sectional curvature $c = 1/r^2$ (example 4.3.4), and let γ be a unit-speed geodesic, a great circle traversed at unit speed. Fix $p = \gamma(0)$. By example 5.2.5, applied with $c = 1/r^2$, the normal Jacobi fields along γ with $J(0) = 0$ are exactly

$$J(t) = f(t)\,E(t), \qquad f(t) = \frac{\sin(t/r)}{1/r} = r\sin\Big(\frac{t}{r}\Big),$$

where E is any parallel unit field along γ orthogonal to $\dot\gamma$ (the coefficient function solves $f''+cf = 0$, $f(0) = 0$, $f'(0) = 1$). Choose a parallel orthonormal frame $(E_1, \ldots, E_{n-1})$ of the normal bundle along γ, obtained by parallel transport of an orthonormal basis of $\dot\gamma(0)^\perp$; each E_i is parallel and unit, and

$$J_i(t) = r\sin\Big(\frac{t}{r}\Big)E_i(t), \qquad i = 1, \ldots, n-1,$$

are $n-1$ linearly independent normal Jacobi fields vanishing at $t = 0$.

Location of the conjugate points. The common coefficient $r\sin(t/r)$ vanishes precisely when t/r is an integer multiple of π, that is, at $t = \pi r,\ 2\pi r,\ 3\pi r, \ldots$. At each such t_0 all $n-1$ fields J_i vanish, so each is a nonzero Jacobi field vanishing at 0 and at t_0, and $\gamma(t_0)$ is conjugate to p. The *first* conjugate point is at

$$t_0 = \pi r,$$

the point $\gamma(\pi r)$, which is the antipode $-p$ of p: the great circle of length $2\pi r$ reaches the point diametrically opposite p after arc length πr. This is exactly the refocusing described at the start of the section, where all great circles from p meet again at $-p$.

Multiplicity. At $t_0 = \pi r$ the space $\mathcal{J}_{0,t_0}$ of Jacobi fields vanishing at both ends contains the $n-1$ independent fields $J_1, \ldots, J_{n-1}$, so its dimension is at least $n-1$. It is at most $n-1$ by the general bound following definition 5.3.1: every field in $\mathcal{J}_{0,t_0}$ is normal, and the normal Jacobi fields vanishing at 0 form an $(n-1)$-dimensional space. Hence the multiplicity of the first conjugate point $\gamma(\pi r)$ is exactly

$$\dim \mathcal{J}_{0,\pi r} = n-1$$

The same count holds at every conjugate point $t_0 = k\pi r$, $k \geq 1$. For the two-sphere S^2_r the multiplicity is 1, the single normal direction; for S^n_r it is $n-1$, the full transverse space, which is why the antipode is a point of total refocusing.

The exponential-map reading. By theorem 5.3.2, the differential $d(\exp_p)_v$ at $|v| = \pi r$ is singular with $(n-1)$-dimensional kernel: the exponential map on the sphere of radius πr in $T_pS^n_r$ collapses the entire coordinate sphere to the single point $-p$, matching the direct computation $\exp_p(v) =$

$(\cos(|v|/r))p + \cdots$ of example 3.2.4, which sends every v with $|v| = \pi r$ to $-p$. Below the first conjugate radius, on $|v| < \pi r$, the differential is everywhere nonsingular and $\exp_p$ is a diffeomorphism, matching the Jacobi-field computation that no J_i vanishes on $(0, \pi r)$.

The contrast with the other two model space forms completes the trichotomy. Hyperbolic space H^n has $c = -1 < 0$ and Euclidean space $\mathbb{R}^n$ has $c = 0$, so both satisfy $K \leq 0$ and have *no* conjugate points along any geodesic, by proposition 5.3.3; explicitly, the Jacobi coefficient of example 5.2.5 is $\sinh(t)$ for H^n and t for $\mathbb{R}^n$, neither of which returns to zero for $t > 0$. Positive curvature refocuses the geodesics at finite distance, zero curvature lets them spread linearly, and negative curvature drives them apart exponentially; the sphere, the plane, and the hyperbolic space realize the three cases, with the location of the first conjugate point $\pi r = \pi/\sqrt{c}$ on the sphere decreasing as the curvature grows.

5.4 The Index Form and the Index Lemma

The second variation of energy in section 5.1 attached to every geodesic a quadratic form on the vector fields that deform it: the value $E''(0)$ produced by a proper variation with variation field V. That form is the true measure of whether a geodesic minimizes. A geodesic is a critical point of the energy, so its first variation vanishes for every proper variation; what separates a minimum from a saddle is the sign of the second variation, and a single field V with $E''(0) < 0$ is enough to show that the geodesic is beaten by a nearby competitor. This section isolates the quadratic form itself, the *index form* I, studies it as an object in its own right, and proves the two results that convert Jacobi-field information into minimization conclusions.

The center of the analysis is a comparison. Among all fields V with prescribed endpoint values, which one makes $I(V, V)$ smallest? The answer, when the geodesic carries no interior conjugate point, is the Jacobi field with those endpoint values, and it wins strictly against every other competitor. This is the index lemma, and it is the exact statement needed to run the two directions of the minimization theory. In one direction it shows that before the first conjugate point a geodesic cannot be shortened by a small deformation that keeps its endpoints; in the other it shows that past the first conjugate point the geodesic is no longer even a local minimum, because a deformation exists that strictly decreases the energy. The Jacobi fields of section 5.2 and the conjugate points of section 5.3 thereby acquire their decisive variational meaning. Throughout, (M, g) is a Riemannian manifold with Levi-Civita connection ∇, $\gamma\colon [0, b] \to M$ is a unit-speed geodesic, D_t is the covariant derivative along γ, and R is the curvature endomorphism of definition 4.1.2; the curvature sign convention of box 4.1.1 is in force.

Definition 5.4.1: The index form

Let $\gamma\colon [0,b] \to M$ be a unit-speed geodesic. Denote by $\mathcal{V}(\gamma)$ the real vector space of piecewise-smooth vector fields V along γ with $V(0) = V(b) = 0$: fields that are continuous on $[0,b]$, smooth on each subinterval of some partition $0 = t_0 < t_1 < \cdots < t_m = b$, and vanishing at the endpoints. The *index form* of γ is the symmetric bilinear form $I\colon \mathcal{V}(\gamma) \times \mathcal{V}(\gamma) \to \mathbb{R}$ given by

$$I(V,W) = \int_0^b \Big(\langle D_t V, D_t W\rangle - \langle R(V,\dot\gamma)\dot\gamma, W\rangle \Big)\, dt$$

the integral being taken over each smooth subinterval and summed, since D_tV and D_tW may jump at the finitely many breakpoints while the integrand stays bounded.

The index form is exactly the second variation of energy, now regarded as a bilinear pairing rather than a value on a single deformation. Its diagonal reproduces the formula of section 5.1:

$$I(V,V) = \int_0^b \Big(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle \Big)\, dt = E''(0)$$

For a smooth field V vanishing at the endpoints this is precisely the second-variation formula of theorem 5.1.1. For a merely piecewise-smooth field the identity still holds, but theorem 5.1.1 does not deliver it directly, since that theorem is stated for a smooth variation with a smooth variation field; the extension to the piecewise-smooth fields of $\mathcal{V}(\gamma)$ is recorded in lemma 5.4.2 below, whose proof splits the variation over the smooth subintervals and checks that the corners contribute no boundary term. Symmetry in V and W is visible in the first term and, for the second, follows from the pair symmetry of the curvature tensor: $\langle R(V,\dot\gamma)\dot\gamma, W\rangle = \mathrm{Riem}(V,\dot\gamma,\dot\gamma,W) = \mathrm{Riem}(W,\dot\gamma,\dot\gamma,V) = \langle R(W,\dot\gamma)\dot\gamma, V\rangle$ by proposition 4.2.1. Bilinearity is immediate from the linearity of D_t and of R in the relevant slots. The two terms pull in opposite directions. The first, the derivative energy $\int |D_tV|^2$, is a nonnegative penalty on how fast V turns, and it alone would make I positive definite. The second subtracts a curvature term whose sign is that of the sectional curvature of the plane spanned by V and $\dot\gamma$, since $\langle R(V,\dot\gamma)\dot\gamma, V\rangle = K(V,\dot\gamma)\ |V \wedge \dot\gamma|^2$; positive curvature therefore lowers I and can make it negative, while nonpositive curvature only raises it. The whole minimization theory is the contest between these two terms.

The identification of $I(V,V)$ with a second variation of energy is needed below precisely for the piecewise-smooth fields of $\mathcal{V}(\gamma)$, since the field that defeats minimization past a conjugate point has a corner. The following lemma supplies that identification. Its content is that the corner costs nothing: the second-variation computation is carried out on each smooth piece, and the boundary terms produced at the breakpoints cancel.

Lemma 5.4.2: Second variation for a piecewise-smooth field

Let $\gamma\colon [0,b] \to M$ be a geodesic and let $V \in \mathcal{V}(\gamma)$ be a piecewise-smooth field vanishing at the endpoints, smooth on each subinterval of a partition $0 = t_0 < t_1 < \cdots < t_m = b$. Let $\Gamma(s,t) = \exp_{\gamma(t)}(sV(t))$ be the proper variation built from V as in example 3.5.2, defined for $|s|$ small. Then $s \mapsto E(\gamma_s)$ is twice differentiable at $s = 0$ and

$$E''(0) = \int_0^b \Big(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle \Big)\, dt = I(V,V)$$

***Proof*:**
The map Γ is smooth on each strip $(-\varepsilon, \varepsilon) \times [t_{i-1}, t_i]$, being the composition of the exponential map with the smooth field $sV(t)$ on that strip, and continuous across the breakpoints because V is continuous. Write $T = \partial_t \Gamma$ and $S = \partial_s \Gamma$ for the coordinate velocity fields; at $s = 0$ they restrict to $T = \dot\gamma$ and $S = V$ along γ. The energy decomposes over the smooth pieces,

$$E(\gamma_s) = \sum_{i=1}^{m} E_i(s), \qquad E_i(s) = \tfrac{1}{2} \int_{t_{i-1}}^{t_i} \langle T, T \rangle \, dt$$

and each E_i is a smooth function of s because Γ is smooth on the closed strip over $[t_{i-1}, t_i]$.

Fix one subinterval and differentiate E_i twice in s, exactly as in the proof of theorem 5.1.1, whose steps use only the smoothness of Γ on that strip and not any endpoint condition. Metric compatibility of ∇, the symmetry lemma $D_s T = D_t S$ (lemma 3.5.3), and the definition of R (definition 4.1.2) give, at $s = 0$ where $S = V$, $T = \dot\gamma$, and $D_t \dot\gamma = 0$,

$$E_i''(0) = \int_{t_{i-1}}^{t_i} \Big(|D_t V|^2 - \langle R(V, \dot\gamma)\dot\gamma, V \rangle \Big) \, dt + \Big[\langle D_s S, \dot\gamma \rangle \Big]_{t_{i-1}}^{t_i}$$

the boundary term arising from the same integration by parts as in Step 3 there, namely $\int_{t_{i-1}}^{t_i} \langle D_t D_s S, \dot\gamma \rangle \, dt = \int_{t_{i-1}}^{t_i} \frac{d}{dt} \langle D_s S, \dot\gamma \rangle \, dt$ once $\langle D_s S, D_t \dot\gamma \rangle = 0$ is used along the geodesic γ. Unlike the smooth proper case, this boundary term is not discarded by an endpoint condition; instead it vanishes for a structural reason. For each fixed t the curve $s \mapsto \Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ is a geodesic, so its covariant acceleration vanishes for all s: $D_s S = D_s \, \partial_s \Gamma \equiv 0$ on the whole rectangle. Hence $\langle D_s S, \dot\gamma \rangle \equiv 0$ at $s = 0$, and the boundary term $[\langle D_s S, \dot\gamma \rangle]_{t_{i-1}}^{t_i}$ is zero on every subinterval.

Summing over $i = 1, \ldots, m$, the interval integrals combine into a single integral over $[0, b]$, whose integrand $|D_t V|^2 - \langle R(V, \dot\gamma)\dot\gamma, V \rangle$ is bounded and has at most the finitely many jumps of $D_t V$ at the breakpoints, so the sum is exactly $\int_0^b (|D_t V|^2 - \langle R(V, \dot\gamma)\dot\gamma, V \rangle) \, dt$. Therefore

$$E''(0) = \sum_{i=1}^{m} E_i''(0) = \int_0^b \Big(|D_t V|^2 - \langle R(V, \dot\gamma)\dot\gamma, V \rangle \Big) \, dt = I(V, V)$$

the last equality by definition 5.4.1, as required.

A useful first computation records what the index form does to a field that is proportional to a fixed profile in a parallel direction, on a space of constant curvature. This is the setting in which every quantity is explicit, and it will let the abstract comparison below be checked against a known answer.

Example 5.4.3: The index form on a constant-curvature space
Let (M, g) have constant sectional curvature c, and let $\gamma \colon [0, b] \to M$ be a unit-speed geodesic. Fix a parallel unit field E along γ with $E \perp \dot\gamma$, which exists because parallel transport preserves both length and the angle with $\dot\gamma$ (the latter is parallel along γ). For a scalar function $f \colon [0, b] \to \mathbb{R}$ with $f(0) = f(b) = 0$, smooth, consider the normal field $V = fE \in \mathcal{V}(\gamma)$. Since E is parallel, $D_t V = f'E$, so $|D_t V|^2 = (f')^2$. For the curvature term, the constant-curvature form of proposition 4.5.3 gives

$$R(V, \dot\gamma)\dot\gamma = c\big(\langle \dot\gamma, \dot\gamma \rangle \, V - \langle V, \dot\gamma \rangle \, \dot\gamma \big) = c \, f \, E$$

using $\langle \dot\gamma, \dot\gamma \rangle = 1$ and $\langle V, \dot\gamma \rangle = f \langle E, \dot\gamma \rangle = 0$, so $\langle R(V, \dot\gamma)\dot\gamma, V \rangle = cf^2$. Hence

$$I(V, V) = \int_0^b \left((f')^2 - cf^2\right) dt$$

On the round sphere S_r^n, where $c = 1/r^2$, take $b = \pi r$ and $f(t) = \sin(t/r)$, which vanishes at $t = 0$ and $t = \pi r$. Then $f'(t) = \frac{1}{r}\cos(t/r)$, and

$$I(V, V) = \int_0^{\pi r} \left(\tfrac{1}{r^2}\cos^2(t/r) - \tfrac{1}{r^2}\sin^2(t/r)\right) dt = \tfrac{1}{r^2}\int_0^{\pi r} \cos(2t/r)\, dt = 0$$

so this particular field lies in the null space of I. Indeed $V = \sin(t/r)E$ is the Jacobi field of example 5.2.5 vanishing at both ends of the arc to the antipode, and the vanishing of $I(V, V)$ is the borderline case the index lemma describes at a conjugate point. On hyperbolic space, where $c = -1$, the curvature term $-cf^2 = f^2$ adds to the derivative energy, so $I(V, V) = \int_0^b ((f')^2 + f^2)\, dt > 0$ for every nonzero f vanishing at the endpoints; the index form is positive definite and there is no competing minimizer, in agreement with the absence of conjugate points in nonpositive curvature.

The comparison at the heart of the section rests on an integration by parts that rewrites $I(V, W)$ so that the Jacobi operator appears. For a field V that is smooth on a subinterval $[t_{i-1}, t_i]$, metric compatibility of ∇ gives $\frac{d}{dt}\langle D_tV, W \rangle = \langle D_t^2V, W \rangle + \langle D_tV, D_tW \rangle$, so

$$\int_{t_{i-1}}^{t_i} \langle D_tV, D_tW \rangle\, dt = \Big[\langle D_tV, W \rangle\Big]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle D_t^2V, W \rangle\, dt$$

Summing over the subintervals and using $W(0) = W(b) = 0$, the interior boundary terms collect into a sum over breakpoints, and the index form becomes

$$I(V, W) = -\int_0^b \langle D_t^2V + R(V, \dot\gamma)\dot\gamma, W \rangle\, dt - \sum_{i=1}^{m-1} \langle \Delta_i D_tV, W(t_i) \rangle \tag{5.7}$$

where $\Delta_i D_tV = D_tV(t_i^+) - D_tV(t_i^-)$ is the jump in the covariant derivative of V at the interior breakpoint t_i. Formula (5.7) holds whenever V is piecewise smooth and $W \in \mathcal{V}(\gamma)$; it makes the connection between the index form and the Jacobi operator $J \mapsto D_t^2J + R(J, \dot\gamma)\dot\gamma$ of definition 5.2.1 explicit. A Jacobi field is smooth, so it has no jumps, and the operator applied to it vanishes; both terms on the right then drop, a fact used repeatedly below.

Theorem 5.4.4: The Index Lemma

Let $\gamma\colon [0, b] \to M$ be a unit-speed geodesic with no interior point conjugate to $\gamma(0)$ along γ; that is, $\gamma(t)$ is not conjugate to $\gamma(0)$ for any $t \in (0, b]$. Fix $w \in T_{\gamma(b)}M$ with $w \perp \dot\gamma(b)$, and let J be the unique Jacobi field along γ with $J(0) = 0$ and $J(b) = w$. Then for every piecewise-smooth normal field V along γ with $V(0) = 0$ and $V(b) = w$,

$$I(J, J) \leq I(V, V)$$

with equality if and only if $V = J$ on $[0, b]$.

Before the proof, the hypotheses deserve comment. The absence of interior conjugate points is what guarantees both that the boundary-value Jacobi field J with $J(0) = 0$, $J(b) = w$ exists and is unique, and that the Jacobi fields vanishing at 0 furnish a moving frame in which any competitor V can be expanded. The restriction to normal fields (perpendicular to $\dot{\gamma}$) costs nothing: the tangential part of any field in $\mathcal{V}(\gamma)$ contributes to I exactly the derivative energy of a scalar function vanishing at both ends, which is strictly positive unless that function is zero, so tangential components only increase I and the sharpest comparison is the normal one. The conclusion says that the Jacobi field is the unique minimizer of the index form subject to its boundary values: no other admissible field bends as economically.

***Proof*:**
The proof proceeds in three steps: first the existence of a Jacobi frame adapted to the no-conjugate-point hypothesis, then the expansion of an arbitrary competitor in that frame, and finally the algebraic identity that exhibits the difference $I(V,V) - I(J,J)$ as a nonnegative quantity vanishing only when $V = J$.

Step 1: a Jacobi frame with no interior zeros. Let $\{e_1, \dots, e_{n-1}\}$ be an orthonormal basis of the orthogonal complement of $\dot{\gamma}(0)$ in $T_{\gamma(0)}M$, and extend each e_α to a parallel field E_α along γ; the E_α remain orthonormal and normal to $\dot{\gamma}$ for all t, since parallel transport is a linear isometry preserving the angle with the parallel field $\dot{\gamma}$. For each $\alpha = 1, \dots, n-1$ let J_α be the Jacobi field with $J_\alpha(0) = 0$ and $D_t J_\alpha(0) = E_\alpha(0) = e_\alpha$. Each J_α is normal to $\dot{\gamma}$: a Jacobi field with $J_\alpha(0) = 0$ and $D_t J_\alpha(0) \perp \dot{\gamma}$ stays normal by the tangential-normal splitting of proposition 5.2.4. The $n-1$ fields $J_1, \dots, J_{n-1}$ are linearly independent as solutions of the Jacobi equation, because their initial data $(0, e_\alpha)$ are linearly independent.

The no-conjugate-point hypothesis makes these fields a frame away from $t = 0$. Fix $t \in (0, b]$. A relation $\sum_\alpha c_\alpha J_\alpha(t) = 0$ would exhibit the nonzero normal Jacobi field $\sum_\alpha c_\alpha J_\alpha$ (nonzero because the c_α are not all zero and the fields are independent) as vanishing at both 0 and t, making $\gamma(t)$ conjugate to $\gamma(0)$; this is excluded for $t \in (0, b]$ by hypothesis. Hence $\{J_1(t), \dots, J_{n-1}(t)\}$ is a basis of the $(n-1)$-dimensional space of normal vectors at $\gamma(t)$ for every $t \in (0, b]$. In particular the target w is in their span at $t = b$, so there are constants $a_1, \dots, a_{n-1}$ with $w = \sum_\alpha a_\alpha J_\alpha(b)$, and $J := \sum_\alpha a_\alpha J_\alpha$ is the Jacobi field with $J(0) = 0$, $J(b) = w$. This proves existence; uniqueness holds because the difference of two such fields is a Jacobi field vanishing at 0 and b, hence zero by the no-conjugate-point hypothesis.

Step 2: expansion of a competitor. Let V be any piecewise-smooth normal field with $V(0) = 0$, $V(b) = w$. On the half-open interval $(0, b]$ the frame $\{J_\alpha\}$ is defined and pointwise a basis of the normal space, so V can be written

$$V(t) = \sum_{\alpha=1}^{n-1} \varphi^\alpha(t)\, J_\alpha(t), \qquad t \in (0, b]$$

for scalar coefficient functions φ^α. Each φ^α is piecewise smooth on $(0, b]$ wherever V is smooth, being obtained from V and the smooth invertible frame $\{J_\alpha(t)\}$ by Cramer's rule. Two facts about the behavior at the endpoints are needed. At $t = b$, the equality $V(b) = w = \sum_\alpha a_\alpha J_\alpha(b)$ and the independence of the $J_\alpha(b)$ force $\varphi^\alpha(b) = a_\alpha$. At $t = 0$, the coefficients extend continuously with a finite limit: expanding $J_\alpha(t) = tE_\alpha(0) + O(t^2)$ near 0 (since $J_\alpha(0) = 0$, $D_t J_\alpha(0) = e_\alpha$) and $V(t) = O(t)$ (since $V(0) = 0$ and V is Lipschitz near 0, being piecewise smooth), the matrix relating V to the J_α shows that each $\varphi^\alpha(t)$ has a finite limit as $t \to 0^+$, and moreover $t\,(\varphi^\alpha)'(t) \to 0$; these are recorded now and used in the boundary evaluation of Step 3. The field $J = \sum_\alpha a_\alpha J_\alpha$ corresponds to the constant coefficients $\varphi^\alpha \equiv a_\alpha$.

Step 3: the key identity. The computation rests on two auxiliary quantities built from the frame.

Define, on any subinterval where V is smooth, the normal fields

$$A = \sum_\alpha (\varphi^\alpha)'\, J_\alpha, \qquad B = \sum_\alpha \varphi^\alpha\, D_t J_\alpha$$

so that $D_t V = A + B$ because $D_t V = \sum_\alpha \big((\varphi^\alpha)' J_\alpha + \varphi^\alpha D_t J_\alpha\big)$. Two identities among the J_α are used. First, the *Lagrange identity* for Jacobi fields: for any two Jacobi fields J_α, J_β the quantity $\langle D_t J_\alpha, J_\beta\rangle - \langle J_\alpha, D_t J_\beta\rangle$ is constant in t, because its derivative is

$$\left\langle D_t^2 J_\alpha, J_\beta\right\rangle - \left\langle J_\alpha, D_t^2 J_\beta\right\rangle = -\left\langle R(J_\alpha,\dot\gamma)\dot\gamma, J_\beta\right\rangle + \left\langle J_\alpha, R(J_\beta,\dot\gamma)\dot\gamma\right\rangle = 0$$

the last equality by the pair symmetry $\langle R(J_\alpha,\dot\gamma)\dot\gamma, J_\beta\rangle = \langle R(J_\beta,\dot\gamma)\dot\gamma, J_\alpha\rangle$ from proposition 4.2.1. Evaluating the constant at $t = 0$, where every $J_\alpha(0) = 0$, gives

$$\langle D_t J_\alpha, J_\beta\rangle = \langle J_\alpha, D_t J_\beta\rangle \qquad \text{for all } t \tag{5.8}$$

Second, $\langle A, B\rangle$ is symmetric under this identity:

$$\langle A, B\rangle = \sum_{\alpha,\beta} (\varphi^\alpha)'\varphi^\beta\, \langle J_\alpha, D_t J_\beta\rangle = \sum_{\alpha,\beta} (\varphi^\alpha)'\varphi^\beta\, \langle D_t J_\alpha, J_\beta\rangle$$

by (5.8), so $\langle A, B\rangle = \sum_{\alpha,\beta} \varphi^\beta (\varphi^\alpha)'\, \langle D_t J_\alpha, J_\beta\rangle$.

Now expand $|D_t V|^2 = |A|^2 + 2\langle A, B\rangle + |B|^2$ and treat the curvature term. Using the Jacobi equation $D_t^2 J_\alpha = -R(J_\alpha,\dot\gamma)\dot\gamma$ and metric compatibility,

$$\langle R(V,\dot\gamma)\dot\gamma, V\rangle = \sum_{\alpha,\beta} \varphi^\alpha\varphi^\beta\, \langle R(J_\alpha,\dot\gamma)\dot\gamma, J_\beta\rangle = -\sum_{\alpha,\beta} \varphi^\alpha\varphi^\beta\, \left\langle D_t^2 J_\alpha, J_\beta\right\rangle$$

The claim is that the integrand of I satisfies, on each smooth subinterval,

$$|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle = |A|^2 + \frac{d}{dt}\langle B, V\rangle \tag{5.9}$$

To verify (5.9), compute the total derivative on the right using metric compatibility:

$$\frac{d}{dt}\langle B, V\rangle = \langle D_t B, V\rangle + \langle B, D_t V\rangle = \langle D_t B, V\rangle + \langle B, A\rangle + |B|^2$$

Here $D_t B = \sum_\alpha \big((\varphi^\alpha)' D_t J_\alpha + \varphi^\alpha D_t^2 J_\alpha\big)$, so

$$\langle D_t B, V\rangle = \sum_{\alpha,\beta} \varphi^\beta (\varphi^\alpha)'\, \langle D_t J_\alpha, J_\beta\rangle + \sum_{\alpha,\beta} \varphi^\alpha\varphi^\beta\, \left\langle D_t^2 J_\alpha, J_\beta\right\rangle = \langle A, B\rangle - \langle R(V,\dot\gamma)\dot\gamma, V\rangle$$

the first sum being $\langle A, B\rangle$ by the symmetric form derived above and the second being $-\langle R(V,\dot\gamma)\dot\gamma, V\rangle$ by the curvature computation. Substituting,

$$\frac{d}{dt}\langle B, V\rangle = \langle A, B\rangle - \langle R(V,\dot\gamma)\dot\gamma, V\rangle + \langle B, A\rangle + |B|^2 = 2\langle A, B\rangle + |B|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle$$

Adding $|A|^2$ to both sides yields $|A|^2 + \frac{d}{dt}\langle B, V\rangle = |A|^2 + 2\langle A, B\rangle + |B|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle = |D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle$, which is (5.9).

Integrate (5.9) over $[0, b]$. The field $\langle B, V\rangle$ is continuous across the interior breakpoints of V: V is continuous there by definition of $\mathcal{V}(\gamma)$, and $B = \sum_\alpha \varphi^\alpha D_t J_\alpha$ is continuous because the J_α are smooth and the φ^α are continuous (they are recovered from the continuous field V by the smooth invertible frame). Hence the total-derivative term telescopes across breakpoints, contributing only the endpoint values:

$$I(V, V) = \int_0^b |A|^2 \, dt + \Big[\langle B, V\rangle\Big]_0^b$$

At $t = b$, $V(b) = w$; at $t = 0$, $V(0) = 0$ and $B(0) = \sum_\alpha \varphi^\alpha(0) D_t J_\alpha(0) = \sum_\alpha \varphi^\alpha(0) e_\alpha$ is finite by Step 2, so $\langle B, V\rangle \to 0$ as $t \to 0^+$. Therefore the boundary term is $\langle B(b), w\rangle$, and

$$I(V, V) = \int_0^b |A|^2 \, dt + \langle B(b), w\rangle$$

The boundary term depends only on the endpoint data. At $t = b$ one has $\varphi^\alpha(b) = a_\alpha$ from Step 2, so $B(b) = \sum_\alpha a_\alpha D_t J_\alpha(b) = D_t J(b)$, and $\langle B(b), w\rangle = \langle D_t J(b), J(b)\rangle$, a quantity fixed by w alone and independent of which competitor V realizes w. Consequently

$$I(V, V) - I(J, J) = \int_0^b |A|^2 \, dt - \int_0^b |A_J|^2 \, dt$$

where A_J is the field A built from the coefficients of J. But J has constant coefficients $\varphi^\alpha \equiv a_\alpha$, so $(\varphi^\alpha)' \equiv 0$ and $A_J \equiv 0$; hence $I(J, J) = \langle D_t J(b), J(b)\rangle$ and

$$I(V, V) - I(J, J) = \int_0^b |A|^2 \, dt = \int_0^b \Big| \sum_\alpha (\varphi^\alpha)' J_\alpha \Big|^2 dt \geq 0$$

This proves $I(J, J) \leq I(V, V)$. Equality holds if and only if $A \equiv 0$ on $[0, b]$, that is $\sum_\alpha (\varphi^\alpha)' J_\alpha \equiv 0$; since the $J_\alpha(t)$ are independent for $t \in (0, b]$, this forces $(\varphi^\alpha)' \equiv 0$, so each φ^α is constant, equal to its value a_α at b. Then $V = \sum_\alpha a_\alpha J_\alpha = J$ on $(0, b]$, and by continuity at 0 on all of $[0, b]$. Thus equality holds exactly when $V = J$, completing the proof.

The index lemma is the technical engine of the minimization theory, and its two consequences run in opposite directions. Read one way, it says that on a geodesic without interior conjugate points the index form is as small as the endpoint data allow when the field is Jacobi, and strictly larger otherwise; this positivity underlies the fact that such geodesics do minimize among nearby curves. Read the other way, it isolates the failure mode: a Jacobi field vanishing at both endpoints is a nonzero element on which the two competing terms of the index form balance exactly, $I(J, J) = \langle D_t J(b), J(b)\rangle = 0$, and this borderline field is the seed of a strictly energy-decreasing deformation once the geodesic is pushed past the conjugate point. That second reading is the next theorem.

The construction that produces a negative direction of the index form past the first conjugate point is a broken-field argument. A Jacobi field J that vanishes at the conjugate parameter t_0 is truncated there and continued by zero; the resulting field lies in $\mathcal{V}(\gamma)$ but is not smooth at t_0, and its index-form value is exactly zero. Adding a small multiple of a smooth field concentrated at t_0, in the direction dictated by the jump in $D_t J$, then lowers the value below zero. The resulting competitor is still a piecewise-smooth field, so its second variation is computed through lemma 5.4.2. The following theorem carries this out.

Theorem 5.4.5: No minimization past a conjugate point

Let $\gamma\colon [0,b] \to M$ be a unit-speed geodesic, and suppose $\gamma(t_0)$ is conjugate to $\gamma(0)$ along γ for some $t_0 \in (0,b)$ strictly interior. Then there is a proper variation of γ whose variation field $W \in \mathcal{V}(\gamma)$ satisfies $I(W,W) = E''(0) < 0$. Consequently γ is not a local minimum of the energy among fixed-endpoint curves, and γ does not minimize length from $\gamma(0)$ to $\gamma(b)$.

***Proof*:**
Let J be a nonzero Jacobi field along γ with $J(0) = 0$ and $J(t_0) = 0$, which exists by the definition of conjugate point (definition 5.3.1). Its derivative at t_0 is nonzero: were $D_t J(t_0) = 0$ as well, then J would be the solution of the Jacobi equation with vanishing data $J(t_0) = 0$, $D_t J(t_0) = 0$, hence identically zero by the uniqueness in theorem 5.2.3, contradicting $J \neq 0$. Put $u := D_t J(t_0) \neq 0$. The field J is normal to $\dot\gamma$: it vanishes at 0 with $D_t J(0)$ necessarily normal, since a tangential Jacobi field has the form $(a+bt)\dot\gamma$ by proposition 5.2.4 and cannot vanish twice unless it is zero, so the tangential part of J is zero and J, together with u, is normal to $\dot\gamma$.

Step 1: the broken field has zero index. Define $Z \in \mathcal{V}(\gamma)$ by truncating J at t_0 and extending by zero:

$$Z(t) = \begin{cases} J(t), & 0 \le t \le t_0 \\ 0, & t_0 \le t \le b \end{cases}$$

This Z is continuous (both pieces give 0 at t_0 since $J(t_0) = 0$), piecewise smooth with a single breakpoint at t_0, normal to $\dot\gamma$, and vanishing at 0 and b; so $Z \in \mathcal{V}(\gamma)$. Its index-form value is computed from (5.7). On $[0,t_0]$ the field $Z = J$ is a smooth Jacobi field, so $D_t^2 Z + R(Z,\dot\gamma)\dot\gamma = 0$ there; on $[t_0,b]$ it is identically zero, so the operator vanishes there too. The integral term in (5.7) thus vanishes. The only breakpoint is t_0, where $D_t Z(t_0^-) = D_t J(t_0) = u$ and $D_t Z(t_0^+) = 0$, giving the jump $\Delta Z := D_t Z(t_0^+) - D_t Z(t_0^-) = -u$. Since $Z(t_0) = J(t_0) = 0$, the breakpoint term $\langle \Delta Z, Z(t_0)\rangle$ also vanishes. Hence

$$I(Z,Z) = 0$$

The field Z is nonzero, so this exhibits a nonzero element of $\mathcal{V}(\gamma)$ on which the index form is zero; the geodesic γ sits exactly at the boundary between minimizing and non-minimizing, and the next step tips it over.

Step 2: a rounding field that lowers the index. Choose a smooth field $Y \in \mathcal{V}(\gamma)$ concentrated near t_0 with

$$\langle Y(t_0), u\rangle = 1, \qquad Y(0) = Y(b) = 0$$

Such a Y exists: let U be the parallel field along γ with $U(t_0) = u$, let $\rho\colon [0,b] \to \mathbb{R}$ be a smooth bump with $\rho(t_0) = 1$ and $\rho(0) = \rho(b) = 0$, and set $Y = \rho U$; then $\langle Y(t_0), u\rangle = \rho(t_0)\,|u|^2 = |u|^2 > 0$, and rescaling Y by $|u|^{-2}$ arranges $\langle Y(t_0), u\rangle = 1$. Now Y is smooth, hence has no jump in $D_t Y$, so by (5.7) the mixed value $I(Z,Y)$ receives no contribution from the integral over the pieces where Z satisfies the Jacobi equation or is zero, and picks up only the breakpoint term of Z at t_0:

$$I(Z,Y) = -\langle \Delta Z, Y(t_0)\rangle = -\langle -u, Y(t_0)\rangle = \langle u, Y(t_0)\rangle = 1$$

Here (5.7) is applied with $V = Z$ (piecewise smooth, with its single jump at t_0) and $W = Y \in \mathcal{V}(\gamma)$; the integral term vanishes as in Step 1 and only the t_0 jump survives. By symmetry of I, $I(Y,Z) = 1$ as well.

Consider the one-parameter family of fields $W_\varepsilon := Z + \varepsilon Y \in \mathcal{V}(\gamma)$. By bilinearity and Step 1,

$$I(W_\varepsilon, W_\varepsilon) = I(Z,Z) + 2\varepsilon\, I(Z,Y) + \varepsilon^2 I(Y,Y) = 2\varepsilon + \varepsilon^2 I(Y,Y)$$

Since $I(Y,Y)$ is a fixed finite number, choosing $\varepsilon < 0$ with $|\varepsilon|$ small makes the linear term dominate: for $\varepsilon < 0$ small enough that $|\varepsilon|\ |I(Y,Y)| < 2$, one has $I(W_\varepsilon, W_\varepsilon) = \varepsilon\big(2+\varepsilon I(Y,Y)\big) < 0$. Fix such an ε and set $W := W_\varepsilon$, a nonzero normal field in $\mathcal{V}(\gamma)$ with $I(W,W) < 0$.

Step 3: conclusion. The field W vanishes at both endpoints, so the construction of example 3.5.2 makes it the variation field of the proper variation $\Gamma(s,t) = \exp_{\gamma(t)}(sW(t))$ of γ. This W is piecewise smooth with a single corner at t_0, coming from the corner of Z; it is not a smooth field, so the smooth second-variation theorem does not apply to it directly. The bridge is lemma 5.4.2, which gives the second variation of energy of this variation as $E''(0) = I(W,W) < 0$. Because γ is a geodesic, the first variation vanishes for every proper variation (theorem 3.5.4), so along the variation Γ the energy $s \mapsto E(\gamma_s)$ has vanishing first derivative and negative second derivative at $s = 0$; hence $E(\gamma_s) < E(\gamma)$ for all small $s \neq 0$, and γ is not a local minimum of the energy among fixed-endpoint curves. Finally, minimizing length would force minimizing energy: for a fixed-endpoint competitor σ on $[0,b]$ one has $L(\sigma)^2 \leq 2b\, E(\sigma)$ with equality for constant speed, and the unit-speed geodesic γ satisfies $L(\gamma)^2 = 2b\, E(\gamma)$, so a curve γ_s with $E(\gamma_s) < E(\gamma)$ has $L(\gamma_s)^2 \leq 2b\, E(\gamma_s) < 2b\, E(\gamma) = L(\gamma)^2$, whence $L(\gamma_s) < L(\gamma)$. Thus γ does not minimize length either, as required.

The two theorems complete what section 5.1 began. Together with the local minimization of theorem 3.4.3 and the identification of conjugate points with singularities of the exponential map in theorem 5.3.2, they give a clean dichotomy along a geodesic emanating from $p = \gamma(0)$: up to the first conjugate point the geodesic is a local minimizer of energy and length, and immediately past it the geodesic fails to minimize, beaten by an explicit deformation. The constant-curvature computation of example 5.2.5 makes this quantitative on the round sphere S^n_r, where the first conjugate point sits at arc length πr: a great-circle arc shorter than πr minimizes, and one longer than πr does not, recovering the elementary fact that the long way around a great circle is not the shortest path. On a manifold of nonpositive curvature there are no conjugate points at all (proposition 5.3.3), the index form is positive definite on every geodesic segment, and no geodesic ever loses its minimizing property to a conjugate point.

The count of exactly how negative the index form becomes, past several conjugate points, is the content of a theorem too large to prove here.

5.4.6 Foreshadowing: The Morse Index Theorem The index form I of a geodesic $\gamma\colon [0,b] \to M$ is a symmetric bilinear form on the infinite-dimensional space $\mathcal{V}(\gamma)$, and its *index* is the maximal dimension of a subspace of $\mathcal{V}(\gamma)$ on which I is negative definite. The Morse index theorem asserts that this index is finite and equals the number of interior points $\gamma(t)$, $t \in (0,b)$, conjugate to $\gamma(0)$ along γ, each counted with its multiplicity as in definition 5.3.1. The theorem contains theorem 5.4.5 as its first increment: the appearance of one conjugate point in $(0,b)$ raises the index from 0 to at least 1, so I acquires a negative direction, which is exactly the field W constructed above. It also refines proposition 5.3.3: when there are no interior conjugate points the index is 0, so I is positive semidefinite, and by the index lemma theorem 5.4.4 in fact positive definite on normal fields, which is the analytic form of local minimization.
The proof organizes $\mathcal{V}(\gamma)$ by cutting $[0,b]$ into subintervals short enough to carry no conjugate points, reducing I on each piece to a finite-dimensional form via the index lemma, and tracking how the index jumps as b increases through each conjugate value; the jump at a conjugate point equals its multiplicity. A complete account is given in John Milnor, *Morse Theory*, and in John M. Lee, *Introduction to Riemannian Manifolds*, where the theorem is proved from the index lemma established above. It is the bridge from this chapter to Morse theory proper, in which the conjugate points of geodesics become the critical-point data of the energy functional on the path space.

6

Completeness and Comparison Theorems

The local theory of the previous chapters becomes global here. Jacobi fields measured how neighboring geodesics spread, and conjugate points marked where they refocus; the present chapter turns those infinitesimal statements into theorems about the manifold as a whole, about its diameter, its topology, and its diffeomorphism type. The bridge is completeness, the hypothesis under which geodesics run forever and any two points can be joined by a shortest one.

The chapter opens with the Hopf–Rinow theorem, which reconciles the two meanings completeness could have, that geodesics extend indefinitely and that the distance function makes M a complete metric space, and shows they coincide and guarantee minimizing geodesics. This is a place where the Riemannian hypothesis cannot be dropped: the theorem fails for Lorentzian metrics, and the failure is flagged where it occurs. With minimizers in hand, the cut locus records exactly how far a geodesic stays shortest, and the injectivity radius quantifies it. The curvature then reasserts control, now globally and through comparison. Positive Ricci curvature bounds the diameter and forces compactness, by Bonnet–Myers; nonpositive sectional curvature unwraps the manifold to Euclidean space, by Cartan–Hadamard, completing the noncompact space forms left open in chapter 4. The Rauch comparison theorem is the general engine behind these results, comparing the growth of Jacobi fields under one- sided curvature bounds against the constant-curvature models, and it returns the conjugate-point statements of chapter 5 as special cases. Throughout, the recurring benchmark is the trichotomy of model spaces: the sphere refocuses and is compact, Euclidean space is flat and open, and hyperbolic space spreads and is a Cartan–Hadamard manifold.

6.1 Geodesic Completeness and the Hopf–Rinow Theorem

The exponential map of section 3.2 was defined on a domain $\mathcal{E}_p \subseteq T_pM$ that need not be all of T_pM: a geodesic may run off the manifold in finite time, as the geodesic aimed at the deleted origin does on $\mathbb{R}^n \setminus \{0\}$, reaching the puncture at a finite parameter and having nowhere to continue. This possibility has been flagged since definition 3.1.1 and set aside; the present section confronts it. The manifolds on which geodesics never fall off the edge are the *geodesically complete* ones, and they are the manifolds on which the local geometry of the preceding chapters becomes global. Only when geodesics run forever can one ask whether *every* pair of points is joined by a shortest one, and only then do curvature bounds control the manifold as a whole rather than a neighborhood at a time.

There is a second sense in which a manifold can be complete, older and purely metric. The distance function d_g of definition 3.4.1 makes M a metric space, and a metric space is complete when every Cauchy sequence converges. These two completeness conditions, geodesic and metric, are stated in different languages, one about the domain of an ODE and one about the convergence of sequences, and there is no obvious reason they should agree. The Hopf–Rinow theorem is the statement that they do agree, and that either of them supplies the conclusion one actually wants: that any two points of a connected complete manifold are joined by a minimizing geodesic, a global shortest path where the local theory of section 3.4 produced only local ones. The proof turns on a single construction, growing a minimizing geodesic outward from a point and showing it cannot stop short of its target, and this construction is where positive-definiteness of g is used decisively. The theorem is false for Lorentzian metrics, and the failure is not a technicality; it is recorded at the end of the section.

Throughout, (M, g) is a connected Riemannian manifold with its Levi-Civita connection, and d_g is the Riemannian distance of definition 3.4.1. Positive-definiteness is in force from the outset and will be flagged where it is the hypothesis in use.

Definition 6.1.1: Geodesic Completeness

A Riemannian manifold (M, g) is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$; equivalently, if for every $p \in M$ the exponential map $\exp_p$ of definition 3.2.1 is defined on all of T_pM, that is $\mathcal{E}_p = T_pM$.

The two formulations agree by the rescaling identity of proposition 3.2.2. If every maximal geodesic is defined on all of $\mathbb{R}$, then for any p and any $v \in T_pM$ the geodesic γ_v is defined at $t = 1$, so $v \in \mathcal{E}_p$; conversely, if $\mathcal{E}_p = T_pM$ for every p, then given a maximal geodesic γ and any time t_1 in its domain, the vector $t_1\dot\gamma(0) \in T_pM$ lies in $\mathcal{E}_p$, so $\gamma_{t_1\dot\gamma(0)}(1) = \gamma(t_1)$ is defined, and letting t_1 range over all of $\mathbb{R}$ shows the maximal geodesic extends to all of $\mathbb{R}$. Completeness is thus a property one can test one point at a time, through the domain of a single exponential map, or globally, through the extension of every geodesic; the equivalence of these tests with metric completeness is the theorem below, and the further equivalence with testing $\exp_p$ at a *single* point is one of its more useful consequences.

The three model spaces sort themselves immediately, and they exhibit completeness alongside the three curvature signs.

Example 6.1.2: Completeness of the model spaces

Euclidean space $(\mathbb{R}^n, \bar g)$ of example 1.1.2 is geodesically complete: its geodesics are the affine lines $\gamma_v(t) = p + tv$ of example 3.2.3, defined for all $t \in \mathbb{R}$, so $\mathcal{E}_p = T_p\mathbb{R}^n$ for every p. The round sphere S^n_r of example 1.3.4 is geodesically complete: its geodesics are the great circles traversed at constant speed, periodic and hence defined for all $t \in \mathbb{R}$, as recorded in example 3.2.4, so $\mathcal{E}_p = T_p S^n_r$. Hyperbolic space H^n is geodesically complete as well: in the hyperboloid model of example 1.3.5 its geodesics are the constant-speed intersections with 2-planes through the origin of $\mathbb{R}^{1,n}$, unbounded curves defined for all real time. Thus all three space forms, of curvature $+1/r^2$, 0, and -1, are complete, and none exhibits the finite-time breakdown of a punctured or open example.

By contrast, the open unit ball $B \subseteq \mathbb{R}^n$ with the Euclidean metric is *not* complete: a geodesic aimed at the boundary sphere reaches $|x| = 1$ at a finite parameter and cannot continue, so its maximal domain is a bounded interval and $\mathcal{E}_p \neq T_p B$. The same manifold with the hyperbolic (Poincaré ball) metric of example 1.3.5 *is* complete, the metric stretching the radial distance to the boundary to infinity so that no geodesic reaches it in finite time. Completeness is a joint property of the smooth manifold and the metric, not of the manifold alone.

The mismatch between the ball and hyperbolic space warns that the definition of completeness has content only relative to g. The distance d_g of example 3.4.4 and its Euclidean counterpart provide the metric side of the story. On $\mathbb{R}^n$ the Riemannian distance is the ordinary Euclidean distance, and $(\mathbb{R}^n, d_{\bar{g}})$ is a complete metric space in the classical sense; on the round sphere the distance is bounded by πr, yet (S^n_r, d_{g_r}) is again complete because it is compact. The open ball with the Euclidean metric fails metric completeness too: a sequence marching toward the boundary is Cauchy in $d_{\bar{g}}$ but has no limit in B. That the geodesic and metric failures coincide in every one of these examples is the pattern the theorem explains.

The proof of Hopf–Rinow rests on one local fact, that geodesics can always be continued for a short additional time from any point they reach, together with one global construction, that a minimizing geodesic can be grown outward from a point until it arrives at any prescribed target. The following theorem assembles these into the equivalence of every reasonable notion of completeness and, as its operative conclusion, the existence of minimizing geodesics between arbitrary points.

Theorem 6.1.3: Hopf–Rinow

Let (M, g) be a connected Riemannian manifold and let d_g be its Riemannian distance (definition 3.4.1). The following are equivalent.

1. The metric space (M, d_g) is complete: every d_g-Cauchy sequence converges.
2. M is geodesically complete (definition 6.1.1): every maximal geodesic is defined on all of $\mathbb{R}$.
3. There is a point $p \in M$ for which $\exp_p$ is defined on all of $T_p M$.
4. Every closed and d_g-bounded subset of M is compact.

Moreover, if any one of these holds, then for every pair of points $p, q \in M$ there is a minimizing geodesic from p to q: a geodesic γ with $\gamma(0) = p$, $\gamma(1) = q$, and $L(\gamma) = d_g(p, q)$.

***Proof*:**
The scheme is (iii) $\Rightarrow$ minimizers from p $\Rightarrow$ (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), with the last implication trivial and the minimizing-geodesic conclusion produced along the way. The heart of the argument is the first implication, a construction due to Hopf and Rinow that builds a minimizing

geodesic from p to an arbitrary target assuming only that geodesics from p run forever.

Step 1: from (iii), any point is joined to p by a minimizing geodesic. Suppose $\exp_p$ is defined on all of T_pM, so every geodesic from p is defined for all time. Fix $q \in M$ with $q \neq p$ and set $\rho = d_g(p,q) > 0$; the goal is a geodesic of length ρ from p to q. Choose a geodesic sphere about p of a small radius δ on which theorem 3.4.3 and corollary 3.4.5 apply, with $0 < \delta < \rho$. The geodesic sphere $S_\delta = \exp_p(\left\{v \in T_pM \;\middle|\; |v|_g = \delta\right\})$ is the continuous image of a Euclidean sphere, hence compact, and the function $x \mapsto d_g(x,q)$ is continuous on it, so it attains a minimum at some point $x_0 \in S_\delta$. Write $x_0 = \exp_p(\delta u)$ for a unit vector $u \in T_pM$, and let

$$\gamma(s) = \exp_p(su), \qquad s \in [0,\infty)$$

be the unit-speed geodesic from p in the direction u, defined for all $s \geq 0$ by hypothesis (iii). The claim is that γ reaches q at parameter $s = \rho$, that is, $\gamma(\rho) = q$; since γ is unit-speed, its length from 0 to ρ is then $\rho = d_g(p,q)$, so $\gamma|_{[0,\rho]}$ is the desired minimizing geodesic (reparametrized to $[0,1]$ if one insists on the endpoint convention).

First, the point x_0 realizes the distance from p to q through the sphere: every curve from p to q must cross S_δ, because q lies outside the geodesic ball of radius δ (as $\delta < \rho = d_g(p,q)$ and corollary 3.4.5 identifies that ball with the metric ball of radius δ), so

$$d_g(p,q) = \min_{x \in S_\delta}\big(d_g(p,x) + d_g(x,q)\big) = \delta + \min_{x \in S_\delta} d_g(x,q) = \delta + d_g(x_0,q) \tag{6.1}$$

Here $d_g(p,x) = \delta$ for every $x \in S_\delta$ by corollary 3.4.5; the first equality holds because a shortest route to q passes through the sphere and its length is at least the distance to the crossing point plus the distance from there to q, while conversely concatenating a radial geodesic to any $x \in S_\delta$ with a near-minimizing curve from x to q realizes the right-hand side up to any ε. Rearranging (6.1) gives $d_g(x_0,q) = \rho - \delta$, and since $x_0 = \gamma(\delta)$ this reads

$$d_g\big(\gamma(\delta),q\big) = \rho - \delta \tag{6.2}$$

Now consider the set of parameters along γ at which this exact accounting persists:

$$A = \left\{s \in [\delta,\rho] \;\middle|\; d_g\big(\gamma(s),q\big) = \rho - s\right\}$$

The identity $d_g(\gamma(s),q) = \rho - s$ says that the geodesic has spent length s getting from p to $\gamma(s)$ and has exactly $\rho - s$ of length left to reach q, with no waste: it stays on a shortest route. By (6.2), $\delta \in A$, so A is nonempty. It is closed in $[\delta,\rho]$, being defined by the vanishing of the continuous function $s \mapsto d_g(\gamma(s),q) - (\rho - s)$. Let $s_0 = \sup A$; by closedness $s_0 \in A$, so

$$d_g\big(\gamma(s_0),q\big) = \rho - s_0 \tag{6.3}$$

The claim is $s_0 = \rho$. Suppose instead $s_0 < \rho$, and write $p_0 = \gamma(s_0)$. Then $p_0 \neq q$, since $d_g(p_0,q) = \rho - s_0 > 0$. Repeat the sphere construction at p_0: choose a geodesic sphere $S_{\delta'} = \exp_{p_0}(\left\{w \;\middle|\; |w|_g = \delta'\right\})$ of radius δ' small enough that theorem 3.4.3 and corollary 3.4.5 hold at p_0 and $\delta' < \rho - s_0$. As before, $x \mapsto d_g(x,q)$ attains its minimum on the compact sphere $S_{\delta'}$ at some x_0', and the analogue of (6.1) gives

$$d_g(p_0,q) = \delta' + d_g(x_0',q), \qquad \text{hence} \qquad d_g(x_0',q) = (\rho - s_0) - \delta' \tag{6.4}$$

using (6.3). Combining (6.4) with the triangle inequality,

$$d_g(p, x_0') \geq d_g(p, q) - d_g(x_0', q) = \rho - \big((\rho - s_0) - \delta'\big) = s_0 + \delta'$$

On the other hand, the concatenation of $\gamma|_{[0,s_0]}$ (length s_0, a curve from p to p_0) with the radial geodesic of length δ' from p_0 to x_0' is a curve from p to x_0' of length $s_0 + \delta'$, so $d_g(p, x_0') \leq s_0 + \delta'$. Therefore $d_g(p, x_0') = s_0 + \delta'$, and this concatenated curve is length-minimizing from p to x_0'. By proposition 3.4.7, a length-minimizing piecewise-smooth curve, once reparametrized by arc length, is a smooth geodesic with no corner; in particular the concatenation has no corner at p_0, so the outgoing radial geodesic from p_0 to x_0' continues γ smoothly, with the same unit velocity. By uniqueness of geodesics from given initial data (theorem 3.1.3), the radial geodesic from p_0 to x_0' is exactly γ continued, so $x_0' = \gamma(s_0 + \delta')$. Substituting into (6.4),

$$d_g\big(\gamma(s_0 + \delta'), q\big) = d_g(x_0', q) = (\rho - s_0) - \delta' = \rho - (s_0 + \delta')$$

which says $s_0 + \delta' \in A$, contradicting $s_0 = \sup A$ since $\delta' > 0$. Hence $s_0 = \rho$, and (6.3) becomes $d_g(\gamma(\rho), q) = 0$, so $\gamma(\rho) = q$ by theorem 3.4.6. The geodesic $\gamma|_{[0,\rho]}$ is unit-speed of length $\rho = d_g(p, q)$, hence a minimizing geodesic from p to q. This proves the minimizing-geodesic conclusion from the base point p whose exponential map is everywhere defined.

Step 2: (iii) implies (iv). Let p be as in (iii), so every point is joined to p by a minimizing geodesic by Step 1. Let $C \subseteq M$ be closed and d_g-bounded, say $C \subseteq \{x \mid d_g(p, x) \leq R\}$ for some R. Every $x \in C$ satisfies $x = \exp_p(v)$ for a vector $v = v(x)$ with $|v|_g = d_g(p, x) \leq R$, namely the initial velocity of the minimizing geodesic from p to x produced in Step 1. Hence

$$C \subseteq \exp_p\big(\overline{B_R(0)}\big), \qquad \overline{B_R(0)} = \Big\{ v \in T_pM \;\Big|\; |v|_g \leq R \Big\}$$

The closed ball $\overline{B_R(0)}$ is a compact subset of T_pM, and $\exp_p$ is continuous and defined on all of T_pM by hypothesis (iii), so $\exp_p(\overline{B_R(0)})$ is compact. Thus C is a closed subset of a compact set, hence compact. This proves (iv).

Step 3: (iv) implies (i). Let (x_k) be a d_g-Cauchy sequence. A Cauchy sequence is bounded: there is N with $d_g(x_k, x_N) \leq 1$ for all $k \geq N$, so the whole sequence lies within d_g-distance $\max_{k<N} d_g(x_k, x_N) + 1$ of x_N, a bounded set. Its closure C is then closed and bounded, hence compact by (iv). A Cauchy sequence in a compact metric space has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges to the same limit. Therefore (x_k) converges in M, proving (i).

Step 4: (i) implies (ii). Assume (M, d_g) is complete, and let $\gamma\colon [0, b) \to M$ be a maximal unit-speed geodesic with $b < \infty$; a contradiction will show $b = \infty$, and rescaling the velocity then handles geodesics of any speed. Take any sequence $t_k \uparrow b$. For j, k, the geodesic segment of γ between t_j and t_k has length $|t_j - t_k|$, so

$$d_g\big(\gamma(t_j), \gamma(t_k)\big) \leq |t_j - t_k|$$

As (t_k) is Cauchy in $\mathbb{R}$, the sequence $(\gamma(t_k))$ is d_g-Cauchy, hence converges to a point $x_* \in M$ by (i). This limit does not depend on the chosen sequence: any two sequences increasing to b interleave, so their images share the limit x_*, and $\gamma(t) \to x_*$ as $t \uparrow b$. Now extend γ past b. Choose a *uniformly normal* neighborhood of x_*: a neighborhood W and a radius $\eta > 0$ such that for every $x \in W$ the exponential map $\exp_x$ is a diffeomorphism on the ball $\Big\{ w \in T_xM \;\Big|\; |w|_g < \eta \Big\}$, and any two points of W within distance η are joined by a unique minimizing geodesic lying in W. Such neighborhoods exist by the smooth dependence of geodesics on their initial data (theorem 3.1.3), applied to the exponential map of the whole tangent bundle near $(x_*, 0)$; this is a standard consequence of section 3.2 used here as a local input. For t close enough to b, $\gamma(t) \in W$ and $b - t < \eta$, so the geodesic γ restricted to $[t, b)$ is a geodesic in W of length $b - t < \eta$; being a unit-speed geodesic starting at $\gamma(t)$ with initial velocity $\dot{\gamma}(t)$, it equals $s \mapsto \exp_{\gamma(t)}\big((s-t)\dot{\gamma}(t)\big)$, which is defined for s in an interval of radius η about t, in particular past b. This extends γ smoothly to a geodesic on an interval strictly containing $[0, b]$, contradicting the maximality of $b < \infty$. Hence every maximal geodesic has domain all of $\mathbb{R}$, which is (ii).

Step 5: (ii) implies (iii). If every maximal geodesic is defined on all of $\mathbb{R}$, then in particular for a fixed p every geodesic from p is defined at time 1, so $\mathcal{E}_p = T_pM$; this is (iii) for that (indeed any) p.

The five implications close the cycle (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), so all four conditions are equivalent. Finally, when they hold, (iii) holds at every point p, so Step 1 applies with any p as base point and produces a minimizing geodesic from p to any q. This establishes the minimizing-geodesic conclusion in full, completing the proof.

The theorem does two jobs, and it is worth separating them. As an equivalence it collapses a list of a priori distinct hypotheses into one, so that "complete" may be used unambiguously: a Riemannian manifold is complete when any one, hence all, of (i) through (iv) holds. The reduction to (iii), completeness of a single exponential map, is the sharpest of these, and it is what makes completeness checkable in examples: to know that a manifold is complete it is enough to fix one point and verify that geodesics from it run forever. The second job is the minimizing-geodesic conclusion, which is the geometric payoff. The local theory of section 3.4 guaranteed shortest paths only inside a normal ball; Hopf–Rinow removes the locality, at the cost of the completeness hypothesis, and asserts that on a complete manifold the infimum defining $d_g(p, q)$ is attained by an actual geodesic, for every pair p, q. The construction that delivers it, growing γ outward and tracking the identity $d_g(\gamma(s), q) = \rho - s$, is the geometric content: at each stage the point of a small sphere closest to q continues the geodesic without a corner, and proposition 3.4.7 is exactly what forbids the corner and forces the continuation to lie along γ itself.

Two consequences record the conclusions that are used most often downstream, and they are separated out here because later chapters will cite them by name.

Corollary 6.1.4: Minimizing Geodesics on Complete Manifolds

If (M, g) is a connected complete Riemannian manifold, then any two points $p, q \in M$ are joined by a minimizing geodesic: a geodesic γ with endpoints p and q and $L(\gamma) = d_g(p, q)$.

***Proof*:**
Completeness is any of the equivalent conditions of theorem 6.1.3, so in particular (iii) holds at p, and the final assertion of that theorem produces the minimizing geodesic from p to q, as claimed.

Corollary 6.1.5: Compact Manifolds Are Complete

Every compact connected Riemannian manifold is geodesically complete, and hence any two of its points are joined by a minimizing geodesic.

***Proof*:**
Let M be compact and connected. A Cauchy sequence in the metric space (M, d_g) is in particular a sequence in the compact space M, so it has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges; hence (M, d_g) is complete, condition (i) of theorem 6.1.3. By the equivalence, M is geodesically complete, and by corollary 6.1.4 any two points are joined by a minimizing geodesic, as required.

The compactness corollary is the reason the round sphere and the flat torus need no separate completeness check: both are compact, so both are complete, and every pair of points on either is joined by a shortest geodesic. On the sphere this recovers, without any new work, the fact that the distance $d_{g_r}(p,q) = r\arccos(\langle p,q\rangle / r^2)$ of example 3.4.4 is realized by a great-circle arc for every pair p, q, including the antipodal pair, where infinitely many half great circles tie for the shortest. Noncompact complete examples, such as Euclidean and hyperbolic space, require condition (iii) instead, but there the single-exponential-map criterion makes the verification immediate: one point suffices, and every geodesic from it runs forever.

Every step of the theorem used the Riemannian, positive-definite, structure of g, and the reliance is not removable. The construction of Step 1 minimized the continuous function $d_g(-, q)$ over a compact geodesic sphere, which presumes d_g is a genuine distance, and the vanishing of d_g only on the diagonal (theorem 3.4.6) rests on positive-definiteness, as recorded there. The following warning makes precise that the theorem is not simply proved using positive-definiteness but is actually false without it.

6.1.6 Warning: Hopf–Rinow fails for Lorentzian metrics The equivalence and the minimizing-geodesic conclusion of theorem 6.1.3 are special to the Riemannian case; for a pseudo-Riemannian metric of indefinite signature (definition 1.1.3), in particular a Lorentzian metric of signature $(-, +, \ldots, +)$, they all fail.

The obstruction is present already in the statement. An indefinite metric assigns negative and null values of $g(v, v)$, so the length functional $L(\gamma) = \int |\dot{\gamma}|_g \, dt$ is not defined as a real number for curves whose velocity is timelike or null, and there is no distance function d_g of the kind built in definition 3.4.1. Conditions (i) and (iv) of the theorem, which are statements about the metric space (M, d_g), therefore have no meaning to begin with, so the equivalence with geodesic completeness cannot even be posed in the same terms. Geodesic completeness, condition (ii), remains meaningful, since geodesics are defined by the connection alone; but it no longer forces the existence of "shortest" geodesics, because there is nothing to minimize.

The divergence is concrete. Compactness no longer implies geodesic completeness: the Clifton–Pohl torus, the quotient of $\mathbb{R}^2 \setminus \{0\}$ with the Lorentzian metric $\frac{2\,dx\,dy}{x^2+y^2}$ by the multiplication $(x, y) \mapsto (2x, 2y)$, is a compact Lorentzian manifold that is geodesically incomplete, so the Lorentzian analogue of corollary 6.1.5 is false. The Riemannian proof of that corollary passed through metric completeness of (M, d_g), exactly the object an indefinite metric fails to provide, and no replacement holds. Even where geodesics are complete, two points of a Lorentzian manifold causally related by a timelike curve need not be joined by a length-*maximizing* geodesic, the Lorentzian counterpart of a minimizer, without a further global hypothesis (global hyperbolicity), which has no Riemannian analogue because on a Riemannian manifold completeness alone suffices. The single hypothesis of completeness carrying so much, four equivalent formulations and a global existence statement, is thus a Riemannian phenomenon, resting on positive-definiteness at every step, and it is one of the places where the passage from the Riemannian to the Lorentzian setting is a change of substance rather than of sign.

6.2 The Cut Locus and the Injectivity Radius

The Hopf–Rinow theorem of the previous section guarantees, on a complete manifold, that any two points are joined by a minimizing geodesic. It says nothing about how long that geodesic stays minimizing if extended past its endpoint. A geodesic leaving p is shortest near p, by the local minimizing theorem of section 3.4, but the property is local: continued far enough, a geodesic can be overtaken by a competitor and cease to realize the distance from p. The round sphere already displays this. A great-circle arc from a point p is the shortest path to each point it meets, until it reaches the antipode; the moment it passes the antipode it is longer than the arc coming the other way around, and it has stopped minimizing. The antipode is the exact instant of transition, and the goal of this section is to identify and study that instant on an arbitrary manifold.

Two mechanisms can end a geodesic's run as a minimizer, and the sphere shows both at once. The first is refocusing: the geodesic reaches a conjugate point, where the exponential map degenerates and nearby geodesics from p collide, after which minimization is impossible by the second-variation obstruction of section 5.4. The second is competition: a second minimizing geodesic of the same length arrives from p, so the point is reached two ways and neither extends as a strict minimizer. The central theorem of the section shows that at the cut point, the place where minimization fails, at least one of these two must occur. The distance from p to the nearest cut point is the injectivity radius, the exact radius out to which the exponential map is a genuine chart, and it is the scale on which the manifold looks like its tangent space. Throughout, (M, g) is a connected, complete Riemannian manifold; completeness is used to secure minimizers, and the positive-definiteness of g is what makes distances, and hence the cut locus, well behaved. Both hypotheses fail to have their consequences in the Lorentzian setting, where Hopf–Rinow is false and the cut-locus theory takes a different form.

The starting point is to measure, for a fixed direction, exactly how long the geodesic in that direction minimizes.

Definition 6.2.1: Cut Point and Cut Time

Let (M, g) be a complete, connected Riemannian manifold, let $p \in M$, and let $v \in T_pM$ be a unit vector, $|v|_g = 1$. Consider the unit-speed geodesic $\gamma_v(t) = \exp_p(tv)$ (definition 3.2.1), defined for all $t \geq 0$ by completeness (definition 6.1.1). The *cut time* of p in the direction v is

$$t_{\text{cut}}(v) = \sup \left\{ t > 0 \;\middle|\; d_g\big(p, \gamma_v(t)\big) = t \right\}$$

the supremum of times up to which γ_v minimizes the distance from p. If $t_{\text{cut}}(v) < \infty$, the point $\gamma_v\big(t_{\text{cut}}(v)\big)$ is the *cut point* of p along γ_v. If $t_{\text{cut}}(v) = \infty$, the geodesic γ_v minimizes from p forever and there is no cut point in the direction v.

The definition records the largest arc on which a given geodesic is a shortest path. The condition $d_g(p, \gamma_v(t)) = t$ says exactly that the arc $\gamma_v|_{[0,t]}$, of length t since γ_v has unit speed, realizes the distance from p to its endpoint, and so is minimizing. The set of such t is nonempty and contains an interval $(0, \varepsilon)$: inside a geodesic ball the radial geodesic is the unique shortest path by theorem 3.4.3, so γ_v minimizes for small t. It is also an interval starting at 0, by the following monotonicity, which is the first structural fact and is used repeatedly below.

A geodesic that minimizes up to time t_1 minimizes on every shorter initial arc. Concretely, if $d_g(p, \gamma_v(t_1)) = t_1$ then $d_g(p, \gamma_v(t)) = t$ for every $t \in [0, t_1]$. Indeed the restriction $\gamma_v|_{[0,t]}$ is a path of length t from p to $\gamma_v(t)$, so $d_g(p, \gamma_v(t)) \leq t$; if strict inequality held, a shorter path from p to $\gamma_v(t)$ followed by $\gamma_v|_{[t,t_1]}$ would join p to $\gamma_v(t_1)$ in length less than $t + (t_1 - t) = t_1$, contradicting $d_g(p, \gamma_v(t_1)) = t_1$. Hence the set $\{t > 0 \mid d_g(p, \gamma_v(t)) = t\}$ is an interval of the form $(0, t_{\text{cut}}(v)]$ or $(0, \infty)$, and γ_v minimizes on $[0, t_{\text{cut}}(v)]$ and fails to minimize strictly beyond it, as the next paragraph confirms. The cut point is therefore the last point out to which the geodesic is a shortest path.

Two facts sharpen the picture and are recorded now for use throughout. First, minimization holds up to and including the cut time: at $t = t_{\text{cut}}(v)$ one has $d_g(p, \gamma_v(t)) = t$ by continuity of d_g (theorem 3.4.6) applied to the supremum, so the geodesic arc *to* the cut point is still minimizing, and the cut point is the last point with this property. Second, minimization fails immediately past the cut time: for $t > t_{\text{cut}}(v)$ one has $d_g(p, \gamma_v(t)) < t$, for otherwise t would belong to the defining set and exceed its supremum. Thus the cut point is exactly the transition from minimizing to non-minimizing along γ_v.

Collecting the cut points over all directions gives a subset of the manifold that encodes the global geometry of geodesics from p.

Definition 6.2.2: Cut Locus

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$. The *cut locus* of p is the set of all cut points of p,

$$\mathrm{Cut}(p) = \left\{ \gamma_v\big(t_{\mathrm{cut}}(v)\big) \;\middle|\; v \in T_pM,\ |v|_g = 1,\ t_{\mathrm{cut}}(v) < \infty \right\} \subseteq M$$

Equivalently, writing $\widetilde{\mathrm{Cut}}(p) = \left\{ t_{\mathrm{cut}}(v)\, v \;\middle|\; |v|_g = 1,\ t_{\mathrm{cut}}(v) < \infty \right\} \subseteq T_pM$ for the *tangent cut locus*, one has $\mathrm{Cut}(p) = \exp_p\big(\widetilde{\mathrm{Cut}}(p)\big)$.

The cut locus is the boundary, seen from p, between the region where the geodesics from p still minimize and the region where they no longer do. Points strictly inside are reached by a unique minimizing geodesic that extends past them; points of the cut locus are reached by a minimizing geodesic that goes no farther as a minimizer; points beyond are reached only by geodesics that have already given up minimizing. The tangent cut locus $\widetilde{\mathrm{Cut}}(p)$ is the graph, in polar form, of the function $v \mapsto t_{\mathrm{cut}}(v)$ on the unit sphere of T_pM; its image under $\exp_p$ is $\mathrm{Cut}(p)$, and the open region inside it, the set of tv with $|v| = 1$ and $t < t_{\mathrm{cut}}(v)$, is the *segment domain*, mapped diffeomorphically to $M \setminus \mathrm{Cut}(p)$ by the exponential map, as proposition 6.2.6 will confirm on the ball where t_{cut} is bounded below. On a compact manifold every $t_{\mathrm{cut}}(v)$ is finite, the segment domain is a bounded star-shaped set, and M is the disjoint union of the open segment domain's image with the cut locus.

The distance from p to its cut locus is the single number that controls how large a normal ball can be, and it deserves a name.

Definition 6.2.3: Injectivity Radius

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$. The *injectivity radius* of M at p is

$$\mathrm{inj}(p) = d_g\big(p, \mathrm{Cut}(p)\big) = \inf\{d_g(p, q) \mid q \in \mathrm{Cut}(p)\}$$

with the convention $\mathrm{inj}(p) = \infty$ when $\mathrm{Cut}(p) = \varnothing$. The *injectivity radius* of M is

$$\mathrm{inj}(M) = \inf_{p \in M} \mathrm{inj}(p)$$

Equivalently, since the distance from p to a cut point $\gamma_v(t_{\mathrm{cut}}(v))$ is $t_{\mathrm{cut}}(v)$ (the arc to the cut point being minimizing),

$$\mathrm{inj}(p) = \inf \left\{ t_{\mathrm{cut}}(v) \;\middle|\; v \in T_pM,\ |v|_g = 1 \right\}$$

The two expressions for $\mathrm{inj}(p)$ agree because the geodesic arc from p to a cut point $\gamma_v(t_{\mathrm{cut}}(v))$ is minimizing, so its length $t_{\mathrm{cut}}(v)$ equals $d_g(p, \gamma_v(t_{\mathrm{cut}}(v)))$; taking the infimum over unit v of these distances is the same as taking the infimum of $d_g(p, q)$ over $q \in \mathrm{Cut}(p)$. The quantity $\mathrm{inj}(p)$ is the radius of the largest metric ball about p inside which the exponential map is a diffeomorphism, so that geodesic polar coordinates and normal coordinates are valid; below that radius the manifold is a faithful copy of a ball in T_pM, and $\mathrm{inj}(M)$ is the uniform such radius valid at every point. It appears as a hypothesis or a controlled quantity throughout global Riemannian geometry, and a positive lower bound on $\mathrm{inj}(M)$ is one of the standard ways of preventing a manifold from degenerating.

A first computation on the model spaces fixes the meaning of all three definitions before the general theory. The three space forms of constant curvature realize the three possibilities: a finite cut locus that is a single point, and no cut locus at all.

Example 6.2.4: Cut locus and injectivity radius of the round sphere

Let S^n_r be the round sphere of radius r (example 1.3.4), which is compact, hence complete (corollary 6.1.5). Fix $p \in S^n_r$ and a unit vector $v \in T_pS^n_r$. The unit-speed geodesic in the direction v is the great-circle arc computed in example 3.4.4,

$$\gamma_v(t) = \cos\Big(\frac{t}{r}\Big)p + \sin\Big(\frac{t}{r}\Big)(rv)$$

traced at unit speed, closing up after arc length $2\pi r$ and passing through the antipode $-p$ at arc length πr.

The cut time is πr in every direction. By the distance formula of example 3.4.4, for any point q on this great circle at arc length $t \in [0, \pi r]$ the distance $d_{g_r}(p, q)$ equals the length t of the short arc, so γ_v minimizes on $[0, \pi r]$ and $d_{g_r}(p, \gamma_v(t)) = t$ there. For $t \in (\pi r, 2\pi r)$ the point $\gamma_v(t)$ is reached from p by the complementary arc of the same great circle, of length $2\pi r - t < t$, so $d_{g_r}(p, \gamma_v(t)) \leq 2\pi r - t < t$ and γ_v no longer minimizes. Hence

$$t_{\text{cut}}(v) = \pi r \qquad \text{for every unit } v \in T_pS^n_r$$

independent of the direction, by the homogeneity and isotropy of the sphere.

The cut locus is the antipode. The cut point in the direction v is $\gamma_v(\pi r) = -p$, the antipode, for every v. As v ranges over the unit sphere the cut point is always the same point $-p$, so

$$\text{Cut}(p) = \{-p\}$$

a single point. The tangent cut locus $\widetilde{\text{Cut}}(p) = \Big\{\pi r\, v \;\Big|\; |v|_{g_r} = 1\Big\}$ is the entire sphere of radius πr in $T_pS^n_r$, and $\exp_p$ collapses this whole sphere to $-p$, matching the direct formula $\exp_p(v) = (\cos(|v|/r))p + \cdots$ of example 3.2.4, which sends every v of length πr to $-p$.

The injectivity radius is πr. By definition 6.2.3,

$$\text{inj}(p) = d_{g_r}(p, \text{Cut}(p)) = d_{g_r}(p, -p) = \pi r$$

and since the sphere is homogeneous this holds at every point, so $\text{inj}(S^n_r) = \pi r$.

Both mechanisms coincide here. The cut point $-p$ is simultaneously the first conjugate point of p, at distance πr with multiplicity $n-1$ by example 5.3.4, and a point reached by infinitely many distinct minimizing geodesics, the infinitely many half great circles from p to $-p$ recorded in example 3.4.4. On the sphere the two alternatives of the cut-point characterization below occur together. For the two remaining model spaces the situation is opposite: hyperbolic space H^n has curvature -1 and Euclidean space $\mathbb{R}^n$ has curvature 0, both satisfying $K \leq 0$, so by proposition 5.3.3 there are no conjugate points, and, as the Cartan–Hadamard theorem of section 6.4 will show, $\exp_p$ is a global diffeomorphism and every geodesic minimizes forever. Hence for H^n and $\mathbb{R}^n$ one has $t_{\text{cut}}(v) = \infty$ in every direction, $\text{Cut}(p) = \varnothing$, and $\text{inj}(p) = \infty$ at every point.

The sphere shows the two ways a geodesic can stop minimizing occurring simultaneously; on a general manifold they need not, but one of them must. This is the central theorem of the section. It identifies the geometric event at the cut point: either the exponential map has degenerated, so the point is conjugate, or two minimizing geodesics have arrived, so the point is reached ambiguously. The proof is a limiting argument. Approach the cut point along the geodesic from just beyond it, take minimizing geodesics to those approaching points, and extract a limit; the limit is either a second minimizing geodesic distinct from γ, or, if it coincides with γ, the coincidence forces the differential of the exponential map to be singular.

Theorem 6.2.5: Characterization of the Cut Point

Let (M, g) be a complete, connected Riemannian manifold, $p \in M$, and $v \in T_pM$ a unit vector with finite cut time $t_c = t_{\text{cut}}(v) < \infty$. Let $q = \gamma_v(t_c)$ be the cut point of p along γ_v. Then at least one of the following holds:

1. q is the first conjugate point of p along γ_v; equivalently, $d(\exp_p)_{t_c v}$ is singular.
2. There exist two distinct minimizing geodesics from p to q; that is, there is a unit vector $w \neq v$ in T_pM with $\exp_p(t_c w) = q$ and $d_g(p, q) = t_c$.

Conversely, if at time t_c either $q = \gamma_v(t_c)$ is conjugate to p along γ_v or there are two distinct minimizing geodesics of length t_c from p to q, and γ_v minimizes on $[0, t_c]$, then $t_c = t_{\text{cut}}(v)$.

***Proof*:**
Write $\gamma = \gamma_v$ throughout and set $q = \gamma(t_c)$, so $d_g(p, q) = t_c$ because the arc to the cut point is minimizing (the discussion after definition 6.2.1).

The direct implication. Choose a strictly decreasing sequence $t_k \downarrow t_c$ with $t_k > t_c$, and set $q_k = \gamma(t_k)$. By completeness and the Hopf–Rinow theorem (theorem 6.1.3, corollary 6.1.4) there is, for each k, a minimizing geodesic from p to q_k; write it as $\sigma_k(t) = \exp_p(t\, w_k)$ for a unit vector $w_k \in T_pM$, defined on $[0, \ell_k]$ with $\ell_k = d_g(p, q_k)$, so that σ_k has unit speed, $\sigma_k(0) = p$, and $\sigma_k(\ell_k) = q_k$. Because $t_k > t_c = t_{\text{cut}}(v)$, the original geodesic γ does not minimize to q_k, so

$$\ell_k = d_g(p, q_k) < t_k$$

The lengths ℓ_k are bounded: $\ell_k = d_g(p, q_k) \leq d_g(p, q) + d_g(q, q_k) = t_c + (t_k - t_c) = t_k$ by the triangle inequality, and $q_k \to q$ gives $\ell_k = d_g(p, q_k) \to d_g(p, q) = t_c$ by continuity of the distance (theorem 3.4.6). The unit vectors w_k lie on the unit sphere of T_pM, which is compact, so after passing to a subsequence (not relabeled) $w_k \to w$ for some unit vector $w \in T_pM$. Passing to the limit in $\exp_p(\ell_k w_k) = q_k$, using $\ell_k \to t_c$, $w_k \to w$, $q_k \to q$, and continuity of $\exp_p$ (which is defined on all of T_pM by completeness),

$$\exp_p(t_c w) = q$$

Thus $\sigma(t) = \exp_p(tw)$, $t \in [0, t_c]$, is a unit-speed geodesic from p to q of length $t_c = d_g(p, q)$, so σ is a minimizing geodesic from p to q. Two cases arise according to whether σ differs from γ.

Case A: $w \neq v$. Then σ and γ are two distinct minimizing geodesics from p to q, since geodesics are determined by their initial velocity (theorem 3.1.3) and $\dot\sigma(0) = w \neq v = \dot\gamma(0)$. This is alternative (2).

Case B: $w = v$. Then $\sigma = \gamma$, and it is shown that q is conjugate to p along γ, giving alternative (1). Suppose, toward a contradiction, that q is *not* conjugate to p along γ. By theorem 5.3.2, this means $d(\exp_p)_{t_c v}$ is nonsingular, hence a linear isomorphism, so by the inverse function theorem $\exp_p$ is a diffeomorphism from an open neighborhood $\mathcal{U}$ of $t_c v$ in T_pM onto an open neighborhood $\mathcal{V}$ of q in M. For k large the vectors $t_k v$ (which converge to $t_c v$) lie in $\mathcal{U}$, and the endpoints $q_k = \gamma(t_k) = \exp_p(t_k v)$ lie in $\mathcal{V}$. The vectors $\ell_k w_k$ converge to $t_c v$ as well (as $\ell_k \to t_c$ and $w_k \to w = v$), so for k large they too lie in $\mathcal{U}$, and they satisfy $\exp_p(\ell_k w_k) = q_k = \exp_p(t_k v)$. Since $\exp_p$ is injective on $\mathcal{U}$, this forces

$$\ell_k w_k = t_k v$$

for all large k. Taking norms, $\ell_k = |\ell_k w_k| = |t_k v| = t_k$ (both w_k and v are unit), contradicting $\ell_k < t_k$. Therefore q is conjugate to p along γ. It is moreover the *first* conjugate point: no interior point $\gamma(t_0)$ with $0 < t_0 < t_c$ can be conjugate to p, because on $[0, t_c]$ the geodesic γ is minimizing, and a geodesic does not minimize past its first conjugate point (theorem 5.4.5); a conjugate point at $t_0 < t_c$ would make $\gamma|_{[0,t_c]}$ non-minimizing. Hence $q = \gamma(t_c)$ is the first conjugate point, which is alternative (1).

In either case at least one of (1), (2) holds, proving the direct implication.

The converse. Assume $\gamma = \gamma_v$ minimizes on $[0, t_c]$, so $d_g(p, q) = t_c$ and $t_{\mathrm{cut}}(v) \geq t_c$. It remains to show $t_{\mathrm{cut}}(v) \leq t_c$, that is, γ fails to minimize past t_c, under either hypothesis.

Suppose first there are two distinct minimizing geodesics from p to q, namely $\gamma|_{[0,t_c]}$ and a second unit-speed geodesic $\sigma|_{[0,t_c]}$ with $\dot\sigma(0) = w \neq v$, both of length t_c. For $\varepsilon > 0$ small, consider the point $\gamma(t_c + \varepsilon)$ on the extension of γ. The broken path formed by following σ from p to q and then the arc of γ from $q = \gamma(t_c)$ to $\gamma(t_c + \varepsilon)$ has length $t_c + \varepsilon = d_g(p, q) + \varepsilon$. This broken path has a corner at q: its incoming velocity there is $\dot\sigma(t_c)$ and its outgoing velocity is $\dot\gamma(t_c)$, and these differ, for if $\dot\sigma(t_c) = \dot\gamma(t_c)$ then σ and γ would be the same geodesic by uniqueness (theorem 3.1.3), contradicting $w \neq v$. A piecewise-smooth path with a genuine corner is not a geodesic and can be strictly shortened by rounding the corner: replacing the two segments near q by the minimizing geodesic between two nearby interior points (which is strictly shorter than going through q, since inside a normal ball about q the unique shortest path between two points off the two segments is the radial construction of theorem 3.4.3, and the broken path is not that geodesic) yields a strictly shorter path from p to $\gamma(t_c + \varepsilon)$. Hence

$$d_g\big(p, \gamma(t_c + \varepsilon)\big) < t_c + \varepsilon$$

so γ does not minimize on $[0, t_c + \varepsilon]$, whence $t_{\mathrm{cut}}(v) \leq t_c$.

Suppose instead that $q = \gamma(t_c)$ is conjugate to p along γ. A geodesic does not minimize past its first conjugate point (theorem 5.4.5): for every $t_c' > t_c$ there is a proper variation of $\gamma|_{[0,t_c']}$ with fixed endpoints and strictly negative second variation of energy, so $\gamma|_{[0,t_c']}$ is not a local minimizer of energy, hence not a minimizer of length among fixed-endpoint curves; consequently $d_g(p, \gamma(t_c')) < t_c'$. Thus γ fails to minimize on $[0, t_c']$ for every $t_c' > t_c$, and again $t_{\mathrm{cut}}(v) \leq t_c$.

In both cases $t_{\mathrm{cut}}(v) = t_c$, completing the converse and the proof.

The theorem is the exact dictionary the section was built to establish. It says that a geodesic stops minimizing for one of exactly two reasons, and both are geometrically legible: refocusing of nearby geodesics (a conjugate point, where $\exp_p$ folds) or the arrival of a competing shortest path (two minimizing geodesics, where $\exp_p$ overlaps). On the sphere the two reasons coincide at the antipode, as example 6.2.4 recorded, but the theorem does not require them to; a flat cylinder, obtained by rolling up the plane, has no conjugate points at all yet a nonempty cut locus, so that there the second alternative always occurs alone. The converse half is equally usable: to locate the cut point in a given direction, it suffices to find the first time at which either a conjugate point occurs or a second minimizing geodesic appears, and the smaller of the two candidate times is the cut time. This is the computational content behind the model-space calculations and behind the estimates of the next sections.

A symmetry of the cut relation is worth extracting from the proof, as it is used in the injectivity-radius arguments below. If q is the cut point of p along a minimizing geodesic γ, then p is the cut point of q along the reversed geodesic. Both alternatives of the theorem are symmetric in p and q: a conjugate pair is a conjugate pair for the reversed geodesic with the same multiplicity (the observation in definition 5.3.1), and two distinct minimizing geodesics from p to q are two distinct minimizing geodesics from q to p when reversed. Consequently $p \in \mathrm{Cut}(q)$ whenever $q \in \mathrm{Cut}(p)$, so the cut relation is symmetric. In particular $d_g(p, \mathrm{Cut}(p))$ and $d_g(q, \mathrm{Cut}(q))$ are linked, and inj is controlled by the shortest closed geodesic and the shortest geodesic loop, a point returned to in the compact case.

With the cut point understood, the geometric meaning of the injectivity radius becomes a theorem: it is exactly the radius out to which the exponential map is a chart.

Proposition 6.2.6: The Exponential Map is a Diffeomorphism Within the Injectivity Radius

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$ with injectivity radius $\mathrm{inj}(p) > 0$. Then $\exp_p$ restricts to a diffeomorphism from the open ball

$$B_{\mathrm{inj}(p)}(0) = \left\{ u \in T_pM \;\middle|\; |u|_g < \mathrm{inj}(p) \right\}$$

onto the open metric ball $B_g\big(p, \mathrm{inj}(p)\big) = \{x \in M \mid d_g(p,x) < \mathrm{inj}(p)\}$, and this ball contains no cut point of p. The radius $\mathrm{inj}(p)$ is the largest for which this holds: $\exp_p$ is not injective on any open ball of radius exceeding $\mathrm{inj}(p)$.

***Proof*:**
Write $\rho = \mathrm{inj}(p)$; the case $\rho = \infty$ is read with $B_\rho(0) = T_pM$ and $B_g(p,\rho) = M$ throughout. Set $D = \left\{ u \in T_pM \;\middle|\; |u|_g < \rho \right\}$, the open ball of radius ρ.

Step 1: no geodesic from p has a cut point or conjugate point before radius ρ. By the second form of definition 6.2.3, $\rho = \inf_{|v|=1} t_{\mathrm{cut}}(v)$, so $t_{\mathrm{cut}}(v) \ge \rho$ for every unit v. Hence for every unit v the geodesic γ_v minimizes on $[0, \rho)$, and in particular has no cut point with parameter less than ρ. Moreover no γ_v has a conjugate point at parameter $t < \rho$: a conjugate point at $t < \rho \le t_{\mathrm{cut}}(v)$ would, by theorem 5.4.5, make γ_v non-minimizing on $[0, t']$ for $t < t' < \rho$, contradicting $t_{\mathrm{cut}}(v) \ge \rho$. By theorem 5.3.2, the absence of conjugate points at parameters below ρ means $d(\exp_p)_u$ is nonsingular for every $u \in D$: writing $u = tv$ with $|v| = 1$ and $t = |u| < \rho$, the differential $d(\exp_p)_{tv}$ is nonsingular because $\gamma_v(t)$ is not conjugate to p. Thus $\exp_p$ is a local diffeomorphism at every point of D.

Step 2: $\exp_p$ is injective on D. Suppose $u_1, u_2 \in D$ with $\exp_p(u_1) = \exp_p(u_2) = x$. Write $u_i = t_i v_i$ with $|v_i| = 1$ and $t_i = |u_i| < \rho$. Each arc $\gamma_{v_i}|_{[0,t_i]}$ has length $t_i < \rho \le t_{\mathrm{cut}}(v_i)$, so it is minimizing, giving $d_g(p,x) = t_i$; hence $t_1 = t_2 =: t < \rho$. If $v_1 \ne v_2$, then $\gamma_{v_1}|_{[0,t]}$ and $\gamma_{v_2}|_{[0,t]}$ are two distinct minimizing geodesics from p to x of common length t. By the converse in theorem 6.2.5, the existence of two distinct minimizing geodesics of length t from p to x, together with γ_{v_1} minimizing on $[0,t]$, forces $t_{\mathrm{cut}}(v_1) = t < \rho$, contradicting $t_{\mathrm{cut}}(v_1) \geq \rho$. Therefore $v_1 = v_2$, and with $t_1 = t_2$ this gives $u_1 = u_2$. So $\exp_p$ is injective on D.

Step 3: $\exp_p$ *maps* D *diffeomorphically onto* $B_g(p,\rho)$. By Steps 1 and 2, $\exp_p|_D$ is an injective local diffeomorphism, hence a diffeomorphism from D onto its image $\exp_p(D)$, an open subset of M. Identify the image. If $u \in D$ then, as in Step 2, $d_g(p, \exp_p(u)) = |u| < \rho$, so $\exp_p(D) \subseteq B_g(p,\rho)$. Conversely let $x \in B_g(p,\rho)$, so $d_g(p,x) < \rho$. By corollary 6.1.4 there is a minimizing geodesic from p to x; write it $\exp_p(u) = x$ with $u = d_g(p,x)\,v$, $|v| = 1$, so $|u| = d_g(p,x) < \rho$ and $u \in D$. Hence $x \in \exp_p(D)$. Therefore $\exp_p(D) = B_g(p,\rho)$, and $\exp_p \colon D \to B_g(p,\rho)$ is a diffeomorphism.

Step 4: the ball contains no cut point, and ρ *is optimal.* Every point $x = \exp_p(u) \in B_g(p,\rho)$ satisfies $d_g(p,x) = |u| < \rho \leq \inf_v t_{\mathrm{cut}}(v)$, so x is reached before any cut time and is not a cut point; equivalently $\mathrm{Cut}(p) \cap B_g(p,\rho) = \varnothing$, consistent with $\rho = d_g(p, \mathrm{Cut}(p))$. For optimality, suppose $\rho < \infty$ and let $\rho' > \rho$. By definition of ρ as the infimum of the cut times there is a unit v with $t_{\mathrm{cut}}(v) < \rho'$; set $t_c = t_{\mathrm{cut}}(v)$ and $q = \gamma_v(t_c)$. By theorem 6.2.5, either $d(\exp_p)_{t_c v}$ is singular, so $\exp_p$ is not a local diffeomorphism at $t_c v \in B_{\rho'}(0)$, or there is a unit $w \neq v$ with $\exp_p(t_c w) = q = \exp_p(t_c v)$, and then $t_c v$, $t_c w$ are two distinct points of $B_{\rho'}(0)$ (both of norm $t_c < \rho'$) with the same image. In either case $\exp_p$ fails to be a diffeomorphism onto its image on the open ball $B_{\rho'}(0)$. Hence no open ball of radius exceeding ρ is carried diffeomorphically by $\exp_p$, proving optimality and completing the proof.

The proposition is the payoff of the whole section. It says that the injectivity radius is precisely the size of the largest normal ball at p: inside $B_g(p, \mathrm{inj}(p))$ the exponential map is a chart, normal coordinates are valid, every point is reached by a unique minimizing geodesic, and the local geometry is that of a ball in the tangent space; at radius $\mathrm{inj}(p)$ the chart breaks down, for the reason the cut-point characterization supplies, either a fold of $\exp_p$ (conjugate point) or an overlap (two minimizers). On the sphere S^n_r this recovers the concrete picture of example 6.2.4: the exponential map is a diffeomorphism on the open ball of radius πr in $T_pS^n_r$, onto the sphere minus the antipode, and no larger ball works because the sphere of radius πr collapses to the single point $-p$. For $\mathbb{R}^n$ and H^n, where $\mathrm{inj}(p) = \infty$, the proposition reduces to the global statement that $\exp_p$ is a diffeomorphism of all of T_pM onto M, which is the Cartan–Hadamard conclusion proved in section 6.4; here it is anticipated as the case $\rho = \infty$, in which no cut point and no conjugate point ever occur.

A quantitative consequence for compact manifolds is immediate and closes the section. On a compact M every cut time is finite and the injectivity radius is a positive number attained as a distance, so it can be estimated by the two obstructions of the cut-point theorem: the length of the shortest closed geodesic and the distance to the first conjugate point along any geodesic. The distance to the first conjugate point is controlled by curvature through the Rauch comparison theorem of section 6.5, and the length of the shortest closed geodesic is a global invariant; together they give the standard lower bounds for $\mathrm{inj}(M)$. That the infimum defining $\mathrm{inj}(M)$ is positive on a compact manifold follows because $p \mapsto \mathrm{inj}(p)$ is continuous and positive, so it attains a positive minimum on the compact M; the continuity is a consequence of the two mechanisms in theorem 6.2.5 varying continuously with the base point and direction. The upshot is that a compact Riemannian manifold has a well-defined positive scale $\mathrm{inj}(M)$ below which it is uniformly modeled on Euclidean balls, and this scale is the quantitative form of the local-to-global principle that opened the chapter.

6.3 Bonnet–Myers

The second variation of energy showed that a great-circle arc on the round sphere stops minimizing once it passes the antipode: past that point the positive curvature, accumulated over a long enough geodesic, overwhelms the bending term and produces a deformation that shortens the arc. The distance to the antipode is πr on S^n_r, and the sphere has no pair of points farther apart than that. A lower bound on curvature, imposed uniformly over the manifold, caps how long a geodesic can minimize, and therefore caps the distance between any two points. The Bonnet–Myers theorem is the precise form of this principle, and it needs only a lower bound on the Ricci curvature, an average of sectional curvatures rather than a bound on each one.

The mechanism is the one already assembled. Along a minimizing geodesic the index form is nonnegative, by corollary 5.1.2; if the geodesic is too long, a family of explicit test fields makes the index form negative in aggregate, so at least one of them makes it negative, contradicting minimization. Summing over $n-1$ orthonormal normal fields converts the curvature terms into a single Ricci curvature, which is where the hypothesis enters at exactly the right strength. The conclusion is a bound on the diameter, and from a finite diameter on a complete manifold the topology follows at once: the manifold is compact, and, applying the same bound to its universal cover, its fundamental group is finite. Throughout, (M, g) is a connected Riemannian manifold of dimension n with Levi-Civita connection ∇, curvature endomorphism R of definition 4.1.2 in the sign convention of box 4.1.1, and Ricci curvature Ric of definition 4.4.1; D_t is the covariant derivative along a curve, and the index form I is that of definition 5.4.1. The Einstein summation convention is not needed here; sums are written explicitly.

Theorem 6.3.1: Bonnet–Myers

Let (M, g) be a complete, connected Riemannian manifold of dimension $n \geq 2$, and suppose there is a constant $k > 0$ with

$$\mathrm{Ric} \geq (n-1)k\, g,$$

meaning $\mathrm{Ric}(X, X) \geq (n-1)k\, |X|^2$ for every $X \in T_pM$ and every $p \in M$. Then the diameter of M satisfies

$$\mathrm{diam}(M) = \sup_{p,q\in M} d_g(p, q) \leq \frac{\pi}{\sqrt{k}}$$

Consequently M is compact, and its fundamental group $\pi_1(M)$ is finite.

The proof of the diameter bound occupies the bulk of the argument; compactness and the finiteness of π_1 are drawn out from it afterward and are separated into corollary 6.3.2. The heart of the matter is the following: a minimizing geodesic cannot be longer than $\pi/\sqrt{k}$, for if it were, the second variation would detect that it can be shortened.

***Proof*:**
The claim is the diameter bound $\mathrm{diam}(M) \leq \pi/\sqrt{k}$; the topological consequences are deferred to corollary 6.3.2. Suppose, for contradiction, that there are points $p, q \in M$ with $d_g(p, q) = L > \pi/\sqrt{k}$. Because M is complete and connected, corollary 6.1.4 (a consequence of the Hopf–Rinow theorem theorem 6.1.3) provides a minimizing geodesic from p to q; parametrize it by arc length as $\gamma\colon [0, L] \to M$, so that γ has unit speed, $\gamma(0) = p$, $\gamma(L) = q$, and $L(\gamma) = d_g(p, q) = L$. Being minimizing, γ minimizes length, hence energy among fixed-endpoint curves of the same speed,

so corollary 5.1.2 applies: for every smooth field V along γ with $V(0) = V(L) = 0$,

$$I(V,V) = \int_0^L \Big(|D_t V|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle \Big)\, dt \geq 0 \tag{6.5}$$

The strategy is to construct fields V that violate (6.5) once $L > \pi/\sqrt{k}$.

Step 1: a parallel orthonormal normal frame. The velocity $\dot\gamma$ is a parallel unit field along γ, since γ is a geodesic (definition 2.4.1). Choose an orthonormal basis $\{e_1, \dots, e_{n-1}\}$ of the orthogonal complement of $\dot\gamma(0)$ in T_pM, and let E_i be the parallel field along γ with $E_i(0) = e_i$, obtained by parallel transport (definition 2.4.1). Parallel transport is a linear isometry (proposition 2.4.4) and preserves the inner product with the parallel field $\dot\gamma$, so for every t the vectors $E_1(t), \dots, E_{n-1}(t)$ are orthonormal and orthogonal to $\dot\gamma(t)$. Together with $\dot\gamma(t)$ they form an orthonormal basis of $T_{\gamma(t)}M$.

Step 2: the test fields and their index-form values. For each $i = 1, \dots, n-1$ define the field

$$V_i(t) = \sin\!\Big(\frac{\pi t}{L}\Big)\, E_i(t)$$

along γ. Each V_i is smooth and vanishes at $t = 0$ and $t = L$, since $\sin 0 = \sin\pi = 0$; thus each V_i is admissible in (6.5). Because E_i is parallel, $D_t E_i = 0$, and

$$D_t V_i = \frac{\pi}{L}\cos\!\Big(\frac{\pi t}{L}\Big) E_i, \qquad |D_t V_i|^2 = \frac{\pi^2}{L^2}\cos^2\!\Big(\frac{\pi t}{L}\Big),$$

using $|E_i| \equiv 1$. For the curvature term, $V_i = \sin(\pi t/L) E_i$ gives

$$\langle R(V_i,\dot\gamma)\dot\gamma, V_i\rangle = \sin^2\!\Big(\frac{\pi t}{L}\Big)\, \langle R(E_i,\dot\gamma)\dot\gamma, E_i\rangle = \sin^2\!\Big(\frac{\pi t}{L}\Big)\, \mathrm{Riem}(E_i,\dot\gamma,\dot\gamma,E_i),$$

by the bilinearity of R and the definition of the $(0,4)$-tensor (definition 4.1.4). Substituting into the index form,

$$I(V_i,V_i) = \int_0^L \Big(\frac{\pi^2}{L^2}\cos^2\!\Big(\frac{\pi t}{L}\Big) - \sin^2\!\Big(\frac{\pi t}{L}\Big)\, \mathrm{Riem}(E_i,\dot\gamma,\dot\gamma,E_i) \Big)\, dt$$

Step 3: summing over the frame produces the Ricci curvature. Add the $n-1$ values. The bending terms are identical, so they contribute $(n-1)\frac{\pi^2}{L^2}\cos^2(\pi t/L)$, and the curvature terms combine into a trace:

$$\sum_{i=1}^{n-1} \mathrm{Riem}(E_i,\dot\gamma,\dot\gamma,E_i) = \sum_{i=1}^{n-1} \langle R(E_i,\dot\gamma)\dot\gamma, E_i\rangle$$

This sum is exactly the Ricci curvature in the direction $\dot\gamma$. By definition 4.4.1, in the sign convention of box 4.1.1,

$$\mathrm{Ric}(\dot\gamma,\dot\gamma) = \sum_{a=1}^{n} \langle R(f_a,\dot\gamma)\dot\gamma, f_a\rangle$$

for any orthonormal basis $\{f_a\}$ of $T_{\gamma(t)}M$; take the basis $\{f_a\} = \{\dot\gamma, E_1, \dots, E_{n-1}\}$ of Step 1. The term with $f_a = \dot\gamma$ vanishes, since $\langle R(\dot\gamma,\dot\gamma)\dot\gamma,\dot\gamma\rangle = \mathrm{Riem}(\dot\gamma,\dot\gamma,\dot\gamma,\dot\gamma) = 0$ by the antisymmetry of the curvature tensor in its first pair (proposition 4.2.1). The remaining $n-1$ terms are precisely the sum above, so

$$\sum_{i=1}^{n-1} \mathrm{Riem}(E_i,\dot\gamma,\dot\gamma,E_i) = \mathrm{Ric}(\dot\gamma,\dot\gamma)$$

Therefore
$$\sum_{i=1}^{n-1} I(V_i, V_i) = \int_0^L \Big((n-1)\frac{\pi^2}{L^2}\cos^2\Big(\frac{\pi t}{L}\Big) - \sin^2\Big(\frac{\pi t}{L}\Big)\ \mathrm{Ric}(\dot\gamma,\dot\gamma)\Big)\,dt$$

Step 4: the curvature hypothesis makes the sum negative. The geodesic has unit speed, $|\dot\gamma| = 1$, so the hypothesis $\mathrm{Ric} \geq (n-1)k\, g$ gives

$$\mathrm{Ric}(\dot\gamma,\dot\gamma) \geq (n-1)k\ |\dot\gamma|^2 = (n-1)k$$

at every point of γ. Since $\sin^2(\pi t/L) \geq 0$, this bound gives $-\sin^2(\pi t/L)\ \mathrm{Ric}(\dot\gamma,\dot\gamma) \leq -(n-1)k\ \sin^2(\pi t/L)$, so the sum is bounded above by

$$\sum_{i=1}^{n-1} I(V_i, V_i) \leq (n-1)\int_0^L \Big(\frac{\pi^2}{L^2}\cos^2\Big(\frac{\pi t}{L}\Big) - k\ \sin^2\Big(\frac{\pi t}{L}\Big)\Big)\,dt$$

The two elementary integrals over a half-period are equal,

$$\int_0^L \cos^2\Big(\frac{\pi t}{L}\Big)\,dt = \int_0^L \sin^2\Big(\frac{\pi t}{L}\Big)\,dt = \frac{L}{2},$$

so
$$\sum_{i=1}^{n-1} I(V_i, V_i) \leq (n-1)\cdot\frac{L}{2}\Big(\frac{\pi^2}{L^2} - k\Big) = \frac{(n-1)L}{2}\Big(\frac{\pi^2}{L^2} - k\Big)$$

By assumption $L > \pi/\sqrt{k}$, so $L^2 > \pi^2/k$, that is $\pi^2/L^2 < k$, and the factor $\pi^2/L^2 - k$ is strictly negative. Hence
$$\sum_{i=1}^{n-1} I(V_i, V_i) < 0$$

Step 5: contradiction. A sum of $n-1$ real numbers is negative only if at least one summand is negative, so $I(V_{i_0}, V_{i_0}) < 0$ for some index i_0. But V_{i_0} is a smooth field along the minimizing geodesic γ vanishing at both endpoints, so (6.5) forces $I(V_{i_0}, V_{i_0}) \geq 0$. This contradiction shows that no two points can be at distance greater than $\pi/\sqrt{k}$; that is, $d_g(p,q) \leq \pi/\sqrt{k}$ for all $p, q \in M$, and therefore
$$\mathrm{diam}(M) = \sup_{p,q\in M} d_g(p,q) \leq \frac{\pi}{\sqrt{k}},$$

as we sought to show. The topological conclusions follow in corollary 6.3.2.

The proof isolates exactly where the hypothesis is used, and shows why Ricci curvature, not sectional curvature, is the right object. A single test field V_i sees only the sectional curvature of the plane spanned by E_i and $\dot\gamma$; a bound on that one plane would be enough to make one $I(V_i, V_i)$ negative, but that is a bound in a single direction and cannot be assumed. Summing over the whole orthonormal frame $E_1, \dots, E_{n-1}$ averages these planes, and the average is precisely $\mathrm{Ric}(\dot\gamma,\dot\gamma)$, so the argument needs only a lower bound on this average. This is a weaker hypothesis than a lower bound on every sectional curvature, and it is the natural one: the pigeonhole step at the end, that a negative sum has a negative summand, is what lets an averaged bound do the work of a directional one. The number $\pi/\sqrt{k}$ is the half-period of the coefficient equation $f'' + kf = 0$, the same equation that governs Jacobi fields on a space of constant curvature k (example 5.2.5), which is why the threshold coincides with the first conjugate distance $\pi r = \pi/\sqrt{k}$ on the sphere S^n_r with $r = 1/\sqrt{k}$.

With the diameter bounded, the topology of M is constrained. A complete manifold of finite diameter is compact, and a compact manifold satisfying the same Ricci bound has, after passing to its universal cover, a finite fundamental group.

Corollary 6.3.2: Compactness and Finite Fundamental Group

Let (M, g) be a complete, connected Riemannian manifold of dimension $n \geq 2$ with $\mathrm{Ric} \geq (n-1)k\, g$ for a constant $k > 0$. Then:

1. M is compact.
2. The fundamental group $\pi_1(M)$ is finite.

***Proof*:**

Both statements rest on the diameter bound $\mathrm{diam}(M) \leq \pi/\sqrt{k}$ of theorem 6.3.1.

1. Fix $p \in M$ and set $\rho = \pi/\sqrt{k}$. Every point $q \in M$ satisfies $d_g(p, q) \leq \mathrm{diam}(M) \leq \rho$, so M equals the closed metric ball $\overline{B}_\rho(p) = \{q \in M \mid d_g(p, q) \leq \rho\}$. This set is closed and bounded, so by the Hopf–Rinow theorem (theorem 6.1.3, the equivalence of completeness with the compactness of closed bounded sets) it is compact. Hence M is compact.
2. The manifold M is a connected smooth manifold, hence locally path-connected and semilocally simply connected, so it admits a universal covering map $\pi\colon \tilde{M} \to M$ with $\tilde{M}$ connected and simply connected; this is the standard existence theorem for the universal cover of a manifold (see [6]). Equip $\tilde{M}$ with the pullback metric $\tilde{g} = \pi^* g$. Since π is a local diffeomorphism, $\tilde{g}$ is a Riemannian metric and $\pi\colon (\tilde{M}, \tilde{g}) \to (M, g)$ is a local isometry, indeed a Riemannian covering: each point of M has an evenly covered neighborhood on which π restricts to an isometry with each sheet.

   Three properties of $\tilde{M}$ are needed. First, $\tilde{M}$ is complete. A maximal geodesic $\tilde{\sigma}$ of $\tilde{M}$ projects to a geodesic $\sigma = \pi \circ \tilde{\sigma}$ of M, because π is a local isometry and local isometries carry geodesics to geodesics; conversely, since M is complete, σ extends to all of $\mathbb{R}$, and by the unique path-lifting property of the covering π its extension lifts back through $\tilde{\sigma}$'s initial data to an extension of $\tilde{\sigma}$ over all of $\mathbb{R}$. Thus every maximal geodesic of $\tilde{M}$ is defined on all of $\mathbb{R}$, so $\tilde{M}$ is geodesically complete, hence complete by theorem 6.1.3 (the equivalence of geodesic completeness with metric completeness). Second, $\tilde{M}$ satisfies the same Ricci bound. Curvature is a local isometry invariant, and π is a local isometry, so the curvature tensor of $\tilde{g}$ at a point $\tilde{x}$ is the pullback of the curvature tensor of g at $\pi(\tilde{x})$; the Ricci tensor is a contraction of the curvature tensor with the metric, so likewise $\mathrm{Ric}_{\tilde{g}} = \pi^* \mathrm{Ric}_g$. Because π is a local isometry, $\mathrm{Ric}_g \geq (n-1)k\, g$ pulls back to $\mathrm{Ric}_{\tilde{g}} \geq (n-1)k\, \tilde{g}$. Third, $\tilde{M}$ has the same dimension n.

   Therefore $(\tilde{M}, \tilde{g})$ is a complete, connected Riemannian n-manifold with $\mathrm{Ric}_{\tilde{g}} \geq (n-1)k\, \tilde{g}$, and theorem 6.3.1 together with part (1) applies to it: $\tilde{M}$ is compact. Now consider a fiber $\pi^{-1}(p)$ over a point $p \in M$. It is a discrete closed subset of the compact space $\tilde{M}$, hence finite. For the universal cover, the fiber $\pi^{-1}(p)$ is in bijection with the deck transformation group, which is isomorphic to $\pi_1(M, p)$; this is the standard correspondence between the fiber of the universal cover and the fundamental group (see [6]). Hence $\pi_1(M)$ is finite, completing the proof.

The two consequences are of a different character from the diameter bound but follow from it with no further geometry. Compactness is immediate: a complete manifold whose points are all within a fixed distance of a base point is a closed bounded set, which Hopf–Rinow makes compact. The finiteness of π_1 is subtler only in that it passes to the universal cover, where the same curvature bound holds because curvature is local and the covering is a local isometry; the cover is then compact for the same reason M is, and a compact cover has finite fibers, which for the universal cover count the fundamental group. The contrapositive is a source of nonexistence results: a manifold with infinite fundamental group, such as a compact hyperbolic manifold or a flat torus, admits no metric with Ricci curvature bounded below by a positive constant. The flat torus $\mathbb{R}^n/\mathbb{Z}^n$ has $\mathrm{Ric} \equiv 0$ and infinite fundamental group $\mathbb{Z}^n$, consistent with the theorem, since 0 is not bounded below by any $(n-1)k$ with $k > 0$; and although the torus is compact, its noncompact universal cover $\mathbb{R}^n$ shows directly that no positive Ricci lower bound can hold.

It remains to see that the diameter bound cannot be improved: the sphere of the matching radius attains it. This is the exact case in which every inequality in the proof becomes an equality.

Example 6.3.3: Sharpness on the round sphere

Fix $k > 0$ and let $M = S^n(1/\sqrt{k})$ be the round sphere of radius $r = 1/\sqrt{k}$ in $\mathbb{R}^{n+1}$, with the induced metric. It is compact and complete (a compact manifold is complete, and the sphere is compact), and its curvature was computed in Chapter 4. By example 4.4.9, the sphere S^n_r of radius r has

$$\mathrm{Ric} = \frac{n-1}{r^2}\, g$$

With $r = 1/\sqrt{k}$ this is $\mathrm{Ric} = (n-1)k\, g$, so the hypothesis of theorem 6.3.1 holds with equality: the Ricci curvature equals its lower bound in every direction, at every point. The theorem then gives $\mathrm{diam}(M) \le \pi/\sqrt{k}$.

This bound is attained. The distance function on S^n_r was computed in example 3.4.4: for $p, q \in S^n_r \subset \mathbb{R}^{n+1}$, with $\langle p, q\rangle$ the ambient inner product of $\mathbb{R}^{n+1}$,

$$d_g(p,q) = r \arccos\Big(\frac{\langle p, q\rangle}{r^2}\Big),$$

the length of the shorter great-circle arc joining them. The largest possible value of d_g occurs when $\langle p, q\rangle = -r^2$, that is when $q = -p$ is the antipode of p; then $\arccos(-1) = \pi$ and

$$d_g(p, -p) = r\pi = \frac{\pi}{\sqrt{k}}$$

Since no pair of points is farther apart than an antipodal pair, $\mathrm{diam}(S^n(1/\sqrt{k})) = \pi/\sqrt{k}$, exactly the Bonnet–Myers bound. The estimate of theorem 6.3.1 is therefore sharp, and the sphere is the extremal manifold.

The sharpness is legible inside the proof of theorem 6.3.1. On S^n_r the sectional curvature is the constant $1/r^2 = k$ in every plane (example 4.3.4), so $\mathrm{Riem}(E_i, \dot\gamma, \dot\gamma, E_i) = k$ for each parallel normal field E_i along a unit-speed great circle, and $\mathrm{Ric}(\dot\gamma,\dot\gamma) = (n-1)k$ with no slack in the inequality of Step 4. Taking a minimizing arc of length exactly $L = \pi/\sqrt{k} = \pi r$ to the antipode, each test field $V_i = \sin(\pi t/L)E_i$ gives

$$I(V_i, V_i) = \int_0^{\pi r} \Big(\frac{\pi^2}{(\pi r)^2}\cos^2\Big(\frac{t}{r}\Big) - k\sin^2\Big(\frac{t}{r}\Big)\Big)\, dt = \int_0^{\pi r} \Big(\frac{1}{r^2}\cos^2\Big(\frac{t}{r}\Big) - \frac{1}{r^2}\sin^2\Big(\frac{t}{r}\Big)\Big)\, dt = 0,$$

using $\pi/L = 1/r$ and $k = 1/r^2$. The value is zero, not negative: at the threshold length the second variation degenerates rather than turning negative, which is the borderline signalled at the antipode in example 5.1.3. Indeed each such $V_i = \sin(t/r)E_i$ is precisely the Jacobi field of example 5.2.5 vanishing at both ends of the arc to the first conjugate point, so the antipode is simultaneously the diameter-realizing point, the first conjugate point (example 5.3.4), and the cut point of p. Any length $L > \pi r$ pushes the arc past the antipode, makes the sum of Step 3 strictly negative as in the proof, and the arc ceases to minimize. This is the same phenomenon recorded from the second-variation side in example 5.1.3, now read as the extremal case of the diameter theorem.

Positive Ricci curvature thus imposes a hard ceiling on size and forces both compactness and near-triviality of the fundamental group, with the round sphere as the extremal model. The two remaining model spaces sit on the other side of the hypothesis. Euclidean space has $\mathrm{Ric} \equiv 0$ and hyperbolic space has $\mathrm{Ric} = -(n-1)g$ (example 4.4.9), so neither meets a positive Ricci lower bound, and neither is compact: both are diffeomorphic to $\mathbb{R}^n$, of infinite diameter. That noncompactness is not accidental but forced by the sign of the curvature is the content of the Cartan–Hadamard theorem of the next section, which unwraps a complete simply connected manifold of nonpositive curvature onto all of $\mathbb{R}^n$.

6.4 Cartan–Hadamard

Bonnet–Myers read one sign of the curvature and returned compactness: positive Ricci curvature refocuses geodesics, bounds the diameter, and closes the manifold up. The opposite sign runs the other way. On a manifold of nonpositive sectional curvature the geodesics from a point spread at least as fast as in Euclidean space and never come back together, so the exponential map develops no singularities. The proposition of section 5.3 already recorded the infinitesimal statement, that $d(\exp_p)_v$ is nonsingular everywhere; what remains is to make it global. The Cartan–Hadamard theorem does exactly that: it upgrades the everywhere-local diffeomorphism to a genuine diffeomorphism of the whole tangent space onto the manifold, provided the manifold is complete and simply connected. The conclusion is that a complete simply connected manifold with $K \le 0$ is diffeomorphic to $\mathbb{R}^n$ and carries a unique geodesic between any two of its points.

Three ingredients combine, and each is worth isolating. Nonpositive curvature supplies the absence of conjugate points, hence the local diffeomorphism. Completeness supplies enough room for geodesics to run forever, which is what lets a local diffeomorphism be promoted to a covering map. Simple connectivity then forces the covering to be a single-sheeted one, that is, a diffeomorphism. Removing any one of the three destroys the conclusion: the flat torus has $K \equiv 0$ and is complete but not simply connected, and it is not diffeomorphic to $\mathbb{R}^n$; an open geodesic ball in $\mathbb{R}^n$ has $K \equiv 0$ and is simply connected but incomplete, and its exponential map is defined only on a bounded set. The argument that follows is where the completeness hypothesis, and with it the Riemannian setting through Hopf–Rinow, does its work; the Lorentzian analogue fails for the same reason Hopf–Rinow does.

Throughout, (M, g) is a connected Riemannian manifold of dimension n with Levi-Civita connection ∇ (definition 2.3.2), curvature endomorphism R (definition 4.1.2), and sectional curvature K (definition 4.3.1); γ denotes a geodesic (definition 3.1.1) and D_t the covariant derivative along it. The curvature and second-variation sign conventions of Chapters 4 and 5 are in force. The Einstein summation convention is used where indices appear.

The metric estimate underlying the theorem is a quantitative version of the convexity argument that excluded conjugate points. There it was enough that $|J|^2$ be convex; here the same convexity is pushed to a lower bound on the growth of a Jacobi field, and thereby on the differential of the exponential map.

Proposition 6.4.1: Nonpositive Curvature Makes the Exponential Map Expanding

Let (M, g) have sectional curvature $K \leq 0$. Let $\gamma\colon [0, \infty) \to M$ be a geodesic and J a Jacobi field along γ with $J(0) = 0$. Then

$$|J(t)| \geq t\,|D_t J(0)| \qquad \text{for all } t \geq 0 \tag{6.6}$$

Consequently, if $p \in M$ and $v \in T_pM$ lies in the domain of $\exp_p$, then for every $w \in T_pM$ (identifying $T_v(T_pM)$ with T_pM as usual)

$$\left|d(\exp_p)_v(w)\right| \geq |w| \tag{6.7}$$

In particular $d(\exp_p)_v$ is injective at every v in the domain, and $\exp_p$ is distance nondecreasing wherever it is defined.

***Proof*:**
The bound (6.6) is proved first, then transferred to the differential of $\exp_p$ through the Jacobi-field description of theorem 5.3.2.

Step 1: the lower bound on the Jacobi field. Write $f(t) = |J(t)|^2 = \langle J, J\rangle$. Differentiating twice with metric compatibility (definition 2.2.4), and using the Jacobi equation (5.1) in the form $D_t^2 J = -R(J, \dot\gamma)\dot\gamma$ exactly as in the proof of proposition 5.3.3,

$$f''(t) = 2\left\langle D_t^2 J, J\right\rangle + 2\,|D_t J|^2 = -2\,\mathrm{Riem}(J, \dot\gamma, \dot\gamma, J) + 2\,|D_t J|^2$$

The term $\mathrm{Riem}(J, \dot\gamma, \dot\gamma, J) = K(\Pi)\big(|J|^2\,|\dot\gamma|^2 - \langle J, \dot\gamma\rangle^2\big)$ is ≤ 0, since $K \leq 0$ and the Gram determinant is nonnegative by Cauchy–Schwarz (definition 4.3.1); this identity holds also where J and $\dot\gamma$ are dependent, both sides then vanishing. Hence

$$f''(t) \geq 2\,|D_t J|^2 \geq 0 \tag{6.8}$$

so $f = |J|^2$ is convex on $[0, \infty)$, with $f(0) = 0$.

Set $a = |D_t J(0)|$. If $a = 0$ then $J(0) = 0$ and $D_t J(0) = 0$, so $J \equiv 0$ by uniqueness of Jacobi fields from initial data (theorem 5.2.3) and (6.6) reads $0 \geq 0$. Assume $a > 0$; then $J \not\equiv 0$, so by proposition 5.3.3 the field J has no zero on $(0, \infty)$, and the norm $g(t) = |J(t)|$ is smooth and positive there. The claim (6.6) is that $g(t) \geq at$, and it will follow from convexity of g. Consider the ratio

$$u(t) = \frac{g(t)}{t} = \frac{|J(t)|}{t} \qquad (t > 0)$$

Because $J(0) = 0$ and J is smooth with initial derivative $D_t J(0)$, one has $|J(t)| = at + o(t)$ as $t \downarrow 0$, so $u(t) \to a$ as $t \downarrow 0$. It suffices to show g is convex on $(0, \infty)$; then, as shown below, u is nondecreasing, so $u(t) \geq \lim_{s\downarrow 0} u(s) = a$, which is the claim. Where $J \neq 0$ the identity $f = g^2$ gives $f' = 2gg'$ and $f'' = 2g'^2 + 2gg''$, whence $g\,g'' = \frac{1}{2}f'' - g'^2$. By (6.8), $\frac{1}{2}f'' \geq |D_t J|^2$; and by

Cauchy–Schwarz $g' = \langle D_t J, J\rangle \,/\, |J|$ satisfies $g'^2 \le |D_t J|^2$. Therefore

$$g\,g'' = \tfrac{1}{2}f'' - g'^2 \ge |D_t J|^2 - |D_t J|^2 \cdot \frac{\langle D_t J, J\rangle^2}{|D_t J|^2\,|J|^2} \ge 0$$

whenever $D_t J \neq 0$, and $g\,g'' \ge 0$ trivially where $D_t J = 0$ (there $g'' \ge 0$ from (6.8)). Since $g > 0$ on $(0,\infty)$ this gives $g'' \ge 0$, so $g = |J|$ is convex on $(0,\infty)$; and g extends continuously to $t = 0$ with $g(0) = 0$, so it is convex on $[0,\infty)$. A convex function with $g(0) = 0$ has $g(t)/t$ nondecreasing: for $0 < s < t$, convexity with $g(0) = 0$ gives $g(s) = g\big(\frac{s}{t}\cdot t + (1-\frac{s}{t})\cdot 0\big) \le \frac{s}{t}g(t)$, that is $g(s)/s \le g(t)/t$. Letting $s \downarrow 0$ yields $u(t) = g(t)/t \ge a$ for every $t > 0$, which is (6.6); at $t = 0$ the inequality is the equality $0 = 0$.

Step 2: the differential of $\exp_p$. Fix $p \in M$, let $v \in T_pM$ be in the domain of $\exp_p$, and let $w \in T_pM$. If $v = 0$ then $d(\exp_p)_0 = \mathrm{id}$ (proposition 3.2.5) and (6.7) is an equality. Assume $v \neq 0$ and set $\gamma(t) = \exp_p(tv)$, the geodesic with $\dot\gamma(0) = v$, defined for t in an interval containing $[0,1]$. By part (1) of theorem 5.3.2 the field

$$J_w(t) = d(\exp_p)_{tv}(t\,w)$$

is the Jacobi field along γ with $J_w(0) = 0$ and $D_t J_w(0) = w$. Applying (6.6) at $t = 1$,

$$\big|d(\exp_p)_v(w)\big| = |J_w(1)| \ge 1 \cdot |D_t J_w(0)| = |w|$$

which is (6.7). In particular $d(\exp_p)_v(w) = 0$ forces $w = 0$, so $d(\exp_p)_v$ is injective, and being a linear map between spaces of equal dimension n it is a linear isomorphism. That $\exp_p$ is distance nondecreasing is the infinitesimal statement integrated: for a smooth curve σ in the domain, the velocity of $\exp_p \circ \sigma$ is $d(\exp_p)_\sigma(\dot\sigma)$, whose length is $\ge |\dot\sigma|$ by (6.7), so $L(\exp_p \circ \sigma) \ge L(\sigma)$, as claimed.

The inequality (6.7) is the metric heart of the theorem. It says that under nonpositive curvature the exponential map, viewed as a map from the flat tangent space into M, never contracts: distances measured in T_pM are lower bounds for the distances of their images. The model case makes the estimate transparent. On Euclidean space $\mathbb{R}^n$, where $K \equiv 0$, the exponential map at the origin is the identity of $\mathbb{R}^n$ and (6.7) is an equality, with the Jacobi field $J_w(t) = tw$ satisfying $|J_w(t)| = t\,|w|$ exactly. On hyperbolic space H^n, where $K \equiv -1$, the normal Jacobi field with $J(0) = 0$ and $|D_t J(0)| = 1$ is $J(t) = \sinh(t)\,E(t)$ (example 5.2.5), so $|J(t)| = \sinh t \ge t$ for $t \ge 0$: the exponential map strictly expands, and the more so as t grows, which is the analytic form of hyperbolic space being larger than Euclidean space at every radius. The estimate degenerates for positive curvature, and indeed must: on S^n_r the coefficient $\sin(t/r)$ returns to zero at $t = \pi r$, so no lower bound of the form $t\,|D_t J(0)|$ can hold, and the exponential map is singular there.

With the differential of $\exp_p$ under control, the main theorem is a matter of promoting a local diffeomorphism to a global one. The step that requires completeness is the covering-map property; the following lemma isolates it.

Lemma 6.4.2: A Distance-Nondecreasing Local Diffeomorphism from a Complete Manifold is a Covering

Let $\widetilde{N}$ and N be connected Riemannian manifolds, with $\widetilde{N}$ complete, and let $\Phi\colon \widetilde{N} \to N$ be a smooth surjective local diffeomorphism such that $|d\Phi_x(u)| \geq |u|$ for every $x \in \widetilde{N}$ and every $u \in T_x\widetilde{N}$. Then Φ is a covering map.

***Proof*:**
The idea is the path-lifting criterion: a local diffeomorphism Φ under which every geodesic of N lifts, from any chosen starting preimage, to a curve defined for all parameter values, is a covering map. Completeness of $\widetilde{N}$ together with the expanding estimate supplies exactly this lifting. The argument is given in three parts.

1. Φ *does not decrease lengths, hence does not decrease distances.* For any piecewise-smooth curve σ in $\widetilde{N}$, the hypothesis $|d\Phi(u)| \geq |u|$ gives $L(\Phi \circ \sigma) \geq L(\sigma)$ pointwise on velocities, so $L(\Phi \circ \sigma) \geq L(\sigma)$. Taking infima over curves joining two points $x, y \in \widetilde{N}$, and noting $\Phi \circ \sigma$ joins $\Phi(x)$ to $\Phi(y)$,
$$d_N\big(\Phi(x), \Phi(y)\big) \leq d_{\widetilde{N}}(x, y)$$
Reversing the estimate is not needed; this direction alone is used below to keep lifts within controlled distance.

2. *Geodesics of N lift for all time.* Let $c\colon [0, \ell] \to N$ be a geodesic (in particular a smooth curve) and let $\tilde{x}_0 \in \Phi^{-1}(c(0))$. Because Φ is a local diffeomorphism, there is a unique lift $\tilde{c}$ of c with $\tilde{c}(0) = \tilde{x}_0$ on some maximal subinterval $[0, \beta) \subset [0, \ell]$ (uniqueness and local existence of the lift are the standard consequence of Φ being a local diffeomorphism: on a neighborhood where Φ inverts, set $\tilde{c} = \Phi^{-1} \circ c$, and patch). Suppose $\beta \leq \ell$. The lift is a curve into the *complete* manifold $\widetilde{N}$; its length is controlled by part (1) applied to $\Phi \circ \tilde{c} = c$: for $0 \leq s < t < \beta$,
$$d_{\widetilde{N}}\big(\tilde{c}(s), \tilde{c}(t)\big) \leq L\big(\tilde{c}|_{[s,t]}\big) \leq L\big(c|_{[s,t]}\big) = \int_s^t |\dot{c}|$$
the middle inequality because the expanding estimate gives $\left|\dot{\tilde{c}}\right| = \left|(\Phi^{-\dot{1}} \circ c)\right| \leq |\dot{c}|$ (from $|d\Phi(u)| \geq |u|$, the inverse contracts). Since $|\dot{c}|$ is bounded on $[0, \ell]$ (a geodesic has constant speed), the right-hand side tends to 0 as $s, t \uparrow \beta$, so $\tilde{c}(t)$ is a Cauchy net in $\widetilde{N}$ as $t \uparrow \beta$. By completeness of $\widetilde{N}$ (theorem 6.1.3, in the equivalence of metric completeness with geodesic completeness) the limit $\tilde{x}_\beta = \lim_{t\uparrow\beta} \tilde{c}(t)$ exists in $\widetilde{N}$, and by continuity $\Phi(\tilde{x}_\beta) = c(\beta)$. Now Φ is a local diffeomorphism at $\tilde{x}_\beta$, so the lift extends past β (again by inverting Φ near $\tilde{x}_\beta$), contradicting maximality unless $\beta = \ell$. Hence $\beta = \ell$ and every geodesic of N lifts on its full parameter interval, from any prescribed starting preimage.

3. *The lifting property gives evenly covered neighborhoods.* Fix $q \in N$. Since N is a manifold, choose $\varepsilon > 0$ with $\exp_q$ a diffeomorphism of the ball $B(0, \varepsilon) \subset T_qN$ onto a geodesic ball $U = B(q, \varepsilon)$ (theorem 3.4.3 provides such a normal ball). Every point of U is joined to q by a unique radial geodesic inside U. Let $\tilde{q} \in \Phi^{-1}(q)$ be one preimage. For $x \in U$, let c_x be the radial geodesic from q to x; by part (2) it lifts uniquely to a curve $\tilde{c}_x$ starting at $\tilde{q}$,

and define $\lambda_{\tilde{q}}(x) = \tilde{c}_x(1)$ (parametrizing c_x on $[0,1]$). This $\lambda_{\tilde{q}}\colon U \to \widetilde{N}$ is smooth (lifts depend smoothly on the smoothly varying geodesics), satisfies $\Phi \circ \lambda_{\tilde{q}} = \mathrm{id}_U$, and maps q to $\tilde{q}$; so it is a smooth section of Φ over U through $\tilde{q}$, and being a right inverse of the local diffeomorphism Φ it is a diffeomorphism of U onto its image $V_{\tilde{q}} \subset \widetilde{N}$, an open set. The sets $V_{\tilde{q}}$, one for each $\tilde{q} \in \Phi^{-1}(q)$, are pairwise disjoint: if $V_{\tilde{q}} \cap V_{\tilde{q}'} \neq \emptyset$, two lifts of some radial geodesic c_x would agree at their common endpoint and hence, by uniqueness of lifts (part (2)), everywhere, forcing $\tilde{q} = \tilde{q}'$. Their union is all of $\Phi^{-1}(U)$: given $\tilde{x} \in \Phi^{-1}(U)$, let $x = \Phi(\tilde{x}) \in U$ and lift the *reversed* radial geodesic c_x (from x back to q) starting at $\tilde{x}$; its endpoint is a preimage $\tilde{q}$ of q, and running the lift forward exhibits $\tilde{x} \in V_{\tilde{q}}$. Finally Φ restricts to a diffeomorphism $V_{\tilde{q}} \to U$ for each $\tilde{q}$, being the inverse of the diffeomorphism $\lambda_{\tilde{q}}$. Thus U is evenly covered. As q was arbitrary and Φ is surjective, Φ is a covering map, as we sought to show.

The lemma is the point at which completeness is spent, and it is worth naming what it gives. A local diffeomorphism need not be a covering: the exponential chart of an incomplete manifold, or the restriction of a covering to a proper open subset, is a local diffeomorphism that fails to cover. What rules out such pathologies is that geodesics downstairs can always be lifted upstairs for all time, and it is completeness of the source, delivered by Hopf–Rinow (theorem 6.1.3) in its guise as the equivalence of metric and geodesic completeness, that guarantees the lift never escapes to a boundary. This is the sole ingredient of the whole development for which the Riemannian hypothesis, rather than mere smoothness, is used, and it is the ingredient that has no Lorentzian counterpart.

Theorem 6.4.3: Cartan–Hadamard

Let (M, g) be a complete, connected, simply connected Riemannian manifold of dimension n with sectional curvature $K \leq 0$. Then for every $p \in M$ the exponential map

$$\exp_p\colon T_pM \longrightarrow M$$

is a diffeomorphism. In particular M is diffeomorphic to $\mathbb{R}^n$, and M has no conjugate points along any geodesic.

***Proof*:**
Fix $p \in M$. The proof assembles three facts, in the order local diffeomorphism, then covering map, then diffeomorphism.

Step 1: $\exp_p$ is defined on all of T_pM and is a local diffeomorphism. Completeness means geodesics are defined for all time (definition 6.1.1), and by Hopf–Rinow (theorem 6.1.3) this is equivalent to $\exp_p$ being defined on the whole of T_pM. Since $K \leq 0$, proposition 5.3.3 shows $d(\exp_p)_v$ is nonsingular at every $v \in T_pM$; equivalently, by proposition 6.4.1, it satisfies $\left|d(\exp_p)_v(w)\right| \geq |w|$ and is a linear isomorphism at each v. By the inverse function theorem $\exp_p$ is a local diffeomorphism at each point of T_pM.

Step 2: transporting the metric and applying the covering lemma. Equip the vector space T_pM with the pullback metric $\hat{g} = \exp_p^* g$, and write $\Phi = \exp_p\colon (T_pM, \hat{g}) \to (M, g)$. This is a genuine Riemannian metric on T_pM, because $\exp_p$ is a local diffeomorphism (Step 1), so each $d\Phi_v$ is a linear isomorphism and $\hat{g}_v(u,u) = g\big(d\Phi_v u, d\Phi_v u\big) > 0$ for $u \neq 0$. By construction Φ is then a local isometry, so $|d\Phi_v(u)|_g = |u|_{\hat{g}}$ for all u, and in particular the hypothesis $|d\Phi(u)| \geq |u|$ of lemma 6.4.2 holds with equality.

The two remaining hypotheses of the lemma are surjectivity of Φ and completeness of $(T_pM, \hat{g})$. Surjectivity: M is complete and connected, so by Hopf–Rinow (theorem 6.1.3, corollary 6.1.4) every point $q \in M$ is joined to p by a minimizing geodesic $t \mapsto \exp_p(tw)$, whence $q = \exp_p(w)$ lies in the image. Completeness: a straight line $t \mapsto tu$ through the origin of T_pM is mapped by Φ to the curve $t \mapsto \exp_p(tu)$, which is a geodesic of M; since Φ is a local isometry and carries this line to a geodesic, and a local isometry sends geodesics to geodesics and back, the line $t \mapsto tu$ is a $\hat{g}$-geodesic. Its velocity at $t = 0$ is u (identifying $T_0(T_pM)$ with T_pM), so the exponential map of $(T_pM, \hat{g})$ at the origin, $\exp_0^{\hat{g}}(u) =$ (the $\hat{g}$-geodesic from 0 with velocity u at time 1), equals u and is defined for every $u \in T_pM$. Thus $\exp_0^{\hat{g}}$ is defined on all of $T_0(T_pM)$, which by criterion (iii) of Hopf–Rinow (theorem 6.1.3) makes $(T_pM, \hat{g})$ complete. All hypotheses of lemma 6.4.2 are met, so $\Phi = \exp_p$ is a covering map.

Step 3: simple connectivity forces a single sheet. The base M is simply connected. A covering map onto a simply connected, connected space, with connected total space, has exactly one sheet: the number of sheets equals the index of $\Phi_*\pi_1(T_pM)$ in $\pi_1(M)$, and $\pi_1(M) = 1$, so the covering is a homeomorphism onto M (the covering criterion and this sheet count are standard covering-space theory, [4]). The total space T_pM is connected (indeed contractible), which is required for the single-sheet conclusion. Being a covering map that is bijective, $\Phi = \exp_p$ is a homeomorphism, and being in addition a local diffeomorphism it is a diffeomorphism $T_pM \to M$.

The stated consequences follow. Since $T_pM \cong \mathbb{R}^n$ as a smooth manifold and $\exp_p$ is a diffeomorphism, M is diffeomorphic to $\mathbb{R}^n$. That M has no conjugate points is proposition 5.3.3 again, now recorded as a feature of the diffeomorphism: $\exp_p$ nonsingular everywhere means no geodesic through p meets a conjugate point, and p was arbitrary. This completes the proof.

The theorem says that nonpositive curvature, imposed on a complete simply connected manifold, forces the underlying topology to be that of $\mathbb{R}^n$ and nothing else: whatever the manifold looked like, it is diffeomorphic to a vector space, and its geometry is displayed globally in a single exponential chart. This is the promised complement to Bonnet–Myers. There, positive Ricci curvature made the manifold small and compact; here, nonpositive sectional curvature makes it, topologically, as large and open as possible. The two model spaces sit at the extremes: the round sphere realizes the Bonnet–Myers conclusion, and hyperbolic space realizes the Cartan–Hadamard conclusion, with Euclidean space marking the flat boundary between them. It is instructive to see why each hypothesis is needed exactly where it is used. If M is not simply connected, the covering map of Step 2 can genuinely have several sheets, and M is a quotient $\mathbb{R}^n/\Gamma$ rather than $\mathbb{R}^n$ itself; the flat torus $\mathbb{R}^n/\mathbb{Z}^n$, with $K \equiv 0$, is the first example, complete and nonpositively curved but not simply connected and not diffeomorphic to $\mathbb{R}^n$. If M is not complete, the lifting argument of lemma 6.4.2 breaks down at the boundary; an open ball in $\mathbb{R}^n$ shows the exponential map need not even be surjective.

A word on the standing theme of the chapter. The one place completeness entered, the covering lemma, is precisely the place where the Riemannian hypothesis is indispensable. Completeness was used through Hopf–Rinow (theorem 6.1.3), whose equivalence of metric and geodesic completeness fails for Lorentzian metrics; there a geodesically complete manifold need not be metrically complete, Cauchy sequences of lifts need not converge, and geodesics need not lift for all time. The Cartan–Hadamard conclusion has no Lorentzian analogue in this form, and its collapse can be traced to exactly this failure of Hopf–Rinow, flagged when that theorem was proved. Nonpositive curvature by itself, the ingredient that gave the local diffeomorphism, transfers to the pseudo-Riemannian setting; it is the global promotion that does not.

Two consequences record the geometric and topological content of the diffeomorphism.

Corollary 6.4.4: Unique Geodesics and Asphericity

Let M be as in theorem 6.4.3. Then any two points of M are joined by a unique geodesic, and M is contractible; in particular all higher homotopy groups vanish, so M is aspherical (a $K(\pi,1)$ when regarded as a universal cover). Any complete connected manifold of nonpositive curvature has $\mathbb{R}^n$ as its universal cover.

***Proof*:**
Fix $p, q \in M$. By theorem 6.4.3 the map $\exp_p : T_pM \to M$ is a diffeomorphism, so there is a unique $w \in T_pM$ with $\exp_p(w) = q$. The geodesics from p to q are exactly the curves $t \mapsto \exp_p(tv)$ with $\exp_p(v) = q$, one for each preimage of q under $\exp_p$; since there is exactly one such preimage w, there is exactly one geodesic from p to q, namely $t \mapsto \exp_p(tw)$ on $[0, 1]$. As p, q were arbitrary, any two points are joined by a unique geodesic.

Contractibility is immediate from the diffeomorphism $M \cong T_pM \cong \mathbb{R}^n$: a vector space is contractible via the straight-line homotopy $H(x, s) = (1-s)x$ to its origin, transported through the diffeomorphism. A contractible space has all homotopy groups trivial, in particular $\pi_k(M) = 0$ for all $k \geq 1$, so M is aspherical. For a complete connected manifold M_0 of nonpositive curvature that is not assumed simply connected, its universal cover $\widetilde{M}_0$ carries the pulled-back metric, is again complete (the covering is a local isometry and completeness lifts along it, by the lifting argument of lemma 6.4.2 applied to the covering projection), is connected and simply connected by construction, and has the same sectional curvature $K \leq 0$ pointwise since the covering is a local isometry. Theorem 6.4.3 applies to $\widetilde{M}_0$ and gives $\widetilde{M}_0 \cong \mathbb{R}^n$, as claimed.

The corollary is the statement usually attached to the phrase *Cartan–Hadamard manifold*: a complete simply connected manifold of nonpositive curvature, in which geodesics are unique globally and the space is topologically trivial. The uniqueness of geodesics is the global upgrade of the local uniqueness inside a normal ball; on a general manifold two points can be joined by many geodesics, as antipodal points on the sphere are joined by infinitely many great semicircles, but nonpositive curvature suppresses all such refocusing. That every complete nonpositively curved manifold is covered by $\mathbb{R}^n$ is the reason such manifolds are the natural home of geometric group theory: their fundamental groups act freely on a contractible space, so the manifold is a classifying space for the group.

Specializing to constant curvature closes a case left open two chapters earlier.

Corollary 6.4.5: Killing–Hopf for Nonpositive Curvature

A complete, connected, simply connected Riemannian manifold of constant sectional curvature $c \leq 0$ is isometric to Euclidean space $\mathbb{R}^n$ (if $c = 0$) or to the rescaled hyperbolic space H^n of curvature c (if $c < 0$).

***Proof*:**
Let M have constant sectional curvature $c \leq 0$; then $K \leq 0$, so theorem 6.4.3 applies and $\exp_p : T_pM \to M$ is a diffeomorphism for each p. Fix p and let M_c denote the model space of curvature c: Euclidean space $\mathbb{R}^n$ for $c = 0$, and hyperbolic space of curvature c for $c < 0$. The model M_c is itself complete, simply connected, and of constant curvature $c \leq 0$, so $\exp_{q_0} : T_{q_0}M_c \to M_c$ is also a diffeomorphism for a chosen basepoint q_0.

By the local classification of space forms (theorem 4.5.5), applied about p and q_0, both metrics take, in geodesic polar coordinates on any normal ball, the common radial normal form $d\rho^2 + \psi_c(\rho)^2 g_\Sigma$, with the identical radial factor ψ_c determined by c alone ($\psi_0(\rho) = \rho$ for $c = 0$; $\psi_c(\rho) = \frac{1}{\sqrt{-c}} \sinh(\sqrt{-c}\,\rho)$ for $c < 0$). Choose a linear isometry $\iota\colon (T_pM, g_p) \to (T_{q_0}M_c, (g_{M_c})_{q_0})$ and define

$$\Psi = \exp_{q_0} \circ\, \iota \circ (\exp_p)^{-1} \colon M \longrightarrow M_c$$

This is a diffeomorphism, being a composite of the diffeomorphism $(\exp_p)^{-1}$ (from theorem 6.4.3), the linear isomorphism ι, and the diffeomorphism $\exp_{q_0}$ (from the same theorem applied to M_c). It is defined on all of M precisely because both exponential maps are global diffeomorphisms; this is exactly the input the local classification could not provide on its own. In geodesic polar coordinates Ψ is $(\rho, \theta) \mapsto (\rho, \iota\theta)$: it fixes the radial coordinate and applies the linear isometry ι on the sphere of directions. Because both metrics equal $d\rho^2 + \psi_c(\rho)^2 g_\Sigma$ with the same ψ_c, and ι carries the unit sphere $\Sigma \subset T_pM$ isometrically to the unit sphere in $T_{q_0}M_c$ (so $\iota^* g_\Sigma = g_\Sigma$), the pullback $\Psi^* g_{M_c} = g$. Hence Ψ is an isometry (definition 1.3.1) of M onto M_c, which is the assertion. For $c = 0$ this identifies M with $\mathbb{R}^n$; for $c < 0$ with rescaled hyperbolic space, matching the models of example 1.3.5.

This completes the nonpositive half of the Killing–Hopf theorem, whose statement was recorded but deferred in box 4.5.6: the global rigidity of complete simply connected space forms of nonpositive curvature is now a theorem, the completeness machinery of the present chapter supplying the global exponential-chart diffeomorphism that the local classification of theorem 4.5.5 lacked. The positive-curvature case $c > 0$ is different in character: there conjugate points appear, $\exp_p$ is no longer a diffeomorphism, and the sphere is compact, so the argument by a global normal chart cannot run. That case is handled separately, and its proof does not follow from Cartan–Hadamard; the round sphere S^n, with cut locus a single antipodal point and injectivity radius π (example 5.3.4 and the cut-locus discussion of section 6.2), is the model whose global geometry the positive Killing–Hopf statement pins down.

6.5 The Rauch Comparison Theorem

The three chapters behind this one have measured how geodesics spread through a single quantity: the norm of a Jacobi field. On a space of constant curvature that norm is a trigonometric or hyperbolic function whose behavior is transparent, and the trichotomy of the model spaces is read off from it directly, refocusing on the sphere, linear growth in Euclidean space, exponential growth in hyperbolic space. On a general manifold the Jacobi equation $D_t^2 J + R(J, \dot\gamma)\dot\gamma = 0$ still governs the spreading, but its curvature coefficient varies from point to point and direction to direction, and the norm $|J|$ is no longer a function one can write down. What remains available is a comparison. A one-sided bound on the sectional curvature does not determine $|J|$, but it does trap it between the model values, and the direction of the trapping is dictated by the direction of the curvature bound.

The Rauch comparison theorem is the exact statement of this trapping, and it is the engine behind the global results of the chapter. Its content is a monotonicity: where the curvature is smaller, Jacobi fields grow faster, so geodesics spread more and refocus later; where the curvature is larger, they grow more slowly and refocus sooner. Cast against the constant-curvature models it recovers, in a single argument, that positive curvature forces a conjugate point at a computable distance (the mechanism behind the diameter bound of Bonnet–Myers) and that nonpositive curvature forbids conjugate points entirely (the mechanism behind Cartan–Hadamard). The proof is a comparison of index forms, run infinitesimally: the logarithmic derivative of $|J|$ is expressed as a ratio built from the index form on a shrinking interval, and the index lemma of section 5.4 makes that ratio monotone in the curvature. Throughout, M and $\tilde{M}$ are Riemannian manifolds, $\gamma\colon [0,b] \to M$ and $\tilde{\gamma}\colon [0,b] \to \tilde{M}$ are unit-speed geodesics with covariant derivatives D_t and $\tilde{D}_t$ and curvature endomorphisms R and $\tilde{R}$, and the curvature sign convention of box 4.1.1 is in force. That both manifolds are Riemannian, and not of mixed signature, is used at the one point where positive-definiteness is needed, and is flagged there.

The comparison rests on rewriting the index form of a Jacobi field vanishing at the initial point as a boundary term. This is the same integration by parts that drove the index lemma, specialized to a Jacobi field and to a variable upper endpoint.

Lemma 6.5.1: The index form of a Jacobi field as a boundary term

Let $\gamma\colon [0,b] \to M$ be a unit-speed geodesic and J a Jacobi field along γ with $J(0) = 0$. For $t \in (0,b]$ define the index form of J over the subinterval $[0,t]$ by

$$I_t(J,J) = \int_0^t \Big(|D_s J|^2 - \langle R(J,\dot{\gamma})\dot{\gamma}, J\rangle \Big)\, ds$$

Then

$$I_t(J,J) = \langle D_t J(t), J(t)\rangle$$

Consequently, wherever $J(t) \neq 0$,

$$\frac{d}{dt} \log |J(t)| = \frac{\langle D_t J(t), J(t)\rangle}{|J(t)|^2} = \frac{I_t(J,J)}{|J(t)|^2}$$

***Proof*:**
On $[0,t]$ the field J is smooth, so metric compatibility of ∇ (definition 2.2.4) gives $\frac{d}{ds}\langle D_s J, J\rangle = \big\langle D_s^2 J, J\big\rangle + \langle D_s J, D_s J\rangle$, that is $\langle D_s J, D_s J\rangle = \frac{d}{ds}\langle D_s J, J\rangle - \big\langle D_s^2 J, J\big\rangle$. Integrating over $[0,t]$,

$$\int_0^t |D_s J|^2\, ds = \Big[\langle D_s J, J\rangle\Big]_0^t - \int_0^t \big\langle D_s^2 J, J\big\rangle\, ds = \langle D_t J(t), J(t)\rangle - \int_0^t \big\langle D_s^2 J, J\big\rangle\, ds$$

the boundary contribution at $s = 0$ vanishing because $J(0) = 0$. The Jacobi equation (5.1) gives $D_s^2 J = -R(J,\dot{\gamma})\dot{\gamma}$, so $-\big\langle D_s^2 J, J\big\rangle = \langle R(J,\dot{\gamma})\dot{\gamma}, J\rangle$ and

$$\int_0^t |D_s J|^2\, ds = \langle D_t J(t), J(t)\rangle + \int_0^t \langle R(J,\dot{\gamma})\dot{\gamma}, J\rangle\, ds$$

Rearranging is exactly $I_t(J,J) = \langle D_t J(t), J(t)\rangle$. For the last statement, metric compatibility gives $\frac{d}{dt}|J|^2 = 2\langle D_t J, J\rangle$, so $\frac{d}{dt}\log|J| = \frac{1}{2}\frac{d}{dt}\log|J|^2 = \frac{\langle D_t J, J\rangle}{|J|^2}$, and substituting $\langle D_t J, J\rangle = I_t(J,J)$ yields the claim, as required.

The identity turns a statement about the growth rate of $|J|$ into a statement about the index form, and the index form is where the curvature bound bites. The index lemma of section 5.4 says that on a geodesic without interior conjugate points the Jacobi field minimizes the index form among all fields with the same endpoint values. Comparing the two manifolds through this minimizing property, with a transplant of one field onto the other manifold via parallel frames, is the substance of the theorem. The transplant is set up now.

Theorem 6.5.2: The Rauch Comparison Theorem

Let $\gamma\colon [0,b] \to M$ and $\tilde{\gamma}\colon [0,b] \to \tilde{M}$ be unit-speed geodesics in Riemannian manifolds of the same or different dimension, and let J and $\tilde{J}$ be Jacobi fields along γ and $\tilde{\gamma}$ respectively, normal to the geodesics, with

$$J(0) = 0, \qquad \tilde{J}(0) = 0, \qquad |D_t J(0)| = \left|\tilde{D}_t \tilde{J}(0)\right|$$

Suppose that $\tilde{\gamma}$ has no point conjugate to $\tilde{\gamma}(0)$ on $(0,b]$, and that the sectional curvatures satisfy

$$K_M(\Pi) \leq K_{\tilde{M}}(\tilde{\Pi})$$

for every $t \in [0,b]$, every 2-plane $\Pi \subseteq T_{\gamma(t)}M$ containing $\dot{\gamma}(t)$, and every 2-plane $\tilde{\Pi} \subseteq T_{\tilde{\gamma}(t)}\tilde{M}$ containing $\dot{\tilde{\gamma}}(t)$. Then

$$|J(t)| \geq \left|\tilde{J}(t)\right| \qquad \text{for all } t \in [0,b]$$

The hypotheses require a word before the proof. The curvature comparison is between *all* planes tangent to γ on the M side and *all* planes tangent to $\tilde{\gamma}$ on the $\tilde{M}$ side, so the statement is that the largest sectional curvature encountered along γ is no larger than the smallest encountered along $\tilde{\gamma}$; in the applications one manifold is a constant-curvature model and the bound reduces to a single number. The assumption $|D_t J(0)| = \left|\tilde{D}_t \tilde{J}(0)\right|$ normalizes the two fields to leave the initial point at the same infinitesimal rate, which is what makes their subsequent growth comparable; without it the inequality could be scaled either way. The no-conjugate-point hypothesis is imposed on the *comparison* manifold $\tilde{M}$, the one whose Jacobi field is the smaller, and it is exactly what the index lemma needs. The conclusion says that the field on the less curved manifold is at least as long throughout: less curvature, more spreading.

***Proof*:**
If $D_t J(0) = 0$ then, since $J(0) = 0$ as well, $J \equiv 0$ by the uniqueness of Jacobi fields from initial data (theorem 5.2.3); the hypothesis then forces $\tilde{D}_t \tilde{J}(0) = 0$ and $\tilde{J} \equiv 0$, and the inequality holds trivially. Assume henceforth $\lambda := |D_t J(0)| = \left|\tilde{D}_t \tilde{J}(0)\right| > 0$.

The proof proceeds in four steps: first the behavior of the two norms near $t = 0$, then the reduction of the inequality to a monotonicity of the ratio, then the index-form comparison that drives the monotonicity, and finally the assembly.

Step 1: initial behavior and non-vanishing of $\tilde{J}$. Because $\tilde{\gamma}$ has no conjugate point on $(0,b]$ and $\tilde{J}$ is a nonzero Jacobi field with $\tilde{J}(0) = 0$, the field $\tilde{J}$ does not vanish on $(0,b]$: a zero at $t_0 \in (0,b]$ would make $\tilde{J}$ a nonzero Jacobi field vanishing at 0 and t_0, so $\tilde{\gamma}(t_0)$ would be conjugate to $\tilde{\gamma}(0)$ (definition 5.3.1), contrary to hypothesis. Thus $\left|\tilde{J}(t)\right| > 0$ for $t \in (0,b]$, and the ratio $|J(t)| \,/\, \left|\tilde{J}(t)\right|$ is defined there wherever the denominator is positive.

Near $t = 0$ both fields grow linearly. Expanding a Jacobi field with $J(0) = 0$ in a Taylor expansion, using $D_t^2 J(0) = -R(J(0), \dot{\gamma})\dot{\gamma} = 0$ from the Jacobi equation and $J(0) = 0$, gives $J(t) = t\, P_0 D_t J(0) + O(t^3)$, where P_0 denotes parallel transport from $\gamma(0)$; hence $|J(t)| =$

$\lambda t + O(t^3)$, and likewise $|\tilde{J}(t)| = \lambda t + O(t^3)$. Therefore

$$\lim_{t\to 0^+} \frac{|J(t)|}{|\tilde{J}(t)|} = 1 \tag{6.9}$$

and both $|J|$ and $|\tilde{J}|$ are positive on some interval $(0, \varepsilon)$.

Step 2: reduction to a pointwise index inequality. Let

$$c^* = \sup\{t \in (0, b] \mid |J(s)| > 0 \text{ for all } s \in (0, t]\}$$

which is positive because $|J| > 0$ on $(0, \varepsilon)$ by Step 1. On the open interval $(0, c^*)$ both $|J|$ and $|\tilde{J}|$ are positive, so the ratio

$$\rho(t) = \frac{|J(t)|^2}{|\tilde{J}(t)|^2}$$

is smooth and positive there, with $\rho(0^+) = 1$ by (6.9). Its logarithmic derivative is

$$\tfrac{1}{2}\frac{\rho'(t)}{\rho(t)} = \frac{d}{dt}\log|J(t)| - \frac{d}{dt}\log|\tilde{J}(t)| \tag{6.10}$$

and by lemma 6.5.1, applied on each manifold, the two logarithmic derivatives are the index-form ratios

$$\frac{d}{dt}\log|J(t)| = \frac{I_t(J,J)}{|J(t)|^2}, \qquad \frac{d}{dt}\log|\tilde{J}(t)| = \frac{\tilde{I}_t(\tilde{J},\tilde{J})}{|\tilde{J}(t)|^2} \tag{6.11}$$

where I_t and $\tilde{I}_t$ are the index forms over $[0,t]$ on M and $\tilde{M}$. Combining (6.10) and (6.11), the sign of $\rho'(t)$ is the sign of

$$\frac{I_t(J,J)}{|J(t)|^2} - \frac{\tilde{I}_t(\tilde{J},\tilde{J})}{|\tilde{J}(t)|^2}$$

so ρ is nondecreasing on $(0, c^*)$ once it is shown that, for every fixed $c \in (0, c^*)$,

$$\frac{I_c(J,J)}{|J(c)|^2} \geq \frac{\tilde{I}_c(\tilde{J},\tilde{J})}{|\tilde{J}(c)|^2} \tag{6.12}$$

Step 3 proves (6.12) at each such c.

Step 3: the transplant and the index-form comparison. Fix $c \in (0, c^*)$. Rescale both fields to unit length at $t = c$: put $J^* = J/|J(c)|$ and $\tilde{J}^* = \tilde{J}/|\tilde{J}(c)|$, which are Jacobi fields (the Jacobi equation is linear) with $J^*(0) = \tilde{J}^*(0) = 0$ and $|J^*(c)| = |\tilde{J}^*(c)| = 1$. Because I_c and $\tilde{I}_c$ are quadratic, $I_c(J^*, J^*) = I_c(J,J)/|J(c)|^2$ and $\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) = \tilde{I}_c(\tilde{J},\tilde{J})/|\tilde{J}(c)|^2$, so (6.12) is the normalized inequality

$$I_c(J^*, J^*) \geq \tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \tag{6.13}$$

Transplant J^* from γ onto $\tilde{\gamma}$ by matching parallel frames. Both geodesics are unit speed, so $\dot{\gamma}$ and $\dot{\tilde{\gamma}}$ are parallel unit fields and their normal bundles $\dot{\gamma}^\perp$, $\dot{\tilde{\gamma}}^\perp$ over $[0,c]$ carry parallel orthonormal frames obtained by parallel-transporting orthonormal bases of the normal spaces at $t = 0$. Let $n = \dim M$ and $\tilde{n} = \dim \tilde{M}$, and choose a parallel orthonormal frame $(E_2, \ldots, E_n)$ of $\dot{\gamma}^\perp$ along γ. Write the normal field J^* in this frame,

$$J^*(t) = \sum_{a=2}^{n} \varphi^a(t)\, E_a(t)$$

with scalar coefficients φ^a, smooth on $[0, c]$ and vanishing at $t = 0$. Fix a linear isometry $L\colon \dot\gamma(0)^\perp \to \dot{\tilde\gamma}(0)^\perp$ into the target normal space, which exists because $\dim \dot\gamma(0)^\perp = n - 1 \le \tilde n - 1 = \dim \dot{\tilde\gamma}(0)^\perp$ and a linear map preserving inner products can be defined on an orthonormal basis; in the equidimensional case $n = \tilde n$ relevant to every application below, L is a linear isometry of the two normal spaces. Let (LE_a) denote the parallel field along $\tilde\gamma$ with initial value $L(E_a(0))$, and define the transplanted field along $\tilde\gamma$ by

$$\tilde V(t) = \sum_{a=2}^{n} \varphi^a(t)\,\big(LE_a\big)(t)$$

Since L is a linear isometry and parallel transport preserves inner products on both manifolds, at every t the vectors $\{(LE_a)(t)\}$ are orthonormal in $\dot{\tilde\gamma}(t)^\perp$ with the same Gram relations as $\{E_a(t)\}$; hence the transplant is norm-preserving pointwise,

$$\big|\tilde V(t)\big|^2 = \sum_a (\varphi^a(t))^2 = |J^*(t)|^2, \qquad \big|\tilde D_t \tilde V(t)\big|^2 = \sum_a (\dot\varphi^a(t))^2 = |D_t J^*(t)|^2 \tag{6.14}$$

the derivative identity holding because both frames are parallel, so covariant differentiation acts only on the coefficients φ^a. Thus $\tilde V$ is a smooth normal field along $\tilde\gamma$ with $\tilde V(0) = 0$ and $\big|\tilde V(c)\big| = |J^*(c)| = 1$. Choose L so that in addition $\tilde V(c) = \tilde J^*(c)$; this is arranged by composing L with a rotation of the target normal space carrying the unit vector $\tilde V(c)/\big|\tilde V(c)\big|$ to $\tilde J^*(c)$, a rotation that preserves all the inner-product relations used above.

The comparison now runs in two links.

Link (a): the index lemma on $\tilde M$. The comparison manifold $\tilde\gamma$ has no point conjugate to $\tilde\gamma(0)$ on $(0, c]$, since $c < c^* \le b$ and no conjugate point exists on $(0, b]$ by hypothesis. The field $\tilde J^*$ is the Jacobi field along $\tilde\gamma$ with $\tilde J^*(0) = 0$ and endpoint value $\tilde J^*(c) = \tilde V(c)$, and $\tilde V$ is a normal field along $\tilde\gamma$ with the same endpoint data $\tilde V(0) = 0$, $\tilde V(c) = \tilde J^*(c)$. The index lemma (theorem 5.4.4) applies verbatim on $\tilde M$ and gives

$$\tilde I_c(\tilde J^*, \tilde J^*) \le \tilde I_c(\tilde V, \tilde V) \tag{6.15}$$

with equality iff $\tilde V = \tilde J^*$.

Link (b): the curvature term is larger on $\tilde M$. Compute the two index forms in their frames, abbreviating $\Pi_t = \operatorname{span}(J^*(t), \dot\gamma(t))$ and $\tilde\Pi_t = \operatorname{span}(\tilde V(t), \dot{\tilde\gamma}(t))$,

$$I_c(J^*, J^*) = \int_0^c \Big(|D_t J^*|^2 - \langle R(J^*, \dot\gamma)\dot\gamma, J^*\rangle\Big)\,dt, \qquad \tilde I_c(\tilde V, \tilde V) = \int_0^c \Big(\big|\tilde D_t \tilde V\big|^2 - \big\langle \tilde R(\tilde V, \dot{\tilde\gamma})\dot{\tilde\gamma}, \tilde V\big\rangle\Big)\,dt$$

The derivative-energy integrands agree pointwise by (6.14): $\big|\tilde D_t \tilde V\big|^2 = |D_t J^*|^2$. For the curvature integrands, the definition of sectional curvature (definition 4.3.1) gives, where J^* and $\dot\gamma$ are independent,

$$\langle R(J^*, \dot\gamma)\dot\gamma, J^*\rangle = K_M(\Pi_t)\big(|J^*|^2 - \langle J^*, \dot\gamma\rangle^2\big) = K_M(\Pi_t)\,|J^*|^2$$

since $J^* \perp \dot\gamma$ and $|\dot\gamma| = 1$; where J^* and $\dot\gamma$ are dependent the Gram determinant $|J^*|^2 - \langle J^*, \dot\gamma\rangle^2$ vanishes and so does the pairing, so the identity $\langle R(J^*, \dot\gamma)\dot\gamma, J^*\rangle = K_M(\Pi_t)\,|J^*|^2$ still holds with $K_M(\Pi_t)$ read as any value. The identical computation on $\tilde M$ gives $\big\langle \tilde R(\tilde V, \dot{\tilde\gamma})\dot{\tilde\gamma}, \tilde V\big\rangle = K_{\tilde M}(\tilde\Pi_t)\,\big|\tilde V\big|^2$.

By (6.14) the two lengths agree, $\left|\tilde{V}(t)\right|^2 = |J^*(t)|^2$, and by the curvature hypothesis $K_M(\Pi_t) \leq K_{\tilde{M}}(\tilde{\Pi}_t)$ for the two tangent-plane families. Therefore, pointwise in t,

$$\langle R(J^*, \dot{\gamma})\dot{\gamma}, J^*\rangle = K_M(\Pi_t)\,|J^*|^2 \leq K_{\tilde{M}}(\tilde{\Pi}_t)\left|\tilde{V}\right|^2 = \left\langle \tilde{R}(\tilde{V}, \dot{\tilde{\gamma}})\dot{\tilde{\gamma}}, \tilde{V}\right\rangle$$

Here the Riemannian hypothesis enters: $|J^*|^2 = \left|\tilde{V}\right|^2 \geq 0$ is a genuine nonnegative length, so multiplying the curvature inequality $K_M \leq K_{\tilde{M}}$ by it preserves the direction. In a mixed-signature setting a normal field can have $\langle J^*, J^*\rangle$ of either sign, this step fails, and indeed Hopf–Rinow and the whole comparison theory break down; this is the analytic reason the theorem is confined to Riemannian metrics. Subtracting the larger curvature term on $\tilde{M}$ leaves the smaller integrand, so

$$\tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*) \tag{6.16}$$

Chaining (6.15) and (6.16),

$$\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \leq \tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*)$$

which is the normalized inequality (6.13), hence (6.12).

Step 4: integration and the boundary case. By Step 3 the inequality (6.12) holds for every $c \in (0, c^*)$, so by Step 2 the ratio ρ is nondecreasing on $(0, c^*)$. Since $\rho(0^+) = 1$, this gives $\rho(t) \geq 1$, that is

$$|J(t)| \geq \left|\tilde{J}(t)\right| \qquad \text{for } t \in (0, c^*)$$

It remains to show $c^* = b$, so that the inequality holds on all of $(0, b]$. Suppose $c^* \leq b$. On $(0, c^*)$ one has $|J(t)| \geq \left|\tilde{J}(t)\right|$, and $\left|\tilde{J}\right|$ is continuous and positive on $(0, b] \supseteq (0, c^*]$ by Step 1; letting $t \uparrow c^*$ and using continuity of $|J|$ gives $|J(c^*)| \geq \left|\tilde{J}(c^*)\right| > 0$. Thus $|J|$ is positive at c^* and, being continuous, positive on a neighborhood of c^*, contradicting the definition of c^* as the supremum of parameters up to which $|J|$ stays positive unless $c^* = b$. Hence $c^* = b$, and the inequality $|J(t)| \geq \left|\tilde{J}(t)\right|$ holds for all $t \in (0, b]$; at $t = 0$ both sides vanish. This is the assertion, completing the proof.

The proof deserves a plain reading, because its structure is the template for every comparison argument to come. The quantity that carries the geometry is the logarithmic derivative of $|J|$, which lemma 6.5.1 identifies with the index form of the Jacobi field over a shrinking interval, divided by $|J|^2$. Comparing that ratio across the two manifolds is done by transplanting the field J^* from the less-curved manifold M onto the comparison geodesic $\tilde{\gamma}$, where two facts then push the index form in the same direction. The index lemma says the Jacobi field $\tilde{J}^*$ is the most economical field on $\tilde{M}$ with its endpoint value, so the transplant $\tilde{V}$ has at least as large an index form; and the curvature bound says that, term by term, the larger curvature of $\tilde{M}$ subtracts more, making the index form of the same field smaller on $\tilde{M}$ than on M. Reading the two facts along the chain $\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \leq \tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*)$ produces the normalized inequality that is the differential form of the conclusion. Integrating the resulting monotonicity from the common linear start at $t = 0$ delivers $|J| \geq \left|\tilde{J}\right|$. The transplant is the only nontrivial device, and it works precisely because parallel frames along a unit-speed geodesic carry the metric structure faithfully, so the derivative-energy term is transported unchanged and only the curvature term is allowed to differ.

The first use of the theorem is to convert one-sided curvature bounds into conjugate-point conclusions, by comparing against the constant-curvature models whose Jacobi fields are the explicit functions of example 5.2.5.

Corollary 6.5.3: Curvature bounds and conjugate points

Let (M,g) be a Riemannian manifold and $\gamma\colon [0,\infty) \to M$ a unit-speed geodesic issuing from $p = \gamma(0)$.

1. If the sectional curvature satisfies $K_M \le \kappa$ along γ for a constant κ, then γ has no point conjugate to p before parameter $\pi/\sqrt{\kappa}$ when $\kappa > 0$, and no conjugate point at all when $\kappa \le 0$. In particular $K_M \le 0$ forces no conjugate points, recovering proposition 5.3.3.
2. If the sectional curvature satisfies $K_M \ge \kappa$ along γ for a constant $\kappa > 0$, and M has dimension at least two, then γ has a point conjugate to p at some parameter $t \le \pi/\sqrt{\kappa}$.

***Proof*:**
Let M_κ denote the simply connected model space of constant curvature κ, and $\tilde\gamma$ a unit-speed geodesic in it. Fix a normal Jacobi field J along γ with $J(0) = 0$ and $|D_t J(0)| = 1$, and let $\tilde J$ be the normal Jacobi field along $\tilde\gamma$ with $\tilde J(0) = 0$ and $\left|\tilde D_t \tilde J(0)\right| = 1$; by example 5.2.5, $\left|\tilde J(t)\right| = s_\kappa(t)$, the solution of $s'' + \kappa s = 0$ with $s(0) = 0$, $s'(0) = 1$, that is $s_\kappa(t) = \sin(\sqrt{\kappa}\,t)/\sqrt{\kappa}$ for $\kappa > 0$, t for $\kappa = 0$, and $\sinh(\sqrt{-\kappa}\,t)/\sqrt{-\kappa}$ for $\kappa < 0$.

1. Suppose $K_M \le \kappa$ along γ. Fix any $b > 0$ with $b < \pi/\sqrt{\kappa}$ when $\kappa > 0$, and any $b > 0$ when $\kappa \le 0$; then on $(0,b]$ the model function s_κ is positive, so $\tilde\gamma$ has no conjugate point on $(0,b]$. The Rauch theorem theorem 6.5.2, with the roles $M \to M$ and $\tilde M \to M_\kappa$ (the curvature bound $K_M \le \kappa = K_{M_\kappa}$ is the required $K_M \le K_{\tilde M}$), gives $|J(t)| \ge \left|\tilde J(t)\right| = s_\kappa(t) > 0$ on $(0,b]$. Hence J does not vanish on $(0,b]$; as b is arbitrary below $\pi/\sqrt{\kappa}$ (or arbitrary when $\kappa \le 0$), J does not vanish on $(0,\pi/\sqrt{\kappa})$ (respectively on $(0,\infty)$). The same holds for every normal Jacobi field vanishing at 0, by taking it as J, so no such field vanishes before $\pi/\sqrt{\kappa}$, and γ has no conjugate point of p there. When $\kappa \le 0$ this is the absence of conjugate points of proposition 5.3.3, now obtained as a special case of Rauch.
2. Suppose $K_M \ge \kappa > 0$ along γ, and $n = \dim M \ge 2$. The comparison runs with the roles reversed: the less-curved manifold is now the model M_κ, and the more-curved comparison manifold is M, whose geodesic γ must be shown conjugate-point-free before Rauch can be applied to it. Argue by contradiction: suppose γ has no conjugate point of p on $(0,\pi/\sqrt{\kappa}]$. Then theorem 6.5.2 applies with base geodesic $\tilde\gamma \subset M_\kappa$ and comparison geodesic $\gamma \subset M$, the curvature hypothesis $\kappa \le K_M$ being the required $K_{M_\kappa} \le K_M$ and the comparison manifold M being conjugate-point-free on $(0,\pi/\sqrt{\kappa}]$ by assumption; taking the model field $\tilde J$ along $\tilde\gamma$ (norm s_κ) and the field J along γ with the matched initial derivative $|D_t J(0)| = 1 = \left|\tilde D_t \tilde J(0)\right|$, its conclusion is that the model field is the longer, $s_\kappa(t) = \left|\tilde J(t)\right| \ge |J(t)|$ on $(0,\pi/\sqrt{\kappa}]$. But $s_\kappa(\pi/\sqrt{\kappa}) = \sin(\pi)/\sqrt{\kappa} = 0$, so $|J(\pi/\sqrt{\kappa})| \le 0$, forcing $J(\pi/\sqrt{\kappa}) = 0$. Since $n \ge 2$ there is a nonzero normal Jacobi field J with $J(0) = 0$, and it vanishes at $\pi/\sqrt{\kappa}$, so $\gamma(\pi/\sqrt{\kappa})$ is conjugate to p, contradicting the assumption. Hence γ has a conjugate point at some $t \le \pi/\sqrt{\kappa}$, as claimed.

Both parts follow, completing the proof.

The two halves of the corollary are the quantitative skeleton of the two global theorems of the chapter. Part (2), that curvature bounded below by $\kappa > 0$ forces a conjugate point by parameter $\pi/\sqrt{\kappa}$, is the geodesic-focusing mechanism behind Bonnet–Myers: past a conjugate point a geodesic no longer minimizes (theorem 5.4.5), so no minimizing geodesic can be longer than $\pi/\sqrt{\kappa}$, and the diameter is bounded. Bonnet–Myers in fact needs only a lower bound on the Ricci curvature rather than on all sectional curvatures, so it is proved by a direct second-variation computation rather than by this corollary, but the scale $\pi/\sqrt{\kappa}$ is the same and its origin is the model conjugate distance seen here. Part (1), that curvature bounded above by 0 forbids conjugate points, is the local input to Cartan–Hadamard, giving a second proof of proposition 5.3.3 that exhibits it as one end of a continuum: as the upper bound κ decreases through 0, the first possible conjugate distance $\pi/\sqrt{\kappa}$ increases to infinity and then disappears.

Against a constant-curvature model the theorem sharpens to a two-sided estimate that pins the growth of any Jacobi field between the two model solutions.

Example 6.5.4: Two-sided comparison against the constant-curvature models
Let (M, g) be a Riemannian manifold whose sectional curvature along a unit-speed geodesic γ is pinched between two constants,

$$\delta \leq K_M \leq \Delta$$

for every tangent 2-plane containing $\dot{\gamma}$. Let J be a normal Jacobi field along γ with $J(0) = 0$ and $|D_t J(0)| = 1$. Write s_κ for the model solution of $s'' + \kappa s = 0$ with $s(0) = 0$, $s'(0) = 1$, namely

$$s_\kappa(t) = \begin{cases} \dfrac{\sin(\sqrt{\kappa}\, t)}{\sqrt{\kappa}} & \kappa > 0 \\ t & \kappa = 0 \\ \dfrac{\sinh(\sqrt{-\kappa}\, t)}{\sqrt{-\kappa}} & \kappa < 0 \end{cases}$$

as in example 5.2.5. Then, on the interval $[0, b]$ up to the first zero of s_Δ (which is $\pi/\sqrt{\Delta}$ if $\Delta > 0$ and $+\infty$ otherwise),

$$s_\Delta(t) \leq |J(t)| \leq s_\delta(t) \tag{6.17}$$

The two inequalities are two applications of theorem 6.5.2 with opposite role assignments.

For the lower bound $s_\Delta(t) \leq |J(t)|$, compare M against the model M_Δ of constant curvature Δ. The hypothesis $K_M \leq \Delta = K_{M_\Delta}$ is the required $K_M \leq K_{\tilde{M}}$ with $\tilde{M} = M_\Delta$; the model geodesic $\tilde{\gamma}$ has no conjugate point on $(0, b]$ because $s_\Delta > 0$ there; and the model Jacobi field $\tilde{J}$ with matched initial data has $|\tilde{J}| = s_\Delta$ by example 5.2.5. Rauch gives $|J(t)| \geq |\tilde{J}(t)| = s_\Delta(t)$.

For the upper bound $|J(t)| \leq s_\delta(t)$, compare the model M_δ of constant curvature δ against M. Now the base manifold is M_δ and the comparison manifold is M: the hypothesis $\delta \leq K_M$ is $K_{M_\delta} \leq K_M$, the required inequality with the model in the role of the less-curved manifold; the comparison manifold M has no conjugate point on $(0, b]$ because $b \leq \pi/\sqrt{\Delta}$ and part (1) of corollary 6.5.3 (using $K_M \leq \Delta$) excludes conjugate points before $\pi/\sqrt{\Delta}$; and the model field on M_δ with matched initial data has norm s_δ. Rauch gives $s_\delta(t) \geq |J(t)|$, which is the upper bound.

The two model space forms make (6.17) concrete. On the round sphere S^n_r, where $K \equiv 1/r^2$, both bounds coincide and force $|J(t)| = r\sin(t/r)$, in agreement with example 5.2.5; the pinching is trivial because the space is already a model. For a genuine pinch, take a metric with $1/4 \leq K \leq 1$ on the unit-normalized scale, the setting of the sphere theorem. Then (6.17) reads $\sin t \leq |J(t)| \leq 2\sin(t/2)$ on $[0, \pi]$, so a Jacobi field vanishing at 0 with unit initial derivative cannot vanish again before $t = \pi$ (the lower bound $\sin t$ is positive on $(0, \pi)$) and must vanish by $t = 2\pi$ (the upper bound $2\sin(t/2)$ vanishes at 2π); the first conjugate point lies in $[\pi, 2\pi]$, trapping the injectivity-radius scale between the two models. On hyperbolic space H^n, where $K \equiv -1$, both bounds again coincide and give $|J(t)| = \sinh t$, and for a manifold with $-4 \leq K \leq -1$ the estimate becomes $\sinh t \leq |J(t)| \leq \frac{1}{2}\sinh(2t)$, both sides growing without bound, so no conjugate point ever appears, consistent with corollary 6.5.3(1).

The comparison theorem and its corollary complete the pattern promised at the start of the chapter. Curvature, an infinitesimal quantity attached to each 2-plane, controls the global spreading of geodesics through the single monotone comparison of Rauch, and the two model-space bounds bracket every intermediate behavior. Where the curvature is large the geodesics refocus early, capping the diameter and the injectivity radius; where it is small they spread freely, and in nonpositive curvature they never refocus at all. The conjugate-point statements of section 5.3, proved there by direct constant-curvature computation and by the convexity of $|J|^2$, reappear here as the two endpoints of a single inequality, and the same machinery, applied with a Ricci bound in place of a sectional one, drives the diameter and topology theorems that the models exemplify: the sphere compact and refocusing, hyperbolic space open and expanding, Euclidean space the flat boundary between them.

7

Submanifold Geometry

A Riemannian submanifold carries two geometries at once. From the induced metric of section 1.5 it inherits an intrinsic geometry of its own, with a Levi-Civita connection, geodesics, and curvature computed without reference to anything outside it. At the same time it sits inside an ambient manifold, curving through it, and this bending is extrinsic data invisible to the intrinsic metric alone. The organizing question of the chapter is how the two are related: how much of a submanifold's shape in the ambient space is already determined by its intrinsic metric, and how much is genuinely additional. The instrument that measures the difference is the second fundamental form, the normal component of the ambient covariant derivative, and the structural equations of Gauss, Codazzi, and Ricci express the ambient curvature in terms of the intrinsic curvature and this one extra tensor. The chapter's central result, Gauss's Theorema Egregium, extracts from these equations the fact that for a surface in Euclidean space the Gaussian curvature, defined extrinsically as the product of principal curvatures, is in truth intrinsic: it survives any bending that preserves the metric, and a flat sheet of paper can never be wrapped isometrically onto a sphere.

The second half of the chapter takes up the dual construction. Where a submanifold is the image of an isometric immersion, a Riemannian submersion presents its base as a metric quotient of the total space, the horizontal directions carrying the geometry and the vertical fibers collapsed away. The two settings are mirror images, and the mirror carries across: O'Neill's formula is the submersion counterpart of the Gauss equation, computing the curvature of the base from the curvature above together with a tensor measuring the failure of horizontal fields to stay horizontal. Its first consequence is that submersions cannot decrease horizontal sectional curvature, and its first examples are the ones that matter downstream. The Hopf fibration exhibits the round two-sphere as a quotient of the three-sphere, and the same construction produces the Fubini–Study metric on complex projective space, the running example through which Part II will realize characteristic classes as curvature.

7.1 The Second Fundamental Form and the Shape Operator

A Riemannian submanifold $M \subseteq \tilde{M}$ carries two kinds of geometry at once. The induced metric g of definition 1.5.1 makes M a Riemannian manifold in its own right, with its own Levi-Civita connection, geodesics, and curvature, all computed intrinsically from lengths and angles measured inside M. But M also sits inside the ambient manifold $\tilde{M}$, and how it curves *within* $\tilde{M}$ is extra information, invisible to a resident of M who can only measure inside the submanifold. A great circle on the round sphere and a straight line in the plane are both geodesics of their intrinsic geometry, yet the circle bends in $\mathbb{R}^3$ while the line does not; that bending is the ambient datum this chapter is built to capture. The tool that captures it is the *second fundamental form*, the normal component of the ambient derivative of tangent fields, and its metric transpose the *shape operator*. This section defines both, proves that the tangential part of the ambient connection is exactly the intrinsic Levi-Civita connection of g, and computes the model case of the sphere, where every principal curvature is $-1/r$.

The standing setup is fixed once here and used throughout the chapter. The ambient manifold is $(\tilde{M}, \tilde{g})$, with Levi-Civita connection $\tilde{\nabla}$ and curvature $\widetilde{\mathrm{Riem}}$; a tilde marks every ambient object. The submanifold $M \subseteq \tilde{M}$ is an immersed Riemannian submanifold carrying the induced metric g of definition 1.5.1, positive definite by proposition 1.5.2, with its own Levi-Civita connection ∇ and curvature Riem. Throughout, X, Y, Z, W denote vector fields tangent to M and ξ, η denote fields normal to M. The computations are local, so on a neighborhood of any point M may be treated as embedded and its tangent fields extended to ambient fields where a derivative $\tilde{\nabla}_X$ requires it; the results will be seen not to depend on the extension.

Before differentiating along M, the ambient tangent space must be split into the part along M and the part transverse to it. Positive-definiteness of $\tilde{g}$ on T_pM makes this splitting orthogonal and unambiguous.

Definition 7.1.1: Normal space and normal bundle

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold with induced metric $g = \tilde{g}|_{TM}$. For $p \in M$ the *normal space* to M at p is the $\tilde{g}$-orthogonal complement of T_pM inside $T_p\tilde{M}$,

$$N_pM = \left\{ v \in T_p\tilde{M} \;\middle|\; \tilde{g}(v, w) = 0 \text{ for all } w \in T_pM \right\}$$

so that $T_p\tilde{M} = T_pM \oplus N_pM$ is an orthogonal direct sum. The *normal bundle* $NM = \bigsqcup_{p \in M} N_pM$ is the disjoint union of the normal spaces; it is a smooth vector bundle over M of rank $\dim \tilde{M} - \dim M$ ([6], where it is constructed as the quotient bundle $\iota^* T\tilde{M}/TM$ and identified with the orthogonal complement, and where its total space furnishes a tubular neighborhood of M). For a vector $v \in T_p\tilde{M}$ write $v = v^\top + v^\perp$ for the resulting decomposition, with $v^\top \in T_pM$ the *tangential part* and $v^\perp \in N_pM$ the *normal part*; the projections $(\cdot)^\top : T_p\tilde{M} \to T_pM$ and $(\cdot)^\perp : T_p\tilde{M} \to N_pM$ are the associated orthogonal projections.

The splitting exists because $\tilde{g}$ restricts to a positive-definite form on T_pM, so its orthogonal complement is a genuine complement and the projections are well defined; this is where the Riemannian hypothesis on the induced metric enters, and it is exactly the content of proposition 1.5.2. The projections are pointwise linear and vary smoothly with p, so applying $(\cdot)^\top$ or $(\cdot)^\perp$ to a smooth ambient field defined along M yields a smooth tangent field or a smooth section of NM. Two conventions are worth stating at the outset. First, a field along M that is everywhere tangent has

$v^\perp = 0$, and a field that is everywhere normal has $v^\top = 0$; the projections are what they should be on the summands. Second, because the decomposition is orthogonal, for any $u, v \in T_p\tilde{M}$ one has $\tilde{g}(u^\top, v^\perp) = 0$, and the Pythagorean identity $\tilde{g}(v, v) = \tilde{g}(v^\top, v^\top) + \tilde{g}(v^\perp, v^\perp)$ holds; both are used repeatedly below without further comment.

For a concrete instance, take the round sphere. The normal line at each point is spanned by the position vector, a picture that will drive the sphere computation at the end of the section.

Example 7.1.2: The normal bundle of the sphere

Let $S^n_r = \{x \in \mathbb{R}^{n+1} \mid |x| = r\}$ with the induced metric of example 1.5.3, sitting inside $\mathbb{R}^{n+1}$ with its Euclidean metric $\bar{g}$. Identify $T_x\mathbb{R}^{n+1}$ with $\mathbb{R}^{n+1}$ in the standard way. A tangent vector to S^n_r at x is the velocity $\dot{c}(0)$ of a curve c in the sphere with $c(0) = x$; differentiating $\bar{g}(c(t), c(t)) = r^2$ at $t = 0$ gives $2\bar{g}(x, \dot{c}(0)) = 0$, so every tangent vector is $\bar{g}$-orthogonal to x. The tangent space $T_xS^n_r = x^\perp$ is therefore the n-dimensional hyperplane orthogonal to x, and its orthogonal complement in $\mathbb{R}^{n+1}$ is the line $\mathbb{R}x$ spanned by the position vector. Hence

$$N_xS^n_r = \mathbb{R}x$$

a line bundle over S^n_r, and $N = x/r$ is a smooth unit normal field, the outward one. The tangential and normal projections of a vector $v \in \mathbb{R}^{n+1}$ at x are

$$v^\top = v - \frac{\bar{g}(v, x)}{r^2}\, x, \qquad v^\perp = \frac{\bar{g}(v, x)}{r^2}\, x$$

the second being the orthogonal projection onto the line $\mathbb{R}x$. This unit normal field and these projections are the data fed into the shape-operator computation of example 7.1.8.

With the splitting in hand, the ambient connection can be applied to a tangent field and its output decomposed. The tangential part of $\tilde{\nabla}_X Y$ turns out to be an intrinsic object: it is the Levi-Civita connection of the induced metric, so nothing new is needed to differentiate tangent fields along M from the inside. The point of the following proposition is that the two connections, one defined extrinsically by projecting the ambient derivative and one defined intrinsically from g alone, coincide.

Proposition 7.1.3: The tangential connection is the Levi-Civita connection

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold with induced metric g, and for tangent fields X, Y on M define

$$\nabla^\top_X Y = (\tilde{\nabla}_X Y)^\top$$

the tangential part of the ambient covariant derivative. Then $\nabla^\top$ is an affine connection on M that is torsion-free and compatible with g. Consequently, by the uniqueness clause of theorem 2.3.1, $\nabla^\top$ equals the Levi-Civita connection ∇ of (M, g) from definition 2.3.2.

***Proof*:**

The argument verifies the four defining properties of a connection, then torsion-freeness and metric compatibility, and finally invokes uniqueness. Throughout, X, Y, Z are tangent to M and $f \in C^\infty(M)$; ambient fields extending them are used only to form $\tilde{\nabla}$, and every quantity produced is intrinsic to M because it is expressed through g and the bracket of tangent fields.

Step 1: $\nabla^\top$ is a connection. The ambient connection $\tilde{\nabla}$ is $\mathbb{R}$-bilinear in (X, Y), C^∞-linear in

X, and Leibniz in Y, by definition 2.3.2 applied to $\tilde{M}$. Composing with the pointwise linear projection $(\cdot)^\top$ preserves each of these. Additivity in X and Y is immediate. For $C^\infty(M)$-linearity in the differentiating slot, extend f to a neighborhood in $\tilde{M}$; then

$$\nabla^\top_{fX}Y = (\tilde{\nabla}_{fX}Y)^\top = (f\,\tilde{\nabla}_X Y)^\top = f\,(\tilde{\nabla}_X Y)^\top = f\,\nabla^\top_X Y$$

using that $(\cdot)^\top$ is $C^\infty(M)$-linear. For the Leibniz rule in Y,

$$\nabla^\top_X(fY) = \big(\tilde{\nabla}_X(fY)\big)^\top = \big((Xf)Y + f\,\tilde{\nabla}_X Y\big)^\top = (Xf)\,Y^\top + f\,(\tilde{\nabla}_X Y)^\top = (Xf)Y + f\,\nabla^\top_X Y$$

where $Y^\top = Y$ because Y is tangent to M. Thus $\nabla^\top$ satisfies the axioms of definition 2.1.1, and its output $(\tilde{\nabla}_X Y)^\top$ is a tangent field, so it is a connection on TM.

Step 2: $\nabla^\top$ is torsion-free. The ambient connection is torsion-free, so $\tilde{\nabla}_X Y - \tilde{\nabla}_Y X = [X,Y]$ by definition 2.2.1 applied to $\tilde{M}$. For X, Y tangent to M their Lie bracket $[X,Y]$ is again tangent to M: on the embedded model $[X,Y]$ is the commutator of derivations that preserve the ideal of functions vanishing on M, hence itself preserves that ideal, so $[X,Y]$ is tangent, and $[X,Y]^\top = [X,Y]$. Taking the tangential part of the torsion identity,

$$\nabla^\top_X Y - \nabla^\top_Y X = (\tilde{\nabla}_X Y)^\top - (\tilde{\nabla}_Y X)^\top = (\tilde{\nabla}_X Y - \tilde{\nabla}_Y X)^\top = [X,Y]^\top = [X,Y]$$

so the torsion of $\nabla^\top$ vanishes.

Step 3: $\nabla^\top$ is compatible with g. The induced metric is the restriction $g(Y,Z) = \tilde{g}(Y,Z)$ for tangent fields Y, Z, and along M a tangent vector X differentiates $g(Y,Z)$ as the ambient function $\tilde{g}(Y,Z)$. Metric compatibility of $\tilde{\nabla}$ on $\tilde{M}$, from definition 2.2.4, gives

$$X\,\tilde{g}(Y,Z) = \tilde{g}(\tilde{\nabla}_X Y, Z) + \tilde{g}(Y, \tilde{\nabla}_X Z)$$

Because Z is tangent to M, $\tilde{g}(\tilde{\nabla}_X Y, Z) = \tilde{g}((\tilde{\nabla}_X Y)^\top, Z)$: the normal part of $\tilde{\nabla}_X Y$ pairs to zero against the tangent vector Z by the orthogonality of the splitting in definition 7.1.1. The same holds for the second term with the roles of Y and Z reversed. Hence

$$X\,g(Y,Z) = \tilde{g}\big((\tilde{\nabla}_X Y)^\top, Z\big) + \tilde{g}\big(Y, (\tilde{\nabla}_X Z)^\top\big) = g(\nabla^\top_X Y, Z) + g(Y, \nabla^\top_X Z)$$

which is the compatibility identity of definition 2.2.4 for $\nabla^\top$ and g.

Step 4: uniqueness. By Steps 1 through 3, $\nabla^\top$ is a torsion-free connection on M compatible with the induced metric g. The fundamental theorem theorem 2.3.1 asserts that such a connection is unique; it is by definition the Levi-Civita connection ∇ of (M, g) named in definition 2.3.2. Therefore $\nabla^\top = \nabla$, completing the proof.

This identification is what makes submanifold geometry tractable: to covariantly differentiate a tangent field along M using the intrinsic geometry, one may compute the ambient derivative and drop its normal component, and the answer is exactly the intrinsic $\nabla_X Y$. The extrinsic recipe and the intrinsic definition never disagree. Equivalently, $\nabla_X Y$ is characterized as the unique tangent field whose difference from $\tilde{\nabla}_X Y$ is normal. The leftover normal component is not intrinsic, and it is the whole subject of what follows: it records how M bends inside $\tilde{M}$, information that the tangential projection discards.

That leftover is the second fundamental form. Writing the ambient derivative as its tangential part, now known to be $\nabla_X Y$, plus its normal part gives the decomposition on which the entire theory rests.

Definition 7.1.4: Second fundamental form and the Gauss formula

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold. The *second fundamental form* of M is the normal-valued map

$$\mathrm{II}(X, Y) = (\tilde{\nabla}_X Y)^{\perp}$$

assigning to tangent fields X, Y the normal part of the ambient covariant derivative. Combining with proposition 7.1.3, the ambient derivative splits into its tangential and normal parts as the *Gauss formula*

$$\tilde{\nabla}_X Y = \nabla_X Y + \mathrm{II}(X, Y) \tag{7.1}$$

where $\nabla_X Y = (\tilde{\nabla}_X Y)^{\top}$ is the induced Levi-Civita connection and $\mathrm{II}(X, Y) = (\tilde{\nabla}_X Y)^{\perp}$ is normal to M.

The Gauss formula (7.1) is the organizing identity of the chapter, and its two halves have complementary meanings. The tangential half is the intrinsic derivative, the one a resident of M can compute; the normal half $\mathrm{II}(X, Y)$ measures the failure of the intrinsic derivative to reproduce the ambient derivative, that is, the amount by which a tangent field, transported along X, is pushed out of M. When $\mathrm{II} \equiv 0$ the two connections agree and M does not bend at all inside $\tilde{M}$; the submanifolds with this property are the totally geodesic ones, studied later in the chapter. For a curve γ in M with velocity $\dot{\gamma}$, the specialization $\tilde{\nabla}_{\dot{\gamma}} \dot{\gamma} = \nabla_{\dot{\gamma}} \dot{\gamma} + \mathrm{II}(\dot{\gamma}, \dot{\gamma})$ says that the ambient acceleration of γ splits into an intrinsic acceleration, tangent to M, and a normal acceleration $\mathrm{II}(\dot{\gamma}, \dot{\gamma})$; a curve that is a geodesic of M has $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$, so its ambient acceleration is purely normal, exactly the sense in which a great circle on the sphere accelerates only toward the center.

The name "form" anticipates the next proposition: despite being defined through a connection, II is not a differential operator but a tensor, symmetric and bilinear over functions, so its value $\mathrm{II}(X, Y)_p$ depends only on the vectors X_p, Y_p. The symmetry is inherited from the vanishing of the ambient torsion.

Proposition 7.1.5: Symmetry and tensoriality of the second fundamental form

The second fundamental form II of definition 7.1.4 is symmetric,

$$\mathrm{II}(X, Y) = \mathrm{II}(Y, X)$$

and $C^{\infty}(M)$-bilinear. Consequently its value $\mathrm{II}(X, Y)_p$ at a point p depends only on X_p and Y_p, and II is a well-defined smooth section of $\mathrm{Sym}^2 T^* M \otimes NM$, a symmetric normal-bundle-valued bilinear form on TM.

***Proof*:**
Step 1: symmetry. Subtract the Gauss formula (7.1) from itself with X and Y interchanged:

$$\tilde{\nabla}_X Y - \tilde{\nabla}_Y X = (\nabla_X Y - \nabla_Y X) + (\mathrm{II}(X, Y) - \mathrm{II}(Y, X))$$

The left-hand side is $[X, Y]$ by torsion-freeness of $\tilde{\nabla}$ (definition 2.2.1 on $\tilde{M}$), and the first parenthesized term on the right is $[X, Y]$ by torsion-freeness of ∇, established in proposition 7.1.3. These two brackets are equal (both are the Lie bracket of the tangent fields X, Y), so subtracting gives

$$\mathrm{I\!I}(X,Y) - \mathrm{I\!I}(Y,X) = 0$$

Equivalently, $\mathrm{I\!I}(X,Y) - \mathrm{I\!I}(Y,X) = [X,Y]^{\perp}$, which vanishes because $[X,Y]$ is tangent to M (Step 2 of proposition 7.1.3), so its normal part is zero. Either reading gives $\mathrm{I\!I}(X,Y) = \mathrm{I\!I}(Y,X)$.

Step 2: bilinearity over $C^\infty(M)$. Additivity in each argument is immediate from that of $\tilde{\nabla}$ and $(\cdot)^{\perp}$. For homogeneity in the first argument, $\tilde{\nabla}_{fX}Y = f\,\tilde{\nabla}_X Y$ gives

$$\mathrm{I\!I}(fX,Y) = (\tilde{\nabla}_{fX}Y)^{\perp} = f\,(\tilde{\nabla}_X Y)^{\perp} = f\,\mathrm{I\!I}(X,Y)$$

Homogeneity in the second argument then follows from symmetry, $\mathrm{I\!I}(X,fY) = \mathrm{I\!I}(fY,X) = f\,\mathrm{I\!I}(Y,X) = f\,\mathrm{I\!I}(X,Y)$. Alternatively, computing directly with the Leibniz rule,

$$\mathrm{I\!I}(X,fY) = (\tilde{\nabla}_X(fY))^{\perp} = \big((Xf)Y + f\,\tilde{\nabla}_X Y\big)^{\perp} = (Xf)\,Y^{\perp} + f\,(\tilde{\nabla}_X Y)^{\perp} = f\,\mathrm{I\!I}(X,Y)$$

since $Y^{\perp} = 0$ for the tangent field Y; the derivative term $(Xf)Y$ falls entirely in TM and is annihilated by $(\cdot)^{\perp}$. This is the mechanism by which the normal projection converts a differential operator into a tensor: every term carrying a derivative of f lands in the tangent bundle and is projected away.

Step 3: from bilinearity to a tensor. A map that is $C^\infty(M)$-bilinear in its two tangent arguments and takes values in NM is induced by a smooth section of $T^*M \otimes T^*M \otimes NM$, by the tensor characterization of [6]; concretely, $\mathrm{I\!I}(X,Y)_p$ depends only on X_p, Y_p. Symmetry from Step 1 places this section in the symmetric subbundle, so $\mathrm{I\!I}$ is a smooth section of $\mathrm{Sym}^2 T^*M \otimes NM$, as we sought to show.

Tensoriality is the license, exactly as for the curvature tensor of proposition 4.1.3, to evaluate $\mathrm{I\!I}$ on vectors at a point by extending them to any convenient fields; the result is independent of the extension, which retroactively justifies treating M as embedded and extending fields for the sake of computing $\tilde{\nabla}$. Symmetry says $\mathrm{I\!I}$ is a quadratic datum: it is determined by its diagonal values $\mathrm{I\!I}(X,X)$, the normal accelerations $\mathrm{I\!I}(\dot{\gamma},\dot{\gamma})$ of curves, through the polarization $2\,\mathrm{I\!I}(X,Y) = \mathrm{I\!I}(X+Y,X+Y) - \mathrm{I\!I}(X,X) - \mathrm{I\!I}(Y,Y)$. The classical "first fundamental form" is the induced metric g itself, a symmetric 2-tensor; $\mathrm{I\!I}$ is the second such tensor attached to M, now vector-valued in the normal directions, and the pair $(g,\mathrm{I\!I})$ is the data of the local extrinsic geometry.

It is often more convenient to pair $\mathrm{I\!I}$ against a fixed normal field and record the result as an operator on the tangent bundle. Doing so trades the normal-valued form for a family of self-adjoint operators, one for each normal direction, whose eigenvalues will be the principal curvatures. The operator is obtained by differentiating the normal field instead of the tangent field, and taking the tangential part.

Definition 7.1.6: Shape operator and the Weingarten formula

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold and ξ a normal field on M. The *shape operator* (or *Weingarten map*) in the direction ξ is the endomorphism $S_\xi \colon TM \to TM$ of the tangent bundle defined by

$$S_\xi X = -(\tilde{\nabla}_X \xi)^\top \tag{7.2}$$

the tangential part of the ambient derivative of ξ along X, with a sign. Writing $\nabla^\perp_X \xi = (\tilde{\nabla}_X \xi)^\perp$ for the normal part, the ambient derivative of a normal field splits as the *Weingarten formula*

$$\tilde{\nabla}_X \xi = -S_\xi X + \nabla^\perp_X \xi \tag{7.3}$$

where $-S_\xi X$ is tangent to M and $\nabla^\perp_X \xi$ is normal. The operation $\xi \mapsto \nabla^\perp_X \xi$ is the *normal connection*, a connection on the normal bundle NM developed in section 7.2.

The Weingarten formula (7.3) is the normal-field companion of the Gauss formula (7.1): where Gauss splits the ambient derivative of a tangent field, Weingarten splits that of a normal field, the tangential half being $-S_\xi X$ and the normal half defining $\nabla^\perp$. The shape operator therefore measures how the normal field ξ tips back toward M as one moves along X; for a unit normal to a hypersurface, $S_\xi X$ is the classical rate of turning of the normal, the extrinsic curvature of M read through the motion of the field perpendicular to it. The sign in (7.2) is the binding convention of this chapter and is chosen precisely so that S_ξ pairs against $\mathrm{I\!I}$ through the metric, the identity recorded next.

That S_ξ carries the same information as $\mathrm{I\!I}$, repackaged, is the content of the pairing identity: for tangent X, Y and normal ξ,

$$g(S_\xi X, Y) = \tilde{g}(\mathrm{I\!I}(X, Y), \xi) \tag{7.4}$$

The proof is a differentiation of orthogonality. Since ξ is normal and Y is tangent, $\tilde{g}(\xi, Y) = 0$ as a function on M; differentiating along X and using metric compatibility of $\tilde{\nabla}$ (definition 2.2.4 on $\tilde{M}$),

$$0 = X\,\tilde{g}(\xi, Y) = \tilde{g}(\tilde{\nabla}_X \xi, Y) + \tilde{g}(\xi, \tilde{\nabla}_X Y)$$

In the first pairing Y is tangent, so only the tangential part of $\tilde{\nabla}_X \xi$ contributes: $\tilde{g}(\tilde{\nabla}_X \xi, Y) = \tilde{g}((\tilde{\nabla}_X \xi)^\top, Y) = -\tilde{g}(S_\xi X, Y) = -g(S_\xi X, Y)$. In the second pairing ξ is normal, so only the normal part of $\tilde{\nabla}_X Y$ contributes: $\tilde{g}(\xi, \tilde{\nabla}_X Y) = \tilde{g}(\xi, (\tilde{\nabla}_X Y)^\perp) = \tilde{g}(\xi, \mathrm{I\!I}(X, Y))$. Substituting,

$$0 = -g(S_\xi X, Y) + \tilde{g}(\mathrm{I\!I}(X, Y), \xi)$$

which rearranges to (7.4). The identity shows that $\mathrm{I\!I}$ and the family $\{S_\xi\}$ determine each other: $\mathrm{I\!I}$ is recovered from all the S_ξ by (7.4) together with nondegeneracy of $\tilde{g}$ on NM, and each S_ξ is recovered from $\mathrm{I\!I}$ by the same equation with nondegeneracy of g on TM. In particular S_ξ is $C^\infty(M)$-linear in X and $C^\infty(M)$-linear in ξ, since $\mathrm{I\!I}$ is bilinear by proposition 7.1.5 and $\tilde{g}$ is bilinear.

The pairing (7.4) has the symmetry of $\mathrm{I\!I}$ built into it, and reading it with X and Y exchanged shows the shape operator is self-adjoint.

Proposition 7.1.7: Self-adjointness of the shape operator

For every normal field ξ, the shape operator S_ξ of definition 7.1.6 is g-self-adjoint: for all tangent fields X, Y,

$$g(S_\xi X, Y) = g(X, S_\xi Y)$$

***Proof*:**
By the pairing identity (7.4),

$$g(S_\xi X, Y) = \tilde{g}(\mathrm{I\!I}(X,Y), \xi)$$

The second fundamental form is symmetric by proposition 7.1.5, so $\mathrm{I\!I}(X,Y) = \mathrm{I\!I}(Y,X)$, and therefore

$$g(S_\xi X, Y) = \tilde{g}(\mathrm{I\!I}(Y,X), \xi) = g(S_\xi Y, X) = g(X, S_\xi Y)$$

the last step by symmetry of g. Hence S_ξ is self-adjoint, as claimed.

Self-adjointness is the structural fact that unlocks the eigentheory of S_ξ. A g-self-adjoint endomorphism of a Euclidean tangent space is diagonalizable with real eigenvalues in a g-orthonormal basis, by the spectral theorem; for a hypersurface with unit normal these eigenvalues are the principal curvatures and the eigenvectors the principal directions, developed in section 7.3. Everything quantitative about how M curves in $\tilde{M}$, from mean curvature to the Gauss–Kronecker curvature to the sign of sectional curvature through the Gauss equation, is extracted from the spectrum of these self-adjoint operators, and it is available only because $\mathrm{I\!I}$ is symmetric.

The model computation is the round sphere, where a single differentiation identifies every shape operator at once. The key is that the position vector field is its own ambient derivative.

Example 7.1.8: The second fundamental form of the sphere
Let $S^n_r = \left\{x \in \mathbb{R}^{n+1} \;\middle|\; |x| = r\right\}$ carry the induced metric of example 1.5.3, embedded in $\mathbb{R}^{n+1}$ with its flat Euclidean connection $\tilde{\nabla} = \bar{\nabla}$, the ordinary directional derivative. By example 7.1.2 the outward unit normal is $N = x/r$, where $x = (x^1, \ldots, x^{n+1})$ denotes the position vector field, whose components are the ambient coordinate functions.

Step 1: differentiate the position vector. The Euclidean connection differentiates a vector field componentwise: for a tangent field X to S^n_r at a point, with ambient components X^a,

$$\bar{\nabla}_X x = \sum_a (X\, x^a)\, \partial_a = \sum_a X^a\, \partial_a = X$$

since $X\, x^a = X^a$ is the derivative of the a-th coordinate function in the direction X, and ∂_a is the standard frame. Thus the position vector field satisfies $\bar{\nabla}_X x = X$: differentiating x in any direction returns that direction. Consequently

$$\bar{\nabla}_X N = \bar{\nabla}_X\left(\tfrac{1}{r}x\right) = \tfrac{1}{r}\,\bar{\nabla}_X x = \tfrac{1}{r}X$$

because r is constant on S^n_r.

Step 2: the shape operator. The vector X is tangent to S^n_r, so $(\bar{\nabla}_X N)^\top = \frac{1}{r}X^\top = \frac{1}{r}X$. By the defining formula (7.2),

$$S_N X = -(\bar{\nabla}_X N)^\top = -\tfrac{1}{r}X$$

for every tangent X, so the shape operator in the outward normal direction is the scalar operator

$$S_N = -\tfrac{1}{r}\,\mathrm{id}_{TS^n_r}$$

It is g-self-adjoint, as proposition 7.1.7 requires of any shape operator, being a real scalar multiple of the identity; every tangent vector is a principal direction, and the single eigenvalue $-1/r$ is the principal curvature repeated n times.

Step 3: the second fundamental form. From the pairing identity (7.4), for tangent X, Y,

$$\tilde{g}(\mathrm{II}(X,Y), N) = g(S_N X, Y) = g\big(-\tfrac{1}{r}X, Y\big) = -\tfrac{1}{r}\, g(X,Y)$$

Since N spans the one-dimensional normal space $N_x S^n_r = \mathbb{R}x$ (example 7.1.2) and is a unit vector, a normal vector is determined by its $\tilde{g}$-pairing with N, so

$$\mathrm{II}(X,Y) = \tilde{g}(\mathrm{II}(X,Y), N)\, N = -\tfrac{1}{r}\, g(X,Y)\, N = -\frac{1}{r}\, \langle X, Y \rangle\, N$$

the second fundamental form is $-\frac{1}{r}$ times the induced metric, valued in the outward normal. The normal acceleration of a unit-speed curve γ on the sphere is $\mathrm{II}(\dot{\gamma}, \dot{\gamma}) = -\frac{1}{r}\, |\dot{\gamma}|^2\, N = -\frac{1}{r} N = -x/r^2$, pointing inward toward the center with magnitude $1/r$, which is the centripetal acceleration of a great circle of radius r traversed at unit speed.

These values feed the mandatory consistency check of the chapter. In section 7.2 the Gauss equation converts $\mathrm{II}(X,Y) = -\frac{1}{r}\, \langle X, Y \rangle\, N$ into the intrinsic curvature: for g-orthonormal tangent X, Y, using the flatness $\widetilde{\mathrm{Riem}} \equiv 0$ of $\mathbb{R}^{n+1}$ (example 4.1.6),

$$K(X,Y) = \langle \mathrm{II}(X,X), \mathrm{II}(Y,Y) \rangle - |\mathrm{II}(X,Y)|^2 = \big(-\tfrac{1}{r}\big)\big(-\tfrac{1}{r}\big)\, \langle X, X \rangle\, \langle Y, Y \rangle - 0 = \frac{1}{r^2}$$

recovering $K_{S^n_r} = +1/r^2$ in agreement with the value fixed in box 4.1.1 and computed intrinsically in section 4.3. The extrinsic datum II thus reproduces the intrinsic sectional curvature of the sphere.

7.2 The Gauss, Codazzi, and Ricci Equations

The second fundamental form of section 7.1 records how a submanifold M bends inside its ambient manifold $\tilde{M}$, and the shape operator packages the same information as a self-adjoint endomorphism of each tangent space. What has not yet been extracted is the relation between the two curvatures in play: the intrinsic curvature Riem of (M, g), computed from the induced metric alone, and the ambient curvature $\widetilde{\mathrm{Riem}}$ of $(\tilde{M}, \tilde{g})$. These are not independent. A submanifold cannot bend arbitrarily and still carry a metric whose curvature is unconstrained; the two are locked together by the geometry of the embedding, and the equations that state the lock are the subject of this section.

The mechanism is uniform. The Gauss formula splits an ambient covariant derivative $\tilde{\nabla}_X Y$ into a tangential part $\nabla_X Y$ and a normal part $\mathrm{II}(X,Y)$, and the Weingarten formula splits $\tilde{\nabla}_X \xi$ into a tangential part $-S_\xi X$ and a normal part $\nabla^\perp_X \xi$. Applying the ambient curvature operator $\widetilde{R}(X,Y)$ to a tangent field and to a normal field and then projecting onto the tangent and normal bundles produces three identities. The tangential projection acting on a tangent field is the *Gauss equation*, which expresses Riem through $\widetilde{\mathrm{Riem}}$ and II; the normal projection acting on a tangent field is the *Codazzi equation*, a first-order condition on II; and the normal projection acting on a normal field is the *Ricci equation*, which governs the curvature of the normal connection. Together they are the integrability conditions of submanifold theory, and each is derived below by the same bookkeeping of tangential and normal components.

Throughout, the standing setup of the chapter is in force: $(\tilde{M}, \tilde{g})$ is the ambient (pseudo-)Riemannian manifold with Levi-Civita connection $\tilde{\nabla}$ and curvature $\widetilde{R}$, $M \subseteq \tilde{M}$ is a Riemannian submanifold with the induced metric g of definition 1.5.1, its own Levi-Civita connection ∇, and curvature R; at each $p \in M$ the ambient tangent space splits g-orthogonally $T_p\tilde{M} = T_pM \oplus N_pM$, with $(\cdot)^\top$ and $(\cdot)^\perp$ the tangential and normal projections. Latin capitals X, Y, Z, W denote fields tangent to M and ξ, η fields normal to M. The two structural formulas that drive every computation are the Gauss formula

$$\tilde{\nabla}_X Y = \nabla_X Y + \mathrm{I\!I}(X, Y), \qquad \nabla_X Y = (\tilde{\nabla}_X Y)^\top, \quad \mathrm{I\!I}(X, Y) = (\tilde{\nabla}_X Y)^\perp \tag{7.5}$$

of definition 7.1.4, and the Weingarten decomposition

$$\tilde{\nabla}_X \xi = -S_\xi X + \nabla^\perp_X \xi, \qquad S_\xi X = -(\tilde{\nabla}_X \xi)^\top, \quad \nabla^\perp_X \xi = (\tilde{\nabla}_X \xi)^\perp \tag{7.6}$$

of definition 7.1.6, with the adjunction $\langle S_\xi X, Y\rangle = \langle \mathrm{I\!I}(X, Y), \xi\rangle$ tying them together and S_ξ self-adjoint (proposition 7.1.7). The curvature sign convention is the one fixed in box 4.1.1, applied both on M and on $\tilde{M}$.

Theorem 7.2.1: The Gauss equation

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold with second fundamental form $\mathrm{I\!I}$ of definition 7.1.4. For all vector fields X, Y, Z, W tangent to M,

$$\mathrm{Riem}(X, Y, Z, W) = \widetilde{\mathrm{Riem}}(X, Y, Z, W) + \langle \mathrm{I\!I}(X, W), \mathrm{I\!I}(Y, Z)\rangle - \langle \mathrm{I\!I}(X, Z), \mathrm{I\!I}(Y, W)\rangle$$

where Riem is the $(0, 4)$ curvature tensor of the induced metric g and $\widetilde{\mathrm{Riem}}$ is that of the ambient metric $\tilde{g}$, both taken in the convention of box 4.1.1.

***Proof*:**
Fix tangent fields X, Y, Z; extend them to fields on $\tilde{M}$ near M if needed, the tensorial character of both curvatures making the choice of extension immaterial by proposition 4.1.3. The plan is to compute the ambient curvature operator $\tilde{R}(X, Y)Z$ by two applications of the Gauss and Weingarten formulas, then pair it with a tangent field W; the normal pieces will drop out of that pairing and leave the intrinsic curvature together with two second-fundamental-form terms.

Step 1: expand $\tilde{\nabla}_X\tilde{\nabla}_Y Z$. By the Gauss formula (7.5), $\tilde{\nabla}_Y Z = \nabla_Y Z + \mathrm{I\!I}(Y, Z)$, a sum of a tangent field and a normal field. Apply $\tilde{\nabla}_X$ to each. On the tangent field $\nabla_Y Z$ the Gauss formula applies again,

$$\tilde{\nabla}_X(\nabla_Y Z) = \nabla_X\nabla_Y Z + \mathrm{I\!I}(X, \nabla_Y Z)$$

while on the normal field $\mathrm{I\!I}(Y, Z)$ the Weingarten formula (7.6) applies,

$$\tilde{\nabla}_X\big(\mathrm{I\!I}(Y, Z)\big) = -S_{\mathrm{I\!I}(Y,Z)}X + \nabla^\perp_X\big(\mathrm{I\!I}(Y, Z)\big)$$

Adding,

$$\tilde{\nabla}_X\tilde{\nabla}_Y Z = \underbrace{\nabla_X\nabla_Y Z - S_{\mathrm{I\!I}(Y,Z)}X}_{\text{tangential}} + \underbrace{\mathrm{I\!I}(X, \nabla_Y Z) + \nabla^\perp_X\big(\mathrm{I\!I}(Y, Z)\big)}_{\text{normal}} \tag{7.7}$$

Step 2: pair with a tangent field W. Take the $\tilde{g}$-inner product of (7.7) with a tangent field W. The two normal terms are g-orthogonal to W and contribute nothing, so

$$\big\langle \tilde{\nabla}_X\tilde{\nabla}_Y Z, W\big\rangle = \langle \nabla_X\nabla_Y Z, W\rangle - \big\langle S_{\mathrm{I\!I}(Y,Z)}X, W\big\rangle$$

The Weingarten adjunction $\langle S_\xi X, W\rangle = \langle \mathrm{II}(X,W), \xi\rangle$ of definition 7.1.6, applied with the normal field $\xi = \mathrm{II}(Y,Z)$, rewrites the last term as $\langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle$, giving

$$\left\langle \tilde{\nabla}_X \tilde{\nabla}_Y Z, W\right\rangle = \langle \nabla_X \nabla_Y Z, W\rangle - \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle \tag{7.8}$$

Interchanging X and Y throughout Steps 1 and 2 gives the companion identity

$$\left\langle \tilde{\nabla}_Y \tilde{\nabla}_X Z, W\right\rangle = \langle \nabla_Y \nabla_X Z, W\rangle - \langle \mathrm{II}(Y,W), \mathrm{II}(X,Z)\rangle \tag{7.9}$$

Step 3: the bracket term. The Lie bracket $[X,Y]$ of two fields tangent to M is again tangent to M, since M is a submanifold and the flows of X and Y preserve it. Hence the Gauss formula gives $\tilde{\nabla}_{[X,Y]} Z = \nabla_{[X,Y]} Z + \mathrm{II}([X,Y], Z)$, whose second summand is normal, and pairing with the tangent field W leaves

$$\left\langle \tilde{\nabla}_{[X,Y]} Z, W\right\rangle = \left\langle \nabla_{[X,Y]} Z, W\right\rangle \tag{7.10}$$

Step 4: assemble. By the sign convention of box 4.1.1, $\widetilde{R}(X,Y)Z = \tilde{\nabla}_X \tilde{\nabla}_Y Z - \tilde{\nabla}_Y \tilde{\nabla}_X Z - \tilde{\nabla}_{[X,Y]} Z$, so

$$\widetilde{\mathrm{Riem}}(X,Y,Z,W) = \left\langle \tilde{\nabla}_X \tilde{\nabla}_Y Z, W\right\rangle - \left\langle \tilde{\nabla}_Y \tilde{\nabla}_X Z, W\right\rangle - \left\langle \tilde{\nabla}_{[X,Y]} Z, W\right\rangle$$

Substituting (7.8), (7.9), and (7.10),

$$\widetilde{\mathrm{Riem}}(X,Y,Z,W) = \left\langle \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z, W\right\rangle - \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle + \langle \mathrm{II}(Y,W), \mathrm{II}(X,Z)\rangle$$

The first inner product is $\langle R(X,Y)Z, W\rangle = \mathrm{Riem}(X,Y,Z,W)$, the intrinsic curvature of the induced metric in the same convention. Solving for Riem,

$$\mathrm{Riem}(X,Y,Z,W) = \widetilde{\mathrm{Riem}}(X,Y,Z,W) + \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle - \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W)\rangle$$

which is the stated identity, completing the proof.

The Gauss equation is the statement that intrinsic curvature is ambient curvature corrected by a purely algebraic expression in the second fundamental form. It is the exact form of the observation that a submanifold's own geometry is determined jointly by where it sits and how it bends. Two special cases organize its use. When the ambient manifold is flat, $\widetilde{\mathrm{Riem}} \equiv 0$ and the entire intrinsic curvature of M is manufactured by the bending term $\langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle - \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W)\rangle$; this is the case of surfaces in $\mathbb{R}^3$ and is the source of Gauss's Theorema Egregium, taken up in section 7.4. When instead $\mathrm{II} \equiv 0$, the correction vanishes and M inherits the ambient curvature verbatim; such totally geodesic submanifolds are studied in the same later section. The right-hand side mixes an ambient quantity with a submanifold quantity, yet the left-hand side depends only on the induced metric, so the equation is also a constraint: the bending term is forced to reproduce whatever intrinsic curvature the induced metric already has.

For a 2-plane the equation reduces to a scalar relation between sectional curvatures, and this is the form in which it is most often applied.

Corollary 7.2.2: Gauss equation in sectional form

Let X, Y be tangent to M and g-orthonormal at a point p, spanning a 2-plane $\Pi \subseteq T_pM$. Writing K for the sectional curvature of (M, g) and $\tilde{K}$ for that of $(\tilde{M}, \tilde{g})$ (definition 4.3.1),

$$K(X,Y) = \tilde{K}(X,Y) + \langle \mathrm{II}(X,X), \mathrm{II}(Y,Y)\rangle - |\mathrm{II}(X,Y)|^2$$

where $\tilde{K}(X,Y)$ is the ambient sectional curvature of the plane Π regarded as a 2-plane in $T_p\tilde{M}$.

***Proof*:**

For a g-orthonormal pair the Gram determinant $|X|^2\,|Y|^2 - \langle X, Y\rangle^2$ equals 1, so by definition 4.3.1 the sectional curvatures are the curvature-tensor values $K(X,Y) = \mathrm{Riem}(X,Y,Y,X)$ and $\tilde{K}(X,Y) = \widetilde{\mathrm{Riem}}(X,Y,Y,X)$; the pair is orthonormal for g and hence for $\tilde{g}$, since $\tilde{g}$ restricts to g on T_pM, so the ambient plane spanned by X, Y has the same unit Gram determinant. Apply theorem 7.2.1 with $Z = Y$ and $W = X$:

$$\mathrm{Riem}(X,Y,Y,X) = \widetilde{\mathrm{Riem}}(X,Y,Y,X) + \langle \mathrm{II}(X,X), \mathrm{II}(Y,Y)\rangle - \langle \mathrm{II}(X,Y), \mathrm{II}(Y,X)\rangle$$

The second fundamental form is symmetric (proposition 7.1.5), so $\mathrm{II}(Y,X) = \mathrm{II}(X,Y)$ and the last inner product is $\langle \mathrm{II}(X,Y), \mathrm{II}(X,Y)\rangle = |\mathrm{II}(X,Y)|^2$. Substituting the curvature-tensor values for the sectional curvatures gives the claim, as required.

The sectional form isolates the geometric content: a plane's curvature inside M exceeds its curvature inside $\tilde{M}$ by the difference $\langle \mathrm{II}(X,X), \mathrm{II}(Y,Y)\rangle - |\mathrm{II}(X,Y)|^2$ of the normal-valued second fundamental form. The first term compares the two normal accelerations along the orthonormal directions and the second penalizes the mixed bending. The round sphere is the test case, and it is arranged so that the extrinsic computation of section 7.1 reproduces the intrinsic curvature already fixed for the model spaces.

Example 7.2.3: The Gauss equation returns $K = +1/r^2$ for the sphere

Take $M = S^n_r \subseteq \mathbb{R}^{n+1}$ with the outward unit normal $N = x/r$, ambient $(\mathbb{R}^{n+1}, \bar{g})$ flat. From example 7.1.8 the second fundamental form is

$$\mathrm{II}(X,Y) = -\tfrac{1}{r}\,\langle X, Y\rangle\, N$$

for tangent fields X, Y. Let X, Y be g-orthonormal and tangent to S^n_r at a point, so $\langle X, X\rangle = \langle Y, Y\rangle = 1$ and $\langle X, Y\rangle = 0$. Then

$$\mathrm{II}(X,X) = -\tfrac{1}{r}N, \qquad \mathrm{II}(Y,Y) = -\tfrac{1}{r}N, \qquad \mathrm{II}(X,Y) = 0$$

The ambient manifold $\mathbb{R}^{n+1}$ is flat by example 4.1.6, so $\tilde{K}(X,Y) = 0$. Substituting into corollary 7.2.2 and using $|N| = 1$,

$$K(X,Y) = 0 + \left\langle -\tfrac{1}{r}N, -\tfrac{1}{r}N\right\rangle - 0 = \tfrac{1}{r^2}\,\langle N, N\rangle = \tfrac{1}{r^2}$$

Every 2-plane in $T_pS^n_r$ is spanned by such an orthonormal pair, so $K \equiv +1/r^2$ on the round sphere of radius r. This is computed here from the extrinsic data II and a flat ambient, and it agrees with the value fixed intrinsically in box 4.1.1 and recovered in example 4.3.4. The sign convention was arranged precisely so that a sphere, which curls toward its center, records positive sectional curvature; the two computations meeting on $+1/r^2$ is the consistency check the Gauss equation is meant to pass.

The tangential projection of $\widetilde{R}(X,Y)Z$ produced the Gauss equation. The normal projection produces the Codazzi equation, and stating it requires a derivative of the second fundamental form. The second fundamental form is a section of $\mathrm{Sym}^2T^*M \otimes NM$, taking values in the normal bundle, and differentiating it calls for a connection on that bundle. The normal connection, named in definition 7.1.6 through the Weingarten decomposition, is that connection, and it is recorded now together with the induced covariant derivative of $\mathrm{I\!I}$.

Definition 7.2.4: The normal connection and the covariant derivative of $\mathrm{I\!I}$

The *normal connection* $\nabla^\perp$ on the normal bundle NM of $M \subseteq \tilde{M}$ is defined for a tangent field X and a normal field ξ by

$$\nabla^\perp_X \xi = (\tilde{\nabla}_X \xi)^\perp$$

the normal part of the ambient covariant derivative, as in (7.6). Its curvature endomorphism is

$$R^\perp(X,Y) = \nabla^\perp_X\nabla^\perp_Y - \nabla^\perp_Y\nabla^\perp_X - \nabla^\perp_{[X,Y]}$$

acting on sections of NM, in the sign convention of box 4.1.1. The *covariant derivative of the second fundamental form* is the map $(X,Y,Z) \mapsto (\nabla^\perp_X \mathrm{I\!I})(Y,Z)$, a normal field, given by

$$(\nabla^\perp_X \mathrm{I\!I})(Y,Z) = \nabla^\perp_X\big(\mathrm{I\!I}(Y,Z)\big) - \mathrm{I\!I}(\nabla_X Y, Z) - \mathrm{I\!I}(Y, \nabla_X Z)$$

for tangent fields X, Y, Z, where ∇ is the induced Levi-Civita connection on M.

That $\nabla^\perp$ is a connection on NM is a routine consequence of the axioms for $\tilde{\nabla}$: it is $C^\infty(M)$-linear in the differentiating slot X and satisfies the Leibniz rule $\nabla^\perp_X(f\xi) = (Xf)\xi + f\nabla^\perp_X\xi$ in ξ, because $\tilde{\nabla}$ has these properties and the normal projection $(\cdot)^\perp$ is $C^\infty(M)$-linear and fixes the term $(Xf)\xi$, which is already normal. Metric compatibility with the induced fiber metric on NM is inherited the same way: for normal fields ξ, η, differentiating $\langle \xi, \eta\rangle$ along the tangent field X and using compatibility of $\tilde{\nabla}$ gives $X\langle\xi,\eta\rangle = \langle\tilde{\nabla}_X\xi,\eta\rangle + \langle\xi,\tilde{\nabla}_X\eta\rangle = \langle\nabla^\perp_X\xi,\eta\rangle + \langle\xi,\nabla^\perp_X\eta\rangle$, the tangential parts of $\tilde{\nabla}_X\xi$ and $\tilde{\nabla}_X\eta$ being g-orthogonal to the normal fields η and ξ. The definition of $(\nabla^\perp_X\mathrm{I\!I})(Y,Z)$ is the standard product-rule extension of $\nabla^\perp$ to the bundle $\mathrm{Sym}^2T^*M\otimes NM$: the whole of $\nabla^\perp_X(\mathrm{I\!I}(Y,Z))$ is corrected by the two terms accounting for the differentiation of the arguments Y and Z, so that $(\nabla^\perp_X\mathrm{I\!I})$ is again tensorial in Y and Z and normal-valued.

Example 7.2.5: The normal connection and $\nabla^\perp\mathrm{I\!I}$ for the sphere

For a hypersurface such as $S^n_r \subseteq \mathbb{R}^{n+1}$ the normal bundle has rank 1, spanned at each point by the unit normal $N = x/r$ of example 7.1.8. Any normal field is $\xi = fN$ for a function f on the sphere, and the normal connection acts by $\nabla^\perp_X(fN) = (Xf)N + f\,\nabla^\perp_X N$. To evaluate $\nabla^\perp_X N$, differentiate $N = x/r$ in $\mathbb{R}^{n+1}$: for a tangent field X, the ambient derivative $\bar{\nabla}_X N = \frac{1}{r}\bar{\nabla}_X x = \frac{1}{r}X$ is tangent to the sphere, so its normal part vanishes, $\nabla^\perp_X N = (\bar{\nabla}_X N)^\perp = 0$. Thus N is parallel for $\nabla^\perp$, and $\nabla^\perp_X(fN) = (Xf)N$: the normal connection on a rank-one normal bundle with a parallel unit section is just directional differentiation of the coefficient. Consequently $R^\perp \equiv 0$ for the sphere, the normal bundle being flat.

The covariant derivative of the second fundamental form is equally explicit. Writing $\mathrm{I\!I}(Y,Z) =$

$-\frac{1}{r}\langle Y, Z\rangle N$ with the scalar coefficient $h(Y,Z) = -\frac{1}{r}\langle Y, Z\rangle$, and using $\nabla^\perp N = 0$,

$$(\nabla^\perp_X \mathrm{I\!I})(Y,Z) = \big(X\,h(Y,Z) - h(\nabla_X Y, Z) - h(Y, \nabla_X Z)\big)N = (\nabla_X h)(Y,Z)\,N$$

Here $\nabla_X h$ is the ordinary covariant derivative of the symmetric 2-tensor $h = -\frac{1}{r}g$. Since g is parallel for its own Levi-Civita connection ∇ (definition 2.2.4), $\nabla_X h = -\frac{1}{r}\nabla_X g = 0$, and therefore $(\nabla^\perp_X \mathrm{I\!I})(Y,Z) = 0$ identically. The second fundamental form of the round sphere is parallel; the Codazzi equation below will confirm that this is forced, the ambient being flat.

With the normal connection in hand, the normal projection of the ambient curvature operator can be read off from the same expansion (7.7) that produced the Gauss equation.

Theorem 7.2.6: The Codazzi equation

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold with second fundamental form $\mathrm{I\!I}$ and normal connection $\nabla^\perp$ of definition 7.2.4. For all vector fields X, Y, Z tangent to M,

$$\big(\widetilde{R}(X,Y)Z\big)^\perp = (\nabla^\perp_X \mathrm{I\!I})(Y,Z) - (\nabla^\perp_Y \mathrm{I\!I})(X,Z)$$

the normal part of the ambient curvature operator applied to a tangent field.

***Proof*:**

Return to the expansion (7.7) of $\tilde{\nabla}_X \tilde{\nabla}_Y Z$ and retain its normal part, which by that display is

$$\big(\tilde{\nabla}_X \tilde{\nabla}_Y Z\big)^\perp = \mathrm{I\!I}(X, \nabla_Y Z) + \nabla^\perp_X\big(\mathrm{I\!I}(Y,Z)\big)$$

Interchanging X and Y,

$$\big(\tilde{\nabla}_Y \tilde{\nabla}_X Z\big)^\perp = \mathrm{I\!I}(Y, \nabla_X Z) + \nabla^\perp_Y\big(\mathrm{I\!I}(X,Z)\big)$$

and, as in Step 3 of theorem 7.2.1, the bracket $[X,Y]$ is tangent to M, so

$$\big(\tilde{\nabla}_{[X,Y]} Z\big)^\perp = \mathrm{I\!I}\big([X,Y], Z\big)$$

the tangential part $\nabla_{[X,Y]}Z$ contributing nothing to the normal projection. By the sign convention of box 4.1.1, $\widetilde{R}(X,Y)Z = \tilde{\nabla}_X\tilde{\nabla}_Y Z - \tilde{\nabla}_Y\tilde{\nabla}_X Z - \tilde{\nabla}_{[X,Y]}Z$, and projecting onto NM term by term gives

$$\big(\widetilde{R}(X,Y)Z\big)^\perp = \nabla^\perp_X\big(\mathrm{I\!I}(Y,Z)\big) - \nabla^\perp_Y\big(\mathrm{I\!I}(X,Z)\big) + \mathrm{I\!I}(X, \nabla_Y Z) - \mathrm{I\!I}(Y, \nabla_X Z) - \mathrm{I\!I}\big([X,Y], Z\big) \quad (7.11)$$

Now convert the two undifferentiated $\nabla^\perp$ terms into covariant derivatives of $\mathrm{I\!I}$ using definition 7.2.4. From the definition,

$$\nabla^\perp_X\big(\mathrm{I\!I}(Y,Z)\big) = (\nabla^\perp_X \mathrm{I\!I})(Y,Z) + \mathrm{I\!I}(\nabla_X Y, Z) + \mathrm{I\!I}(Y, \nabla_X Z)$$

and likewise, interchanging X and Y,

$$\nabla^\perp_Y\big(\mathrm{I\!I}(X,Z)\big) = (\nabla^\perp_Y \mathrm{I\!I})(X,Z) + \mathrm{I\!I}(\nabla_Y X, Z) + \mathrm{I\!I}(X, \nabla_Y Z)$$

Substitute both into (7.11). The terms $\mathrm{I\!I}(Y, \nabla_X Z)$ and $\mathrm{I\!I}(X, \nabla_Y Z)$ introduced by these substitutions cancel against the like terms already present in (7.11), leaving

$$\big(\widetilde{R}(X,Y)Z\big)^{\perp} = (\nabla^{\perp}_X \mathrm{I\!I})(Y,Z) - (\nabla^{\perp}_Y \mathrm{I\!I})(X,Z) + \mathrm{I\!I}(\nabla_X Y, Z) - \mathrm{I\!I}(\nabla_Y X, Z) - \mathrm{I\!I}\big([X,Y], Z\big)$$

The induced connection ∇ is torsion-free (theorem 2.3.1), so $\nabla_X Y - \nabla_Y X = [X,Y]$, and by bilinearity of $\mathrm{I\!I}$ the three trailing terms combine to $\mathrm{I\!I}(\nabla_X Y - \nabla_Y X - [X,Y],\, Z) = \mathrm{I\!I}(0, Z) = 0$. What remains is

$$\big(\widetilde{R}(X,Y)Z\big)^{\perp} = (\nabla^{\perp}_X \mathrm{I\!I})(Y,Z) - (\nabla^{\perp}_Y \mathrm{I\!I})(X,Z)$$

as we sought to show.

The Codazzi equation is a first-order compatibility condition: it relates the normal component of the ambient curvature to the antisymmetric part of $\nabla^{\perp}\mathrm{I\!I}$ in its first two slots. Where the Gauss equation was an algebraic identity in $\mathrm{I\!I}$, this one differentiates $\mathrm{I\!I}$ and constrains how it may vary across M. Its most-used consequence is the case of a constant-curvature ambient, where the constraint becomes a symmetry of $\nabla^{\perp}\mathrm{I\!I}$ outright.

Corollary 7.2.7: $\mathrm{I\!I}$ is a Codazzi tensor in a constant-curvature ambient

Suppose the ambient manifold $\tilde{M}$ has constant sectional curvature c. Then for all tangent fields X, Y, Z the ambient curvature operator $\widetilde{R}(X,Y)Z$ is tangent to M, and consequently

$$(\nabla^{\perp}_X \mathrm{I\!I})(Y,Z) = (\nabla^{\perp}_Y \mathrm{I\!I})(X,Z)$$

so that $\nabla^{\perp}\mathrm{I\!I}$ is symmetric in its first two arguments; the second fundamental form is a Codazzi tensor.

***Proof*:**

By proposition 4.5.3, constant sectional curvature c means $\widetilde{R}(X,Y)Z = c\big(\langle Y,Z\rangle X - \langle X,Z\rangle Y\big)$ for all ambient fields, where the inner products are those of $\tilde{g}$. When X, Y, Z are tangent to M, the right-hand side is a linear combination of the tangent fields X and Y, hence tangent to M, so its normal part vanishes: $(\widetilde{R}(X,Y)Z)^{\perp} = 0$. Theorem 7.2.6 then reads

$$0 = (\nabla^{\perp}_X \mathrm{I\!I})(Y,Z) - (\nabla^{\perp}_Y \mathrm{I\!I})(X,Z)$$

which is the asserted symmetry, completing the proof.

Euclidean space, the round sphere, and hyperbolic space all have constant sectional curvature (example 4.3.4, example 4.3.5, example 4.1.6), so every submanifold of a model space has a Codazzi second fundamental form. The parallel $\mathrm{I\!I}$ of the round sphere in example 7.2.5 is a limiting instance: there $\nabla^{\perp}\mathrm{I\!I}$ vanished entirely, and a fortiori its first two slots are symmetric, exactly as the corollary requires for the flat ambient $\mathbb{R}^{n+1}$.

One projection of $\widetilde{R}$ remains. The Gauss and Codazzi equations came from applying $\widetilde{R}(X,Y)$ to a tangent field and reading its tangential and normal parts. Applying $\widetilde{R}(X,Y)$ instead to a *normal* field and pairing with a normal field measures the curvature of the normal connection, and the result is the Ricci equation.

Theorem 7.2.8: The Ricci equation

Let $M \subseteq \tilde{M}$ be a Riemannian submanifold, with shape operators S_ξ of definition 7.1.6 and normal curvature $R^\perp$ of definition 7.2.4. For all tangent fields X, Y and all normal fields ξ, η,

$$\widetilde{\mathrm{Riem}}(X, Y, \xi, \eta) = \left\langle R^\perp(X, Y)\xi, \eta \right\rangle - \langle [S_\xi, S_\eta]X, Y \rangle$$

where $\widetilde{\mathrm{Riem}}(X, Y, \xi, \eta) = \left\langle \tilde{R}(X, Y)\xi, \eta \right\rangle$ in the convention of box 4.1.1 and $[S_\xi, S_\eta] = S_\xi S_\eta - S_\eta S_\xi$ is the commutator of the two shape operators.

***Proof*:**
Fix tangent fields X, Y and normal fields ξ, η. The computation runs parallel to that of theorem 7.2.1, but with the Weingarten formula (7.6) in place of the Gauss formula at the first differentiation, since the field being differentiated is now normal. Throughout, only the normal part of each doubly-differentiated field survives the pairing with the normal field η.

Step 1: expand $\tilde{\nabla}_X \tilde{\nabla}_Y \xi$. By the Weingarten formula, $\tilde{\nabla}_Y \xi = -S_\xi Y + \nabla^\perp_Y \xi$, a sum of a tangent field and a normal field. Differentiate each along X. On the tangent field $-S_\xi Y$ the Gauss formula (7.5) gives

$$\tilde{\nabla}_X(-S_\xi Y) = -\nabla_X(S_\xi Y) - \mathrm{I\!I}(X, S_\xi Y)$$

and on the normal field $\nabla^\perp_Y \xi$ the Weingarten formula gives

$$\tilde{\nabla}_X \left(\nabla^\perp_Y \xi\right) = -S_{\nabla^\perp_Y \xi} X + \nabla^\perp_X \nabla^\perp_Y \xi$$

Adding, the normal part is

$$\left(\tilde{\nabla}_X \tilde{\nabla}_Y \xi\right)^\perp = -\mathrm{I\!I}(X, S_\xi Y) + \nabla^\perp_X \nabla^\perp_Y \xi \tag{7.12}$$

the two tangential terms $-\nabla_X(S_\xi Y)$ and $-S_{\nabla^\perp_Y \xi} X$ being discarded.

Step 2: pair with η. Taking the $\tilde{g}$-inner product of (7.12) with the normal field η,

$$\left\langle \tilde{\nabla}_X \tilde{\nabla}_Y \xi, \eta \right\rangle = -\left\langle \mathrm{I\!I}(X, S_\xi Y), \eta \right\rangle + \left\langle \nabla^\perp_X \nabla^\perp_Y \xi, \eta \right\rangle$$

The Weingarten adjunction of definition 7.1.6, $\langle \mathrm{I\!I}(X, V), \eta \rangle = \langle S_\eta X, V \rangle$ for a tangent field V, applies with $V = S_\xi Y$, and self-adjointness of S_η (proposition 7.1.7) moves it across:

$$\langle \mathrm{I\!I}(X, S_\xi Y), \eta \rangle = \langle S_\eta X, S_\xi Y \rangle = \langle X, S_\eta S_\xi Y \rangle$$

Hence

$$\left\langle \tilde{\nabla}_X \tilde{\nabla}_Y \xi, \eta \right\rangle = -\langle X, S_\eta S_\xi Y \rangle + \left\langle \nabla^\perp_X \nabla^\perp_Y \xi, \eta \right\rangle \tag{7.13}$$

Interchanging X and Y,

$$\left\langle \tilde{\nabla}_Y \tilde{\nabla}_X \xi, \eta \right\rangle = -\langle Y, S_\eta S_\xi X \rangle + \left\langle \nabla^\perp_Y \nabla^\perp_X \xi, \eta \right\rangle \tag{7.14}$$

Step 3: the bracket term. By the Weingarten formula, $\tilde{\nabla}_{[X,Y]} \xi = -S_\xi [X, Y] + \nabla^\perp_{[X,Y]} \xi$, whose first summand is tangent to M and so orthogonal to η; thus

$$\left\langle \tilde{\nabla}_{[X,Y]} \xi, \eta \right\rangle = \left\langle \nabla^\perp_{[X,Y]} \xi, \eta \right\rangle \tag{7.15}$$

Step 4: assemble. With $\widetilde{R}(X,Y)\xi = \tilde{\nabla}_X\tilde{\nabla}_Y\xi - \tilde{\nabla}_Y\tilde{\nabla}_X\xi - \tilde{\nabla}_{[X,Y]}\xi$ from box 4.1.1, subtract (7.14) and (7.15) from (7.13):

$$\widetilde{\mathrm{Riem}}(X,Y,\xi,\eta) = \left\langle \nabla^\perp_X\nabla^\perp_Y\xi - \nabla^\perp_Y\nabla^\perp_X\xi - \nabla^\perp_{[X,Y]}\xi, \eta\right\rangle - \langle X, S_\eta S_\xi Y\rangle + \langle Y, S_\eta S_\xi X\rangle$$

The first inner product is $\langle R^\perp(X,Y)\xi,\eta\rangle$ by definition 7.2.4. It remains to identify the two shape-operator terms with the commutator. Using self-adjointness of S_η and then of S_ξ,

$$\langle X, S_\eta S_\xi Y\rangle = \langle S_\eta X, S_\xi Y\rangle = \langle S_\xi S_\eta X, Y\rangle, \qquad \langle Y, S_\eta S_\xi X\rangle = \langle S_\eta S_\xi X, Y\rangle$$

so the two terms combine to

$$-\langle X, S_\eta S_\xi Y\rangle + \langle Y, S_\eta S_\xi X\rangle = -\langle S_\xi S_\eta X, Y\rangle + \langle S_\eta S_\xi X, Y\rangle = -\langle (S_\xi S_\eta - S_\eta S_\xi)X, Y\rangle = -\langle [S_\xi, S_\eta]X, Y\rangle$$

Therefore

$$\widetilde{\mathrm{Riem}}(X,Y,\xi,\eta) = \langle R^\perp(X,Y)\xi,\eta\rangle - \langle [S_\xi,S_\eta]X,Y\rangle$$

which is the stated equation, completing the proof.

The Ricci equation completes the triple. It measures the normal curvature $R^\perp$ against two other quantities: the ambient curvature paired across a tangent and a normal plane, and the commutator of the shape operators of ξ and η. Two readings are useful. First, the normal bundle is flat, $R^\perp \equiv 0$, exactly when the ambient mixed curvature $\widetilde{\mathrm{Riem}}(X,Y,\xi,\eta)$ equals the commutator term $\langle [S_\xi,S_\eta]X,Y\rangle$ for all arguments; when the ambient is flat this reduces to the statement that $R^\perp$ vanishes precisely when the shape operators of any two normal directions commute. Second, for a hypersurface the normal bundle has rank one, so ξ and η are proportional, their shape operators commute trivially, and $[S_\xi,S_\eta]=0$; the equation then forces $\langle R^\perp(X,Y)\xi,\eta\rangle = \widetilde{\mathrm{Riem}}(X,Y,\xi,\eta)$, and in a flat ambient the rank-one normal bundle is automatically flat, as example 7.2.5 found directly for the sphere. The Ricci equation carries independent content only in higher codimension, where two shape operators need not commute and the normal connection can itself curve.

7.3 Hypersurfaces, Principal Curvatures, and Mean Curvature

The second fundamental form and shape operator of section 7.1 were built for a submanifold of any codimension, where $\mathrm{I\!I}$ takes values in a normal bundle that can have several dimensions and the shape operator depends on a choice of normal vector. When the codimension drops to one, the normal bundle becomes a line, and a choice of unit normal reduces all of this normal-valued data to a single self-adjoint operator on each tangent space. That operator is diagonalizable, its eigenvalues are the principal curvatures, and every extrinsic invariant of the hypersurface is a symmetric function of them: their sum gives the mean curvature, their product the Gauss–Kronecker curvature. This is the classical setting in which the geometry of surfaces was first developed, and it is where the abstract structural equations of section 7.2 become concrete numbers.

The section fixes the hypersurface notation, defines the principal curvatures and the two curvature functions read off from the shape operator, specializes the Gauss equation to codimension one, and singles out the minimal hypersurfaces where the mean curvature vanishes. Two examples carry the computation through. The round sphere reappears with all principal curvatures equal, its Gaussian curvature recovered from the ambient-flat Gauss equation; the cylinder appears with one principal curvature zero, intrinsically flat while extrinsically curved, and it is set aside as the object on which the next section will read off Gauss's Theorema Egregium.

Throughout, $\tilde{M}^{n+1}$ is the ambient (pseudo-)Riemannian manifold with Levi-Civita connection $\tilde{\nabla}$ and curvature $\widetilde{\mathrm{Riem}}$, and $M^n \subseteq \tilde{M}^{n+1}$ is a Riemannian hypersurface: a submanifold of codimension one, carrying the induced metric g of definition 1.5.1, its Levi-Civita connection ∇, and its curvature Riem. At each $p \in M$ the ambient tangent space splits g-orthogonally as $T_p\tilde{M} = T_pM \oplus N_pM$ with N_pM one-dimensional. The hypersurface is assumed *oriented* through a choice of smooth unit normal field N along M, so that $\langle N, N\rangle = 1$ and $N_pM = \mathbb{R}\, N_p$ at every point; the second fundamental form $\mathrm{I\!I}$, the shape operator S_ξ of definition 7.1.6, and the sign conventions are those of section 7.1.

Definition 7.3.1: Principal Curvatures and Principal Directions

Let $M^n \subseteq \tilde{M}^{n+1}$ be an oriented Riemannian hypersurface with unit normal field N. The *shape operator of the hypersurface* is the Weingarten map in the normal direction N,

$$S := S_N, \qquad SX = -(\tilde{\nabla}_X N)^\top$$

a field of g-self-adjoint endomorphisms of TM (proposition 7.1.7). At each $p \in M$ the operator $S_p\colon T_pM \to T_pM$ is self-adjoint with respect to the inner product g_p, so it has n real eigenvalues, counted with multiplicity. These eigenvalues $\kappa_1(p), \ldots, \kappa_n(p)$ are the *principal curvatures* of M at p. A unit eigenvector of S_p is a *principal direction*, and its eigenvalue is the principal curvature in that direction.

The reality of the principal curvatures is forced by the self-adjointness of S, which is itself forced by the symmetry of the second fundamental form (proposition 7.1.5), through the identity $\langle S_\xi X, Y\rangle = \langle \mathrm{I\!I}(X,Y), \xi\rangle$ of definition 7.1.6. Because S_p is g_p-self-adjoint, the spectral theorem in the form valid for a self-adjoint operator on a finite-dimensional real inner-product space applies: there is a g_p-orthonormal basis of T_pM consisting of eigenvectors of S_p, and the eigenvalues are real. The principal directions at p therefore form an orthonormal basis, and along each of them the shape operator acts by multiplication by the corresponding principal curvature. Geometrically, a principal curvature κ_i measures how fast the chosen unit normal N tips over as one moves along the i-th principal direction, and its sign records which way: reversing the orientation $N \mapsto -N$ sends $S \mapsto -S$ and hence $\kappa_i \mapsto -\kappa_i$, so the principal curvatures are extrinsic invariants only up to a global sign fixed by the orientation.

Definition 7.3.2: Mean Curvature, Gauss–Kronecker Curvature, Scalar Second Fundamental Form

Let $M^n \subseteq \tilde{M}^{n+1}$ be an oriented Riemannian hypersurface with unit normal N, shape operator $S = S_N$, and principal curvatures $\kappa_1, \dots, \kappa_n$. The *mean curvature* of M is the normalized trace of the shape operator,

$$H := \frac{1}{n}\mathrm{tr}S = \frac{1}{n}\sum_{i=1}^{n}\kappa_i$$

the *Gauss–Kronecker curvature* is the determinant of the shape operator,

$$\det S = \prod_{i=1}^{n}\kappa_i$$

and the *scalar second fundamental form* is the symmetric 2-tensor

$$h(X, Y) := \langle \mathrm{I\!I}(X, Y), N\rangle = \langle SX, Y\rangle$$

on TM. In terms of h the full second fundamental form is $\mathrm{I\!I}(X, Y) = h(X, Y)\, N$.

The three quantities package the same shape-operator data at three levels of resolution. The scalar second fundamental form h retains everything: it is the bilinear form whose matrix in a g-orthonormal basis is the matrix of S, so its eigenvalues are the principal curvatures, and the two equalities in its definition are the two ways of reading $\langle S_\xi X, Y\rangle = \langle \mathrm{I\!I}(X, Y), \xi\rangle$ at $\xi = N$. The mean curvature and the Gauss–Kronecker curvature are the first and last elementary symmetric functions of the principal curvatures, up to the normalizing factor $1/n$ in the mean; both are independent of the orthonormal basis in which one computes because trace and determinant are. The normalization $H = \frac{1}{n}\mathrm{tr}S$, dividing by the dimension, is chosen so that a totally umbilic hypersurface, one with $S = \kappa\,\mathrm{id}$ and hence $\kappa_1 = \cdots = \kappa_n = \kappa$, has $H = \kappa$: the mean curvature is the average of the principal curvatures, and for the round sphere it will read out as a single number, not n times one. The trace normalization is the source of the name and is the one under which H is the quantity whose vanishing marks a critical point of the area functional; that variational meaning is recorded below and belongs to geometric analysis further downstream. Reversing the orientation sends $H \mapsto -H$ and $\det S \mapsto (-1)^n \det S$, so neither is orientation-independent; their vanishing, however, is.

A worked instance grounds the definitions before the structural equation is specialized. On the round sphere, and again on the cylinder below, the shape operator was computed in section 7.1, and the numbers above are read directly off it.

Example 7.3.3: Principal curvatures of the round sphere

Let $S^n_r \subseteq \mathbb{R}^{n+1}$ be the round sphere of radius r, oriented by the outward unit normal $N = x/r$ at the point x. In example 7.1.8 the shape operator was computed from the flat ambient connection $\tilde{\nabla}_X x = X$ to be

$$S = S_N = -\frac{1}{r}\,\mathrm{id}_{TM}, \qquad \mathrm{I\!I}(X, Y) = -\frac{1}{r}\,\langle X, Y\rangle\; N$$

so S is a multiple of the identity at every point. Every unit tangent vector is therefore a principal direction, and all n principal curvatures are equal:

$$\kappa_1 = \cdots = \kappa_n = -\frac{1}{r}$$

The mean curvature is their average,

$$H = \frac{1}{n}\sum_{i=1}^{n}\Big(-\frac{1}{r}\Big) = -\frac{1}{r}$$

the Gauss–Kronecker curvature is their product,

$$\det S = \prod_{i=1}^{n}\Big(-\frac{1}{r}\Big) = \Big(-\frac{1}{r}\Big)^n = \frac{(-1)^n}{r^n}$$

and the scalar second fundamental form is $h(X,Y) = \langle SX, Y\rangle = -\frac{1}{r}\langle X, Y\rangle$, the induced metric scaled by $-1/r$. The sphere is totally umbilic, with S a scalar operator, and its mean curvature $-1/r$ is a single number as the normalization intends. The sign is that of the outward normal: pointing N inward would flip every κ_i to $+1/r$ and H to $+1/r$, while leaving $\det S$ multiplied by $(-1)^n$.

These principal curvatures recover the intrinsic curvature of the sphere through the hypersurface Gauss equation of the next result. The ambient space $\mathbb{R}^{n+1}$ is flat, and every unit tangent vector is a principal direction with principal curvature $-1/r$, so for any orthonormal tangent pair e_i, e_j formula (7.17) gives

$$K(e_i, e_j) = \kappa_i\kappa_j = \Big(-\frac{1}{r}\Big)\Big(-\frac{1}{r}\Big) = \frac{1}{r^2}$$

The sectional curvature is $+1/r^2$ on every tangent 2-plane, so S^n_r has constant sectional curvature $1/r^2$. This matches the value computed intrinsically in section 4.3: the outward-normal shape operator $-\frac{1}{r}\mathrm{id}$ feeds through the Gauss equation to the positive intrinsic curvature $+1/r^2$. The sign of the individual principal curvatures depended on the choice of normal, but their product did not, and it is the product that the intrinsic curvature sees.

The Gauss equation of theorem 7.2.1 relates the intrinsic curvature of a submanifold to the ambient curvature and the second fundamental form in any codimension. For a hypersurface the second-fundamental-form terms collapse: $\mathrm{I\!I}$ is N-valued, so each inner product $\langle \mathrm{I\!I}(-,-), \mathrm{I\!I}(-,-)\rangle$ becomes a product of two scalar quantities $h(-,-)$, and the equation is expressed entirely through the single tensor h and the shape operator. The specialization is the hypersurface form of the Gauss equation, and in a flat ambient it computes sectional curvatures as products of principal curvatures.

Proposition 7.3.4: The Gauss Equation for a Hypersurface

Let $M^n \subseteq \tilde{M}^{n+1}$ be an oriented Riemannian hypersurface with unit normal N, shape operator S, and scalar second fundamental form $h(X,Y) = \langle SX, Y\rangle$. For all tangent fields X, Y, Z, W on M,

$$\mathrm{Riem}(X,Y,Z,W) = \widetilde{\mathrm{Riem}}(X,Y,Z,W) + h(X,W)h(Y,Z) - h(X,Z)h(Y,W) \tag{7.16}$$

If in addition the ambient manifold is flat, $\widetilde{\mathrm{Riem}} \equiv 0$, and e_i, e_j are orthonormal principal directions with principal curvatures κ_i, κ_j (so $i \neq j$), then the sectional curvature of the plane they span is

$$K(e_i, e_j) = \kappa_i\kappa_j \tag{7.17}$$

***Proof*:**
Step 1: reduce the second fundamental form to the scalar form. For a hypersurface the normal bundle is spanned by the unit field N, so by definition 7.1.4 and the definition of h in definition 7.3.2,

$$\mathrm{I\!I}(X,Y) = h(X,Y)\,N$$

Since $\langle N,N\rangle = 1$, the pairing of any two values of $\mathrm{I\!I}$ is

$$\langle \mathrm{I\!I}(A,B), \mathrm{I\!I}(C,D)\rangle = h(A,B)h(C,D)\,\langle N,N\rangle = h(A,B)h(C,D)$$

for tangent fields A,B,C,D.

Step 2: specialize the Gauss equation. The Gauss equation theorem 7.2.1 in the binding form reads

$$\mathrm{Riem}(X,Y,Z,W) = \widetilde{\mathrm{Riem}}(X,Y,Z,W) + \langle \mathrm{I\!I}(X,W), \mathrm{I\!I}(Y,Z)\rangle - \langle \mathrm{I\!I}(X,Z), \mathrm{I\!I}(Y,W)\rangle$$

Substituting the Step 1 identity into each of the two inner products gives

$$\langle \mathrm{I\!I}(X,W), \mathrm{I\!I}(Y,Z)\rangle = h(X,W)h(Y,Z), \qquad \langle \mathrm{I\!I}(X,Z), \mathrm{I\!I}(Y,W)\rangle = h(X,Z)h(Y,W)$$

and hence (7.16).

Step 3: read off sectional curvatures in a flat ambient. Assume $\widetilde{\mathrm{Riem}} \equiv 0$, and let e_i, e_j be orthonormal principal directions, $Se_i = \kappa_i e_i$ and $Se_j = \kappa_j e_j$, with $i \neq j$. Because these vectors are orthonormal, the denominator in the sectional-curvature formula of definition 4.3.1 is

$$|e_i|^2\,|e_j|^2 - \langle e_i, e_j\rangle^2 = 1\cdot 1 - 0 = 1$$

so $K(e_i,e_j) = \mathrm{Riem}(e_i,e_j,e_j,e_i)$. The scalar second fundamental form on principal directions is diagonal:

$$h(e_i,e_j) = \langle Se_i, e_j\rangle = \kappa_i\,\langle e_i,e_j\rangle = \kappa_i\,\delta_{ij}$$

so $h(e_i,e_i) = \kappa_i$, $h(e_j,e_j) = \kappa_j$, and $h(e_i,e_j) = 0$ for $i \neq j$. Applying (7.16) with $X = e_i$, $Y = e_j$, $Z = e_j$, $W = e_i$ and $\widetilde{\mathrm{Riem}} \equiv 0$,

$$K(e_i,e_j) = \mathrm{Riem}(e_i,e_j,e_j,e_i) = h(e_i,e_i)h(e_j,e_j) - h(e_i,e_j)h(e_j,e_i) = \kappa_i\kappa_j - 0 = \kappa_i\kappa_j$$

which is (7.17), completing the proof.

The hypersurface Gauss equation is the mechanism that turns extrinsic bending into intrinsic curvature. The left side, Riem, is computed inside M from the induced metric alone, with no reference to the ambient space; the right side is built from ambient data, the ambient curvature and the shape operator. Their equality says that the intrinsic curvature is determined by the ambient curvature and the second fundamental form together, and (7.17) isolates the sharpest case: when the ambient space is flat, the sectional curvature of a coordinate plane spanned by principal directions is simply the product of the two principal curvatures in those directions. For a surface $M^2 \subseteq \mathbb{R}^3$ this reads $K = \kappa_1\kappa_2$, the product of the two principal curvatures, which is the classical Gaussian curvature; the observation that this extrinsically defined product is in fact intrinsic is Gauss's Theorema Egregium, proved in section 7.4. The examples of this section supply its two poles, the sphere with $\kappa_1\kappa_2 > 0$ (worked in example 7.3.3) and the cylinder with $\kappa_1\kappa_2 = 0$.

Among all hypersurfaces the ones with vanishing mean curvature form a class of their own, distinguished by a variational property that will not be developed here but is the reason for their name.

Definition 7.3.5: Minimal Hypersurface

An oriented Riemannian hypersurface $M^n \subseteq \tilde{M}^{n+1}$ is *minimal* if its mean curvature vanishes identically,

$$H \equiv 0, \qquad \text{equivalently} \qquad \operatorname{tr} S \equiv 0, \qquad \text{equivalently} \qquad \sum_{i=1}^{n} \kappa_i \equiv 0$$

at every point.

The condition $H \equiv 0$ does not require the individual principal curvatures to vanish; it requires only that they cancel, summing to zero at each point. A minimal hypersurface therefore bends, but with its bending balanced across its principal directions, positive principal curvatures offset by negative ones. The term *minimal* records that these are exactly the hypersurfaces that are critical points of the area functional: a compactly supported normal deformation changes the area of M at first order by a multiple of $\int_M H$ against the deformation, so the area is stationary under every such deformation precisely when $H \equiv 0$. That first-variation-of-area computation, and the sense in which minimal hypersurfaces model soap films spanning a fixed boundary, belong to geometric analysis and are taken up downstream; the definition is recorded here because it names the vanishing of the invariant just introduced. The n principal curvatures of the sphere are all $-1/r \neq 0$ and sum to $-n/r \neq 0$, so no round sphere is minimal; the cylinder below, with principal curvatures -1 and 0, has $H = -\frac{1}{2} \neq 0$ and is not minimal either, though it comes closer to the balance the condition demands.

The second example is the one the next section is built on. Where the sphere had all principal curvatures nonzero and equal, the cylinder has one of them zero, and the contrast between its vanishing Gaussian curvature and its nonzero mean curvature is exactly the phenomenon Theorema Egregium explains.

Example 7.3.6: The flat cylinder

Let $M = S^1 \times \mathbb{R} \subseteq \mathbb{R}^3$ be the unit cylinder, the set of points $(\cos\theta, \sin\theta, z)$ with $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ and $z \in \mathbb{R}$, oriented by the outward unit normal $N = (\cos\theta, \sin\theta, 0)$, which is the horizontal radial direction and satisfies $\langle N, N\rangle = 1$ with N tangent to $\mathbb{R}^3$ but normal to M. At the point $x = (\cos\theta, \sin\theta, z)$ the tangent space T_xM is spanned by the two orthonormal vectors

$$e_1 = (-\sin\theta, \cos\theta, 0), \qquad e_2 = (0, 0, 1)$$

the first running around the circle factor and the second along the axis. These are the principal directions, as the following computation shows.

Step 1: compute the shape operator. The ambient connection on $\mathbb{R}^3$ is the flat directional derivative, $\tilde{\nabla}_X Y = X(Y)$ acting componentwise. To apply $SX = -(\tilde{\nabla}_X N)^\top$, extend N to the normal field $N(\theta, z) = (\cos\theta, \sin\theta, 0)$ along M. Differentiating along e_1, which is ∂_θ normalized (here $|\partial_\theta| = 1$ since the circle has unit radius),

$$\tilde{\nabla}_{e_1} N = \frac{d}{d\theta}(\cos\theta, \sin\theta, 0) = (-\sin\theta, \cos\theta, 0) = e_1$$

This vector is tangent to M, so $(\tilde{\nabla}_{e_1} N)^\top = e_1$ and

$$Se_1 = -(\tilde{\nabla}_{e_1} N)^\top = -e_1$$

Differentiating along $e_2 = \partial_z$, the normal N does not depend on z, so

$$\tilde{\nabla}_{e_2} N = \frac{\partial}{\partial z}(\cos\theta, \sin\theta, 0) = 0, \qquad Se_2 = -(\tilde{\nabla}_{e_2} N)^\top = 0$$

Thus S is diagonal in the basis (e_1, e_2) with

$$Se_1 = -e_1, \qquad Se_2 = 0$$

Step 2: read off the invariants. The principal directions are e_1 (around the circle) and e_2 (along the axis), with principal curvatures

$$\kappa_1 = -1, \qquad \kappa_2 = 0$$

The circle direction bends, tracking the unit circle of radius 1; the axis direction is a straight line in $\mathbb{R}^3$ and does not bend at all. The mean curvature is their average,

$$H = \frac{1}{2}(\kappa_1 + \kappa_2) = \frac{1}{2}(-1 + 0) = -\frac{1}{2} \neq 0$$

so the cylinder is not minimal, and the Gauss–Kronecker curvature is their product,

$$\det S = \kappa_1 \kappa_2 = (-1) \cdot 0 = 0$$

Step 3: the intrinsic curvature is zero. The ambient space $\mathbb{R}^3$ is flat, so (7.17) applies to the orthonormal principal directions e_1, e_2:

$$K(e_1, e_2) = \kappa_1 \kappa_2 = (-1) \cdot 0 = 0$$

The cylinder has a single tangent 2-plane at each point, spanned by e_1 and e_2, so its sectional curvature vanishes identically and M is intrinsically flat, $K \equiv 0$. This is visible without the Gauss equation as well: the map $(\theta, z) \mapsto (\cos\theta, \sin\theta, z)$ pulls the Euclidean metric of $\mathbb{R}^3$ back to $d\theta^2 + dz^2$ on M, the flat metric of example 1.1.2, so (M, g) is locally isometric to the Euclidean plane and $K \equiv 0$ by section 4.3. The two computations agree.

The cylinder is the pointed contrast that motivates the next section. Extrinsically it is curved, its mean curvature $H = -\frac{1}{2}$ is nonzero, and the circle direction bends with principal curvature -1; intrinsically it is flat, indistinguishable by any measurement within its own metric from the Euclidean plane, since the plane rolls onto it without stretching. What survives the rolling is the Gaussian curvature $\det S = \kappa_1 \kappa_2 = 0$, which vanishes on both the cylinder and the plane, while the mean curvature H, nonzero on the cylinder and zero on the plane, does not survive: it is extrinsic through and through. The Gauss equation already forced this outcome, since (7.17) computes the intrinsic K as the product $\kappa_1 \kappa_2$ and one factor is zero. That the product, alone among the symmetric functions of the principal curvatures, is intrinsic is the content of Gauss's Theorema Egregium.

7.4 Gauss's Theorema Egregium; Totally Geodesic Submanifolds

The Gauss equation of section 7.2 relates three curvatures at once: the intrinsic curvature of a submanifold M, the curvature of the ambient manifold $\tilde{M}$, and the second fundamental form II, which measures how M bends inside $\tilde{M}$. For a surface in Euclidean space the ambient curvature is zero, and the equation degenerates into a single scalar identity. The extrinsic product of principal curvatures, $\det S$, is on its face an artifact of the embedding, computed from how the surface curls away from its tangent plane in the surrounding space. Gauss's discovery is that this product is not an artifact at all: it equals the intrinsic sectional curvature of the induced metric, a quantity a two-dimensional inhabitant of the surface could measure without ever leaving it. A being confined to the surface, with access only to lengths and angles along it, can compute a number that a distant observer would read off from the surface's shape in space. This is the Theorema Egregium, and it is where the two-page apparatus of the Gauss equation pays for itself.

The second theme of the section is the opposite extreme of bending. A submanifold whose second fundamental form vanishes identically is *totally geodesic*: it sits inside the ambient manifold as flatly as possible, curving only insofar as the ambient forces it to. These are the submanifolds on which the induced and ambient geometries agree most completely, and they are characterized by the property that suggests the name, that geodesics of the submanifold are geodesics of the ambient space. The model spaces of section 1.3 supply the guiding examples: linear subspaces of Euclidean space, great subspheres of a sphere, and totally geodesic copies of hyperbolic space inside hyperbolic space. A single mechanism, the fixed-point set of an isometry, produces all of them at once.

Throughout, $(\tilde{M}, \tilde{g})$ is the ambient manifold with Levi-Civita connection $\tilde{\nabla}$ and curvature $\widetilde{\mathrm{Riem}}$, and $M \subseteq \tilde{M}$ is a Riemannian submanifold with the induced metric g (definition 1.5.1), Levi-Civita connection ∇, and curvature Riem; the second fundamental form II is that of definition 7.1.4, and the curvature sign convention is the one fixed in section 4.1.

Theorem 7.4.1: Gauss's Theorema Egregium

Let $M^2 \subseteq \mathbb{R}^3$ be a surface with the induced metric, unit normal N, and shape operator $S = S_N$ (definition 7.3.1), with principal curvatures κ_1, κ_2. Then the Gauss–Kronecker curvature

$$\det S = \kappa_1 \kappa_2$$

equals the sectional curvature K of the induced metric g at each point of M. Consequently $\det S$ depends only on the induced metric g, and not on the embedding of M into $\mathbb{R}^3$: it is invariant under local isometries. In particular, if $\varphi\colon U \to V$ is an isometry between open subsets of two surfaces in $\mathbb{R}^3$, then the Gaussian curvatures at p and $\varphi(p)$ agree.

The claim splits into two assertions of unequal difficulty. The first, that the extrinsic $\det S$ equals the intrinsic K, is a one-line specialization of the hypersurface Gauss equation once the ambient curvature is set to zero. The second, that $\det S$ is therefore an invariant of the induced metric alone, is a consequence of the first together with the fact that sectional curvature is built from g through the Levi-Civita connection and the curvature tensor. It is the second assertion that carries the weight: $\det S$ is defined by data external to the surface, the eigenvalues of an operator recording how the surface tilts in $\mathbb{R}^3$, yet the theorem locates it entirely within the surface's own metric geometry.

***Proof*:**
Step 1: the extrinsic curvature equals the intrinsic curvature. The surface M^2 is a hypersurface in $\mathbb{R}^3$, and $\mathbb{R}^3$ with its Euclidean metric is flat, so $\widetilde{\mathrm{Riem}} \equiv 0$ (example 4.1.6). Let $p \in M$ and let $\{e_1, e_2\}$ be an orthonormal basis of T_pM, which is the unique tangent 2-plane of the surface. The hypersurface Gauss equation (proposition 7.3.4) reads

$$\mathrm{Riem}(X, Y, Z, W) = \widetilde{\mathrm{Riem}}(X, Y, Z, W) + h(X, W)\, h(Y, Z) - h(X, Z)\, h(Y, W)$$

where $h(X, Y) = \langle SX, Y \rangle$ is the scalar second fundamental form. Applying it to $X = W = e_1$, $Y = Z = e_2$ and using $\widetilde{\mathrm{Riem}} = 0$ gives

$$\mathrm{Riem}(e_1, e_2, e_2, e_1) = h(e_1, e_1)\, h(e_2, e_2) - h(e_1, e_2)^2 \tag{7.18}$$

The right-hand side is the determinant of the symmetric matrix $\big(h(e_i, e_j)\big)$ in the orthonormal basis $\{e_1, e_2\}$. Since $h(e_i, e_j) = \langle Se_i, e_j \rangle$ is the matrix of the self-adjoint operator S in that orthonormal basis (proposition 7.1.7), its determinant is the determinant of S:

$$h(e_1, e_1)\, h(e_2, e_2) - h(e_1, e_2)^2 = \det S = \kappa_1 \kappa_2$$

the last equality because $\det S$ is the product of the eigenvalues of S, which are the principal curvatures. On the left-hand side of (7.18), the tangent plane $\Pi = T_pM$ is spanned by the orthonormal pair $\{e_1, e_2\}$, so its sectional curvature (definition 4.3.1) is

$$K = \frac{\mathrm{Riem}(e_1, e_2, e_2, e_1)}{|e_1|^2\, |e_2|^2 - \langle e_1, e_2 \rangle^2} = \mathrm{Riem}(e_1, e_2, e_2, e_1)$$

the denominator being 1 for an orthonormal pair. Combining the two computations with (7.18) yields

$$K = \det S = \kappa_1 \kappa_2 \tag{7.19}$$

which is the first assertion.

Step 2: the extrinsic curvature is an intrinsic invariant. The sectional curvature K on the left of (7.19) is defined from the induced metric g alone: g determines its Levi-Civita connection ∇ by the uniqueness in theorem 2.3.1, ∇ determines the curvature endomorphism R and hence Riem by definition 4.1.2, and K is the ratio built from Riem in definition 4.3.1. None of these constructions refers to the ambient $\mathbb{R}^3$ or to the normal N. Therefore K is a function of g, and by (7.19) so is $\det S$, even though $\det S$ was defined through the shape operator, an object visible only from the ambient space.

Invariance under local isometries follows. Let $\varphi \colon U \to V$ be an isometry between open subsets of two surfaces in $\mathbb{R}^3$, so $\varphi^* g_V = g_U$ (definition 1.3.1). An isometry pulls back the Levi-Civita connection to the Levi-Civita connection and the curvature tensor to the curvature tensor, since both are determined by the metric; concretely, for tangent vectors u, v, w, x at $p \in U$,

$$\mathrm{Riem}^U(u, v, w, x) = \mathrm{Riem}^V\big(d\varphi_p u,\, d\varphi_p v,\, d\varphi_p w,\, d\varphi_p x\big)$$

and $d\varphi_p$ carries a g_U-orthonormal basis of T_pU to a g_V-orthonormal basis of $T_{\varphi(p)}V$. Hence the sectional curvatures satisfy $K_U(p) = K_V(\varphi(p))$, and by (7.19) the Gaussian curvatures agree:

$$\det S_U(p) = K_U(p) = K_V(\varphi(p)) = \det S_V(\varphi(p))$$

This establishes the invariance and completes the proof.

The name Gauss gave the theorem, *egregium* ("outstanding"), records his own estimate of it. The content is that curvature, which one would expect to be a feature of how a surface is placed in space, is instead a feature of the surface's internal metric. Bending a sheet of paper does not change the lengths of curves drawn on it, so it does not change the induced metric, so by the theorem it cannot change the Gaussian curvature. Paper has $K \equiv 0$ in its flat state and must keep $K \equiv 0$ however it is rolled or folded without stretching: this is why a flat sheet rolls into a cylinder or a cone but never wraps onto a sphere, and why a bent sheet always has at least one principal curvature equal to zero. The mean curvature $H = \frac{1}{2}(\kappa_1 + \kappa_2)$ enjoys no such invariance, and the contrast is exactly the cylinder of example 7.3.6: its principal curvatures are -1 and 0, so $\det S = 0$ and it is intrinsically flat, isometric in patches to the plane, while $H = -\frac{1}{2} \neq 0$ registers the extrinsic bend that the intrinsic geometry cannot detect.

Example 7.4.2: The sphere admits no flat chart

The round sphere $S^2_r \subseteq \mathbb{R}^3$ has, by the sphere computation of example 7.3.3, both principal curvatures equal to $-1/r$ with respect to the outward normal, so its Gaussian curvature is

$$\det S = \left(-\frac{1}{r}\right)\left(-\frac{1}{r}\right) = \frac{1}{r^2}$$

consistent with the intrinsic value $K \equiv 1/r^2$ from example 4.3.4, exactly as theorem 7.4.1 demands. The plane $\mathbb{R}^2$ has $K \equiv 0$. If some open piece of S^2_r were isometric to an open piece of the plane, the Theorema Egregium would force $1/r^2 = 0$, which is impossible for $r < \infty$. Hence no neighborhood of any point of the sphere is isometric to a region of the plane: there is no distance-preserving planar chart, however small the patch.

This is the mathematical content behind the impossibility of a perfect flat map of the earth. Any chart of a spherical region onto the plane must distort lengths, because a length-preserving one would carry $K = 1/r^2$ to $K = 0$. Every cartographic projection trades one kind of distortion for another for this reason, and the trade cannot be avoided: the obstruction is the curvature, an invariant of the sphere's metric that no choice of projection can remove.

The Theorema Egregium is the statement that a submanifold can be forced to curve. The remainder of the section concerns the opposite: submanifolds that curve as little as the ambient geometry permits, those with $\mathrm{I\!I} \equiv 0$.

Definition 7.4.3: Totally geodesic submanifold

A Riemannian submanifold $M \subseteq \tilde{M}$ is *totally geodesic* if its second fundamental form vanishes identically, $\mathrm{I\!I} \equiv 0$; equivalently (definition 7.1.6), if the shape operator $S_\xi = 0$ for every normal field ξ.

The condition $\mathrm{I\!I} \equiv 0$ says that the ambient covariant derivative $\tilde{\nabla}_X Y$ of two tangent fields is again tangent, with no normal component: by the Gauss formula $\tilde{\nabla}_X Y = \nabla_X Y + \mathrm{I\!I}(X, Y)$, vanishing $\mathrm{I\!I}$ means $\tilde{\nabla}_X Y = \nabla_X Y \in TM$. Differentiating along M never pushes a tangent vector out of TM. A totally geodesic submanifold is thus one whose tangent distribution is preserved by ambient parallel transport in tangential directions, the flattest way a submanifold can sit inside its ambient. The name is justified by the next proposition, which shows the condition is the same as asking that the submanifold's geodesics be geodesics of the ambient space.

Proposition 7.4.4: Characterizations of totally geodesic submanifolds

Let $M \subseteq \tilde{M}$ be a connected Riemannian submanifold. The following are equivalent.

1. M is totally geodesic, $\mathrm{I\!I} \equiv 0$.
2. Every geodesic of (M, g) is a geodesic of $(\tilde{M}, \tilde{g})$.
3. For each $p \in M$ and each $v \in T_pM$, the ambient geodesic $\tilde{\gamma}$ of $\tilde{M}$ with $\tilde{\gamma}(0) = p$ and $\dot{\tilde{\gamma}}(0) = v$ satisfies $\tilde{\gamma}(t) \in M$ for all t in some interval $(-\varepsilon, \varepsilon)$.

The equivalence rests on one identity. For a curve γ lying in M, the Gauss formula relates the ambient acceleration $\tilde{D}_t\dot{\gamma}$ (covariant derivative of $\dot{\gamma}$ along γ in $\tilde{M}$) to the intrinsic acceleration $D_t\dot{\gamma}$ (in M) by a normal correction that is exactly $\mathrm{I\!I}$ evaluated on the velocity. A curve is a geodesic of M when its intrinsic acceleration vanishes and a geodesic of $\tilde{M}$ when its ambient acceleration vanishes; the two conditions differ by the normal term $\mathrm{I\!I}(\dot{\gamma}, \dot{\gamma})$, and controlling that term is the whole content.

***Proof*:**
The proof establishes the identity

$$\tilde{D}_t\dot{\gamma} = D_t\dot{\gamma} + \mathrm{I\!I}(\dot{\gamma}, \dot{\gamma}) \tag{7.20}$$

for any smooth curve γ in M, and then reads off the three implications from it. To derive (7.20), extend $\dot{\gamma}$ locally to a tangent vector field Y on M with $Y \circ \gamma = \dot{\gamma}$; the Gauss formula $\tilde{\nabla}_X Y = \nabla_X Y + \mathrm{I\!I}(X, Y)$ applied along γ with $X = \dot{\gamma}$ gives, since the induced covariant derivatives along the curve are the restrictions of $\tilde{\nabla}$ and ∇ to γ,

$$\tilde{D}_t\dot{\gamma} = \tilde{\nabla}_{\dot{\gamma}} Y = \nabla_{\dot{\gamma}} Y + \mathrm{I\!I}(\dot{\gamma}, \dot{\gamma}) = D_t\dot{\gamma} + \mathrm{I\!I}(\dot{\gamma}, \dot{\gamma})$$

The tangential part $\nabla_{\dot{\gamma}} Y = D_t\dot{\gamma}$ is the acceleration of γ in M, and the normal part $\mathrm{I\!I}(\dot{\gamma}, \dot{\gamma})$ lies in NM; (7.20) is thus the decomposition of the ambient acceleration into its tangential and normal components, valid for every curve in M.

(i) $\Rightarrow$ (ii): Suppose $\mathrm{I\!I} \equiv 0$ and let γ be a geodesic of M, so $D_t\dot{\gamma} = 0$. By (7.20), $\tilde{D}_t\dot{\gamma} = D_t\dot{\gamma} + \mathrm{I\!I}(\dot{\gamma}, \dot{\gamma}) = 0 + 0 = 0$, so γ is a geodesic of $\tilde{M}$ as well.

(ii) $\Rightarrow$ (i): Suppose every geodesic of M is a geodesic of $\tilde{M}$. Fix $p \in M$ and a unit vector $v \in T_pM$, and let γ be the geodesic of M with $\gamma(0) = p$, $\dot{\gamma}(0) = v$, which exists by theorem 3.1.3. Then $D_t\dot{\gamma} = 0$, and by hypothesis $\tilde{D}_t\dot{\gamma} = 0$; (7.20) forces $\mathrm{I\!I}(\dot{\gamma}, \dot{\gamma}) = 0$ at every point of γ, in particular $\mathrm{I\!I}_p(v, v) = 0$. Since $v \in T_pM$ was an arbitrary unit vector and $\mathrm{I\!I}_p$ is 2-homogeneous, $\mathrm{I\!I}_p(w, w) = 0$ for every $w \in T_pM$. The form $\mathrm{I\!I}_p$ is symmetric (proposition 7.1.5), so polarization recovers it from its diagonal values:

$$\mathrm{I\!I}_p(u, w) = \tfrac{1}{2}\big(\mathrm{I\!I}_p(u + w, u + w) - \mathrm{I\!I}_p(u, u) - \mathrm{I\!I}_p(w, w)\big) = 0$$

for all $u, w \in T_pM$. As $p \in M$ was arbitrary, $\mathrm{I\!I} \equiv 0$, so M is totally geodesic.

(ii) $\Rightarrow$ (iii): Assume (ii), and let $p \in M$, $v \in T_pM$. Let γ be the geodesic of M with $\gamma(0) = p$, $\dot{\gamma}(0) = v$, defined on some $(-\varepsilon, \varepsilon)$ and taking values in M by construction. By (ii), γ is also a geodesic of $\tilde{M}$ with the same initial data (p, v). The ambient geodesic $\tilde{\gamma}$ with $\tilde{\gamma}(0) = p$, $\dot{\tilde{\gamma}}(0) = v$ is unique with these initial conditions (theorem 3.1.3), so $\tilde{\gamma} = \gamma$ on $(-\varepsilon, \varepsilon)$. Hence $\tilde{\gamma}(t) = \gamma(t) \in M$ for $|t| < \varepsilon$, which is (iii).

(iii) ⇒ (ii): Assume (iii), and let γ be a geodesic of M with $\gamma(0) = p$, $\dot\gamma(0) = v$. Let $\tilde\gamma$ be the ambient geodesic with the same initial data. By (iii), $\tilde\gamma(t) \in M$ for $|t| < \varepsilon$, so $\tilde\gamma|_{(-\varepsilon,\varepsilon)}$ is a curve in M; being a geodesic of $\tilde M$, its ambient acceleration vanishes, and (7.20) then reads $0 = D_t\dot{\tilde\gamma} + \mathrm{II}(\dot{\tilde\gamma}, \dot{\tilde\gamma})$. Taking tangential parts gives $D_t\dot{\tilde\gamma} = 0$, so $\tilde\gamma$ is a geodesic of M through p with velocity v; by uniqueness of geodesics in M (theorem 3.1.3) it coincides with γ near 0. Thus $\gamma = \tilde\gamma$ is a geodesic of $\tilde M$ on $(-\varepsilon, \varepsilon)$. A geodesic of M that is a geodesic of $\tilde M$ on every subinterval is a geodesic of $\tilde M$ throughout its domain, since the geodesic equations are local; hence every geodesic of M is a geodesic of $\tilde M$, which is (ii).

The implications (i) ⇒ (ii) ⇒ (i) and (ii) ⇒ (iii) ⇒ (ii) together give the equivalence of all three conditions, as we sought to show.

The proposition explains the terminology and gives the working definition. A totally geodesic submanifold is one inside which geodesics need no correction to remain geodesics of the ambient space: a shortest path along M is, at least locally, a shortest path in $\tilde M$ restricted to M. Characterization (iii) is the operational test. To find the totally geodesic submanifolds through a point p with a prescribed tangent space $W \subseteq T_p\tilde M$, one shoots ambient geodesics in every direction of W and checks whether they sweep out a submanifold; when they do, that submanifold is totally geodesic. The next proposition turns this into a construction that produces totally geodesic submanifolds in abundance, from the symmetries of the ambient space.

Proposition 7.4.5: Fixed-point sets of isometries are totally geodesic

Let $\varphi \in \mathrm{Isom}(\tilde M, \tilde g)$ be an isometry (definition 1.3.3) and let $F = \mathrm{Fix}(\varphi) = \{q \in \tilde M \mid \varphi(q) = q\}$ be its fixed-point set. Then each connected component of F is a totally geodesic submanifold of $\tilde M$, and its tangent space at a point $p \in F$ is the $+1$-eigenspace

$$T_pF = \{v \in T_p\tilde M \mid d\varphi_p(v) = v\}$$

of the differential $d\varphi_p$.

An isometry moves the ambient manifold rigidly; its fixed points are the locus it leaves in place. The proposition says this locus is not just a closed subset but a submanifold, and one of the flattest kind. The mechanism is geodesic uniqueness: an isometry sends geodesics to geodesics with the transported initial data, so a geodesic launched from a fixed point in a fixed direction is mapped to itself, hence stays fixed pointwise, hence stays in F. Characterization (iii) of proposition 7.4.4 then applies verbatim.

***Proof*:**

Step 1: F is a submanifold with the stated tangent space. Fix $p \in F$, so $\varphi(p) = p$ and $d\varphi_p\colon T_p\tilde M \to T_p\tilde M$ is a linear isometry of the inner product space $(T_p\tilde M, \tilde g_p)$. Let $W = \ker(d\varphi_p - \mathrm{id})$ be its $+1$-eigenspace. The exponential map $\exp_p$ of $\tilde M$ is a diffeomorphism from a ball $B \subseteq T_p\tilde M$ onto a normal neighborhood U of p (proposition 3.2.5). Because φ is an isometry fixing p, it commutes with the exponential map: for $v \in B$,

$$\varphi\big(\exp_p(v)\big) = \exp_p\big(d\varphi_p(v)\big) \tag{7.21}$$

This holds because $t \mapsto \varphi(\exp_p(tv))$ and $t \mapsto \exp_p(t\, d\varphi_p(v))$ are both geodesics of $\tilde M$ (an isometry carries geodesics to geodesics, as φ preserves $\tilde\nabla$) with the same initial point p and the same initial velocity $d\varphi_p(v)$, so they coincide by theorem 3.1.3; evaluating at $t = 1$ gives (7.21). Now a point $q = \exp_p(v) \in U$ is fixed by φ if and only if $\exp_p(d\varphi_p(v)) = \exp_p(v)$, and since $\exp_p$ is injective on B this holds if and only if $d\varphi_p(v) = v$, that is, $v \in W$. Therefore

$$F \cap U = \exp_p(B \cap W)$$

which is the image under the diffeomorphism $\exp_p$ of the open subset $B \cap W$ of the linear subspace W. Hence $F \cap U$ is an embedded submanifold of U of dimension $\dim W$, and its tangent space at p is $d(\exp_p)_0(W) = W$ (using $d(\exp_p)_0 = \mathrm{id}$ from proposition 3.2.5). As $p \in F$ was arbitrary, every component of F is a submanifold with $T_pF = W = \ker(d\varphi_p - \mathrm{id})$.

Step 2: F is totally geodesic. Let $p \in F$ and $v \in T_pF = W$, so $d\varphi_p(v) = v$. Let $\tilde\gamma$ be the ambient geodesic with $\tilde\gamma(0) = p$, $\dot{\tilde\gamma}(0) = v$. The image curve $\varphi \circ \tilde\gamma$ is again a geodesic of $\tilde M$, because φ preserves the Levi-Civita connection, and its initial data are

$$(\varphi \circ \tilde\gamma)(0) = \varphi(p) = p, \qquad \frac{d}{dt}\Big|_0 (\varphi \circ \tilde\gamma) = d\varphi_p\big(\dot{\tilde\gamma}(0)\big) = d\varphi_p(v) = v$$

Thus $\varphi \circ \tilde\gamma$ and $\tilde\gamma$ are geodesics with identical initial point and velocity, so by uniqueness (theorem 3.1.3), $\varphi \circ \tilde\gamma = \tilde\gamma$. This says $\varphi(\tilde\gamma(t)) = \tilde\gamma(t)$ for every t in the domain, so $\tilde\gamma(t) \in F$. Restricting to the component of F containing p, the ambient geodesic $\tilde\gamma$ with tangent $v \in T_pF$ stays in F for a short time, which is condition (iii) of proposition 7.4.4 for the submanifold F. By that proposition, F is totally geodesic, completing the proof.

The construction is generative: every isometry of the ambient manifold contributes its fixed-point set to the stock of totally geodesic submanifolds, and the symmetric model spaces have isometries in abundance. The prototype is a reflection. A reflection is an involutive isometry, $\varphi^2 = \mathrm{id}$, and its fixed-point set is the locus it flips across, a hyperplane of symmetry; the proposition guarantees that this hyperplane sits totally geodesically in the ambient space. The example below runs this argument through the three space forms and recovers the expected list of flat subspaces, great subspheres, and hyperbolic subspaces.

Example 7.4.6: Totally geodesic submanifolds of the model spaces

The three constant-curvature models of section 1.3 each have a family of totally geodesic submanifolds, one for each dimension, and in every case they are cut out as fixed-point sets of reflections.

1. *Euclidean space.* In $(\mathbb{R}^n, \bar g)$ the geodesics are the affine lines $t \mapsto p + tv$. An affine subspace $P = p_0 + V$, with $V \subseteq \mathbb{R}^n$ a linear subspace, contains the whole line through any of its points in any of its own directions, so by proposition 7.4.4(iii) it is totally geodesic. Conversely these are the only connected totally geodesic submanifolds: a totally geodesic submanifold through p_0 with tangent space V must, by (iii), contain the ambient geodesic (the affine line) in each direction of V, hence contains $p_0 + V$, and equals it near p_0. Each P arises as a fixed-point set: for a k-dimensional $P = p_0 + V$, the orthogonal reflection $\varphi(x) = p_0 + 2\,\mathrm{proj}_V(x - p_0) - (x - p_0)$ across P is a Euclidean isometry whose fixed set is exactly P, and $d\varphi_{p_0}$ is the reflection of $\mathbb{R}^n$ fixing V, whose $+1$-eigenspace is $V = T_{p_0}P$, matching proposition 7.4.5.
2. *The sphere.* In $S^n = S^n(1) \subseteq \mathbb{R}^{n+1}$ (example 1.3.4) a *great subsphere* is the intersection $S^k = S^n \cap V$ with a linear subspace $V \subseteq \mathbb{R}^{n+1}$ of dimension $k + 1$; it is a round k-sphere. Let $\rho_V \colon \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be the orthogonal reflection fixing V and negating $V^\perp$. As a linear orthogonal map ρ_V preserves the round metric of S^n (it preserves the ambient inner

product and the unit sphere), so $\rho_V \in \mathrm{Isom}(S^n)$, and its fixed-point set on the sphere is $\{x \in S^n \mid \rho_V(x) = x\} = S^n \cap V = S^k$. By proposition 7.4.5, S^k is totally geodesic in S^n, with tangent space at p the $+1$-eigenspace $T_pS^n \cap V$. In particular, for $k = 1$ this recovers that the great circles $S^n \cap V$ ($\dim V = 2$) are the geodesics of the sphere, and for $k = n-1$ that the equatorial $(n-1)$-spheres are totally geodesic hypersurfaces. A latitude circle $\{x_{n+1} = c\}$ with $c \neq 0$ is, by contrast, not totally geodesic: it is not a great circle, so its ambient acceleration has a nonzero component, and indeed theorem 7.4.1 and the sphere's constant K leave the great subspheres as the only totally geodesic submanifolds.

3. *Hyperbolic space.* In the hyperboloid model $H^n \subseteq \mathbb{R}^{1,n}$ of example 1.3.5, with the Minkowski form $\langle -, - \rangle_{1,n}$, a totally geodesic $H^k \subseteq H^n$ is the intersection $H^n \cap V$ with a linear subspace $V \subseteq \mathbb{R}^{1,n}$ of dimension $k+1$ on which the Minkowski form has signature $(1, k)$ (a "timelike" subspace, so that $H^n \cap V$ is a k-dimensional hyperboloid, itself a copy of H^k). The Minkowski-orthogonal reflection ρ_V fixing V and negating the $\langle -, - \rangle_{1,n}$-orthogonal complement $V^\perp$ preserves the Minkowski form, hence lies in $\mathrm{Isom}(H^n)$ (the subgroup $O^+(1,n)$ of $O(1,n)$ preserving the upper sheet, to which ρ_V belongs since the timelike V meets that sheet), and its fixed-point set on H^n is $H^n \cap V = H^k$. By proposition 7.4.5 this H^k is totally geodesic. For $k = 1$ these are the geodesics of hyperbolic space, the intersections of H^n with timelike 2-planes through the origin; for general k they are the totally geodesic hyperbolic subspaces, appearing in the upper-half-space model of example 1.3.5 as the vertical half-planes and the hemispheres centered on the boundary.

The three cases share one pattern: a totally geodesic submanifold of dimension k is the trace on the model of a linear subspace of the surrounding coordinate space, and it is pinned down as the fixed set of the reflection across that subspace. The reflection supplies the isometry, proposition 7.4.5 supplies the total geodesy, and the ambient linear structure supplies the tangent space.

7.5 Riemannian Submersions

An isometric immersion presents a manifold as a submanifold sitting inside a larger one, and the geometry of the preceding sections measured how it bends there. This section takes up the opposite arrow. A submersion presents its base as a quotient of a larger total space: points of the base are whole fibers of the map, and a metric on the base is recovered from a metric above by remembering only the directions transverse to the fibers. When the transverse directions carry the metric faithfully, the map is a Riemannian submersion, and the relationship between the curvature of the total space and the curvature of the base is the submersion counterpart of the Gauss equation. Where the Gauss equation of theorem 7.2.1 corrected the ambient curvature by a term built from the second fundamental form, the correction here is built from a single tensor A measuring the failure of the transverse distribution to be integrable. The resulting identity, O'Neill's formula, will be derived in full and then run on the two examples that carry the most weight downstream: the Hopf fibration, which exhibits the round two-sphere as a quotient of the three-sphere, and the Fubini–Study metric on complex projective space, the running example of Part II.

The plan is as follows. After fixing the definition and the horizontal lift, geodesics come first: a horizontal geodesic upstairs projects to a geodesic downstairs, which is what makes the base metric geometrically meaningful and supplies the tool for later computations. O'Neill's tensor A is then isolated and shown to be, on the transverse directions, exactly one half of the Lie bracket, hence a true obstruction to integrability. The curvature formula follows, and its immediate reading is that a Riemannian submersion never decreases horizontal sectional curvature. The Hopf and Fubini–Study examples close the section.

Throughout, $\pi\colon (\tilde{M}, \tilde{g}) \to (B, g_B)$ is a smooth surjective submersion between Riemannian manifolds, $\tilde{\nabla}$ and ∇^B are the Levi-Civita connections of definition 2.3.2, and the curvature sign convention is that of box 4.1.1. The Einstein summation convention is not needed here; all computations are frame-free.

Definition 7.5.1: Riemannian submersion

Let $\pi\colon \tilde{M} \to B$ be a smooth submersion, so that $d\pi_p\colon T_p\tilde{M} \to T_{\pi(p)}B$ is surjective at every $p \in \tilde{M}$ ([6]). For each p the *vertical space* is the kernel

$$\mathcal{V}_p = \ker d\pi_p \subseteq T_p\tilde{M}$$

tangent to the fiber $\pi^{-1}(\pi(p))$, and the *horizontal space* is its $\tilde{g}$-orthogonal complement

$$\mathcal{H}_p = \mathcal{V}_p^{\perp}$$

so that $T_p\tilde{M} = \mathcal{V}_p \oplus \mathcal{H}_p$ orthogonally. Write $(\cdot)^{\mathcal{V}}$ and $(\cdot)^{\mathcal{H}}$ for the corresponding projections. The map π is a *Riemannian submersion* if for every p the restriction

$$d\pi_p|_{\mathcal{H}_p}\colon \mathcal{H}_p \longrightarrow T_{\pi(p)}B$$

is a linear isometry onto $(T_{\pi(p)}B, g_B)$.
For a vector field X on B, the *horizontal lift* $\tilde{X}$ is the unique horizontal vector field on $\tilde{M}$ with $d\pi_p(\tilde{X}_p) = X_{\pi(p)}$ for all p; it exists and is smooth because $d\pi|_{\mathcal{H}}$ is a fiberwise isomorphism. A horizontal vector field of the form $\tilde{X}$ for some X on B is called *basic*.

The isometry condition is the entire content of the word "Riemannian" here: a bare submersion carries no metric on the base at all, and even with metrics fixed on both ends the differential $d\pi$ generally distorts lengths. Requiring $d\pi_p|_{\mathcal{H}_p}$ to be an isometry says that the base metric at $\pi(p)$ is the pushforward of the total-space metric restricted to the horizontal space, with the vertical directions discarded. Two vectors of $\mathcal{H}_p$ then have the same inner product as their images downstairs, so horizontal geometry above and geometry below agree pointwise; all the information lost in passing to the quotient is vertical. The horizontal lift makes this precise at the level of vector fields: it selects, out of every tangent vector on B, the one horizontal vector above each point that projects to it, and by construction $\langle \tilde{X}, \tilde{Y} \rangle_{\tilde{g}} = \langle X, Y \rangle_{g_B} \circ \pi$ for all X, Y on B. Basic fields are the tool for transferring computations between the two manifolds: a bracket, a covariant derivative, or a curvature computed among basic fields upstairs projects to the corresponding object downstairs, up to the vertical corrections this section is built to track.

The simplest Riemannian submersion is the projection off a product, and it fixes the picture before the curved examples arrive.

Example 7.5.2: Projection of a Riemannian product

Let (B, g_B) and (F, g_F) be Riemannian manifolds and give the product $\tilde{M} = B \times F$ the product metric $\tilde{g} = g_B \oplus g_F$ of definition 1.6.1. The projection $\pi\colon B \times F \to B$, $\pi(b, f) = b$, is a smooth submersion whose fiber over b is $\{b\} \times F$. At a point (b, f) the vertical space is $\mathcal{V}_{(b,f)} = \{0\} \oplus T_fF$, the tangent space to the fiber, and by the orthogonal splitting of the product metric the horizontal space is $\mathcal{H}_{(b,f)} = T_bB \oplus \{0\}$. The differential $d\pi_{(b,f)}$ is the projection $T_bB \oplus T_fF \to T_bB$ onto the first factor, so its restriction to $\mathcal{H}_{(b,f)} = T_bB \oplus \{0\}$ is the identification $T_bB \oplus \{0\} \cong T_bB$, which carries $\tilde{g}|_{\mathcal{H}} = g_B$ to g_B isometrically. Hence π is a Riemannian submersion. The horizontal lift of a field X on B is $\tilde{X}_{(b,f)} = (X_b, 0)$, independent of the fiber coordinate; two such basic lifts commute exactly as the fields on B do, and their bracket is again basic. This last property will fail in the curved examples, and its failure is precisely what O'Neill's tensor records.

For a product projection the horizontal distribution $\mathcal{H}$ is the tangent distribution to the slices $B \times \{f\}$, which are embedded submanifolds; the distribution is integrable, and its leaves are copies of the base sitting horizontally in the product. Nothing forces this in general. A Riemannian submersion may have a horizontal distribution that no submanifold is tangent to, and then there is no way to lift the base into the total space as a family of horizontal leaves. The Hopf fibration below is exactly this: its horizontal planes twist as one moves along a fiber, and the twisting is what makes the base curve more than the total space does. Before measuring the twist, geodesics establish that the base metric deserves its name.

The following proposition is the reason a Riemannian submersion is a geometric object rather than a metric bookkeeping device. It says that horizontal geodesics upstairs and geodesics downstairs are the same curves, so distances and shortest paths on the base are computed by lifting horizontally and running geodesics in the total space.

Proposition 7.5.3: Horizontal geodesics project to geodesics

Let $\pi\colon (\tilde{M}, \tilde{g}) \to (B, g_B)$ be a Riemannian submersion.

1. If $\tilde{\gamma}$ is a geodesic of $\tilde{M}$ whose velocity $\dot{\tilde{\gamma}}(t_0)$ is horizontal at one parameter t_0, then $\dot{\tilde{\gamma}}(t)$ is horizontal at every t, the projection $\gamma = \pi \circ \tilde{\gamma}$ is a geodesic of B, and $|\dot{\gamma}|_{g_B} = |\dot{\tilde{\gamma}}|_{\tilde{g}}$ is constant. In particular $\tilde{\gamma}$ and γ have the same speed and the same length.
2. Conversely, let γ be a geodesic of B and let $\tilde{p} \in \pi^{-1}(\gamma(0))$. Then there is a unique horizontal vector $v \in \mathcal{H}_{\tilde{p}}$ with $d\pi_{\tilde{p}}(v) = \dot{\gamma}(0)$, and the geodesic $\tilde{\gamma}$ of $\tilde{M}$ with $\tilde{\gamma}(0) = \tilde{p}$, $\dot{\tilde{\gamma}}(0) = v$ stays horizontal and satisfies $\pi \circ \tilde{\gamma} = \gamma$ on the common domain. It is called the *horizontal lift* of γ through $\tilde{p}$.

The proof rests on one lemma about how the connections of the two manifolds relate on basic fields, which is worth stating on its own because it recurs in the curvature computation. For basic horizontal fields X, Y (lifts of X', Y' on B) the horizontal part of $\tilde{\nabla}_X Y$ is the basic lift of $\nabla^B_{X'} Y'$.

Lemma 7.5.4: The connection lemma for basic fields

Let $\pi\colon \tilde{M} \to B$ be a Riemannian submersion and let X, Y be basic horizontal fields, the lifts of X', Y' on B. Then the horizontal part of $\tilde{\nabla}_X Y$ is the basic lift of $\nabla^B_{X'} Y'$:

$$(\tilde{\nabla}_X Y)^{\mathcal{H}} = \widetilde{\nabla^B_{X'} Y'} \tag{7.22}$$

In particular $d\pi\big((\tilde{\nabla}_X Y)^{\mathcal{H}}\big) = \nabla^B_{X'} Y' \circ \pi$.

***Proof*:**
Both connections are computed from their Koszul formulas (eq. (2.3)). For $\tilde{\nabla}$ and horizontal fields X, Y, Z,

$$2\left\langle \tilde{\nabla}_X Y, Z\right\rangle_{\tilde{g}} = X\left\langle Y, Z\right\rangle_{\tilde{g}} + Y\left\langle Z, X\right\rangle_{\tilde{g}} - Z\left\langle X, Y\right\rangle_{\tilde{g}} + \left\langle [X,Y], Z\right\rangle_{\tilde{g}} - \left\langle [Y,Z], X\right\rangle_{\tilde{g}} + \left\langle [Z,X], Y\right\rangle_{\tilde{g}}$$

Take $Z = \tilde{W}$ basic as well, the lift of W' on B. Because X, Y, Z are basic, each inner product $\langle X, Y\rangle_{\tilde{g}} = \langle X', Y'\rangle_{g_B} \circ \pi$ is a function pulled back from B, and a basic field X differentiates such a pullback by $X(h \circ \pi) = (X'h) \circ \pi$, since $d\pi(X) = X' \circ \pi$. Hence the first three terms above equal the corresponding three terms of the Koszul formula for $2\left\langle \nabla^B_{X'} Y', W'\right\rangle_{g_B}$, pulled back to $\tilde{M}$. For the bracket terms, the projectability of basic fields gives $d\pi([X,Y]) = [X', Y'] \circ \pi$, so the horizontal part of $[X, Y]$ is the basic lift $\widetilde{[X', Y']}$; the vertical part of $[X, Y]$ is $\tilde{g}$-orthogonal to the horizontal field Z and drops out of $\langle [X,Y], Z\rangle_{\tilde{g}}$. Thus $\langle [X,Y], Z\rangle_{\tilde{g}} = \langle [X', Y'], W'\rangle_{g_B} \circ \pi$, and likewise for the other two bracket terms. Every term on the right therefore equals the pullback of the matching term in the Koszul formula for ∇^B, so

$$\left\langle \tilde{\nabla}_X Y, Z\right\rangle_{\tilde{g}} = \left\langle \nabla^B_{X'} Y', W'\right\rangle_{g_B} \circ \pi = \left\langle \widetilde{\nabla^B_{X'} Y'}, Z\right\rangle_{\tilde{g}}$$

for every basic Z. Since $d\pi|_{\mathcal{H}}$ is an isometry, the basic fields span $\mathcal{H}$ at each point, so the horizontal parts of $\tilde{\nabla}_X Y$ and of $\widetilde{\nabla^B_{X'} Y'}$ have the same inner product with every horizontal vector; being horizontal, they are equal. This is (7.22), and applying $d\pi$ gives the last claim, completing the proof.

***Proof of proposition 7.5.3*:**
The two parts are proved in turn.

1. *Horizontality is preserved.* Let $\tilde{\gamma}$ be a geodesic with $\dot{\tilde{\gamma}}(t_0)$ horizontal. Decompose the velocity as $\dot{\tilde{\gamma}} = H + V$ with $H = \dot{\tilde{\gamma}}^{\mathcal{H}}$ and $V = \dot{\tilde{\gamma}}^{\mathcal{V}}$ along $\tilde{\gamma}$. The claim is that $V \equiv 0$. The clean route is through the projected curve. Set $\gamma = \pi \circ \tilde{\gamma}$; then $\dot{\gamma} = d\pi(\dot{\tilde{\gamma}}) = d\pi(H)$, since $V \in \ker d\pi$. Because $d\pi|_{\mathcal{H}}$ is an isometry, $|\dot{\gamma}|_{g_B} = |H|_{\tilde{g}}$, and $\left|\dot{\tilde{\gamma}}\right|^2_{\tilde{g}} = |H|^2_{\tilde{g}} + |V|^2_{\tilde{g}}$ is constant along the geodesic $\tilde{\gamma}$ by proposition 3.1.6. It suffices to prove γ is a geodesic of B; then $|\dot{\gamma}|_{g_B} = |H|_{\tilde{g}}$ is constant, and evaluating the constant $|H|^2_{\tilde{g}} + |V|^2_{\tilde{g}}$ and the constant $|H|^2_{\tilde{g}}$ at t_0, where $V(t_0) = 0$, forces $|V|^2_{\tilde{g}} \equiv 0$, hence $V \equiv 0$ and $\dot{\tilde{\gamma}}$ is horizontal throughout.

   To see that γ is a geodesic, compute $\nabla^B_{\dot{\gamma}} \dot{\gamma}$ using that $\tilde{\gamma}$ is a geodesic. Extend $\dot{\gamma}$ to a field near the curve and lift horizontally; the point is local and it suffices to work with the horizontal lift of $\dot{\gamma}$, which is H. By lemma 7.5.4 the horizontal part of the total-space covariant derivative of a horizontal lift along the curve projects to the base covariant derivative: $d\pi\big((\tilde{D}_t \dot{\tilde{\gamma}})^{\mathcal{H}}\big) = D^B_t \dot{\gamma}$, where $\tilde{D}_t$ and D^B_t are the covariant derivatives along $\tilde{\gamma}$ and γ of definition 2.1.6. Since $\tilde{\gamma}$ is a geodesic, $\tilde{D}_t \dot{\tilde{\gamma}} = 0$, so its horizontal part vanishes and $D^B_t \dot{\gamma} = d\pi(0) = 0$. Thus γ is a geodesic of B, and by the paragraph above $\tilde{\gamma}$ is horizontal with $|\dot{\gamma}|_{g_B} = \left|\dot{\tilde{\gamma}}\right|_{\tilde{g}}$ constant. Equal constant speeds on a common interval give equal lengths.

2. *Existence and uniqueness of the lift.* Given the geodesic γ of B and $\tilde{p} \in \pi^{-1}(\gamma(0))$, the isometry $d\pi_{\tilde{p}}|_{\mathcal{H}_{\tilde{p}}}$ has a unique preimage $v \in \mathcal{H}_{\tilde{p}}$ of $\dot{\gamma}(0)$. Let $\tilde{\gamma}$ be the geodesic of $\tilde{M}$ with

> initial data $(\tilde{p}, v)$, which exists and is unique by theorem 3.1.3. Its initial velocity v is horizontal, so by part (i) it stays horizontal and its projection $\pi \circ \tilde{\gamma}$ is a geodesic of B. That projection has the same initial point $\gamma(0)$ and the same initial velocity $d\pi_{\tilde{p}}(v) = \dot{\gamma}(0)$ as γ, so by the uniqueness half of theorem 3.1.3 applied in B, $\pi \circ \tilde{\gamma} = \gamma$ on the common domain. Uniqueness of $\tilde{\gamma}$ as a horizontal lift is uniqueness of the geodesic with initial data $(\tilde{p}, v)$, which is the only horizontal initial vector over $\dot{\gamma}(0)$. This is the horizontal lift, as claimed.

The proposition converts the base geometry into a horizontal slice of the total-space geometry. Distances downstairs are computed upstairs: the length of a base geodesic equals the length of its horizontal lift, so a shortest path on B can be found by lifting horizontally and minimizing among horizontal geodesics of $\tilde{M}$. It also explains why the two examples of this section are computable at all. Great circles of the three-sphere that start horizontally project to geodesics of the round two-sphere, and this is what identifies the base metric of the Hopf fibration as a round metric of a definite radius; the same mechanism produces the geodesics of the Fubini–Study metric from great circles of the odd sphere. What the proposition does not do is match curvatures, because curvature involves second derivatives of the metric and the horizontal distribution can rotate along a fiber. That rotation is the tensor A.

Definition 7.5.5: O'Neill's integrability tensor

Let $\pi\colon \tilde{M} \to B$ be a Riemannian submersion. For horizontal vector fields X, Y on $\tilde{M}$, O'Neill's *integrability tensor* is the vertical field

$$A_X Y = (\tilde{\nabla}_X Y)^{\mathcal{V}}$$

the vertical part of the total-space covariant derivative of one horizontal field along another.

The tensor is named for its geometric meaning, which the next proposition makes exact: A measures precisely the failure of the horizontal distribution $\mathcal{H}$ to be integrable. A distribution is integrable when the Lie bracket of any two of its sections stays inside it; for $\mathcal{H}$ this would say $[X, Y]^{\mathcal{V}} = 0$ for horizontal X, Y. The proposition computes A on basic fields as one half of that vertical bracket, so $A \equiv 0$ on horizontal vectors exactly when $\mathcal{H}$ is integrable, exactly when the base lifts locally to horizontal leaves as in the product example of example 7.5.2.

Proposition 7.5.6: ***A* as half the vertical bracket**

Let X, Y be basic horizontal fields on the Riemannian submersion $\pi\colon \tilde{M} \to B$. Then

$$A_X Y = \tfrac{1}{2}[X, Y]^{\mathcal{V}} \tag{7.23}$$

Consequently $A_X Y = -A_Y X$ on horizontal vectors: the tensor A is alternating. Moreover $A_X Y$ depends only on the values of X and Y at the point, so it defines a tensor, and $A_X Y$ is $\tilde{g}$-orthogonal to the horizontal directions, taking values in $\mathcal{V}$.

***Proof*:**
Since $\tilde{\nabla}$ is torsion-free (theorem 2.3.1), $\tilde{\nabla}_X Y - \tilde{\nabla}_Y X = [X, Y]$. Taking vertical parts,

$$A_X Y - A_Y X = (\tilde{\nabla}_X Y)^{\mathcal{V}} - (\tilde{\nabla}_Y X)^{\mathcal{V}} = [X, Y]^{\mathcal{V}}$$

This already gives the alternating property once symmetry is ruled out, so it remains to show $A_X Y + A_Y X = 0$, equivalently $A_X Y = -A_Y X$; combined with the display, $2A_X Y = [X, Y]^{\mathcal{V}}$ follows. For the symmetric part, let V be any vertical field and pair with the metric:

$$\langle A_X Y + A_Y X, V \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_X Y, V \right\rangle_{\tilde{g}} + \left\langle \tilde{\nabla}_Y X, V \right\rangle_{\tilde{g}}$$

using that V is vertical so only the vertical parts of the derivatives contribute. Since X, Y are horizontal, $\langle X, V \rangle_{\tilde{g}} = 0$ and $\langle Y, V \rangle_{\tilde{g}} = 0$ as functions on $\tilde{M}$; differentiating the first along Y and the second along X and using metric compatibility (definition 2.2.4),

$$0 = Y \langle X, V \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_Y X, V \right\rangle_{\tilde{g}} + \left\langle X, \tilde{\nabla}_Y V \right\rangle_{\tilde{g}}, \qquad 0 = X \langle Y, V \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_X Y, V \right\rangle_{\tilde{g}} + \left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}}$$

Adding,

$$\langle A_X Y + A_Y X, V \rangle_{\tilde{g}} = -\left\langle X, \tilde{\nabla}_Y V \right\rangle_{\tilde{g}} - \left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}}$$

Now $\tilde{\nabla}_Y V - \tilde{\nabla}_V Y = [Y, V]$ and $\tilde{\nabla}_X V - \tilde{\nabla}_V X = [X, V]$ by torsion-freeness. For basic X, Y and vertical V tangent to the fibers, the bracket $[X, V]$ is vertical: a basic field is π-related to a field on B and V is π-related to the zero field, so $d\pi([X, V]) = [X', 0] = 0$, whence $[X, V] \in \mathcal{V}$, and likewise $[Y, V] \in \mathcal{V}$. Being orthogonal to the horizontal X, Y, these bracket terms drop out of the inner products, leaving

$$\langle A_X Y + A_Y X, V \rangle_{\tilde{g}} = -\left\langle X, \tilde{\nabla}_V Y \right\rangle_{\tilde{g}} - \left\langle Y, \tilde{\nabla}_V X \right\rangle_{\tilde{g}} = -V \langle X, Y \rangle_{\tilde{g}}$$

by metric compatibility read in reverse. But $\langle X, Y \rangle_{\tilde{g}} = \langle X', Y' \rangle_{g_B} \circ \pi$ is a function pulled back from B, and a vertical field kills such a pullback: $V(h \circ \pi) = (dh)(d\pi\, V) = 0$ since $d\pi\, V = 0$. Therefore $\langle A_X Y + A_Y X, V \rangle_{\tilde{g}} = 0$ for every vertical V, and as $A_X Y + A_Y X$ is itself vertical, it vanishes. This yields (7.23) and the alternating property.
Tensoriality is a consequence of (7.23) together with the general theory of A. From the display $2A_X Y = [X, Y]^{\mathcal{V}}$, the value at a point depends on X, Y only through the point-values needed to evaluate the vertical bracket among basic representatives; more directly, A extends to all horizontal vector fields (not only basic ones) by the formula $A_X Y = (\tilde{\nabla}_X Y)^{\mathcal{V}}$, and one checks $A_{fX} Y = f\, A_X Y$ and $A_X(fY) = f\, A_X Y$ for $f \in C^\infty(\tilde{M})$: the first is immediate from C^∞-linearity of $\tilde{\nabla}$ in the lower slot, and the second holds because $\tilde{\nabla}_X(fY) = (Xf)Y + f\tilde{\nabla}_X Y$ and the vertical part of the horizontal term $(Xf)Y$ vanishes. Hence $A_X Y$ depends only on the values of X and Y at the point and is a tensor, valued in $\mathcal{V}$, as claimed.

The tensor A is one of two O'Neill tensors. The other, denoted T, is the vertical analogue: $T_U W = (\tilde{\nabla}_U W)^{\mathcal{H}}$ for vertical fields U, W, and it is the second fundamental form of the fibers as submanifolds of $\tilde{M}$ (in the sense of definition 7.1.4, with each fiber $\pi^{-1}(b)$ the submanifold). A submersion has *totally geodesic fibers* when $T \equiv 0$, meaning each fiber is a totally geodesic submanifold in the sense of definition 7.4.3. Both examples below have totally geodesic fibers: the fibers of the Hopf fibration are great circles of S^3, and the fibers of $S^{2n+1} \to \mathbb{C}\mathrm{P}^n$ are great circles of S^{2n+1}, and a great circle is a geodesic of the round sphere, hence a totally geodesic one-dimensional submanifold. With $T = 0$ the general O'Neill machinery simplifies, and the curvature of a horizontal plane depends on the geometry only through A. The following theorem records that dependence.

O'Neill's formula is the submersion counterpart of the Gauss equation: it computes the sectional curvature of a plane in the base from the curvature of its horizontal lift in the total space, corrected by a term quadratic in A. The correction has a fixed sign, and reading it off is the payoff.

Theorem 7.5.7: O'Neill's curvature formula

Let $\pi\colon (\tilde{M}, \tilde{g}) \to (B, g_B)$ be a Riemannian submersion, and let X, Y be horizontal vectors at a point $\tilde{p}$ that are $\tilde{g}$-orthonormal. Then the sectional curvature of the plane $\operatorname{span}(d\pi X, d\pi Y) \subseteq T_{\pi(\tilde{p})}B$ is

$$K_B(d\pi X, d\pi Y) = K_{\tilde{M}}(X, Y) + 3\,|A_X Y|^2_{\tilde{g}} \tag{7.24}$$

where $K_{\tilde{M}}(X, Y)$ is the sectional curvature of the horizontal plane $\operatorname{span}(X, Y)$ in $\tilde{M}$. In particular $K_B(d\pi X, d\pi Y) \geq K_{\tilde{M}}(X, Y)$: a Riemannian submersion does not decrease horizontal sectional curvature.

***Proof*:**
Fix $\tilde{g}$-orthonormal horizontal vectors at $\tilde{p}$ and extend them to basic horizontal fields X, Y, the lifts of g_B-orthonormal fields X', Y' near $\pi(\tilde{p})$; this is possible because $d\pi|_{\mathcal{H}}$ is an isometry, so orthonormality of X', Y' is equivalent to orthonormality of their lifts. With the sign convention of box 4.1.1, the sectional curvature of orthonormal vectors is

$$K_{\tilde{M}}(X, Y) = \left\langle \tilde{R}(X, Y)Y, X \right\rangle_{\tilde{g}}, \qquad K_B(X', Y') = \left\langle R^B(X', Y')Y', X' \right\rangle_{g_B}$$

The proof computes the base curvature $\left\langle R^B(X', Y')Y', X' \right\rangle_{g_B} \circ \pi$ by expanding it through the Koszul-derived connection and comparing with the total-space curvature. It proceeds in three steps.

Step 1: the horizontal correction to the connection. By definition 7.5.5 and proposition 7.5.6, for basic horizontal X, Y,

$$\tilde{\nabla}_X Y = (\tilde{\nabla}_X Y)^{\mathcal{H}} + A_X Y, \qquad A_X Y = \tfrac{1}{2}[X, Y]^{\mathcal{V}} \tag{7.25}$$

and by lemma 7.5.4 the horizontal part is the basic lift of the base connection: $(\tilde{\nabla}_X Y)^{\mathcal{H}} = \widetilde{\nabla^B_{X'} Y'}$. Thus $\tilde{\nabla}_X Y$ differs from the lift of $\nabla^B_{X'} Y'$ by exactly the vertical field $A_X Y$.

Step 2: expanding the base curvature. Write R^B for the base curvature endomorphism and abbreviate $\nabla' = \nabla^B$. Then

$$\left\langle R^B(X', Y')Y', X' \right\rangle_{g_B} = \langle \nabla'_{X'} \nabla'_{Y'} Y', X' \rangle_{g_B} - \langle \nabla'_{Y'} \nabla'_{X'} Y', X' \rangle_{g_B} - \left\langle \nabla'_{[X', Y']} Y', X' \right\rangle_{g_B}$$

Each term is a base object; lift it to $\tilde{M}$ using (7.25) and lemma 7.5.4. Because horizontal inner products of basic lifts equal the pullbacks of base inner products, $\left\langle \widetilde{\nabla'_{U'} W'}, \tilde{Z} \right\rangle_{\tilde{g}} = \langle \nabla'_{U'} W', Z' \rangle_{g_B} \circ \pi$ for any basic $\tilde{Z}$. Now compute the total-space curvature term by term and subtract.

Consider first the total-space expression $\left\langle \tilde{\nabla}_X \tilde{\nabla}_Y Y, X \right\rangle_{\tilde{g}}$. Using (7.25) on $\tilde{\nabla}_Y Y$,

$$\tilde{\nabla}_Y Y = \widetilde{\nabla'_{Y'} Y'} + A_Y Y = \widetilde{\nabla'_{Y'} Y'}$$

since $A_Y Y = 0$ by the alternating property of proposition 7.5.6. Then $\tilde{\nabla}_X \tilde{\nabla}_Y Y = \tilde{\nabla}_X \big(\widetilde{\nabla'_{Y'} Y'}\big)$, whose horizontal part is $\widetilde{\nabla'_{X'} \nabla'_{Y'} Y'}$ and whose inner product with the horizontal field X is

$$\left\langle \tilde{\nabla}_X \tilde{\nabla}_Y Y, X \right\rangle_{\tilde{g}} = \langle \nabla'_{X'} \nabla'_{Y'} Y', X' \rangle_{g_B} \circ \pi \tag{7.26}$$

The vertical part $A_X(\widetilde{\nabla'_{Y'} Y'})$ contributes nothing here because it is orthogonal to the horizontal X. The same computation applies to $\left\langle \tilde{\nabla}_Y \tilde{\nabla}_X Y, X \right\rangle_{\tilde{g}}$, except that $\tilde{\nabla}_X Y = \widetilde{\nabla'_{X'} Y'} + A_X Y$ now carries a nonzero vertical part $A_X Y$, so

$$\tilde{\nabla}_Y \tilde{\nabla}_X Y = \tilde{\nabla}_Y \big(\widetilde{\nabla'_{X'} Y'}\big) + \tilde{\nabla}_Y (A_X Y)$$

The first summand contributes $\langle \nabla'_{Y'} \nabla'_{X'} Y', X' \rangle_{g_B} \circ \pi$ to the inner product with X, as before. For the second, $A_X Y$ is vertical; pair its derivative with the horizontal X and use that $X \langle A_X Y, X \rangle_{\tilde{g}}$ splits by metric compatibility. Since $\langle A_X Y, X \rangle_{\tilde{g}} = 0$ (vertical against horizontal),

$$0 = Y \langle A_X Y, X \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_Y (A_X Y), X \right\rangle_{\tilde{g}} + \left\langle A_X Y, \tilde{\nabla}_Y X \right\rangle_{\tilde{g}}$$

so $\left\langle \tilde{\nabla}_Y (A_X Y), X \right\rangle_{\tilde{g}} = -\left\langle A_X Y, \tilde{\nabla}_Y X \right\rangle_{\tilde{g}} = -\langle A_X Y, A_Y X \rangle_{\tilde{g}}$, the last equality because only the vertical part $A_Y X$ of $\tilde{\nabla}_Y X$ pairs with the vertical $A_X Y$. By the alternating property $A_Y X = -A_X Y$, so

$$\left\langle \tilde{\nabla}_Y \tilde{\nabla}_X Y, X \right\rangle_{\tilde{g}} = \langle \nabla'_{Y'} \nabla'_{X'} Y', X' \rangle_{g_B} \circ \pi + |A_X Y|^2_{\tilde{g}} \tag{7.27}$$

Step 3: the bracket term and assembly. It remains to handle $\left\langle \tilde{\nabla}_{[X,Y]} Y, X \right\rangle_{\tilde{g}}$. Split the bracket into horizontal and vertical parts, $[X, Y] = [X, Y]^{\mathcal{H}} + [X, Y]^{\mathcal{V}}$. The horizontal part is the basic lift $\widetilde{[X', Y']}$ by projectability of basic fields, and it contributes

$$\left\langle \tilde{\nabla}_{[X,Y]^{\mathcal{H}}} Y, X \right\rangle_{\tilde{g}} = \left\langle \nabla'_{[X',Y']} Y', X' \right\rangle_{g_B} \circ \pi$$

again by lemma 7.5.4. The vertical part is $[X, Y]^{\mathcal{V}} = 2A_X Y$ by (7.25), so its contribution is $\left\langle \tilde{\nabla}_{2A_X Y} Y, X \right\rangle_{\tilde{g}} = 2\left\langle \tilde{\nabla}_{A_X Y} Y, X \right\rangle_{\tilde{g}}$. Let $V = A_X Y$, a vertical field. Since $\langle Y, X \rangle_{\tilde{g}} = 0$ (orthonormal X, Y, so their inner product is the constant 0) and V kills pullbacks,

$$0 = V \langle Y, X \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_V Y, X \right\rangle_{\tilde{g}} + \left\langle Y, \tilde{\nabla}_V X \right\rangle_{\tilde{g}}$$

so $\left\langle \tilde{\nabla}_V Y, X \right\rangle_{\tilde{g}} = -\left\langle Y, \tilde{\nabla}_V X \right\rangle_{\tilde{g}}$. Both $\tilde{\nabla}_V X$ and $\tilde{\nabla}_V Y$ have horizontal parts governed by the fiber tensor T, but $T \equiv 0$ is not assumed; instead pair through A directly. Write $\tilde{\nabla}_V X = (\tilde{\nabla}_X V) - [X, V]$. As shown in the proof of proposition 7.5.6, $[X, V]$ is vertical for basic X and vertical V, so its horizontal part vanishes and $(\tilde{\nabla}_V X)^{\mathcal{H}} = (\tilde{\nabla}_X V)^{\mathcal{H}}$. Now $\left\langle Y, \tilde{\nabla}_V X \right\rangle_{\tilde{g}} = \left\langle Y, (\tilde{\nabla}_X V)^{\mathcal{H}} \right\rangle_{\tilde{g}} = \left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}}$, and since $\langle Y, V \rangle_{\tilde{g}} = 0$ with V vertical,

$$0 = X \langle Y, V \rangle_{\tilde{g}} = \left\langle \tilde{\nabla}_X Y, V \right\rangle_{\tilde{g}} + \left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}}$$

gives $\left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}} = -\left\langle \tilde{\nabla}_X Y, V \right\rangle_{\tilde{g}} = -\langle A_X Y, V \rangle_{\tilde{g}} = -|A_X Y|^2_{\tilde{g}}$, the middle equality because only the vertical part $A_X Y$ of $\tilde{\nabla}_X Y$ pairs with the vertical $V = A_X Y$. Chaining the last three displays,

$$\left\langle \tilde{\nabla}_V Y, X \right\rangle_{\tilde{g}} = -\left\langle Y, \tilde{\nabla}_V X \right\rangle_{\tilde{g}} = -\left\langle Y, \tilde{\nabla}_X V \right\rangle_{\tilde{g}} = |A_X Y|^2_{\tilde{g}}$$

so the vertical bracket contributes $2\left\langle \tilde{\nabla}_V Y, X\right\rangle_{\tilde{g}} = 2\left|A_X Y\right|^2_{\tilde{g}}$, and

$$\left\langle \tilde{\nabla}_{[X,Y]} Y, X\right\rangle_{\tilde{g}} = \left\langle \nabla'_{[X',Y']} Y', X'\right\rangle_{g_B} \circ \pi + 2\left|A_X Y\right|^2_{\tilde{g}} \tag{7.28}$$

Now assemble. By the curvature convention,

$$K_{\tilde{M}}(X,Y) = \left\langle \tilde{\nabla}_X \tilde{\nabla}_Y Y, X\right\rangle_{\tilde{g}} - \left\langle \tilde{\nabla}_Y \tilde{\nabla}_X Y, X\right\rangle_{\tilde{g}} - \left\langle \tilde{\nabla}_{[X,Y]} Y, X\right\rangle_{\tilde{g}}$$

Substituting (7.26), (7.27), and (7.28), the three base terms reassemble into $K_B(X',Y')\circ\pi$, and the A-terms combine as $0 - \left|A_X Y\right|^2 - 2\left|A_X Y\right|^2 = -3\left|A_X Y\right|^2$:

$$K_{\tilde{M}}(X,Y) = K_B(d\pi X, d\pi Y) - 3\left|A_X Y\right|^2_{\tilde{g}}$$

evaluated at $\tilde{p}$, where $K_B(d\pi X, d\pi Y) = K_B(X',Y')\circ\pi$ since $d\pi X = X'$, $d\pi Y = Y'$. Rearranging gives (7.24). Because $\left|A_X Y\right|^2_{\tilde{g}} \geq 0$, the base curvature is at least the horizontal total-space curvature, as we sought to show.

The formula reads in the direction one expects from the sign of the correction. Quotienting by fibers cannot make a horizontal plane less curved; the twisting of the horizontal distribution along the fibers, measured by $\left|A_X Y\right|^2$, can only add curvature to the base. When $A \equiv 0$ the horizontal distribution is integrable and the base is locally a horizontal leaf, so its curvature agrees with the horizontal curvature above, as it did for the product projection of example 7.5.2; there the horizontal leaves are the slices $B \times \{f\}$ and A vanishes identically. The interest is in the opposite case, where $A \neq 0$ forces the base to curve strictly more than the total space along horizontal planes. The two examples that follow are of this kind, and in both the total space is a round sphere of curvature 1 while the base acquires curvature up to 4 from a nonvanishing A.

The Hopf fibration is the first. It presents the round two-sphere of radius $\frac{1}{2}$ as the base of a Riemannian submersion from the unit three-sphere, with circle fibers, and O'Neill's formula recovers the base curvature 4 from the total-space curvature 1 and a computation of $\left|A_X Y\right|^2 = 1$.

Example 7.5.8: The Hopf fibration

Regard the unit sphere $S^3(1)$ as the unit sphere in $\mathbb{C}^2$,

$$S^3(1) = \left\{(z_1, z_2) \in \mathbb{C}^2 \;\middle|\; \left|z_1\right|^2 + \left|z_2\right|^2 = 1\right\}$$

with the round metric of example 1.3.4, curvature $K_{S^3} \equiv 1$. Multiplication by the unit complex scalar $e^{i\theta}$ acts on $\mathbb{C}^2$ and preserves $S^3(1)$; the orbits of this circle action are the great circles $\theta \mapsto (e^{i\theta} z_1, e^{i\theta} z_2)$, and the quotient by the action is the Hopf map

$$\pi\colon S^3(1) \longrightarrow \mathbb{CP}^1 = S^2\left(\tfrac{1}{2}\right), \qquad \pi(z_1, z_2) = [z_1 : z_2]$$

realizing complex projective space $\mathbb{CP}^1$ as a two-sphere. As a fiber bundle $S^1 \hookrightarrow S^3 \to S^2$ this is the Hopf fibration of [4], the generator of $\pi_3(S^2)$; here it is treated metrically. The fibers are the circle orbits, and their common tangent direction at $p = (z_1, z_2)$ is the velocity of the orbit,

$$V_p = \left.\frac{d}{d\theta}\right|_{\theta=0} (e^{i\theta} z_1, e^{i\theta} z_2) = (iz_1, iz_2) = ip = Jp$$

where J denotes multiplication by i on $\mathbb{C}^2 \cong \mathbb{R}^4$, an orthogonal complex structure. Since $|Jp| = |p| = 1$, the field $V = Jp$ is a unit vertical field, and $\mathcal{V}_p = \mathbb{R}\cdot Jp$.

The submersion metric. Give $S^2(\frac{1}{2})$ the metric that makes π a Riemannian submersion: at each point declare $d\pi_p|_{\mathcal{H}_p}$ an isometry, where $\mathcal{H}_p = (\mathbb{R} \cdot Jp)^\perp \cap T_pS^3$. This determines g_B uniquely, and the claim to be verified through O'Neill is that g_B is the round metric on the two-sphere of radius $\frac{1}{2}$, that is, $K_B \equiv 1/(\frac{1}{2})^2 = 4$. The circle action is by isometries of $S^3(1)$ (multiplication by $e^{i\theta}$ is a complex-linear isometry of $\mathbb{C}^2$ restricting to S^3), so the metric is well defined on the quotient and π is a Riemannian submersion.

Computing $|A_XY|^2$. Work in the left-invariant frame of $S^3(1) = SU(2)$. Identify $\mathbb{C}^2$-multiplication by i, j, k (unit quaternions) with three orthonormal left-invariant fields e_1, e_2, e_3 on $S^3(1)$, orthonormal for the round unit metric, with Lie brackets

$$[e_1, e_2] = 2e_3, \qquad [e_2, e_3] = 2e_1, \qquad [e_3, e_1] = 2e_2$$

The vertical field $V = Jp$ is multiplication by i, which is e_1; hence $\mathcal{V} = \mathbb{R} \cdot e_1$ and $\mathcal{H} = \operatorname{span}(e_2, e_3)$ at each point. Take the orthonormal horizontal pair $X = e_2$, $Y = e_3$. Because $d\pi_p|_{\mathcal{H}_p}$ is an isometry, the images $d\pi_p(e_2)$ and $d\pi_p(e_3)$ are g_B-orthonormal at $\pi(p)$, and O'Neill's formula requires only this pointwise orthonormality of a horizontal pair.

The left-invariant fields e_2, e_3 are not basic (they are not invariant along the fibers, since $[e_1, e_2] = 2e_3$ is horizontal, so $d\pi([e_1, e_2]) \neq 0$), so proposition 7.5.6 does not apply to them; instead compute $A_{e_2}e_3$ directly from definition 7.5.5 as $A_{e_2}e_3 = (\tilde{\nabla}_{e_2}e_3)^{\mathcal{V}}$. The round unit metric on $S^3(1) = SU(2)$ is bi-invariant, and for left-invariant fields U, W on a Lie group with bi-invariant metric the Levi-Civita connection is $\tilde{\nabla}_U W = \frac{1}{2}[U, W]$. This is read off the Koszul formula (eq. (2.3)): the inner products $\langle U, W\rangle_{\tilde{g}}$ of left-invariant fields are constant, so the first three terms vanish, and bi-invariance makes ad_U skew-adjoint, $\langle [U, Z], W\rangle_{\tilde{g}} = -\langle Z, [U, W]\rangle_{\tilde{g}}$, which collapses the three bracket terms to $2\left\langle \tilde{\nabla}_U W, Z\right\rangle_{\tilde{g}} = \langle [U, W], Z\rangle_{\tilde{g}}$ for every left-invariant Z. Applying this to $U = e_2$, $W = e_3$,

$$\tilde{\nabla}_{e_2}e_3 = \tfrac{1}{2}[e_2, e_3] = \tfrac{1}{2}(2e_1) = e_1$$

which is vertical, so $A_{e_2}e_3 = (e_1)^{\mathcal{V}} = e_1$ and therefore

$$|A_XY|^2_{\tilde{g}} = |e_1|^2 = 1$$

Applying O'Neill. The total space is $S^3(1)$ with $K_{\tilde{M}}(X, Y) = 1$ for every orthonormal horizontal pair. By theorem 7.5.7,

$$K_B(d\pi X, d\pi Y) = K_{\tilde{M}}(X, Y) + 3\,|A_XY|^2_{\tilde{g}} = 1 + 3 \cdot 1 = 4$$

The base is two-dimensional, so this single sectional curvature is its entire curvature: $K_B \equiv 4$. A two-sphere of constant curvature 4 is the round sphere of radius $\frac{1}{2}$ (example 1.3.4), confirming $\mathbb{CP}^1 = S^2(\frac{1}{2})$ with its round metric. The base curves four times as sharply as the total space along horizontal planes, and the extra factor is entirely the twisting of the Hopf circles, recorded by $|A_XY|^2 = 1$. The fibers themselves are great circles, hence geodesics of S^3 and totally geodesic (definition 7.4.3), so the fiber tensor T vanishes and the only correction is through A, exactly as O'Neill's formula was applied.

The Fubini–Study metric is the higher-dimensional continuation of the same construction and the running example of Part II. Complex projective space $\mathbb{CP}^n$ is the base of the Riemannian submersion from the unit odd sphere $S^{2n+1}(1)$ by the circle action, and O'Neill's formula computes its sectional curvature explicitly, exhibiting the pinching $1 \leq K \leq 4$ that makes $\mathbb{CP}^n$ a compact rank-one symmetric space.

Example 7.5.9: The Fubini–Study metric on $\mathbb{C}\mathrm{P}^n$

Regard $S^{2n+1}(1)$ as the unit sphere in $\mathbb{C}^{n+1}$ with the round metric of curvature 1, and let J be multiplication by i on $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$, an orthogonal complex structure. The circle action $e^{i\theta} \cdot (z_0, \ldots, z_n)$ has orbits the great circles $\theta \mapsto e^{i\theta} p$, with quotient the Hopf-type submersion

$$\pi \colon S^{2n+1}(1) \longrightarrow \mathbb{C}\mathrm{P}^n, \qquad \pi(z_0, \ldots, z_n) = [z_0 : \cdots : z_n]$$

At $p \in S^{2n+1}$ the vertical space is $\mathcal{V}_p = \mathbb{R} \cdot Jp$, the tangent line to the orbit, and the horizontal space is $\mathcal{H}_p = (\mathbb{R}p \oplus \mathbb{R}Jp)^\perp$, the real orthogonal complement in $\mathbb{C}^{n+1}$ of the complex line $\mathbb{C}p$. Since the action is by isometries, π is a Riemannian submersion onto $\mathbb{C}\mathrm{P}^n$ with the induced metric; this metric is the *Fubini–Study metric*, normalized here so that the total space is the unit sphere. It is the metric under which $\mathbb{C}\mathrm{P}^n$ arises in algebraic geometry as the base of the tautological bundle ([1]), and it is the running example through which Part II realizes Chern classes as curvature.

The tensor A on horizontal fields. The horizontal space $\mathcal{H}_p$ is J-invariant: if $X \perp p$ and $X \perp Jp$, then $JX \perp Jp$ (as J is orthogonal, $\langle JX, Jp \rangle = \langle X, p \rangle = 0$) and $JX \perp p$ (since $\langle JX, p \rangle = -\langle X, Jp \rangle = 0$), so $JX \in \mathcal{H}_p$. Thus J restricts to a complex structure on each horizontal space, and it descends to the complex structure of $\mathbb{C}\mathrm{P}^n$. For horizontal X, Y, O'Neill's tensor is computed from the flat ambient connection D of $\mathbb{C}^{n+1}$. The vertical part of $\tilde{\nabla}_X Y$ is its component along the unit vertical Jp:

$$A_X Y = (\tilde{\nabla}_X Y)^{\mathcal{V}} = \left\langle \tilde{\nabla}_X Y, Jp \right\rangle_{\tilde{g}} Jp$$

Since Jp is tangent to S^{2n+1} (it is orthogonal to p), the tangential-to-sphere derivative agrees with the ambient one against Jp: $\left\langle \tilde{\nabla}_X Y, Jp \right\rangle_{\tilde{g}} = \langle D_X Y, Jp \rangle$. Extend Y to a horizontal field; then $\langle Y, Jp \rangle = 0$ identically, so differentiating along X,

$$0 = X \langle Y, Jp \rangle = \langle D_X Y, Jp \rangle + \langle Y, D_X(Jp) \rangle$$

Now $D_X(Jp) = J\, D_X p = JX$, because $D_X p = X$ (the position field p has ambient derivative the identity) and J is a constant linear map on $\mathbb{C}^{n+1}$. Hence $\langle D_X Y, Jp \rangle = -\langle Y, JX \rangle = \langle X, JY \rangle$, the last step by skew-symmetry of J ($\langle Y, JX \rangle = -\langle JY, X \rangle = -\langle X, JY \rangle$). Therefore

$$A_X Y = \langle X, JY \rangle\ Jp, \qquad |A_X Y|^2_{\tilde{g}} = \langle X, JY \rangle^2 \tag{7.29}$$

using that Jp is a unit vector.

The curvature formula. Let X, Y be a g_B-orthonormal horizontal pair, so their images $d\pi X, d\pi Y$ are orthonormal on $\mathbb{C}\mathrm{P}^n$. By theorem 7.5.7 and (7.29),

$$K(d\pi X, d\pi Y) = K_{\tilde{M}}(X, Y) + 3\,|A_X Y|^2_{\tilde{g}} = 1 + 3\,\langle X, JY \rangle^2 \tag{7.30}$$

since the total space $S^{2n+1}(1)$ has constant curvature 1. The pinching is immediate. By the Cauchy–Schwarz inequality with $|X| = |Y| = 1$ and $|JY| = |Y| = 1$,

$$0 \le \langle X, JY \rangle^2 \le |X|^2\, |JY|^2 = 1$$

so (7.30) gives

$$1 \le K \le 4$$

for every orthonormal pair of tangent vectors on $\mathbb{C}\mathrm{P}^n$. Both extremes are attained. The upper bound $K = 4$ occurs on a *holomorphic* plane $\mathrm{span}(X, JX)$: taking $Y = JX$ (which is horizontal, orthonormal to X, since $\langle X, JX \rangle = 0$), one has $\langle X, JY \rangle = \langle X, J(JX) \rangle = \langle X, -X \rangle = -1$, so $\langle X, JY \rangle^2 = 1$ and $K = 1 + 3 = 4$. The plane $\mathrm{span}(X, JX)$ is the tangent plane of a complex line, and its curvature 4 is the *holomorphic sectional curvature* of the Fubini–Study metric. The lower bound $K = 1$ occurs on a *totally real* plane, one with $Y \perp JX$, so that $\langle X, JY \rangle = -\langle JX, Y \rangle = 0$ and $K = 1$.

Cross-check with $\mathbb{C}\mathrm{P}^1$. For $n = 1$ the horizontal space is two-dimensional and J-invariant, so it equals $\mathrm{span}(X, JX)$: every horizontal plane is holomorphic, and (7.30) gives $K \equiv 4$ with no other option. This recovers $\mathbb{C}\mathrm{P}^1 = S^2(\frac{1}{2})$ of curvature 4 from example 7.5.8, as it must. For $n \geq 2$ the totally real planes appear and the curvature ranges over the whole interval $[1, 4]$, so $\mathbb{C}\mathrm{P}^n$ is not of constant curvature; it is instead the model of a compact rank-one symmetric space, curvature-pinched between the round metrics of curvature 1 and 4.

8

Curvature and Topology

The comparison theorems of chapter 6 turned curvature bounds into metric conclusions: a diameter estimate, a diffeomorphism to Euclidean space, a comparison of Jacobi fields. This chapter turns curvature into topology. The quantity that mediates the passage is the total curvature, the integral of a curvature function over the whole manifold, and the recurring discovery is that such an integral, defined by the metric, in fact computes a number that depends only on the underlying smooth manifold and not on the metric at all. Curvature is free to vary from point to point and from metric to metric, but its integral is pinned to a topological invariant.

The prototype is the Gauss–Bonnet theorem, with which the chapter opens. For a closed surface the integral of the Gaussian curvature is exactly 2π times the Euler characteristic, so a sphere and a torus are distinguished by their total curvature no matter how either is bent. The generalization to higher even dimensions is the Gauss–Bonnet–Chern theorem, stated here and proved later in the book once the Chern–Weil machinery is available; it expresses the Euler characteristic of any closed even-dimensional manifold as the integral of a curvature form. The last two sections run the implication in the opposite direction, from a curvature sign to a topological constraint. Synge's theorem shows that positive sectional curvature, on a compact manifold, forces simple connectivity in the even orientable case and orientability in the odd case, through a second variation applied to a closed geodesic. The sphere theorem, the deepest result named in the chapter, shows that a manifold whose curvature is pinched strictly between $\frac{1}{4}$ and 1 must be a sphere; its proof lies beyond this book, but the pinching constant is sharp, and the Fubini–Study metric of chapter 7 is exactly the example that makes it so. With this passage from local curvature to global shape, the metric theory of curvature has reached its topological conclusions; one chapter of Part I remains, turning from manifolds in general to the most symmetric ones.

8.1 The Gauss–Bonnet Theorem for Surfaces

The sectional curvature of section 4.3 is a pointwise quantity: it reads off, plane by plane, how the manifold bends near a point. Nothing in its definition suggests that its values are constrained from afar. Yet on a closed surface they are, and the constraint is the sharpest possible. Integrate the Gaussian curvature over the whole surface and the answer is not a number that varies as the metric is deformed but a fixed integer multiple of 2π, namely $2\pi\chi(M)$, determined by the topology alone. A sphere admits metrics of wildly varying curvature, dented here, bulging there, but the total curvature is always 4π; a torus, whatever metric it carries, has total curvature exactly 0. This chapter opens with the theorem that first tied curvature to topology, and it remains the cleanest instance of the theme that occupies the rest of the chapter: local differential-geometric data, integrated, produce a global topological invariant.

The theorem comes in a local form and a global form. The local Gauss–Bonnet theorem accounts for the total curvature of a bounded region in terms of the turning of its boundary and the angles at its corners; it is a statement about a single triangle or polygon, provable by Stokes' theorem once the curvature is written as an exact form on a frame. The global theorem is assembled from the local one by triangulating the surface and summing: the boundary contributions cancel in pairs across shared edges, the corner angles reorganize into the Euler characteristic through the combinatorics of the triangulation, and $\int_M K\,dA = 2\pi\chi(M)$ falls out. This section proves both in full. The one external input is that the tangent direction of a simple closed curve turns through a total angle of exactly 2π, the theorem of turning tangents (Hopf's Umlaufsatz), whose proof is a degree computation belonging to topology and is cited to [4].

Throughout, (M, g) is an oriented Riemannian surface with Levi-Civita connection ∇ (definition 2.3.2), and K denotes its sectional curvature (definition 4.3.1), which on a surface is a single function on M since there is one tangent 2-plane at each point; it is the classical Gaussian curvature. The curvature sign convention is the one fixed in box 4.1.1, so that the round sphere S^2_r has $K = +1/r^2$ (example 4.3.4). The area form dA is the Riemannian volume form dV_g of definition 1.4.5 in dimension 2; it is the oriented 2-form assigning the value 1 to every positively oriented orthonormal frame.

Definition 8.1.1: Geodesic curvature

Let $\gamma\colon I \to M$ be a unit-speed curve on the oriented Riemannian surface M, and let n be the *leftward unit normal* to γ: the unique unit vector field along γ orthogonal to $\dot\gamma$ such that the ordered pair $(\dot\gamma, n)$ is a positively oriented orthonormal frame at each point. The *geodesic curvature* of γ is the function

$$\kappa_g = \langle D_t\dot\gamma, n\rangle$$

where D_t is the covariant derivative along γ (definition 2.1.6). It is the signed length of the tangential acceleration of γ, positive when γ bends toward its left.

The geodesic curvature isolates the part of a curve's acceleration that the surface itself feels. A unit-speed curve has $|\dot\gamma| \equiv 1$, so differentiating $\langle\dot\gamma, \dot\gamma\rangle = 1$ along γ gives $\langle D_t\dot\gamma, \dot\gamma\rangle = 0$: the acceleration $D_t\dot\gamma$ is everywhere orthogonal to the velocity. On the surface, orthogonal to $\dot\gamma$ means proportional to the leftward normal n, so $D_t\dot\gamma = \kappa_g\, n$ and the single number κ_g records the acceleration completely. A curve with $\kappa_g \equiv 0$ has $D_t\dot\gamma \equiv 0$, which is precisely the geodesic equation of definition 3.1.1: geodesics are the curves of vanishing geodesic curvature, the ones that do not bend within the surface. The sign is a genuine sign, fixed by the orientation, and it is what will let the boundary integrals of the local theorem combine correctly around a polygon.

Example 8.1.2: Geodesic curvature of a latitude circle

On the unit sphere S^2 with coordinates (θ, φ) and metric $g = d\theta^2 + \sin^2\theta\, d\varphi^2$ (example 4.3.4), fix a latitude $\theta = \theta_0$ and let γ trace the circle $\varphi \mapsto (\theta_0, \varphi)$. Its coordinate velocity is ∂_φ, of length $|\partial_\varphi| = \sin\theta_0$, so the unit-speed curve has velocity $\dot\gamma = (\sin\theta_0)^{-1}\partial_\varphi$. The covariant acceleration is computed from the Christoffel symbols $\Gamma^\theta_{\varphi\varphi} = -\sin\theta\cos\theta$ and $\Gamma^\varphi_{\theta\varphi} = \cot\theta$ of example 4.3.4. Along γ the velocity has constant components, so

$$D_t\dot\gamma = (\sin\theta_0)^{-2}\,\nabla_{\partial_\varphi}\partial_\varphi = (\sin\theta_0)^{-2}\,\Gamma^\theta_{\varphi\varphi}\,\partial_\theta = (\sin\theta_0)^{-2}(-\sin\theta_0\cos\theta_0)\,\partial_\theta = -\frac{\cos\theta_0}{\sin\theta_0}\,\partial_\theta$$

The leftward unit normal to $\dot\gamma$ (with $(\dot\gamma, n)$ positively oriented, and $\partial_\theta, \partial_\varphi$ ordered to match the orientation) is $n = \pm\partial_\theta$; taking the northward normal $n = -\partial_\theta$ for a curve traversed so that the enclosed cap lies to its left, the geodesic curvature is

$$\kappa_g = \langle D_t\dot\gamma, n\rangle = \left\langle -\frac{\cos\theta_0}{\sin\theta_0}\,\partial_\theta,\ -\partial_\theta\right\rangle = \frac{\cos\theta_0}{\sin\theta_0} = \cot\theta_0$$

a constant along the circle. At the equator $\theta_0 = \pi/2$ this is $\cot(\pi/2) = 0$: the equator is a geodesic, consistent with its being a great circle. As $\theta_0 \to 0$ the latitude shrinks toward the north pole and $\kappa_g = \cot\theta_0 \to +\infty$, the circle curving ever more sharply, as a small circle should.

To integrate curvature over a region the curvature must first be written in a form Stokes' theorem can act on. The device is a local orthonormal frame and its connection 1-form, which repackage the Levi-Civita connection into a single 1-form whose exterior derivative is the curvature. On a surface this is one equation, and it is the analytic engine of the whole section.

Proposition 8.1.3: The connection 1-form and the structure equation

Let $U \subseteq M$ be an open set on which there is a positively oriented local orthonormal frame (e_1, e_2), with dual coframe (θ^1, θ^2) (so $\theta^i(e_j) = \delta^i_j$). Define the *connection 1-form* ω_{12} on U by

$$\omega_{12}(X) = \langle \nabla_X e_1, e_2\rangle, \qquad X \in T_pM,\ p \in U$$

Then

1. $\nabla_X e_1 = \omega_{12}(X)\, e_2$ and $\nabla_X e_2 = -\omega_{12}(X)\, e_1$ for every X;
2. the area form on U is $dA = \theta^1 \wedge \theta^2$;
3. the structure equation holds:
$$d\omega_{12} = -K\, dA$$

The connection 1-form encodes how the frame rotates as one moves across U. Because (e_1, e_2) is orthonormal at every point, the covariant derivative of the frame in any direction is an infinitesimal rotation, and a rotation in two dimensions is a single number: the rate $\omega_{12}(X)$ at which e_1 turns toward e_2. The structure equation says that the failure of this rotation rate to be a gradient, its exterior derivative $d\omega_{12}$, is exactly the curvature. Curvature is the obstruction to the frame being locally constant, and $d\omega_{12} = -K\, dA$ makes that obstruction quantitative and, being an exact form's differential, ready for Stokes.

***Proof*:**
Step 1: the frame derivatives. Fix $X \in T_pM$. Since (e_1, e_2) is orthonormal, $\nabla_X e_1$ expands in this basis as $\nabla_X e_1 = \langle \nabla_X e_1, e_1 \rangle e_1 + \langle \nabla_X e_1, e_2 \rangle e_2$. Metric compatibility of the Levi-Civita connection (definition 2.2.4) gives $2 \langle \nabla_X e_1, e_1 \rangle = X \langle e_1, e_1 \rangle = X(1) = 0$, so the e_1-component vanishes and $\nabla_X e_1 = \langle \nabla_X e_1, e_2 \rangle e_2 = \omega_{12}(X) e_2$. Likewise $\langle \nabla_X e_2, e_2 \rangle = 0$, and differentiating $\langle e_1, e_2 \rangle = 0$ gives $\langle \nabla_X e_1, e_2 \rangle + \langle e_1, \nabla_X e_2 \rangle = 0$, so $\langle \nabla_X e_2, e_1 \rangle = - \langle \nabla_X e_1, e_2 \rangle = -\omega_{12}(X)$. Hence $\nabla_X e_2 = -\omega_{12}(X) e_1$, which is (i). That $X \mapsto \omega_{12}(X)$ is a 1-form (that is, $C^\infty(U)$-linear in X) is immediate from the C^∞-linearity of ∇_X in the direction X.

Step 2: the area form. The frame (e_1, e_2) is positively oriented and orthonormal, so its dual coframe (θ^1, θ^2) satisfies $(\theta^1 \wedge \theta^2)(e_1, e_2) = 1$. The Riemannian area form dA of definition 1.4.5 is the unique positively oriented 2-form assigning 1 to every positively oriented orthonormal frame, hence $dA(e_1, e_2) = 1$ as well. Two 2-forms on a surface agreeing on the single (up to positive scalar) frame (e_1, e_2) agree everywhere, so $dA = \theta^1 \wedge \theta^2$, which is (ii).

Step 3: reduce the curvature to $d\omega_{12}(e_1, e_2)$. The curvature endomorphism of definition 4.1.2 applied to the frame is

$$R(e_1, e_2)e_1 = \nabla_{e_1} \nabla_{e_2} e_1 - \nabla_{e_2} \nabla_{e_1} e_1 - \nabla_{[e_1, e_2]} e_1$$

Compute each term with (i). Writing $\omega = \omega_{12}$ for brevity, $\nabla_{e_2} e_1 = \omega(e_2) e_2$, and applying (i) once more,

$$\nabla_{e_1} \nabla_{e_2} e_1 = \nabla_{e_1} \big(\omega(e_2) e_2\big) = \big(e_1\, \omega(e_2)\big) e_2 + \omega(e_2)\, \nabla_{e_1} e_2 = \big(e_1\, \omega(e_2)\big) e_2 - \omega(e_2) \omega(e_1) e_1$$

Symmetrically,

$$\nabla_{e_2} \nabla_{e_1} e_1 = \nabla_{e_2} \big(\omega(e_1) e_2\big) = \big(e_2\, \omega(e_1)\big) e_2 - \omega(e_1) \omega(e_2) e_1$$

and, with $[e_1, e_2]$ a tangent vector,

$$\nabla_{[e_1, e_2]} e_1 = \omega([e_1, e_2])\, e_2$$

Subtracting, the e_1-terms cancel $(-\omega(e_2)\omega(e_1) + \omega(e_1)\omega(e_2) = 0)$ and the e_2-terms combine into

$$R(e_1, e_2)e_1 = \Big(e_1\, \omega(e_2) - e_2\, \omega(e_1) - \omega([e_1, e_2])\Big) e_2$$

The coefficient is exactly the invariant formula for the exterior derivative of a 1-form on the pair (e_1, e_2), namely $d\omega(e_1, e_2) = e_1(\omega(e_2)) - e_2(\omega(e_1)) - \omega([e_1, e_2])$. Therefore

$$R(e_1, e_2)e_1 = d\omega_{12}(e_1, e_2)\, e_2 \tag{8.1}$$

Step 4: identify the coefficient with $-K$. Taking the inner product of (8.1) with e_2 gives $\langle R(e_1, e_2)e_1, e_2 \rangle = d\omega_{12}(e_1, e_2)$. The left side is the $(0, 4)$-curvature $\operatorname{Riem}(e_1, e_2, e_1, e_2)$ of definition 4.1.4. Its relation to the sectional curvature runs through the antisymmetry of Riem in its last two slots (proposition 4.2.1): $\operatorname{Riem}(e_1, e_2, e_1, e_2) = -\operatorname{Riem}(e_1, e_2, e_2, e_1)$. Since (e_1, e_2) is orthonormal, the Gram determinant $G(e_1, e_2) = 1$, and the sectional curvature of the tangent plane (definition 4.3.1) is $K = \operatorname{Riem}(e_1, e_2, e_2, e_1)$. Hence

$$d\omega_{12}(e_1, e_2) = \operatorname{Riem}(e_1, e_2, e_1, e_2) = -\operatorname{Riem}(e_1, e_2, e_2, e_1) = -K$$

Both $d\omega_{12}$ and $-K\, dA$ are 2-forms on the surface U; they agree on the frame (e_1, e_2), since $(-K\, dA)(e_1, e_2) = -K\, dA(e_1, e_2) = -K$ by (ii), and two 2-forms on a surface agreeing on a frame are equal. Therefore $d\omega_{12} = -K\, dA$, completing the proof.

The connection 1-form is defined only on the frame's domain U, and on the overlap of two frames the two connection 1-forms differ by the differential of the angle between the frames. This is why ω_{12} is not a globally defined object on a surface that admits no global orthonormal frame, such as S^2; but the equation $d\omega_{12} = -K\,dA$ is local and correct on each U, and that is all the local Gauss–Bonnet theorem needs. The global theorem will circumvent the absence of a global frame by triangulating.

With the curvature written as $d\omega_{12}$, Stokes' theorem converts an area integral of K into a boundary integral of ω_{12}, and the boundary integral of ω_{12} measures the turning of the tangent. This is the local Gauss–Bonnet theorem. The setting is a region small enough to lie in a single frame's domain and to be, topologically, a disk with a polygonal boundary.

Theorem 8.1.4: Local Gauss–Bonnet

Let $R \subseteq M$ be a compact region lying in the domain U of a positively oriented orthonormal frame, homeomorphic to a closed disk, whose boundary ∂R is a piecewise-regular simple closed curve: a finite concatenation of regular unit-speed arcs $\gamma_1, \ldots, \gamma_k$, positively oriented (so that R lies to the left), meeting at vertices where the exterior angles are $\epsilon_1, \ldots, \epsilon_k \in (-\pi, \pi]$. Then

$$\iint\limits_R K\,dA + \int_{\partial R} \kappa_g\,ds + \sum_{i=1}^{k} \epsilon_i = 2\pi$$

where κ_g is the geodesic curvature of definition 8.1.1 along the smooth arcs and ds is arc length. In particular, if every boundary arc is a geodesic, so $\kappa_g \equiv 0$, then $\iint_R K\,dA + \sum_i \epsilon_i = 2\pi$.

The three terms measure three sources of turning as the boundary is traversed once. The corner angles ϵ_i are the abrupt turns at the vertices; the integral of κ_g is the continuous turning of the tangent within each smooth arc, as the arc bends inside the surface; and the curvature integral $\iint_R K\,dA$ is the turning contributed by the surface's own curvature, the amount by which parallel transport around ∂R fails to be the identity. Their sum is the total turning of the tangent direction, and for a simple closed curve bounding a disk that total is 2π: the tangent makes exactly one full revolution. The theorem is the accounting identity that equates the geometric decomposition on the left with the topological total on the right.

***Proof*:**
The region R lies in the domain U of a fixed positively oriented orthonormal frame (e_1, e_2), with connection 1-form ω_{12} and dual coframe as in proposition 8.1.3.

Step 1: Stokes converts the curvature integral into a boundary integral of ω_{12}. By the structure equation (proposition 8.1.3), $K\,dA = -d\omega_{12}$ on U, so

$$\iint\limits_R K\,dA = -\iint\limits_R d\omega_{12} = -\int_{\partial R} \omega_{12}$$

the last equality by Stokes' theorem ([6]), the boundary ∂R carrying its induced (positive) orientation. It remains to evaluate $\int_{\partial R} \omega_{12}$ in terms of the geodesic curvature and the turning of the tangent.

Step 2: the boundary integral of ω_{12} in terms of κ_g and the turning angle. Along a smooth boundary arc γ_i, parametrized by arc length, write the unit tangent in the frame as

$$\dot\gamma_i = \cos\varphi_i\, e_1 + \sin\varphi_i\, e_2 \tag{8.2}$$

where φ_i is a smooth angle function along γ_i measuring the tangent's direction relative to e_1; such a lift exists and is smooth because γ_i is a regular arc in the frame's domain. The leftward unit normal of definition 8.1.1 is then $n_i = -\sin\varphi_i\, e_1 + \cos\varphi_i\, e_2$, so that $(\dot\gamma_i, n_i)$ is a positively oriented orthonormal frame. Differentiate (8.2) covariantly along γ_i, using $D_t e_j = \nabla_{\dot\gamma_i} e_j$ and part (i) of proposition 8.1.3, and abbreviate $\omega = \omega_{12}(\dot\gamma_i)$:

$$D_t\dot\gamma_i = \dot\varphi_i(-\sin\varphi_i\, e_1 + \cos\varphi_i\, e_2) + \cos\varphi_i\,(\omega\, e_2) + \sin\varphi_i\,(-\omega\, e_1) = (\dot\varphi_i + \omega)\, n_i$$

Taking the inner product with n_i gives $\kappa_g = \langle D_t\dot\gamma_i, n_i\rangle = \dot\varphi_i + \omega_{12}(\dot\gamma_i)$, that is,

$$\omega_{12}(\dot\gamma_i) = \kappa_g - \dot\varphi_i \tag{8.3}$$

Integrating (8.3) over γ_i and summing over the arcs,

$$\int_{\partial R}\omega_{12} = \sum_i\int_{\gamma_i}\omega_{12}(\dot\gamma_i)\,ds = \sum_i\int_{\gamma_i}\kappa_g\,ds - \sum_i\int_{\gamma_i}\dot\varphi_i\,ds = \int_{\partial R}\kappa_g\,ds - \sum_i\Delta\varphi_i$$

where $\Delta\varphi_i$ is the net change of the angle φ_i along the arc γ_i, the turning of the tangent within that arc.

Step 3: the total turning is 2π (theorem of turning tangents). As the boundary is traversed once, the tangent direction turns by the sum of the continuous turnings $\sum_i\Delta\varphi_i$ along the arcs and the jumps at the vertices. At the i-th vertex the tangent jumps by the exterior angle ϵ_i, the signed angle from the incoming tangent to the outgoing tangent. The theorem of turning tangents, Hopf's Umlaufsatz, states that for a positively oriented, piecewise-regular *simple* closed curve bounding a disk, the tangent direction makes exactly one positive revolution, so the total turning is 2π:

$$\sum_i\Delta\varphi_i + \sum_i\epsilon_i = 2\pi \tag{8.4}$$

This is the one input taken from outside; it is a statement about the degree of the tangent map $\partial R \to S^1$ of a simple closed plane-domain curve, proved as a winding-number (degree) computation in [4], with the winding-number formulation available in [5]. That the angle functions φ_i measured relative to the frame e_1 differ from the plane-tangent angles only by the frame's own rotation, which contributes no net turning around the closed loop, is what allows the flat statement to be applied verbatim here; the discrepancy is a d(frame angle) term that integrates to zero around ∂R.

Step 4: assemble. From (8.4), $\sum_i\Delta\varphi_i = 2\pi - \sum_i\epsilon_i$. Substituting into the Step 2 identity,

$$\int_{\partial R}\omega_{12} = \int_{\partial R}\kappa_g\,ds - \Big(2\pi - \sum_i\epsilon_i\Big)$$

and combining with Step 1, $\iint_R K\,dA = -\int_{\partial R}\omega_{12}$, gives

$$\iint_R K\,dA = -\int_{\partial R}\kappa_g\,ds + 2\pi - \sum_i\epsilon_i$$

Rearranging yields $\iint_R K\,dA + \int_{\partial R}\kappa_g\,ds + \sum_i\epsilon_i = 2\pi$, as we sought to show.

The local theorem already has geometric consequences one can test. Take R a geodesic triangle, three geodesic arcs meeting at three vertices. Then $\kappa_g \equiv 0$ on the edges, and the exterior angles are $\epsilon_i = \pi - \alpha_i$ in terms of the interior angles α_i. The identity becomes $\iint_R K\, dA + \sum_i(\pi - \alpha_i) = 2\pi$, that is, $\sum_i \alpha_i - \pi = \iint_R K\, dA$. The interior angles of a geodesic triangle sum to π plus the total curvature it encloses: more than π where $K > 0$, exactly π where $K = 0$, less than π where $K < 0$. This is the sharp form of the ancient observation that the angle sum of a triangle reflects the curvature of the space, and it is worked out numerically below.

Example 8.1.5: The angle excess of a geodesic triangle

Let T be a geodesic triangle on the round sphere S^2_r of example 1.3.4, with interior angles α, β, γ; its edges are arcs of great circles, hence geodesics with $\kappa_g \equiv 0$ (definition 8.1.1). Apply theorem 8.1.4 with the exterior angles $\epsilon_1 = \pi - \alpha$, $\epsilon_2 = \pi - \beta$, $\epsilon_3 = \pi - \gamma$:

$$\iint_T K\, dA + \big((\pi - \alpha) + (\pi - \beta) + (\pi - \gamma)\big) = 2\pi$$

The sphere has constant $K = 1/r^2$ (example 4.3.4), so $\iint_T K\, dA = \frac{1}{r^2}\,\mathrm{Area}(T)$, and the equation rearranges to the spherical excess formula

$$\alpha + \beta + \gamma - \pi = \iint_T K\, dA = \frac{\mathrm{Area}(T)}{r^2}$$

The angle sum of a spherical triangle exceeds π by exactly the area divided by r^2, a positive amount. A triangle with three right angles, one eighth of the unit sphere (a vertex at the north pole and two on the equator a quarter-turn apart), has $\alpha = \beta = \gamma = \pi/2$, angle sum $3\pi/2$, excess $\pi/2$; on the unit sphere its area is indeed $\frac{1}{8} \cdot 4\pi = \pi/2$, matching the formula exactly. The two other model geometries realize the two other signs. In the Euclidean plane $K \equiv 0$, so the excess is zero and the angle sum is exactly π, the classical theorem. In the hyperbolic plane $K \equiv -1$ (example 4.3.4 records the model curvatures), the same identity gives $\alpha + \beta + \gamma - \pi = -\,\mathrm{Area}(T) < 0$: hyperbolic triangles have an angle *defect* equal to their area, and every hyperbolic triangle has angle sum strictly less than π.

The global theorem is obtained by cutting a closed surface into geodesic-friendly pieces, applying the local theorem to each, and adding. The interior-edge contributions cancel because each interior edge is traversed twice, once in each direction, and the geodesic-curvature integrand $\kappa_g\, ds$ changes sign under reversal. The exterior angles, summed over all faces at all vertices, reorganize by the combinatorics of the triangulation into $2\pi(V - E + F) = 2\pi\chi(M)$. What remains on the left is the total curvature.

Theorem 8.1.6: Global Gauss–Bonnet

Let M be a closed (compact, without boundary) oriented Riemannian surface with Gaussian curvature K. Then

$$\int_M K\, dA = 2\pi\chi(M)$$

where $\chi(M)$ is the Euler characteristic of M. In particular the total curvature $\int_M K\, dA$ depends only on the topology of M and not on the choice of Riemannian metric.

***Proof*:**
Step 1: triangulate. A smooth closed surface admits a smooth triangulation ([6]): a finite collection of curved triangles $T_1, \ldots, T_F$ covering M, meeting only along whole edges and at vertices, with each T_f contained in the domain of a positively oriented orthonormal frame and small enough to be a piecewise-regular disk to which theorem 8.1.4 applies. Let V, E, F be the numbers of vertices, edges, and faces (triangles). Each triangle has three edges and each edge is shared by exactly two triangles, so counting edge-incidences two ways gives $3F = 2E$. Orient each triangle by the orientation of M, so that its boundary is positively oriented; then each interior edge is traversed once in each direction by the two triangles sharing it.

Step 2: sum the local theorem over the faces. Apply theorem 8.1.4 to each triangle T_f, whose three interior angles $\alpha_{f,1}, \alpha_{f,2}, \alpha_{f,3}$ give exterior angles $\pi - \alpha_{f,j}$:

$$\iint_{T_f} K\,dA + \int_{\partial T_f} \kappa_g\,ds + \sum_{j=1}^{3}(\pi - \alpha_{f,j}) = 2\pi$$

Sum over all F faces. The three groups of terms are handled in Steps 3–5.

Step 3: the curvature integrals. The triangles cover M and overlap only on edges and vertices, sets of area zero, so $\sum_f \iint_{T_f} K\,dA = \int_M K\,dA$.

Step 4: the boundary integrals cancel. Each interior edge e is shared by two triangles, and by Step 1 they induce opposite orientations on e: one traverses e from a vertex a to a vertex b, the other from b to a. Along a fixed smooth edge the geodesic curvature computed with respect to the reversed orientation is the negative of that computed with the original orientation, because reversing the parametrization fixes the acceleration $D_t\dot{\gamma}$ while flipping the leftward normal $n \mapsto -n$, so $\kappa_g = \langle D_t\dot{\gamma}, n\rangle$ reverses sign, and with ds positive either way the line integral $\int_e \kappa_g\,ds$ reverses sign accordingly. Thus the two contributions $\int_e \kappa_g\,ds$ from the two adjacent triangles are equal in magnitude and opposite in sign, and cancel. Every edge is interior (the surface is closed, without boundary), so every boundary integral cancels in its pair:

$$\sum_f \int_{\partial T_f} \kappa_g\,ds = 0$$

Step 5: the angle terms give $2\pi\chi(M)$. Expand the angle sum:

$$\sum_f \sum_{j=1}^{3}(\pi - \alpha_{f,j}) = \sum_f 3\pi - \sum_f \sum_{j=1}^{3} \alpha_{f,j} = 3\pi F - \sum_{\text{all angles}} \alpha$$

The angles at each vertex, taken over all triangles meeting there, fill out the full turn around that vertex on the surface, summing to 2π (the surface is smooth, so a small circle about each vertex has total angle 2π). Summing over the V vertices, the total of all interior angles is $\sum_{\text{all angles}} \alpha = 2\pi V$. Using $3F = 2E$ from Step 1, $3\pi F = 2\pi \cdot \frac{3F}{2} = 2\pi E$, hence

$$\sum_f \sum_{j=1}^{3}(\pi - \alpha_{f,j}) = 2\pi E - 2\pi V = -2\pi(V - E)$$

Step 6: assemble. Summing the local identities over the F faces, the right-hand side is $2\pi F$, and by Steps 3–5 the left-hand side is

$$\int_M K\,dA + 0 - 2\pi(V - E) = 2\pi F$$

Solving, $\int_M K\,dA = 2\pi F + 2\pi(V-E) = 2\pi(V-E+F) = 2\pi\chi(M)$, where $\chi(M) = V - E + F$ is the Euler characteristic, an invariant of M independent of the triangulation and equal to $\sum_i (-1)^i \beta_i$ in terms of the Betti numbers ([4]). The metric entered only through K and dA on the left, while the right-hand side is topological; since the two are equal, the total curvature is independent of the metric, completing the proof.

The theorem is an equality between a quantity that has nothing to do with topology, the integral of a curvature that can be made large and positive here and large and negative there, and a quantity that has nothing to do with geometry, an alternating count of cells in a triangulation. That the two must agree constrains each by the other. Deform the metric on a closed surface however one likes and the pointwise curvature changes everywhere, but the changes conspire to leave $\int_M K\,dA$ fixed; there is a conservation law for total curvature, and its conserved value is $2\pi\chi(M)$. Read the other way, the theorem computes a topological invariant by an integral: to find $\chi(M)$ one may forgo counting cells and instead integrate the curvature of any convenient metric. The sphere is the first check.

Example 8.1.7: Total curvature of the sphere

The round sphere S^2_r of example 1.3.4 is closed and orientable, with constant curvature $K \equiv 1/r^2$ (example 4.3.4) and total area $4\pi r^2$. Its total curvature is therefore

$$\int_{S^2_r} K\,dA = \frac{1}{r^2}\cdot \mathrm{Area}(S^2_r) = \frac{1}{r^2}\cdot 4\pi r^2 = 4\pi$$

independent of the radius r, as theorem 8.1.6 requires: the radius is a metric datum, and total curvature sees only topology. Comparing with $2\pi\chi(S^2)$, this reads off the Euler characteristic of the 2-sphere,

$$\int_{S^2_r} K\,dA = 4\pi = 2\pi\cdot 2 = 2\pi\chi(S^2)$$

so $\chi(S^2) = 2$, recovered from geometry alone. The cancellation of r is the content of the theorem in this instance: as the sphere is enlarged its curvature $1/r^2$ falls while its area $4\pi r^2$ grows, and the product is pinned. Any other metric on S^2, however dented, gives the same total 4π: a metric with a region of negative curvature must compensate with enough positive curvature elsewhere to keep the integral at 4π, which is why no metric on the sphere can have $K \le 0$ everywhere.

The impossibility just noted is one of a family of topological restrictions that the global theorem places on which curvature signs a surface can carry. They follow by inspecting the sign of $\chi(M)$ forced by a sign condition on K.

Corollary 8.1.8: Curvature signs and topology

Let M be a closed oriented Riemannian surface.

1. If $K > 0$ everywhere, then $\chi(M) > 0$; a closed oriented surface with $\chi > 0$ is the sphere S^2 (and the sphere is the only closed surface, orientable or not, besides the projective plane $\mathbb{R}\mathrm{P}^2$, admitting $K > 0$).
2. If $K < 0$ everywhere, then $\chi(M) < 0$; in particular M has genus at least 2, and no metric of everywhere-negative curvature exists on the sphere or the torus.
3. The torus T^2 ($\chi = 0$) admits no Riemannian metric whose Gaussian curvature is everywhere positive or everywhere negative; any metric on it has curvature that changes sign or vanishes identically.

***Proof*:**

1. If $K > 0$ everywhere on the compact surface M, then $\int_M K\,dA > 0$, since the integrand is positive and dA is a positive area form of positive total mass. By theorem 8.1.6, $2\pi\chi(M) = \int_M K\,dA > 0$, so $\chi(M) > 0$. The closed oriented surfaces are classified by genus $h \geq 0$ with $\chi = 2 - 2h$ ([4]); $\chi > 0$ forces $h = 0$ and $\chi = 2$, the sphere. That $\mathbb{R}\mathrm{P}^2$ (nonorientable, $\chi = 1$) also admits $K > 0$ is realized by the round metric descended from S^2, but the corollary as stated concerns the oriented case, where S^2 is the only possibility.
2. If $K < 0$ everywhere, then $\int_M K\,dA < 0$, so $2\pi\chi(M) < 0$ and $\chi(M) < 0$. Among closed oriented surfaces $\chi = 2 - 2h < 0$ forces $h \geq 2$, genus at least two. Since $\chi(S^2) = 2 > 0$ and $\chi(T^2) = 0$, neither admits an everywhere-negative metric.
3. The torus has $\chi(T^2) = 0$ ([4]), so theorem 8.1.6 gives $\int_{T^2} K\,dA = 2\pi \cdot 0 = 0$ for every Riemannian metric. A continuous function with vanishing integral over a connected surface is either identically zero or takes both signs; hence K cannot be everywhere positive nor everywhere negative. The flat metric on $\mathbb{R}^2/\mathbb{Z}^2$ realizes $K \equiv 0$, consistent with $\int K\,dA = 0$.

In every case the sign of the total curvature, fixed by χ, obstructs a global sign condition on K, completing the proof.

The corollary is the prototype of an obstruction theorem: a purely local condition, a sign on the curvature at every point, is incompatible with a purely global datum, the Euler characteristic, unless the two signs match. The torus is the boundary case where the obstruction is total, forbidding any definite sign. A doughnut sitting in space has an outer rim where it curves like a sphere ($K > 0$) and an inner throat where it curves like a saddle ($K < 0$), and the theorem says the two regions must exactly balance, $\int K\,dA = 0$, no matter how the doughnut is shaped. The higher-genus surfaces admit metrics of constant negative curvature (hyperbolic metrics), and the sphere admits its round metric of constant positive curvature; only the torus is caught between, admitting a flat metric and nothing of constant nonzero sign. This trichotomy of the closed oriented surfaces into spherical, flat, and hyperbolic types, read off here from the sign of χ, is the two-dimensional case of the uniformization of Riemann surfaces and the first appearance of curvature as an arbiter of global geometry.

8.2 The Generalized Gauss–Bonnet–Chern Theorem

The Gauss–Bonnet theorem of section 8.1 equates two quantities of entirely different provenance. On one side stands $\int_M K\,dA$, an integral of the metric's curvature, a real number that a priori depends on the geometry of (M, g) in every detail. On the other stands $2\pi\chi(M)$, a multiple of the Euler characteristic, an integer read off from the topology of the underlying surface with no reference to any metric. That these agree for every metric on a closed oriented surface is the first instance of a pattern that recurs throughout geometry: a curvature integral, seemingly a soft geometric invariant, is forced by a hidden rigidity to take a value fixed by topology alone. The present section states the theorem that carries this pattern into all even dimensions.

The obstacle to generalization is that in higher dimensions there is no single scalar curvature that plays the role of the Gaussian curvature K. A four-dimensional manifold has a full curvature tensor with many independent components, and no obvious scalar built from it is destined to integrate to the Euler characteristic. The resolution, found by Chern in 1944, is that the correct integrand is not a scalar at all but a differential form of top degree, assembled from the curvature by an algebraic operation, the Pfaffian, that has no counterpart for odd-order matrices. This is why the theorem lives in even dimensions, and why the two-dimensional case, where the Pfaffian of a 2×2 antisymmetric matrix is a single entry, looks deceptively like a statement about a scalar. This section states the theorem, records the dimension-two reduction, and works out what it says on spheres, tori, and products; the construction of the Pfaffian curvature form and the proof that its integral computes χ are deferred to the Chern–Weil chapter of Part II, where the necessary machinery of curvature on vector bundles is available.

Throughout, M is a closed (compact, without boundary) oriented smooth manifold of even dimension $2n$, equipped with a Riemannian metric g and its Levi-Civita connection ∇, and the curvature sign convention is the one fixed in section 4.1. The Euler characteristic $\chi(M) = \sum_i (-1)^i \dim H^i(M; \mathbb{R})$ is the topological invariant of [4], independent of the metric.

The statement below packages the theorem in the form that makes its structure visible: there is a curvature form, canonically attached to the metric, whose integral is the Euler characteristic. The precise formula for that form in dimension $2n$ is the Pfaffian expression, whose definition belongs to Part II; here it is named and its dimension-two value is pinned down, which is all that is needed to see that the theorem contains section 8.1.

Theorem 8.2.1: The Generalized Gauss–Bonnet–Chern Theorem

Let (M, g) be a closed oriented Riemannian manifold of even dimension $2n$, with Levi-Civita connection ∇. There is a smooth differential form $e(\nabla)$ of top degree $2n$ on M, the *Euler form* of the metric, built algebraically and pointwise from the curvature tensor of ∇, such that

$$\int_M e(\nabla) = \chi(M)$$

the Euler characteristic of M. The Euler form is a polynomial in the components of the curvature, given in general by the Pfaffian of the curvature 2-form; in dimension $2n = 2$ it is

$$e(\nabla) = \frac{1}{2\pi} K \, dA$$

where K is the sectional (Gaussian) curvature (definition 4.3.1) and dA the Riemannian area form. In particular the integral $\int_M e(\nabla)$ is independent of the metric g: every Riemannian metric on the fixed closed oriented M^{2n} gives the Euler form the same integral, namely the integer $\chi(M)$.

The theorem asserts two things at once, and separating them clarifies the content. The first is the existence of a distinguished top-degree form $e(\nabla)$ constructed from the curvature, local data at each point. The second is that its integral, a real number that the local construction gives no reason to be special, equals the integer $\chi(M)$. The tension between these two is the entire content. Locally the Euler form varies with the metric, bunching curvature where the metric bends sharply and thinning it where the metric flattens; globally these local variations must cancel exactly, so that the total is pinned to $\chi(M)$ no matter how the curvature is redistributed. A metric of large positive curvature somewhere is compensated by curvature elsewhere; the ledger always balances to the same integer.

The mechanism behind this rigidity is that $e(\nabla)$, while it depends on the metric, represents a fixed de Rham cohomology class, the *Euler class* of the tangent bundle, which is a topological invariant independent of the connection used to compute it. Changing the metric changes $e(\nabla)$ by an exact form, whose integral over the closed manifold M vanishes by Stokes' theorem, so the integral is unchanged. This is the Chern–Weil principle: a curvature polynomial of the right kind is closed, and its cohomology class does not see the connection. The construction of $e(\nabla)$ from the curvature, the proof that it is closed, the identification of its class as the Euler class, and the theorem that this class pairs with the fundamental class to give $\chi(M)$, all belong to Chern–Weil theory and are carried out in Part II, in the chapter on Chern–Weil theory of vector bundles. The Pfaffian, the algebraic operation that produces $e(\nabla)$ from the curvature 2-form, is defined there; naming it here is a forward reference in prose, not an appeal to an object already in hand.

The dimension-two case is where the general statement meets the concrete theorem of section 8.1, and the reduction is exact.

Example 8.2.2: Recovering the Gauss–Bonnet theorem for surfaces

Take $2n = 2$, so M is a closed oriented surface. The theorem gives the Euler form

$$e(\nabla) = \frac{1}{2\pi} K \, dA$$

and asserts $\int_M e(\nabla) = \chi(M)$. Multiplying through by 2π,

$$\int_M K\, dA = 2\pi\, \chi(M)$$

which is precisely the global Gauss–Bonnet theorem theorem 8.1.6. The Gauss–Bonnet–Chern theorem is thus a true extension: the surface case is not an analogy but the literal specialization to $2n = 2$.

The round sphere makes the numbers explicit. On S^2_r the sectional curvature is constant, $K \equiv 1/r^2$ (example 4.3.4), and the total area is $4\pi r^2$, so

$$\int_{S^2_r} K\, dA = \frac{1}{r^2} \cdot 4\pi r^2 = 4\pi$$

independent of the radius r. Dividing by 2π,

$$\int_{S^2_r} e(\nabla) = \frac{1}{2\pi} \int_{S^2_r} K\, dA = \frac{4\pi}{2\pi} = 2 = \chi(S^2)$$

The Euler characteristic of the 2-sphere is 2, and the curvature integral returns exactly this integer for every r. Shrinking the sphere multiplies K by r^{-2} and divides the area by the same factor; the two changes cancel, and the total is fixed. This is the metric-independence of the theorem in its simplest form: a sphere of any radius, and indeed the surface S^2 carried by any Riemannian metric whatever, integrates its Euler form to 2.

The reduction to section 8.1 settles what the theorem says on surfaces. In higher even dimensions the same accounting holds, with χ now the alternating sum of all the Betti numbers rather than the surface formula $2 - 2\gamma$. Before turning to the even-dimensional examples, one structural point deserves emphasis: the theorem has nothing to say in odd dimensions, and the reason is topological, not a gap in the construction.

Corollary 8.2.3: Vanishing Euler characteristic in odd dimensions

A closed orientable manifold M of odd dimension has $\chi(M) = 0$. Consequently the Gauss–Bonnet–Chern theorem theorem 8.2.1 carries content only in even dimensions: in odd dimensions its assertion $\int_M e(\nabla) = \chi(M)$ would read $0 = 0$, and the curvature form of even degree $2n$ cannot even be the top-degree form on an odd-dimensional manifold.

***Proof*:**
Let $\dim M = m$ be odd. By Poincaré duality on the closed orientable manifold M ([4]), the Betti numbers satisfy $\beta_i = \dim H^i(M;\mathbb{R}) = \dim H^{m-i}(M;\mathbb{R}) = \beta_{m-i}$ for every i. The Euler characteristic is the alternating sum

$$\chi(M) = \sum_{i=0}^{m} (-1)^i \beta_i$$

Pair the term in degree i with the term in degree $m - i$. Because m is odd, i and $m - i$ have opposite parity, so $(-1)^i = -(-1)^{m-i}$; since $\beta_i = \beta_{m-i}$, the two paired terms are $(-1)^i\beta_i$ and $(-1)^{m-i}\beta_{m-i} = -(-1)^i\beta_i$, which cancel. As m is odd there is no self-paired middle term

($i = m - i$ would force $m = 2i$, even), so the degrees pair off completely and the whole sum vanishes. Hence $\chi(M) = 0$, as we sought to show.

The corollary explains a division that runs through the geometry of curvature and topology. In even dimensions the Euler characteristic is an integer that curvature can detect, and the Gauss–Bonnet–Chern theorem is the instrument that reads it. In odd dimensions the Euler characteristic is uniformly zero, so there is no such invariant for curvature to compute, and the theorem is silent. This is not a defect of the theory but a feature of it: the top-degree curvature form $e(\nabla)$ has even degree $2n$ by construction, and a form of even degree can be a volume-type integrand only on an even-dimensional manifold. The parity of the dimension governs both sides of the equation simultaneously, and they agree, trivially, in the odd case.

The requirement that M be orientable in the corollary is present because the cleanest proof uses Poincaré duality with real coefficients, which is stated for oriented manifolds. The conclusion $\chi(M) = 0$ in odd dimensions holds without orientability as well, since the orientation double cover $\widetilde{M} \to M$ is a closed orientable manifold of the same odd dimension, and Euler characteristics of a double cover and its base are related by $\chi(\widetilde{M}) = 2\chi(M)$; with $\chi(\widetilde{M}) = 0$ from the oriented case this forces $\chi(M) = 0$. The Gauss–Bonnet–Chern theorem itself does require orientability, both to integrate the top form over M and to make the Euler class of the tangent bundle available, so the oriented statement is the one that matters here.

The examples that carry content are therefore the even-dimensional ones. The prototype is the even-dimensional sphere, whose Euler characteristic is 2 in every even dimension, matching the surface case; the odd spheres, by the corollary, have Euler characteristic 0. The product formula for the Euler characteristic then generates a further supply of examples, the tori among them.

Example 8.2.4: Spheres, tori, and products

The Euler characteristics of the standard closed manifolds follow from their cohomology, and the Gauss–Bonnet–Chern theorem then predicts the value of the curvature integral $\int_M e(\nabla)$ for any Riemannian metric on each.

1. *Spheres.* The sphere S^m has $H^0(S^m;\mathbb{R}) = H^m(S^m;\mathbb{R}) = \mathbb{R}$ and all other real cohomology zero ([4]), so
$$\chi(S^m) = (-1)^0 \cdot 1 + (-1)^m \cdot 1 = 1 + (-1)^m$$
For $m = 2n$ even this is $\chi(S^{2n}) = 2$, and for m odd it is $\chi(S^{2n+1}) = 0$, consistent with corollary 8.2.3. The even-dimensional case says that for the round metric on S^{2n}, and for any Riemannian metric on S^{2n}, the Euler form integrates to 2; in dimension 2 this is the value $\chi(S^2) = 2$ already computed in example 8.2.2, and the pattern continues unchanged to S^4, S^6, and every even sphere.

2. *Products.* For closed oriented manifolds M and N, the Künneth theorem gives $H^k(M \times N;\mathbb{R}) = \bigoplus_{i+j=k} H^i(M;\mathbb{R}) \otimes H^j(N;\mathbb{R})$ ([4]), and taking the alternating sum of dimensions turns the direct sum into a product:
$$\chi(M \times N) = \chi(M)\,\chi(N) \tag{8.5}$$
A product of an even-dimensional and an odd-dimensional closed manifold therefore has Euler characteristic zero, since one factor already contributes a zero by corollary 8.2.3; a product of two even-dimensional manifolds multiplies their Euler characteristics.

3. *Tori.* The m-torus is the m-fold product $T^m = (S^1)^{\times m}$. Since $\chi(S^1) = 1 + (-1)^1 = 0$, the product formula (8.5) gives

$$\chi(T^m) = \chi(S^1)^m = 0^m = 0$$

for every $m \geq 1$, in particular $\chi(T^{2n}) = 0$ in every even dimension. The Gauss–Bonnet–Chern theorem then forces $\int_{T^{2n}} e(\nabla) = 0$ for any Riemannian metric on the torus. The flat metric makes this transparent: a flat metric has vanishing curvature, so its Euler form $e(\nabla)$, a curvature polynomial, is identically zero and integrates to zero. The theorem says more, that even a curved metric on T^{2n}, with regions of positive and negative curvature, must have its Euler form integrate to the same zero; the positive and negative contributions cancel exactly, enforced by the topology.

The three cases illustrate the range of the theorem. The even spheres carry the maximal-symmetry constant-curvature metric and integrate to 2; the tori carry a flat metric and integrate to 0; the products interpolate between these through the multiplicativity (8.5). In each case the curvature integral is decided before any metric is chosen, by the topology of the manifold alone.

The torus example marks the boundary of what curvature can and cannot prescribe. On a surface, the Gauss–Bonnet theorem forbids a metric of everywhere-positive curvature on the torus, because such a metric would force $\int_{T^2} K\, dA > 0$ against $2\pi\chi(T^2) = 0$; the vanishing Euler characteristic obstructs constant positive curvature. In higher even dimensions the Euler form is a more intricate curvature polynomial than K, and the sign of the curvature no longer controls the sign of $e(\nabla)$ pointwise, so the direct obstruction weakens. The theorem still pins the integral to $\chi(M)$, but reading off a pointwise curvature restriction from that integral is a subtler matter, one that the sphere theorem of section 8.4 and the Bochner technique of Part II take up by different means. What the Gauss–Bonnet–Chern theorem provides, in every even dimension, is the exact bridge with which the chapter opened: a curvature integral, computed from the metric, that is forced to equal a topological integer, computed from the manifold.

8.3 Synge's Theorem

Bonnet–Myers (theorem 6.3.1) extracted a topological conclusion from a curvature bound by pushing the second variation of energy far enough to shorten an over-long minimizing geodesic: a positive lower bound on Ricci curvature forces the manifold to be compact with finite fundamental group. Synge's theorem runs the same engine on a different fuel. Where Bonnet–Myers minimized among curves with fixed endpoints and detected an interior conjugate point, Synge minimizes among *closed* curves in a homotopy class and detects that a closed geodesic can be shortened by sliding it sideways. The variation field that does the sliding is not built by hand, as the sinusoidal fields of example 5.1.3 were; it is produced by parallel transport around the loop, and whether such a field exists at all is decided by a parity count that mixes the dimension of the manifold with its orientability. This is why the conclusion, unlike Bonnet–Myers, is sensitive to dimension parity and to orientation.

The theorem has two faces. In even dimensions, positive curvature and orientability together force simple connectivity, a conclusion far stronger than the finite fundamental group of Bonnet–Myers. In odd dimensions, positive curvature alone forces orientability. Both faces are proved by the same contradiction: assume the missing structure (a nontrivial loop, or an orientation-reversing loop), promote it to a shortest closed geodesic, and show that the second variation along a parallel normal field is strictly negative, so the geodesic was not shortest after all. The two ingredients are therefore a source of shortest closed geodesics and a source of parallel normal fields; the first is a minimization argument, the second a linear-algebra count. The section assembles them in that order.

Throughout, (M, g) is a Riemannian manifold with its Levi-Civita connection ∇, curvature endomorphism R in the sign convention of box 4.1.1, and D_t the covariant derivative along a curve. Sectional curvature K is that of definition 4.3.1. The positive-definiteness of g is used decisively, in the minimization and in the eigenvalue count alike.

The minimization step produces the object every subsequent argument acts on: a closed geodesic that is shortest in its free homotopy class. A free homotopy class of loops in M is a path component of the space of continuous maps $S^1 \to M$, equivalently a conjugacy class of elements of $\pi_1(M)$ once a basepoint is fixed ([4]); it is the natural home for closed curves, since a closed geodesic carries no preferred basepoint. On a compact manifold every nontrivial such class contains a shortest smooth closed geodesic, and this is the direct-method statement that opens the door.

Lemma 8.3.1: Shortest closed geodesic in a homotopy class

Let (M, g) be a compact Riemannian manifold and let $\mathcal{C}$ be a nontrivial free homotopy class of loops in M (a free homotopy class not containing the constant loops). Then $\mathcal{C}$ contains a closed geodesic γ of minimal length among all piecewise-smooth loops in $\mathcal{C}$, and this minimal length is positive.

The proof is the direct method of the calculus of variations: take a sequence of loops in the class whose lengths approach the infimum, reparametrize them to unit speed so they cannot oscillate wildly, extract a uniformly convergent subsequence by Arzelà–Ascoli, and identify the limit as a closed geodesic using the local minimization theory of section 3.4. Compactness of M enters twice: to bound the loops in a fixed compact set for Arzelà–Ascoli, and to supply, through corollary 6.1.5, a uniform radius below which geodesic balls are convex and shortest paths are unique.

***Proof*:**
Write $\ell = \inf \{L(c) \mid c \text{ a piecewise-smooth loop in } \mathcal{C}\}$ for the infimal length in the class, where L is the length functional of definition 3.4.1.

Step 1: the infimum is positive. Since M is compact it is complete by corollary 6.1.5, and there is a positive number ρ, a lower bound for the injectivity radius over the compact M, such that for every $p \in M$ the geodesic ball $B(p, \rho)$ is a normal ball in which the radial geodesic from p is the unique shortest path to each of its points (theorem 3.4.3). A loop of length less than ρ lies inside a single such ball $B(p, \rho)$ about its basepoint p, since every point of the loop is within distance $L(c) < \rho$ of p along the loop, hence within d_g-distance $< \rho$; a loop contained in a normal ball is null-homotopic in that ball, by the straight-line homotopy in normal coordinates, and therefore null-homotopic in M, contradicting the nontriviality of $\mathcal{C}$. Thus every loop in $\mathcal{C}$ has length at least ρ, and $\ell \geq \rho > 0$.

Step 2: a minimizing sequence, reparametrized to constant speed. Choose loops $c_k \in \mathcal{C}$ with $L(c_k) \to \ell$. Free homotopy is insensitive to reparametrization, so each c_k may be taken to have constant speed on $S^1 = \mathbb{R}/\mathbb{Z}$, that is $|\dot{c}_k| \equiv L(c_k)$; then $c_k \colon S^1 \to M$ is Lipschitz with constant $L(c_k)$, and since $L(c_k) \to \ell$ these Lipschitz constants are bounded, say by Λ, uniformly in k. The maps c_k all take values in the compact space M, and they are uniformly Lipschitz, hence equicontinuous. By the Arzelà–Ascoli theorem a subsequence, still written c_k, converges uniformly to a limit $\gamma\colon S^1 \to M$, which is again Λ-Lipschitz and satisfies $L(\gamma) \le \liminf_k L(c_k) = \ell$ by lower semicontinuity of length under uniform convergence.

Step 3: the limit lies in $\mathcal{C}$. Uniformly close loops are freely homotopic. Concretely, once $\sup_t d_g(c_k(t), \gamma(t)) < \rho$ for k large, the geodesic joining $c_k(t)$ to $\gamma(t)$ inside the normal ball of radius ρ is unique and depends continuously on t; the map $(s,t) \mapsto \exp_{\gamma(t)}\bigl(s\, w_k(t)\bigr)$, where $w_k(t)$ is the initial velocity of that short geodesic from $\gamma(t)$ to $c_k(t)$, is a free homotopy from γ to c_k. Hence γ lies in the same class $\mathcal{C}$ as c_k, so $\gamma \in \mathcal{C}$ and in particular γ is nonconstant; therefore $L(\gamma) \ge \ell$ by definition of ℓ. Combined with $L(\gamma) \le \ell$ from Step 2, $L(\gamma) = \ell$: the limit realizes the infimal length.

Step 4: the minimizer is a smooth closed geodesic. It remains to show γ, a length-minimizer in its class, is a smooth geodesic with no corners, including across the basepoint $t = 0 \sim 1$. Fix any $t_0 \in S^1$ and two nearby parameters $t_- < t_0 < t_+$ with $d_g(\gamma(t_-), \gamma(t_+)) < \rho$, so that $\gamma(t_-)$ and $\gamma(t_+)$ lie in a common normal ball. Were the restriction $\gamma|_{[t_-,t_+]}$ not the unique shortest path between its endpoints, replacing it by that shortest radial geodesic would produce a loop in $\mathcal{C}$ (the replacement is homotopic rel endpoints inside the ball, so the free homotopy class is unchanged) of strictly smaller length, contradicting $L(\gamma) = \ell$. Hence $\gamma|_{[t_-,t_+]}$ is, after unit-speed reparametrization, the minimizing radial geodesic, which is smooth by theorem 3.4.3 and proposition 3.4.7. As t_0 was arbitrary, including $t_0 = 0$, the loop γ is a smooth geodesic on all of S^1 with matching velocities at $t = 0$ and $t = 1$: a closed geodesic. Its length is $\ell > 0$, as we sought to show.

The lemma is where compactness pays for the existence of the geodesic; from here on the argument is local to a single loop and linear. The closed geodesic γ it produces has one property that drives everything: parallel transport once around γ is a linear isometry of the tangent space at the basepoint, and it fixes the velocity $\dot\gamma$. Whether it fixes anything *orthogonal* to $\dot\gamma$ is the question the next lemma answers, and the answer turns entirely on the parity of the dimension and on whether the isometry preserves or reverses orientation.

An orientation-preserving isometry of the plane is a rotation, which fixes no nonzero vector unless it is the identity; an orientation-reversing isometry of the plane is a reflection, which fixes an entire axis. In three dimensions the roles switch: a rotation of $\mathbb{R}^3$ always fixes its axis, while an orientation-reversing isometry (a rotary reflection) may fix nothing. The pattern is that a fixed vector is forced when the parity of the dimension and the orientation behavior are matched in the following precise way.

Lemma 8.3.2: An isometry with a matched parity fixes a vector

Let W be a finite-dimensional real inner-product space and $P\colon W \to W$ a linear isometry. Suppose that either

1. $\dim W$ is even and P reverses orientation ($\det P = -1$), or
2. $\dim W$ is odd and P preserves orientation ($\det P = +1$).

Then P has $+1$ as an eigenvalue; that is, P fixes some nonzero vector of W.

The claim is that under either parity condition $\ker(P - I) \ne 0$. The proof reads the eigenvalue structure of an orthogonal map, whose eigenvalues are ± 1 together with conjugate pairs on the unit circle, and shows that ruling out the eigenvalue $+1$ would put the number of (-1)-eigenvalues into a parity that contradicts $\det P$. The determinant and the dimension count together leave no room to avoid a fixed vector.

***Proof*:**
Set $m = \dim W$, and argue by contradiction: assume 1 is *not* an eigenvalue of P, and derive that the parity hypothesis fails. The contradiction gives $\ker(P - I) \neq 0$, hence a nonzero fixed vector.

Step 1: the eigenvalue tally. Since P is a linear isometry of an inner-product space, its matrix in any orthonormal basis is orthogonal, so every complex eigenvalue of P has modulus 1, and the non-real eigenvalues occur in conjugate pairs $\lambda, \bar{\lambda}$ with $\lambda\bar{\lambda} = |\lambda|^2 = 1$. The real eigenvalues of an orthogonal map are ± 1. By the assumption that 1 is not an eigenvalue, every real eigenvalue equals -1; let r be their number (with multiplicity) and let q be the number of conjugate pairs of non-real eigenvalues. Counting all m eigenvalues,

$$m = r + 2q \tag{8.6}$$

so m and r have the same parity.

Step 2: the determinant. The determinant of P is the product of all its eigenvalues. Each conjugate pair contributes $\lambda\bar{\lambda} = 1$, and the r real eigenvalues each contribute -1, so

$$\det P = (-1)^r \tag{8.7}$$

It remains to check that (8.7) and (8.6) are incompatible with the parity hypothesis in each case.

In case (i), $\dim W = m$ is even and $\det P = -1$. Then (8.7) gives $(-1)^r = -1$, so r is odd, while (8.6) with m even gives $r = m - 2q$ even. The number r cannot be both odd and even, a contradiction. In case (ii), $\dim W = m$ is odd and $\det P = +1$. Then (8.7) gives $(-1)^r = +1$, so r is even, and (8.6) gives $m = r + 2q$ even, contradicting m odd. In both cases the assumption that 1 is not an eigenvalue is untenable. Hence 1 is an eigenvalue of P, so $\ker(P - I)$ contains a nonzero vector, which P fixes, as we sought to show.

The two lemmas supply the two halves of the argument, and their combination is the theorem. The dimension parity that appears in the eigenvalue lemma is exactly the dimension of the space $\dot{\gamma}(0)^\perp$ that parallel transport acts on, which is $\dim M - 1$; whether that transport preserves or reverses orientation is dictated by the orientability of M and the class chosen. Reconciling the two parity conditions is the bookkeeping that splits Synge into its even and odd cases.

Theorem 8.3.3: Synge's Theorem

Let (M, g) be a compact Riemannian manifold with strictly positive sectional curvature, $K > 0$.

1. If $\dim M$ is even and M is orientable, then M is simply connected.

2. If $\dim M$ is odd, then M is orientable.

The theorem is proved by contradiction in both parts, and the two proofs differ only in which structure is assumed to fail and how the orientation behavior of parallel transport is arranged. The common core is a second-variation computation along a shortest closed geodesic with a parallel normal variation field. That computation is the closed-curve counterpart of theorem 5.1.1, and its derivation must account for the absence of fixed endpoints; the boundary terms that vanished there by the proper-variation hypothesis now vanish instead by periodicity, and this is checked once, inside the proof, before either case is addressed.

***Proof*:**
Both parts assume the desired conclusion fails, produce a shortest closed geodesic γ of some length $L > 0$ from lemma 8.3.1, and derive a contradiction by shortening γ through a closed-curve variation. The shared apparatus is set up first.

Setup: parallel transport around γ and the second variation of a closed-curve variation. Let $\gamma\colon \mathbb{R}/L\mathbb{Z} \to M$ be a shortest closed geodesic in a nontrivial (case a) or orientation-reversing (case b) free homotopy class, parametrized by arc length, $|\dot\gamma| \equiv 1$, and let $p = \gamma(0) = \gamma(L)$. Let $P\colon T_pM \to T_pM$ be parallel transport once around γ, from $t = 0$ to $t = L$ along γ. By proposition 2.4.4, P is a linear isometry of T_pM. The velocity $\dot\gamma$ is a parallel field along γ, since $D_t\dot\gamma = 0$ for a geodesic (definition 3.1.1, definition 2.4.1); as γ is a smooth closed geodesic its velocity matches at the join, $\dot\gamma(0) = \dot\gamma(L)$, so parallel transport of $\dot\gamma(0)$ around γ returns $\dot\gamma(L) = \dot\gamma(0)$, that is

$$P\big(\dot\gamma(0)\big) = \dot\gamma(0)$$

Since P is an isometry fixing $\dot\gamma(0)$, it preserves the orthogonal complement $W := \dot\gamma(0)^\perp \subseteq T_pM$, a subspace of dimension $\dim M - 1$, and restricts there to a linear isometry $P' = P|_W\colon W \to W$.

Suppose $v \in W$ is a nonzero vector fixed by P', so $P'v = v$ and $v \perp \dot\gamma(0)$. Let $V(t)$ be the parallel field along γ with $V(0) = v$; parallel transport around the loop returns $V(L) = Pv = P'v = v = V(0)$, so V is a *periodic* field, closing up smoothly across the join. It is parallel, $D_tV \equiv 0$, and normal to $\dot\gamma$ throughout: the function $\langle V, \dot\gamma\rangle$ has derivative $\langle D_tV, \dot\gamma\rangle + \langle V, D_t\dot\gamma\rangle = 0$ by metric compatibility, so $\langle V, \dot\gamma\rangle$ is constant, equal to its value $\langle v, \dot\gamma(0)\rangle = 0$ at $t = 0$. Thus V is a parallel, periodic, unit-length field normal to $\dot\gamma$.

This periodic field defines a closed-curve variation $\Gamma(s,t) = \exp_{\gamma(t)}\big(sV(t)\big)$, smooth on $(-\varepsilon, \varepsilon) \times (\mathbb{R}/L\mathbb{Z})$ for small ε, whose longitudinal curves $\gamma_s = \Gamma(s, \cdot)$ are smooth loops with variation field $\partial_s\Gamma(0, \cdot) = V$; each γ_s is freely homotopic to γ through the variation itself, hence lies in the same class. The second derivative $E''(0) = \frac{d^2}{ds^2}\big|_0 E(\gamma_s)$ is computed exactly as in theorem 5.1.1, with one change at the boundary. The derivation there reaches

$$E''(0) = \int_0^L \Big(\langle D_tD_sS, T\rangle + \langle R(S,T)S, T\rangle + |D_tS|^2\Big)\,dt$$

at $s = 0$ with $S = V$, $T = \dot\gamma$, and the first integrand equals $\frac{d}{dt}\langle D_sS, \dot\gamma\rangle$ along the geodesic base curve. Its integral is the boundary difference $\big[\langle D_sS, \dot\gamma\rangle\big]_0^L$. For a proper variation this vanished because the endpoints were pinned; here it vanishes by periodicity instead. The variation is t-periodic, $\Gamma(s,0) = \Gamma(s,L)$ for all s, so $\partial_s\Gamma$ and its covariant t-derivatives agree at $t = 0$ and $t = L$, and $\dot\gamma(0) = \dot\gamma(L)$; hence the integrand $\langle D_sS, \dot\gamma\rangle$ takes equal values at the two endpoints and the boundary difference is 0. Therefore, exactly as in theorem 5.1.1,

$$E''(0) = \int_0^L \Big(|D_tV|^2 - \langle R(V,\dot\gamma)\dot\gamma, V\rangle\Big)\,dt$$

Now specialize to the parallel normal field V just constructed. Its bending term vanishes, $D_tV \equiv 0$, so $|D_tV|^2 = 0$; and since $V \perp \dot\gamma$ with $|V| = 1$ and $|\dot\gamma| = 1$, the curvature term is the sectional curvature of the plane they span,

$$\langle R(V,\dot\gamma)\dot\gamma, V\rangle = \mathrm{Riem}(V,\dot\gamma,\dot\gamma,V) = K\big(V,\dot\gamma\big)$$

by definition 4.3.1, the Gram determinant being $|V|^2\,|\dot\gamma|^2 - \langle V,\dot\gamma\rangle^2 = 1$. Hence

$$E''(0) = -\int_0^L K\big(V(t),\dot\gamma(t)\big)\,dt \tag{8.8}$$

By hypothesis $K > 0$ everywhere, so the integrand is strictly positive and $E''(0) < 0$. The first variation vanishes as well: the first-variation computation gives $E'(0) = \big[\langle V,\dot\gamma\rangle\big]_0^L - \int_0^L \langle V, D_t\dot\gamma\rangle\,dt$, whose integral term is 0 because γ is a geodesic ($D_t\dot\gamma = 0$) and whose boundary term is 0 because V and $\dot\gamma$ are periodic, agreeing at the two ends. Thus $s = 0$ is a critical point of the smooth function $s \mapsto E(\gamma_s)$ with $E''(0) < 0$, hence a strict local maximum, and for small $s \neq 0$ the loop γ_s has $E(\gamma_s) < E(\gamma)$. By the length–energy inequality $L(\gamma_s)^2 \le 2L\,E(\gamma_s)$ of theorem 3.5.6, with equality for the constant-speed γ (so $L(\gamma)^2 = 2L\,E(\gamma)$), the length also strictly drops: $L(\gamma_s)^2 \le 2L\,E(\gamma_s) < 2L\,E(\gamma) = L(\gamma)^2$, hence $L(\gamma_s) < L(\gamma)$ for small $s \neq 0$. But γ_s lies in the same class as γ, contradicting that γ has minimal length in that class. Producing a fixed vector $v \in W$ for P' in each case therefore yields the desired contradiction, and this is where the two lemmas on parity enter.

1. *Even dimension, orientable: M is simply connected.* Suppose not, so $\pi_1(M) \neq 0$ and there is a nontrivial free homotopy class $\mathcal{C}$. Lemma 8.3.1 gives a shortest closed geodesic γ in $\mathcal{C}$, and the setup above produces $P' = P|_W$ on $W = \dot\gamma(0)^\perp$, of dimension $\dim M - 1$, which is odd since $\dim M$ is even. Because M is orientable and P arises from parallel transport of the Levi-Civita connection, P preserves orientation: parallel transport is a linear isometry (proposition 2.4.4) depending continuously on the path, and on an oriented manifold with a metric connection it maps an oriented orthonormal frame to an oriented orthonormal frame, so $\det P = +1$. As P fixes $\dot\gamma(0)$ and splits $T_pM = \mathbb{R}\dot\gamma(0) \oplus W$ orthogonally, $\det P = (+1)\cdot\det P'$, whence $\det P' = +1$: the restriction P' preserves orientation of W. Now $\dim W = \dim M - 1$ is odd and P' preserves orientation, which is case (ii) of lemma 8.3.2. That lemma provides a nonzero $v \in W$ with $P'v = v$, and the setup then produces the parallel periodic normal field V with $E''(0) < 0$ by (8.8), contradicting minimality of γ. Hence no nontrivial class exists, so $\pi_1(M) = 0$ and M is simply connected.

2. *Odd dimension: M is orientable.* Suppose not, so M is nonorientable. Nonorientability supplies a loop around which parallel transport reverses orientation: the orientation double cover $\pi\colon \widehat{M} \to M$ ([4]) is a connected two-sheeted cover, and a loop γ_0 whose lift to $\widehat{M}$ fails to close up is orientation-reversing, meaning that transporting a frame once around γ_0 flips its orientation. Such a loop is not null-homotopic (a contractible loop lifts to a loop), so it represents a nontrivial free homotopy class $\mathcal{C}$; moreover every loop in $\mathcal{C}$ is orientation-reversing, since the orientation behavior is the homomorphism $w_1\colon \pi_1(M) \to \mathbb{Z}/2$ ($\gamma_0 \mapsto 1$) that is constant on a free homotopy class. Lemma 8.3.1 gives a shortest closed geodesic γ in $\mathcal{C}$; being orientation-reversing, its total parallel transport P has $\det P = -1$. As before P fixes $\dot\gamma(0)$ and restricts to the isometry P' of $W = \dot\gamma(0)^\perp$, with $\det P = (+1)\cdot\det P'$, so $\det P' = -1$: P' reverses orientation of W. Here $\dim W = \dim M - 1$ is even, since $\dim M$ is odd, and P' reverses orientation, which is case (i) of lemma 8.3.2. Again P' fixes a nonzero $v \in W$, the parallel periodic normal field V gives $E''(0) < 0$ by (8.8), and γ is not shortest, a contradiction. Hence M admits no orientation-reversing loop, so M is orientable.

In each part the parity of $\dim W = \dim M - 1$ and the orientation behavior of P' meet the matched-parity hypothesis of lemma 8.3.2, yielding the fixed vector, the negative second variation (8.8), and the contradiction, completing the proof.

The two parts share one mechanism and diverge only in bookkeeping, and the bookkeeping is what makes the hypotheses non-negotiable. In part (a) the space W is odd-dimensional and P' must preserve orientation, which orientability of M guarantees; in part (b) W is even-dimensional and P' must reverse orientation, which nonorientability supplies. In both the fixed vector v launches a parallel normal field along which the geodesic can be pushed sideways at strictly negative energy cost, because $D_t V = 0$ removes the bending term entirely and $K > 0$ leaves the curvature term alone with its minus sign. The formula (8.8) is the whole theorem in one line: a shortest closed geodesic on a positively curved space is second-order unstable along any parallel normal field, so if such a field exists the geodesic cannot have been shortest. Synge's insight is that the parity count decides existence.

The conclusion of part (a) is markedly stronger than what Bonnet–Myers delivers, and the contrast is worth stating precisely, since both theorems take a positive curvature bound to a statement about π_1.

Corollary 8.3.4: Fundamental group under positive curvature

Let (M, g) be a compact, orientable, even-dimensional Riemannian manifold with sectional curvature $K > 0$. Then $\pi_1(M)$ is trivial. In particular M is simply connected, and its fundamental group is not just finite but zero.

***Proof*:**
This is precisely the conclusion of theorem 8.3.3(a): under the stated hypotheses M is simply connected, so $\pi_1(M) = 0$, as required.

Bonnet–Myers (corollary 6.3.2) assumes only a positive lower Ricci bound, weaker than a positive sectional bound, and concludes that $\pi_1(M)$ is finite. It says nothing about the order of that finite group, and it applies in every dimension and without an orientability hypothesis. Synge's corollary trades the weaker curvature hypothesis for a stronger one, adds parity and orientability, and in return collapses the finite group all the way to the trivial group. The gap between "finite" and "trivial" is real: there exist compact manifolds with $K > 0$ and nontrivial finite π_1, so Bonnet–Myers cannot be improved to Synge's conclusion without Synge's extra hypotheses. The odd-dimensional real projective spaces are exactly such manifolds, and the following example lays out how each hypothesis of the theorem is used.

Example 8.3.5: Spheres and projective spaces
The round sphere and the real projective spaces test each hypothesis of theorem 8.3.3 against a curvature that is constant and positive, so the sectional condition $K > 0$ holds throughout. Recall that $\mathbb{RP}^n = S^n/\{\pm 1\}$ carries the metric descended from the round metric of example 1.2.3, for which the antipodal double cover $S^n \to \mathbb{RP}^n$ is a local isometry; hence $\mathbb{RP}^n$ has constant sectional curvature $K \equiv 1 > 0$, inherited from S^n (example 4.3.4), and $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2$ for $n \geq 2$, generated by a loop that lifts to a path from a point of S^n to its antipode.

1. *The even sphere S^{2n}*. It is compact, orientable, even-dimensional, with $K \equiv 1 > 0$, so part (a) applies and predicts simple connectivity. This is consistent: S^{2n} is simply connected for $n \geq 1$, indeed $\pi_1(S^m) = 0$ for all $m \geq 2$. The theorem here confirms a known fact rather

than producing a new one, and it is the model case in which all hypotheses of (a) are met.

2. *Even projective space* $\mathbb{R}\mathrm{P}^{2n}$. It is compact, even-dimensional, with $K \equiv 1 > 0$, but it is *nonorientable* (real projective space is orientable exactly in odd dimensions, since the antipodal map of S^m preserves orientation iff m is odd). Its fundamental group is $\mathbb{Z}/2 \neq 0$, so it is not simply connected. This does not contradict part (a): the orientability hypothesis fails, and $\mathbb{R}\mathrm{P}^{2n}$ shows that this hypothesis cannot be dropped. A positively curved compact even-dimensional manifold can have nontrivial π_1 if it is nonorientable.

3. *Odd projective space* $\mathbb{R}\mathrm{P}^{2n+1}$. It is compact, odd-dimensional, with $K \equiv 1 > 0$, so part (b) applies and predicts orientability. This is consistent: $\mathbb{R}\mathrm{P}^{2n+1}$ is orientable, the antipodal map of S^{2n+1} preserving orientation. Note what part (b) does *not* claim: it says nothing about π_1, and indeed $\pi_1(\mathbb{R}\mathrm{P}^{2n+1}) = \mathbb{Z}/2 \neq 0$, so $\mathbb{R}\mathrm{P}^{2n+1}$ is a compact positively curved manifold that is orientable yet not simply connected. This is exactly the "finite but not trivial" case that Bonnet–Myers permits and that Synge's part (a) forbids only under the even-plus-orientable hypotheses, which fail here on the parity.

The three cases exhibit the theorem's sensitivity to parity and orientation as a single coherent pattern. Even and orientable forces triviality of π_1; dropping orientability (case 2) or changing parity to odd (case 3) permits $\pi_1 = \mathbb{Z}/2$, and in the odd case the theorem retreats to the weaker orientability conclusion, which the odd projective spaces satisfy. The odd projective spaces also mark the boundary of Synge's reach: $K > 0$ and compactness alone, without the even-orientable pairing, cannot force simple connectivity, and $\mathbb{R}\mathrm{P}^{2n+1}$ is the witness.

8.4 The Sphere Theorem and Pinching

The comparison theorems of the previous chapter extract metric and topological control from a curvature bound: Bonnet–Myers (theorem 6.3.1) turns a positive lower bound on Ricci curvature into a diameter bound and compactness, and Rauch (theorem 6.5.2) turns a sectional-curvature bound into control on the spreading of geodesics. Each of these constrains the manifold without pinning down its diffeomorphism type. The sphere theorem is the sharpest result of this kind: a two-sided sectional-curvature bound of a specific numerical strength does determine the manifold, forcing it to be a sphere. The strength required is a *pinching* condition, a bound of the form $\delta \leq K \leq 1$ with the ratio δ close enough to 1 that the manifold cannot deviate far from constant curvature 1, whose complete simply connected model is the round sphere.

The subject sits at the boundary of what this book proves. The topological sphere theorem, which concludes that a strictly $\frac{1}{4}$-pinched manifold is homeomorphic to a sphere, assembles exactly the comparison machinery already in hand, together with one further estimate on the injectivity radius; its proof is attributed here and its ingredients named, but written out in full it would occupy a chapter of its own. The differentiable sphere theorem, which upgrades the conclusion to a diffeomorphism, was open for half a century and was settled only in 2009 by an argument through the Ricci flow, a tool developed in the closing chapter of this book. This section states both theorems precisely, fixes the pinching terminology they rely on, and records the example that shows the constant $\frac{1}{4}$ cannot be improved.

Throughout, (M, g) is a Riemannian manifold with the sectional curvature K of definition 4.3.1, and the Ch. 4 sign convention is in force, so the round sphere S^n_r of example 1.3.4 has constant

$K = 1/r^2 > 0$.

Definition 8.4.1: Pinched curvature

Let (M, g) be a Riemannian manifold with sectional curvature bounded between two positive constants, and rescale the metric so that $\max K = 1$. For $\delta \in (0, 1]$, the manifold is *δ-pinched* if every sectional curvature satisfies

$$\delta \leq K \leq 1$$

and *strictly δ-pinched* if the lower bound is strict, $\delta < K \leq 1$. The number δ is a *pinching constant*; the largest δ for which the bound holds is the *pinching* of the metric. When the two-sided bound is required only at each point separately, with the same δ but no fixed scale relating different points, the manifold is *pointwise δ-pinched*: for every $p \in M$ and all 2-planes $\Pi, \Pi' \subseteq T_pM$, $K(\Pi) \geq \delta\, K(\Pi')$; it is *strictly pointwise δ-pinched* if this inequality is strict, $K(\Pi) > \delta\, K(\Pi')$ for all such Π, Π'.

The normalization $\max K = 1$ removes the ambiguity of overall scale: multiplying g by a positive constant c multiplies every sectional curvature by $1/c$, so the ratio of the smallest to the largest curvature is the invariant content of a two-sided bound, and fixing $\max K = 1$ selects one representative from each scaling family. A δ-pinched metric is one whose curvatures all lie in the band $[\delta, 1]$; the closer δ is to 1, the narrower the band, and $\delta = 1$ forces constant curvature $K \equiv 1$. The pointwise version is weaker in one respect and stronger in another: it does not compare curvatures at different points to a common scale, but it does demand the ratio bound at every point, which is the hypothesis under which the differentiable sphere theorem is proved. For a manifold that is δ-pinched in the global sense, the pointwise ratio bound holds automatically with the same δ, since all curvatures already lie in $[\delta, 1]$.

Example 8.4.2: The round sphere and the pinching scale

The round sphere S^n_r has constant sectional curvature $K \equiv 1/r^2$. Rescaling to $r = 1$ gives $K \equiv 1$, so after the normalization $\max K = 1$ every sectional curvature equals 1 and the sphere is δ-pinched for *every* $\delta \leq 1$, including $\delta = 1$: it is the extreme case in which the pinching band collapses to the single value 1. Constant curvature is the tightest possible pinching, and the round sphere realizes it.

At the opposite end, a manifold with sectional curvatures filling the whole interval $[\delta, 1]$ for some δ close to 0 is only weakly pinched. The interest of the sphere theorem is the threshold in between: there is a definite value of δ, namely $\frac{1}{4}$, above which the pinching band is narrow enough to force the manifold to be a sphere, and at or below which it is not.

The choice of threshold is not arbitrary, and the reason it is exactly $\frac{1}{4}$ is best seen by first stating the theorem and then exhibiting the manifold that sits precisely at the boundary. The topological sphere theorem is the statement that strict $\frac{1}{4}$-pinching suffices to determine the homeomorphism type.

Theorem 8.4.3: Topological Sphere Theorem

Let M be a complete, simply connected Riemannian manifold that is strictly $\frac{1}{4}$-pinched: after rescaling so that $\max K = 1$, every sectional curvature satisfies

$$\tfrac{1}{4} < K \leq 1$$

Then M is homeomorphic to the sphere S^n, where $n = \dim M$.

None of the hypotheses can be dropped. Completeness is needed to bring Hopf–Rinow and Bonnet–Myers to bear; without it the manifold could be an open piece of a sphere, which is not homeomorphic to S^n. Simple connectivity excludes the quotients of the sphere: real projective space $\mathbb{R}\mathrm{P}^n = S^n/\{\pm 1\}$ inherits from S^n the constant curvature 1, hence is $\frac{1}{4}$-pinched (indeed 1-pinched), yet it is not homeomorphic to a sphere for $n \geq 2$, its fundamental group being $\mathbb{Z}/2$. The strictness of the lower bound is the subtlest hypothesis and is the point of the sharpness discussion below: the weakly $\frac{1}{4}$-pinched manifolds include examples that are not spheres, so the inequality $\frac{1}{4} < K$ cannot be relaxed to $\frac{1}{4} \leq K$. Positive-definiteness is used everywhere, and the theorem has no Lorentzian analogue.

The proof is not carried out here; the result is attributed to Harry Rauch, who proved a version with a pinching constant near 0.75 in 1951, and to Marcel Berger and Wilhelm Klingenberg, who independently brought the constant down to the optimal $\frac{1}{4}$ around 1960. The argument combines exactly the tools of chapter 6 with one estimate this book has not proved. The ingredients are the following. Bonnet–Myers (theorem 6.3.1) applied to the pinched bound $K > \frac{1}{4}$, hence $\mathrm{Ric} \geq (n-1)\frac{1}{4}\, g$, gives $\mathrm{diam}(M) \leq 2\pi$, so M is compact. Rauch comparison (theorem 6.5.2) against the model spheres of curvature $\frac{1}{4}$ and 1 controls the geometry of geodesic balls from both sides: geodesics spread at most as slowly as on the sphere of curvature 1 (so the first conjugate point occurs no sooner than distance π) and at least as slowly as on the sphere of curvature $\frac{1}{4}$ (so the first conjugate point occurs no later than distance 2π). The missing estimate is *Klingenberg's injectivity-radius lemma*: on an even-dimensional, simply connected, strictly $\frac{1}{4}$-pinched manifold, the injectivity radius is at least π, matching the conjugate-radius bound so that no short geodesic loop can form. With these in place, one covers M by two geodesic balls, one about a point p and one about a point q nearly antipodal to it, each an image of a Euclidean ball under an exponential map that is a diffeomorphism on the ball of radius π; the two balls overlap in an annular region, and a gluing map built from the geodesics between them exhibits M as two disks glued along their boundary spheres, which is the standard construction of a topological sphere. The strict pinching enters precisely in Klingenberg's lemma and in keeping the two centers far enough apart that the balls cover M; weakening it to $\frac{1}{4} \leq K$ breaks the injectivity-radius estimate.

The conclusion is a homeomorphism, not a diffeomorphism, and the gap between the two is genuine. The construction above glues two disks along a homeomorphism of their boundary spheres, and such a gluing always yields a manifold homeomorphic to S^n; but whether it is diffeomorphic to the standard S^n depends on the isotopy class of the gluing map, and in dimensions where exotic spheres exist the gluing can produce a smooth manifold homeomorphic but not diffeomorphic to the standard sphere. Closing this gap, and proving that strict $\frac{1}{4}$-pinching forces the standard smooth structure, required a different method entirely.

Theorem 8.4.4: Differentiable Sphere Theorem

Let M be a compact Riemannian manifold that is strictly pointwise $\frac{1}{4}$-pinched: for every $p \in M$ and all 2-planes $\Pi, \Pi' \subseteq T_pM$, the sectional curvatures satisfy $K(\Pi) > \frac{1}{4}\,K(\Pi')$, with $K > 0$. Then M is diffeomorphic to a spherical space form S^n/Γ, where Γ is a finite subgroup of $O(n+1)$ acting freely and orthogonally on S^n. In particular, if M is simply connected, then M is diffeomorphic to the standard sphere S^n.

The differentiable sphere theorem strengthens the conclusion from homeomorphism to diffeomorphism and simultaneously relaxes the hypothesis from global to pointwise pinching, dropping the requirement of simple connectivity in exchange for the more general spherical-space-form conclusion. It was proved by Simon Brendle and Richard Schoen in 2009, resolving a problem that had been open since the topological version was completed around 1960. The method is not comparison geometry but the Ricci flow, the evolution of the metric g by $\partial_t g = -2\,\mathrm{Ric}$, which deforms a pinched metric toward one of constant curvature; Brendle and Schoen showed that strict pointwise $\frac{1}{4}$-pinching is preserved by the flow and improves under it, so that the metric converges, after rescaling, to a round metric, whose underlying manifold is a spherical space form. The Ricci flow is developed in the closing chapter of this book, and the argument is deferred to that chapter in prose; it uses no machinery from the present one beyond the definition of pinching. The pointwise hypothesis is what the flow requires, and it is weaker than the global $\frac{1}{4}$-pinching of theorem 8.4.3: a metric can be pointwise $\frac{1}{4}$-pinched without any global scale relating the curvatures at distant points.

The two theorems together settle the diffeomorphism type of a strictly $\frac{1}{4}$-pinched manifold completely. What remains is to see that $\frac{1}{4}$ is the exact threshold, neither too large nor capable of being lowered. The example that forces $\frac{1}{4}$ is the Fubini–Study metric on complex projective space, computed in example 7.5.9.

8.4.5 Note: sharpness of the constant $\frac{1}{4}$ The constant $\frac{1}{4}$ in theorem 8.4.3 cannot be weakened to a non-strict bound. The Fubini–Study metric on complex projective space $\mathbb{C}\mathrm{P}^n$, derived in example 7.5.9 as the base of the Riemannian submersion $S^{2n+1}(1) \to \mathbb{C}\mathrm{P}^n$, has sectional curvatures filling the interval

$$1 \leq K \leq 4$$

the value 4 attained on holomorphic 2-planes $\mathrm{span}(X, JX)$ and the value 1 on totally real planes. Rescaling so that $\max K = 1$, that is, replacing the metric g by $4g$ (which divides every curvature by 4), the curvatures fill $[\frac{1}{4}, 1]$: the Fubini–Study metric is *exactly* $\frac{1}{4}$-pinched, weakly but not strictly, since the value $\frac{1}{4}$ is attained on the totally real planes.
For $n \geq 2$ the manifold $\mathbb{C}\mathrm{P}^n$ is not homeomorphic to any sphere. Its homology is nontrivial in every even degree up to $2n$: the cohomology ring is $\mathbb{Z}[h]/(h^{n+1})$ with h in degree 2, so $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}$ for $0 \leq k \leq n$, whereas a sphere S^m has nonzero cohomology only in degrees 0 and m. Thus $\mathbb{C}\mathrm{P}^n$ is a compact, simply connected, weakly $\frac{1}{4}$-pinched manifold that is not a topological sphere, and the strict inequality in theorem 8.4.3 is necessary: replacing $\frac{1}{4} < K$ by $\frac{1}{4} \leq K$ would admit $\mathbb{C}\mathrm{P}^n$ as a counterexample. The threshold $\frac{1}{4}$ is the pinching of the Fubini–Study metric, and the sphere theorem is sharp against it.

The Fubini–Study metric is not an isolated example. It belongs to a short list of homogeneous spaces, the compact rank-one symmetric spaces, whose pinchings cluster exactly at the boundary of the sphere theorem and explain why $\frac{1}{4}$ appears.

Example 8.4.6: The weakly $\frac{1}{4}$-pinched symmetric spaces

The compact rank-one symmetric spaces are the round sphere S^n, complex projective space $\mathbb{C}\mathrm{P}^n$, quaternionic projective space $\mathbb{HP}^n$, and the Cayley plane $\mathbb{OP}^2$. With their symmetric metrics, each is homogeneous, so its curvatures are the same at every point, and each is weakly $\frac{1}{4}$-pinched after the normalization $\max K = 1$. They form the family sitting on the closed boundary of the sphere-theorem hypothesis.

1. The round sphere S^n has constant curvature, and after scaling $\max K = 1$ every sectional curvature equals 1; it is 1-pinched, the tightest possible, as recorded in example 8.4.2. It is the one member of the family that is a sphere.
2. Complex projective space $\mathbb{C}\mathrm{P}^n$ with the Fubini–Study metric of example 7.5.9 has $K \in [1, 4]$, so after scaling $\max K = 1$ it is exactly $\frac{1}{4}$-pinched with $K \in [\frac{1}{4}, 1]$: the holomorphic planes attain $\frac{1}{4}$ and the totally real planes attain 1. For $n \geq 2$ it is not a sphere, as shown in box 8.4.5.
3. Quaternionic projective space $\mathbb{HP}^n$ and the Cayley plane $\mathbb{OP}^2$ carry symmetric metrics whose sectional curvatures likewise fill $[1, 4]$ before normalization, with the analogous submersion normalization inherited from the round sphere in place of the Hopf submersion of example 7.5.9, so after scaling they too are exactly $\frac{1}{4}$-pinched with $K \in [\frac{1}{4}, 1]$. Neither is a sphere: $\mathbb{HP}^n$ has nontrivial cohomology in degrees $0, 4, 8, \ldots, 4n$, and $\mathbb{OP}^2$ is a 16-dimensional manifold with nontrivial cohomology in degrees $0, 8, 16$.

Every one of these spaces except the sphere is weakly $\frac{1}{4}$-pinched and fails to be a topological sphere, so the family exhibits, in each dimension where it appears, a manifold that meets the pinching bound $\frac{1}{4} \leq K \leq 1$ non-strictly and is not a sphere. This is the structural reason the sphere theorem's constant is $\frac{1}{4}$ and its lower bound must be strict: the projective spaces are precisely the obstruction that a strict bound excludes and a non-strict bound would admit.

The pattern that emerges is a clean dichotomy. A complete simply connected manifold whose curvature is confined to the band $[\frac{1}{4}, 1]$ is either a sphere or, if it touches the lower end $\frac{1}{4}$ on some plane, one of the projective models $\mathbb{C}\mathrm{P}^n$, $\mathbb{HP}^n$, $\mathbb{OP}^2$. The strict inequality $\frac{1}{4} < K$ rules out the second alternative and leaves only the sphere. This dichotomy is made precise and proved by the *rigidity* refinement of the sphere theorem: Berger's rigidity theorem states that a complete simply connected manifold that is weakly $\frac{1}{4}$-pinched, $\frac{1}{4} \leq K \leq 1$, is either homeomorphic to a sphere or isometric to one of the compact rank-one symmetric spaces $\mathbb{C}\mathrm{P}^n$, $\mathbb{HP}^n$, $\mathbb{OP}^2$ with a symmetric metric. Its proof, like those of the two sphere theorems, lies outside this book and is due to Berger in the topological case and to Brendle and Schoen in the differentiable case, where the weakly pinched conclusion is that M is either a spherical space form or isometric to a rank-one symmetric space.

The comparison with the constant-curvature situation returns to the earlier chapters. A manifold of constant curvature is 1-pinched, and the Killing–Hopf classification identifies the complete simply connected ones as the model spaces exactly. The sphere theorem asks how far the curvature may be allowed to vary from that constant value while still forcing the sphere, and the answer is that the curvature may drop as low as, but not including, one quarter of its maximum. Below that ratio the projective spaces intervene, and the connection between curvature and topology, tight enough through Gauss–Bonnet and Synge to constrain a manifold, becomes tight enough here to determine it.

9

Homogeneous and Symmetric Spaces

Part I closes with the most uniform Riemannian manifolds, the ones whose geometry looks the same at every point. Every construction so far has allowed the metric to vary freely: curvature was permitted to change from point to point, and the theorems measured what that variation forces. This chapter turns to the opposite extreme, the spaces so symmetric that a single tangent space determines the whole geometry, and asks what such uniformity requires and what it produces.

The organizing notion is the isometry group. A manifold is homogeneous when its isometries act transitively, so that any point can be carried to any other; it is symmetric when, in addition, each point is the center of an isometric reflection that reverses geodesics. The chapter builds these ideas in four steps. It opens with the isometry group itself and the Myers–Steenrod theorem, which makes it a Lie group and bounds its size, with the constant-curvature models of chapter 1 realizing the maximum. Homogeneous manifolds are then presented as quotients G/H of a Lie group by an isotropy subgroup, and their invariant metrics are matched with invariant inner products on a single tangent space, reducing global geometry to linear algebra. The Lie groups themselves, carrying bi-invariant metrics, supply the richest examples: their Levi-Civita connection is the bracket, their geodesics are the one-parameter subgroups, and their curvature is nonnegative and computed by the Jacobi identity, so that $SU(2)$ reappears as the round three-sphere already met in the Hopf fibration of chapter 7. The chapter ends with the symmetric spaces, where the curvature tensor is parallel and the geometry is rigid enough to be classified; the classification itself, Cartan's, belongs to the theory of Lie groups and is stated with its proof deferred there. The model spaces, the projective spaces, and the compact Lie groups all fall into place as symmetric spaces of the three types, and the curvature that has governed all of Part I is here as constrained as it can be.

9.1 Isometry Groups and the Myers–Steenrod Theorem

The isometry group $\mathrm{Isom}(M, g)$ was introduced in definition 1.3.3 as an abstract group and used only through the size of its orbits, which recorded homogeneity and isotropy of the model spaces. This chapter takes the isometry group seriously as a geometric object, and its first task is to give the group its own geometry. The organizing principle is rigidity: an isometry of a connected Riemannian manifold is not free to be prescribed pointwise but is pinned down by a finite amount of data at a single point, its value and its differential there. This is a metric phenomenon with no counterpart for diffeomorphisms, where a map can be altered on any open set without disturbing it elsewhere. The rigidity has two consequences developed here. It bounds the isometry group in size, forcing $\dim \mathrm{Isom}(M) \le \binom{n+1}{2}$, and it is the local input behind the theorem of Myers and Steenrod, promised back in definition 1.3.3, that $\mathrm{Isom}(M, g)$ is a Lie group.

The mechanism is that an isometry commutes with the exponential map. An isometry carries geodesics to geodesics, because it preserves the Levi-Civita connection that defines them, so it intertwines the two exponential maps in a single identity. From that identity the rigidity proposition follows by an open-and-closed argument on the connected manifold, and the dimension bound follows by counting the data $(\varphi(p), d\varphi_p)$. The Myers–Steenrod theorem, whose proof rests on further Lie-theoretic machinery, is then stated and its proof deferred to [10]; the section closes by identifying the isometry groups of the three constant-curvature models and checking that each attains the maximal dimension.

Throughout, (M, g) is a connected Riemannian manifold of dimension n with Levi-Civita connection ∇ (definition 2.3.2), and $\mathrm{Isom}(M) = \mathrm{Isom}(M, g)$ is its isometry group in the sense of definition 1.3.3. For a point $p \in M$ the exponential map $\exp_p$ is that of definition 3.2.1, defined on the star-shaped domain $\mathcal{E}_p \subseteq T_pM$.

The starting observation is that an isometry, being a metric map, respects every object the metric manufactures: the connection, its geodesics, and hence the exponential map. Making this precise fixes the identity on which the whole section turns.

Lemma 9.1.1: Isometries intertwine the exponential map

Let $\varphi\colon (M, g) \to (\tilde{M}, \tilde{g})$ be an isometry, and let $p \in M$. Then φ carries the geodesic γ_v with $\gamma_v(0) = p$, $\dot{\gamma}_v(0) = v$ to the geodesic $\tilde{\gamma}_{d\varphi_p(v)}$ of $\tilde{M}$ with initial point $\varphi(p)$ and initial velocity $d\varphi_p(v)$, and consequently

$$\varphi \circ \exp_p = \exp_{\varphi(p)} \circ d\varphi_p \tag{9.1}$$

as maps on the domain $\mathcal{E}_p \subseteq T_pM$ (whose image under $d\varphi_p$ is the domain of $\exp_{\varphi(p)}$ on $\tilde{M}$).

***Proof*:**
An isometry preserves the Levi-Civita connection: since $\varphi^*\tilde{g} = g$ and the connection of definition 2.3.2 is the unique torsion-free metric-compatible connection, φ intertwines ∇ on M with $\tilde{\nabla}$ on $\tilde{M}$, that is $d\varphi(\nabla_X Y) = \tilde{\nabla}_{d\varphi(X)}(d\varphi(Y))$ for vector fields X, Y (this transport of ∇ under an isometry was recorded after definition 2.3.2). Let $\gamma = \gamma_v$ be the geodesic with $\gamma(0) = p$, $\dot{\gamma}(0) = v$, and set $\sigma = \varphi \circ \gamma$, a smooth curve in $\tilde{M}$ with $\sigma(0) = \varphi(p)$ and $\dot{\sigma}(0) = d\varphi_p(v)$. Because φ intertwines the connections, the covariant derivative of $\dot{\sigma}$ along σ is the image of the covariant derivative of $\dot{\gamma}$ along γ:

$$\tilde{D}_t\dot{\sigma} = d\varphi\big(D_t\dot{\gamma}\big) = d\varphi(0) = 0$$

using $D_t\dot\gamma = 0$ for the geodesic γ (definition 3.1.1). Hence σ is a geodesic of $\tilde M$ with the stated initial data, so $\sigma = \tilde\gamma_{d\varphi_p(v)}$ by uniqueness of geodesics (theorem 3.1.3). Evaluating at $t = 1$ and using $\exp_p(v) = \gamma_v(1)$ from definition 3.2.1,

$$\varphi\big(\exp_p(v)\big) = \varphi\big(\gamma_v(1)\big) = \sigma(1) = \tilde\gamma_{d\varphi_p(v)}(1) = \exp_{\varphi(p)}\big(d\varphi_p(v)\big)$$

which is (9.1). The domain statement holds because $v \in \mathcal{E}_p$ exactly when γ_v reaches time 1, and then so does its image $\sigma = \tilde\gamma_{d\varphi_p(v)}$; thus $d\varphi_p$ carries $\mathcal{E}_p$ onto the domain of $\exp_{\varphi(p)}$, completing the proof.

The identity (9.1) says that an isometry, read in normal coordinates at p and at $\varphi(p)$, is exactly the linear map $d\varphi_p$. The exponential map trades the nonlinear map φ near p for its differential near 0, and the two are conjugate through $\exp_p$ and $\exp_{\varphi(p)}$. This is already the source of the rigidity: on the normal neighborhood where $\exp_p$ is a diffeomorphism (proposition 3.2.5), the value $\varphi(p)$ and the differential $d\varphi_p$ recover φ completely, by the formula $\varphi = \exp_{\varphi(p)} \circ d\varphi_p \circ \exp_p^{-1}$. The next proposition upgrades this local determination to a global one, propagating it from one normal neighborhood across the whole connected manifold.

Proposition 9.1.2: An isometry is determined by its 1-jet at one point

Let (M, g) be a connected Riemannian manifold and let $\varphi, \psi\colon M \to M$ be isometries. Suppose there is a point $p \in M$ with

$$\varphi(p) = \psi(p) \qquad \text{and} \qquad d\varphi_p = d\psi_p$$

Then $\varphi = \psi$ on all of M. In particular, an isometry φ of a connected Riemannian manifold is determined by the pair $(\varphi(p), d\varphi_p)$ at a single point p.

***Proof*:**
Consider the set

$$A = \{q \in M \mid \varphi(q) = \psi(q) \text{ and } d\varphi_q = d\psi_q\}$$

of points where φ and ψ agree to first order. The claim is that $A = M$; since M is connected, it suffices to show A is nonempty, open, and closed.

Nonempty. By hypothesis $p \in A$.

Open. Let $q \in A$. Choose a normal neighborhood U of q on which $\exp_q$ is a diffeomorphism from a star-shaped $\mathcal{U} \subseteq T_qM$ onto U (proposition 3.2.5). By lemma 9.1.1 applied to φ and to ψ, for every $v \in \mathcal{U}$,

$$\varphi\big(\exp_q(v)\big) = \exp_{\varphi(q)}\big(d\varphi_q(v)\big), \qquad \psi\big(\exp_q(v)\big) = \exp_{\psi(q)}\big(d\psi_q(v)\big)$$

Because $q \in A$, the right-hand sides coincide: $\varphi(q) = \psi(q)$ makes the base points equal, and $d\varphi_q = d\psi_q$ makes the arguments equal. Hence $\varphi = \psi$ on the entire neighborhood $U = \exp_q(\mathcal{U})$. Two smooth maps that agree on an open set have equal differentials there, so $d\varphi_r = d\psi_r$ for all $r \in U$ as well, and therefore $U \subseteq A$. Thus A is open.

Closed. Let q be a limit point of A, and take $q_k \in A$ with $q_k \to q$. The maps φ and ψ are continuous, so $\varphi(q) = \lim_k \varphi(q_k) = \lim_k \psi(q_k) = \psi(q)$; the differentials $d\varphi$ and $d\psi$ vary continuously, so passing to the limit in $d\varphi_{q_k} = d\psi_{q_k}$ gives $d\varphi_q = \lim_k d\varphi_{q_k} = \lim_k d\psi_{q_k} = d\psi_q$. Hence $q \in A$, so A is closed.

Being nonempty, open, and closed in the connected space M, the set A equals M. In particular $\varphi = \psi$ everywhere. For the final statement, if φ_1, φ_2 are isometries with $\varphi_1(p) = \varphi_2(p)$ and $d(\varphi_1)_p = d(\varphi_2)_p$, the argument gives $\varphi_1 = \varphi_2$, so the pair $(\varphi(p), d\varphi_p)$ determines φ, as we sought to show.

The proposition is the metric analogue of the identity theorem for analytic functions, or of the fact that a Lie-group homomorphism is determined by its differential at the identity: a small amount of infinitesimal data forces the whole map. A diffeomorphism enjoys no such rigidity, since it can be modified inside any coordinate ball while agreeing with a given map to all orders on the boundary. What breaks this freedom is that an isometry must move along geodesics in the prescribed way, and the geodesics emanating from p sweep out a neighborhood, then, by connectedness, the entire manifold. The data $(\varphi(p), d\varphi_p)$ live in $M \times O(T_pM)$, and the count of that data is the source of the dimension bound extracted next.

The immediate structural payoff concerns the isotropy group at a point, the isometries fixing that point. Rigidity says such an isometry is recorded faithfully by its differential, which is an orthogonal transformation of the tangent space.

Corollary 9.1.3: The isotropy representation is faithful

Let (M, g) be a connected Riemannian manifold of dimension n and let $p \in M$. Write $\mathrm{Isom}_p(M) = \{\varphi \in \mathrm{Isom}(M) \mid \varphi(p) = p\}$ for the isotropy group at p. The map

$$\iota_p \colon \mathrm{Isom}_p(M) \longrightarrow O(T_pM), \qquad \iota_p(\varphi) = d\varphi_p$$

is an injective group homomorphism. Consequently $\mathrm{Isom}_p(M)$ is isomorphic to a subgroup of the orthogonal group $O(n)$, and

$$\dim \mathrm{Isom}(M) \leq n + \binom{n}{2} = \binom{n+1}{2}$$

***Proof*:**
The map lands in $O(T_pM)$. For $\varphi \in \mathrm{Isom}_p(M)$ the differential $d\varphi_p \colon T_pM \to T_{\varphi(p)}M = T_pM$ is a linear map of T_pM to itself, and it preserves the inner product g_p: from $\varphi^*g = g$ one reads $g_p(d\varphi_p u, d\varphi_p w) = g_{\varphi(p)}(d\varphi_p u, d\varphi_p w) = (\varphi^*g)_p(u, w) = g_p(u, w)$ for all $u, w \in T_pM$. Thus $d\varphi_p$ is a linear isometry of (T_pM, g_p), an element of $O(T_pM)$.

Homomorphism. For $\varphi, \psi \in \mathrm{Isom}_p(M)$ the chain rule at the fixed point $p = \psi(p)$ gives $d(\varphi \circ \psi)_p = d\varphi_{\psi(p)} \circ d\psi_p = d\varphi_p \circ d\psi_p$, so $\iota_p(\varphi \circ \psi) = \iota_p(\varphi)\iota_p(\psi)$.

Injective. Suppose $\iota_p(\varphi) = \iota_p(\psi)$, that is $d\varphi_p = d\psi_p$. Both maps also fix p, so $\varphi(p) = p = \psi(p)$. By proposition 9.1.2, $\varphi = \psi$. Hence $\ker \iota_p$ is trivial and ι_p is injective.

The injection identifies $\mathrm{Isom}_p(M)$ with a subgroup of $O(T_pM) \cong O(n)$. For the dimension bound, fix p and consider the orbit map $\mathrm{Isom}(M) \to M$, $\varphi \mapsto \varphi(p)$, whose fiber over p is exactly $\mathrm{Isom}_p(M)$. Its image is the orbit of p, a subset of the n-dimensional manifold M, so it contributes at most n to the dimension; its fiber $\mathrm{Isom}_p(M)$ embeds in $O(n)$, of dimension $\binom{n}{2}$. Once the Lie-group structure of $\mathrm{Isom}(M)$ is in hand (theorem 9.1.4 below), these combine to

$$\dim \mathrm{Isom}(M) \leq \dim(\text{orbit of } p) + \dim \mathrm{Isom}_p(M) \leq n + \binom{n}{2} = \binom{n+1}{2}$$

which is the asserted bound, completing the proof.

The corollary is the quantitative form of rigidity. An isometry fixing p is nothing more than an orthogonal transformation of T_pM that happens to extend to a global isometry, and different orthogonal transformations extend to different isometries when they extend at all. The bound $\binom{n+1}{2} = \frac{1}{2}n(n+1)$ counts the maximal possible degrees of freedom: n to move the basepoint anywhere in M, and $\binom{n}{2} = \dim O(n)$ to rotate the frame there. The bound is sharp only in the presence of a very large symmetry supply, and identifying exactly when it is attained is the content of the closing example. The reasoning also explains why the isotropy group is compact: it is a closed subgroup of the compact group $O(n)$, a point recorded in the Myers–Steenrod statement.

The dimension count above borrowed the Lie-group structure of $\mathrm{Isom}(M)$, which has not yet been established. That structure is the theorem of Myers and Steenrod. Its statement makes precise the sense in which the isometry group is not only an abstract group but a finite-dimensional Lie group acting smoothly, and it packages the rigidity results above into a single conclusion. The proof requires machinery beyond the scope of this book, principally the closed-subgroup theorem for Lie groups and the construction of a chart on $\mathrm{Isom}(M)$ from the intertwining identity (9.1), and it is deferred to the analytic theory of Lie groups.

Theorem 9.1.4: Myers–Steenrod

Let (M, g) be a connected Riemannian manifold of dimension n. Then the isometry group $\mathrm{Isom}(M, g)$, with the compact-open topology, is a Lie group, and its action

$$\mathrm{Isom}(M, g) \times M \longrightarrow M, \qquad (\varphi, q) \mapsto \varphi(q)$$

is smooth and proper. For each $p \in M$ the isotropy group $\mathrm{Isom}_p(M)$ is compact, and its isotropy representation $\iota_p \colon \mathrm{Isom}_p(M) \to O(T_pM)$ of corollary 9.1.3 is a faithful representation onto a closed subgroup of $O(n)$. The dimension satisfies

$$\dim \mathrm{Isom}(M, g) \leq \binom{n+1}{2}$$

and if equality holds then (M, g) has constant sectional curvature and M is a quotient of one of the model spaces $\mathbb{R}^n$, S^n, or H^n. Conversely, each of the complete simply connected models $\mathbb{R}^n$, S^n, H^n (up to homothety) attains the bound. The converse implication runs only one way: a constant-curvature manifold need not attain the bound, the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ having zero curvature yet only the isometry dimension n of its translation group.

The proof, due to Myers and Steenrod in 1939, is deferred to the analytic theory of Lie groups; see [10]. Two features of the argument are worth naming, since they are exactly the results proved above in miniature. The rigidity proposition proposition 9.1.2 supplies the injectivity that keeps the group finite-dimensional: a chart near a given isometry φ_0 is built by recording, for φ near φ_0, the pair $(\varphi(p), d\varphi_p)$, and rigidity guarantees this record is faithful, so the group embeds locally in the finite-dimensional manifold $M \times O(T_pM)$. The intertwining identity (9.1) supplies the smoothness: it expresses φ near any point through the exponential maps and the linear datum $d\varphi_p$, so smooth dependence of the exponential map on its arguments (theorem 3.1.3) transfers to smooth dependence of the action. What the Lie-group theory adds is the closed-subgroup theorem, which turns the closed image in $M \times O(T_pM)$ into an embedded Lie subgroup, and the properness of the action, which secures compactness of the isotropy groups. For the constant-curvature equality clause the space-form classification (theorem 4.5.5), globally the Killing–Hopf theorem (box 4.5.6), identifies the maximally symmetric spaces as quotients of the models.

Beyond the group being Lie, the theorem carries a rigidity of its own that is easy to overlook: the topology on $\mathrm{Isom}(M)$ is the compact-open topology, so a sequence of isometries that converges uniformly on compact sets converges to an isometry, and the limit is again smooth. A uniform limit of isometries cannot degenerate into a map that is only continuous. This is a metric analogue of Weierstrass's theorem that a locally uniform limit of holomorphic functions is holomorphic, and it is what makes the isometry group closed in $\mathrm{Diff}(M)$ as a topological group. The properness of the action is the statement that the set of isometries moving a fixed compact set to overlap another fixed compact set is itself compact; on a homogeneous space this is what makes orbits closed and isotropy groups compact simultaneously.

The theorem asserts that the dimension bound $\binom{n+1}{2}$ is attained by the complete simply connected constant-curvature models $\mathbb{R}^n$, S^n, H^n, and that attaining it forces constant curvature. It does not assert that every constant-curvature space attains it: the flat torus is a standing counterexample. The following example computes the three model isometry groups explicitly and verifies the dimensions, closing the loop with the model spaces of section 1.3 and confirming the attainment half of the equality clause of theorem 9.1.4.

Example 9.1.5: Isometry groups of the model spaces

The three simply connected constant-curvature models each realize the maximal isometry dimension $\binom{n+1}{2} = \frac{1}{2}n(n+1)$. The isometry group and its dimension are computed for each in turn; in every case the group is exhibited by producing enough isometries to attain the bound of corollary 9.1.3, which proposition 9.1.2 then shows is the whole group.

1. *Euclidean space* $(\mathbb{R}^n, \bar{g})$. The maps

$$x \longmapsto Ax + b, \qquad A \in O(n),\ b \in \mathbb{R}^n$$

are isometries: a translation $x \mapsto x + b$ has differential the identity, and a linear map A has $A^*\bar{g} = \bar{g}$ exactly when $A \in O(n)$, so the composite preserves $\bar{g}$. These form the group

$$\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$$

the semidirect product in which $O(n)$ acts on the translation subgroup $\mathbb{R}^n$ in the standard way, $(A, b) \cdot (A', b') = (AA', Ab' + b)$. Conversely, any isometry φ is of this form: set $b = \varphi(0)$ and $A = d\varphi_0$, which lies in $O(n)$ by corollary 9.1.3; the affine map $x \mapsto Ax + b$ is an isometry with the same value and differential at 0 as φ, so $\varphi(x) = Ax + b$ by proposition 9.1.2. The dimension is

$$\dim \mathrm{Isom}(\mathbb{R}^n) = \underbrace{n}_{\text{translations}} + \underbrace{\binom{n}{2}}_{\dim O(n)} = \binom{n+1}{2}$$

so $\mathbb{R}^n$ attains the maximal dimension.

2. *The round sphere* S^n. Realize $S^n \subset \mathbb{R}^{n+1}$ with its round metric (example 1.3.4). Every $A \in O(n+1)$ preserves the Euclidean metric of $\mathbb{R}^{n+1}$ and the sphere $\{x \mid |x| = 1\}$, hence restricts to an isometry of S^n for the induced metric. This gives an injective homomorphism $O(n+1) \to \mathrm{Isom}(S^n)$, and it is surjective: given an isometry φ of S^n, fix $p \in S^n$ and complete p with an orthonormal basis of T_pS^n to an orthonormal basis of $\mathbb{R}^{n+1}$; the orthogonal matrix

$A \in O(n+1)$ sending this frame to the corresponding frame at $\varphi(p)$ restricts to an isometry of S^n agreeing with φ in value and differential at p, so $A|_{S^n} = \varphi$ by proposition 9.1.2. Therefore

$$\mathrm{Isom}(S^n) = O(n+1), \qquad \dim \mathrm{Isom}(S^n) = \binom{n+1}{2}$$

the last equality because $\dim O(n+1) = \binom{n+1}{2}$. The sphere too is maximally symmetric.

3. *Hyperbolic space H^n.* Realize H^n as the upper sheet of the hyperboloid in Minkowski space $\mathbb{R}^{1,n}$ (example 1.3.5), with the metric induced by the Lorentzian form m of signature $(-,+,\ldots,+)$. The Lorentz group $O(1,n)$ of linear maps preserving m acts on the two-sheeted hyperboloid $\{x \mid m(x,x) = -1\}$; the index-two subgroup

$$O^+(1,n) = \left\{A \in O(1,n) \;\middle|\; A \text{ preserves the upper sheet } x^0 > 0\right\}$$

preserves H^n and restricts to isometries of it, since $A^*m = m$ and the induced metric is the restriction of m. As in the two previous cases, every isometry of H^n arises this way: a hyperbolic isometry φ is matched by the unique $A \in O^+(1,n)$ sending an m-orthonormal frame $(p, e_1, \ldots, e_n)$ at p, with p timelike and the e_i spanning T_pH^n, to the corresponding frame at $\varphi(p)$, and $A|_{H^n} = \varphi$ by proposition 9.1.2. Hence

$$\mathrm{Isom}(H^n) = O^+(1,n), \qquad \dim \mathrm{Isom}(H^n) = \binom{n+1}{2}$$

since $\dim O^+(1,n) = \dim O(1,n) = \binom{n+1}{2}$, the Lorentz group having the same dimension as $O(n+1)$. Hyperbolic space is maximally symmetric as well.

The three groups $\mathbb{R}^n \rtimes O(n)$, $O(n+1)$, and $O^+(1,n)$ all have dimension $\binom{n+1}{2}$, so the three simply connected models $\mathbb{R}^n$, S^n, and H^n attain the maximal isometry dimension, one precise formulation of their maximal symmetry. They are the complete simply connected attainers; a quotient of a model, such as the real projective space $\mathbb{RP}^n = S^n/\{\pm 1\}$ with $\mathrm{Isom}(\mathbb{RP}^n) = \mathrm{PO}(n+1)$ of the same dimension $\binom{n+1}{2}$, can attain it too, while a quotient such as the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ does not. What theorem 9.1.4 records is the reverse: any space attaining the bound has constant curvature. In each of the three model cases the isotropy group at a point is a full copy of $O(n)$: for $\mathbb{R}^n$ it is the stabilizer $O(n)$ of the origin, for S^n it is the stabilizer $O(n) \subset O(n+1)$ of the point p (block $\mathrm{diag}(1, R)$ fixing the line $\mathbb{R}p$ pointwise), and for H^n it is the stabilizer $O(n) \subset O^+(1,n)$ of the timelike line through p. The full orthogonal isotropy is what makes each model isotropic in the sense of definition 1.3.3, its isotropy group acting transitively on the unit sphere of every tangent space, and this maximal isotropy is what the dimension count of corollary 9.1.3 detects.

9.2 Homogeneous Riemannian Manifolds and Invariant Metrics

The previous section identified the isometry group of a Riemannian manifold as a Lie group (theorem 9.1.4) and bounded its dimension by $\binom{n+1}{2}$. This section takes the group as the primary object and reads the geometry off it. A manifold whose isometry group acts transitively looks the same at every point: any point can be moved to any other by an isometry, so no point carries a geometric feature that another lacks. Such a manifold is homogeneous, and the entire section is organized around a single principle, that all of its geometry is concentrated at one point and spread everywhere else by the group. A metric on a homogeneous space is determined by its value at a base point, and that value is constrained only by the subgroup of isometries fixing the point. The section makes this principle exact and extracts from it both a completeness theorem, obtained at no cost from transitivity, and a dictionary that converts the search for invariant metrics into a finite-dimensional problem in linear algebra on a Lie algebra.

The Lie theory needed here is imported. Lie groups and their Lie algebras, left-invariant vector fields and the identification $\mathfrak{g} \cong T_eG$, the exponential map, the adjoint representations Ad and ad, and the presentation of a homogeneous space as a coset manifold G/H are developed in [6] and are used without reproof; the analytic construction of $\mathrm{Isom}(M, g)$ as a Lie group is theorem 9.1.4. The reader who wants the group theory in its own right will find it in [10]. What this section adds is the Riemannian layer: how a metric interacts with the transitive action, and what its value at one point must satisfy.

A homogeneous manifold is one on which the isometry group leaves no point distinguished.

Definition 9.2.1: Homogeneous Riemannian manifold

A Riemannian manifold (M, g) is *(Riemannian-)homogeneous* if its isometry group $\mathrm{Isom}(M, g)$ (definition 1.3.3) acts transitively on M: for every pair of points $p, q \in M$ there is an isometry $\varphi \in \mathrm{Isom}(M, g)$ with $\varphi(p) = q$.
Fix a base point $o \in M$. Let $G \subseteq \mathrm{Isom}(M, g)$ be a Lie subgroup acting transitively on M (for instance $G = \mathrm{Isom}(M, g)^0$, the identity component, or the full isometry group, both Lie groups by theorem 9.1.4), and let

$$H = G_o = \{\varphi \in G \mid \varphi(o) = o\}$$

be the *isotropy subgroup* at o. Then H is a closed subgroup of G, and the orbit map $G \to M$, $\varphi \mapsto \varphi(o)$, descends to a G-equivariant diffeomorphism

$$G/H \xrightarrow{\sim} M, \qquad \varphi H \longmapsto \varphi(o)$$

This is the *presentation* $M \cong G/H$ of the homogeneous manifold, with o corresponding to the base coset eH.

The presentation $M \cong G/H$ is the working form of homogeneity, and each of its ingredients earns its place. Transitivity of G makes the orbit map surjective, so every point of M is $\varphi(o)$ for some φ; two isometries φ, ψ send o to the same point exactly when $\psi^{-1}\varphi \in H$, so the fibers of the orbit map are the left cosets of H, and the induced map on cosets is a bijection. That it is a diffeomorphism, and that H is closed, are the coset-manifold facts imported from [6]: the isotropy subgroup of a smooth action is closed, and the quotient of a Lie group by a closed subgroup is a smooth manifold on which G acts smoothly and transitively. The isotropy group H is compact, because it injects into the orthogonal group $O(T_oM)$ by the isotropy representation $\varphi \mapsto d\varphi_o$ of corollary 9.1.3, a closed subgroup of the compact $O(n)$, and a closed subgroup that is also a subgroup of a compact group is compact. Compactness of H is the hypothesis behind every averaging argument later in the section.

The three constant-curvature model spaces are homogeneous, and each is the archetype of a case that recurs throughout the chapter.

Example 9.2.2: The model spaces as homogeneous manifolds
Each of the three space forms is homogeneous, presented as a coset of its isometry group from example 9.1.5.

1. Euclidean space $\mathbb{R}^n$ has isometry group $\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$, the group of rigid motions $x \mapsto Ax + b$ with $A \in O(n)$, $b \in \mathbb{R}^n$. The translation subgroup $\mathbb{R}^n$ already acts transitively, since $x \mapsto x + (q - p)$ carries p to q. Taking $G = \mathbb{R}^n \rtimes O(n)$ and base point $o = 0$, the isotropy subgroup is $H = O(n)$, the motions fixing the origin, and $\mathbb{R}^n \cong (\mathbb{R}^n \rtimes O(n))/O(n)$.
2. The round sphere S^n (example 1.3.4) has isometry group $O(n+1)$ acting by the standard linear action on the unit sphere of $\mathbb{R}^{n+1}$. This action is transitive: an orthonormal frame at one point is carried to an orthonormal frame at any other by an element of $O(n+1)$. With base point $o = e_{n+1}$ the north pole, the isotropy subgroup is the copy of $O(n)$ acting on the equatorial $\mathbb{R}^n = e_{n+1}^{\perp}$, so $S^n \cong O(n+1)/O(n)$.
3. Hyperbolic space H^n (example 1.3.5), in the hyperboloid model inside $\mathbb{R}^{1,n}$, has isometry group the orthochronous Lorentz group $O^+(1,n)$ acting linearly, transitively on the hyperboloid. With base point the vertex, the isotropy subgroup is $O(n)$ acting on the spacelike hyperplane through the vertex, so $H^n \cong O^+(1,n)/O(n)$.

In all three the isotropy group is $O(n)$, acting on the tangent space at the base point as the full orthogonal group. This is the strongest possible isotropy, and it forces constant curvature, as the next examples and example 9.2.6 make precise. The dimensions match the maximal count $\binom{n+1}{2}$ of corollary 9.1.3: $\dim O(n+1) = \binom{n+1}{2}$, and $\dim(\mathbb{R}^n \rtimes O(n)) = n + \binom{n}{2} = \binom{n+1}{2}$, so each model attains the isometry-dimension bound, exactly as theorem 9.1.4 predicts for a maximally symmetric space.

Homogeneity is a strong hypothesis, and its first consequence costs nothing beyond the definition. A geodesic that runs off the edge of its interval of existence would witness a point near which geometry degenerates; on a homogeneous manifold there is no such point, because the group carries any small piece of the manifold to any other. Transitivity therefore converts a local existence time for geodesics near one point into a uniform existence time everywhere, and a uniform existence time is completeness.

Proposition 9.2.3: Homogeneous manifolds are complete

Every homogeneous Riemannian manifold (M, g) is geodesically complete: every geodesic extends to all of $\mathbb{R}$. In particular any two points are joined by a minimizing geodesic.

***Proof*:**
The strategy is to produce a single $\varepsilon > 0$, uniform over M, such that every geodesic of unit speed extends for at least an additional time ε from any point it reaches, and then to iterate. Homogeneity supplies the uniformity, and the isometry action supplies the extension.

Step 1: a uniform existence time from one point. Fix the base point $o \in M$. By existence and smooth dependence of geodesics on their initial data (theorem 3.1.3), the exponential map $\exp_o$ (definition 3.2.1) is defined on a neighborhood of 0 in T_oM; in particular there is an $\varepsilon > 0$ such that for every unit vector $u \in T_oM$ the geodesic γ_u with $\gamma_u(0) = o$, $\dot\gamma_u(0) = u$ is defined at least on the interval $[0, \varepsilon]$. Since the unit sphere in $T_o M$ is compact and the domain of existence is open in the tangent bundle, one ε works for all unit u at o simultaneously. By the rescaling identity $\gamma_{\lambda u}(t) = \gamma_u(\lambda t)$ of proposition 3.2.2, every unit-speed geodesic issuing from o is defined on $[0, \varepsilon]$.

Step 2: transporting the existence time by isometries. Let $p \in M$ be arbitrary and let $u \in T_p M$ be a unit vector. By homogeneity there is an isometry $\varphi \in \mathrm{Isom}(M, g)$ with $\varphi(o) = p$. The differential $d\varphi_o \colon T_o M \to T_p M$ is a linear isometry, so $w := (d\varphi_o)^{-1} u \in T_o M$ is again a unit vector, and by Step 1 the geodesic γ_w from o in direction w is defined on $[0, \varepsilon]$. An isometry carries geodesics to geodesics, because $\varphi^* g = g$ preserves the Levi-Civita connection and hence solutions of the geodesic equation; concretely $\varphi \circ \gamma_w$ is a geodesic with initial point $\varphi(o) = p$ and initial velocity $d\varphi_o(w) = u$. By uniqueness (theorem 3.1.3) it is the geodesic γ_u from p in direction u, and it is defined wherever γ_w is, namely on all of $[0, \varepsilon]$. Thus every unit-speed geodesic, from every point p, extends over a time interval of length at least ε, with the same ε throughout.

Step 3: iteration. Let $\gamma \colon [0, b) \to M$ be a maximal unit-speed geodesic, and suppose for contradiction that $b < \infty$. Choose $t_0 \in (b - \varepsilon, b)$, and set $p = \gamma(t_0)$, $u = \dot{\gamma}(t_0)$, a unit vector by constant speed (proposition 3.1.6). By Step 2 the geodesic from p in direction u extends over $[0, \varepsilon]$, so γ extends past $t_0 + \varepsilon > b$, contradicting maximality. Hence $b = \infty$, and the same argument in reverse gives extension to $-\infty$; every geodesic is defined on all of $\mathbb{R}$, so M is geodesically complete.

The final assertion follows because a geodesically complete Riemannian manifold has any two points joined by a minimizing geodesic, by corollary 6.1.4, completing the proof.

Completeness of a homogeneous space is a structural fact, not a curvature bound: it holds for the flat model $\mathbb{R}^n$, the positively curved sphere, and the negatively curved hyperbolic space alike, and for every homogeneous space in between, with no hypothesis beyond transitivity of the isometry group. The proof is worth reading as a template. Anything a homogeneous space enjoys at one point, it enjoys everywhere, and the mechanism is always the same: a local statement at the base point o is proved once, and the isometry moving o to an arbitrary p transports it, because isometries preserve every metric object in sight, including geodesics and their domains of existence. The same template will manufacture invariant metrics below, this time transporting an inner product at o rather than an existence time.

To make the linear-algebra reduction precise, the algebra of the group must be split so that one summand models the base point's tangent space. This is the reductive condition, and it is the standing hypothesis for the rest of the section.

Definition 9.2.4: Reductive decomposition

Let $M \cong G/H$ be a homogeneous manifold as in definition 9.2.1, with $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras of G and H. The presentation is *reductive* if there is a vector-space complement $\mathfrak{m}$ to $\mathfrak{h}$ in $\mathfrak{g}$,

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$$

that is invariant under the adjoint action of H:

$$\mathrm{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m} \qquad \text{for all } h \in H$$

The complement $\mathfrak{m}$ is a *reductive complement*, and $(G/H, \mathfrak{m})$ is a *reductive homogeneous space*.
Let $o = eH$ be the base point and $\pi\colon G \to G/H$ the projection. The differential $d\pi_e\colon \mathfrak{g} \to T_oM$ has kernel $\mathfrak{h}$ (the tangent space to the fiber H), so it restricts to a linear isomorphism

$$d\pi_e|_{\mathfrak{m}}\colon \mathfrak{m} \xrightarrow{\sim} T_oM$$

This isomorphism identifies the reductive complement $\mathfrak{m}$ with the tangent space at the base point. Under it, the isotropy action of H on T_oM, $h \mapsto d(L_h)_o$ where L_h is the action of h on M, corresponds to the adjoint action $\mathrm{Ad}(h)|_{\mathfrak{m}}$ on $\mathfrak{m}$.

The reductive decomposition splits the Lie algebra of the group into a piece $\mathfrak{h}$ that generates the isotropy (motions fixing o) and a complementary piece $\mathfrak{m}$ that generates the motion of o itself. The isomorphism $\mathfrak{m} \cong T_oM$ makes $\mathfrak{m}$ the algebraic stand-in for the tangent space at the base point, and the $\mathrm{Ad}(H)$-invariance is precisely the condition that this stand-in respects the isotropy action: the way H rotates the tangent space T_oM is read off inside $\mathfrak{m}$ as the restricted adjoint representation. The final clause of the definition is the linchpin of everything that follows, converting a geometric action of H on T_oM into an algebraic action of H on $\mathfrak{m}$. Not every homogeneous presentation is reductive, but a large class is: whenever H is compact, a reductive complement exists, because an $\mathrm{Ad}(H)$-invariant inner product on $\mathfrak{g}$ (built by averaging any inner product over the compact H, exactly as in the metric construction below) makes $\mathfrak{m} = \mathfrak{h}^{\perp}$ an $\mathrm{Ad}(H)$-invariant complement. Since the isotropy group of a homogeneous Riemannian manifold is compact, every such manifold admits a reductive presentation, and the reductive hypothesis is not a restriction in the Riemannian setting.

With the reductive complement in hand, the search for invariant metrics becomes a search for invariant inner products on a single vector space. This is the central result of the section: it identifies the geometry of a homogeneous space with a piece of linear algebra at the base point.

Theorem 9.2.5: Invariant metrics and $\mathrm{Ad}(H)$-invariant inner products

Let $(G/H, \mathfrak{m})$ be a reductive homogeneous space with base point $o = eH$, and let $d\pi_e|_{\mathfrak{m}} \colon \mathfrak{m} \to T_oM$ be the identification of definition 9.2.4. Then the assignment

$$g \longmapsto g_o \circ (d\pi_e|_{\mathfrak{m}} \times d\pi_e|_{\mathfrak{m}})$$

sending a G-invariant Riemannian metric g on $M = G/H$ to its value at o, pulled back to $\mathfrak{m}$, is a bijection

$$\{\, G\text{-invariant Riemannian metrics on } G/H \,\} \xrightarrow{\ \sim\ } \{\, \mathrm{Ad}(H)\text{-invariant inner products on } \mathfrak{m} \,\}$$

onto the set of inner products $\langle -, - \rangle$ on $\mathfrak{m}$ satisfying $\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle = \langle X, Y \rangle$ for all $h \in H$ and $X, Y \in \mathfrak{m}$. Here a metric g is *G-invariant* if $L_a^* g = g$ for every $a \in G$, where $L_a \colon M \to M$ is the action of a.

***Proof*:**
Write $\iota = d\pi_e|_{\mathfrak{m}} \colon \mathfrak{m} \xrightarrow{\sim} T_oM$ for the identification, and for $a \in G$ write $L_a \colon M \to M$ for the diffeomorphism $x \mapsto a \cdot x$, so that $L_a(o) = a \cdot o$ and $L_h(o) = o$ for $h \in H$. The proof establishes that the value map is well defined into $\mathrm{Ad}(H)$-invariant inner products, that it is injective, and that it is surjective.

Step 1: the value at o of a G-invariant metric is $\mathrm{Ad}(H)$*-invariant.* Let g be a G-invariant Riemannian metric. Its value g_o is an inner product on T_oM, and $b(X, Y) := g_o(\iota X, \iota Y)$ is an inner product on $\mathfrak{m}$. For $h \in H$ the map L_h fixes o, so its differential $d(L_h)_o$ is a linear map $T_oM \to T_oM$; G-invariance $L_h^* g = g$ evaluated at o reads

$$g_o\big(d(L_h)_o\, v,\ d(L_h)_o\, w\big) = g_o(v, w) \qquad (v, w \in T_oM)$$

so $d(L_h)_o$ is a g_o-isometry of T_oM. By the last clause of definition 9.2.4, $d(L_h)_o$ corresponds under ι to $\mathrm{Ad}(h)|_{\mathfrak{m}}$, that is $d(L_h)_o \circ \iota = \iota \circ \mathrm{Ad}(h)|_{\mathfrak{m}}$. Substituting $v = \iota X$, $w = \iota Y$ and using this intertwining,

$$b(\mathrm{Ad}(h)X, \mathrm{Ad}(h)Y) = g_o\big(\iota\ \mathrm{Ad}(h)X, \iota\ \mathrm{Ad}(h)Y\big) = g_o\big(d(L_h)_o\, \iota X,\ d(L_h)_o\, \iota Y\big) = g_o(\iota X, \iota Y) = b(X, Y)$$

Thus b is $\mathrm{Ad}(H)$-invariant, and the value map lands where claimed.

Step 2: injectivity. Suppose two G-invariant metrics g, g' have the same value at o, so $g_o = g'_o$. Any point of M is $a \cdot o$ for some $a \in G$ by transitivity, and $L_a \colon M \to M$ carries o to $a \cdot o$ with differential the linear isomorphism $d(L_a)_o \colon T_oM \to T_{a \cdot o}M$. For tangent vectors $v, w \in T_{a \cdot o}M$, write $v = d(L_a)_o\, \bar{v}$, $w = d(L_a)_o\, \bar{w}$ with $\bar{v}, \bar{w} \in T_oM$; then G-invariance $L_a^* g = g$ at o gives

$$g_{a \cdot o}(v, w) = g_{a \cdot o}\big(d(L_a)_o\, \bar{v},\ d(L_a)_o\, \bar{w}\big) = g_o(\bar{v}, \bar{w})$$

and the identical computation holds for g'. Since $g_o = g'_o$, the right-hand sides agree, so $g_{a \cdot o} = g'_{a \cdot o}$ at every point $a \cdot o$, that is $g = g'$. The value at o determines the metric, so the value map is injective.

Step 3: surjectivity. Let b be an $\mathrm{Ad}(H)$-invariant inner product on $\mathfrak{m}$, and let $g_o^b := b(\iota^{-1}-, \iota^{-1}-)$ be the corresponding inner product on T_oM. The task is to spread g_o^b to a G-invariant metric on M whose value at o is g_o^b. For $p = a \cdot o \in M$ define

$$g_p(v, w) := g_o^b\big(d(L_a)_o^{-1}\, v,\ d(L_a)_o^{-1}\, w\big) \qquad (v, w \in T_pM)$$

transporting g_o^b from o to p by the differential of any a with $a \cdot o = p$.

The only issue is well-definedness, since the choice of a with $a \cdot o = p$ is unique only up to right multiplication by H: if $a \cdot o = a' \cdot o$ then $a^{-1}a' =: h \in H$, so $a' = ah$. The claim is that the transported inner product does not depend on this choice, and this is exactly where $\mathrm{Ad}(H)$-invariance of b is used. With $a' = ah$ one has $L_{a'} = L_a \circ L_h$, hence $d(L_{a'})_o = d(L_a)_o \circ d(L_h)_o$, because L_h fixes o. Therefore, for $v \in T_pM$,

$$d(L_{a'})_o^{-1}\, v = d(L_h)_o^{-1}\, d(L_a)_o^{-1}\, v$$

Write $\bar{v} = d(L_a)_o^{-1}v \in T_oM$ and $\bar{w} = d(L_a)_o^{-1}w$. Using this and the intertwining $d(L_h)_o = \iota \circ \mathrm{Ad}(h)|_{\mathfrak{m}} \circ \iota^{-1}$ from definition 9.2.4,

$$g_o^b\big(d(L_{a'})_o^{-1}v,\, d(L_{a'})_o^{-1}w\big) = g_o^b\big(d(L_h)_o^{-1}\bar{v},\, d(L_h)_o^{-1}\bar{w}\big) = b\big(\mathrm{Ad}(h)^{-1}\iota^{-1}\bar{v},\, \mathrm{Ad}(h)^{-1}\iota^{-1}\bar{w}\big)$$

and $\mathrm{Ad}(H)$-invariance of b (applied to $h^{-1} \in H$) collapses the right-hand side to $b(\iota^{-1}\bar{v}, \iota^{-1}\bar{w}) = g_o^b(\bar{v}, \bar{w}) = g_o^b\big(d(L_a)_o^{-1}v, d(L_a)_o^{-1}w\big)$. The two choices a and $a' = ah$ therefore give the same value, so g_p is well defined.

It remains to check that $g = (g_p)_{p \in M}$ is a smooth G-invariant Riemannian metric with the prescribed value at o. Each g_p is an inner product, being the pullback of the inner product g_o^b under the linear isomorphism $d(L_a)_o^{-1}$, so g is a symmetric positive-definite 2-tensor at every point. Smoothness holds because the action $G \times M \to M$ is smooth ([6]) and g is obtained from the fixed smooth tensor g_o^b by the action of local smooth sections $a(p)$ of the orbit map, so its components are smooth functions of p. At the base point, taking $a = e$ gives $g_o = g_o^b$, the prescribed value. Finally g is G-invariant: for $c \in G$ and $p = a \cdot o$, the point $c \cdot p = (ca) \cdot o$ is realized by ca, so by the defining formula

$$(L_c^*g)_p(v, w) = g_{c \cdot p}\big(d(L_c)_p\, v,\, d(L_c)_p\, w\big) = g_o^b\big(d(L_{ca})_o^{-1} d(L_c)_p\, v,\, \dots\big)$$

and $d(L_{ca})_o^{-1} d(L_c)_p = d(L_a)_o^{-1} d(L_c)_p^{-1} d(L_c)_p = d(L_a)_o^{-1}$ since $L_{ca} = L_c \circ L_a$ and $d(L_c)_p$ is the differential of L_c at $p = a \cdot o$; hence $(L_c^*g)_p(v, w) = g_o^b(d(L_a)_o^{-1}v, d(L_a)_o^{-1}w) = g_p(v, w)$. Thus $L_c^*g = g$ for all $c \in G$, so g is G-invariant with value g_o^b at o, and its image under the value map is b. The value map is therefore surjective.

Steps 1 through 3 show the value map is a well-defined bijection onto the $\mathrm{Ad}(H)$-invariant inner products on $\mathfrak{m}$, as we sought to show.

The theorem is the reason homogeneous geometry is tractable. A G-invariant metric is an infinite amount of data, an inner product at every point of M, but the theorem compresses it to a single inner product on the finite-dimensional vector space $\mathfrak{m} \cong T_oM$, subject to one linear constraint: invariance under the isotropy representation $\mathrm{Ad}(H)|_{\mathfrak{m}}$. Two consequences are immediate. First, existence of a G-invariant metric is equivalent to existence of an $\mathrm{Ad}(H)$-invariant inner product on $\mathfrak{m}$, which is guaranteed whenever H is compact by averaging, so every homogeneous Riemannian manifold does carry a G-invariant metric, as it must. Second, the classification of G-invariant metrics is the classification of $\mathrm{Ad}(H)$-invariant inner products, a representation-theoretic problem. When the isotropy representation on $\mathfrak{m}$ is irreducible, an $\mathrm{Ad}(H)$-invariant inner product is unique up to a positive scalar, by Schur's lemma over the reals for a compact group; the G-invariant metric is then unique up to scaling, and the geometry is rigid. The maximal-isotropy models of example 9.2.2 are exactly this rigid case, and the next example works one of them out in full.

Example 9.2.6: The round sphere as a reductive homogeneous space

The sphere $S^n = SO(n+1)/SO(n)$ carries a unique $SO(n+1)$-invariant metric up to scale, and it is the round metric. Take $G = SO(n+1)$, the orientation-preserving isometries of the round sphere, which act transitively on $S^n \subset \mathbb{R}^{n+1}$; the identity component suffices, and using SO in place of O streamlines the Lie algebra without changing the invariant metrics. Fix the base point $o = e_{n+1} = (0, \dots, 0, 1)$, the north pole. The isotropy subgroup is

$$H = SO(n) = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} \,\middle|\, A \in SO(n) \right\}$$

the rotations fixing e_{n+1}, acting on the equatorial hyperplane $e_{n+1}^{\perp} = \mathbb{R}^n$.

The reductive decomposition. The Lie algebra $\mathfrak{g} = \mathfrak{so}(n+1)$ consists of the antisymmetric $(n+1) \times (n+1)$ real matrices, and $\mathfrak{h} = \mathfrak{so}(n)$ is the block sitting in the upper-left $n \times n$ corner (with last row and column zero). A vector-space complement is the set of antisymmetric matrices whose upper-left $n \times n$ block vanishes,

$$\mathfrak{m} = \left\{ \begin{pmatrix} 0 & -\xi \\ \xi^{\top} & 0 \end{pmatrix} \,\middle|\, \xi \in \mathbb{R}^n \right\}$$

an n-dimensional space, with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ since an antisymmetric matrix decomposes uniquely into its upper-left block and its last row/column. To see that this complement is reductive, compute the adjoint action of $h = \left(\begin{smallmatrix} A & 0 \\ 0 & 1 \end{smallmatrix}\right) \in SO(n)$ on $X_\xi = \begin{pmatrix} 0 & -\xi \\ \xi^{\top} & 0 \end{pmatrix} \in \mathfrak{m}$. Since $\mathrm{Ad}(h)X = hXh^{-1}$ for a matrix group,

$$\mathrm{Ad}(h)X_\xi = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -\xi \\ \xi^{\top} & 0 \end{pmatrix} \begin{pmatrix} A^{\top} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -A\xi \\ (A\xi)^{\top} & 0 \end{pmatrix} = X_{A\xi}$$

which again lies in $\mathfrak{m}$. Hence $\mathrm{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m}$ for every $h \in SO(n)$, so $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ is a reductive decomposition, and the isotropy representation of $SO(n)$ on $\mathfrak{m} \cong \mathbb{R}^n$ (via $\xi \mapsto X_\xi$) is the standard representation $\xi \mapsto A\xi$.

The invariant inner products. By theorem 9.2.5, the $SO(n+1)$-invariant metrics on S^n correspond to the $\mathrm{Ad}(SO(n))$-invariant inner products on $\mathfrak{m} \cong \mathbb{R}^n$, that is, the inner products $\langle -, - \rangle$ on $\mathbb{R}^n$ satisfying $\langle A\xi, A\eta \rangle = \langle \xi, \eta \rangle$ for all $A \in SO(n)$. For $n \geq 2$ the standard representation of $SO(n)$ on $\mathbb{R}^n$ is irreducible over $\mathbb{R}$, so by Schur's lemma any invariant inner product is a positive multiple of the Euclidean one: an invariant symmetric bilinear form is represented by a symmetric matrix S with $A^{\top} S A = S$ for all $A \in SO(n)$, and the only such matrices are scalar multiples of the identity, $S = cI$ with $c > 0$ by positive-definiteness. Thus the $SO(n+1)$-invariant metric on S^n is unique up to a positive scalar, matching the intuition that a maximally symmetric space has essentially one metric.

Identifying the round metric. It remains to see that the invariant metric attached to the Euclidean inner product on $\mathfrak{m}$ is the round metric of radius 1. Under the isomorphism $d\pi_e|_{\mathfrak{m}} \colon \mathfrak{m} \to T_o S^n$, the generator $X_\xi \in \mathfrak{m}$ maps to the velocity at o of the curve $t \mapsto \exp(tX_\xi) \cdot o$ in S^n. Compute this velocity directly: $\exp(tX_\xi) \cdot e_{n+1}$ is the geodesic of the round sphere leaving the north pole, and its derivative at $t = 0$ is $X_\xi \, e_{n+1} = \begin{pmatrix} 0 & -\xi \\ \xi^{\top} & 0 \end{pmatrix} \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -\xi \\ 0 \end{smallmatrix}\right) = -\xi \in \mathbb{R}^n = e_{n+1}^{\perp} = T_o S^n$. So $d\pi_e|_{\mathfrak{m}}$ sends $X_\xi \mapsto -\xi$, a linear isometry from $(\mathfrak{m}, \text{Euclidean})$ to $(T_o S^n, \text{induced round metric})$, since $|X_\xi| = |\xi| = |-\xi|$ and the induced metric on $T_o S^n \subset \mathbb{R}^{n+1}$ is the restriction of the Euclidean inner product. The $SO(n+1)$-invariant metric built from the Euclidean inner product on $\mathfrak{m}$ therefore has value at o equal to the induced round metric, and being $SO(n+1)$-invariant it agrees with the round metric everywhere, since the round metric is itself $SO(n+1)$-invariant and both are determined by their common value at o by the injectivity in theorem 9.2.5. Thus the correspondence recovers S^n with its round metric from the pair $\big(SO(n+1)/SO(n),\ \text{Euclidean inner product on } \mathfrak{m}\big)$, and the maximal isotropy $SO(n)$ forces this metric to be unique up to scale, consistent with the constant curvature and the maximal isometry dimension $\binom{n+1}{2}$ recorded in example 9.2.2.

9.3 Bi-invariant Metrics on Lie Groups

The homogeneous spaces of the previous section were quotients G/H, and their invariant metrics were governed by the isotropy action of H on a complement $\mathfrak{m}$. A Lie group G is the extreme case $H = \{e\}$: it acts on itself, transitively and freely, by translations. There are two families of translations, left and right, and each can be asked to be an isometry. A metric invariant under both is *bi-invariant*, and this section develops its Riemannian geometry from the algebra of $\mathfrak{g} = T_eG$ alone. The payoff is that every metric quantity becomes a bracket computation. The Levi-Civita connection of two left-invariant fields is half their Lie bracket; the geodesics through the identity are the one-parameter subgroups $t \mapsto \exp(tX)$; and the curvature is a triple bracket, forcing sectional curvature to be nonnegative, and to vanish exactly along the abelian directions. The unit three-sphere $SU(2)$ is the running example, and it recovers, from pure Lie algebra, the round metric of curvature 1 that the Hopf submersion of example 7.5.8 produced by a transverse computation.

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$, identified with T_eG and with the space of left-invariant vector fields ([6]). For $a \in G$ write $L_a\colon G \to G$, $L_a(x) = ax$, and $R_a(x) = xa$ for the left and right translations, both diffeomorphisms. The exponential $\exp\colon \mathfrak{g} \to G$, the one-parameter subgroups $t \mapsto \exp(tX)$, the adjoint representations $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ and $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ with $\mathrm{ad}_X Y = [X, Y]$, and Haar measure on a compact group are all imported from [6] and used without reproof. The curvature sign convention is that of box 4.1.1, so $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$, and ∇ is the Levi-Civita connection of definition 2.3.2. No coordinate frames appear, and the Einstein convention is not needed.

A left-invariant metric is cheap: pick any inner product on $\mathfrak{g} = T_eG$ and slide it around by left translation. Requiring right translations to be isometries as well is a genuine constraint, and the definition and the following proposition pin down exactly what that constraint says on the Lie algebra.

Definition 9.3.1: Left-, right-, and bi-invariant metrics

Let G be a Lie group with a Riemannian metric g. The metric is *left-invariant* if every left translation $L_a\colon G \to G$ is an isometry, $L_a^* g = g$ for all $a \in G$; *right-invariant* if every right translation R_a is an isometry, $R_a^* g = g$; and *bi-invariant* if it is both left- and right-invariant. A left-invariant metric is determined by its value at the identity: the inner product $\langle -, - \rangle_e$ on $T_eG = \mathfrak{g}$, transported to T_aG by $d(L_{a^{-1}})_a$. Conversely any inner product on $\mathfrak{g}$ arises this way from a unique left-invariant metric, since L_a acts simply transitively. Under this correspondence, left-invariant vector fields have constant inner products: if $X, Y \in \mathfrak{g}$ are viewed as left-invariant fields, then $\langle X, Y \rangle_g$ is the constant function on G equal to $\langle X_e, Y_e \rangle_e$.

The word "bi-invariant" encodes a symmetry that the left-invariant machinery does not see. A left-invariant metric remembers only an inner product on $\mathfrak{g}$; bi-invariance demands that this inner product interact correctly with the group's inner automorphisms. The precise translation is the following, and it is the algebraic engine of the entire section.

Proposition 9.3.2: Bi-invariance and Ad-invariance

Let G be a connected Lie group. A left-invariant metric g on G, with corresponding inner product $\langle -, - \rangle$ on $\mathfrak{g}$, is bi-invariant if and only if the inner product is $\mathrm{Ad}(G)$-invariant:

$$\langle \mathrm{Ad}(a)X, \mathrm{Ad}(a)Y \rangle = \langle X, Y \rangle \qquad \text{for all } a \in G,\ X, Y \in \mathfrak{g} \tag{9.2}$$

Equivalently, differentiating at $a = e$, the operators ad_X are skew-adjoint:

$$\langle [X, Z], Y \rangle = -\langle Z, [X, Y] \rangle \qquad \text{for all } X, Y, Z \in \mathfrak{g} \tag{9.3}$$

***Proof*:**
The argument has two stages: relate right-invariance to $\mathrm{Ad}(G)$-invariance, then differentiate the group identity to the Lie-algebra identity.

Step 1: right-invariance is $\mathrm{Ad}(G)$*-invariance.* Assume g is left-invariant. For $a \in G$ the inner automorphism $C_a = L_a \circ R_{a^{-1}}$ satisfies $C_a(x) = axa^{-1}$ and fixes the identity, with differential $d(C_a)_e = \mathrm{Ad}(a)$ on $\mathfrak{g}$ ([6]). Right-invariance of a left-invariant metric is equivalent to invariance under all C_a: indeed $R_{a^{-1}} = L_{a^{-1}} \circ C_a$, and since $L_{a^{-1}}$ is already an isometry, $R_{a^{-1}}^* g = g$ holds if and only if $C_a^* g = g$. As a ranges over G, so does a^{-1}, so g is right-invariant precisely when every C_a is an isometry.

It remains to see that C_a is an isometry for all a if and only if (9.2) holds. Suppose every C_a is an isometry. Both g and C_a are left-invariant objects fixing e ($C_a(e) = e$), so $C_a^* g = g$ is determined by its value at e: it says $d(C_a)_e = \mathrm{Ad}(a)$ preserves the inner product $\langle -, - \rangle_e$ on $\mathfrak{g}$, which is exactly (9.2). Conversely, if (9.2) holds then $\mathrm{Ad}(a)$ preserves $\langle -, - \rangle_e$; because g is left-invariant and C_a intertwines left translations ($C_a \circ L_x = L_{C_a(x)} \circ C_a$), the equality of inner products at e propagates to every point, so $C_a^* g = g$. Thus g is bi-invariant if and only if (9.2) holds.

Step 2: differentiating to ad*-skew-symmetry.* Fix $X \in \mathfrak{g}$ and set $a = \exp(tX)$, so that $\mathrm{Ad}(\exp(tX)) = e^{t\,\mathrm{ad}_X}$ as a curve in $\mathrm{GL}(\mathfrak{g})$ ([6]). Then (9.2) reads $\left\langle e^{t\,\mathrm{ad}_X} Z, e^{t\,\mathrm{ad}_X} Y \right\rangle = \langle Z, Y \rangle$ for all t. Differentiating in t at $t = 0$, with $\frac{d}{dt}\big|_0 e^{t\,\mathrm{ad}_X} = \mathrm{ad}_X$,

$$0 = \langle \mathrm{ad}_X Z, Y \rangle + \langle Z, \mathrm{ad}_X Y \rangle = \langle [X, Z], Y \rangle + \langle Z, [X, Y] \rangle$$

which is (9.3). Conversely, (9.3) says each ad_X is skew-adjoint, so $e^{t\,\mathrm{ad}_X}$ is orthogonal for every t; the orthogonal operators of the form $\mathrm{Ad}(a)$ for $a \in \exp(\mathfrak{g})$ therefore preserve the inner product, and since G is connected it is generated by $\exp(\mathfrak{g})$, so $\mathrm{Ad}(a)$ preserves the inner product for all $a \in G$. This is (9.2), completing the proof.

The equivalence is the reason bi-invariant metrics are computable. Condition (9.3) says that with respect to a bi-invariant inner product each bracket operator $\mathrm{ad}_X = [X, -]$ is a skew-symmetric endomorphism of $\mathfrak{g}$, so the metric and the bracket are compatible in the way an inner product is compatible with the rotation generators of $\mathrm{SO}(n)$. Every computation below reduces to sliding a bracket from one slot of the inner product to the other, at the cost of a sign, and (9.3) is what licenses that move. The two forms are used interchangeably: the group form (9.2) for existence, the infinitesimal form (9.3) for connection and curvature.

Not every Lie group admits a bi-invariant metric. A noncompact simple group such as $\mathrm{SL}(2, \mathbb{R})$ does not, because its Killing form is indefinite and no Ad-invariant inner product exists. Compactness removes the obstruction entirely: any inner product can be averaged over the group to make it Ad-invariant, and the averaging integral converges precisely because the group is compact.

Theorem 9.3.3: Existence of bi-invariant metrics

Every compact Lie group G admits a bi-invariant Riemannian metric.

***Proof*:**
The strategy is to produce an $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$ by averaging an arbitrary one against Haar measure, then invoke proposition 9.3.2. A compact Lie group need not be connected, but connectedness is not needed here: the correspondence of proposition 9.3.2 used it only to pass from (9.3) back to (9.2), whereas below the group form (9.2) is established directly.

Step 1: averaging an inner product. Let $\langle -, - \rangle_0$ be any inner product on $\mathfrak{g}$; one exists because $\mathfrak{g}$ is a finite-dimensional real vector space. Since G is compact it carries a bi-invariant Haar measure μ, which may be normalized to total mass $\mu(G) = 1$ ([6]). Define a new symmetric bilinear form on $\mathfrak{g}$ by

$$\langle X, Y \rangle = \int_G \langle \mathrm{Ad}(a)X,\ \mathrm{Ad}(a)Y \rangle_0 \, d\mu(a)$$

The integrand $a \mapsto \langle \mathrm{Ad}(a)X, \mathrm{Ad}(a)Y \rangle_0$ is continuous in a (the map $a \mapsto \mathrm{Ad}(a)$ is smooth and $\langle -, - \rangle_0$ is bilinear), and G is compact, so the integral converges and defines a real number for each X, Y. Bilinearity and symmetry pass through the integral. For positive-definiteness, take $Y = X \neq 0$: each integrand $\langle \mathrm{Ad}(a)X, \mathrm{Ad}(a)X \rangle_0 = |\mathrm{Ad}(a)X|_0^2 \geq 0$ is nonnegative, and it is strictly positive at $a = e$ where $\mathrm{Ad}(e)X = X \neq 0$; by continuity it is positive on a neighborhood of e of positive measure, so the integral is strictly positive. Hence $\langle -, - \rangle$ is an inner product on $\mathfrak{g}$.

Step 2: Ad*-invariance.* Fix $b \in G$. Since Ad is a homomorphism, $\mathrm{Ad}(a)\,\mathrm{Ad}(b) = \mathrm{Ad}(ab)$, so

$$\langle \mathrm{Ad}(b)X, \mathrm{Ad}(b)Y \rangle = \int_G \langle \mathrm{Ad}(a)\,\mathrm{Ad}(b)X,\ \mathrm{Ad}(a)\,\mathrm{Ad}(b)Y \rangle_0 \, d\mu(a) = \int_G \langle \mathrm{Ad}(ab)X,\ \mathrm{Ad}(ab)Y \rangle_0 \, d\mu(a)$$

Now replace the integration variable a by ab^{-1}; by right-invariance of Haar measure ($d\mu(ab^{-1}) = d\mu(a)$, from bi-invariance of μ),

$$\int_G \langle \mathrm{Ad}(ab)X,\ \mathrm{Ad}(ab)Y \rangle_0 \, d\mu(a) = \int_G \langle \mathrm{Ad}(a)X,\ \mathrm{Ad}(a)Y \rangle_0 \, d\mu(a) = \langle X, Y \rangle$$

so $\langle \mathrm{Ad}(b)X, \mathrm{Ad}(b)Y \rangle = \langle X, Y \rangle$ for every $b \in G$. This is (9.2).

Step 3: transport to a metric. The $\mathrm{Ad}(G)$-invariant inner product $\langle -, - \rangle$ on $\mathfrak{g}$ defines, by left translation, a left-invariant Riemannian metric on G as in definition 9.3.1. By the group form (9.2) of proposition 9.3.2, which as noted does not require connectedness, this left-invariant metric is bi-invariant. Thus G carries a bi-invariant metric, completing the proof.

The averaging argument is the same device that produced invariant metrics on homogeneous spaces, specialized to $H = \{e\}$ acting through the adjoint representation. Compactness cannot be dropped: it is exactly the condition that makes the Haar integral finite, and dropping it can destroy the conclusion, as the indefinite Killing form of $\mathrm{SL}(2,\mathbb{R})$ shows. On a compact group the bi-invariant metric need not be unique, but on a compact *simple* group it is unique up to scale, because the $\mathrm{Ad}(G)$-invariant inner products then form a one-dimensional cone spanned by the negative of the Killing form. The torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ is the abelian extreme: every inner product on $\mathfrak{g} = \mathbb{R}^n$ is $\mathrm{Ad}(G)$-invariant, since Ad is trivial, and each yields a flat bi-invariant metric.

With existence in hand, the Riemannian geometry follows from a single computation. The Koszul formula, fed left-invariant fields, collapses under the two forces that bi-invariance supplies: the inner products of left-invariant fields are constant, killing the derivative terms, and ad-skew-symmetry collapses the three bracket terms into one.

Proposition 9.3.4: The connection of a bi-invariant metric

Let G be a Lie group with a bi-invariant metric, and let $X, Y \in \mathfrak{g}$ be viewed as left-invariant vector fields. Then

$$\nabla_X Y = \tfrac{1}{2}[X, Y] \tag{9.4}$$

In particular $\nabla_X X = 0$ for every left-invariant X, and the left-invariant fields form a family closed under ∇.

***Proof*:**
Apply the Koszul formula (2.3) to left-invariant fields X, Y and an arbitrary left-invariant field Z:

$$2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle X, [Y, Z]\rangle + \langle Y, [Z, X]\rangle + \langle Z, [X, Y]\rangle$$

Each of the first three terms is the derivative of a function of the form $\langle U, V\rangle$ for left-invariant U, V, and by definition 9.3.1 such a function is constant on G; hence $X\langle Y, Z\rangle = Y\langle Z, X\rangle = Z\langle X, Y\rangle = 0$. The formula reduces to the three bracket terms:

$$2\langle \nabla_X Y, Z\rangle = -\langle X, [Y, Z]\rangle + \langle Y, [Z, X]\rangle + \langle Z, [X, Y]\rangle$$

Now use ad-skew-symmetry (9.3) to rewrite the first two terms so that $[X, Y]$ appears in every slot paired against Z. For the first, (9.3) with the roles $\langle [Y, Z], X\rangle = -\langle Z, [Y, X]\rangle$ gives

$$-\langle X, [Y, Z]\rangle = -\langle [Y, Z], X\rangle = \langle Z, [Y, X]\rangle = -\langle Z, [X, Y]\rangle$$

For the second term, move the bracket ad_Z off X and then the bracket ad_Y off X, applying (9.3) twice:

$$\langle Y, [Z, X]\rangle = \langle [Z, X], Y\rangle = -\langle X, [Z, Y]\rangle = \langle X, [Y, Z]\rangle = \langle [Y, Z], X\rangle = -\langle Z, [Y, X]\rangle = \langle Z, [X, Y]\rangle$$

Substituting both,

$$2\langle \nabla_X Y, Z\rangle = -\langle Z, [X, Y]\rangle + \langle Z, [X, Y]\rangle + \langle Z, [X, Y]\rangle = \langle Z, [X, Y]\rangle = \langle [X, Y], Z\rangle$$

This holds for every left-invariant Z, hence for every tangent vector at e, and since the metric is nondegenerate it determines $\nabla_X Y$ at e. Both $\nabla_X Y$ and $\frac{1}{2}[X, Y]$ are left-invariant fields (the connection of a left-invariant metric carries left-invariant fields to left-invariant fields, because L_a is an isometry and so intertwines the Levi-Civita connection with itself), and they agree at e; therefore they agree on all of G. This is (9.4). Setting $Y = X$ gives $\nabla_X X = \frac{1}{2}[X, X] = 0$, as we sought to show.

The formula $\nabla_X Y = \frac{1}{2}[X, Y]$ is the computation example 7.5.8 of section 7.5 took on faith. There the round unit metric on $S^3 = SU(2)$ was declared bi-invariant and $\tilde{\nabla}_U W = \frac{1}{2}[U, W]$ was read off the Koszul formula to compute O'Neill's tensor for the Hopf fibration; (9.4) is that identity, now proved for every bi-invariant metric. The antisymmetry of the right-hand side, $\nabla_X Y = -\nabla_Y X$ for left-invariant fields, records that the symmetric part of ∇ on left-invariant fields vanishes, which is the connection-level form of the fact that one-parameter subgroups are geodesics. That is the next result.

The geodesics fall out of (9.4) at once. A one-parameter subgroup has velocity a left-invariant field along itself, and $\nabla_X X = 0$ says such a field is autoparallel.

Theorem 9.3.5: Geodesics and curvature of a bi-invariant metric

Let G be a Lie group with a bi-invariant metric, and regard $X, Y, Z \in \mathfrak{g}$ as left-invariant vector fields.

1. The geodesics of G through the identity are exactly the one-parameter subgroups: for $X \in \mathfrak{g}$, the curve $\gamma(t) = \exp(tX)$ is the geodesic with $\gamma(0) = e$ and $\dot\gamma(0) = X$. The geodesic through an arbitrary $a \in G$ with initial velocity $d(L_a)_e X$ is $t \mapsto a\exp(tX)$, so G is geodesically complete.

2. The curvature endomorphism is

$$R(X, Y)Z = -\tfrac{1}{4}\big[[X, Y], Z\big] \tag{9.5}$$

and consequently, for g-orthonormal X, Y, the sectional curvature is

$$K(X, Y) = \tfrac{1}{4}\,\|[X, Y]\|^2 \;\geq\; 0 \tag{9.6}$$

A bi-invariant metric has nonnegative sectional curvature, and $K(X, Y) = 0$ precisely when $[X, Y] = 0$, that is, along the abelian directions.

***Proof*:**
The two parts use proposition 9.3.4 in turn.

1. *Geodesics.* Let $X \in \mathfrak{g}$ and $\gamma(t) = \exp(tX)$. By construction of the one-parameter subgroup, γ is an integral curve of the left-invariant field X, so $\dot\gamma(t) = X_{\gamma(t)}$; in particular $\gamma(0) = e$ and $\dot\gamma(0) = X_e = X$ ([6]). The covariant acceleration is the covariant derivative of the velocity field along γ. Since the velocity is the restriction of the left-invariant field X to γ, and γ is an integral curve of X, the covariant derivative along γ agrees with $\nabla_X X$ evaluated along γ:
$$D_t\dot\gamma = (\nabla_X X) \circ \gamma = 0$$
by proposition 9.3.4. Thus γ satisfies the geodesic equation $D_t\dot\gamma = 0$ of definition 3.1.1 with initial data (e, X), and by the uniqueness of geodesics it is *the* geodesic with that data. For a general starting point $a \in G$, the left translation L_a is an isometry, hence carries geodesics to geodesics, so $L_a \circ \gamma$, that is $t \mapsto a\exp(tX)$, is the geodesic with $(L_a \circ \gamma)(0) = a$ and velocity $d(L_a)_e\dot\gamma(0) = d(L_a)_e X$. Every such geodesic is defined for all $t \in \mathbb{R}$, because $\exp(tX)$ is; since every geodesic of G has this form (each initial velocity at a is $d(L_a)_e X$ for a unique $X \in \mathfrak{g}$), all geodesics extend to all of $\mathbb{R}$ and G is geodesically complete.

2. *Curvature.* Compute $R(X,Y)Z$ for left-invariant X,Y,Z from the sign convention of box 4.1.1 and definition 4.1.2:

$$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$$

Each covariant derivative of left-invariant fields is given by (9.4), and $[X,Y]$ is again left-invariant, so (9.4) applies to $\nabla_{[X,Y]}Z$ as well:

$$\nabla_X\nabla_Y Z = \nabla_X\big(\tfrac{1}{2}[Y,Z]\big) = \tfrac{1}{4}\big[X,[Y,Z]\big], \qquad \nabla_Y\nabla_X Z = \tfrac{1}{4}\big[Y,[X,Z]\big]$$

$$\nabla_{[X,Y]}Z = \tfrac{1}{2}\big[[X,Y],Z\big]$$

Assembling,

$$R(X,Y)Z = \tfrac{1}{4}\big[X,[Y,Z]\big] - \tfrac{1}{4}\big[Y,[X,Z]\big] - \tfrac{1}{2}\big[[X,Y],Z\big]$$

The first two terms are simplified by the Jacobi identity. Writing the Jacobi identity as $\big[X,[Y,Z]\big] + \big[Y,[Z,X]\big] + \big[Z,[X,Y]\big] = 0$ and using $[Y,[X,Z]] = -[Y,[Z,X]]$,

$$\big[X,[Y,Z]\big] - \big[Y,[X,Z]\big] = \big[X,[Y,Z]\big] + \big[Y,[Z,X]\big] = -\big[Z,[X,Y]\big] = \big[[X,Y],Z\big]$$

Substituting this into the curvature expression,

$$R(X,Y)Z = \tfrac{1}{4}\big[[X,Y],Z\big] - \tfrac{1}{2}\big[[X,Y],Z\big] = -\tfrac{1}{4}\big[[X,Y],Z\big]$$

which is (9.5). For the sectional curvature of an orthonormal pair X,Y, the convention of definition 4.3.1 gives $K(X,Y) = \langle R(X,Y)Y, X\rangle$. By (9.5) with $Z=Y$,

$$K(X,Y) = \big\langle -\tfrac{1}{4}\big[[X,Y],Y\big], X\big\rangle = -\tfrac{1}{4}\big\langle\big[[X,Y],Y\big], X\big\rangle$$

Move the outer bracket ad_Y across the inner product by ad-skew-symmetry (9.3). Since $\big[[X,Y],Y\big] = -\big[Y,[X,Y]\big]$,

$$\big\langle\big[[X,Y],Y\big],X\big\rangle = -\big\langle\big[Y,[X,Y]\big],X\big\rangle = \langle[X,Y],[Y,X]\rangle = -\langle[X,Y],[X,Y]\rangle = -\,|[X,Y]|^2$$

where the second equality is (9.3) applied to ad_Y (moving it from the first slot onto X, producing $[Y,X]$ in the second slot), and the third uses $[Y,X] = -[X,Y]$. Therefore

$$K(X,Y) = -\tfrac{1}{4}\big(-\,|[X,Y]|^2\big) = \tfrac{1}{4}\,|[X,Y]|^2$$

which is (9.6). The right-hand side is nonnegative, and it vanishes exactly when $[X,Y]=0$; for an orthonormal pair this is the statement that the plane they span is flat precisely when they commute, that is, along the abelian directions of $\mathfrak{g}$. This completes the proof.

The three formulas of proposition 9.3.4 and theorem 9.3.5 turn Riemannian geometry on G into bracket algebra. The connection is half a bracket, the geodesics are the exponential's one-parameter subgroups, and the curvature is a quarter of a double bracket with the sign fixed so that $K \geq 0$. The nonnegativity has a clean reading: a bi-invariant metric can never be negatively curved, and it is flat exactly where the group is locally abelian. An abelian Lie group with a bi-invariant metric, such as the torus $\mathbb{T}^n$, has $[\mathfrak{g},\mathfrak{g}] = 0$, so (9.6) gives $K \equiv 0$ and the metric is flat, matching the flat torus of example 1.3.6. At the opposite extreme, a compact simple group is curved in every nonabelian direction. The curvature formula also explains why bi-invariant metrics do not exist on noncompact simple groups without violating the sign: negative sectional curvatures there would contradict (9.6), and indeed the required Ad-invariant inner product fails to be positive-definite.

The identification of geodesics with one-parameter subgroups says that on a bi-invariant metric the Riemannian exponential $\exp_e$ at the identity coincides with the Lie-group exponential $\exp\colon \mathfrak{g} \to G$. This is the geometric meaning of the name shared by the two maps, and it holds precisely for bi-invariant metrics; a left-invariant metric that fails to be right-invariant generally has $\exp_e \neq \exp$.

The running example is the unit three-sphere, presented as $SU(2)$. Its Lie algebra $\mathfrak{su}(2)$ carries a bi-invariant inner product in which the standard brackets are $[e_1, e_2] = 2e_3$ and cyclic, and the curvature formula returns $K \equiv 1$, identifying the metric as the round $S^3(1)$. This is the mandatory reconciliation with the Hopf computation of example 7.5.8.

Example 9.3.6: $SU(2)$ as the round three-sphere

Let $G = SU(2)$, the group of 2×2 complex unitary matrices of determinant 1. Its Lie algebra $\mathfrak{su}(2)$ consists of the trace-free skew-Hermitian matrices, a three-dimensional real vector space with basis

$$e_1 = \tfrac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad e_2 = \tfrac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad e_3 = \tfrac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

A direct matrix computation of commutators gives

$$[e_1, e_2] = e_3, \qquad [e_2, e_3] = e_1, \qquad [e_3, e_1] = e_2 \tag{9.7}$$

so $\mathfrak{su}(2)$ is the standard three-dimensional cross-product algebra in this basis.

The bi-invariant metric. Since $SU(2)$ is compact, theorem 9.3.3 provides a bi-invariant metric. The Killing form of $\mathfrak{su}(2)$ is negative-definite and Ad-invariant, so any positive multiple of its negative is an Ad-invariant inner product, and by simplicity of $\mathfrak{su}(2)$ these are all of them up to scale. The remaining freedom is the scale, which is fixed once an orthonormal basis is declared. The bracket structure constants (9.7) are fixed by the algebra, but the constant relating a bracket to the basis vectors in an *orthonormal* basis depends on the scale: if $\hat{e}_k = \lambda\, e_k$ is orthonormal for a chosen inner product, then $[\hat{e}_1, \hat{e}_2] = \lambda^2 [e_1, e_2] = \lambda^2 e_3 = \lambda\, \hat{e}_3$, and cyclically, so the orthonormal-bracket constant is λ. Choosing $\lambda = 2$, that is taking the inner product for which $\hat{e}_k = 2e_k$ is orthonormal, gives

$$[\hat{e}_1, \hat{e}_2] = 2\hat{e}_3, \qquad [\hat{e}_2, \hat{e}_3] = 2\hat{e}_1, \qquad [\hat{e}_3, \hat{e}_1] = 2\hat{e}_2 \tag{9.8}$$

This is the normalization of example 7.5.8, in which the left-invariant fields e_1, e_2, e_3 of $S^3(1) = SU(2)$ are orthonormal for the round *unit* metric and satisfy exactly (9.8). The scale $\lambda = 2$ selects the round metric of the unit three-sphere, the cross-check to be confirmed below.

The sectional curvature. The basis $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$ is orthonormal, so theorem 9.3.5 (ii) applies directly. For the pair $(\hat{e}_1, \hat{e}_2)$, by (9.6) and (9.8),

$$K(\hat{e}_1, \hat{e}_2) = \tfrac{1}{4}\, |[\hat{e}_1, \hat{e}_2]|^2 = \tfrac{1}{4}\, |2\hat{e}_3|^2 = \tfrac{1}{4} \cdot 4\, |\hat{e}_3|^2 = 1$$

and by the cyclic symmetry of (9.8) the same holds for $(\hat{e}_2, \hat{e}_3)$ and $(\hat{e}_3, \hat{e}_1)$. The value 1 in fact holds for *every* 2-plane, not only the coordinate ones. In the orthonormal basis $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$ the brackets (9.8) are twice the Euclidean cross product, $[X, Y] = 2\,(X \times Y)$, since both sides are bilinear and agree on the basis pairs. Hence for any g-orthonormal X, Y orthonormality gives $|X \times Y| = |X|\,|Y| = 1$, so $|[X, Y]| = 2\,|X \times Y| = 2$ and (9.6) yields

$$K(X, Y) = \tfrac{1}{4}\, |[X, Y]|^2 = \tfrac{1}{4} \cdot 4 = 1$$

for the plane spanned by any orthonormal pair. Homogeneity then carries the value 1 to every point; the metric therefore has constant sectional curvature $K \equiv 1$.
Identification with $S^3(1)$. The group $SU(2)$ is diffeomorphic to S^3: writing a matrix of $SU(2)$ as $\left(\begin{smallmatrix} \alpha & -\bar\beta \\ \beta & \bar\alpha \end{smallmatrix}\right)$ with $|\alpha|^2 + |\beta|^2 = 1$ identifies $SU(2)$ with the unit sphere in $\mathbb{C}^2 = \mathbb{R}^4$, so it is compact, connected, and simply connected. With the bi-invariant metric normalized by (9.8) it has constant curvature $K \equiv 1$ and is geodesically complete by theorem 9.3.5 (i). A complete, simply connected Riemannian 3-manifold of constant sectional curvature 1 is globally isometric to the round unit sphere $S^3(1)$, by the Killing–Hopf theorem (box 4.5.6); the local classification theorem 4.5.5 already gives the local model, and completeness together with simple connectivity upgrades it to a global isometry. Therefore $(SU(2), \text{bi-invariant}) = S^3(1)$, the round unit three-sphere.
Reconciliation with the Hopf submersion. This is the total space of the Hopf fibration $\pi\colon S^3(1) \to S^2(\frac{1}{2})$ of example 7.5.8. There the same fields e_1, e_2, e_3 with brackets (9.8) were used, the round unit metric was invoked as bi-invariant, and $\tilde\nabla_{e_2} e_3 = \frac{1}{2}[e_2, e_3] = e_1$ was computed from (9.4). The present example supplies the two facts taken as given there: that the round unit metric on $S^3 = SU(2)$ is bi-invariant, so $\tilde\nabla_U W = \frac{1}{2}[U, W]$ holds, and that its sectional curvature is $K_{\tilde M} \equiv 1$. O'Neill's formula then produced the base curvature $K_B = 1 + 3\,|A_{e_2} e_3|^2 = 1 + 3 = 4$, exhibiting $\mathbb{C}\mathrm{P}^1$ as $S^2(\frac{1}{2})$. The intrinsic computation of this example and the transverse computation of example 7.5.8 agree: the bi-invariant S^3 has curvature 1, and quotienting by the Hopf circles multiplies horizontal curvature by four.

9.4 Locally Symmetric and Symmetric Spaces

The homogeneous spaces of the previous sections carry the same geometry at every point because a group of isometries moves any point to any other. Symmetric spaces demand more: at each point there is a single isometry that reverses direction, fixing the point and sending every geodesic through it to its own reversal. In a normal ball this reflection is written down immediately, as $\exp_p \circ (-\mathrm{id}) \circ \exp_p^{-1}$; the content of the theory is deciding when this local reflection is an isometry, and when it extends to the whole manifold. The first question has a clean answer that ties the notion back to the object that has governed all of Part I: the reflection is a local isometry exactly when the curvature tensor is parallel, $\nabla R \equiv 0$. The second question, once the first is answered, upgrades the local symmetry to a transitive group of global isometries, so that a symmetric space is homogeneous and complete, and the whole geometry is encoded in the curvature tensor at one point together with the constraint that it be parallel.

This section proves the two structural facts in full. The parallel-curvature characterization comes first: it is the mechanism that makes symmetric spaces computable, since $\nabla R = 0$ turns the curvature into an algebraic datum on a single tangent space. Then a globally symmetric space is shown to be homogeneous, hence complete. The final classification, Cartan's, sorts the complete simply connected symmetric spaces into a Euclidean factor and irreducible pieces of compact and noncompact type dual to one another and pinned down by the simple real Lie algebras; that classification is the province of Lie theory and is stated here with its proof deferred to [10]. The section closes by placing every model space of the book into the scheme.

Throughout, (M, g) is a Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), curvature endomorphism R of definition 4.1.2 in the sign convention of box 4.1.1, and associated $(0,4)$ tensor $\mathrm{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W\rangle$ of definition 4.1.4; sectional curvature K is that of definition 4.3.1. The exponential map is $\exp_p$ of definition 3.2.1. Positive-definiteness of g is used where geodesic completeness and normal balls are invoked.

Definition 9.4.1: Geodesic symmetry and symmetric space

Let (M, g) be a Riemannian manifold and $p \in M$. Choose a normal neighborhood U of p (definition 3.2.6), on which $\exp_p \colon V \to U$ is a diffeomorphism from a star-shaped $V \subseteq T_pM$. The *geodesic symmetry* at p is the map

$$s_p \colon U \longrightarrow U, \qquad s_p = \exp_p \circ(-\mathrm{id}) \circ \exp_p^{-1}$$

where $-\mathrm{id}\colon T_pM \to T_pM$ is $v \mapsto -v$. Concretely, if $q = \exp_p(v)$ then $s_p(q) = \exp_p(-v)$: along each geodesic through p, the point at parameter distance r is sent to the point at distance r on the opposite side. In particular $s_p(p) = p$, $d(s_p)_p = -\mathrm{id}_{T_pM}$, and $s_p \circ s_p = \mathrm{id}$, so s_p is an involutive diffeomorphism of U with p as isolated fixed point.
The manifold M is a *(globally) symmetric space* if for every $p \in M$ the geodesic symmetry s_p extends to a global isometry of M, necessarily an involution fixing p with differential $-\mathrm{id}$ at p.

The map s_p exists on some normal ball around any point of any Riemannian manifold, purely from the exponential map; it is a diffeomorphism there because it is conjugate to the linear involution $-\mathrm{id}$ by the diffeomorphism $\exp_p$. What is special is whether it preserves the metric. Requiring $s_p^* g = g$ near p is a strong constraint, and requiring it at every point, together with the extension of each s_p to all of M, is the definition of a symmetric space. The differential $d(s_p)_p = -\mathrm{id}$ records the defining behavior: s_p reverses every tangent direction at p, so it reverses the velocity of every geodesic through p and interchanges the two ends of such a geodesic. The condition $d(s_p)_p = -\mathrm{id}$ is also what forces p to be an isolated fixed point, since a fixed point $q \neq p$ near p would lie on a geodesic from p that s_p reverses, moving q to the antipodal point rather than fixing it.

The round sphere already exhibits the picture, and it is worth seeing the reflection explicitly before extracting the general criterion.

Example 9.4.2: The geodesic symmetry of the round sphere

Take $S^n \subseteq \mathbb{R}^{n+1}$ with the round metric of example 1.3.4 and a point $p \in S^n$. The geodesics through p are the great circles, and the geodesic through p with unit velocity $v \in T_pS^n$ is $\gamma_v(r) = \cos(r)\, p + \sin(r)\, v$. At parameter r it reaches $q = \cos(r)p + \sin(r)v$; the geodesic symmetry sends this to $\gamma_{-v}(r) = \cos(r)p - \sin(r)v$. In ambient terms, writing a point $q = \cos(r)p + \sin(r)v$ of the normal ball, the reflection is

$$s_p(q) = \cos(r)\, p - \sin(r)\, v = 2\,\langle q, p\rangle\; p - q$$

the last equality because $\langle q, p\rangle = \cos(r)$ and $q - \langle q, p\rangle\, p = \sin(r)v$. The right-hand side $q \mapsto 2\,\langle q, p\rangle\, p - q$ is the restriction to S^n of the linear reflection of $\mathbb{R}^{n+1}$ fixing the line $\mathbb{R}p$ and negating its orthogonal complement. That linear map is orthogonal, hence an isometry of $\mathbb{R}^{n+1}$ preserving S^n, so it restricts to a global isometry of S^n that agrees with the local s_p. Thus every s_p extends to a global isometry, and S^n is a symmetric space. The extension is defined on all of S^n, not just a normal ball, which is exactly the global condition the definition asks for.

The sphere computation succeeded because the local reflection happened to be the restriction of a linear isometry. On a general manifold there is no ambient linear structure to borrow, and whether s_p preserves g near p must be read off the metric itself. The reading is governed by the Taylor expansion of g in normal coordinates, whose even-order terms are built from R and its even covariant derivatives, and whose odd-order terms are built from the odd covariant derivatives, the first of which is ∇R. Since s_p negates the normal coordinates, it preserves the even part and negates the odd part, so it is an isometry exactly when the odd part vanishes, and the leading odd obstruction is ∇R. This is the mechanism behind the following characterization, which also defines the local notion.

Definition 9.4.3: Locally symmetric space

A Riemannian manifold (M, g) is *locally symmetric* if its curvature tensor is parallel:

$$\nabla R \equiv 0$$

equivalently $\nabla \operatorname{Riem} \equiv 0$, where ∇R is the total covariant derivative of the $(1, 3)$ curvature endomorphism (definition 4.1.2) and $\nabla \operatorname{Riem}$ that of the $(0, 4)$ tensor (definition 4.1.4); the two vanish together since $\nabla g = 0$.

The name anticipates the theorem: $\nabla R \equiv 0$ is precisely the condition that makes each geodesic symmetry a local isometry, so a locally symmetric space is one that is symmetric in the infinitesimal, reflection-preserving sense near every point, without any claim that the reflections extend globally. Before the theorem, the definition already covers a large stock of examples for free.

Example 9.4.4: Constant-curvature manifolds are locally symmetric

A manifold of constant sectional curvature c has, by proposition 4.5.3,

$$R(X, Y)Z = c\big(\langle Y, Z\rangle\, X - \langle X, Z\rangle\, Y\big)$$

The right-hand side is built entirely from the metric g and the constant c. Since $\nabla g = 0$ (metric compatibility, definition 2.2.4) and c is constant, the covariant derivative of this expression vanishes: for any W,

$$(\nabla_W R)(X, Y)Z = c\big((\nabla_W g)(Y, Z)X - (\nabla_W g)(X, Z)Y\big) = 0$$

Hence $\nabla R \equiv 0$, and every constant-curvature manifold is locally symmetric. In particular the model spaces $\mathbb{R}^n$ ($c = 0$), S^n ($c > 0$, example 1.3.4), and H^n ($c < 0$, example 1.3.5) are locally symmetric, as is any manifold locally isometric to one of them. This is the flat, spherical, and hyperbolic core of the theory; the constraint $\nabla R = 0$ is strictly weaker than constant curvature, and admits also the projective spaces and the compact Lie groups, whose curvature varies with the plane but not with the point along the parallel-transport of the tensor.

The central result identifies the two notions of local symmetry: the geometric one (every geodesic symmetry is a local isometry) and the analytic one ($\nabla R = 0$). The proof turns on how the metric grows away from a point, encoded by the Jacobi fields along radial geodesics, which are what the differential of $\exp_p$ transports.

Theorem 9.4.5: Geodesic symmetries and parallel curvature

Let (M, g) be a Riemannian manifold. Then the geodesic symmetry s_p is a local isometry near p for every $p \in M$ if and only if M is locally symmetric, that is $\nabla R \equiv 0$.

The two directions are read off a single computation. The differential of s_p along a radial geodesic is expressed through the Jacobi fields that carry the differential of $\exp_p$; requiring s_p to preserve inner products of tangent vectors becomes an equation on those Jacobi fields, and expanding the equation in the radial parameter isolates its content order by order. The zeroth and quadratic orders are automatic; the cubic order is exactly $\nabla R = 0$; and once $\nabla R = 0$ holds, every higher odd order vanishes as well because each odd covariant derivative of R is expressible through ∇R and its iterates, all of which vanish. The proof makes this precise by comparing the pulled-back metric $s_p^* g$ with g through the exponential chart.

***Proof*:**
Fix $p \in M$ and work in a normal ball $U = \exp_p(V)$ about p, with $V \subseteq T_pM$ star-shaped. The map $\sigma := -\mathrm{id}_{T_pM}$ is a linear isometry of (T_pM, g_p), and by definition $s_p = \exp_p \circ \sigma \circ \exp_p^{-1}$. Since $\exp_p$ is a diffeomorphism of V onto U, the map s_p is a diffeomorphism of U, and

$$s_p^* g = g \text{ on } U \iff (\exp_p \circ \sigma)^* g = \exp_p^* g \text{ on } V$$

because $\exp_p$ is a common diffeomorphism to be undone on both sides. Both sides are metrics on the domain $V \subseteq T_pM$; the claim reduces to comparing the pullback of g under $\exp_p$ with its pullback under $\exp_p \circ \sigma$.

Step 1: the pulled-back metric through Jacobi fields. For $v \in V$ and $w \in T_v(T_pM) \cong T_pM$, the differential $d(\exp_p)_v(w)$ is computed by a Jacobi field. Consider the radial geodesic $\gamma(r) = \exp_p(r\hat{v})$ for $\hat{v} = v/|v|$, so that $\exp_p(v) = \gamma(|v|)$. The variation $\alpha(s, r) = \exp_p\big(r(\hat{v} + s\hat{w})\big)$ through geodesics from p has variation field $J_w(r) = \partial_s \alpha(0, r)$, the Jacobi field along γ with $J_w(0) = 0$ and $D_r J_w(0) = w$ (theorem 5.2.2, definition 5.2.1). By the standard identity for the differential of the exponential map, $d(\exp_p)_v(w) = \frac{1}{|v|} J_w(|v|)$ up to the reparametrization by $|v|$; more precisely, writing $t = |v|$ and letting J be the Jacobi field along the unit-speed γ with $J(0) = 0$, $D_r J(0) = w$, one has $d(\exp_p)_v(w) = \frac{1}{t} J(t)$, and hence

$$\big(\exp_p^* g\big)_v(w, w) = \big|d(\exp_p)_v(w)\big|^2 = \tfrac{1}{t^2} \langle J(t), J(t) \rangle_{\gamma(t)}$$

Thus the pulled-back metric at v in the direction w is, up to the common positive factor $1/t^2$, the squared length of a Jacobi field vanishing at $r = 0$ with initial derivative w, evaluated at $r = t$. This is the pullback under $\exp_p$.
Now compute the pullback under $\exp_p \circ \sigma$. Since $\sigma(v) = -v$ and σ is linear with $d\sigma_v = -\mathrm{id}$, the chain rule gives $d(\exp_p \circ \sigma)_v(w) = d(\exp_p)_{-v}(-w)$, so

$$\big((\exp_p \circ \sigma)^* g\big)_v(w, w) = \big(\exp_p^* g\big)_{-v}(-w, -w) = \tfrac{1}{t^2} \big\langle \bar{J}(t), \bar{J}(t) \big\rangle_{\bar{\gamma}(t)}$$

where $\bar{\gamma}(r) = \exp_p(-r\hat{v})$ is the radial geodesic in the opposite direction and $\bar{J}$ is the Jacobi field along $\bar{\gamma}$ with $\bar{J}(0) = 0$, $D_r\bar{J}(0) = -w$; the sign on $-w$ is immaterial since only $\big|\bar{J}\big|^2$ enters. Both pullbacks carry the same positive factor $1/t^2$, which cancels in the comparison $(\exp_p \circ \sigma)^* g = \exp_p^* g$. Therefore $s_p^* g = g$ on U if and only if, for every unit $\hat{v} \in T_pM$, every $w \in T_pM$, and every t with $t\hat{v} \in V$,

$$|J(t)|^2 = \big|\bar{J}(t)\big|^2 \tag{9.9}$$

the squared length of the Jacobi field along γ agreeing with that along the reversed geodesic $\bar{\gamma}$, at equal parameter.

Step 2: the Taylor expansion in t. Both γ and $\bar{\gamma}$ start at p with $\gamma(0) = \bar{\gamma}(0) = p$ and velocities $\dot{\gamma}(0) = \hat{v}$, $\dot{\bar{\gamma}}(0) = -\hat{v}$. The Jacobi equation along γ is $D_r^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$ (definition 5.2.1), with $J(0) = 0$, $D_r J(0) = w$. Expand J in a Taylor series in r about 0, differentiating the Jacobi equation and using parallel transport along γ to identify the tangent spaces $T_{\gamma(r)}M$ with T_pM; write $u = \hat{v}$ for the initial velocity. From $J(0) = 0$, $D_r J(0) = w$, and the Jacobi equation the successive covariant derivatives at $r = 0$ are

$$D_r^2 J(0) = 0, \qquad D_r^3 J(0) = -R(w,u)u, \qquad D_r^4 J(0) = -2(\nabla_u R)(w,u)u$$

the second from evaluating the Jacobi equation at $r = 0$ where $J(0) = 0$, and the third from differentiating $D_r^2 J = -R(J,\dot{\gamma})\dot{\gamma}$ twice: writing $\mathcal{R}(r)X = R(X,\dot{\gamma})\dot{\gamma}$ so that $D_r^2 J = -\mathcal{R}J$, the Leibniz rule gives $D_r^4 J = -(D_r^2\mathcal{R})J - 2(D_r\mathcal{R})(D_r J) - \mathcal{R}(D_r^2 J)$, and at $r = 0$, using $J(0) = 0$, $D_r^2 J(0) = 0$, $D_r J(0) = w$, $D_r\dot{\gamma} = 0$ (geodesic), and $(D_r\mathcal{R})(0)w = (\nabla_u R)(w,u)u$ with $\dot{\gamma}(0) = u$, only the middle term survives, giving the factor 2. Hence, transporting to T_pM,

$$J(r) = rw - \tfrac{r^3}{6}R(w,u)u - \tfrac{r^4}{12}(\nabla_u R)(w,u)u + O(r^5)$$

Taking squared lengths against the parallel-transported metric,

$$|J(r)|^2 = r^2|w|^2 - \tfrac{r^4}{3}\langle R(w,u)u,w\rangle - \tfrac{r^5}{6}\langle(\nabla_u R)(w,u)u,w\rangle + O(r^6)$$

where the r^4 cross term $2\left\langle rw, -\tfrac{r^3}{6}R(w,u)u\right\rangle = -\tfrac{r^4}{3}\langle R(w,u)u,w\rangle$ and the r^5 cross term $2\left\langle rw, -\tfrac{r^4}{12}(\nabla_u R)(w,u)u\right\rangle = -\tfrac{r^5}{6}\langle(\nabla_u R)(w,u)u,w\rangle$ collect the leading contributions; the $\langle D_r^2 J(0), -\rangle$ contribution at order r^4 vanishes because $D_r^2 J(0) = 0$.

Step 3: comparison with the reversed geodesic. The Jacobi field $\bar{J}$ along $\bar{\gamma}$ satisfies the same expansion with u replaced by $-u$ (the initial velocity of $\bar{\gamma}$) and w replaced by $-w$, but as noted only $|\bar{J}|^2$ enters and it is even in the initial derivative, so

$$|\bar{J}(r)|^2 = r^2|w|^2 - \tfrac{r^4}{3}\langle R(w,-u)(-u),w\rangle - \tfrac{r^5}{6}\langle(\nabla_{-u}R)(w,-u)(-u),w\rangle + O(r^6)$$

Now $R(w,-u)(-u) = R(w,u)u$ is even in u: replacing u by $-u$ leaves it unchanged, being quadratic in the second and third slots. Thus the r^4 terms of $|J|^2$ and $|\bar{J}|^2$ agree automatically, for any manifold. The r^5 term, however, is *odd* in u: $(\nabla_{-u}R)(w,-u)(-u) = -(\nabla_u R)(w,u)u$, the three sign changes in the derivative direction and the two curvature slots giving a net $(-1)^3 = -1$. Therefore

$$|J(r)|^2 - |\bar{J}(r)|^2 = -\tfrac{r^5}{3}\langle(\nabla_u R)(w,u)u,w\rangle + O(r^6) \tag{9.10}$$

the two r^5 contributions adding rather than cancelling. The order-5 coefficient is the first possible obstruction to (9.9); all lower orders match identically.

Step 4: both directions. Suppose first $\nabla R \equiv 0$. Rather than compare Taylor coefficients, which for a metric that is only smooth would not force equality of the two functions in (9.9), solve the Jacobi equation exactly. Fix the unit-speed geodesic γ with $\dot{\gamma}(0) = u$ and let $\mathcal{R}(r)\colon T_{\gamma(r)}M \to T_{\gamma(r)}M$, $\mathcal{R}(r)X = R(X,\dot{\gamma})\dot{\gamma}$, be the associated curvature operator along γ. Since $\nabla R \equiv 0$ and $\dot{\gamma}$ is parallel ($D_r\dot{\gamma} = 0$, as γ is a geodesic), the operator $\mathcal{R}$ is parallel along γ: $D_r\mathcal{R} = 0$. Choose a parallel orthonormal frame $\{E_i(r)\}$ along γ and write $J(r) = \sum_i y_i(r)E_i(r)$; then $D_r^2 J = \sum_i \ddot{y}_i E_i$, and because $\mathcal{R}$ is parallel its matrix $A = (\langle\mathcal{R}E_j, E_i\rangle)$ in this frame is constant, symmetric (a curvature symmetry, proposition 4.2.1), and independent of r. The Jacobi equation $D_r^2 J = -\mathcal{R}J$ becomes the constant-coefficient linear system $\ddot{y} = -A\,y$ with initial data $y(0) = 0$, $\dot{y}(0) = w$ (the components of w in the frame). Along the reversed geodesic $\bar{\gamma}$, with velocity $\dot{\bar{\gamma}}(0) = -u$, the curvature operator is $X \mapsto R(X, -u)(-u) = R(X, u)u$, the same operator, so parallel-transporting the same frame gives the same constant matrix A; the data are $\bar{y}(0) = 0$, $\dot{\bar{y}}(0) = -w$. By linearity and uniqueness for $\ddot{y} = -A\,y$, the solution with data $(0, -w)$ is exactly the negative of the one with data $(0, w)$, so $\bar{y}(r) = -y(r)$ for all r. Since the frame is orthonormal, $\left|\bar{J}(r)\right|^2 = |\bar{y}(r)|^2 = |y(r)|^2 = |J(r)|^2$ for all r, with no analyticity assumption. Thus (9.9) holds; polarizing in w recovers the full inner product $\left(\exp_p^* g\right)_v(w, w')$, so $s_p^* g = g$ on U and s_p is a local isometry. Conversely, suppose s_p is a local isometry near every point. Then (9.9) holds identically, so the order-5 coefficient in (9.10) vanishes for all unit $u \in T_pM$ and all w:

$$\langle (\nabla_u R)(w, u)u, w \rangle = 0 \qquad \text{for all } u, w \in T_pM$$

This is a symmetric constraint whose vanishing forces $\nabla R = 0$. Fix p and set $\Phi(u, w) = \langle (\nabla_u R)(w, u)u, w \rangle$. It vanishes for all u, w, and it is a homogeneous polynomial of degree 3 in u and degree 2 in w. Polarizing in u (replacing u by $u + u'$ and extracting the multilinear part) and in w, and using the algebraic symmetries of R and the second Bianchi identity (theorem 4.2.3, proposition 4.2.1), the full component $\langle (\nabla_a R)(b, c)d, e \rangle$ is recovered from the symmetrized vanishing of Φ. Concretely, the pairing $\langle (\nabla_u R)(w, u)u, w \rangle$ is, after symmetrization, the complete symmetrization of $\nabla\,\mathrm{Riem}$ in the appropriate slots, and the second Bianchi identity together with the pair and antisymmetry relations leaves the symmetric part determining $\nabla\,\mathrm{Riem}$ entirely. Its vanishing for all u, w therefore gives $\nabla\,\mathrm{Riem} = 0$ at p, hence $\nabla R = 0$ at p; as p was arbitrary, $\nabla R \equiv 0$, completing the proof.

The theorem is the technical heart of the subject. It converts a geometric symmetry condition into the vanishing of a tensor, and thereby makes the class of locally symmetric spaces amenable to algebra. Once $\nabla R = 0$, the curvature endomorphism is invariant under parallel transport, so its value at a single point together with the metric there determines it everywhere; the geometry is then encoded in the pair (T_pM, R_p) subject to the curvature symmetries, a purely algebraic object. The proof also shows why the obstruction is ∇R and not R itself: the reflection s_p negates tangent directions, so it preserves the even (curvature) part of the metric's growth and can only be spoiled by the odd (first-derivative-of-curvature) part. A space can be as curved as one likes and still admit reflections, provided the curvature does not *change* along geodesics in the parallel sense.

Local symmetry is a pointwise-infinitesimal condition and does not by itself produce global reflections. The global condition is stronger, and it carries a structural consequence: a globally symmetric space is homogeneous. The mechanism is that composing two geodesic symmetries produces a *translation* along a geodesic, and translations move any point to any nearby point, hence, by connectedness, to any point at all.

Proposition 9.4.6: A symmetric space is homogeneous and complete

Let (M, g) be a connected globally symmetric space (definition 9.4.1). Then the isometry group $\mathrm{Isom}(M)$ acts transitively on M, so M is homogeneous in the sense of definition 9.2.1, and M is geodesically complete.

The argument is in two moves. First, a geodesic symmetry s_p, being a global isometry, extends every geodesic through p indefinitely, which already forces completeness. Second, the composition $s_q \circ s_p$ for q the midpoint of a short geodesic segment translates that geodesic by a fixed amount, carrying p to a point beyond q; sliding the midpoint along a geodesic then moves p to any point of the geodesic, and connectedness spreads transitivity to all of M.

***Proof*:**
Both conclusions follow from the global isometries s_p.

Step 1: completeness. Let $\gamma\colon [0,a) \to M$ be a geodesic parametrized by arc length, maximal to the right, and suppose for contradiction $a < \infty$. Pick r with $\frac{a}{2} < r < a$ and set $q = \gamma(r)$. The geodesic symmetry s_q is a global isometry of M by hypothesis, and isometries carry geodesics to geodesics with $s_q \circ \exp_q = \exp_q \circ d(s_q)_q$ (the intertwining of section 9.1); since $d(s_q)_q = -\mathrm{id}$, the isometry s_q reverses the geodesic γ about q, sending $\gamma(r+u) \mapsto \gamma(r-u)$ wherever both are defined. But s_q is defined on all of M, so $s_q(\gamma(r-u)) = \gamma(r+u)$ defines γ for parameters $r+u$ up to $u = r$, that is up to parameter $2r > a$, extending γ past a. This contradicts maximality, so $a = \infty$; the same argument extends γ to the left, and γ is defined on all of $\mathbb{R}$. Every geodesic extends indefinitely, so M is geodesically complete.

Step 2: local transitivity by translation. Fix $p \in M$ and a unit-speed geodesic γ with $\gamma(0) = p$, defined on all of $\mathbb{R}$ by Step 1. For a parameter c, let $q = \gamma(c/2)$ be the midpoint of the segment from $\gamma(0)$ to $\gamma(c)$, and consider the *transvection*

$$T_c := s_{\gamma(c/2)} \circ s_{\gamma(0)}$$

a composition of two global isometries, hence a global isometry. Each $s_{\gamma(t_0)}$ reverses γ about $\gamma(t_0)$: $s_{\gamma(t_0)}(\gamma(t_0+u)) = \gamma(t_0-u)$. Composing, $s_{\gamma(0)}$ sends $\gamma(u) \mapsto \gamma(-u)$, and then $s_{\gamma(c/2)}$ sends $\gamma(-u) = \gamma(c/2 - (c/2+u)) \mapsto \gamma(c/2 + (c/2+u)) = \gamma(c+u)$. Therefore

$$T_c(\gamma(u)) = \gamma(u+c) \qquad \text{for all } u \in \mathbb{R}$$

so T_c translates γ by arc length c along itself, and in particular $T_c(p) = T_c(\gamma(0)) = \gamma(c)$. As c ranges over $\mathbb{R}$, the point $\gamma(c)$ ranges over the entire geodesic γ, so the isometries T_c carry p to every point on any geodesic through p.

Step 3: from geodesics to all of M. The set $\mathcal{O}$ of points reachable from p by isometries, the orbit $\mathrm{Isom}(M) \cdot p$, contains, by Step 2, every point on every geodesic issuing from p. Since M is complete and connected, the exponential map $\exp_p$ is surjective by theorem 6.1.3: every point of M lies on a geodesic from p. Hence $\mathcal{O} = M$, and $\mathrm{Isom}(M)$ acts transitively. A group of isometries acting transitively presents M as homogeneous in the sense of definition 9.2.1, and homogeneity gives completeness again through proposition 9.2.3, consistent with Step 1. Thus M is homogeneous and complete, as we sought to show.

The transvections $T_c = s_{\gamma(c/2)} \circ s_{\gamma(0)}$ are the analogue, for a general symmetric space, of translation along a straight line in Euclidean space: they slide the whole manifold along a geodesic, and their one-parameter families are exactly the geodesics viewed as orbits. The proposition places symmetric spaces inside the homogeneous framework of the previous sections, so that a symmetric space is a quotient G/H with G a transitive group of isometries and H the isotropy at a base point, and the extra involution s_p makes H the fixed-point set of a Lie-group involution of G. That involutive structure is what the classification exploits.

With completeness and homogeneity in hand, the local invariant R (constrained by $\nabla R = 0$) determines the geometry, and Cartan's theorem sorts the possibilities. The statement below is the endpoint of the theory; its proof and the accompanying classification tables are matters of the structure theory of real Lie algebras and are developed in [10].

Theorem 9.4.7: Cartan's classification of symmetric spaces

Let (M, g) be a complete, simply connected Riemannian symmetric space. Then M splits as a Riemannian product
$$M \cong M_0 \times M_+ \times M_-$$
where M_0 is a Euclidean space $\mathbb{R}^k$ (the *flat* factor), M_+ is a product of irreducible symmetric spaces of *compact type* (nonnegative sectional curvature, associated to compact simple Lie groups), and M_- is a product of irreducible symmetric spaces of *noncompact type* (nonpositive sectional curvature, associated to noncompact real forms). The irreducible factors are classified by the simple real Lie algebras: each irreducible symmetric space corresponds to an involutive automorphism of a simple Lie algebra, and the compact-type and noncompact-type factors occur in dual pairs, one obtained from the other by the sign change $R \mapsto -R$ on the curvature at the base point.

The classification is the reason symmetric spaces are a closed and computable class rather than an open-ended one. It says that, up to taking Riemannian products, the complete simply connected symmetric spaces are exhausted by a Euclidean factor together with a finite list of irreducible pieces, and that list is in bijection with the list of simple real Lie algebras produced by the Cartan classification of Lie theory. The duality between compact and noncompact type is the geometric form of an algebraic operation: negating the curvature at one point turns a compact-type space (curving toward its geodesics, like a sphere) into its noncompact-type dual (curving away, like hyperbolic space), and the two are related by complexifying the Lie algebra and choosing a different real form. The sphere S^n and hyperbolic space H^n are the rank-one, constant-curvature instance of this duality; the complex projective space $\mathbb{C}\mathrm{P}^n$ and its noncompact dual the complex hyperbolic space are the next. The full proof requires the structure theory of semisimple real Lie algebras, the Cartan decomposition, and the correspondence between symmetric spaces and orthogonal symmetric Lie algebras, all of which belong to [10]; the theorem is stated here to place the geometry of Part I in its final structural frame.

Cartan's list is not needed to recognize the examples that have run through the book. Each model space and each running example is a symmetric space, and its type is read off its curvature.

Example 9.4.8: The model spaces, Lie groups, and projective spaces as symmetric spaces

The book's recurring manifolds are all symmetric spaces, distributed across the three Cartan types.

1. *Euclidean space* $\mathbb{R}^n$. The geodesic symmetry at p is $s_p(x) = 2p - x$, the point reflection through p, which is an affine isometry of $(\mathbb{R}^n, \bar{g})$ defined on all of $\mathbb{R}^n$. Thus $\mathbb{R}^n$ is a symmetric space, of flat type: $K \equiv 0$, and it is the M_0 factor of theorem 9.4.7. Its curvature is zero, so trivially parallel, consistent with local symmetry.

2. *The sphere S^n (compact type).* As computed in example 9.4.2, the geodesic symmetry at p extends to the linear reflection $x \mapsto 2\langle x, p\rangle p - x$ of $\mathbb{R}^{n+1}$ restricted to S^n, a global isometry. With $K \equiv 1/r^2 > 0$ (example 1.3.4), S^n is an irreducible symmetric space of compact type, of rank one. Its noncompact dual is H^n.

3. *Hyperbolic space H^n (noncompact type).* The hyperboloid model in $\mathbb{R}^{1,n}$ (example 1.3.5) carries geodesic symmetries extending to the Lorentz reflections fixing a timelike line and negating its spacelike orthogonal complement, global isometries of H^n. With $K \equiv -1 <$

0, H^n is an irreducible symmetric space of noncompact type; it is a Cartan–Hadamard manifold (theorem 6.4.3), complete, simply connected, of nonpositive curvature, and it is the noncompact dual of S^n under $R \mapsto -R$.

4. *A compact Lie group with a bi-invariant metric.* Let G be a compact Lie group with a bi-invariant metric (definition 9.3.1); the running instance is $SU(2)$ with the metric of example 9.3.6. The geodesic symmetry at the identity is the inversion

$$s_e(g) = g^{-1}$$

which reverses each one-parameter subgroup $t \mapsto \exp(tX)$, the geodesics through e, since $(\exp tX)^{-1} = \exp(-tX)$. Inversion is a diffeomorphism, and for a bi-invariant metric it is an isometry: bi-invariance makes $d(\mathrm{inv})_e = -\mathrm{id}$ and the inversion intertwines with left and right translations, both isometries, so s_e is a global isometry with $d(s_e)_e = -\mathrm{id}$. Translating s_e by the transitive isometry group produces s_g at every g, so G is a symmetric space. For $SU(2) = S^3(1)$ the sectional curvature is $K \equiv 1$ (example 9.3.6), of compact type, agreeing with its identification as the round three-sphere; in general a compact Lie group with bi-invariant metric has $K \geq 0$ and is of compact type.

5. *Complex projective space* $\mathbb{C}\mathrm{P}^n$ *(compact type, rank one).* The Fubini–Study metric of example 7.5.9 makes $\mathbb{C}\mathrm{P}^n$ a symmetric space: the geodesic symmetry at a point extends to the isometry induced by the unitary reflection of $\mathbb{C}^{n+1}$ fixing the complex line of the point and negating its Hermitian orthogonal complement, descending through the Hopf submersion. Its curvature is pinched, $1 \leq K \leq 4$, positive but nonconstant, so $\mathbb{C}\mathrm{P}^n$ is a symmetric space of compact type that is not of constant curvature; it is the rank-one compact-type space beyond the sphere, its noncompact dual being complex hyperbolic space. This is the example in which parallel curvature $\nabla R = 0$ holds with R varying over the 2-planes at a point, illustrating that local symmetry is strictly weaker than constant curvature.

The five cases populate the classification: $\mathbb{R}^n$ is the flat factor, S^n and $\mathbb{C}\mathrm{P}^n$ and the compact Lie groups are of compact type, and H^n is of noncompact type, dual to S^n. Every one has $\nabla R \equiv 0$, and each geodesic symmetry extends globally, so the parallel-curvature criterion of theorem 9.4.5 and the homogeneity of proposition 9.4.6 are visible in each.

Part II

Analysis and Curvature on Manifolds

The second Part develops the theme that curvature governs analysis and characteristic classes. It begins by freeing the connection story of Part I from the tangent bundle. Chapter 10 places a connection on an arbitrary vector bundle and studies its curvature as an endomorphism-valued two-form, recovering the Riemann tensor as the special case of the tangent bundle and the Levi-Civita connection. Chapter 11 feeds that curvature through invariant polynomials to manufacture the characteristic classes, the Chern and Pontryagin and Euler classes, whose cohomology classes see only the bundle and not the connection; the Part's first debt to Part I is discharged there, when the Euler form of the tangent bundle is integrated to prove the generalized Gauss–Bonnet–Chern theorem stated in chapter 8.

The metric then becomes an analytic operator. Chapter 12 builds the Hodge Laplacian from the metric and proves, through elliptic theory, that every de Rham cohomology class has a unique harmonic representative, so that the topology of the manifold is computed by a space of solutions to a differential equation. Chapter 13 closes the Part, and the book, by coupling the two threads: the Weitzenböck formula writes the Hodge Laplacian on forms as the connection Laplacian plus a curvature term, and Bochner's technique turns a sign on that curvature into the vanishing of harmonic fields, returning Part II's analysis to Part I's curvature. This general analysis-and-curvature layer is upstream infrastructure for the whole geometry track, shared by geometric analysis, spin geometry and index theory, gauge theory, and the complex and Kähler refinement of Hodge theory, and it is developed here as a self-contained unit that a reader may take up on its own once the metric foundations of Part I are in place.

10

Connections and Curvature on Vector Bundles

Part I built its entire theory on one connection, the Levi-Civita connection of a metric, living on one bundle, the tangent bundle. Its curvature governed geodesics, comparison, topology, and symmetry. Part II keeps curvature at the center but frees it from the tangent bundle: a connection is placed on an arbitrary vector bundle, and its curvature is shown to produce, on the one hand, topological invariants through Chern–Weil theory, and on the other, the analytic operators whose kernels are computed by the Hodge and Bochner theorems. This chapter lays the common foundation, the differential geometry of a connection on a general vector bundle.

The generalization is immediate but its payoff is large. A connection is exactly what is needed to differentiate a section of a bundle, and the failure of second covariant derivatives to commute is again the curvature, now an endomorphism-valued two-form rather than the tangent-vector-valued tensor of Part I. Written in a local frame, the connection becomes a matrix of one-forms and the curvature a matrix of two-forms, related by Cartan's structure equation, and the whole apparatus of moving frames becomes available. Two refinements carry the chapter toward Part II's applications. When the bundle carries a metric, the compatible connections have curvature valued in the skew-symmetric endomorphisms, the algebraic feature that will make invariant polynomials of the curvature closed forms in the next chapter. When the bundle is holomorphic and Hermitian, there is a distinguished connection, the Chern connection, whose curvature is of type $(1,1)$; on a line bundle its curvature form, suitably normalized, is the first Chern form, a closed two-form whose cohomology class does not depend on the metric. That last fact, worked out on the tautological and hyperplane bundles over complex projective space, is the Chern–Weil theorem in its simplest instance, and it returns the Fubini–Study geometry of chapter 7 as the curvature of a line bundle. The general theory that promotes it to all bundles and all degrees is the subject of chapter 11.

10.1 Connections on a Vector Bundle

Part I studied one connection on one bundle: the Levi-Civita connection on the tangent bundle of a (pseudo-)Riemannian manifold, singled out among all affine connections by torsion-freeness and metric compatibility. That specialization was the right one for metric geometry, but the notion of covariant differentiation is far more general. A connection is a rule for differentiating sections of any vector bundle $E \to M$, and nothing in the rule refers to a metric on M or to the tangent bundle at all. This chapter, which opens Part II, develops that general theory: connections and their curvature on an arbitrary vector bundle, the structure and Bianchi equations in a frame, metric and Chern connections, and the curvature of line bundles. The payoff, realized here and in the two chapters that follow, is Chern–Weil theory, which reads topological invariants off the curvature of a connection.

The present section fixes the object and its local description. A connection is defined coordinate-free as an $\mathbb{R}$-linear (or $\mathbb{C}$-linear) operator satisfying a Leibniz rule; connections are shown to exist and to form an affine space modeled on the endomorphism-valued 1-forms; and the local picture in a frame is recorded, where a connection becomes a matrix of ordinary 1-forms. The frame description is where computation happens, and the transformation law of the connection matrix under a change of frame is the first place the theory diverges from the naive expectation that a connection "is" a matrix of forms: the matrix transforms inhomogeneously, by conjugation plus a correction. That inhomogeneity is not a defect but the source of the theory's content, and it is proved in full below.

Throughout, $\pi\colon E \to M$ is a smooth real or complex vector bundle of rank r over a smooth n-manifold M; the theory of vector bundles, local frames, differential forms, and partitions of unity is imported from [6]. Write $\Gamma(E)$ for the space of smooth sections of E, and

$$\Omega^k(M;E) = \Gamma\big(\Lambda^k T^*M \otimes E\big)$$

for the E-valued k-forms, with $\Omega^k(M) = \Omega^k(M;\underline{\mathbb{R}})$ the ordinary scalar k-forms. A section of E is thus an element of $\Omega^0(M;E) = \Gamma(E)$. The scalars are $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ according as E is real or complex; $C^\infty(M)$ denotes the $\mathbb{K}$-valued smooth functions in the complex case, and Einstein's summation convention (sum over any index appearing once up and once down) is in force wherever frames are used.

Definition 10.1.1: Connection on a vector bundle

A *connection* on the vector bundle $E \to M$ is a $\mathbb{K}$-linear map

$$\nabla\colon \Gamma(E) \longrightarrow \Omega^1(M;E)$$

satisfying the *Leibniz rule*

$$\nabla(fs) = df \otimes s + f\,\nabla s$$

for every $f \in C^\infty(M)$ and $s \in \Gamma(E)$. For a vector field $X \in \Gamma(TM)$, the *covariant derivative of s in the direction X* is the section

$$\nabla_X s := (\nabla s)(X) \in \Gamma(E)$$

obtained by evaluating the E-valued 1-form ∇s on X. In these terms $\nabla_X s$ is $C^\infty(M)$-linear in X, additive in s, and satisfies $\nabla_X(fs) = (Xf)\,s + f\,\nabla_X s$.

The two descriptions are equivalent. Giving the full 1-form ∇s is the same as giving $\nabla_X s$ for every

X, because a 1-form is determined by its values on vector fields; the $C^\infty(M)$-linearity in X is exactly what makes $X \mapsto \nabla_X s$ the evaluation of a form rather than a mere pointwise-in-X derivative. The Leibniz rule is the single axiom that distinguishes a connection from an arbitrary linear operator: it forces ∇ to behave as a derivation with respect to multiplication by functions, and its failure to be $C^\infty(M)$-linear in s is precisely the term $df \otimes s$. The affine connections of definition 2.1.1 are the case $E = TM$ of this definition, phrased there through $\nabla_X Y$; the reformulation as a map $\Gamma(TM) \to \Omega^1(M; TM)$ is the same operator packaged so that both arguments are visible at once.

A connection is a local operator. If s and s' agree on an open set U, then $\nabla s = \nabla s'$ on U: choosing a bump function φ supported in U with $\varphi \equiv 1$ near a point $p \in U$, the section $\varphi(s - s')$ vanishes identically, so $0 = \nabla(\varphi(s-s')) = d\varphi \otimes (s-s') + \varphi\, \nabla(s-s')$, and evaluating at p (where $d\varphi_p = 0$ and $\varphi(p) = 1$) gives $\nabla(s - s')_p = 0$. This is the standard localization argument for a Leibniz operator, imported from the tensor-field discussion of [6], and it is what lets a connection be studied one chart at a time.

Example 10.1.2: The trivial connection on a product bundle

Let $E = M \times \mathbb{K}^r$ be the trivial bundle of rank r, whose sections are smooth maps $s\colon M \to \mathbb{K}^r$, that is, r-tuples $s = (s^1, \ldots, s^r)$ of functions. The *trivial connection* d differentiates each component,

$$d\, s = (ds^1, \ldots, ds^r) \in \Omega^1(M; \mathbb{K}^r)$$

The Leibniz rule $d(fs) = df \otimes s + f\, ds$ is the componentwise product rule $d(fs^i) = s^i\, df + f\, ds^i$, so d is a connection. Its covariant derivative $d_X s = (Xs^1, \ldots, Xs^r)$ is the plain directional derivative of the component functions. This is the flat model against which every other connection on the trivial bundle is measured: any connection on $M \times \mathbb{K}^r$ has the form $\nabla = d + A$ for a matrix A of 1-forms, as the affine-space structure below makes precise. On $E = TM = \mathbb{R}^n \times \mathbb{R}^n$ over $M = \mathbb{R}^n$ in Cartesian coordinates, d is the Euclidean connection $\bar{\nabla}$ of Part I, whose Christoffel symbols all vanish.

Two questions organize the general theory at once: do connections exist on every bundle, and how many are there. Both are answered by comparing an arbitrary connection to the trivial one over each trivializing chart and gluing. The comparison is governed by a single algebraic fact, that the *difference* of two connections is a tensor.

Proposition 10.1.3: Connections form an affine space over $\Omega^1(M; \mathrm{End} E)$

Let $E \to M$ be a smooth vector bundle. Then:

1. If ∇ and ∇' are connections on E, their difference $A := \nabla' - \nabla$ is $C^\infty(M)$-linear, hence an element of $\Omega^1(M; \mathrm{End} E) = \Gamma\big(T^*M \otimes \mathrm{End} E\big)$; that is, $(\nabla' - \nabla)s$ depends on s only through its pointwise values, via a bundle map $A(X) \in \mathrm{End}(E)$ for each X.
2. Conversely, if ∇ is a connection and $A \in \Omega^1(M; \mathrm{End} E)$, then $\nabla + A$ is again a connection.
3. Connections on E exist.

Consequently the set of connections on E is an affine space modeled on the vector space $\Omega^1(M; \mathrm{End} E)$: fixing any one connection ∇_0, every connection is $\nabla_0 + A$ for a unique $A \in \Omega^1(M; \mathrm{End} E)$.

***Proof*:**
The three claims are proved in turn.

1. Both ∇ and ∇' are additive, so $A = \nabla' - \nabla$ is additive. For $f \in C^\infty(M)$ and $s \in \Gamma(E)$, the Leibniz terms cancel:

$$A(fs) = \nabla'(fs) - \nabla(fs) = \big(df \otimes s + f\,\nabla' s\big) - \big(df \otimes s + f\,\nabla s\big) = f\,\big(\nabla' s - \nabla s\big) = f\,A(s)$$

   Thus $A\colon \Gamma(E) \to \Omega^1(M;E)$ is $C^\infty(M)$-linear. By the tensor characterization of [6], a $C^\infty(M)$-linear map from sections of E to sections of $T^*M \otimes E$ is induced by a bundle homomorphism, that is, by a section A of $\mathrm{Hom}(E, T^*M \otimes E) = T^*M \otimes \mathrm{End}E$; equivalently $A \in \Omega^1(M;\mathrm{End}E)$, so that $(As)(X) = A(X)s$ with $A(X) \in \mathrm{End}(E)$ acting fiberwise. In particular $(\nabla' s)_p - (\nabla s)_p$ depends on s only through $s(p)$.

2. Additivity is immediate. For the Leibniz rule, with A tensorial (so $A(fs) = fA(s)$),

$$(\nabla + A)(fs) = \big(df \otimes s + f\,\nabla s\big) + f\,A(s) = df \otimes s + f\,(\nabla + A)(s)$$

   Hence $\nabla + A$ is a connection.

3. Existence is by patching the trivial connections of example 10.1.2. Choose an open cover $\{U_\alpha\}$ over which E trivializes, with local frames giving isomorphisms $E|_{U_\alpha} \cong U_\alpha \times \mathbb{K}^r$, and let $\nabla^\alpha = d$ be the trivial connection on U_α in that trivialization, a genuine connection on $E|_{U_\alpha}$. Let $\{\rho_\alpha\}$ be a partition of unity subordinate to $\{U_\alpha\}$ ([6]), so $\rho_\alpha \in C^\infty(M)$, $\operatorname{supp}\rho_\alpha \subseteq U_\alpha$, and $\sum_\alpha \rho_\alpha \equiv 1$ with the sum locally finite. Define

$$\nabla := \sum_\alpha \rho_\alpha\,\nabla^\alpha$$

   where each $\rho_\alpha \nabla^\alpha s$ is extended by zero outside $\operatorname{supp}\rho_\alpha$; the sum is locally finite, hence a well-defined $\mathbb{K}$-linear map $\Gamma(E) \to \Omega^1(M;E)$. It satisfies the Leibniz rule:

$$\begin{aligned}\nabla(fs) &= \sum_\alpha \rho_\alpha\,\nabla^\alpha(fs) = \sum_\alpha \rho_\alpha\big(df \otimes s + f\,\nabla^\alpha s\big)\\ &= \Big(\sum_\alpha \rho_\alpha\Big)\,df \otimes s + f\sum_\alpha \rho_\alpha\,\nabla^\alpha s = df \otimes s + f\,\nabla s\end{aligned}$$

   using $\sum_\alpha \rho_\alpha = 1$ in the last step. A convex combination of Leibniz operators is again a Leibniz operator precisely because the coefficients ρ_α sum to 1, so that the anomalous term $df \otimes s$ is reproduced with coefficient 1 rather than $\sum_\alpha \rho_\alpha \neq 1$. Hence ∇ is a connection on E.

Parts (1) and (2) say that the set of connections is nonempty by (3), and that translation by $\Omega^1(M;\mathrm{End}E)$ acts freely and transitively on it: the difference of any two is a unique element of $\Omega^1(M;\mathrm{End}E)$, and adding any such element yields a connection. This is the assertion that the connections form an affine space over $\Omega^1(M;\mathrm{End}E)$, as we sought to show.

The affine-space structure is the organizing principle of the entire theory. There is no canonical connection on a general bundle, so no "zero": the space of connections has no distinguished origin, only a well-defined difference operation valued in the endomorphism-valued 1-forms. Once one connection is chosen, every other is recorded by a tensor A, and geometric conditions on connections (metric compatibility in section 10.3, compatibility with a holomorphic structure) become affine conditions on A, often uniquely solvable. The count $\dim \Omega^1(M; \mathrm{End} E)$ measures how much freedom remains after such conditions are imposed. For $E = TM$ this recovers proposition 2.1.5: affine connections exist and their differences are $(1,2)$-tensors, which is the present statement with $\mathrm{End}(TM)$-valued 1-forms rewritten as $(1,2)$-tensors.

The affine structure is invisible until a frame is chosen, and it is in a frame that connections become computable. A *local frame* for E over an open set U is an r-tuple $(e_1, \dots, e_r)$ of sections that is a basis of each fiber E_p for $p \in U$; it exists exactly where E trivializes. Every section over U is $s = s^i e_i$ for unique functions $s^i \in C^\infty(U)$, assembled into the column $\sigma = (s^i)$. Applying ∇ to a frame produces the local data that determine it.

Definition 10.1.4: Connection matrix of a frame

Let ∇ be a connection on E and $(e_1, \dots, e_r)$ a local frame over U. Since each $\nabla e_j \in \Omega^1(U; E)$ is an E-valued 1-form, it expands in the frame with 1-form coefficients: there are unique $\omega^i_j \in \Omega^1(U)$ with

$$\nabla e_j = \omega^i_j \otimes e_i$$

The matrix $\omega = (\omega^i_j) \in \Omega^1(U; \mathfrak{gl}_r)$ of ordinary 1-forms is the *connection matrix* (or *connection form*) of ∇ in the frame (e_i). For a section $s = s^i e_i$ with column $\sigma = (s^i)$, the Leibniz rule gives the covariant derivative in the frame as

$$\nabla s = e\,(d\sigma + \omega\sigma), \qquad \text{that is} \qquad \nabla s = \left(ds^i + \omega^i_j\, s^j\right) \otimes e_i$$

where $e\,(d\sigma + \omega\sigma)$ abbreviates $\sum_i \left(ds^i + \omega^i_j s^j\right) \otimes e_i$.

The displayed formula for ∇s is a direct computation from the two axioms. Writing $s = s^j e_j$ and applying additivity and Leibniz,

$$\nabla s = \nabla(s^j e_j) = \sum_j \left(ds^j \otimes e_j + s^j\, \nabla e_j\right) = \sum_j ds^j \otimes e_j + \sum_j s^j\, \omega^i_j \otimes e_i$$

and relabelling the summation index $j \to i$ in the first sum collects the coefficient of e_i into $ds^i + \omega^i_j s^j$, which is the i-th entry of the column $d\sigma + \omega\sigma$. Thus in a frame a connection is completely encoded by the $r \times r$ matrix ω of 1-forms, and covariant differentiation is ordinary differentiation $d\sigma$ corrected by the algebraic term $\omega\sigma$. Comparison with example 10.1.2 identifies ω as the deviation $\nabla - d$ of the connection from the trivial one attached to this frame: the frame trivializes $E|_U$, and $\nabla = d + \omega$ there, matching proposition 10.1.3 with $A = \omega$ read as an $\mathrm{End}(E)$-valued 1-form.

Example 10.1.5: The connection matrix in polar coordinates

Take $E = T\mathbb{R}^2$ with the Euclidean connection $\bar\nabla$ of example 10.1.2, and the coordinate frame $(e_r, e_\theta) = (\partial_r, \partial_\theta)$ of polar coordinates on $U = \mathbb{R}^2 \setminus \{0\}$. The nonzero Christoffel symbols of $\bar\nabla$ in polar coordinates are $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = 1/r$ and $\Gamma^r_{\theta\theta} = -r$ (computed in section 2.1). Reading off $\bar\nabla \partial_j = \omega^i_j \otimes \partial_i$ from $\bar\nabla_{\partial_k} \partial_j = \Gamma^i_{kj} \partial_i$, the entry ω^i_j is the 1-form $\Gamma^i_{kj}\, dx^k$ with $x^1 = r, x^2 = \theta$. Hence

$$\omega = \begin{pmatrix} \omega^r_r & \omega^r_\theta \\ \omega^\theta_r & \omega^\theta_\theta \end{pmatrix} = \begin{pmatrix} 0 & -r\, d\theta \\ \frac{1}{r}\, d\theta & \frac{1}{r}\, dr \end{pmatrix}$$

The flat connection is not represented by the zero matrix in this frame: $\omega \neq 0$ precisely because the polar frame $(\partial_r, \partial_\theta)$ rotates and rescales from point to point. In the Cartesian frame (∂_x, ∂_y) the same connection has $\omega = 0$. The two matrices describe one connection in two frames, and the rule relating them is the content of the next result.

The connection matrix depends on the frame, and the dependence is not by simple conjugation. When the frame changes, ω picks up an inhomogeneous term built from the derivative of the change-of-frame matrix. This inhomogeneity is the reason ω is *not* a tensor, and it is worth isolating before stating the transformation law: a tensor transforms linearly under a change of frame, by conjugation of its matrix; the connection matrix does not, because the trivial connection d hidden inside $\nabla = d + \omega$ differentiates the change-of-frame matrix itself.

Proposition 10.1.6: Transformation of the connection matrix

Let $(e_1, \ldots, e_r)$ and $(\tilde{e}_1, \ldots, \tilde{e}_r)$ be local frames for E over U, related by

$$\tilde{e}_j = e_i\, g^i_j, \qquad \text{that is} \qquad \tilde{e} = e\, g$$

for a smooth map $g = (g^i_j)\colon U \to \mathrm{GL}_r(\mathbb{K})$. Let ω and $\tilde{\omega}$ be the connection matrices of a connection ∇ in the frames (e_i) and $(\tilde{e}_i)$ respectively. Then

$$\tilde{\omega} = g^{-1}\omega\, g + g^{-1}\, dg$$

the matrix product of the GL_r-valued function g (and its inverse and differential) with the 1-form matrices.

***Proof*:**
Apply ∇ to the new frame $\tilde{e}_j = e_i\, g^i_j$ and expand by additivity and the Leibniz rule, treating each entry g^i_j as a function and each e_i as a section:

$$\nabla \tilde{e}_j = \nabla\big(g^i_j\, e_i\big) = dg^i_j \otimes e_i + g^i_j\, \nabla e_i$$

Insert $\nabla e_i = \omega^k_i \otimes e_k$ into the second term:

$$\nabla \tilde{e}_j = dg^i_j \otimes e_i + g^i_j\, \omega^k_i \otimes e_k \tag{10.1}$$

This expresses $\nabla \tilde{e}_j$ in the *old* frame (e_i). To read off $\tilde{\omega}$ the right-hand side must be rewritten in the *new* frame $(\tilde{e}_i)$, using the inverse relation $e_i = \tilde{e}_l\, (g^{-1})^l_i$, which follows from $\tilde{e} = e\, g$ by right-multiplying by g^{-1}. Substituting $e_i = \tilde{e}_l\, (g^{-1})^l_i$ and $e_k = \tilde{e}_l\, (g^{-1})^l_k$ into (10.1),

$$\nabla \tilde{e}_j = (g^{-1})^l_i\, dg^i_j \otimes \tilde{e}_l + (g^{-1})^l_k\, g^i_j\, \omega^k_i \otimes \tilde{e}_l$$

By definition $\nabla \tilde{e}_j = \tilde{\omega}^l_j \otimes \tilde{e}_l$, and matching the coefficient of $\tilde{e}_l$ gives

$$\tilde{\omega}^l_j = (g^{-1})^l_k\, \omega^k_i\, g^i_j + (g^{-1})^l_i\, dg^i_j$$

The first summand is the (l, j) entry of the matrix product $g^{-1}\omega\, g$ and the second is the (l, j) entry of $g^{-1}\, dg$. Hence

$$\tilde{\omega} = g^{-1}\omega\, g + g^{-1}\, dg$$

completing the proof.

The law has two pieces, and the contrast between them is the point. The conjugation term $g^{-1}\omega g$ is what a genuine tensor would produce, the endomorphism-valued 1-form ω rewritten in the new basis. The inhomogeneous term $g^{-1}\,dg$ is extra: it is the connection matrix, in the old frame, of the trivial connection d attached to the new frame, and it records that "constant in the new frame" differs from "constant in the old frame" by exactly the derivative of g. Because of $g^{-1}\,dg$, the connection matrix is not a tensor, and one cannot make ω vanish at a point by a clever choice of frame unless the inhomogeneous term can be arranged to cancel $g^{-1}\omega g$; this is possible pointwise but generally not on a neighborhood, which is the local statement that a connection has genuine content beyond its value at a point. The curvature, built in section 10.2, is the invariant that survives this inhomogeneity: its matrix Ω transforms by pure conjugation $\tilde{\Omega} = g^{-1}\Omega g$, and it is a genuine tensor. The Yang–Mills and gauge-theory literature calls the change of frame a *gauge transformation* and $g^{-1}\,dg$ the associated *gauge term*; the same algebra reappears in section 2.1, where the transformation rule of the Christoffel symbols under a change of chart is (10.1) specialized to $E = TM$ and coordinate frames.

The polar-coordinate matrix of example 10.1.5 illustrates the law directly: passing from the Cartesian frame, where $\omega = 0$, to the polar frame, the conjugation term $g^{-1} \cdot 0 \cdot g$ vanishes and the entire polar connection matrix is the gauge term $g^{-1}\,dg$, with g the Jacobian of the change from Cartesian to polar frame fields. A connection matrix that is pure gauge term, of the form $g^{-1}\,dg$ for a globally defined g, is exactly the trivial connection written in a rotated frame, and this is the local hallmark of flatness.

Differentiating sections is the case $k = 0$ of a construction that differentiates E-valued forms of every degree. An E-valued k-form is a k-form with coefficients in E; the exterior covariant derivative extends both the exterior derivative d on scalar forms and the covariant derivative ∇ on sections into a single degree-raising operator, coupling the exterior derivative of the form part to the connection on the coefficient part.

Definition 10.1.7: Exterior covariant derivative

Let ∇ be a connection on $E \to M$. The *exterior covariant derivative* is the sequence of $\mathbb{K}$-linear maps

$$d^\nabla : \Omega^k(M;E) \longrightarrow \Omega^{k+1}(M;E) \qquad (k \geq 0)$$

defined by $d^\nabla = \nabla$ on $\Omega^0(M;E) = \Gamma(E)$ and, on a decomposable E-valued k-form $\eta = \beta \otimes s$ with $\beta \in \Omega^k(M)$ and $s \in \Gamma(E)$, by

$$d^\nabla(\beta \otimes s) = d\beta \otimes s + (-1)^k\,\beta \wedge \nabla s$$

extended $\mathbb{K}$-linearly to all of $\Omega^k(M;E)$, where $\beta \wedge \nabla s$ wedges the k-form β with the 1-form part of $\nabla s \in \Omega^1(M;E)$ and keeps the E-coefficient. It satisfies the graded Leibniz rule

$$d^\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^{\deg \alpha}\,\alpha \wedge d^\nabla \eta$$

for a scalar form $\alpha \in \Omega^\bullet(M)$ and an E-valued form $\eta \in \Omega^\bullet(M;E)$.

That the formula is well-defined, independent of the decomposition $\eta = \beta \otimes s$, follows because it is $C^\infty(M)$-balanced in the tensor product $\Omega^k(M) \otimes_{C^\infty(M)} \Gamma(E)$: replacing (β, s) by $(f\beta, s)$ or (β, fs) for $f \in C^\infty(M)$ gives the same value, using $d(f\beta) = df \wedge \beta + f\,d\beta$ against $\nabla(fs) = df \otimes s + f\,\nabla s$ and the sign $(-1)^k$, exactly as the graded Leibniz rule requires. The scalar case $E = \mathbb{R}$ with the trivial connection d recovers the ordinary exterior derivative, so d^∇ is the coupling of d to ∇. Unlike the exterior derivative, d^∇ does *not* in general square to zero; the obstruction $(d^\nabla)^2 = d^\nabla \circ d^\nabla$ is the curvature, an $\mathrm{End}(E)$-valued 2-form, taken up in section 10.2. This is the sense in which curvature measures the failure of covariant differentiation to define a cochain complex.

A connection on E induces connections on every bundle built from E by the standard tensorial operations, by imposing the Leibniz rule against the relevant pairing. These induced connections are used throughout the chapter and are recorded now.

Proposition 10.1.8: Induced connections on dual, tensor, and Hom bundles

Let ∇^E and ∇^F be connections on vector bundles $E, F \to M$. There are canonical induced connections:

1. On the dual bundle E^*, the connection ∇^{E^*} determined by requiring the natural pairing $E^* \otimes E \to \underline{\mathbb{K}}$ to be parallel:

$$d\,\langle \xi, s\rangle = \langle \nabla^{E^*}\xi, s\rangle + \langle \xi, \nabla^E s\rangle$$

   for $\xi \in \Gamma(E^*)$ and $s \in \Gamma(E)$.

2. On the tensor product $E \otimes F$, the connection

$$\nabla^{E\otimes F}(s \otimes t) = \nabla^E s \otimes t + s \otimes \nabla^F t$$

   for $s \in \Gamma(E)$, $t \in \Gamma(F)$, extended $C^\infty(M)$-linearly.

3. On $\mathrm{Hom}(E, F) \cong E^* \otimes F$, the connection determined by requiring evaluation to be parallel:

$$(\nabla^{\mathrm{Hom}}\Phi)(s) = \nabla^F\big(\Phi(s)\big) - \Phi\big(\nabla^E s\big)$$

   for $\Phi \in \Gamma(\mathrm{Hom}(E, F))$ and $s \in \Gamma(E)$; in particular $\mathrm{End}E = \mathrm{Hom}(E, E)$ acquires an induced connection.

Each formula defines a connection, and the constructions are compatible with the natural isomorphisms among these bundles.

***Proof*:**
Each construction is checked to be a well-defined connection.

1. For fixed $\xi \in \Gamma(E^*)$, define a 1-form $\nabla^{E^*}\xi$ by declaring its pairing against every $s \in \Gamma(E)$ through the displayed identity, that is, $\langle \nabla^{E^*}\xi, s\rangle := d\langle \xi, s\rangle - \langle \xi, \nabla^E s\rangle$. The right-hand side is $C^\infty(M)$-linear in s: replacing s by fs,

$$d\langle \xi, fs\rangle - \langle \xi, \nabla^E(fs)\rangle = d\big(f\langle \xi, s\rangle\big) - \langle \xi, df \otimes s + f\nabla^E s\rangle = f\big(d\langle \xi, s\rangle - \langle \xi, \nabla^E s\rangle\big)$$

   since the two $df\langle \xi, s\rangle$ terms cancel. Hence the assignment $s \mapsto \langle \nabla^{E^*}\xi, s\rangle$ is induced by an E^*-valued 1-form, defining $\nabla^{E^*}\xi \in \Omega^1(M; E^*)$. The Leibniz rule for ∇^{E^*} follows by pairing $\nabla^{E^*}(f\xi)$ against s and expanding.

2. The formula is $C^\infty(M)$-balanced over $C^\infty(M)$ in the pair (s, t), by the same cancellation of $df \otimes (\cdot)$ terms, so it descends to $E \otimes F = E \otimes_{C^\infty(M)} F$ and defines a $\mathbb{K}$-linear map $\Gamma(E \otimes F) \to \Omega^1(M; E \otimes F)$. Its Leibniz rule reads $\nabla^{E\otimes F}(f(s \otimes t)) = df \otimes (s \otimes t) + f\,\nabla^{E\otimes F}(s \otimes t)$, obtained by applying the product rule to f in either slot and adding.

3. For $\Phi \in \Gamma(\mathrm{Hom}(E,F))$, the section $\nabla^{\mathrm{Hom}}\Phi$ of $\Omega^1(M;\mathrm{Hom}(E,F))$ is defined by the displayed formula for its action on $s \in \Gamma(E)$. The right-hand side $\nabla^F(\Phi(s)) - \Phi(\nabla^E s)$ is $C^\infty(M)$-linear in s: replacing s by fs,

$$\nabla^F\big(f\Phi(s)\big)-\Phi\big(\nabla^E(fs)\big) = df\otimes\Phi(s)+f\nabla^F(\Phi(s))-df\otimes\Phi(s)-f\Phi(\nabla^E s) = f\big(\nabla^F(\Phi(s))-\Phi(\nabla^E s)\big)$$

so $\nabla^{\mathrm{Hom}}\Phi$ is a well-defined $\mathrm{Hom}(E,F)$-valued 1-form, and the Leibniz rule for ∇^{Hom} follows from that of ∇^F. Under the isomorphism $\mathrm{Hom}(E,F) \cong E^* \otimes F$ the connection of (3) matches the tensor-product connection of (2) applied to ∇^{E^*} from (1) and ∇^F, since both are characterized by making evaluation parallel; this is the asserted compatibility.

Each construction produces an additive Leibniz operator into the appropriate space of 1-forms, hence a connection, and the isomorphisms $E^{**} \cong E$, $\mathrm{Hom}(E,F) \cong E^* \otimes F$ intertwine them, completing the proof.

The induced connection on $\mathrm{End}E$ deserves separate mention, because the curvature F^∇ of section 10.2 lives in $\Omega^2(M;\mathrm{End}E)$ and is differentiated by the exterior covariant derivative built from this induced connection; the Bianchi identity is the statement that it is d^∇-closed. For $\Phi \in \Gamma(\mathrm{End}E)$ the formula reads $(\nabla^{\mathrm{End}}\Phi)(s) = \nabla(\Phi s) - \Phi(\nabla s)$, the commutator of ∇ with Φ, which is why in a frame ∇^{End} acts on a matrix-valued form by d plus the bracket $[\omega, -]$. All of these constructions are the bundle-theoretic counterparts of the induced connections on tensor bundles from definition 2.4.6, now applied to an arbitrary E rather than to TM and its tensor powers.

The chapter's running thread ties the general theory back to Part I. The Levi-Civita connection is a connection on the specific bundle $E = TM$, and reading it through the definitions above recovers its Part I description and identifies its connection matrix with the Christoffel symbols, packaged as 1-forms.

Example 10.1.9: The Levi-Civita connection as a bundle connection

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ of definition 2.3.2, and take $E = TM$, a real vector bundle of rank $r = n$. The affine-connection axioms of definition 2.1.1 are exactly the requirements of definition 10.1.1: $\mathbb{R}$-bilinearity and $C^\infty(M)$-linearity in X package $\nabla_X Y$ as the evaluation of an E-valued 1-form $\nabla Y \in \Omega^1(M;TM)$, and the Leibniz rule $\nabla_X(fY) = (Xf)Y + f\nabla_X Y$ is the direction-evaluated form of $\nabla(fY) = df \otimes Y + f\nabla Y$. So the Levi-Civita connection is a connection on TM in the sense of this section.

Its connection matrix in a coordinate frame is the Christoffel data. Take the coordinate frame $(e_1, \ldots, e_n) = (\partial_1, \ldots, \partial_n)$ of a chart $(U, (x^1, \ldots, x^n))$. By definition 2.1.3, $\nabla_{\partial_k}\partial_j = \Gamma^i_{kj}\,\partial_i$, so evaluating the defining relation $\nabla\partial_j = \omega^i_j \otimes \partial_i$ on ∂_k gives $\omega^i_j(\partial_k) = \Gamma^i_{kj}$, that is

$$\omega^i_j = \Gamma^i_{kj}\,dx^k$$

Because the Levi-Civita connection is torsion-free, its Christoffel symbols are symmetric in the lower pair, $\Gamma^i_{kj} = \Gamma^i_{jk}$ (definition 2.2.1), and the connection 1-forms may equally be written

$$\omega^i_j = \Gamma^i_{jk}\,dx^k$$

the *Christoffel 1-forms*. For a vector field $Y = Y^i\partial_i$ with column $\sigma = (Y^i)$, definition 10.1.4 then reproduces the Part I coordinate formula

$$\nabla Y = e\,(d\sigma + \omega\sigma) = \big(dY^i + \Gamma^i_{jk}\,Y^j\,dx^k\big) \otimes \partial_i$$

whose evaluation on $X = X^k\partial_k$ is $\nabla_X Y = \big(X(Y^i) + \Gamma^i_{jk}X^kY^j\big)\partial_i$, matching definition 2.1.3. The transformation law of proposition 10.1.6 specializes here to the change-of-chart law for the Christoffel symbols: for two coordinate frames related by the Jacobian $g = (\partial\tilde{x}/\partial x)$, the inhomogeneous term $g^{-1}\,dg$ is the second-derivative term that makes the Γ^i_{jk} non-tensorial, recovering the transformation rule established in section 2.1. The concrete polar-coordinate matrix of example 10.1.5 is this specialization for the flat connection on $T\mathbb{R}^2$.

The curvature of this connection, computed in section 10.2, will be the Riemann curvature endomorphism R of definition 4.1.2, with the curvature matrix Ω^i_j carrying the components $R^i{}_{jkl}$ of definition 4.1.5; that identification is the cross-check tying the bundle-curvature machinery of the next section to the Part I tensor.

10.2 Curvature, the Structure Equation, and Bianchi

A connection ∇ on a vector bundle E tells a section how to change to first order as it is transported. Iterating the transport around an infinitesimal loop need not return the section to itself, and the failure to do so is curvature. In section 10.1 the connection was extended to the exterior covariant derivative $d^\nabla\colon \Omega^k(M;E) \to \Omega^{k+1}(M;E)$, which reduces to ∇ on 0-forms and satisfies a Leibniz rule against ordinary forms. Unlike the ordinary exterior derivative, d^∇ need not square to zero. The obstruction $(d^\nabla)^2$ is the object this section studies, and its identification with the second-order commutator of covariant derivatives makes it the bundle-level home of everything the Riemann tensor did in Part I.

This section establishes four facts, each valid for an arbitrary connection on an arbitrary vector bundle. The curvature $F^\nabla = (d^\nabla)^2$ is $C^\infty(M)$-linear in every argument, hence a genuine tensorial object, a 2-form valued in the endomorphisms of E. In a local frame it is computed from the connection matrix by the structure equation $\Omega = d\omega + \omega\wedge\omega$, a formula that trades the second-order operator $(d^\nabla)^2$ for a first-order expression in the connection forms. Under a change of frame the curvature matrix transforms by conjugation, $\tilde{\Omega} = g^{-1}\Omega g$, and this is what makes the globally-defined $\mathrm{End}(E)$-valued form possible; the contrast with the inhomogeneous transformation law of the connection matrix is the reason ω is not itself a tensor while Ω is. Finally the curvature obeys the differential Bianchi identity $d^\nabla F^\nabla = 0$, the bundle form of the second Bianchi identity of theorem 4.2.3. The section closes by specializing to $E = TM$ with the Levi-Civita connection, where F^∇ is exactly the Riemann curvature endomorphism of definition 4.1.2 and Ω carries its components.

Throughout, $\pi\colon E \to M$ is a smooth real or complex vector bundle of rank r over the n-manifold M, with $\Gamma(E)$ its smooth sections and $\Omega^k(M;E) = \Gamma(\Lambda^kT^*M\otimes E)$ the E-valued k-forms ([6]). A local frame $(e_1,\ldots,e_r)$ over U writes a section as $s = s^ie_i$ with column $\sigma = (s^i)$, and the connection through its matrix $\omega = (\omega^i_j) \in \Omega^1(U;\mathfrak{gl}_r)$ by $\nabla e_j = \omega^i_je_i$, so that $\nabla s = e\,(d\sigma + \omega\sigma)$ in the notation of definition 10.1.4. The Einstein summation convention (sum over any index appearing once as an upper and once as a lower symbol) is in force, and the sign convention for curvature is that of box 4.1.1.

The definition can be given in two equivalent guises. The first is the pure operator $(d^\nabla)^2$; the second is the second-order commutator that acts on a section along two directions. Both are recorded, and their equality is the first thing to check.

Definition 10.2.1: Curvature of a connection

Let ∇ be a connection on E with exterior covariant derivative d^∇ of definition 10.1.7. The *curvature* of ∇ is the operator

$$F^\nabla = (d^\nabla)^2 = d^\nabla \circ d^\nabla : \Gamma(E) \longrightarrow \Omega^2(M;E)$$

obtained by applying the exterior covariant derivative twice to a section. Equivalently, for vector fields X, Y and a section s,

$$F^\nabla(X,Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s$$

the failure of the covariant derivatives ∇_X and ∇_Y to commute, corrected by the bracket term. In a local frame the *curvature matrix* is the matrix of 2-forms $\Omega = (\Omega^i_j) \in \Omega^2(U;\mathfrak{gl}_r)$ determined by

$$F^\nabla(X,Y)\, e_j = \Omega^i_j(X,Y)\, e_i$$

That the two guises agree is a short computation with the definition of d^∇ from definition 10.1.7. On a section $s \in \Omega^0(M;E)$, $d^\nabla s = \nabla s$ is the E-valued 1-form with $(\nabla s)(X) = \nabla_X s$. Applying d^∇ again and evaluating the resulting E-valued 2-form on X, Y, the graded Leibniz rule for d^∇ against the 1-form part gives

$$\big((d^\nabla)^2 s\big)(X,Y) = \nabla_X\big((\nabla s)(Y)\big) - \nabla_Y\big((\nabla s)(X)\big) - (\nabla s)([X,Y]) = \nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s$$

exactly the operator form, so $F^\nabla = (d^\nabla)^2$ evaluated on (X,Y) is the second-order commutator. The bracket correction is not an afterthought: it is precisely what is needed to cancel the derivatives of the coefficients of X and Y, and its presence is why F^∇ will turn out to be tensorial where $\nabla_X\nabla_Y$ alone is not. The sign convention matches box 4.1.1 verbatim, so that on $E = TM$ the operator $F^\nabla(X,Y)$ will coincide with the Riemann operator $R(X,Y)$ with no sign to reconcile.

Before proving tensoriality, the flat model fixes intuition and exhibits the definition in coordinates.

Example 10.2.2: Curvature of the trivial connection

Let $E = \underline{\mathbb{R}}^r = M \times \mathbb{R}^r$ be the trivial bundle with the global frame $(e_1, \dots, e_r)$ of constant sections, and let $\nabla = d$ be the flat connection $\nabla(s^i e_i) = (ds^i)e_i$, which has vanishing connection matrix $\omega = 0$ (definition 10.1.4). Then d^∇ on this frame is the ordinary exterior derivative applied componentwise: for an E-valued form $\alpha = \alpha^i e_i$ with $\alpha^i \in \Omega^k(M)$, $d^\nabla \alpha = (d\alpha^i)e_i$. Since $d^2 = 0$ on ordinary forms,

$$(d^\nabla)^2(s^i e_i) = \big(d^2 s^i\big)e_i = 0$$

so $F^\nabla \equiv 0$. In operator form the same conclusion reads $\partial_X\partial_Y s - \partial_Y\partial_X s - \partial_{[X,Y]}s = 0$, the equality of mixed partials for a bundle-valued function, since the flat connection differentiates each component with the plain directional derivative. The trivial connection is flat; the substance of the theory is that a nontrivial connection, or a nontrivial bundle carrying only nonflat connections, cannot make F^∇ vanish. On TM this is the statement of example 4.1.6, that Euclidean space has zero curvature, recovered below in the bundle language as example 10.2.8.

The operator $\nabla_X\nabla_Y$ is emphatically not $C^\infty(M)$-linear in X or in s: it involves two derivatives, and each Leibniz rule contributes a first-order term. The content of the next proposition is that in the specific combination $\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]}$ all such first-order terms cancel, leaving an operator that is pointwise-linear in all three arguments. This is what promotes F^∇ from a differential operator to a tensor, a section of $\Lambda^2 T^*M \otimes \mathrm{End}(E)$.

Proposition 10.2.3: Tensoriality of the curvature

Let ∇ be a connection on E. The curvature F^∇ of definition 10.2.1 is $C^\infty(M)$-linear in each of its three arguments: for all $f \in C^\infty(M)$, vector fields X, Y, and sections s,

$$F^\nabla(fX,Y)s = F^\nabla(X,fY)s = f\,F^\nabla(X,Y)s \qquad \text{and} \qquad F^\nabla(X,Y)(fs) = f\,F^\nabla(X,Y)s$$

Consequently F^∇ is a tensorial object: at each $p \in M$ its value $F^\nabla(X,Y)s$ depends only on X_p, Y_p, and s_p, and F^∇ is a smooth section of $\Lambda^2 T^*M \otimes \mathrm{End}(E)$, that is, an element of $\Omega^2(M;\mathrm{End}E)$.

***Proof*:**
The map $(X,Y) \mapsto F^\nabla(X,Y)$ is $\mathbb{R}$-bilinear and alternating: it is $\mathbb{R}$-bilinear because ∇ and the bracket are, and alternating because interchanging X and Y negates the first two terms and, since $[Y,X] = -[X,Y]$, the third as well, so $F^\nabla(Y,X) = -F^\nabla(X,Y)$. It therefore suffices to prove $C^\infty(M)$-linearity in the first vector argument X and in the section s; linearity in Y then follows from linearity in X together with the alternating property.

Step 1: linearity in X. Fix Y, s and let $f \in C^\infty(M)$. Two Leibniz rules and the bracket identity $[fX,Y] = f[X,Y] - (Yf)X$ are the only inputs. Compute each term of $F^\nabla(fX,Y)s$:

$$\nabla_{fX}\nabla_Y s = f\,\nabla_X\nabla_Y s$$

since ∇ is $C^\infty(M)$-linear in its differentiating slot (definition 10.1.1). Next, $\nabla_Y\nabla_{fX}s = \nabla_Y(f\nabla_X s)$, and the Leibniz rule of definition 10.1.1 applied to the function f and the section $\nabla_X s$ gives

$$\nabla_Y\nabla_{fX}s = (Yf)\,\nabla_X s + f\,\nabla_Y\nabla_X s$$

Finally, using $[fX,Y] = f[X,Y] - (Yf)X$ and $C^\infty(M)$-linearity of ∇ in the differentiating slot,

$$\nabla_{[fX,Y]}s = f\,\nabla_{[X,Y]}s - (Yf)\,\nabla_X s$$

Assembling the three,

$$F^\nabla(fX,Y)s = f\,\nabla_X\nabla_Y s - \big((Yf)\nabla_X s + f\nabla_Y\nabla_X s\big) - \big(f\nabla_{[X,Y]}s - (Yf)\nabla_X s\big)$$

The two terms $\pm(Yf)\nabla_X s$ cancel, and what remains is $f\big(\nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s\big) = f\,F^\nabla(X,Y)s$. This proves linearity in X; the alternating property then gives $F^\nabla(X,fY)s = -F^\nabla(fY,X)s = -f\,F^\nabla(Y,X)s = f\,F^\nabla(X,Y)s$, linearity in Y.

Step 2: linearity in s. Fix X, Y and let $f \in C^\infty(M)$. Apply the Leibniz rule twice. First,

$$\nabla_Y(fs) = (Yf)\,s + f\,\nabla_Y s$$

Differentiating along X and using the Leibniz rule again on each summand,

$$\nabla_X\nabla_Y(fs) = \big(XYf\big)s + (Yf)\nabla_X s + (Xf)\nabla_Y s + f\,\nabla_X\nabla_Y s$$

Interchanging X and Y,

$$\nabla_Y \nabla_X(fs) = \big(YXf\big)s + (Xf)\nabla_Y s + (Yf)\nabla_X s + f\,\nabla_Y \nabla_X s$$

The bracket term is a single Leibniz rule,

$$\nabla_{[X,Y]}(fs) = \big([X,Y]f\big)s + f\,\nabla_{[X,Y]}s$$

Now form the alternating combination. The four first-order terms $(Yf)\nabla_X s$, $(Xf)\nabla_Y s$ appear once with each sign in $\nabla_X\nabla_Y(fs)$ and $\nabla_Y\nabla_X(fs)$ and cancel in the difference. The second-order coefficient of s contributes $\big(XYf - YXf - [X,Y]f\big)s = 0$, since $[X,Y]f = XYf - YXf$ for any function. What survives is $f\big(\nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s\big) = f\,F^\nabla(X,Y)s$, proving linearity in s.

Step 3: from linearity to a tensor. A map that is $C^\infty(M)$-linear in each argument is pointwise: the value $F^\nabla(X,Y)s$ at p depends only on X_p, Y_p, s_p, by the standard tensor characterization ([6]). Concretely, if $X_p = 0$ then writing $X = X^i\partial_i$ near p with $X^i(p) = 0$ gives $F^\nabla(X,Y)s = X^i F^\nabla(\partial_i, Y)s$, which vanishes at p; likewise for Y and s. Being alternating and $\mathbb{R}$-multilinear over $\mathbb{R}$, and pointwise in all three slots, F^∇ is at each point an alternating bilinear map $T_pM \times T_pM \to \mathrm{End}(E_p)$, that is, an element of $\Lambda^2 T_p^*M \otimes \mathrm{End}(E_p)$. Smoothness in p follows because F^∇ acts smoothly on smooth inputs. Hence $F^\nabla \in \Omega^2(M; \mathrm{End}E)$, as we sought to show.

The proposition is the bundle-level counterpart of the tensoriality of the Riemann endomorphism (proposition 4.1.3), and the cancellation mechanism is identical: the two second-order operators $\nabla_X\nabla_Y$ and $\nabla_Y\nabla_X$ each fail to be tensorial, but their commutator, once the bracket correction is subtracted, does. The output lives in $\Omega^2(M;\mathrm{End}E)$: a 2-form because F^∇ is alternating in (X,Y), endomorphism-valued because for fixed X,Y the map $s \mapsto F^\nabla(X,Y)s$ is $C^\infty(M)$-linear and hence a bundle endomorphism of E. This is the object that Chern–Weil theory will feed into invariant polynomials in chapter 11 to manufacture characteristic classes; that it is a globally defined tensor, not a chart-dependent expression, is exactly what makes those constructions possible.

Tensoriality guarantees that the curvature matrix $\Omega = (\Omega^i_j)$ of definition 10.2.1 is a well-defined matrix of 2-forms on each frame domain. The next result computes it from the connection matrix, replacing the second-order operator $(d^\nabla)^2$ by a first-order formula. This is the structure equation, the identity the whole theory uses.

Theorem 10.2.4: The structure equation

Let ∇ be a connection on E with connection matrix $\omega = (\omega^i_j)$ in a local frame (e_j) over U (definition 10.1.4). The curvature matrix $\Omega = (\Omega^i_j)$ of definition 10.2.1 is

$$\Omega = d\omega + \omega \wedge \omega$$

where the matrix wedge is $(\omega\wedge\omega)^i_j = \omega^i_k \wedge \omega^k_j$ (summed over k). Equivalently, entry by entry,

$$\Omega^i_j = d\omega^i_j + \omega^i_k \wedge \omega^k_j$$

***Proof*:**
The exterior covariant derivative of definition 10.1.7 acts on the frame section $e_j \in \Omega^0(U;E)$ by $d^\nabla e_j = \nabla e_j = \omega^i_j\, e_i$, the connection matrix acting on the frame by definition of ω. The curvature is $F^\nabla e_j = (d^\nabla)^2 e_j = d^\nabla(\omega^i_j e_i)$, so the task is to apply d^∇ to the E-valued 1-form $\omega^i_j e_i$ and read off the coefficient of e_i.

Step 1: the graded Leibniz rule. The exterior covariant derivative satisfies the Leibniz rule $d^\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^{|\alpha|}\alpha \wedge \nabla s$ for an ordinary form α and a section s (definition 10.1.7). Applied to $\alpha = \omega^i_j$ (a 1-form, so $|\alpha| = 1$) and $s = e_i$,

$$d^\nabla\left(\omega^i_j\, e_i\right) = d\omega^i_j\, e_i - \omega^i_j \wedge \nabla e_i$$

the sign $(-1)^1 = -1$ appearing because ω^i_j is a 1-form.

Step 2: substitute ∇e_i. The connection acts on the frame by $\nabla e_i = \omega^k_i\, e_k$, so

$$\omega^i_j \wedge \nabla e_i = \omega^i_j \wedge \omega^k_i\, e_k = \left(\omega^k_i \wedge \omega^i_j\right) e_k$$

using the sign rule $\omega^i_j \wedge \omega^k_i = -\omega^k_i \wedge \omega^i_j$ for two 1-forms; the two minus signs, one from Step 1 and one from this reordering, combine to a plus. Relabelling the summation indices in the last expression from (i,k) to (k,i) so that the free upper index is i, the coefficient of e_i is $\omega^i_k \wedge \omega^k_j$. Therefore

$$F^\nabla e_j = d^\nabla\left(\omega^i_j e_i\right) = \left(d\omega^i_j + \omega^i_k \wedge \omega^k_j\right) e_i$$

Step 3: read off Ω. By the defining relation $F^\nabla e_j = \Omega^i_j\, e_i$ of definition 10.2.1, matching coefficients of e_i gives $\Omega^i_j = d\omega^i_j + \omega^i_k \wedge \omega^k_j$, which in matrix notation is $\Omega = d\omega + \omega \wedge \omega$, completing the proof.

The structure equation is the exact analogue, one degree up, of the fact that the curvature of a connection is the exterior derivative of the connection form corrected by a quadratic term. The linear part $d\omega$ measures the infinitesimal variation of the connection forms; the quadratic part $\omega \wedge \omega$ measures the failure of parallel transport in the two directions to commute at second order. For a line bundle, $r = 1$, the connection matrix is a single 1-form $\omega = \omega^1_1$ and $\omega \wedge \omega = \omega \wedge \omega = 0$ because a 1-form wedged with itself vanishes; the structure equation collapses to $\Omega = d\omega$, and the curvature of a line-bundle connection is simply the exterior derivative of its connection form. This special case drives the entire line-bundle theory of section 10.4, where $d\omega$ becomes the first Chern form. For higher rank the matrix ω need not commute with itself under the wedge, and $\omega \wedge \omega$ is generally nonzero; it is this noncommutativity that distinguishes bundle curvature from the abelian gradient-and-curl calculus of scalar forms.

A quadratic-term computation makes the wedge concrete and previews how the same expression will reappear for the Levi-Civita connection.

Example 10.2.5: The structure equation for a rank-two connection

Take $r = 2$, so $\omega = \begin{pmatrix} \omega^1_1 & \omega^1_2 \\ \omega^2_1 & \omega^2_2 \end{pmatrix}$ is a matrix of 1-forms. The matrix wedge $\omega \wedge \omega$ has entries

$$(\omega \wedge \omega)^1_1 = \omega^1_1 \wedge \omega^1_1 + \omega^1_2 \wedge \omega^2_1 = \omega^1_2 \wedge \omega^2_1$$

since $\omega^1_1 \wedge \omega^1_1 = 0$, and similarly $(\omega \wedge \omega)^2_2 = \omega^2_1 \wedge \omega^1_2$, while the off-diagonal entries are $(\omega \wedge \omega)^1_2 =$

$\omega^1_1 \wedge \omega^1_2 + \omega^1_2 \wedge \omega^2_2$ and $(\omega \wedge \omega)^2_1 = \omega^2_1 \wedge \omega^1_1 + \omega^2_2 \wedge \omega^2_1$. Hence

$$\Omega^1_1 = d\omega^1_1 + \omega^1_2 \wedge \omega^2_1, \qquad \Omega^1_2 = d\omega^1_2 + \omega^1_1 \wedge \omega^1_2 + \omega^1_2 \wedge \omega^2_2$$

with the other two entries obtained by the symmetric relabelling. When the connection is compatible with a bundle metric in an orthonormal frame, ω becomes skew-symmetric (definition 10.1.4 and the metric theory of section 10.3), so $\omega^1_1 = \omega^2_2 = 0$ and $\omega^2_1 = -\omega^1_2$. The single independent connection form is then ω^1_2, and the structure equation reduces to

$$\Omega^1_2 = d\omega^1_2 + \omega^1_1 \wedge \omega^1_2 + \omega^1_2 \wedge \omega^2_2 = d\omega^1_2$$

the quadratic term dropping out because both surviving diagonal forms vanish. This is precisely the surface curvature computation of proposition 8.1.3, where the oriented orthonormal frame of a Riemannian surface has a single connection 1-form ω_{12} and Gaussian curvature K read off from $d\omega_{12} = -K\,dA$. The rank-two orthonormal case of the structure equation is the moving-frame computation of the Gauss curvature, recovered here as one instance of a formula valid for every connection on every bundle.

The curvature matrix Ω lives on a single frame domain. To assemble it into the global tensor $F^\nabla \in \Omega^2(M; \mathrm{End}E)$ of proposition 10.2.3, its behavior under a change of frame must be known. This is where the curvature parts company with the connection. The connection matrix transforms inhomogeneously (proposition 10.1.6): under $\tilde{e} = e\,g$ one has $\tilde{\omega} = g^{-1}\omega g + g^{-1}dg$, and the $g^{-1}dg$ term is what prevents ω from patching into a global 1-form. The curvature matrix, by contrast, transforms by pure conjugation, and the inhomogeneous term is gone.

Proposition 10.2.6: Transformation of the curvature matrix

Let (e_j) and $(\tilde{e}_j)$ be two local frames over U related by $\tilde{e} = e\,g$ for a smooth GL_r-valued function g on U, so that $\tilde{e}_j = g^i_j\, e_i$. If $\omega, \tilde{\omega}$ are the connection matrices and $\Omega, \tilde{\Omega}$ the curvature matrices in the two frames, then

$$\tilde{\Omega} = g^{-1}\,\Omega\, g$$

Consequently the local expressions $e\,\Omega\, e^{-1}$, read as endomorphism-valued 2-forms, agree on overlaps and define a single global section $F^\nabla \in \Omega^2(M; \mathrm{End}E)$.

***Proof*:**
The proof runs the structure equation of theorem 10.2.4 through the connection transformation law $\tilde{\omega} = g^{-1}\omega g + g^{-1}dg$ of proposition 10.1.6. Write $\tilde{\Omega} = d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega}$ and expand.

Step 1: differentiate $\tilde{\omega}$. From $\tilde{\omega} = g^{-1}\omega g + g^{-1}dg$, using $d(g^{-1}) = -g^{-1}(dg)g^{-1}$ (differentiate $g^{-1}g = \mathrm{id}$) and the graded Leibniz rule for d on matrix products of forms,

$$d\tilde{\omega} = d(g^{-1}) \wedge \omega g + g^{-1}(d\omega)g - g^{-1}\omega \wedge dg + d(g^{-1}) \wedge dg$$

the sign on the third term arising because ω is a 1-form. Substituting $d(g^{-1}) = -g^{-1}(dg)g^{-1}$ into the first and last terms,

$$d\tilde{\omega} = -g^{-1}(dg)g^{-1} \wedge \omega g + g^{-1}(d\omega)g - g^{-1}\omega \wedge dg - g^{-1}(dg)g^{-1} \wedge dg$$

Step 2: expand $\tilde{\omega} \wedge \tilde{\omega}$. With $A = g^{-1}\omega g$ and $B = g^{-1}dg$, so $\tilde{\omega} = A + B$,

$$\tilde{\omega} \wedge \tilde{\omega} = A \wedge A + A \wedge B + B \wedge A + B \wedge B$$

Compute the four pieces. First $A \wedge A = g^{-1}\omega g \wedge g^{-1}\omega g = g^{-1}(\omega \wedge \omega)g$, the interior $g\,g^{-1}$ cancelling. Next

$$A \wedge B = g^{-1}\omega g \wedge g^{-1}dg = g^{-1}\omega \wedge dg, \qquad B \wedge A = g^{-1}dg \wedge g^{-1}\omega g = g^{-1}(dg)g^{-1} \wedge \omega g$$

and finally $B \wedge B = g^{-1}dg \wedge g^{-1}dg = g^{-1}(dg)g^{-1} \wedge dg$.

Step 3: combine. Add the results of Steps 1 and 2 to form $\tilde{\Omega} = d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega}$. The four inhomogeneous terms cancel in pairs:

$$-g^{-1}(dg)g^{-1} \wedge \omega g + B \wedge A = 0, \qquad -g^{-1}\omega \wedge dg + A \wedge B = 0, \qquad -g^{-1}(dg)g^{-1} \wedge dg + B \wedge B = 0$$

each pair matching a term from $d\tilde{\omega}$ against a term from $\tilde{\omega} \wedge \tilde{\omega}$. What survives is the conjugated structure equation,

$$\tilde{\Omega} = g^{-1}(d\omega)g + g^{-1}(\omega \wedge \omega)g = g^{-1}\big(d\omega + \omega \wedge \omega\big)g = g^{-1}\,\Omega\,g$$

by theorem 10.2.4. This proves the transformation law.

Step 4: gluing into a global tensor. Interpret Ω as the matrix of the endomorphism-valued 2-form F^{∇} in the frame (e_j): for a section written $s = \sigma^j e_j$, the value $F^{\nabla}(X,Y)s$ has column $\Omega(X,Y)\sigma$ in the frame (e_j). Under $\tilde{e} = e\,g$, the same section has column $\tilde{\sigma} = g^{-1}\sigma$, and the transformation $\tilde{\Omega} = g^{-1}\Omega g$ gives $\tilde{\Omega}\tilde{\sigma} = g^{-1}\Omega g\, g^{-1}\sigma = g^{-1}(\Omega\sigma)$, which is the column of the *same* endomorphism $F^{\nabla}(X,Y)$ acting on the same section, now expressed in the frame $(\tilde{e}_j)$. The two local descriptions therefore represent one endomorphism of E at each point, independent of the frame, so they glue to a single global $F^{\nabla} \in \Omega^2(M; \mathrm{End} E)$, as required.

The contrast with proposition 10.1.6 is the point, and it deserves to be stated flatly. The connection matrix transforms by $\tilde{\omega} = g^{-1}\omega g + g^{-1}dg$, with an inhomogeneous term $g^{-1}dg$; the curvature matrix transforms by $\tilde{\Omega} = g^{-1}\Omega g$, with no such term. The two laws must not be conflated. The presence of $g^{-1}dg$ in the first is exactly the reason ω is not a tensor: it does not transform linearly, so there is no frame-independent 1-form it represents, and indeed a nonzero connection can have $\omega = 0$ in one frame and $\omega \neq 0$ in another. The absence of any inhomogeneous term in the second is exactly why Ω *is* a tensor: it transforms by conjugation, the coordinate signature of an $\mathrm{End}(E)$-valued object, and so patches into the global F^{∇}. In gauge-theoretic language ω is a gauge potential and Ω is the gauge field strength; a change of frame is a gauge transformation, under which the potential shifts inhomogeneously while the field strength rotates. The mechanism that produces the cancellation is visible in the proof: the $g^{-1}dg$ term contributes to $d\tilde{\omega}$ and to $\tilde{\omega} \wedge \tilde{\omega}$ two expressions that are negatives of each other, and they annihilate.

The last general identity concerns the covariant exterior derivative of the curvature itself. The curvature is an $\mathrm{End}(E)$-valued 2-form, and the connection induces an exterior covariant derivative on $\mathrm{End}(E)$-valued forms (definition 10.1.7). The differential Bianchi identity states that the curvature is covariantly closed for this operator.

Theorem 10.2.7: The differential Bianchi identity

Let ∇ be a connection on E with curvature $F^\nabla \in \Omega^2(M; \mathrm{End} E)$, and let d^∇ denote the induced exterior covariant derivative on $\mathrm{End}(E)$-valued forms (definition 10.1.7). Then

$$d^\nabla F^\nabla = 0$$

In a local frame, with curvature matrix Ω and connection matrix ω, this reads

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$$

entry by entry $d\Omega^i_j = \Omega^i_k \wedge \omega^k_j - \omega^i_k \wedge \Omega^k_j$.

***Proof*:**
The proof is a frame computation from the structure equation, followed by the identification of the resulting frame identity with the coordinate-free statement $d^\nabla F^\nabla = 0$.

Step 1: differentiate the structure equation. By theorem 10.2.4, $\Omega = d\omega + \omega \wedge \omega$. Apply the ordinary exterior derivative d, using $d^2 = 0$ and the graded Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge d\beta$ for matrix products of forms. Since ω is a matrix of 1-forms,

$$d\Omega = d(d\omega) + d(\omega \wedge \omega) = 0 + \big(d\omega \wedge \omega - \omega \wedge d\omega\big) = d\omega \wedge \omega - \omega \wedge d\omega$$

the middle sign being $(-1)^{|\omega|} = -1$.

Step 2: replace $d\omega$ by $\Omega - \omega \wedge \omega$. From the structure equation $d\omega = \Omega - \omega \wedge \omega$. Substitute into both occurrences:

$$d\Omega = \big(\Omega - \omega \wedge \omega\big) \wedge \omega - \omega \wedge \big(\Omega - \omega \wedge \omega\big) = \Omega \wedge \omega - \omega \wedge \omega \wedge \omega - \omega \wedge \Omega + \omega \wedge \omega \wedge \omega$$

The two cubic terms $\mp \omega \wedge \omega \wedge \omega$ cancel, since associativity of the matrix wedge makes both the same triple product. Hence

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$$

the asserted frame identity, valid entry by entry as $d\Omega^i_j = \Omega^i_k \wedge \omega^k_j - \omega^i_k \wedge \Omega^k_j$.

Step 3: identify with $d^\nabla F^\nabla$. It remains to see that the frame identity $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is exactly the coordinate expression of $d^\nabla F^\nabla = 0$. The induced connection on $\mathrm{End}(E)$ acts on an endomorphism-valued form by the commutator of the connection matrix: for a $\Phi \in \Omega^k(U; \mathrm{End} E)$ with matrix $\Phi = (\Phi^i_j)$, the exterior covariant derivative on $\mathrm{End}(E)$-valued forms is

$$d^\nabla \Phi = d\Phi + \omega \wedge \Phi - (-1)^k \, \Phi \wedge \omega \tag{10.2}$$

This is the matrix form of the $\mathrm{End}(E)$ connection of definition 10.1.7: the induced connection on $\mathrm{End}(E) = E \otimes E^*$ has connection matrix acting by $\Psi \mapsto \omega\Psi - \Psi\omega$ (the term ω from the E factor, the term $-\omega$ transposed from the E^* factor), and the exterior covariant derivative wedges these against forms with the Koszul sign $(-1)^k$ on the right-hand factor. Applying (10.2) to $\Phi = F^\nabla$, whose frame matrix is Ω, a 2-form so $k = 2$ and $(-1)^k = +1$,

$$d^\nabla F^\nabla = d\Omega + \omega \wedge \Omega - \Omega \wedge \omega$$

Substituting $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ from Step 2,

$$d^\nabla F^\nabla = \big(\Omega \wedge \omega - \omega \wedge \Omega\big) + \omega \wedge \Omega - \Omega \wedge \omega = 0$$

every term cancelling against its opposite. Thus the frame identity is equivalent to $d^\nabla F^\nabla = 0$, and since it holds on every frame domain, $d^\nabla F^\nabla = 0$ globally, completing the proof.

The Bianchi identity is the bundle statement that curvature, however it varies from point to point, does so subject to one differential constraint: it is covariantly closed. On $E = TM$ with the Levi-Civita connection it is the second Bianchi identity of theorem 4.2.3, and the cyclic-sum form there is the coordinate transcription of $d^\nabla F^\nabla = 0$ on the tangent bundle. Its consequences run throughout Part II. In Chern–Weil theory (chapter 11) it is what guarantees that the invariant polynomials of Ω are *closed* forms, so that they define de Rham cohomology classes; without $d^\nabla F^\nabla = 0$ the characteristic forms would not be closed and there would be no characteristic classes. For a line bundle it degrades to the plain statement $d\Omega = 0$, since $\Omega \wedge \omega = \omega \wedge \Omega$ when both are scalar-valued forms of degrees 2 and 1 (they commute up to the sign $(-1)^{2\cdot 1} = +1$), so the two terms cancel; the closedness of the first Chern form in section 10.4 is this degenerate Bianchi identity.

Everything above holds for an arbitrary connection on an arbitrary bundle. The chapter's organizing claim is that the tangent bundle with its Levi-Civita connection is one instance, and that the bundle curvature reproduces the Riemann tensor of Part I with no adjustment. This is the mandatory cross-check, and it closes the loop between the two Parts.

Example 10.2.8: Bundle curvature recovers the Riemann tensor

Let (M, g) be a (pseudo-)Riemannian manifold and $E = TM$ its tangent bundle, with ∇ the Levi-Civita connection. As recorded in example 10.1.9, in a coordinate frame $(e_j) = (\partial_j)$ the connection matrix has entries the Christoffel 1-forms $\omega^i_j = \Gamma^i_{jk}\, dx^k$, so $\nabla \partial_j = \omega^i_j\, \partial_i = \Gamma^i_{jk}\, dx^k \otimes \partial_i$ agrees with $\nabla_{\partial_k} \partial_j = \Gamma^i_{jk} \partial_i$.

The operator agrees with R. The curvature operator of definition 10.2.1 is, by definition,

$$F^\nabla(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

for vector fields X, Y, Z regarded as sections of $E = TM$. This is verbatim the Riemann curvature endomorphism $R(X, Y)Z$ of definition 4.1.2 in the sign convention of box 4.1.1; the two definitions are the same formula, since the sign convention for F^∇ was chosen in definition 10.2.1 to match box 4.1.1. Hence $F^\nabla = R$ as $\mathrm{End}(TM)$-valued 2-forms, with no sign to reconcile.

The curvature matrix carries the Riemann components. The curvature matrix $\Omega = (\Omega^i_j)$ is the matrix of 2-forms with $F^\nabla(X, Y)\partial_j = \Omega^i_j(X, Y)\, \partial_i$. Evaluate on the coordinate pair (∂_k, ∂_l). Since $[\partial_k, \partial_l] = 0$,

$$F^\nabla(\partial_k, \partial_l)\partial_j = R(\partial_k, \partial_l)\partial_j = R^i{}_{klj}\, \partial_i$$

using the component convention $R(\partial_k, \partial_l)\partial_j = R^i{}_{klj}\, \partial_i$ of definition 4.1.5, in which the first two lower indices are the operator slots and the third is the vector acted on. Comparing with $F^\nabla(\partial_k, \partial_l)\partial_j = \Omega^i_j(\partial_k, \partial_l)\, \partial_i$, the 2-form Ω^i_j has coordinate values $\Omega^i_j(\partial_k, \partial_l) = R^i{}_{klj}$, that is

$$\Omega^i_j = \tfrac{1}{2}\, R^i{}_{klj}\, dx^k \wedge dx^l$$

the factor $\frac{1}{2}$ being the standard normalization for the components of a 2-form under the antisymmetry $R^i{}_{klj} = -R^i{}_{lkj}$ in (k, l) of definition 4.1.5. Thus Ω^i_j carries exactly the Riemann components

${R^i}_{klj}$ of eq. (4.1), with the antisymmetric operator-slot indices (k, l) housed in the 2-form and the endomorphism indices (i, j) carried by the matrix position.

The structure equation reproduces the component formula. The entry of the structure equation theorem 10.2.4 is $\Omega^i_j = d\omega^i_j + \omega^i_m \wedge \omega^m_j$. Substituting $\omega^i_j = \Gamma^i_{jk}\, dx^k$,

$$d\omega^i_j = \partial_l \Gamma^i_{jk}\, dx^l \wedge dx^k, \qquad \omega^i_m \wedge \omega^m_j = \Gamma^i_{ml}\Gamma^m_{jk}\, dx^l \wedge dx^k$$

Adding and extracting the coefficient of $dx^k \wedge dx^l$ (antisymmetrizing in k, l) returns

$${R^i}_{klj} = \partial_k \Gamma^i_{lj} - \partial_l \Gamma^i_{kj} + \Gamma^i_{km}\Gamma^m_{lj} - \Gamma^i_{lm}\Gamma^m_{kj}$$

which is the component formula eq. (4.1) of definition 4.1.5, with the operator indices in the first two slots. The moving-frame computation and the Christoffel computation of Part I are one and the same identity, seen from two sides. For Euclidean space all Christoffel symbols vanish (example 4.1.6), so $\omega = 0$, and the structure equation gives $\Omega = 0$, recovering the flat trivial connection of example 10.2.2; the bundle curvature of TM is zero exactly when the Riemann tensor is, as it must be.

The tangent bundle is the first of two decisive instances. The second, deferred to section 10.3 and section 10.4, is the holomorphic line bundle, where the same structure equation, specialized to the Chern connection, produces the first Chern form and the topological invariants of the hyperplane and tautological bundles on $\mathbb{CP}^n$; there the integral $\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = 1$ will match the Euler number of the Hopf bundle of example 7.5.8. The general apparatus assembled here, tensoriality, the structure equation, the conjugation transformation law, and the Bianchi identity, is what both instances specialize.

10.3 Metric Connections and the Chern Connection

A connection on a bare vector bundle is a large object: the space of connections is an affine space over the endomorphism-valued one-forms (proposition 10.1.3), so there is a whole $\Omega^1(M; \mathrm{End}E)$ worth of them and no distinguished choice. Extra structure on the bundle cuts this space down. This section imposes two kinds of structure. The first is a metric: a Euclidean inner product on each real fiber, or a Hermitian inner product on each complex fiber. Requiring the connection to differentiate the metric by the Leibniz rule, exactly as the Levi-Civita connection differentiates g, singles out the *metric connections*. There are still many of them, but each has curvature valued in the skew endomorphisms, and that algebraic constraint is what makes invariant polynomials of the curvature into closed forms in chapter 11. The second structure is holomorphic. On a complex manifold a holomorphic bundle carries an operator $\bar\partial_E$ that says what it means for a section to be holomorphic, and pairing a Hermitian metric with this operator pins the connection down completely: there is exactly one connection compatible with the metric whose $(0, 1)$-part is $\bar\partial_E$. That connection is the Chern connection, and its curvature is a two-form of type $(1, 1)$ whose normalization is the first Chern form.

The two constructions run in parallel. Metric compatibility is the condition that parallel transport preserves the fiber inner product, and in an orthonormal frame it forces the connection matrix to be skew. The holomorphic condition adds a splitting of the exterior derivative into ∂ and $\bar\partial$, and the Chern connection is the unique way to reconcile the metric with the splitting. Throughout, $\pi\colon E \to M$

is a smooth vector bundle of rank r; the bundle-theoretic input (frames, forms, the operators ∂ and $\bar\partial$ on a complex manifold, holomorphic bundles) is imported from [6], [5], and [1] rather than rebuilt. The curvature conventions are those fixed in section 10.2: in a local frame $(e_1, \ldots, e_r)$ the connection matrix $\omega = (\omega^i_j)$ satisfies $\nabla e_j = \omega^i_j e_i$, the curvature matrix is $\Omega = d\omega + \omega \wedge \omega$ (theorem 10.2.4), and Ω transforms by conjugation $\tilde\Omega = g^{-1}\Omega g$ under a frame change $\tilde e = eg$ (proposition 10.2.6), while ω transforms inhomogeneously by $\tilde\omega = g^{-1}\omega g + g^{-1}dg$ (proposition 10.1.6).

Definition 10.3.1: Bundle metric and metric connection

A *bundle metric* h on a real vector bundle E is a smooth field of inner products: a section of $\mathrm{Sym}^2 E^*$ that is positive definite on each fiber, giving each E_p a Euclidean inner product $h_p(-,-)$. On a complex vector bundle a bundle metric (a *Hermitian metric*) is a smooth field of Hermitian inner products $h_p \colon E_p \times E_p \to \mathbb{C}$, $\mathbb{C}$-linear in the first slot, conjugate-linear in the second, with $h_p(u,u) > 0$ for $u \neq 0$. In a local frame $(e_1, \ldots, e_r)$ its components are the functions $h_{ij} = h(e_i, e_j)$ (real symmetric in the Euclidean case, Hermitian $h_{ij} = \overline{h_{ji}}$ in the complex case), forming a positive-definite matrix $H = (h_{ij})$.
A connection ∇ on E is *compatible with* h (a *metric connection*) if for all sections $s, t \in \Gamma(E)$

$$d\,h(s,t) = h(\nabla s, t) + h(s, \nabla t)$$

as an identity of one-forms, where in the complex case $h(\nabla s, t)$ is taken $\mathbb{C}$-linear in ∇s and $h(s, \nabla t)$ conjugate-linear in ∇t. Equivalently, for every vector field X,

$$X\,h(s,t) = h(\nabla_X s, t) + h(s, \nabla_X t)$$

the exact analogue of metric compatibility for the Levi-Civita connection (definition 2.2.4).

The condition is the Leibniz rule for differentiating the pairing $h(s,t) \in C^\infty(M)$ (or into $\mathbb{C}$): the derivative of an inner product of two sections is the inner product of the derivative of one with the other, summed over the two slots. Geometrically it says parallel transport is an isometry of the fibers, which is the same content proposition 2.4.4 extracted for the Levi-Civita connection. The utility of the condition is visible already in components. Choose a frame and write σ, τ for the coordinate columns of s, t; the compatibility identity becomes a matrix equation among H and ω, and in a frame that is *orthonormal* for h it forces ω into a Lie subalgebra. Recording that is the point of the next observation, but first the identity in components.

For sections $s = s^i e_i$ and $t = t^j e_j$ in a local frame, write $\sigma = (s^i)$ and $\tau = (t^j)$ for the coordinate columns and $H = (h_{ij})$ with $h_{ij} = h(e_i, e_j)$ for the metric matrix, so the Hermitian pairing is $h(s,t) = \tau^* H \sigma$ with $\tau^* = \overline{\tau}^{\mathsf{T}}$ (the Euclidean case is the same with all conjugations dropped and $\tau^* = \tau^{\mathsf{T}}$). With $\nabla s = e(d\sigma + \omega\sigma)$ from definition 10.1.4, expand both sides of the compatibility identity: the left side is $d(\tau^* H \sigma) = d\tau^*\, H\sigma + \tau^*\, dH\, \sigma + \tau^* H\, d\sigma$, and the right side is $\tau^* H(d\sigma + \omega\sigma) + (d\tau + \omega\tau)^* H\sigma$. Cancelling the terms carrying $d\sigma$ and $d\tau^*$ and matching for all σ, τ gives

$$dH = H\omega + \omega^* H \tag{10.3}$$

in the complex case, where $\omega^* = \overline{\omega}^{\mathsf{T}}$ is the conjugate transpose; in the real case $\omega^* = \omega^{\mathsf{T}}$ and (10.3) reads $dH = H\omega + \omega^{\mathsf{T}} H$. This is the coordinate form of definition 10.3.1, and it is the tool for both the skew-symmetry statement and the existence proof.

Proposition 10.3.2: Skew form of a metric connection in an orthonormal frame

Let ∇ be a metric connection on (E, h) and let $(e_1, \ldots, e_r)$ be a local frame that is orthonormal for h, so $H = I$ is the identity matrix. Then the connection matrix ω is skew-symmetric ($\omega^\mathsf{T} = -\omega$, valued in $\mathfrak{so}_r$) in the real case, and skew-Hermitian ($\omega^* := \bar{\omega}^\mathsf{T} = -\omega$, valued in $\mathfrak{u}_r$) in the complex case. Consequently the curvature matrix Ω is skew-symmetric, respectively skew-Hermitian, in every h-orthonormal frame.

***Proof*:**
In an h-orthonormal frame $H \equiv I$, so $dH = 0$, and (10.3) becomes $0 = \omega + \omega^*$ in the complex case, that is $\omega^* = \bar{\omega}^\mathsf{T} = -\omega$, the definition of skew-Hermitian, and $0 = \omega + \omega^\mathsf{T}$ in the real case, the definition of skew-symmetric.

For the curvature, evaluate the structure equation $\Omega = d\omega + \omega \wedge \omega$ of theorem 10.2.4 in the same orthonormal frame and take conjugate transposes. On the exterior-derivative term, $(-)^*$ commutes with d, so $(d\omega)^* = d(\omega^*) = -d\omega$ using $\omega^* = -\omega$. On the quadratic term, conjugate-transposing a product of matrix-valued one-forms reverses the factors and, because the scalar wedge of one-forms anticommutes ($\alpha \wedge \beta = -\beta \wedge \alpha$), introduces a sign: entrywise $((\omega \wedge \omega)^*)^i_j = \overline{(\omega \wedge \omega)^j_i} = \overline{\omega^j_k \wedge \omega^k_i} = (\omega^*)^k_j \wedge (\omega^*)^i_k = -(\omega^*)^i_k \wedge (\omega^*)^k_j = -(\omega^* \wedge \omega^*)^i_j$, so $(\omega \wedge \omega)^* = -\omega^* \wedge \omega^* = -(-\omega) \wedge (-\omega) = -\omega \wedge \omega$. Adding,

$$\Omega^* = (d\omega)^* + (\omega \wedge \omega)^* = -d\omega - \omega \wedge \omega = -\Omega$$

so Ω is skew-Hermitian. The real computation is identical with transpose in place of conjugate transpose, giving $\Omega^\mathsf{T} = -\Omega$, as we sought to show.

The proposition is the algebraic fact that carries the metric hypothesis into chapter 11. Under a change between two orthonormal frames the transition matrix g takes values in the orthogonal group $O(r)$ (real) or the unitary group $U(r)$ (complex), and the conjugation law $\tilde{\Omega} = g^{-1}\Omega g$ of proposition 10.2.6 keeps the skew form: the skew matrices $\mathfrak{so}_r$ and $\mathfrak{u}_r$ are Ad-invariant subspaces. So the curvature of a metric connection is globally a two-form valued in the sub-bundle of skew endomorphisms, $\mathfrak{so}(E)$ or $\mathfrak{u}(E)$. An Ad-invariant polynomial evaluated on such a curvature will be a well-defined global form, and chapter 11 shows it is closed; the skew constraint is what makes those polynomials the Pontryagin and Chern classes. For the tangent bundle with the Levi-Civita connection the statement is already familiar: in an orthonormal frame the connection one-forms ω^i_j are skew in (i, j), which is the frame form of metric compatibility from Part I.

Example 10.3.3: The Levi-Civita connection as a metric connection on TM
Take $E = TM$ with $h = g$ the Riemannian metric and ∇ the Levi-Civita connection. Metric compatibility in the sense of definition 2.2.4 is exactly the bundle-metric condition of definition 10.3.1: the identity $X\, g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$ is $X\, h(s, t) = h(\nabla_X s, t) + h(s, \nabla_X t)$ read on sections $s = Y$, $t = Z$ of TM. So the Levi-Civita connection is a metric connection on (TM, g), and proposition 10.3.2 applies.

Concretely, let $(E_1, \ldots, E_n)$ be a local g-orthonormal frame of TM, so $g(E_i, E_j) = \delta_{ij}$. Write $\nabla E_j = \omega^i_j E_i$. Differentiating $\delta_{ij} = g(E_i, E_j)$ along a vector field X and using compatibility,

$$0 = X\, g(E_i, E_j) = g(\nabla_X E_i, E_j) + g(E_i, \nabla_X E_j) = \omega^k_i(X)\, \delta_{kj} + \omega^k_j(X)\, \delta_{ik} = \omega^j_i(X) + \omega^i_j(X)$$

so $\omega^i_j = -\omega^j_i$, the connection matrix is skew-symmetric, and ω is valued in $\mathfrak{so}_n$. By proposition 10.3.2 the curvature matrix $\Omega = (\Omega^i_j)$ is likewise skew, $\Omega^i_j = -\Omega^j_i$. This is the frame form of the antisymmetry $\mathrm{Riem}(X, Y, Z, W) = -\mathrm{Riem}(X, Y, W, Z)$ of the Riemann tensor established in Part I, now read off the skew connection matrix. The upshot is that the Riemannian holonomy sits inside $O(n)$, the structure group of an orthonormal frame bundle, a point developed for symmetric spaces in chapter 9.

Metric connections are plentiful, which is what makes the affine-space picture of proposition 10.1.3 useful: one produces a single metric connection and the rest differ from it by a one-form valued in the skew endomorphisms. Existence is the next result, and the cleanest route builds the connection frame by frame in orthonormal frames and glues with a partition of unity, the same technique that produced Riemannian metrics in theorem 1.2.2 and connections in proposition 10.1.3.

Proposition 10.3.4: Existence of metric connections

Every vector bundle (E, h) with a bundle metric admits a metric connection. More precisely, the metric connections form a nonempty affine space over $\Omega^1(M; \mathfrak{so}(E))$ in the real case and over $\Omega^1(M; \mathfrak{u}(E))$ in the complex case, where $\mathfrak{so}(E)$, $\mathfrak{u}(E)$ are the sub-bundles of $\mathrm{End}(E)$ consisting of the h-skew endomorphisms.

***Proof*:**
Step 1: a local metric connection in an orthonormal frame. Cover M by open sets U_α over which E is trivial and carries an h-orthonormal frame $(e^\alpha_1, \ldots, e^\alpha_r)$; such frames exist by applying the Gram–Schmidt process fiberwise to any local frame, the smoothness of h making the outcome smooth ([6]). On U_α define a connection ∇^α by declaring the orthonormal frame parallel, $\nabla^\alpha e^\alpha_j = 0$, and extending by additivity and the Leibniz rule: $\nabla^\alpha(s^i e^\alpha_i) = ds^i \otimes e^\alpha_i$. Its connection matrix in this frame is $\omega^\alpha = 0$, which is trivially skew, so by the componentwise criterion (10.3) (with $H = I$, $dH = 0$, $\omega = 0$) the connection ∇^α is compatible with h on U_α.

Step 2: glue by a partition of unity. Let $\{\rho_\alpha\}$ be a partition of unity subordinate to $\{U_\alpha\}$ ([6]), and set

$$\nabla := \sum_\alpha \rho_\alpha \nabla^\alpha$$

interpreted as $\nabla_X s = \sum_\alpha \rho_\alpha \nabla^\alpha_X s$, the sum locally finite. This ∇ is a connection: additivity is clear, and the Leibniz rule holds because

$$\nabla(fs) = \sum_\alpha \rho_\alpha \big(df \otimes s + f\nabla^\alpha s\big) = \Big(\sum_\alpha \rho_\alpha\Big) df \otimes s + f \sum_\alpha \rho_\alpha \nabla^\alpha s = df \otimes s + f\nabla s$$

using $\sum_\alpha \rho_\alpha = 1$. A convex combination of metric connections is again a metric connection: for each α the identity $d\,h(s,t) = h(\nabla^\alpha s, t) + h(s, \nabla^\alpha t)$ holds, so multiplying by ρ_α and summing gives, again by $\sum_\alpha \rho_\alpha = 1$,

$$d\,h(s,t) = \sum_\alpha \rho_\alpha\, d\,h(s,t) = \sum_\alpha \rho_\alpha \big(h(\nabla^\alpha s, t) + h(s, \nabla^\alpha t)\big) = h(\nabla s, t) + h(s, \nabla t)$$

where the Leibniz product-rule term $d\rho_\alpha \otimes s$ never appears because each ρ_α multiplies the whole compatibility identity, not ∇^α alone. Hence ∇ is a metric connection, and the set is nonempty.

Step 3: the affine structure. Let ∇, ∇' be two metric connections. By proposition 10.1.3 their difference $A := \nabla' - \nabla$ is a one-form valued in $\mathrm{End}(E)$: $A \in \Omega^1(M; \mathrm{End}E)$. Subtracting the two compatibility identities,

$$0 = h(As, t) + h(s, At)$$

for all sections s, t, which is exactly the statement that the endomorphism $A(X)$ is h-skew for every X, that is, $A \in \Omega^1(M; \mathfrak{so}(E))$ in the real case and $\Omega^1(M; \mathfrak{u}(E))$ in the complex case. Conversely, for any such skew-valued A the connection $\nabla + A$ satisfies the compatibility identity, since adding $h(As, t) + h(s, At) = 0$ to it changes nothing. Thus the metric connections are exactly the affine translates of one of them by $\Omega^1(M; \mathfrak{so}(E))$, respectively $\Omega^1(M; \mathfrak{u}(E))$, completing the proof.

The proof shows metric connections are as abundant as connections in general, cut down only from the full $\Omega^1(M; \mathrm{End}E)$ to the skew-valued one-forms. There is no canonical metric connection on a bare Hermitian bundle: the choice made in Step 1, declaring a Gram–Schmidt frame parallel, depends on the frame, and different frames glue to different connections. A canonical choice appears only when a further piece of structure removes the remaining freedom. On a Riemannian manifold that structure was torsion-freeness, which uniquely selected the Levi-Civita connection among the metric connections of TM (theorem 2.3.1). On a holomorphic bundle the extra structure is the holomorphic operator $\bar{\partial}_E$, and the connection it selects is the Chern connection.

The setting is a complex manifold M. The complexified cotangent bundle splits into forms of type $(1, 0)$ and $(0, 1)$, and the exterior derivative splits accordingly as $d = \partial + \bar{\partial}$, with ∂ raising the holomorphic degree and $\bar{\partial}$ the antiholomorphic degree; a function is holomorphic exactly when $\bar{\partial} f = 0$ ([5]). A holomorphic vector bundle $E \to M$ carries transition functions that are holomorphic matrix-valued maps, and this equips it with a canonical operator

$$\bar{\partial}_E \colon \Omega^{p,q}(M; E) \to \Omega^{p,q+1}(M; E)$$

extending $\bar{\partial}$ to E-valued forms, whose kernel on $\Gamma(E) = \Omega^{0,0}(M; E)$ is the sheaf of holomorphic sections; in a holomorphic frame $(e_1, \ldots, e_r)$ it acts by $\bar{\partial}_E(s^i e_i) = (\bar{\partial} s^i) e_i$, with no correction term because the frame is holomorphic ([1]). A connection ∇ on E splits its output by form type as $\nabla = \nabla^{1,0} + \nabla^{0,1}$, and the Chern connection is the unique metric connection whose $(0, 1)$-part is this given operator.

Theorem 10.3.5: The Chern connection

Let $E \to M$ be a holomorphic vector bundle over a complex manifold, equipped with a Hermitian metric h. There is a unique connection ∇ on E that is

1. compatible with h in the sense of definition 10.3.1, and
2. compatible with the holomorphic structure, meaning $\nabla^{0,1} = \bar{\partial}_E$.

It is called the *Chern connection* of (E, h). In any local holomorphic frame with Hermitian matrix $H = (h_{ij})$, $h_{ij} = h(e_i, e_j)$, its connection matrix is

$$\omega = H^{-1}\partial H$$

a matrix of $(1,0)$-forms, and its curvature matrix is

$$\Omega = \bar{\partial}\big(H^{-1}\partial H\big)$$

a matrix of $(1,1)$-forms.

***Proof*:**

Fix a local holomorphic frame $(e_1, \ldots, e_r)$, so $\bar{\partial}_E e_j = 0$ and the metric matrix is $H = (h_{ij})$ with $h_{ij} = h(e_i, e_j)$, Hermitian and positive definite. Write the sought connection matrix as ω and split it by type, $\omega = \omega^{1,0} + \omega^{0,1}$, where $\omega^{1,0}$ collects the $(1,0)$-form entries and $\omega^{0,1}$ the $(0,1)$-form entries.

Step 1: the holomorphic condition fixes $\omega^{0,1}$. The $(0,1)$-part of ∇ acts on $s = s^i e_i$ by $\nabla^{0,1}s = e(\bar{\partial}\sigma + \omega^{0,1}\sigma)$, where $\sigma = (s^i)$ is the coordinate column. The condition $\nabla^{0,1} = \bar{\partial}_E$ requires $\nabla^{0,1}s = \bar{\partial}_E s = e\,\bar{\partial}\sigma$ for all σ, since $\bar{\partial}_E$ has no matrix part in a holomorphic frame. Comparing, $\omega^{0,1}\sigma = 0$ for every column σ, hence

$$\omega^{0,1} = 0 \tag{10.4}$$

so the whole connection matrix is of type $(1,0)$: $\omega = \omega^{1,0}$.

Step 2: metric compatibility fixes $\omega^{1,0}$. The compatibility identity in components is (10.3), $dH = H\omega + \omega^* H$, with $\omega^* = \overline{\omega}^{\mathsf{T}}$. Split this equation by form type. Since H consists of functions, $dH = \partial H + \bar{\partial}H$ is the sum of a $(1,0)$-form matrix ∂H and a $(0,1)$-form matrix $\bar{\partial}H$. On the right, $\omega = \omega^{1,0}$ is of type $(1,0)$ by (10.4), so $H\omega$ is of type $(1,0)$; and $\omega^* = \overline{\omega}^{\mathsf{T}}$ is of type $(0,1)$, since conjugation sends $(1,0)$-forms to $(0,1)$-forms, so $\omega^* H$ is of type $(0,1)$. Matching the $(1,0)$-parts of $dH = H\omega + \omega^* H$ gives

$$\partial H = H\omega \tag{10.5}$$

while the $(0,1)$-parts give $\bar{\partial}H = \omega^* H$, which is the conjugate transpose of (10.5) (using $H^* = H$) and so carries no new information. Left-multiplying (10.5) by H^{-1},

$$\omega = H^{-1}\partial H \tag{10.6}$$

a matrix of $(1,0)$-forms. The freedom removed in Step 1 leaves (10.6) as the only solution, so uniqueness follows: any connection satisfying (1) and (2) has $\omega^{0,1} = 0$ by Step 1 and $\omega^{1,0} = H^{-1}\partial H$ by (10.6), hence its matrix is determined in every holomorphic frame, and a connection is determined by its matrices in a covering family of frames (definition 10.1.4).

Step 3: existence. Conversely, cover M by holomorphic-trivialization opens U_α and on each define a local connection ∇^α by declaring its matrix in the holomorphic frame over U_α to be $\omega^\alpha = H_\alpha^{-1}\partial H_\alpha$, with H_α the metric matrix on U_α. This ∇^α satisfies both defining conditions on U_α: it satisfies (2), because $\omega^\alpha = H_\alpha^{-1}\partial H_\alpha$ is of type $(1,0)$ and so its $(0,1)$-part vanishes, giving $\nabla^{0,1} = \bar\partial_E$ as in (10.4); and it satisfies (1), because reversing Step 2 reconstitutes the compatibility identity $dH_\alpha = H_\alpha\omega^\alpha + (\omega^\alpha)^* H_\alpha$: its $(1,0)$-part $\partial H_\alpha = H_\alpha\omega^\alpha$ is (10.5), and its $(0,1)$-part $\bar\partial H_\alpha = (\omega^\alpha)^* H_\alpha$ is the conjugate transpose of the same, so both hold. On an overlap $U_\alpha \cap U_\beta$ the two connections ∇^α and ∇^β each satisfy (1) and (2), so by the uniqueness established in Steps 1–2 (applied on $U_\alpha \cap U_\beta$) they agree there. The local connections therefore patch to a single connection ∇ on E satisfying (1) and (2) globally. Hence the Chern connection exists and is unique.

Step 4: the curvature is of type $(1,1)$. Apply the structure equation $\Omega = d\omega + \omega\wedge\omega$ of theorem 10.2.4 to $\omega = H^{-1}\partial H$. First, $\omega\wedge\omega = (H^{-1}\partial H)\wedge(H^{-1}\partial H)$ is a wedge of two $(1,0)$-form matrices, hence of type $(2,0)$. Next, $d\omega = \partial\omega + \bar\partial\omega$; the $(2,0)$-part $\partial\omega$ and the $(1,1)$-part $\bar\partial\omega$ split $d\omega$ by type. The $(2,0)$-parts combine to

$$\partial\omega + \omega\wedge\omega = \partial(H^{-1}\partial H) + (H^{-1}\partial H)\wedge(H^{-1}\partial H)$$

Compute $\partial(H^{-1}\partial H) = (\partial H^{-1})\wedge\partial H + H^{-1}\partial\partial H$. Since $\partial^2 = 0$ on forms and $\partial H^{-1} = -H^{-1}(\partial H)H^{-1}$ (from $\partial(H^{-1}H) = 0$), this is $-H^{-1}(\partial H)H^{-1}\wedge\partial H = -(H^{-1}\partial H)\wedge(H^{-1}\partial H) = -\omega\wedge\omega$. Therefore the $(2,0)$-part $\partial\omega + \omega\wedge\omega = -\omega\wedge\omega + \omega\wedge\omega = 0$ vanishes, leaving only the $(1,1)$-part:

$$\Omega = \bar\partial\omega = \bar\partial\big(H^{-1}\partial H\big)$$

which is a matrix of $(1,1)$-forms, as claimed. This establishes the formula and the type, completing the proof.

The Chern connection is the holomorphic-Hermitian analogue of the Levi-Civita connection: a metric and a compatible-differentiation condition together pin down a unique connection. Where the Levi-Civita condition was torsion-freeness, here it is that the connection's antiholomorphic part is the intrinsic operator $\bar\partial_E$ of the holomorphic structure, so covariant differentiation of a holomorphic section in an antiholomorphic direction gives zero. The formula $\omega = H^{-1}\partial H$ is explicit in the metric alone, and it makes the curvature computable in any example by two differentiations. The type statement, Ω of type $(1,1)$, is the analytic core of chapter 11: it forces the first Chern form into $H^{1,1}$, the Hodge summand where positivity and Kähler geometry live. The vanishing of the $(2,0)$-part is the algebraic identity $\partial\omega = -\omega\wedge\omega$ that Step 4 extracted from $\omega = H^{-1}\partial H$; it is special to the Chern connection and is what a generic metric connection lacks.

The construction specializes cleanly to line bundles, where the Hermitian matrix H is a single positive function and the matrix operations become ordinary arithmetic. This is the case that carries all the weight of section 10.4, so it is worked in full.

Example 10.3.6: The Chern connection of a Hermitian holomorphic line bundle

Let $L \to M$ be a holomorphic line bundle ($\operatorname{rank} L = 1$) with a Hermitian metric h. In a local holomorphic frame e (a nonvanishing holomorphic section over U) the metric is a single positive smooth function

$$h = h(e,e) > 0$$

and the Hermitian matrix H of theorem 10.3.5 is the 1×1 matrix (h). Matrix multiplication is now commutative, so the Chern connection form is

$$\omega = h^{-1}\partial h = \frac{\partial h}{h} = \partial \log h$$

a single $(1,0)$-form. The curvature form is

$$\Omega = \bar\partial\omega = \bar\partial\,\partial \log h = -\partial\bar\partial \log h$$

the sign coming from anticommuting the two operators, $\bar\partial\partial = -\partial\bar\partial$, on functions. This is a scalar $(1,1)$-form, matching the general type statement, and it depends on the metric only through $\log h$.

Two features are visible. First, Ω is a closed form: $d\Omega = (\partial + \bar\partial)(-\partial\bar\partial \log h) = -\partial^2\bar\partial \log h - \bar\partial\partial\bar\partial \log h = 0$, both terms vanishing by $\partial^2 = \bar\partial^2 = 0$. Second, Ω is imaginary in the sense that $\overline{\Omega} = -\Omega$: conjugating, $\overline{-\partial\bar\partial \log h} = -\bar\partial\partial \log h = \partial\bar\partial \log h = -\Omega$, using that $\log h$ is real. So $i\Omega$ is a real closed $(1,1)$-form. Its normalization

$$\frac{i}{2\pi}\,\Omega = -\frac{i}{2\pi}\,\partial\bar\partial \log h$$

is the first Chern form of (L, h), taken up in section 10.4.

The frame-independence is the transformation check of theorem 10.3.5 in rank one. If $\tilde e = eg$ is another holomorphic frame, g a nonvanishing holomorphic function, then $\tilde h = h(\tilde e, \tilde e) = |g|^2\, h$ and $\tilde\omega = \partial \log \tilde h = \partial \log h + \partial \log |g|^2$. Since $|g|^2 = g\bar g$ gives $\log |g|^2 = \log g + \overline{\log g}$ locally, with $\log g$ holomorphic, $\partial \log |g|^2 = \partial \log g = g^{-1}\partial g = g^{-1} dg$, the inhomogeneous term of the gauge law. The curvature is unchanged: $\tilde\Omega = -\partial\bar\partial \log \tilde h = -\partial\bar\partial \log h - \partial\bar\partial \log |g|^2$, and $\partial\bar\partial \log |g|^2 = \partial\bar\partial(\log g) + \partial\bar\partial(\overline{\log g}) = 0$, the first term vanishing because $\bar\partial \log g = 0$ ($\log g$ holomorphic) and the second because $\partial\bar\partial(\overline{\log g}) = -\bar\partial\partial(\overline{\log g}) = 0$ ($\overline{\log g}$ antiholomorphic, so $\partial\overline{\log g} = 0$). So $\tilde\Omega = \Omega$, confirming that the curvature is a global form on M, valued in the purely imaginary scalars, as proposition 10.3.2 predicts for a line bundle (where $\mathfrak{u}_1 = i\mathbb{R}$).

The line-bundle curvature $\Omega = -\partial\bar\partial \log h$ is the object whose integral computes topological degree, worked out on $\mathbb{CP}^n$ in section 10.4. Before leaving the general theory, one consistency check ties the bundle formalism back to Part I and one anticipates the payoff of the next section.

The first is the tangent-bundle check. When $E = TM$ carries the Levi-Civita connection and g is the metric, the bundle curvature F^∇ of section 10.2 is the Riemann curvature endomorphism: unwinding definition 10.2.1, $F^\nabla(X,Y)s = \nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s$, which is verbatim the definition of $R(X,Y)s$ in definition 4.1.2. In a coordinate frame $\partial_1, \ldots, \partial_n$ the connection matrix is ${\omega^l}_k = \Gamma^l_{ki}\, dx^i$ and the structure equation makes the curvature matrix carry the Riemann components,

$${\Omega^l}_k = \tfrac{1}{2}\, {R^l}_{ijk}\, dx^i \wedge dx^j$$

with ${R^l}_{ijk}$ the components of definition 4.1.5 in the frame convention $R(\partial_i, \partial_j)\partial_k = {R^l}_{ijk}\,\partial_l$; this is the cross-check verified in detail in section 10.2. Example 10.3.3 adds that in a g-orthonormal frame this matrix is skew, which is the frame form of the pair antisymmetry $\operatorname{Riem}(X,Y,Z,W) = -\operatorname{Riem}(X,Y,W,Z)$. The bundle curvature of TM is the Riemann tensor, and the metric-connection theory subsumes Part I with no change of convention.

The second is the projective check, taken up fully in section 10.4 but stated here to fix the target.

On $\mathbb{C}\mathrm{P}^n$ the hyperplane bundle $\mathcal{O}(1)$ with its Fubini–Study metric will have first Chern form

$$\frac{i}{2\pi}\,\Omega = \frac{1}{2\pi}\,\omega_{\mathrm{FS}}$$

a positive multiple of the Fubini–Study Kähler form of example 7.5.9, so $c_1(\mathcal{O}(1)) > 0$; integrating over a projective line gives $\int_{\mathbb{C}\mathrm{P}^1} c_1(\mathcal{O}(1)) = 1$, matching the Euler number $+1$ of the Hopf bundle computed metrically in example 7.5.8 and topologically in [4]. The dual, the tautological bundle $\mathcal{O}(-1)$ whose fiber over a point of $\mathbb{C}\mathrm{P}^n$ is the line it represents, has curvature of the opposite sign and $\int_{\mathbb{C}\mathrm{P}^1} c_1(\mathcal{O}(-1)) = -1$. That the analytic curvature integral returns the integer $+1$ of the topological Hopf invariant is the line-bundle case of Chern–Weil, and the Chern connection of theorem 10.3.5 is the connection that produces it.

10.4 Curvature of Line Bundles; the Example of $\mathbb{C}\mathrm{P}^n$

For a bundle of rank one the curvature apparatus collapses to something scalar. The connection matrix ω and the curvature matrix Ω of definition 10.1.4 and definition 10.2.1 are 1×1, so matrix multiplication is ordinary multiplication of forms, the wedge term $\omega \wedge \omega$ in the structure equation drops out because a one-form wedged with itself vanishes, and the gauge transformations lose their conjugation: the curvature two-form is the same in every local frame. What survives is a single closed two-form attached to the bundle, and its cohomology class is a topological invariant. This section extracts that invariant, the first Chern form, from the Chern connection of a Hermitian holomorphic line bundle, proves that its class does not depend on the metric, and computes it on the one example that anchors the whole theory: the hyperplane bundle over complex projective space, whose curvature is the Fubini–Study form of example 7.5.9 and whose integral over $\mathbb{C}\mathrm{P}^1$ is the integer $+1$.

The line-bundle case is where Chern–Weil theory can be seen in full before any of its machinery is built. A change of Hermitian metric multiplies the metric function h by a positive function, alters $\log h$ by an additive term, and shifts the curvature by $\partial\bar{\partial}$ of that term, an exact form. The class in $H^2(M;\mathbb{R})$ is therefore rigid while the form representing it flexes with the metric. On a compact Riemann surface the integral of the form is forced to be an integer, the degree of the bundle, and the curvature integral computes a topological count. All of this is worked out below for a single line, and chapter 11 promotes it to bundles of every rank and invariants of every degree.

Throughout, $L \to M$ is a holomorphic line bundle over a complex manifold M, equipped with a Hermitian metric h; in a local holomorphic frame e over an open set $U \subseteq M$ the metric is the positive function $h = h(e,e) > 0$, and the Chern connection of theorem 10.3.5 has connection form and curvature

$$\omega = \partial \log h, \qquad \Omega = -\partial\bar{\partial} \log h \tag{10.7}$$

computed in example 10.3.6. Both are scalar-valued forms on U: ω a $(1,0)$-form, Ω a two-form of type $(1,1)$. The operators ∂ and $\bar{\partial}$ and the theory of holomorphic bundles are those of [5] and [1]. Complex conjugation of forms is written $\bar{\cdot}$.

The first order of business is to record how simple the rank-one gauge transformation is, because it is what makes Ω frame-independent.

Lemma 10.4.1: The curvature form of a line bundle is global

Let $L \to M$ be a line bundle with a connection, and let Ω and $\tilde{\Omega}$ be the curvature two-forms of definition 10.2.1 computed in two local frames e and $\tilde{e} = g\,e$ over an overlap, where g is a nowhere-zero $\mathbb{C}$-valued (or $\mathbb{R}$-valued) function. Then $\tilde{\Omega} = \Omega$ on the overlap. Consequently the locally defined curvature forms patch to a single global two-form on M, the *curvature form* of the connection.

***Proof*:**
For a rank-one bundle the transition function g is scalar, so the GL_1-conjugation of the curvature transformation law $\tilde{\Omega} = g^{-1}\Omega g$ (the tensorial gauge law recorded in the binding facts of the chapter, contrasted there with the inhomogeneous law for ω) reads $\tilde{\Omega} = g^{-1}\Omega g = \Omega$, since scalars commute. Thus the two frames assign the same two-form to each point of the overlap. Any two frames of L near a point differ by such a nowhere-zero scalar g, so the assignment $U \mapsto \Omega_U$ is independent of the chosen frame, and the local forms agree on all overlaps. A collection of local two-forms agreeing on overlaps defines a global two-form, as we sought to show.

The lemma is the reason the entire section can speak of "the curvature form" of a line bundle without naming a frame. For a bundle of rank $r \geq 2$ the curvature is a matrix, tensorial but frame-dependent up to conjugation, and only its conjugation-invariant combinations (traces of powers, taken up in chapter 11) descend to global forms. In rank one the curvature is already such an invariant. Normalizing it produces the invariant this section is built around.

Definition 10.4.2: First Chern form of a Hermitian line bundle

Let (L, h) be a Hermitian holomorphic line bundle on a complex manifold M, with Chern connection of theorem 10.3.5 and global curvature form Ω of lemma 10.4.1. The *first Chern form* of (L, h) is the real two-form

$$c_1(L, h) = \frac{i}{2\pi}\,\Omega = -\frac{i}{2\pi}\,\partial\bar{\partial}\log h$$

where $h > 0$ is the metric function of L in any local holomorphic frame.

The factor $i/2\pi$ has two jobs. The i makes the form real, and the 2π makes its periods integers, so that c_1 will integrate to the topological degree rather than to 2π times it. Both claims are proved in the next proposition. The form is of type $(1,1)$ because Ω is, by theorem 10.3.5, and this bidegree is what ties it to the complex geometry of M: a positive $(1,1)$-form is a Kähler form, and a line bundle whose first Chern form can be made positive is exactly a positive line bundle in the sense of complex geometry, the case realized below by the hyperplane bundle.

Before the proof, the reality and closedness are worth isolating, because they are what allow c_1 to define a de Rham class at all.

Proposition 10.4.3: The first Chern class is real, closed, and independent of the metric

Let (L, h) be a Hermitian holomorphic line bundle on a complex manifold M. Then:

1. $c_1(L, h)$ is a real two-form;
2. $c_1(L, h)$ is closed, $d\, c_1(L, h) = 0$;
3. if h' is a second Hermitian metric on L, then $c_1(L, h') - c_1(L, h)$ is exact; consequently the de Rham class
$$c_1(L) := \big[c_1(L, h)\big] \in H^2(M; \mathbb{R})$$
is independent of the choice of Hermitian metric.

***Proof*:**
Work in a local holomorphic frame e over U, in which L has the positive metric function h and curvature $\Omega = -\partial\bar{\partial} \log h$ by (10.7). The three claims are checked in turn.

1. *Reality.* Set $u = \log h$, a real-valued function since $h > 0$. Conjugation of forms interchanges ∂ and $\bar{\partial}$, that is $\overline{\partial\alpha} = \bar{\partial}\,\overline{\alpha}$ and $\overline{\bar{\partial}\alpha} = \partial\,\overline{\alpha}$ for any form α [5]. Applying this twice to the real function u (so $\overline{u} = u$),
$$\overline{\partial\bar{\partial}u} = \bar{\partial}\,\overline{\bar{\partial}u} = \bar{\partial}\,\partial\,\overline{u} = \bar{\partial}\partial u = -\partial\bar{\partial}u$$
the last equality using $\bar{\partial}\partial = -\partial\bar{\partial}$, the anticommutativity of the two operators that comes from $d^2 = 0$ split by bidegree [5]. Therefore $\overline{\Omega} = -\overline{\partial\bar{\partial}u} = \partial\bar{\partial}u = -\Omega$, so Ω is purely imaginary. Multiplying by $i/2\pi$ turns a purely imaginary form into a real one: $\overline{c_1(L, h)} = \overline{(i/2\pi)\Omega} = (-i/2\pi)\overline{\Omega} = (-i/2\pi)(-\Omega) = (i/2\pi)\Omega = c_1(L, h)$. Thus $c_1(L, h)$ is real.

2. *Closedness.* The exterior derivative splits by bidegree as $d = \partial + \bar{\partial}$. Applying it to $\Omega = -\partial\bar{\partial}u$ and using $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial\bar{\partial} = -\bar{\partial}\partial$,
$$d\Omega = (\partial + \bar{\partial})(-\partial\bar{\partial}u) = -\partial\partial\bar{\partial}u - \bar{\partial}\partial\bar{\partial}u = 0 + \partial\bar{\partial}\bar{\partial}u = 0$$
the first term vanishing by $\partial^2 = 0$ and the second because $\bar{\partial}\partial\bar{\partial}u = -\partial\bar{\partial}\bar{\partial}u = 0$ by $\bar{\partial}^2 = 0$. Since $c_1(L, h) = (i/2\pi)\Omega$ is a constant multiple of Ω, it too is closed. The form is globally defined by lemma 10.4.1, so closedness holds on all of M.

3. *Metric independence.* Two Hermitian metrics on the same line bundle differ by a positive function: in the frame e, write $h' = e^{\varphi}h$ for a smooth real function φ on U. The function $\varphi = \log(h'/h)$ is the logarithm of the ratio of the two metric functions, and this ratio is frame-independent (a change of frame multiplies both h and h' by $|g|^2$, which cancels), so φ is a globally defined smooth real function on M. Then $\log h' = \varphi + \log h$, and
$$c_1(L, h') - c_1(L, h) = -\frac{i}{2\pi}\,\partial\bar{\partial}(\log h' - \log h) = -\frac{i}{2\pi}\,\partial\bar{\partial}\varphi$$
It remains to see that $\partial\bar{\partial}\varphi$ is exact. For the global real function φ, the one-form $\bar{\partial}\varphi$ satisfies $\partial\bar{\partial}\varphi = (\partial + \bar{\partial})\bar{\partial}\varphi = d(\bar{\partial}\varphi)$, since $\bar{\partial}\bar{\partial}\varphi = 0$. Hence
$$c_1(L, h') - c_1(L, h) = -\frac{i}{2\pi}\,d(\bar{\partial}\varphi) = d\Big(-\frac{i}{2\pi}\,\bar{\partial}\varphi\Big)$$

is d of a global one-form, hence exact. Two representatives differing by an exact form define the same de Rham class, so $[c_1(L,h)] = [c_1(L,h')]$ in $H^2(M;\mathbb{R})$, completing the proof.

The proposition is the Chern–Weil phenomenon in miniature. The curvature form Ω records the connection, which records the metric; changing the metric changes Ω, but only by an exact form, so the cohomology class is a bundle invariant with no metric attached. The class $c_1(L)$ is the same real cohomology class that algebraic topology attaches to L by classifying complex line bundles up to isomorphism, the image in real coefficients of the integral first Chern class of [4]; the curvature construction realizes that topological invariant by a differential form. The mechanism generalizes without change of idea: for a bundle of higher rank one forms invariant polynomials of the curvature matrix, and the same "exact under change of connection" computation, carried out with the Bianchi identity in place of the line-bundle simplification, shows the resulting forms are closed with metric-independent classes. That is the content of chapter 11; here the single invariant polynomial is the trace, which for a 1×1 matrix is the entry itself.

A remark on the sign of $c_1(L^*)$ will be needed for the projective computation. Dualizing a line bundle inverts its transition functions and, in a holomorphic frame, replaces the metric function h by h^{-1}.

Proposition 10.4.4: First Chern form of the dual line bundle

Let (L,h) be a Hermitian holomorphic line bundle, and give the dual bundle L^* the dual (inverse) metric h^*, whose metric function in the dual holomorphic frame is h^{-1}. Then

$$c_1(L^*,h^*) = -\,c_1(L,h)$$

and hence $c_1(L^*) = -c_1(L)$ in $H^2(M;\mathbb{R})$.

***Proof*:**
If e is a local holomorphic frame of L with metric function $h = h(e,e)$, the dual frame e^* of L^* has metric function h^{-1} by definition of the dual metric. The line-bundle curvature formula (10.7) applied to L^* gives

$$\Omega_{L^*} = -\partial\bar{\partial}\log(h^{-1}) = -\partial\bar{\partial}(-\log h) = \partial\bar{\partial}\log h = -\Omega_L$$

Multiplying by $i/2\pi$, $c_1(L^*,h^*) = -c_1(L,h)$, and passing to classes gives $c_1(L^*) = -c_1(L)$, as claimed.

The example that gives the theory its weight is complex projective space, where two line bundles present themselves with no auxiliary choices. Recall $\mathbb{C}\mathrm{P}^n$ as the space of complex lines through the origin in $\mathbb{C}^{n+1}$, with homogeneous coordinates $[z_0 : \cdots : z_n]$, appearing in example 7.5.9 as the base of the Riemannian submersion from the unit sphere $S^{2n+1}(1)$. The two bundles are the tautological line bundle and its dual.

Theorem 10.4.5: Curvature of the tautological and hyperplane bundles on $\mathbb{C}\mathrm{P}^n$

Let $\mathcal{O}(-1) \to \mathbb{C}\mathrm{P}^n$ be the *tautological line bundle*, whose fiber over a point $\ell \in \mathbb{C}\mathrm{P}^n$ is the line $\ell \subseteq \mathbb{C}^{n+1}$ that the point represents, and let $\mathcal{O}(1) = \mathcal{O}(-1)^*$ be the *hyperplane bundle*. Give $\mathcal{O}(-1)$ the Hermitian metric inherited from the standard Hermitian inner product on $\mathbb{C}^{n+1}$, and $\mathcal{O}(1)$ the dual metric. Then, writing $w = (w_1, \dots, w_n)$ for the affine coordinates $w_j = z_j/z_0$ on the chart $\{z_0 \neq 0\}$:

1. the first Chern form of $\mathcal{O}(1)$ is
$$c_1(\mathcal{O}(1)) = \frac{i}{2\pi}\,\partial\bar\partial \log\big(1 + |w|^2\big) = \frac{1}{2\pi}\,\omega_{FS}$$
where $\omega_{FS} = i\,\partial\bar\partial\log(1 + |w|^2)$ is the Fubini–Study Kähler form; in particular $c_1(\mathcal{O}(1))$ is a positive $(1,1)$-form;
2. $c_1(\mathcal{O}(-1)) = -c_1(\mathcal{O}(1))$ is negative;
3. on $\mathbb{C}\mathrm{P}^1$ the normalization is exact:
$$\int_{\mathbb{C}\mathrm{P}^1} c_1(\mathcal{O}(1)) = +1, \qquad \int_{\mathbb{C}\mathrm{P}^1} c_1(\mathcal{O}(-1)) = -1$$

***Proof*:**
The argument computes the metric function of $\mathcal{O}(-1)$ in a holomorphic frame, dualizes, and integrates. Write $|z|^2 = \sum_{k=0}^n |z_k|^2$ for the standard Hermitian norm on $\mathbb{C}^{n+1}$.

Step 1: a holomorphic frame and the metric of $\mathcal{O}(-1)$. On the affine chart $U_0 = \{[z] \colon z_0 \neq 0\}$, every line $[z]$ has a unique representative with first coordinate 1, namely $\zeta(w) = (1, w_1, \dots, w_n)$ where $w_j = z_j/z_0$. The assignment $[z] \mapsto \zeta(w) \in \mathbb{C}^{n+1}$ is a holomorphic section of $\mathcal{O}(-1)$ over U_0 that is nowhere zero, hence a local holomorphic frame $e_{-1} = \zeta$. The metric on $\mathcal{O}(-1)$ is the restriction of the standard Hermitian form of $\mathbb{C}^{n+1}$, so the metric function of this frame is
$$h_{-1} = h\big(e_{-1}, e_{-1}\big) = |\zeta(w)|^2 = 1 + |w_1|^2 + \cdots + |w_n|^2 = 1 + |w|^2$$
a positive real-analytic function on U_0.

Step 2: the metric of $\mathcal{O}(1)$ and its curvature. The dual frame $e_1 = e_{-1}^*$ of $\mathcal{O}(1)$ has metric function $h_1 = h_{-1}^{-1} = (1+|w|^2)^{-1}$ by definition of the dual metric. By (10.7), the Chern curvature of $\mathcal{O}(1)$ on U_0 is
$$\Omega_1 = -\partial\bar\partial\log h_1 = -\partial\bar\partial\log\big(1 + |w|^2\big)^{-1} = \partial\bar\partial\log\big(1 + |w|^2\big)$$
and therefore
$$c_1(\mathcal{O}(1)) = \frac{i}{2\pi}\,\Omega_1 = \frac{i}{2\pi}\,\partial\bar\partial\log\big(1 + |w|^2\big) = \frac{1}{2\pi}\,\omega_{FS}$$
with $\omega_{FS} = i\,\partial\bar\partial\log(1 + |w|^2)$. This proves the formula in part (1). The form is defined globally on $\mathbb{C}\mathrm{P}^n$ by lemma 10.4.1; the chart U_0 suffices to identify it because $\mathbb{C}\mathrm{P}^n \setminus U_0$ is a proper analytic subset, and the closed form extends across it by continuity.

Step 3: positivity. That ω_{FS}, hence $c_1(\mathcal{O}(1))$, is a positive $(1,1)$-form is the statement that the

Hermitian matrix $\left(\partial_{w_j}\partial_{\bar{w}_k}\log(1+|w|^2)\right)_{j,k}$ is positive definite. Differentiating, with $\rho = 1+|w|^2$,

$$\partial_{\bar{w}_k}\log\rho = \frac{w_k}{\rho}, \qquad \partial_{w_j}\partial_{\bar{w}_k}\log\rho = \frac{\delta_{jk}}{\rho} - \frac{\bar{w}_j w_k}{\rho^2} = \frac{1}{\rho^2}\left(\rho\,\delta_{jk} - \bar{w}_j w_k\right)$$

For a nonzero vector $\xi = (\xi_1, \ldots, \xi_n) \in \mathbb{C}^n$, the Hermitian form evaluates to

$$\sum_{j,k} \overline{\xi_j}\,\xi_k\,\frac{\rho\,\delta_{jk} - \bar{w}_j w_k}{\rho^2} = \frac{1}{\rho^2}\left(\rho\,|\xi|^2 - \Big|\sum_k w_k\xi_k\Big|^2\right)$$

using $\sum_{j,k}\overline{\xi_j}\xi_k\bar{w}_j w_k = \left|\sum_k w_k\xi_k\right|^2$. By the Cauchy–Schwarz inequality $\left|\sum_k w_k\xi_k\right|^2 \le |w|^2\,|\xi|^2$, so the parenthesis is at least $\rho\,|\xi|^2 - |w|^2\,|\xi|^2 = (1+|w|^2-|w|^2)\,|\xi|^2 = |\xi|^2 > 0$. The form is positive definite at every point of U_0, and by homogeneity of $\mathbb{CP}^n$ (the unitary group $U(n+1)$ acts transitively by holomorphic isometries of the Fubini–Study metric, example 7.5.9) it is positive everywhere. Thus $c_1(\mathcal{O}(1))$ is a positive $(1,1)$-form, a positive multiple of the Fubini–Study Kähler form.

Step 4: the dual, part (2). Applying proposition 10.4.4 with $L = \mathcal{O}(1)$ and $L^* = \mathcal{O}(-1)$ gives $c_1(\mathcal{O}(-1)) = -c_1(\mathcal{O}(1))$, which is a negative $(1,1)$-form by Step 3.

Step 5: the integral over $\mathbb{CP}^1$, part (3). Take $n = 1$, so $w = w_1$ is a single affine coordinate on $U_0 = \mathbb{CP}^1 \setminus \{[0:1]\}$, and $U_0 \cong \mathbb{C}$ with coordinate $w = x+iy$. The complement is the single point $[0:1]$, a set of measure zero, so

$$\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = \int_{\mathbb{C}} \frac{i}{2\pi}\,\partial\bar{\partial}\log(1+|w|^2)$$

In one complex variable $\partial\bar{\partial}f = (\partial_w\partial_{\bar{w}}f)\,dw\wedge d\bar{w}$, and $dw\wedge d\bar{w} = (dx+i\,dy)\wedge(dx-i\,dy) = -2i\,dx\wedge dy$, so $\frac{i}{2}\,dw\wedge d\bar{w} = dx\wedge dy$. From Step 3 with $\rho = 1+|w|^2$, the single second derivative is

$$\partial_w\partial_{\bar{w}}\log\rho = \frac{1}{\rho^2}\left(\rho - |w|^2\right) = \frac{1}{(1+|w|^2)^2}$$

Hence

$$\frac{i}{2\pi}\,\partial\bar{\partial}\log(1+|w|^2) = \frac{1}{\pi}\cdot\frac{1}{(1+|w|^2)^2}\,dx\wedge dy$$

Passing to polar coordinates $w = re^{i\theta}$, $dx\wedge dy = r\,dr\,d\theta$ and $|w|^2 = r^2$, so

$$\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = \frac{1}{\pi}\int_0^{2\pi}\!\!\int_0^{\infty}\frac{r}{(1+r^2)^2}\,dr\,d\theta = \frac{1}{\pi}\cdot 2\pi\cdot\left[-\frac{1}{2(1+r^2)}\right]_0^{\infty} = 2\cdot\frac{1}{2} = 1$$

The integral for $\mathcal{O}(-1)$ is the negative of this by part (2), giving $\int_{\mathbb{CP}^1} c_1(\mathcal{O}(-1)) = -1$. This is the value the normalization $i/2\pi$ was chosen to produce, completing the proof.

The computation delivers the cross-check the theory was built to pass. The Hopf fibration $S^3 \to \mathbb{CP}^1$ of example 7.5.8 is the unit-sphere bundle of $\mathcal{O}(-1)$ over $\mathbb{CP}^1$: its fiber circle is exactly the unit circle in the tautological line, and as a topological circle bundle its Euler number is $+1$, the generator of $\pi_3(S^2)$ recorded there and in [4]. The Euler number of an oriented circle bundle over a surface equals the integral of the first Chern class of the associated complex line bundle, so the topological invariant $+1$ must reappear as $\int_{\mathbb{CP}^1} c_1$ of the appropriate bundle. The curvature integral confirms it:

$c_1(\mathcal{O}(1))$ integrates to $+1$ and its dual $c_1(\mathcal{O}(-1))$ to -1, and the sign is pinned by the orientation conventions, with the hyperplane bundle $\mathcal{O}(1)$ the positive generator. That the same $\mathbb{C}\mathrm{P}^1 = S^2(\frac{1}{2})$ whose metric geometry was computed by O'Neill's formula in example 7.5.8 now returns as the base of a line bundle whose curvature integral is an integer is the point of contact between Part I's metric curvature and Part II's characteristic classes.

Positivity of $c_1(\mathcal{O}(1))$ carries independent geometric content. A holomorphic line bundle whose first Chern class is represented by a positive $(1,1)$-form is a *positive* line bundle, and $\mathcal{O}(1)$ is the model: its positivity is what makes $\mathbb{C}\mathrm{P}^n$ projective, since some tensor power of a positive line bundle has enough holomorphic sections to embed the manifold, the content of the Kodaira embedding theorem of [1]. The Fubini–Study form $\omega_{FS} = 2\pi\, c_1(\mathcal{O}(1))$ is simultaneously the Kähler form of the metric of example 7.5.9 and 2π times the curvature of a line bundle. The metric whose sectional curvature was pinched between 1 and 4 in Part I and the curvature two-form of $\mathcal{O}(1)$ in Part II are the same object viewed through the two halves of the book.

The integrality that appeared as a computed $+1$ on $\mathbb{C}\mathrm{P}^1$ is a general theorem, and it identifies the curvature integral with a topological invariant on any compact Riemann surface.

Example 10.4.6: The curvature integral as the degree of a line bundle

Let M be a compact Riemann surface (a compact complex manifold of complex dimension one) and let (L, h) be a Hermitian holomorphic line bundle on M. The first Chern form $c_1(L, h)$ of definition 10.4.2 is a two-form on the real two-dimensional oriented manifold M, and its integral

$$\deg L := \int_M c_1(L, h)$$

is a well-defined real number, independent of the metric h by proposition 10.4.3 together with Stokes's theorem: a change of metric alters the integrand by an exact form $d\alpha$, and $\int_M d\alpha = 0$ on the closed manifold M by [6]. The number $\deg L$ is the *degree* of the line bundle.

The degree is an integer. This is the theorem that the de Rham class $c_1(L)$ is the real image of an integral cohomology class in $H^2(M; \mathbb{Z}) \cong \mathbb{Z}$, whose pairing against the fundamental class of M is by construction an integer; the topological first Chern class and its integrality are established in [4], and the curvature integral computes the pairing $\langle c_1(L), [M]\rangle \in \mathbb{Z}$ against the fundamental class. The projective computation is the base case: for $M = \mathbb{C}\mathrm{P}^1$ and $L = \mathcal{O}(1)$ the integral was found above to be $+1$, so $\deg \mathcal{O}(1) = 1$ and, by additivity of the degree under tensor product (which follows from $c_1(L \otimes L') = c_1(L) + c_1(L')$, itself immediate from $\log h_{L\otimes L'} = \log h_L + \log h_{L'}$ in a product frame), $\deg \mathcal{O}(d) = d$ for every $d \in \mathbb{Z}$.

Two instances anchor the count. Taking L trivial with the constant metric $h \equiv 1$, the curvature $\Omega = -\partial\bar{\partial} \log 1 = 0$ vanishes, so $\deg L = 0$: the trivial bundle has degree zero. Taking $L = T^{1,0}M$ the holomorphic tangent bundle of M, a line bundle in complex dimension one, the curvature of the Chern connection of a Kähler metric is the Ricci form, and $\int_M c_1(T^{1,0}M) = \chi(M)$ is the Euler characteristic by the Gauss–Bonnet theorem; for M of genus γ this reads $\deg T^{1,0}M = 2 - 2\gamma$, negative once $\gamma \geq 2$, matching the value $\int_M c_1$ of the anticanonical bundle. Thus on a Riemann surface the curvature integral of a line bundle is exactly its topological degree, an integer, and the sign and magnitude of that integer read off the twisting of the bundle. This is the line-bundle form of the Gauss–Bonnet theorem; the general statement, that invariant polynomials of curvature integrate to characteristic numbers on manifolds of every dimension, is the subject of chapter 11.

The last cross-check ties the whole chapter to Part I through the tangent bundle. On a compact

Riemann surface the holomorphic tangent bundle $T^{1,0}M$ is a line bundle, its Chern connection for a Kähler metric is the induced connection of the Levi-Civita connection (the two agree on a Kähler manifold), and its curvature two-form is the Riemann curvature endomorphism of definition 4.1.2 restricted to the tangent line. The identity $F^{\nabla} = R$ for $E = TM$, verified in example 10.2.8, is what makes $\int_M c_1(T^{1,0}M)$ equal to the Gauss–Bonnet integral of Part I: the bundle curvature of this section and the Riemann curvature of Chapter 4 are one operator, and integrating it recovers the Euler characteristic either way. The line-bundle degree, the Hopf Euler number, and the Gauss–Bonnet integral are three names for a single integer, and the first Chern form is the two-form whose periods are those integers.

11

Chern–Weil Theory

The previous chapter attached to a connection a curvature two-form and, on a line bundle, extracted from it a closed form whose cohomology class was independent of the metric. Chern–Weil theory is the systematic version of that construction, valid for bundles of every rank. Its input is the curvature of any connection; its output is a collection of closed differential forms whose de Rham classes depend only on the bundle, not on the connection used to compute them. These classes are the characteristic classes, and they are the bridge this Part was built to cross: a differential-geometric object, the curvature, manufactures topological invariants.

The mechanism is a single algebraic observation. The curvature matrix transforms by conjugation under a change of frame, so any polynomial in its entries that is invariant under conjugation produces a globally defined form. The Bianchi identity forces that form to be closed, and a transgression argument forces its cohomology class to be independent of the connection. The resulting map, from invariant polynomials to cohomology, is the Chern–Weil homomorphism, and evaluating it on the standard invariant polynomials yields the standard characteristic classes: the Chern classes and the Chern character from the determinant and the trace of the exponential, the Pontryagin classes from the real theory, and the Euler class from the Pfaffian of a skew-symmetric curvature. The chapter closes by discharging a debt left open in Part I. The Euler form of the tangent bundle is a curvature expression, the Pfaffian of the Riemann curvature, and its integral over a closed manifold is the Euler characteristic. This is the generalized Gauss–Bonnet–Chern theorem, stated without proof in chapter 8; here the Chern–Weil machinery proves it, and the two-dimensional case returns the Gauss–Bonnet theorem of section 8.1 as its lowest instance. Curvature, which in Part I governed the geometry and topology of the manifold directly, is shown in Part II to compute the manifold's topological invariants through the classes it represents.

11.1 Invariant Polynomials and the Chern–Weil Homomorphism

The curvature Ω of a connection is a matrix of 2-forms, and proposition 10.2.6 showed that a change of frame conjugates it, $\tilde{\Omega} = g^{-1}\Omega g$. A single entry of Ω is therefore frame-dependent and carries no invariant meaning. What survives a change of frame is any expression in the entries that is unchanged by conjugation: the trace, the determinant, the elementary symmetric functions of the eigenvalues. These are the invariant polynomials, and evaluating one of them on the curvature produces a differential form that is defined on all of M, independent of any local frame. The Chern–Weil construction turns each such polynomial into a closed differential form, and then into a de Rham cohomology class that does not depend on the connection at all. The output is a homomorphism from the ring of invariant polynomials to the cohomology ring of M, the source of every characteristic class in the differential-geometric approach.

Two facts make the construction work, and both were assembled in section 10.2. The first is that Ω transforms by conjugation, so an Ad-invariant polynomial in Ω patches into a global form. The second is the Bianchi identity $d^\nabla F^\nabla = 0$, which is exactly what forces those global forms to be closed. This section proves closedness (theorem 11.1.6) and connection-independence (theorem 11.1.7) in full, assembles the Chern–Weil homomorphism (definition 11.1.8), and verifies against the line-bundle first Chern form of section 10.4 that the machinery reproduces what was computed by hand there. The named characteristic classes, the Chern class and Chern character, are read off from specific invariant polynomials in the next section.

Throughout, $E \to M$ is a smooth complex vector bundle of rank r over a smooth manifold M, with a connection ∇ and curvature $F^\nabla = \Omega \in \Omega^2(M; \mathrm{End} E)$ in the sense of definition 10.2.1; in a local frame $\Omega = (\Omega^i_j)$ is a matrix of 2-forms valued in $\mathfrak{gl}_r(\mathbb{C})$. The gauge law $\tilde{\Omega} = g^{-1}\Omega g$ of proposition 10.2.6 and the Bianchi identity $d^\nabla F^\nabla = 0$ of theorem 10.2.7 are in force. Because the entries Ω^i_j are 2-forms, they commute with one another under the wedge product ($\alpha \wedge \beta = \beta \wedge \alpha$ for forms of even degree), so a polynomial expression in the matrix Ω is unambiguous: the usual issues of noncommutativity that attend matrices of odd-degree forms do not arise here.

The starting object is algebraic, a polynomial function on the space of matrices that is blind to conjugation. The trace of a power and the determinant are the two examples to keep in view.

Definition 11.1.1: Ad-invariant polynomial

Let $\mathfrak{gl}_r(\mathbb{C})$ denote the space of $r \times r$ complex matrices. A *polynomial function* $P\colon \mathfrak{gl}_r(\mathbb{C}) \to \mathbb{C}$ is a polynomial in the matrix entries $A = (a_{ij})$. It is Ad-*invariant* (or simply *invariant*) if

$$P\big(g^{-1}Ag\big) = P(A) \qquad \text{for all } A \in \mathfrak{gl}_r(\mathbb{C}),\ g \in \mathrm{GL}_r(\mathbb{C})$$

P is *homogeneous of degree* k if $P(tA) = t^k P(A)$ for all $t \in \mathbb{C}$; every polynomial is a finite sum of its homogeneous components, and each component of an invariant polynomial is again invariant. The invariant polynomials form a graded ring under pointwise addition and multiplication, the *ring of invariant polynomials* $I^*(\mathfrak{gl}_r)$. Two families of generators recur:

- the *power sums* $p_k(A) = \mathrm{tr}(A^k)$, homogeneous of degree k;
- the *elementary symmetric polynomials* $\sigma_k(A)$ of the eigenvalues of A, defined by the expansion

$$\det(I + tA) = \sum_{k=0}^{r} \sigma_k(A)\, t^k$$

so that $\sigma_0 = 1$, $\sigma_1 = \mathrm{tr}A$, and $\sigma_r = \det A$; σ_k is homogeneous of degree k.

Invariance under conjugation is invariance under change of basis: $P(A)$ depends only on the linear operator that A represents, not on the coordinates chosen to write it as a matrix. The power sums and the elementary symmetric polynomials are invariant because the eigenvalues are, and $\det(g^{-1}(I+tA)g) = \det(I+tA)$ makes the invariance of the σ_k visible without computing eigenvalues. A theorem of classical invariant theory states that the $\sigma_1, \dots, \sigma_r$ generate $I^*(\mathfrak{gl}_r)$ freely as a polynomial ring, and the power sums $p_1, \dots, p_r$ generate the same ring over $\mathbb{C}$, related to the σ_k by Newton's identities. For the construction ahead only invariance itself is used, not the structure of the ring, so the generation statement is recorded and not proved.

Example 11.1.2: The elementary symmetric and power-sum polynomials in rank two

For $r = 2$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the characteristic expansion is

$$\det(I + tA) = 1 + t\,(a + d) + t^2(ad - bc) = 1 + t\,\mathrm{tr}A + t^2 \det A$$

so $\sigma_1(A) = \mathrm{tr}A = a + d$ and $\sigma_2(A) = \det A = ad - bc$; there are no higher σ_k because $r = 2$. The power sums are $p_1(A) = \mathrm{tr}A = a + d$ and $p_2(A) = \mathrm{tr}(A^2) = a^2 + 2bc + d^2$. Newton's identity $p_2 = \sigma_1^2 - 2\sigma_2$ is verified directly:

$$\sigma_1^2 - 2\sigma_2 = (a + d)^2 - 2(ad - bc) = a^2 + 2ad + d^2 - 2ad + 2bc = a^2 + d^2 + 2bc = p_2$$

Both $\{\sigma_1, \sigma_2\}$ and $\{p_1, p_2\}$ generate the ring of invariant polynomials on $\mathfrak{gl}_2(\mathbb{C})$, and each is a polynomial in the other. Under conjugation $A \mapsto g^{-1}Ag$ the numbers $a + d$ and $ad - bc$ are unchanged, being the trace and determinant, so both polynomials are Ad-invariant, as definition 11.1.1 requires.

To feed the curvature into an invariant polynomial, P must be evaluated on a matrix whose entries are 2-forms rather than numbers. A homogeneous polynomial of degree k is a sum of monomials, each a product of k entries; replacing the entries by 2-forms and the numerical product by the wedge product gives a well-defined $2k$-form. The invariance of P is precisely what makes the result frame-independent.

Definition 11.1.3: Chern–Weil form of an invariant polynomial

Let P be an Ad-invariant polynomial on $\mathfrak{gl}_r(\mathbb{C})$, homogeneous of degree k, and let $\Omega = (\Omega^i_j)$ be the curvature matrix of a connection ∇ on E in a local frame. The *Chern–Weil form* $P(\Omega)$ is the $2k$-form obtained by substituting the 2-forms Ω^i_j for the matrix entries in P, with every product of entries read as a wedge product. Concretely, if $P(A) = \sum c_{i_1 j_1 \cdots i_k j_k}\, a_{i_1 j_1} \cdots a_{i_k j_k}$, then

$$P(\Omega) = \sum c_{i_1 j_1 \cdots i_k j_k}\, \Omega^{i_1}_{j_1} \wedge \cdots \wedge \Omega^{i_k}_{j_k} \in \Omega^{2k}(M)$$

The *normalized Chern–Weil form* of P is $P\big(\frac{i}{2\pi}\Omega\big)$, the same construction applied to the rescaled curvature $\frac{i}{2\pi}\Omega$.

The definition is a local recipe on a frame domain, and the claim that it produces a globally defined form is where Ad-invariance enters. On the overlap of two frames related by $\tilde{e} = e\, g$, the curvature matrices are related by $\tilde{\Omega} = g^{-1}\Omega g$ (proposition 10.2.6), and because the entries of Ω are even-degree forms they multiply exactly as numbers do under the wedge, so the matrix identities of ordinary linear algebra carry over verbatim. Invariance of P then gives $P(\tilde{\Omega}) = P(g^{-1}\Omega g) = P(\Omega)$ on the overlap: the two local forms agree, and they patch into a single global $2k$-form on M. The next proposition records this so it can be cited.

Proposition 11.1.4: The Chern–Weil form is globally defined

Let P be an Ad-invariant polynomial of degree k on $\mathfrak{gl}_r(\mathbb{C})$ and ∇ a connection on E. The locally defined forms $P(\Omega)$ of definition 11.1.3, computed in the various frames of E, agree on overlaps and assemble into a single global $2k$-form on M, also written $P(\Omega)$ or $P(F^\nabla)$.

***Proof*:**
Let (e_j) and $(\tilde{e}_j)$ be two local frames over an overlap U, related by $\tilde{e} = e\, g$ with $g\colon U \to \mathrm{GL}_r(\mathbb{C})$ smooth. By proposition 10.2.6 the curvature matrices satisfy $\tilde{\Omega} = g^{-1}\Omega g$. The entries Ω^i_j are 2-forms, hence commute pairwise under the wedge product, so any polynomial identity valid for numerical matrices holds for the wedge-product algebra generated by the Ω^i_j. In particular the substitution defining $P(\Omega)$ is a homomorphism from the commutative polynomial algebra to the commutative subalgebra of $\Omega^{\text{even}}(U)$ generated by the entries. Applying P to $\tilde{\Omega} = g^{-1}\Omega g$ and using Ad-invariance $P(g^{-1}Ag) = P(A)$ of definition 11.1.1 at each point,

$$P(\tilde{\Omega}) = P\big(g^{-1}\Omega g\big) = P(\Omega)$$

as forms on U. Thus the local forms coincide on every overlap. A family of local forms agreeing on all overlaps of an open cover defines a single global form ([6]), so $P(\Omega) \in \Omega^{2k}(M)$ is well-defined, completing the proof.

The rescaling by $i/2\pi$ in the normalized form is a normalization, not part of the mechanism: it is chosen so that the resulting classes are real and have integer periods, exactly as the factor $i/2\pi$ in definition 10.4.2 was chosen to make the first Chern form real with integral $+1$ over $\mathbb{C}\mathrm{P}^1$. For a homogeneous polynomial of degree k the rescaling multiplies the form by $(i/2\pi)^k$, and since closedness and the cohomology class are unaffected by a constant multiple, the two theorems below are stated for $P(\Omega)$ and apply verbatim to $P(\frac{i}{2\pi}\Omega)$.

To differentiate $P(\Omega)$ the polynomial is first converted to a symmetric multilinear form, its polarization, because the Leibniz rule for d distributes term-by-term across multilinear expressions. This is the standard device, and it is worth stating precisely.

Proposition 11.1.5: Polarization of an invariant polynomial

Let P be a homogeneous polynomial of degree k on $\mathfrak{gl}_r(\mathbb{C})$. There is a unique symmetric k-linear form $\tilde{P}\colon \mathfrak{gl}_r(\mathbb{C})^k \to \mathbb{C}$ with $\tilde{P}(A,\dots,A) = P(A)$, the *polarization* of P, given by

$$\tilde{P}(A_1,\dots,A_k) = \frac{1}{k!}\,\frac{\partial^k}{\partial t_1\cdots\partial t_k}\bigg|_{t=0} P(t_1A_1+\cdots+t_kA_k)$$

If P is Ad-invariant, then $\tilde{P}$ is Ad-invariant in the sense that $\tilde{P}(g^{-1}A_1g,\dots,g^{-1}A_kg) = \tilde{P}(A_1,\dots,A_k)$ for all $g\in \mathrm{GL}_r(\mathbb{C})$; equivalently, differentiating the invariance along a one-parameter subgroup $g=\exp(tB)$ at $t=0$ gives the infinitesimal invariance

$$\sum_{j=1}^{k} \tilde{P}\big(A_1,\dots,[A_j,B],\dots,A_k\big) = 0 \qquad \text{for all } A_1,\dots,A_k,B \tag{11.1}$$

***Proof*:**
Step 1: existence and symmetry. The stated formula is symmetric in $A_1,\dots,A_k$ because mixed partial derivatives commute, and it is k-linear because P is homogeneous of degree k, so each variable t_j enters $P(t_1A_1+\cdots+t_kA_k)$ to total degree k and the coefficient of $t_1\cdots t_k$ is multilinear. Setting all $A_j=A$ gives $P(tA)=t^kP(A)$ with $t=t_1+\cdots+t_k$, and the coefficient of $t_1\cdots t_k$ in $t^kP(A)$, extracted by $\frac{1}{k!}\partial^k/\partial t_1\cdots\partial t_k$, equals $P(A)$ since $\partial^k(t_1+\cdots+t_k)^k/\partial t_1\cdots\partial t_k = k!$. Uniqueness holds because a symmetric k-linear form is determined by its diagonal values $\tilde{P}(A,\dots,A)$ over a field of characteristic zero, by the polarization identity.

Step 2: invariance. Replace each A_j by $g^{-1}A_jg$. Then $t_1g^{-1}A_1g+\cdots+t_kg^{-1}A_kg = g^{-1}(t_1A_1+\cdots+t_kA_k)g$, and Ad-invariance of P gives $P(g^{-1}Xg)=P(X)$ with $X=\sum t_jA_j$. Differentiating in $t_1,\dots,t_k$ shows $\tilde{P}(g^{-1}A_1g,\dots,g^{-1}A_kg) = \tilde{P}(A_1,\dots,A_k)$.

Step 3: infinitesimal invariance. Take $g=\exp(tB)$, so that $g^{-1}A_jg = \exp(-tB)A_j\exp(tB)$ and

$$\frac{d}{dt}\bigg|_0 g^{-1}A_jg = A_jB - BA_j = [A_j,B]$$

Differentiating the invariance $\tilde{P}(g^{-1}A_1g,\dots,g^{-1}A_kg) = \tilde{P}(A_1,\dots,A_k)$ at $t=0$ and using k-linearity to distribute the derivative across the slots,

$$0 = \frac{d}{dt}\bigg|_0 \tilde{P}\big(g^{-1}A_1g,\dots,g^{-1}A_kg\big) = \sum_{j=1}^{k}\tilde{P}\big(A_1,\dots,[A_j,B],\dots,A_k\big)$$

which is (11.1), as we sought to show.

The infinitesimal invariance (11.1) is the algebraic engine of the whole theory. Read with B replaced by the connection matrix and the A_j by the curvature, it says that the terms produced when the exterior derivative hits $P(\Omega)$ and picks up the connection matrix through the Bianchi identity cancel in the alternating sum. This is what makes $P(\Omega)$ closed, and the proof of the next theorem is little more than making that sentence precise.

Theorem 11.1.6: Chern–Weil forms are closed

Let P be an Ad-invariant polynomial of degree k on $\mathfrak{gl}_r(\mathbb{C})$ and ∇ a connection on E with curvature Ω. The Chern–Weil form $P(\Omega)$ of definition 11.1.3 is a closed $2k$-form:

$$d\,P(\Omega) = 0$$

The same holds for the normalized form $P\big(\frac{i}{2\pi}\Omega\big)$.

***Proof*:**
Work in a local frame with curvature matrix Ω and connection matrix ω; closedness is a local condition, and once verified on every frame domain it holds globally because $P(\Omega)$ is the global form of proposition 11.1.4. Let $\tilde{P}$ be the polarization of proposition 11.1.5, so that $P(\Omega) = \tilde{P}(\Omega, \dots, \Omega)$ with Ω appearing k times, the multilinear form evaluated on 2-form arguments with wedge products.

Step 1: differentiate through the polarization. Because $\tilde{P}$ is multilinear and the entries of Ω are 2-forms, the ordinary Leibniz rule for d distributes across the k slots with no sign, each Ω being of even degree:

$$d\,P(\Omega) = d\,\tilde{P}(\Omega, \dots, \Omega) = \sum_{j=1}^{k} \tilde{P}\big(\Omega, \dots, \underbrace{d\Omega}_{j}, \dots, \Omega\big) \tag{11.2}$$

Here $d\Omega$ occupies the jth slot and the remaining $k-1$ slots hold Ω. The signs $(-1)^{\deg}$ that would appear in a graded Leibniz rule are all $+1$ because every Ω preceding the differentiated slot has even degree 2.

Step 2: apply the Bianchi identity. The differential Bianchi identity of theorem 10.2.7 in frame form is $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$, that is, entry by entry $d\Omega = [\Omega, \omega]$ where $[\,\cdot\,, \cdot\,]$ is the graded commutator of the matrix-valued forms Ω (degree 2) and ω (degree 1), which for these degrees is $\Omega \wedge \omega - \omega \wedge \Omega$. Substituting into (11.2),

$$d\,P(\Omega) = \sum_{j=1}^{k} \tilde{P}\big(\Omega, \dots, [\Omega, \omega], \dots, \Omega\big)$$

the commutator sitting in the jth slot.

Step 3: cancel by infinitesimal invariance. This is precisely the sum (11.1) of proposition 11.1.5, with the curvature Ω in the role of every A_j and the connection ω in the role of B. The infinitesimal invariance identity was stated for numerical matrices; it extends to the present setting because $\tilde{P}$ is multilinear over the ring of even-degree forms and the identity (11.1) is a formal consequence of multilinearity and invariance, holding entry by entry when the matrix entries are taken in any commutative ring, here the wedge algebra of even-degree forms on the frame domain. The connection matrix ω has odd degree 1, but each summand

$\tilde{P}(\Omega, \dots, [\Omega, \omega], \dots, \Omega)$ contains ω exactly once and $k-1$ even-degree factors together with one commutator, so the total degree is even and the sign bookkeeping matches the numerical identity. Therefore

$$d\,P(\Omega) = \sum_{j=1}^{k} \tilde{P}\big(\Omega, \dots, [\Omega, \omega], \dots, \Omega\big) = 0$$

Hence $P(\Omega)$ is closed on each frame domain, and being globally defined, closed on M. The normalized form $P(\frac{i}{2\pi}\Omega)$ differs from $P(\Omega)$ by the constant factor $(i/2\pi)^k$, so $d\,P(\frac{i}{2\pi}\Omega) = (i/2\pi)^k\,d\,P(\Omega) = 0$ as well, completing the proof.

Closedness is what promotes $P(\Omega)$ from a differential form to a de Rham cohomology class $[P(\Omega)] \in H^{2k}(M;\mathbb{C})$. The class is where the topological content lives: the form itself remembers the connection, but its class, as the next theorem shows, does not. The proof of closedness has an instructive shape. It never used which invariant polynomial P was chosen, only that P is invariant, and it used the curvature only through the Bianchi identity. The two structural inputs, invariance and Bianchi, are exactly the two facts that survive from the algebra and the geometry, and the theorem is their conjunction. A connection that failed the Bianchi identity, or a polynomial that failed to be invariant, would break the cancellation, and there would be no closed form.

The determinant $\det(I + \frac{i}{2\pi}\Omega)$ that defines the total Chern class in section 11.2 is a sum of the invariant polynomials σ_k, each homogeneous, so theorem 11.1.6 applies to each graded piece and the total Chern form is a sum of closed forms of degrees $0, 2, 4, \dots$. Before turning to the named classes, the stronger fact must be established: the class is independent of the connection.

Two connections ∇_0 and ∇_1 on the same bundle produce two curvatures Ω_0, Ω_1 and two Chern–Weil forms $P(\Omega_0), P(\Omega_1)$. These forms will generally differ, connection by connection. What the transgression argument shows is that their difference is exact, so they represent the same cohomology class. The device is to interpolate linearly between the two connections and integrate the derivative of the Chern–Weil form along the path; the difference of the endpoint forms is d of the integral, an explicit primitive called the transgression form.

Theorem 11.1.7: The Chern–Weil class is independent of the connection

Let P be an Ad-invariant polynomial of degree k on $\mathfrak{gl}_r(\mathbb{C})$, and let ∇_0, ∇_1 be two connections on E with curvatures Ω_0, Ω_1. Then the closed $2k$-forms $P(\Omega_0)$ and $P(\Omega_1)$ differ by an exact form:

$$P(\Omega_1) - P(\Omega_0) = d\,\mathrm{TP}(\nabla_0, \nabla_1)$$

for an explicit $(2k-1)$-form $\mathrm{TP}(\nabla_0, \nabla_1)$, the *transgression form*. Consequently the de Rham class $[P(\Omega)] \in H^{2k}(M;\mathbb{C})$ is the same for every connection ∇; it depends only on P and the bundle E. The same holds for $P\big(\frac{i}{2\pi}\Omega\big)$.

***Proof*:**
Interpolate between the connections and track the Chern–Weil form along the interpolation. Let $\tilde{P}$ be the polarization of proposition 11.1.5.

Step 1: the affine path of connections. For $t \in [0,1]$ set

$$\nabla_t = (1-t)\nabla_0 + t\nabla_1 = \nabla_0 + t\,\alpha, \qquad \alpha := \nabla_1 - \nabla_0$$

The difference $\alpha = \nabla_1 - \nabla_0$ of two connections is $C^\infty(M)$-linear, hence a tensorial 1-form valued in $\mathrm{End}(E)$, that is $\alpha \in \Omega^1(M; \mathrm{End}E)$, by proposition 10.1.3. Thus each ∇_t is a connection, and in a local frame the connection matrices are $\omega_t = \omega_0 + t\,a$ with a the frame matrix of α, a matrix of 1-forms, and $\dot\omega_t := \frac{d}{dt}\omega_t = a$. Let Ω_t be the curvature of ∇_t, with frame matrix $\Omega_t = d\omega_t + \omega_t \wedge \omega_t$ by theorem 10.2.4.

Step 2: the derivative of the curvature. Differentiate the structure equation in t. Since $\dot\omega_t = a$,

$$\dot\Omega_t = \frac{d}{dt}\big(d\omega_t + \omega_t \wedge \omega_t\big) = d\,a + a \wedge \omega_t + \omega_t \wedge a = d^{\nabla_t} a \tag{11.3}$$

where d^{∇_t} is the exterior covariant derivative of ∇_t on $\mathrm{End}(E)$-valued forms, which on a 1-form a acts by $d^{\nabla_t} a = da + \omega_t \wedge a + a \wedge \omega_t$ (the two connection terms coming from the E and E^* factors of $\mathrm{End}E$, as in eq. (10.2)). The intermediate step used $\frac{d}{dt}(\omega_t \wedge \omega_t) = a \wedge \omega_t + \omega_t \wedge a$, valid because the wedge is bilinear and $\dot\omega_t = a$.

Step 3: the derivative of the Chern–Weil form. By multilinearity of $\tilde P$ and (11.3), differentiating $P(\Omega_t) = \tilde P(\Omega_t, \dots, \Omega_t)$ gives, exactly as in (11.2),

$$\frac{d}{dt} P(\Omega_t) = \sum_{j=1}^{k} \tilde P\big(\Omega_t, \dots, \underbrace{\dot\Omega_t}_{j}, \dots, \Omega_t\big) = k\,\tilde P\big(d^{\nabla_t} a, \Omega_t, \dots, \Omega_t\big) \tag{11.4}$$

the last equality using the symmetry of $\tilde P$ to collect the k identical summands, each with $\dot\Omega_t = d^{\nabla_t} a$ in one slot and Ω_t in the other $k-1$ slots.

Step 4: extract an exact form. The claim is that the right-hand side of (11.4) equals d of the $(2k-1)$-form $k\,\tilde P(a, \Omega_t, \dots, \Omega_t)$. Compute d of the latter by the Leibniz rule (11.2), now with a (degree 1) in the first slot and Ω_t (degree 2) in the rest. Because $\tilde P$ is invariant, its covariant derivative agrees with its ordinary derivative on the multilinear expression: the connection terms produced by d^{∇_t} acting through each slot sum to zero by the infinitesimal invariance (11.1) of proposition 11.1.5 applied with $B = \omega_t$, exactly as in the proof of theorem 11.1.6. Concretely,

$$d\,\tilde P(a, \Omega_t, \dots, \Omega_t) = \tilde P(d^{\nabla_t} a, \Omega_t, \dots, \Omega_t) + \sum_{j=2}^{k} \tilde P\big(a, \Omega_t, \dots, d^{\nabla_t}\Omega_t, \dots, \Omega_t\big)$$

where replacing d by d^{∇_t} in each slot is legitimate because the sum of the compensating connection terms vanishes by invariance. The Bianchi identity $d^{\nabla_t}\Omega_t = 0$ of theorem 10.2.7 kills every term in the sum over $j \geq 2$, leaving

$$d\,\tilde P(a, \Omega_t, \dots, \Omega_t) = \tilde P(d^{\nabla_t} a, \Omega_t, \dots, \Omega_t) = \frac{1}{k}\frac{d}{dt} P(\Omega_t) \tag{11.5}$$

by (11.4). Thus $\frac{d}{dt} P(\Omega_t) = d\big(k\,\tilde P(a, \Omega_t, \dots, \Omega_t)\big)$, and the right-hand side is exact for each t.

Step 5: integrate over t. Integrate (11.5) in t from 0 to 1. The left side telescopes:

$$P(\Omega_1) - P(\Omega_0) = \int_0^1 \frac{d}{dt} P(\Omega_t)\,dt = \int_0^1 d\big(k\,\tilde P(a, \Omega_t, \dots, \Omega_t)\big)\,dt = d\,\mathrm{TP}(\nabla_0, \nabla_1)$$

where the exterior derivative d commutes with the t-integral (both are integrations over independent variables, and d acts on M while $\int_0^1$ acts on t), and the transgression form is

$$\mathrm{TP}(\nabla_0, \nabla_1) = k \int_0^1 \tilde P\big(a, \Omega_t, \dots, \Omega_t\big)\,dt \in \Omega^{2k-1}(M)$$

a globally defined $(2k-1)$-form, since $a = \nabla_1 - \nabla_0$ is a global $\mathrm{End}(E)$-valued 1-form and each Ω_t is a global curvature; the invariance of $\tilde{P}$ makes the integrand frame-independent by the argument of proposition 11.1.4. Therefore $P(\Omega_1) - P(\Omega_0)$ is exact, so $[P(\Omega_0)] = [P(\Omega_1)]$ in $H^{2k}(M;\mathbb{C})$. As ∇_0, ∇_1 were arbitrary, the class is independent of the connection. The claim for $P(\frac{i}{2\pi}\Omega)$ follows by the constant factor $(i/2\pi)^k$, completing the proof.

The transgression form $\mathrm{TP}(\nabla_0, \nabla_1)$ is not only an existence device; it is the explicit primitive that will do the concrete work of Chern's proof of the Gauss–Bonnet–Chern theorem in section 11.4, where a connection is compared with one adapted to a vector field and the transgression form is integrated over a sphere around each zero. Here its role is to certify that the assignment $\nabla \mapsto [P(\Omega)]$ lands in a single cohomology class. The mechanism repeats the pattern of the closedness proof: two connections are joined by a path, the derivative along the path is an exact form by the same invariance-plus-Bianchi cancellation, and integrating the exact derivative gives an exact difference. The line-bundle version of this argument was carried out by hand in proposition 10.4.3, where a change of Hermitian metric shifted the curvature by $\partial\bar{\partial}$ of a global function, an exact form; the transgression argument is the same phenomenon for every rank and every invariant polynomial, with $\partial\bar{\partial}\varphi$ replaced by $d\,\mathrm{TP}$.

With both theorems in hand the construction can be packaged as a single map, from the ring of invariant polynomials to the cohomology ring.

Definition 11.1.8: The Chern–Weil homomorphism

Let $E \to M$ be a complex vector bundle of rank r with a connection ∇ of curvature Ω. The *Chern–Weil homomorphism* of E is the map

$$\mathrm{cw}\colon I^*(\mathfrak{gl}_r) \longrightarrow H^{2*}(M;\mathbb{C}), \qquad \mathrm{cw}(P) = \left[P\left(\tfrac{i}{2\pi}\Omega\right)\right]$$

sending an Ad-invariant polynomial P, homogeneous of degree k, to the de Rham class of its normalized Chern–Weil form of definition 11.1.3. By theorem 11.1.6 the form is closed, so the class is defined, and by theorem 11.1.7 the class is independent of the connection ∇, so cw depends only on E. It is a homomorphism of graded rings: it carries the product PQ of invariant polynomials to the cup product $\mathrm{cw}(P) \smile \mathrm{cw}(Q)$, and the sum to the sum.

That cw is a ring homomorphism is a direct consequence of the definitions. Addition is preserved because $(P+Q)(\Omega) = P(\Omega) + Q(\Omega)$ at the level of forms, and passing to classes is additive. Multiplication is preserved because $(PQ)(\Omega) = P(\Omega) \wedge Q(\Omega)$: the product polynomial evaluated on Ω is the wedge of the two evaluated forms, since the substitution of definition 11.1.3 is multiplicative on the commutative wedge algebra of even-degree forms, and the cup product on de Rham cohomology is induced by the wedge ([4]). The normalization is consistent with the grading: if P has degree k and Q degree l, then PQ has degree $k+l$ and the factor $(i/2\pi)^{k+l} = (i/2\pi)^k (i/2\pi)^l$ distributes correctly across the product. The Chern–Weil homomorphism is thus the single algebraic object underlying every characteristic class this chapter constructs: the Chern class is cw of the elementary symmetric polynomials, the Chern character is cw of $\mathrm{tr}\exp$, and the Pontryagin and Euler classes are cw of the corresponding invariant polynomials of a metric curvature. The classes it produces are the images in real (or complex) cohomology of the integral topological characteristic classes constructed by classifying-space methods in [8] and [4]; the identification of the two constructions is stated as theorem 11.2.6 in the next section and proved there by citation, the axiomatic side belonging to the topological development.

The construction closes the loop with the line-bundle case computed directly in section 10.4. In rank one every invariant polynomial is a power of the trace, and the degree-one piece is exactly the first Chern form.

Example 11.1.9: Chern–Weil for a line bundle recovers the first Chern form
Let $L \to M$ be a complex line bundle, $r = 1$, with a connection of curvature Ω, a single 2-form (lemma 10.4.1). The matrix algebra $\mathfrak{gl}_1(\mathbb{C}) = \mathbb{C}$ is one-dimensional, and conjugation is trivial ($g^{-1}Ag = A$ for scalars), so *every* polynomial on $\mathfrak{gl}_1(\mathbb{C})$ is Ad-invariant. The invariant polynomials are the powers $P_k(A) = A^k$, and $P_k(A) = \operatorname{tr}(A^k) = A^k$ coincides with the power sum; the elementary symmetric polynomials degenerate to $\sigma_0 = 1$, $\sigma_1(A) = A$, with no higher σ_k since $r = 1$.

Take the degree-one invariant polynomial $P_1(A) = A = \operatorname{tr} A$. Its Chern–Weil form is $P_1(\Omega) = \Omega$, and the normalized form is

$$P_1\big(\tfrac{i}{2\pi}\Omega\big) = \tfrac{i}{2\pi}\,\Omega = c_1(L, h)$$

which is exactly the first Chern form of definition 10.4.2. The general theory now recovers the two facts established by hand in section 10.4: theorem 11.1.6 gives $d\,c_1(L, h) = 0$, which was part (2) of proposition 10.4.3, and theorem 11.1.7 gives that $[c_1(L, h)] \in H^2(M; \mathbb{C})$ is independent of the connection, hence of the Hermitian metric, which was part (3). In rank one the transgression form of degree $2k - 1 = 1$ specializes to the primitive $-\frac{i}{2\pi}\bar\partial\varphi$ found in the metric-independence proof there, and the abstract $d\,\mathrm{TP}$ is the concrete $d(-\frac{i}{2\pi}\bar\partial\varphi)$. The higher powers $P_k(A) = A^k$ produce the forms $(\frac{i}{2\pi}\Omega)^k = c_1(L, h)^k$, the powers of the first Chern form; these are the only Chern–Weil forms a line bundle carries, reflecting that a line bundle's characteristic data is exhausted by its first Chern class. On a compact Riemann surface the degree-one form integrates to $\int_M c_1(L, h) = \deg L \in \mathbb{Z}$, the computation of example 10.4.6, and the value $+1$ for $\mathcal{O}(1)$ over $\mathbb{CP}^1$ of theorem 10.4.5 is the base case of the whole theory: the normalization $i/2\pi$ in definition 11.1.8 was fixed to make that integral the integer $+1$.

11.2 Chern Classes and the Chern Character

The Chern–Weil homomorphism of section 11.1 turns an Ad-invariant polynomial on $\mathfrak{gl}_r(\mathbb{C})$ into a closed real form whose de Rham class depends only on the bundle, not on the connection used to compute it. Two families of invariant polynomials organize this construction into the characteristic classes that the rest of the chapter uses. The elementary symmetric functions of the curvature eigenvalues assemble into the total Chern class, a graded invariant that is multiplicative under direct sums; the power sums assemble into the Chern character, an invariant that is additive under direct sums and multiplicative under tensor products. This section extracts both from the curvature, establishes their formal properties, and computes them on the one example that drives the theory forward, the tangent bundle of complex projective space.

The two invariants encode complementary information. The total Chern class $c(E) = \det(I + \frac{i}{2\pi}\Omega)$ is the natural home for the Whitney sum formula, and its top component pairs against the fundamental class to give Euler numbers; but it multiplies under $\oplus$, which makes it awkward for the tensor products that appear in K-theory. The Chern character $\operatorname{ch}(E) = \operatorname{tr}\exp(\frac{i}{2\pi}\Omega)$ trades the determinant for the exponential of the trace, and the exchange converts the multiplicative Whitney formula into an additive one while making the tensor product multiplicative. The result is a ring homomorphism from the K-theory of M to its rational cohomology, the differential-geometric form of a purely topological construction. This section builds the curvature side of that homomorphism; the identification with the topological Chern classes of [8] is stated and attributed, since its proof belongs to the axiomatic theory there.

Throughout, $E \to M$ and $F \to M$ are smooth complex vector bundles of ranks r and s over a smooth manifold M, each carrying a connection with curvature $\Omega \in \Omega^2(M; \mathrm{End} E)$ from chapter 10. Invariant polynomials on $\mathfrak{gl}_r(\mathbb{C})$ and the Chern–Weil forms $P(\frac{i}{2\pi}\Omega)$ are those of definition 11.1.3; that these forms are closed with connection-independent classes is theorem 11.1.6 and theorem 11.1.7. The normalized curvature $\frac{i}{2\pi}\Omega$ recurs so often that the abbreviation $A = \frac{i}{2\pi}\Omega \in \Omega^2(M; \mathrm{End} E)$ is used for it. Because A is an $\mathrm{End}(E)$-valued form of even degree, its entries commute under the wedge, and matrix expressions in A, determinants, traces of powers, exponentials, are formed exactly as for a matrix over a commutative ring.

The determinant of $I + A$ is the generating function of the elementary symmetric polynomials in the eigenvalues of A, and this is the invariant that produces the Chern classes.

Definition 11.2.1: Total Chern class and Chern roots

Let $E \to M$ be a complex vector bundle of rank r with a connection of curvature Ω, and set $A = \frac{i}{2\pi}\Omega \in \Omega^2(M; \mathrm{End} E)$. The *total Chern class* of E is the de Rham class of the closed even form

$$c(E) = \det\big(I + A\big) = \det\Big(I + \tfrac{i}{2\pi}\Omega\Big) = 1 + c_1(E) + c_2(E) + \cdots + c_r(E)$$

where $c_k(E) \in H^{2k}(M; \mathbb{R})$ is the class of the degree-$2k$ component of the determinant. The class $c_k(E)$ is the *k-th Chern class* of E; the sum terminates at $k = r$ because $\det(I + A)$ is a polynomial of degree r in the $r \times r$ matrix A. Writing the determinant as $\det(I + A) = \prod_{j=1}^{r}(1 + x_j)$ introduces the *Chern roots* $x_1, \ldots, x_r$, formal degree-2 classes whose elementary symmetric functions are the Chern classes,

$$c_k(E) = \sigma_k(x_1, \ldots, x_r) = \sum_{j_1 < \cdots < j_k} x_{j_1} \cdots x_{j_k}$$

so that $c_1 = x_1 + \cdots + x_r$, $c_2 = \sum_{j<k} x_j x_k$, and $c_r = x_1 \cdots x_r$.

The Chern roots are a bookkeeping device, not eigenvalues of any single matrix: the curvature A need not be diagonalizable, and its entries are 2-forms rather than numbers. What is well-defined is each symmetric function $\sigma_k(x) = c_k(E)$, a closed $2k$-form, because $c(E) = \det(I + A)$ is invariant under conjugation of A and hence a Chern–Weil form of definition 11.1.3: the coefficient of t^k in $\det(I + tA)$ is the invariant polynomial σ_k evaluated at $\frac{i}{2\pi}\Omega$. The splitting principle formalizes the manipulation, letting one compute with the x_j as if the bundle split as a sum of line bundles $\bigoplus L_j$ with $c_1(L_j) = x_j$; every identity among symmetric functions then transfers to an identity among Chern classes. Each $c_k(E)$ is closed by theorem 11.1.6, and its class is independent of the connection by theorem 11.1.7, so $c(E)$ is an invariant of the bundle alone. The class is moreover real, so that $c_k(E) \in H^{2k}(M; \mathbb{R})$ as definition 11.2.1 records. To see this, compute $c(E)$ from a unitary connection, one compatible with a Hermitian metric on E; such connections exist by proposition 10.3.4. In an h-orthonormal frame the curvature matrix Ω is skew-Hermitian by proposition 10.3.2, so $A = \frac{i}{2\pi}\Omega$ satisfies $A^* = -\frac{i}{2\pi}\Omega^* = \frac{i}{2\pi}\Omega = A$ and is Hermitian, where $A^* = \bar{A}^{\mathsf{T}}$ conjugates the 2-form entries and transposes. The symmetric function σ_k has real coefficients and is transpose-invariant, so $\overline{\sigma_k(A)} = \sigma_k(\bar{A}) = \sigma_k\big(A^{\mathsf{T}}\big) = \sigma_k(A)$, using $\bar{A} = A^{\mathsf{T}}$ for the Hermitian matrix A; hence $c_k(E) = \sigma_k(A)$ equals its own conjugate and is a real form for this connection, with de Rham class in $H^{2k}(M; \mathbb{R})$.

By connection-independence the same real class computes $c_k(E)$ from any connection.

The rank-one case is the mandatory cross-check with the line-bundle theory of section 10.4, and it fixes the normalization.

Example 11.2.2: The total Chern class of a line bundle

Let $L \to M$ be a complex line bundle, so $r = 1$ and the curvature Ω is a scalar-valued 2-form (lemma 10.4.1). The matrix $A = \frac{i}{2\pi}\Omega$ is 1×1, and

$$c(L) = \det(I + A) = 1 + \tfrac{i}{2\pi}\Omega = 1 + c_1(L)$$

so the total Chern class has only the two terms 1 and $c_1(L) = \frac{i}{2\pi}\Omega$, and there is a single Chern root $x_1 = c_1(L)$. This is exactly the first Chern form of definition 10.4.2: for a Hermitian holomorphic line bundle with its Chern connection, $c_1(L) = \frac{i}{2\pi}\Omega = -\frac{i}{2\pi}\partial\bar{\partial}\log h$, the real closed 2-form whose class is metric-independent by proposition 10.4.3. The Chern–Weil determinant construction therefore recovers the line-bundle invariant of section 10.4 verbatim, with the same $\frac{i}{2\pi}$ normalization that made $\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = 1$ in theorem 10.4.5. For higher rank the determinant $\det(I + A)$ produces further terms $c_2, \dots, c_r$, but $c_1(E) = \frac{i}{2\pi}\mathrm{tr}\Omega$ remains the trace of the curvature, the sum of the Chern roots.

The structural property that makes the total Chern class computable is its behavior under the two operations that build new bundles from old: pullback along a map, and direct sum. Pullback is a naturality statement, and direct sum is the Whitney formula, which turns the total class into a homomorphism from the semigroup of bundles under $\oplus$ into the multiplicative group of even cohomology classes with constant term 1. Both are proved by choosing the connection that makes the curvature transparent.

Proposition 11.2.3: Naturality and the Whitney sum formula

Let $E, F \to M$ be complex vector bundles.

1. *Naturality.* If $f\colon N \to M$ is smooth, then the pullback bundle satisfies

$$c(f^*E) = f^*c(E) \in H^{2*}(N;\mathbb{R})$$

2. *Whitney sum formula.* The total Chern class of a direct sum is the product of the total Chern classes,

$$c(E \oplus F) = c(E)\,c(F)$$

equivalently $c_k(E \oplus F) = \sum_{i+j=k} c_i(E)\,c_j(F)$ in each degree.

***Proof*:**

Both parts follow from the fact that $c(E)$ is computed from any connection and that convenient connections make the curvature explicit. The proof treats the two parts in turn.

1. *Naturality.* Let ∇ be a connection on E with curvature Ω. The pullback bundle $f^*E \to N$ carries the pullback connection $f^*\nabla$, whose curvature is the pullback form $f^*\Omega$; this is immediate from the structure equation $\Omega = d\omega + \omega \wedge \omega$ of theorem 10.2.4, since pullback commutes with d and with the wedge, so that in a pulled-back frame the connection matrix

of $f^*\nabla$ is $f^*\omega$ and its curvature matrix is $f^*\Omega$ ([6]). Because the determinant is computed entrywise and f^* is an algebra homomorphism on forms,

$$c(f^*E) = \det\Big(I + \tfrac{i}{2\pi} f^*\Omega\Big) = f^* \det\Big(I + \tfrac{i}{2\pi}\Omega\Big) = f^*c(E)$$

at the level of forms; passing to de Rham classes, and using that f^* on forms induces f^* on cohomology, gives $c(f^*E) = f^*c(E)$. Since the class $c(E)$ is connection-independent by theorem 11.1.7, the identity does not depend on the chosen ∇.

2. *Whitney sum formula.* Equip E with a connection ∇^E of curvature Ω^E and F with a connection ∇^F of curvature Ω^F. The direct sum carries the block-diagonal connection $\nabla^{E\oplus F} = \nabla^E \oplus \nabla^F$, defined by $\nabla^{E\oplus F}(s \oplus t) = \nabla^E s \oplus \nabla^F t$. In a frame adapted to the splitting $E \oplus F$, the connection matrix is block-diagonal, $\omega^{E\oplus F} = \begin{pmatrix} \omega^E & 0 \\ 0 & \omega^F \end{pmatrix}$, so the structure equation of theorem 10.2.4 produces a block-diagonal curvature matrix

$$\Omega^{E\oplus F} = \begin{pmatrix} \Omega^E & 0 \\ 0 & \Omega^F \end{pmatrix}$$

the off-diagonal blocks vanishing because both $d\omega^{E\oplus F}$ and $\omega^{E\oplus F} \wedge \omega^{E\oplus F}$ are block-diagonal. The determinant of a block-diagonal matrix is the product of the determinants of its blocks, so

$$c(E \oplus F) = \det\Big(I + \tfrac{i}{2\pi}\Omega^{E\oplus F}\Big) = \det\Big(I + \tfrac{i}{2\pi}\Omega^E\Big) \det\Big(I + \tfrac{i}{2\pi}\Omega^F\Big) = c(E)\, c(F)$$

Extracting the degree-$2k$ component of the product gives $c_k(E \oplus F) = \sum_{i+j=k} c_i(E)\, c_j(F)$. The class $c(E \oplus F)$ is independent of all connection choices by theorem 11.1.7, so the block-diagonal connection computes the correct invariant, completing the proof.

The Whitney formula reads most transparently through the Chern roots of definition 11.2.1. If E has Chern roots $x_1, \ldots, x_r$ and F has Chern roots $y_1, \ldots, y_s$, then the block-diagonal curvature of the proof shows that $E \oplus F$ has Chern roots $x_1, \ldots, x_r, y_1, \ldots, y_s$, the two lists concatenated. The total class is the product $\prod_i (1 + x_i) \prod_j (1 + y_j) = c(E)c(F)$, and the individual formula $c_k(E \oplus F) = \sum_{i+j=k} c_i(E)c_j(F)$ is the statement that σ_k of a concatenated list is the convolution of the σ's of the two sublists. The Whitney formula is what makes the total Chern class computable on any bundle presented as an iterated sum or, through the splitting principle, on any bundle at all: reduce to line bundles, where $c(L) = 1 + c_1(L)$ is known, and multiply.

The determinant is not the only symmetric function of the curvature worth extracting. Replacing $\det(I + A)$ by $\operatorname{tr}\exp(A)$ trades the elementary symmetric polynomials for the power sums, and the exchange has a consequence that the total Chern class does not enjoy: the resulting invariant is additive under direct sum rather than multiplicative, and multiplicative under tensor product. This is the Chern character.

Definition 11.2.4: Chern character

Let $E \to M$ be a complex vector bundle of rank r with a connection of curvature Ω, and set $A = \frac{i}{2\pi}\Omega$. The *Chern character* of E is the de Rham class of the closed even form

$$\mathrm{ch}(E) = \mathrm{tr}\exp(A) = \mathrm{tr}\exp\Big(\tfrac{i}{2\pi}\Omega\Big) = \sum_{m=0}^{\infty} \frac{1}{m!}\mathrm{tr}\big(A^m\big)$$

where the exponential is the matrix exponential and the series terminates because A^m is a $2m$-form, zero for $2m > \dim M$. In terms of the Chern roots $x_1, \ldots, x_r$ of definition 11.2.1,

$$\mathrm{ch}(E) = \sum_{j=1}^{r} e^{x_j} = \sum_{j=1}^{r}\Big(1 + x_j + \tfrac{1}{2}x_j^2 + \cdots\Big)$$

whose homogeneous components are the power sums $\frac{1}{m!}\sum_j x_j^m = \frac{1}{m!}\mathrm{tr}(A^m)$ of the Chern roots.

The Chern character is the exponential generating function of the power sums of the Chern roots, just as the total Chern class is the ordinary generating function of their elementary symmetric functions. Its low-degree terms are expressible in the Chern classes by Newton's identities relating power sums to elementary symmetric polynomials. The degree-zero term is $\sum_j 1 = r = \mathrm{rank}E$. The degree-two term is $\sum_j x_j = c_1(E)$. The degree-four term is $\frac{1}{2}\sum_j x_j^2 = \frac{1}{2}\big((\sum_j x_j)^2 - 2\sum_{j<k} x_j x_k\big) = \frac{1}{2}\big(c_1^2 - 2c_2\big)$, using the same Newton identity that expresses $\sum x_j^2$ through σ_1, σ_2. Assembling the first three terms,

$$\mathrm{ch}(E) = r + c_1(E) + \tfrac{1}{2}\big(c_1(E)^2 - 2c_2(E)\big) + \cdots$$

so the Chern character carries the same information as the Chern classes in low degrees, repackaged as power sums. The advantage of the repackaging is the pair of homomorphism properties proved next.

Proposition 11.2.5: The Chern character is a ring homomorphism

Let $E, F \to M$ be complex vector bundles. Then the Chern character is additive under direct sum and multiplicative under tensor product:

$$\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F), \qquad \mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$$

It is also natural: $\mathrm{ch}(f^*E) = f^*\,\mathrm{ch}(E)$ for every smooth $f\colon N \to M$.

***Proof*:**
Naturality is proved exactly as in proposition 11.2.3(1): the pullback connection $f^*\nabla$ has curvature $f^*\Omega$, and $\mathrm{tr}\exp$ is computed entrywise, so $\mathrm{ch}(f^*E) = \mathrm{tr}\exp(\frac{i}{2\pi}f^*\Omega) = f^*\mathrm{tr}\exp(\frac{i}{2\pi}\Omega) = f^*\,\mathrm{ch}(E)$. The two ring-homomorphism identities are established separately, each using a convenient connection.

Step 1: additivity under direct sum. As in proposition 11.2.3(2), equip $E \oplus F$ with the block-diagonal connection $\nabla^E \oplus \nabla^F$, whose curvature matrix is block-diagonal with blocks Ω^E and Ω^F. Set $A = \frac{i}{2\pi}\Omega^E$ and $B = \frac{i}{2\pi}\Omega^F$, so the normalized curvature of the sum is $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$. The matrix exponential of a block-diagonal matrix is block-diagonal with the exponentials of the blocks,

$$\exp\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \begin{pmatrix} \exp A & 0 \\ 0 & \exp B \end{pmatrix}$$

since each power $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}^m = \begin{pmatrix} A^m & 0 \\ 0 & B^m \end{pmatrix}$ is block-diagonal and the exponential series preserves the block structure. The trace of a block-diagonal matrix is the sum of the traces of the blocks, so

$$\mathrm{ch}(E \oplus F) = \mathrm{tr} \exp\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \mathrm{tr} \exp A + \mathrm{tr} \exp B = \mathrm{ch}(E) + \mathrm{ch}(F)$$

establishing additivity.

Step 2: multiplicativity under tensor product. Equip $E \otimes F$ with the tensor-product connection $\nabla^{E\otimes F} = \nabla^E \otimes \mathrm{id}_F + \mathrm{id}_E \otimes \nabla^F$ of proposition 10.1.8(2), characterized by the Leibniz rule $\nabla^{E\otimes F}(s \otimes t) = \nabla^E s \otimes t + s \otimes \nabla^F t$. Its curvature is

$$\Omega^{E\otimes F} = \Omega^E \otimes \mathrm{id}_F + \mathrm{id}_E \otimes \Omega^F \tag{11.6}$$

an endomorphism-valued 2-form on $E \otimes F$: applying $F^\nabla = (d^\nabla)^2$ to a decomposable section $s \otimes t$ and using the Leibniz rule twice, the mixed terms coupling $\nabla^E s$ to $\nabla^F t$ cancel in the alternating combination that defines the curvature, leaving the sum of the two curvatures acting on the respective factors. Normalizing, $\frac{i}{2\pi}\Omega^{E\otimes F} = A \otimes \mathrm{id}_F + \mathrm{id}_E \otimes B$ with $A = \frac{i}{2\pi}\Omega^E$ and $B = \frac{i}{2\pi}\Omega^F$. The two summands $A\otimes\mathrm{id}_F$ and $\mathrm{id}_E\otimes B$ commute, since $(A\otimes\mathrm{id})(\mathrm{id}\otimes B) = A\otimes B = (\mathrm{id}\otimes B)(A\otimes\mathrm{id})$ (the entries of A and B are even-degree forms, so no wedge sign intervenes). For commuting matrices the exponential of a sum factors, $\exp(P + Q) = \exp(P)\exp(Q)$, so

$$\exp\bigl(A\otimes\mathrm{id}_F + \mathrm{id}_E\otimes B\bigr) = \exp(A\otimes\mathrm{id}_F)\,\exp(\mathrm{id}_E\otimes B) = (\exp A\otimes\mathrm{id}_F)(\mathrm{id}_E\otimes\exp B) = \exp A\otimes\exp B$$

using $\exp(A \otimes \mathrm{id}) = \exp A \otimes \mathrm{id}$ and $\exp(\mathrm{id} \otimes B) = \mathrm{id} \otimes \exp B$, each immediate from the series. Finally the trace of a tensor product of endomorphisms is the product of the traces, $\mathrm{tr}(P \otimes Q) = \mathrm{tr}(P)\mathrm{tr}(Q)$, so

$$\mathrm{ch}(E \otimes F) = \mathrm{tr}\bigl(\exp A \otimes \exp B\bigr) = \mathrm{tr}(\exp A)\,\mathrm{tr}(\exp B) = \mathrm{ch}(E)\,\mathrm{ch}(F)$$

establishing multiplicativity and completing the proof.

The two identities are what the name “ring homomorphism” records. The isomorphism classes of complex vector bundles on M, under $\oplus$ and $\otimes$, generate a commutative ring, the K-theory ring $K^0(M)$ of [8], in which $\oplus$ is addition and $\otimes$ is multiplication. proposition 11.2.5 says precisely that the Chern character descends to a ring homomorphism

$$\mathrm{ch}\colon K^0(M) \longrightarrow H^{2*}(M; \mathbb{R})$$

into the even de Rham cohomology with its cup-product ring structure. Additivity lets ch extend from bundles to formal differences of bundles, the virtual bundles that populate $K^0(M)$; multiplicativity makes the extension respect products. Over the rationals this homomorphism is close to an isomorphism, the content of the Chern character isomorphism $K^0(M) \otimes \mathbb{Q} \cong H^{2*}(M; \mathbb{Q})$ of [8], so the Chern character is not just a ring homomorphism but the comparison map that identifies rationalized K-theory with even cohomology. The total Chern class, by contrast, is only multiplicative under $\oplus$ and does not extend to a homomorphism on the whole ring, which is why the Chern character is the invariant of choice in K-theoretic index theory.

The Chern classes constructed here from curvature carry a name borrowed from topology, and the borrowing is justified: they are the images, under de Rham's identification of cohomology theories, of the topological Chern classes defined for any complex bundle without reference to a connection. That identification is the bridge between the differential-geometric construction of this chapter and the axiomatic theory of characteristic classes, and its proof belongs to the latter.

Theorem 11.2.6: The Chern–Weil classes are the topological Chern classes

Let $E \to M$ be a complex vector bundle over a smooth manifold M. The Chern–Weil classes $c_k(E) \in H^{2k}(M;\mathbb{R})$ of definition 11.2.1, computed from the curvature of any connection, are the images under the natural map $H^{2k}(M;\mathbb{Z}) \to H^{2k}(M;\mathbb{R})$ of the topological Chern classes $c_k^{\mathrm{top}}(E) \in H^{2k}(M;\mathbb{Z})$. In particular the Chern–Weil classes are natural, satisfy the Whitney sum formula, vanish above the rank ($c_k(E) = 0$ for $k > \operatorname{rank} E$), and are normalized by c_1 of the hyperplane bundle on $\mathbb{CP}^1$, the four properties that characterize the topological Chern classes uniquely.

The topological Chern classes are defined either as the pullbacks of the universal classes on the classifying space $BU(r)$ or, axiomatically, as the unique natural classes satisfying the Whitney sum formula and the $\mathbb{CP}^1$ normalization; both constructions, and the theorem that the curvature classes agree with them, are carried out in [8] (with the underlying cohomology of classifying spaces in [4]). The curvature construction of this section verifies the characterizing properties directly: naturality and the Whitney formula are proposition 11.2.3, and the normalization is the computation $\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = 1$ of theorem 10.4.5. By the uniqueness of classes satisfying these axioms, the real de Rham classes $c_k(E)$ must coincide with the real images of the integral topological classes. The consequence used repeatedly below is integrality: although the curvature $c_k(E)$ is a priori only a real class, it is the image of an integer class, so its periods and its pairings against integral cycles are integers. This is what forces the curvature integrals of the chapter to compute integer-valued characteristic numbers, and it is the mechanism behind the integrality of the degree in example 10.4.6 and, in section 11.4, of the Euler number.

The computation that puts the machinery to work is the total Chern class of the tangent bundle of complex projective space. The tangent bundle $T\mathbb{CP}^n$ does not split as a sum of line bundles, so its Chern class is not immediate from the Whitney formula; the Euler sequence provides the missing presentation, expressing $T\mathbb{CP}^n$ in terms of the hyperplane bundle $\mathcal{O}(1)$ whose Chern class was computed in section 10.4.

Example 11.2.7: The Chern class of the tangent bundle of $\mathbb{CP}^n$

The holomorphic tangent bundle of $\mathbb{CP}^n$ fits into the *Euler sequence*, the short exact sequence of holomorphic vector bundles

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1)^{\oplus(n+1)} \longrightarrow T\mathbb{CP}^n \longrightarrow 0 \tag{11.7}$$

in which $\mathcal{O}$ is the trivial line bundle, $\mathcal{O}(1)$ is the hyperplane bundle of theorem 10.4.5, and the middle bundle is its $(n+1)$-fold direct sum; the sequence is derived in [1] from the description of $\mathbb{CP}^n$ as lines in $\mathbb{C}^{n+1}$, the map $\mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)}$ being the tautological inclusion and the quotient being the tangent bundle. Write $u = c_1(\mathcal{O}(1))$ for the first Chern class of the hyperplane bundle, the positive generator of $H^2(\mathbb{CP}^n;\mathbb{R})$ with $\int_{\mathbb{CP}^1} u = 1$ from theorem 10.4.5.

Step 1: the Chern class of the middle bundle. A short exact sequence of vector bundles splits smoothly (choose a Hermitian metric and take the orthogonal complement of the sub-bundle), so as smooth bundles $\mathcal{O}(1)^{\oplus(n+1)} \cong \mathcal{O} \oplus T\mathbb{C}\mathrm{P}^n$. The Whitney sum formula of proposition 11.2.3 applies to both sides. On the left, the total Chern class of an $(n+1)$-fold sum of copies of $\mathcal{O}(1)$ is the $(n+1)$-fold product

$$c\big(\mathcal{O}(1)^{\oplus(n+1)}\big) = c(\mathcal{O}(1))^{n+1} = (1+u)^{n+1}$$

using $c(\mathcal{O}(1)) = 1+u$ from example 11.2.2. On the right, the trivial bundle $\mathcal{O}$ has a flat connection with zero curvature, hence total Chern class $c(\mathcal{O}) = \det(I+0) = 1$, so

$$c\big(\mathcal{O} \oplus T\mathbb{C}\mathrm{P}^n\big) = c(\mathcal{O})\, c(T\mathbb{C}\mathrm{P}^n) = c(T\mathbb{C}\mathrm{P}^n)$$

Step 2: solve for $c(T\mathbb{C}\mathrm{P}^n)$. Equating the two expressions for the total Chern class of the smoothly isomorphic bundles gives

$$c(T\mathbb{C}\mathrm{P}^n) = (1+u)^{n+1} = \sum_{k=0}^{n} \binom{n+1}{k} u^k \tag{11.8}$$

the binomial expansion truncated at $k = n$ because $u^{n+1} = 0$ in $H^{2*}(\mathbb{C}\mathrm{P}^n;\mathbb{R})$, the cohomology ring being $\mathbb{R}[u]/(u^{n+1})$ with u in degree 2 ([4]). Reading off the homogeneous components, the k-th Chern class of the tangent bundle is $c_k(T\mathbb{C}\mathrm{P}^n) = \binom{n+1}{k}u^k$; in particular $c_1(T\mathbb{C}\mathrm{P}^n) = (n+1)u$ and the top class is

$$c_n(T\mathbb{C}\mathrm{P}^n) = \binom{n+1}{n} u^n = (n+1)\, u^n$$

Step 3: the top Chern number equals the Euler characteristic. Integrate the top Chern class over $\mathbb{C}\mathrm{P}^n$. The generator $u = c_1(\mathcal{O}(1))$ satisfies $\int_{\mathbb{C}\mathrm{P}^n} u^n = 1$: this is the normalization of the fundamental class of $\mathbb{C}\mathrm{P}^n$, dual to the hyperplane class, and it follows from $\int_{\mathbb{C}\mathrm{P}^1} u = 1$ together with the ring structure $H^{2*}(\mathbb{C}\mathrm{P}^n;\mathbb{R}) = \mathbb{R}[u]/(u^{n+1})$, in which u^n is the positive generator of the top cohomology $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{R})$ pairing to 1 against $[\mathbb{C}\mathrm{P}^n]$ ([4]). Hence

$$\int_{\mathbb{C}\mathrm{P}^n} c_n(T\mathbb{C}\mathrm{P}^n) = (n+1)\int_{\mathbb{C}\mathrm{P}^n} u^n = (n+1)\cdot 1 = n+1$$

The Euler characteristic of $\mathbb{C}\mathrm{P}^n$ is $\chi(\mathbb{C}\mathrm{P}^n) = n+1$, since its cohomology has one generator in each even degree $0, 2, \ldots, 2n$ and none in odd degrees ([4]), so the alternating sum of Betti numbers is $1 + 1 + \cdots + 1 = n+1$. Therefore

$$\int_{\mathbb{C}\mathrm{P}^n} c_n(T\mathbb{C}\mathrm{P}^n) = n+1 = \chi(\mathbb{C}\mathrm{P}^n)$$

the top Chern number of the tangent bundle equals the Euler characteristic, the mandatory cross-check. This is the complex instance of the identity that section 11.4 proves in general: the top characteristic class of the tangent bundle integrates to $\chi(M)$. For $n = 1$ it reads $\int_{\mathbb{C}\mathrm{P}^1} c_1(T\mathbb{C}\mathrm{P}^1) = 2 = \chi(\mathbb{C}\mathrm{P}^1) = \chi(S^2)$, recovering the sphere value of the Gauss–Bonnet theorem $\int_{S^2} e(TM) = 2$ from example 8.2.2, since $T\mathbb{C}\mathrm{P}^1$ has $c_1 = 2u$ and $\int_{\mathbb{C}\mathrm{P}^1} 2u = 2$.

11.3 Pontryagin Classes, the Euler Class, and the Pfaffian

The Chern classes of section 11.2 are invariants of a complex bundle, extracted from the determinant $\det(I + \frac{i}{2\pi}\Omega)$ of its curvature. A real bundle has no complex structure to feed that determinant, and the naive attempt to imitate the construction on a real curvature matrix fails: the odd-degree symmetric functions of a skew-symmetric matrix vanish, so the surviving real invariants live in degrees $4k$, not $2k$. Two routes lead out of this. The first complexifies the real bundle and reads off the surviving Chern classes; this produces the Pontryagin classes, the real bundle's version of the Chern classes, concentrated in degrees $4k$. The second uses an algebraic operation available only for skew-symmetric matrices of even order, the Pfaffian, a square root of the determinant that sees the orientation the determinant cannot. Applied to the curvature of an oriented rank-$2m$ bundle with a metric connection, the Pfaffian produces the Euler form, the last of the characteristic classes and the one this Part was built to reach: on the tangent bundle its integral is the Euler characteristic, the content of the Gauss–Bonnet–Chern theorem proved in section 11.4.

This section constructs both families. The Pontryagin classes come first, as a direct application of the Chern machinery to the complexification. The Pfaffian is then developed as an algebraic object on skew-symmetric matrices, with its two defining properties, $\mathrm{Pf}(A)^2 = \det A$ and the transformation law $\mathrm{Pf}(g^{\mathsf{T}} A g) = \det(g)\,\mathrm{Pf}(A)$, proved in full; the transformation law is exactly the $SO(2m)$-invariance that lets the Pfaffian pass through the Chern–Weil homomorphism to define the Euler form. The section closes by relating the two families, $e(E)^2 = p_m(E)$ and $e(E) = c_m$ for a complex bundle viewed as real, and by computing the Euler form of a surface's tangent bundle, where $\mathrm{Pf}(\Omega) = K\,dA$ recovers the Gauss–Bonnet integrand of section 8.1.

Throughout, $E \to M$ is a smooth vector bundle of rank r over M, real or complex as stated, with a connection ∇ and curvature $\Omega = F^\nabla \in \Omega^2(M; \mathrm{End} E)$ from chapter 10; the structure equation, the gauge law $\tilde{\Omega} = g^{-1}\Omega g$, and the Bianchi identity $d^\nabla F^\nabla = 0$ of section 10.2 are in force, as is the Chern–Weil machinery of section 11.1: an Ad-invariant polynomial evaluated on the curvature gives a globally defined closed form of connection-independent class (theorem 11.1.6, theorem 11.1.7). Invariant polynomials live on $\mathfrak{gl}_r(\mathbb{C})$, or, for the metric constructions, on the skew-symmetric matrices $\mathfrak{so}(2m)$. The complexification of a real bundle E is $E \otimes \mathbb{C} = E \otimes_{\mathbb{R}} \underline{\mathbb{C}}$, the complex bundle whose fiber is $E_p \otimes_{\mathbb{R}} \mathbb{C}$; a connection on E induces one on $E \otimes \mathbb{C}$ by $\mathbb{C}$-linear extension, with curvature the complexification of Ω.

The first construction extracts real invariants from a real bundle by passing to its complexification and applying the Chern classes already in hand. The odd Chern classes of a complexified real bundle vanish, so only the even ones carry information, and after a sign they are the Pontryagin classes.

Definition 11.3.1: Pontryagin classes

Let $E \to M$ be a real vector bundle, and let $E \otimes \mathbb{C}$ be its complexification, with Chern classes $c_k(E \otimes \mathbb{C}) \in H^{2k}(M;\mathbb{R})$ of definition 11.2.1. The *k-th Pontryagin class* of E is

$$p_k(E) = (-1)^k\, c_{2k}(E \otimes \mathbb{C}) \in H^{4k}(M;\mathbb{R})$$

and the *total Pontryagin class* is $p(E) = 1 + p_1(E) + p_2(E) + \cdots$. Concretely, choosing a metric connection on E with skew-symmetric curvature matrix Ω in a local orthonormal frame, $p_k(E)$ is represented by the degree-$4k$ component of the invariant polynomial

$$\det\Big(I - \frac{1}{2\pi}\Omega\Big) = 1 + p_1(E) + p_2(E) + \cdots$$

the odd-degree components vanishing.

The definition packages a real bundle's curvature invariants in the degrees where they survive. The sign $(-1)^k$ and the concentration in degree $4k$ both come from the skew-symmetry of a real curvature matrix, which the next lemma makes precise: it is what kills the odd Chern classes of $E \otimes \mathbb{C}$ and turns the alternating factor $\frac{i}{2\pi}$ of the Chern construction into the real factor $-\frac{1}{2\pi}$ recorded in the definition. The class $p_k(E)$ lives in $H^{4k}(M;\mathbb{R})$, so the first Pontryagin class p_1 appears in degree 4, the lowest degree in which a real bundle carries a nontrivial characteristic class of this kind, and a rank-r bundle has $p_k(E) = 0$ for $2k > r$ since $c_{2k}(E \otimes \mathbb{C})$ vanishes above the rank of $E \otimes \mathbb{C}$, which is r. The topological identification, that these de Rham classes are the images of the integral Pontryagin classes attached to a real bundle by algebraic topology, rides on the corresponding statement for Chern classes: the Chern–Weil classes match the topological ones by the identification recorded in section 11.2 and proved in [8] and [4].

The vanishing of the odd terms is the computational heart of the definition, and it is a one-line consequence of the reality of the bundle.

Lemma 11.3.2: Odd Chern classes of a complexification vanish

Let $E \to M$ be a real vector bundle with a metric connection, and let $E \otimes \mathbb{C}$ carry the induced connection. Then $c_{2k+1}(E \otimes \mathbb{C}) = 0$ in $H^{4k+2}(M;\mathbb{R})$ for every $k \geq 0$. Consequently the total Chern class of $E \otimes \mathbb{C}$ has only even-degree Chern terms, and the total-Pontryagin representative $\det(I - \frac{1}{2\pi}\Omega)$ of definition 11.3.1 is a polynomial in Ω^2 whose degree-$4k$ component equals $(-1)^k$ times the degree-$4k$ component of the total Chern form $\det(I + \frac{i}{2\pi}\Omega)$ of $E \otimes \mathbb{C}$, so that

$$p_k(E) = (-1)^k\, c_{2k}(E \otimes \mathbb{C})$$

holds at the level of forms, where Ω is the (skew-symmetric) curvature of E in a local orthonormal frame.

***Proof*:**
In a local orthonormal frame the curvature matrix Ω of a metric connection is skew-symmetric, $\Omega^{\mathsf{T}} = -\Omega$ (definition 10.3.1). The complexified bundle $E \otimes \mathbb{C}$ is isomorphic to its own conjugate: complex conjugation on the fibers $E_p \otimes_{\mathbb{R}} \mathbb{C}$ is a real bundle isomorphism $E \otimes \mathbb{C} \to \overline{E \otimes \mathbb{C}}$ commuting with the induced connection, since the connection is the $\mathbb{C}$-linear extension of a real

connection and conjugation fixes the real subbundle E. The curvature of the conjugate bundle is $\overline{\Omega} = \Omega$ (the entries of Ω are real-valued 2-forms in the orthonormal frame), so the Chern form $\det(I + \frac{i}{2\pi}\Omega)$ of $E \otimes \mathbb{C}$ equals the Chern form of its conjugate, which is $\det(I - \frac{i}{2\pi}\Omega)$ by conjugating the defining determinant. Hence

$$\det\Big(I + \frac{i}{2\pi}\Omega\Big) = \det\Big(I - \frac{i}{2\pi}\Omega\Big)$$

The left side is $\sum_k c_k(E \otimes \mathbb{C})$; the right side is $\sum_k (-1)^k c_k(E \otimes \mathbb{C})$, obtained by replacing Ω with $-\Omega$. Matching the two in each degree forces $c_k = (-1)^k c_k$, so $c_k(E \otimes \mathbb{C}) = 0$ whenever k is odd.

For the degree comparison, diagonalize the skew matrix Ω over $\mathbb{C}$: its eigenvalues come in pairs $\pm i\nu_1, \ldots, \pm i\nu_m$ with the ν_j real (2-forms, commuting under the wedge), where $2m = \operatorname{rank} E$ if the rank is even and one eigenvalue is 0 if it is odd. Evaluating the two determinants on these eigenvalues,

$$\det\Big(I + \frac{i}{2\pi}\Omega\Big) = \prod_j \Big(1 + \frac{i}{2\pi}(i\nu_j)\Big)\Big(1 + \frac{i}{2\pi}(-i\nu_j)\Big) = \prod_j \Big(1 - \frac{\nu_j^2}{4\pi^2}\Big)$$

while

$$\det\Big(I - \frac{1}{2\pi}\Omega\Big) = \prod_j \Big(1 - \frac{i\nu_j}{2\pi}\Big)\Big(1 + \frac{i\nu_j}{2\pi}\Big) = \prod_j \Big(1 + \frac{\nu_j^2}{4\pi^2}\Big)$$

Both are polynomials in the ν_j^2, hence sums of forms of degree $4k$, with no odd-degree Chern terms, confirming the vanishing above. Writing σ_k for the k-th elementary symmetric function of the quantities $\frac{\nu_j^2}{4\pi^2}$, the degree-$4k$ component of $\det(I + \frac{i}{2\pi}\Omega)$ is $(-1)^k\sigma_k$, and this is $c_{2k}(E \otimes \mathbb{C})$; the degree-$4k$ component of $\det(I - \frac{1}{2\pi}\Omega)$ is $+\sigma_k$, and this is $p_k(E)$ by definition 11.3.1. Therefore $p_k(E) = \sigma_k = (-1)^k\big((-1)^k\sigma_k\big) = (-1)^k c_{2k}(E \otimes \mathbb{C})$ as forms, completing the proof.

The lemma explains every feature of the Pontryagin definition. The odd Chern classes of $E \otimes \mathbb{C}$ vanish because a real bundle's complexification is conjugate-symmetric, so its characteristic form is even under $\Omega \mapsto -\Omega$; what remains is a polynomial in Ω^2, living in degrees divisible by 4. The sign $(-1)^k$ in $p_k = (-1)^k c_{2k}$ is exactly the sign that appears between the degree-$4k$ parts of the real determinant $\det(I - \frac{1}{2\pi}\Omega)$, with entries $1 + \frac{\nu_j^2}{4\pi^2}$, and the complexified Chern determinant $\det(I + \frac{i}{2\pi}\Omega)$, with entries $1 - \frac{\nu_j^2}{4\pi^2}$; it makes the total Pontryagin class the real, factor-of-i-free polynomial $\det(I - \frac{1}{2\pi}A)$ evaluated on the skew matrix $A = \Omega$. Two instances fix the picture.

Example 11.3.3: Pontryagin classes of the sphere and of a 4-manifold

Take first $E = TS^n$, the tangent bundle of the round sphere. Its complexification $TS^n \otimes \mathbb{C}$ is stably trivial: the sphere sits in $\mathbb{R}^{n+1}$ with $TS^n \oplus \nu \cong \underline{\mathbb{R}}^{n+1}$ where ν is the trivial normal line bundle, so $TS^n \oplus \underline{\mathbb{R}} \cong \underline{\mathbb{R}}^{n+1}$ is trivial, and complexifying, $(TS^n \otimes \mathbb{C}) \oplus \underline{\mathbb{C}} \cong \underline{\mathbb{C}}^{n+1}$. The Whitney sum formula proposition 11.2.3 then gives $c(TS^n \otimes \mathbb{C}) \cdot c(\underline{\mathbb{C}}) = c(\underline{\mathbb{C}}^{n+1}) = 1$, and $c(\underline{\mathbb{C}}) = 1$, so $c(TS^n \otimes \mathbb{C}) = 1$; every Chern class of the complexification vanishes, hence $p_k(TS^n) = 0$ for all $k \geq 1$. The spheres carry no Pontryagin classes.

Take next a closed oriented 4-manifold M and $E = TM$. Here $p_1(TM) \in H^4(M; \mathbb{R})$ is the only Pontryagin class, and its integral is a real number,

$$\int_M p_1(TM) = \int_M \frac{-1}{8\pi^2} \operatorname{tr}(\Omega \wedge \Omega)$$

obtained by expanding $\det(I - \frac{1}{2\pi}\Omega) = 1 - \frac{1}{8\pi^2}\mathrm{tr}(\Omega\wedge\Omega) + \cdots$ to degree 4: the degree-4 part of the determinant of $I - \frac{1}{2\pi}A$ is $\frac{1}{2}\big((\frac{1}{2\pi})^2(\mathrm{tr}A)^2 - (\frac{1}{2\pi})^2\mathrm{tr}(A^2)\big)$, and $\mathrm{tr}\Omega = 0$ for a skew matrix, leaving $-\frac{1}{8\pi^2}\mathrm{tr}(\Omega\wedge\Omega)$. The integer $\langle p_1(TM), [M]\rangle$ is the first Pontryagin number, a real image of the topological invariant of [4] that enters the signature theorem. For $M = \mathbb{C}\mathrm{P}^2$ its value is 3: from $c(T\mathbb{C}\mathrm{P}^2) = (1+u)^3 = 1 + 3u + 3u^2$ of example 11.2.7 one reads $c_1 = 3u$, $c_2 = 3u^2$, and the complexification identity $p_1(TM) = c_1(T\mathbb{C}\mathrm{P}^2)^2 - 2c_2(T\mathbb{C}\mathrm{P}^2)$ (valid because for a complex bundle F one has $F\otimes\mathbb{C} \cong F\oplus\overline{F}$, so the Whitney formula of proposition 11.2.3 gives $c(F\otimes\mathbb{C}) = c(F)c(\overline{F})$ with $c_k(\overline{F}) = (-1)^k c_k(F)$, whence $c_2(F\otimes\mathbb{C}) = 2c_2 - c_1^2$ and $p_1(F) = -c_2(F\otimes\mathbb{C}) = c_1^2 - 2c_2$) yields $p_1 = 9u^2 - 6u^2 = 3u^2$, so $\int_{\mathbb{C}\mathrm{P}^2} p_1(TM) = 3\int_{\mathbb{C}\mathrm{P}^2} u^2 = 3$, the last integral being 1 by example 11.2.7.

The second construction needs an algebraic operation with no analogue in the Chern theory: a square root of the determinant of a skew-symmetric matrix that keeps track of orientation. This is the Pfaffian. It exists only in even order, and its defining algebra is what makes the Euler class possible.

Definition 11.3.4: The Pfaffian

Let $A = (a_{ij})$ be a real skew-symmetric $2m\times 2m$ matrix, $A^{\mathsf{T}} = -A$. Associate to A the 2-form on $\mathbb{R}^{2m}$

$$\omega_A = \sum_{i<j} a_{ij}\, x_i\wedge x_j = \frac{1}{2}\sum_{i,j} a_{ij}\, x_i\wedge x_j$$

in the standard basis $x_1,\ldots,x_{2m}$ of $(\mathbb{R}^{2m})^*$. The *Pfaffian* $\mathrm{Pf}(A)$ is the real number defined by

$$\frac{1}{m!}\,\omega_A^{\wedge m} = \mathrm{Pf}(A)\, x_1\wedge\cdots\wedge x_{2m}$$

that is, $\mathrm{Pf}(A)$ is the coefficient of the volume form $x_1\wedge\cdots\wedge x_{2m}$ in $\frac{1}{m!}\omega_A^{\wedge m}$. Equivalently, in closed form,

$$\mathrm{Pf}(A) = \frac{1}{2^m m!}\sum_{\sigma\in S_{2m}} \mathrm{sgn}(\sigma)\prod_{k=1}^{m} a_{\sigma(2k-1)\,\sigma(2k)}$$

The two descriptions agree because expanding the m-fold wedge $\omega_A^{\wedge m}$ produces exactly the signed sum over permutations, the sign $\mathrm{sgn}(\sigma)$ coming from reordering the wedge factors and the prefactor $2^m m!$ correcting for the $m!$ orderings of the m pairs and the two orderings within each pair. The Pfaffian is a homogeneous polynomial of degree m in the entries of A, so under the substitution $A = \Omega$ it will produce a form of degree $2m$, the top degree on a rank-$2m$ bundle. Its point of contact with the determinant, and its transformation under a change of orthonormal frame, are the two facts that make it a characteristic-class ingredient; both are proved next.

For a 2×2 skew matrix the Pfaffian is a single entry, which is the case that reappears on surfaces.

Example 11.3.5: The Pfaffian in low order

For $m = 1$ the matrix is $A = \begin{pmatrix} 0 & a \\ -a & 0\end{pmatrix}$, and $\omega_A = a\, x_1\wedge x_2$, so $\frac{1}{1!}\omega_A^{\wedge 1} = a\, x_1\wedge x_2$ and $\mathrm{Pf}(A) = a = a_{12}$. The determinant is $\det A = a^2$, so $\mathrm{Pf}(A)^2 = a^2 = \det A$, the identity of definition 11.3.4 in its first instance. Replacing a by $-a$ negates $\mathrm{Pf}(A)$ but not $\det A$: the Pfaffian is the entry, a signed square root of the determinant, and the sign is the orientation datum the determinant discards.

For $m = 2$, writing the six independent entries $a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34}$, the wedge $\omega_A =$

$\sum_{i<j} a_{ij} x_i \wedge x_j$ squares to

$$\omega_A^{\wedge 2} = 2\big(a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23}\big)\, x_1 \wedge x_2 \wedge x_3 \wedge x_4$$

only the three complementary pairings of the four indices surviving, with signs from reordering. Dividing by $2! = 2$,

$$\mathrm{Pf}(A) = a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23}$$

a quadratic polynomial in the entries, and direct expansion of the 4×4 determinant confirms $\det A = (a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23})^2 = \mathrm{Pf}(A)^2$.

The two structural properties of the Pfaffian are the ones the definition was arranged to deliver. The first squares the Pfaffian to the determinant; the second is the transformation law that is, restricted to $SO(2m)$, the invariance that admits the Pfaffian to the Chern–Weil homomorphism.

Proposition 11.3.6: Structural properties of the Pfaffian

Let A be a real skew-symmetric $2m \times 2m$ matrix. Then:

1. $\mathrm{Pf}(A)^2 = \det A$;
2. for every real $2m \times 2m$ matrix g, $\mathrm{Pf}(g^{\mathsf{T}} A g) = \det(g)\ \mathrm{Pf}(A)$;
3. consequently Pf is invariant under $SO(2m)$: if $g \in SO(2m)$, so $g^{\mathsf{T}} g = I$ and $\det g = 1$, then $\mathrm{Pf}(g^{-1}Ag) = \mathrm{Pf}(g^{\mathsf{T}}Ag) = \mathrm{Pf}(A)$; and $g^{\mathsf{T}}Ag$ is again skew-symmetric.

***Proof*:**
The proof of (2) is the substantive one; (1) follows from it by normalization, and (3) is a specialization.

Step 1: the transformation law (2). Regard g as the matrix of a linear map on $\mathbb{R}^{2m}$, and let g^* denote the induced pullback on forms, $g^* x_i = \sum_j g_{ji}\, x_j$ (the transpose acting on the coframe). The 2-form $\omega_A = \frac{1}{2} \sum_{i,j} a_{ij} x_i \wedge x_j$ pulls back to

$$g^*\omega_A = \frac{1}{2}\sum_{i,j} a_{ij}\,(g^*x_i) \wedge (g^*x_j) = \frac{1}{2}\sum_{i,j,k,l} a_{ij}\, g_{ki} g_{lj}\, x_k \wedge x_l = \frac{1}{2}\sum_{k,l}\Big(\sum_{i,j} g_{ki} a_{ij} g_{lj}\Big) x_k \wedge x_l$$

The inner sum is the (k,l)-entry of gAg^{T}. Thus $g^*\omega_A = \omega_{gAg^{\mathsf{T}}}$. Applying this with g replaced by g^{T} (whose transpose is g) gives $(g^{\mathsf{T}})^*\omega_A = \omega_{g^{\mathsf{T}}Ag}$, the form attached to the transformed matrix. Now pull back the defining relation. Taking the m-th wedge power commutes with pullback, and pullback of the top form scales by the determinant: $g^*(x_1 \wedge \cdots \wedge x_{2m}) = \det(g)\, x_1 \wedge \cdots \wedge x_{2m}$. Apply $(g^{\mathsf{T}})^*$ to $\frac{1}{m!}\omega_A^{\wedge m} = \mathrm{Pf}(A)\, x_1 \wedge \cdots \wedge x_{2m}$:

$$\frac{1}{m!}\big((g^{\mathsf{T}})^*\omega_A\big)^{\wedge m} = \mathrm{Pf}(A)\,(g^{\mathsf{T}})^*(x_1 \wedge \cdots \wedge x_{2m})$$

The left side is $\frac{1}{m!}\omega_{g^{\mathsf{T}}Ag}^{\wedge m} = \mathrm{Pf}(g^{\mathsf{T}}Ag)\, x_1 \wedge \cdots \wedge x_{2m}$ by definition 11.3.4, and the right side is $\mathrm{Pf}(A)\det(g^{\mathsf{T}})\, x_1 \wedge \cdots \wedge x_{2m} = \mathrm{Pf}(A)\det(g)\, x_1 \wedge \cdots \wedge x_{2m}$, using $\det(g^{\mathsf{T}}) = \det(g)$. Matching the coefficients of the volume form gives

$$\mathrm{Pf}(g^{\mathsf{T}}Ag) = \det(g)\ \mathrm{Pf}(A)$$

which is (2). That $g^\mathsf{T} A g$ is skew-symmetric is immediate: $(g^\mathsf{T} A g)^\mathsf{T} = g^\mathsf{T} A^\mathsf{T} g = -g^\mathsf{T} A g$.

Step 2: the square is the determinant (1). Every real skew-symmetric $2m \times 2m$ matrix can be brought to a standard block form by an orthogonal change of basis: there is $g \in SO(2m)$ with

$$g^\mathsf{T} A g = \begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_m \end{pmatrix}, \qquad B_k = \begin{pmatrix} 0 & \lambda_k \\ -\lambda_k & 0 \end{pmatrix}$$

a block-diagonal matrix of 2×2 skew blocks, the real normal form of a skew-symmetric operator on a Euclidean space ([6] for the spectral normal form; it is the real analogue of unitary diagonalization). For this block matrix $\Lambda = g^\mathsf{T} A g$, the associated form is $\omega_\Lambda = \sum_k \lambda_k \, x_{2k-1} \wedge x_{2k}$, whose m-th power is $\omega_\Lambda^{\wedge m} = m! \, \lambda_1 \cdots \lambda_m \, x_1 \wedge \cdots \wedge x_{2m}$ (only the term taking one factor from each block survives, and there are $m!$ orderings), so $\operatorname{Pf}(\Lambda) = \lambda_1 \cdots \lambda_m$. On the other hand $\det \Lambda = \prod_k \det B_k = \prod_k \lambda_k^2 = (\lambda_1 \cdots \lambda_m)^2$, so $\operatorname{Pf}(\Lambda)^2 = \det \Lambda$ for the normal form. Transport this to A by (2) and the multiplicativity of the determinant: $\operatorname{Pf}(A)^2 = \big(\det(g)^{-1} \operatorname{Pf}(g^\mathsf{T} A g)\big)^2 = \det(g)^{-2} \operatorname{Pf}(\Lambda)^2$, and since $g \in SO(2m)$ has $\det g = 1$, this is $\operatorname{Pf}(\Lambda)^2 = \det \Lambda = \det(g^\mathsf{T} A g) = \det(g)^2 \det A = \det A$. Hence $\operatorname{Pf}(A)^2 = \det A$, which is (1).

Step 3: $SO(2m)$-invariance (3). For $g \in SO(2m)$, $g^{-1} = g^\mathsf{T}$ and $\det g = 1$, so (2) reads $\operatorname{Pf}(g^{-1} A g) = \operatorname{Pf}(g^\mathsf{T} A g) = \det(g) \operatorname{Pf}(A) = \operatorname{Pf}(A)$. Thus Pf is constant on $SO(2m)$-conjugacy classes of skew matrices, an Ad-invariant polynomial on $\mathfrak{so}(2m)$, as we sought to show.

The three properties partition the Pfaffian's role cleanly. Property (1) makes it a square root of the determinant, so that whatever the Pfaffian of the curvature integrates to, its square is a Pontryagin-type quantity; property (2) shows the square root is not orientation-blind, since it scales by $\det(g)$ rather than $|\det g|$, which is exactly why the Euler class requires an orientation while the Pontryagin classes do not. Property (3), the $SO(2m)$-invariance, is the admission ticket: the Chern–Weil homomorphism of section 11.1 accepts any Ad-invariant polynomial, and for a metric connection the curvature takes values in $\mathfrak{so}(2m)$ with structure group reducing to $SO(2m)$ once an orientation is fixed, so the Pfaffian is precisely such a polynomial. Feeding the curvature to it produces the Euler form.

Definition 11.3.7: The Euler form and Euler class

Let $E \to M$ be an oriented real vector bundle of rank $2m$ with a metric connection ∇, and let $\Omega = (\Omega^i_j)$ be its curvature matrix in a positively oriented local orthonormal frame, a skew-symmetric matrix of 2-forms. The *Euler form* of (E, ∇) is the $2m$-form

$$e(E, \nabla) = \frac{1}{(2\pi)^m} \operatorname{Pf}(\Omega)$$

where $\operatorname{Pf}(\Omega)$ is the Pfaffian of definition 11.3.4 applied entrywise to the matrix Ω of 2-forms, the products in the Pfaffian formula being wedge products. The Euler form is a closed $2m$-form, and its de Rham class

$$e(E) = \big[e(E, \nabla)\big] \in H^{2m}(M; \mathbb{R})$$

the *Euler class* of the oriented bundle E, is independent of the choice of metric connection.

The definition asserts three things beyond the formula: that the Pfaffian of the curvature is a globally defined form, that it is closed, and that its class is connection-independent. All three are instances of the Chern–Weil theory of section 11.1 once the Pfaffian is recognized as an $SO(2m)$-invariant polynomial, which is the content of proposition 11.3.6(3). The proposition that follows records the verification, so that the Euler form takes its place beside the Chern and Pontryagin forms as a bona fide characteristic form. The factor $(2\pi)^{-m}$ is the normalization that will make the integral of $e(TM)$ come out to the integer $\chi(M)$ rather than a multiple of it, exactly as the $\frac{i}{2\pi}$ of definition 10.4.2 normalized the first Chern form; the case $m = 1$ recovers that surface normalization below.

Proposition 11.3.8: The Euler form is a well-defined characteristic form

The Euler form $e(E, \nabla)$ of definition 11.3.7 is a globally defined $2m$-form on M, is closed, and has a de Rham class $e(E) \in H^{2m}(M; \mathbb{R})$ independent of the metric connection ∇. Reversing the orientation of E changes $e(E)$ to $-e(E)$.

***Proof*:**
Step 1: global definition. Under a change of positively oriented orthonormal frame $\tilde{e} = e\,g$, the transition function g takes values in $SO(2m)$: it is orthogonal because both frames are orthonormal, and has $\det g = 1$ because both are positively oriented. The curvature matrix transforms by $\tilde{\Omega} = g^{-1}\Omega g$ (proposition 10.2.6), so $\mathrm{Pf}(\tilde{\Omega}) = \mathrm{Pf}(g^{-1}\Omega g) = \mathrm{Pf}(\Omega)$ by the $SO(2m)$-invariance of proposition 11.3.6(3), the invariance holding verbatim for a matrix of commuting 2-forms since the Pfaffian is a polynomial and 2-forms commute under the wedge. The locally defined forms $\mathrm{Pf}(\Omega)$ therefore agree on overlaps and patch to a single global $2m$-form; this is the general mechanism of definition 11.1.3, applied to the invariant polynomial Pf on $\mathfrak{so}(2m)$.

Step 2: closed, connection-independent class. Since Pf is an Ad-invariant polynomial homogeneous of degree m on $\mathfrak{so}(2m) \subseteq \mathfrak{gl}_{2m}$, the Chern–Weil form $\mathrm{Pf}(\Omega)$ is closed by theorem 11.1.6, whose proof uses only the Bianchi identity $d^{\nabla}F^{\nabla} = 0$ and the infinitesimal invariance of the polynomial, both available here. Its de Rham class is independent of the connection by theorem 11.1.7, whose transgression argument applies to any invariant polynomial and produces, for two metric connections ∇_0, ∇_1, a form TPf with $\mathrm{Pf}(\Omega_1) - \mathrm{Pf}(\Omega_0) = d\,\mathrm{TPf}$; the affine path $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ of metric connections stays within metric connections since metric compatibility is an affine condition, so the transgression stays within the skew-symmetric setting and its output is a global form on M. Multiplying by the constant $(2\pi)^{-m}$ preserves closedness and the class, so $e(E, \nabla)$ is closed with connection-independent class $e(E)$.

Step 3: orientation reversal. Reversing the orientation of E replaces a positively oriented orthonormal frame e by one differing by an orientation-reversing orthogonal transition, that is, by $g \in O(2m)$ with $\det g = -1$. The transformation law proposition 11.3.6(2) gives $\mathrm{Pf}(g^{\mathsf{T}}\Omega g) = \det(g)\,\mathrm{Pf}(\Omega) = -\,\mathrm{Pf}(\Omega)$, and since for an orthogonal g one has $g^{\mathsf{T}} = g^{-1}$ so $g^{\mathsf{T}}\Omega g = g^{-1}\Omega g = \tilde{\Omega}$, the Euler form changes sign. Passing to classes, $e(E) \mapsto -e(E)$ under orientation reversal, completing the proof.

The Euler class is thus the characteristic class of an oriented even-rank real bundle that the Pfaffian, and only the Pfaffian, produces. Its dependence on the orientation, established in Step 3, is what distinguishes it from every class built from the determinant: the determinant is orientation-blind, so the Chern and Pontryagin classes are defined for unoriented bundles, whereas the Euler class flips sign with the orientation and requires one to be fixed. This is the geometric fingerprint of the square root. The Euler class is also the primary invariant among these: its square is the top Pontryagin class, and for an even-rank bundle it refines the determinant-based invariants by remembering a sign the determinant forgot. The precise relations are the next proposition.

The relation to the determinant-based classes runs through the identity $\mathrm{Pf}^2 = \det$, promoted from matrices to characteristic forms, together with the way a complex bundle sits inside its underlying real bundle.

Proposition 11.3.9: The Euler class squares to the top Pontryagin class; the complex case

Let $E \to M$ be an oriented real vector bundle of rank $2m$ with a metric connection.

1. The square of the Euler class is the top Pontryagin class:

$$e(E)^2 = p_m(E) \in H^{4m}(M; \mathbb{R})$$

2. If E is a complex vector bundle of rank m, viewed as an oriented real bundle of rank $2m$ by its underlying real structure and the complex orientation, then the Euler class equals the top Chern class:

$$e(E) = c_m(E) \in H^{2m}(M; \mathbb{R})$$

Proof:

1. Work in a positively oriented orthonormal frame with skew-symmetric curvature matrix Ω. By proposition 11.3.6(1), promoted from scalar matrices to matrices of commuting 2-forms (the identity $\mathrm{Pf}(A)^2 = \det A$ is a polynomial identity in the entries of A, hence holds when the entries are 2-forms, which commute under the wedge),

$$\mathrm{Pf}(\Omega) \wedge \mathrm{Pf}(\Omega) = \det \Omega$$

Multiplying by $(2\pi)^{-2m}$, the left side is $e(E, \nabla) \wedge e(E, \nabla)$, a representative of $e(E)^2$. For the right side, $\det \Omega$ is the top-degree ($4m$) component of the total-Pontryagin representative: expanding $\det(I - \frac{1}{2\pi}\Omega)$, the degree-$4m$ term is $\det(-\frac{1}{2\pi}\Omega) = (-\frac{1}{2\pi})^{2m} \det \Omega = (2\pi)^{-2m} \det \Omega$, and this top component represents $p_m(E)$ by definition 11.3.1, since E has rank $2m$ and p_m is the highest nonvanishing Pontryagin class. Therefore

$$e(E, \nabla) \wedge e(E, \nabla) = (2\pi)^{-2m}\, \mathrm{Pf}(\Omega)^2 = (2\pi)^{-2m} \det \Omega$$

represents both $e(E)^2$ and $p_m(E)$, so $e(E)^2 = p_m(E)$ in $H^{4m}(M; \mathbb{R})$.

2. Let E be a complex bundle of rank m with a Hermitian metric and the Chern connection (theorem 10.3.5), curvature $F \in \Omega^2(M; \mathrm{End}_{\mathbb{C}} E)$ of type $(1,1)$. The top Chern class is the top-degree component of $\det(I + \frac{i}{2\pi} F)$, which for the $m \times m$ matrix F is $c_m(E) = \det(\frac{i}{2\pi} F) = (\frac{i}{2\pi})^m \det F$. At a point, diagonalize the skew-Hermitian curvature: F has purely imaginary eigenvalues $i\mu_1, \ldots, i\mu_m$ with the μ_k real (as 2-forms, commuting under the wedge), so $\det F = \prod_k (i\mu_k) = i^m \prod_k \mu_k$ and

$$c_m(E, \nabla) = \left(\frac{i}{2\pi}\right)^m i^m \prod_k \mu_k = \frac{i^{2m}}{(2\pi)^m} \prod_k \mu_k = \frac{(-1)^m}{(2\pi)^m} \prod_k \mu_k$$

Regard E as a real bundle $E_{\mathbb{R}}$ of rank $2m$: the complex operator F acts on the underlying $\mathbb{R}^{2m}$ as a real $2m \times 2m$ operator $F_{\mathbb{R}}$, and the Hermitian metric gives a real inner product for which the Chern connection is a metric connection, so $F_{\mathbb{R}}$ is skew-symmetric in a real orthonormal frame. The complex orientation of $E_{\mathbb{R}}$ is the one whose positively oriented orthonormal frame is $(f_1, Jf_1, \ldots, f_m, Jf_m)$ for a Hermitian frame $(f_1, \ldots, f_m)$, J the complex structure. On the complex line spanned by an eigenvector f_k with $Ff_k = i\mu_k f_k$, the real operator $F_{\mathbb{R}}$ acts on the ordered real basis (f_k, Jf_k) by $F_{\mathbb{R}} f_k = \mu_k(Jf_k)$ and $F_{\mathbb{R}}(Jf_k) = -\mu_k f_k$, so its 2×2 block is $\begin{pmatrix} 0 & -\mu_k \\ \mu_k & 0 \end{pmatrix}$, with upper-right entry $-\mu_k$. By the block computation of proposition 11.3.6, Step 2, the Pfaffian of the block-diagonal $F_{\mathbb{R}}$ is the product of the upper-right entries,

$$\operatorname{Pf}(F_{\mathbb{R}}) = \prod_k (-\mu_k) = (-1)^m \prod_k \mu_k$$

Therefore the Euler form is

$$e(E_{\mathbb{R}}, \nabla) = \frac{1}{(2\pi)^m} \operatorname{Pf}(F_{\mathbb{R}}) = \frac{(-1)^m}{(2\pi)^m} \prod_k \mu_k = c_m(E, \nabla)$$

the two representatives agreeing. Hence $e(E) = c_m(E)$ in $H^{2m}(M; \mathbb{R})$, as claimed.

This establishes both relations, completing the proof.

The proposition places the Euler class exactly. Squaring it recovers the top Pontryagin class, so the Euler class is a square root, at the level of cohomology, of a Pontryagin invariant; the orientation dependence is the ambiguity of that root, and $e(E)$ resolves it. For a complex bundle the two constructions coincide, $e(E_{\mathbb{R}}) = c_m(E)$, so the top Chern class of a rank-m complex bundle is the Euler class of its underlying oriented rank-$2m$ real bundle. This identification is what closes the computation of example 11.2.7: the integral $\int_{\mathbb{CP}^n} c_n(T\mathbb{CP}^n) = n + 1$ found there is simultaneously $\int_{\mathbb{CP}^n} e(T\mathbb{CP}^n)$, the Euler characteristic $\chi(\mathbb{CP}^n) = n + 1$, so the Gauss–Bonnet–Chern statement for $\mathbb{CP}^n$ is already visible on the complex side. On the tangent bundle of a general even-dimensional manifold there is no complex structure to lean on, and the Euler form must be handled through the Pfaffian directly; the surface case is where that computation is transparent, and it is the cross-check against Part I.

Example 11.3.10: The Euler form of a surface and Gauss–Bonnet

Let (M, g) be a closed oriented Riemannian surface, so $E = TM$ has rank $2m = 2$ with $m = 1$, and take the Levi-Civita connection. Fix a positively oriented local orthonormal frame (e_1, e_2) with dual coframe (θ^1, θ^2), and write ω_{12} for the connection 1-form of proposition 8.1.3, so that $\nabla_X e_1 = \omega_{12}(X)\, e_2$ and $\nabla_X e_2 = -\omega_{12}(X)\, e_1$. In the bundle notation of definition 10.2.1, where $\nabla e_j = \omega^i_j\, e_i$, this reads $\omega^2_1 = \omega_{12}$ and $\omega^1_2 = -\omega_{12}$, and the connection matrix $\omega = (\omega^i_j)$ is skew-symmetric with vanishing diagonal. The curvature matrix is

$$\Omega = \begin{pmatrix} 0 & \Omega^1_2 \\ \Omega^2_1 & 0 \end{pmatrix}, \qquad \Omega^1_2 = d\omega^1_2 = -d\omega_{12}, \qquad \Omega^2_1 = -\Omega^1_2$$

the diagonal vanishing by skew-symmetry and each off-diagonal entry reducing to a single exterior derivative because the quadratic term $\omega \wedge \omega$ of the structure equation drops out in the rank-two orthonormal case (theorem 10.2.4, with the computation of example 10.2.5). The Pfaffian of a

2×2 skew matrix is its upper-right entry (example 11.3.5), so

$$\mathrm{Pf}(\Omega) = \Omega^1_2 = -d\omega_{12}$$

By the structure equation of proposition 8.1.3(iii), $d\omega_{12} = -K\,dA$, where K is the Gaussian curvature and $dA = \theta^1 \wedge \theta^2$ is the area form. Therefore

$$\mathrm{Pf}(\Omega) = -d\omega_{12} = K\,dA$$

The sign is confirmed by evaluating on the frame directly: $\mathrm{Pf}(\Omega)(e_1, e_2) = \Omega^1_2(e_1, e_2) = \langle F^\nabla(e_1, e_2)e_2, e_1\rangle = \langle R(e_1, e_2)e_2, e_1\rangle = \mathrm{Riem}(e_1, e_2, e_2, e_1) = K$, the last equality being the sectional-curvature identification $K = \mathrm{Riem}(e_1, e_2, e_2, e_1)$ of proposition 8.1.3, while $(K\,dA)(e_1, e_2) = K$; the two 2-forms agree on the positively oriented frame and hence everywhere. The Euler form is therefore

$$e(TM, \nabla) = \frac{1}{(2\pi)^1}\,\mathrm{Pf}(\Omega) = \frac{1}{2\pi}\,K\,dA$$

the mandatory cross-check: the Pfaffian of the Riemann curvature of a surface is the Gaussian curvature times the area form, and the Euler form is $\frac{1}{2\pi}K\,dA$. Integrating over the closed surface and applying the Gauss–Bonnet theorem theorem 8.1.6,

$$\int_M e(TM) = \frac{1}{2\pi}\int_M K\,dA = \frac{1}{2\pi}\cdot 2\pi\chi(M) = \chi(M)$$

so the integral of the Euler form is the Euler characteristic, which is the $m = 1$ case of the Gauss–Bonnet–Chern theorem. The general even-dimensional statement, that $\int_M e(TM) = \chi(M)$ for every closed oriented Riemannian M^{2m}, is proved in section 11.4; here the Pfaffian normalization $(2\pi)^{-m}$ is confirmed against Part I in the one dimension where the Pfaffian is a single entry and the answer is already known.

11.4 The Gauss–Bonnet–Chern Theorem; Positivity

The chapter has built, out of the curvature of a connection, a supply of closed forms whose de Rham classes see only the bundle. For the tangent bundle of an oriented Riemannian manifold of even dimension one of these forms carries a name and a debt. The Euler form $e(TM) = \frac{1}{(2\pi)^m}\,\mathrm{Pf}(\Omega)$ of definition 11.3.7 is the Chern–Weil form of the $SO(2m)$-invariant Pfaffian, closed and with connection-independent class by the general theory. What remains is to compute its integral, and the answer is the one promised in Part I: the Euler characteristic of the manifold. This section proves the generalized Gauss–Bonnet–Chern theorem, discharging the statement left unproved in section 8.2, and reads off its two-dimensional and low-dimensional consequences. The proof is Chern's: a vector field with isolated zeros transgresses the Euler form to an exact form away from its zeros, and the topological count of those zeros is the Euler characteristic by the Poincaré–Hopf theorem, the single topological input the argument borrows from [4].

The shape of the argument is worth stating before the details. The integral $\int_M e(TM)$ is metric-independent, so it may be evaluated with any convenient metric and connection. A generic vector field V has finitely many zeros; on the complement the normalized field $V/|V|$ is a section of the unit tangent sphere bundle, and pulling the Euler form back along it exhibits $e(TM)$ as an exact form there. Integrating and passing to the limit as small balls around the zeros shrink turns the integral into a sum of local contributions, one per zero, each equal to the index of V at that zero. Poincaré–Hopf identifies the index sum with $\chi(M)$, and the theorem follows. Every step except the last is a computation with the Euler form; the last is the borrowed topological fact.

Throughout, (M,g) is a closed oriented Riemannian manifold of even dimension $2m$, with Levi-Civita connection ∇ and curvature $\Omega \in \Omega^2(M;\mathfrak{so}(2m))$ skew-symmetric in an oriented orthonormal frame, and $e(TM) = \frac{1}{(2\pi)^m}\operatorname{Pf}(\Omega)$ the Euler form of definition 11.3.7. Its class $[e(TM)] \in H^{2m}(M;\mathbb{R})$ is independent of ∇ by theorem 11.1.7, and the Pfaffian satisfies $\operatorname{Pf}(\Omega)^2 = \det\Omega$ and the $SO(2m)$-invariance $\operatorname{Pf}(A^{\mathsf{T}}\Omega A) = \det(A)\operatorname{Pf}(\Omega)$ of definition 11.3.4. The Euler characteristic $\chi(M) = \sum_i(-1)^i \dim H^i(M;\mathbb{R})$ is the topological invariant of [4].

Two structural facts about the Euler form are needed for the transgression, and both are formal consequences of what the chapter has already established. The first is that the Euler form of an odd-rank orthogonal bundle vanishes, which localizes the argument to even fibers. The second is that on the total space of the tangent sphere bundle the pulled-back Euler form is exact, with a canonical primitive. The second is the technical heart, so it is isolated as a lemma.

Lemma 11.4.1: Transgression of the Euler form

Let (M^{2m},g) be an oriented Riemannian manifold with Levi-Civita connection, Euler form $e(TM) = \frac{1}{(2\pi)^m}\operatorname{Pf}(\Omega)$, and let $\pi\colon SM \to M$ be the unit tangent sphere bundle, whose fiber over p is the unit sphere $\{v \in T_pM\colon\ |v|_g = 1\}$. There is a smooth $(2m-1)$-form Φ on SM, natural in (M,g) and depending only on the connection, with

$$d\Phi = \pi^* e(TM)$$

and with the property that the integral of Φ over a fiber sphere is 1: for each $p \in M$,

$$\int_{\pi^{-1}(p)} \Phi = 1$$

The form Φ is the *transgression form* (or *angular form*) of the Euler form.

***Proof*:**
The construction is the Chern–Weil transgression of theorem 11.1.7 carried out on the sphere bundle, where the tautological unit section supplies a canonical splitting.

Step 1: the tautological section trivializes a summand. Over the total space SM there is a tautological unit vector field: at a point $(p,v) \in SM$, the vector $v \in T_pM$ is a distinguished unit element of the pulled-back bundle π^*TM over SM. Thus π^*TM carries a nowhere-zero section u, namely $u(p,v) = v$, and splits g-orthogonally as

$$\pi^*TM = \mathbb{R}\,u \oplus u^\perp$$

a trivial real line bundle spanned by u and its orthogonal complement $u^\perp$, an oriented Riemannian bundle of rank $2m-1$. The Levi-Civita connection pulls back to a metric connection on π^*TM, and its orthogonal projection to $u^\perp$ is a metric connection there; write $\Omega^\perp$ for the induced skew curvature on $u^\perp$.

Step 2: the pulled-back Euler form comes from an odd-rank bundle. The pullback $\pi^*e(TM) = \frac{1}{(2\pi)^m}\operatorname{Pf}(\pi^*\Omega)$ is the Euler form of π^*TM with the pulled-back connection, computed by definition 11.3.7. The Euler form is multiplicative on orthogonal direct sums of even-rank oriented bundles, and the Euler form of an odd-rank oriented Riemannian bundle vanishes, because the Pfaffian is defined only for even-order skew matrices and the Chern–Weil form built from the $SO(2m-1)$-invariant polynomials of odd degree in a skew $(2m-1)\times(2m-1)$ matrix has no top-degree component of the right parity. The splitting of Step 1 has one factor a trivial line (flat, Euler form 1 in degree 0) and the other the odd-rank bundle $u^\perp$; the naive product would place all of $\pi^*e(TM)$ in the $u^\perp$ factor, whose own Euler class vanishes. Hence $[\pi^*e(TM)] = 0$ in $H^{2m}(SM;\mathbb{R})$: the pulled-back Euler form is exact.

Step 3: a canonical primitive by transgression along the section. Exactness alone gives a primitive; to get a canonical one, apply the transgression of theorem 11.1.7 to the affine family of connections interpolating between the pulled-back Levi-Civita connection $\nabla^0 = \pi^*\nabla$ and the connection ∇^1 that is the pulled-back connection on $u^\perp$ extended by the flat connection du on the line $\mathbb{R}u$. Both connections are metric, so both compute the same class $[\pi^*e(TM)]$; the transgression form

$$\Phi = \int_0^1 \mathrm{TP}_t \, dt, \qquad \nabla^t = (1-t)\nabla^0 + t\nabla^1$$

is a $(2m-1)$-form on SM with $d\Phi = \pi^*e(TM)_{\nabla^0} - \pi^*e(TM)_{\nabla^1}$. Since ∇^1 splits off the odd-rank factor $u^\perp$, its Euler form vanishes by Step 2, so $\pi^*e(TM)_{\nabla^1} = 0$ and $d\Phi = \pi^*e(TM)$. The form is built pointwise from the curvature and the tautological section u, so it is natural in (M, g) and depends only on ∇.

Step 4: the fiber integral. It remains to compute $\int_{\pi^{-1}(p)} \Phi$. On the fiber $\pi^{-1}(p) \cong S^{2m-1}$, the base curvature is frozen and the only variation is angular; the transgression form restricts to the standard rotation-invariant $(2m-1)$-form on the sphere normalized so that its integral is the ratio of the sphere's volume to the round volume. By construction the normalization $\frac{1}{(2\pi)^m}$ of the Euler form is chosen precisely so that this fiber integral is 1: the restriction of Φ to a fiber is the volume form of the unit round $(2m-1)$-sphere divided by its total volume $\mathrm{vol}(S^{2m-1})$, which integrates to 1. This is the same normalization that made $\int_{\mathbb{CP}^1} c_1(\mathcal{O}(1)) = 1$ in theorem 10.4.5: the factor of 2π per complex degree is what turns the angular integral into the integer 1. Thus $\int_{\pi^{-1}(p)} \Phi = 1$ for every p, as we sought to show.

The transgression form is the bundle-theoretic version of the fact that on a surface the curvature integral converts, by the structure equation, into a boundary integral of the connection 1-form; there ω_{12} played the role of Φ, and $d\omega_{12} = -K\,dA$ played the role of $d\Phi = \pi^*e(TM)$. The angular form encodes how the tangent direction rotates as one moves around a loop, and its fiber integral 1 is the statement that going once around the sphere fiber accumulates exactly one unit of Euler form. With the transgression in hand, a vector field turns the top-dimensional integral over M into a sum of contributions localized at the zeros.

The local contribution at a zero is the index, defined next in the form the proof uses.

Definition 11.4.2: Index of a vector field at a zero

Let V be a smooth vector field on M^{2m} with an isolated zero at p, so that $V \neq 0$ on a punctured neighborhood of p. Choose an oriented chart centered at p identifying a neighborhood with a ball in $\mathbb{R}^{2m}$, and let S_ε be a small sphere of radius ε about p. The *index* $\operatorname{ind}_p(V) \in \mathbb{Z}$ is the degree of the Gauss map

$$\frac{V}{|V|} : S_\varepsilon \longrightarrow S^{2m-1}$$

the mapping degree of the normalized field restricted to S_ε into the unit sphere. The zero is *nondegenerate* if the differential $DV_p \colon T_pM \to T_pM$ is invertible, in which case $\operatorname{ind}_p(V) = \operatorname{sign}\det(DV_p) = \pm 1$.

The index counts, with signs, how many times the direction of V winds around the sphere of directions as one circles the zero. A source or a sink has index $+1$ (the field points uniformly outward or inward, a degree-one map), a nondegenerate saddle has index -1. The nondegenerate case is the one the proof uses, because a generic vector field has only nondegenerate zeros, finitely many on a closed manifold, and the sum of their indices is a topological invariant. This last fact is the Poincaré–Hopf theorem, and it is the one ingredient proved elsewhere.

Theorem 11.4.3: The generalized Gauss–Bonnet–Chern theorem

Let (M, g) be a closed oriented Riemannian manifold of even dimension $2m$, with Levi-Civita connection ∇ and curvature Ω. Then the Euler form of definition 11.3.7 integrates to the Euler characteristic:

$$\int_M e(TM) = \chi(M), \qquad e(TM) = \frac{1}{(2\pi)^m}\operatorname{Pf}(\Omega)$$

The integral is independent of the metric g. In dimension $2m = 2$ the identity reads $\frac{1}{2\pi}\int_M K\,dA = \chi(M)$, recovering the global Gauss–Bonnet theorem theorem 8.1.6; and the identity is precisely the statement of theorem 8.2.1, whose proof was deferred in section 8.2.

***Proof*:**
Metric-independence of the integral is immediate: by theorem 11.1.7 the class $[e(TM)] \in H^{2m}(M;\mathbb{R})$ does not depend on the connection, and two metrics give Euler forms differing by an exact form, whose integral over the closed manifold M vanishes by Stokes' theorem ([6]). It therefore suffices to prove $\int_M e(TM) = \chi(M)$ for one convenient metric and connection, and the value found is forced on all others. Fix (g, ∇) as in the statement.

Step 1: choose a vector field with isolated nondegenerate zeros. A closed manifold admits a smooth vector field V whose zeros are isolated and nondegenerate; the gradient of a Morse function is one such field, its zeros the critical points of the function, each nondegenerate by the Morse condition ([4]). Let $Z = \{p_1, \ldots, p_N\}$ be the (finite) zero set. On the open set $M \setminus Z$ the normalized field

$$s = \frac{V}{|V|_g} : M \setminus Z \longrightarrow SM$$

is a smooth section of the unit tangent sphere bundle $\pi\colon SM \to M$ of lemma 11.4.1, satisfying $\pi \circ s = \mathrm{id}$ on $M \setminus Z$.

Step 2: pull the transgression back along the section. Let Φ be the transgression form of lemma 11.4.1, with $d\Phi = \pi^* e(TM)$. On $M \setminus Z$, the pullback $s^*\Phi$ is a $(2m-1)$-form, and since $\pi \circ s = \mathrm{id}$,

$$d(s^*\Phi) = s^*(d\Phi) = s^*\pi^* e(TM) = (\pi \circ s)^* e(TM) = e(TM)$$

on $M \setminus Z$. Thus on the complement of the zeros the Euler form is exact, $e(TM) = d(s^*\Phi)$, with the explicit primitive $s^*\Phi$. This is where the transgression pays off: away from the zeros $s^*\Phi$ is a globally defined primitive of $e(TM)$, so Stokes' theorem can move the integral to the boundary of small balls.

Step 3: excise the zeros and apply Stokes. For small $\varepsilon > 0$ let $B_\varepsilon(p_j)$ be disjoint geodesic balls about the zeros, with boundary spheres $S_\varepsilon(p_j) = \partial B_\varepsilon(p_j)$, and set $M_\varepsilon = M \setminus \bigcup_j B_\varepsilon(p_j)$. On $M_\varepsilon \subset M \setminus Z$ the Euler form is exact by Step 2, and M_ε is a compact manifold with boundary $\partial M_\varepsilon = -\bigcup_j S_\varepsilon(p_j)$ (the sphere orientations induced as the boundary of M_ε are opposite to their orientations as boundaries of the balls). Stokes' theorem gives

$$\int_{M_\varepsilon} e(TM) = \int_{M_\varepsilon} d(s^*\Phi) = \int_{\partial M_\varepsilon} s^*\Phi = -\sum_{j=1}^{N} \int_{S_\varepsilon(p_j)} s^*\Phi$$

The Euler form $e(TM)$ is smooth across the zeros, so as $\varepsilon \to 0$ the left side converges to $\int_M e(TM)$ (the excised balls have volume $O(\varepsilon^{2m}) \to 0$). Therefore

$$\int_M e(TM) = -\lim_{\varepsilon \to 0} \sum_{j=1}^{N} \int_{S_\varepsilon(p_j)} s^*\Phi$$

Step 4: each boundary integral is an index. Fix a zero $p = p_j$ and work in an oriented chart centered at p. Over the chart the sphere bundle is trivialized, $SM|_U \cong U \times S^{2m-1}$, and under this trivialization the section $s = V/|V|$ over the small sphere $S_\varepsilon(p)$ is, up to a base-point factor that contributes nothing in the limit, the Gauss map $g_\varepsilon = V/|V| : S_\varepsilon(p) \to S^{2m-1}$ of definition 11.4.2. As $\varepsilon \to 0$ the restriction of Φ to the sphere fiber dominates, because the base variation of Φ over the shrinking sphere $S_\varepsilon(p)$ is $O(\varepsilon)$ while the fiber term is order one, so in the limit the integral is computed by the degree of g_ε into the fiber sphere. Two orientations of the model sphere S^{2m-1} meet here and must be compared. The degree $\deg(g_\varepsilon) = \mathrm{ind}_p(V)$ of definition 11.4.2 is taken with $S_\varepsilon(p) = \partial B_\varepsilon(p)$ carrying its outward boundary orientation as the sphere of directions; the fiber integral $\int_{\pi^{-1}(p)} \Phi = 1$ of lemma 11.4.1 is taken with $\pi^{-1}(p)$ carrying its orientation as a fiber of SM. These two orientations of S^{2m-1} are opposite, because the fiber orientation of $\pi^{-1}(p)$ (fixed by the orientation of M together with the coorientation of the fiber in SM) is the reverse of the outward-normal boundary orientation of the direction sphere $\partial B_\varepsilon(p)$; write $\varepsilon_0 = -1$ for this comparison sign. The degree formula, applied with the fiber orientation on the target, therefore reads

$$\lim_{\varepsilon \to 0} \int_{S_\varepsilon(p)} s^*\Phi = \int_{S_\varepsilon(p)} g_\varepsilon^*\left(\Phi|_{\mathrm{fiber}}\right) = \varepsilon_0 \deg(g_\varepsilon) \int_{\pi^{-1}(p)} \Phi = -\,\mathrm{ind}_p(V) \cdot 1$$

using that the pullback of a form under a map of closed oriented manifolds of the same dimension multiplies its integral by the degree, that $\deg(g_\varepsilon) = \mathrm{ind}_p(V)$ by definition 11.4.2, and that $\int_{\pi^{-1}(p)} \Phi = 1$ by the fiber-integral normalization of lemma 11.4.1. The comparison sign $\varepsilon_0 = -1$ is the boundary-versus-fiber orientation discrepancy just described, located entirely in the identification of the direction sphere $\partial B_\varepsilon(p)$ with the fiber $\pi^{-1}(p)$; it is not the orientation sign of Step 3, which is the separate explicit $-$ already carried in front of the sum on the display for $\int_M e(TM)$. Consequently

$$-\lim_{\varepsilon\to 0}\int_{S_\varepsilon(p)} s^*\Phi = \mathrm{ind}_p(V)$$

Summing over the zeros,

$$\int_M e(TM) = \sum_{j=1}^{N} \mathrm{ind}_{p_j}(V)$$

the total index of the vector field V.

Step 5: the index sum is $\chi(M)$. The right side is the sum of the indices of V over its zeros. By the Poincaré–Hopf theorem, the total index of a vector field with isolated zeros on a closed manifold equals the Euler characteristic,

$$\sum_{j=1}^{N} \mathrm{ind}_{p_j}(V) = \chi(M)$$

independent of the field chosen ([4]). This is the one topological input the argument borrows; everything preceding it is a computation with the Euler form and its transgression. Combining the last two displays,

$$\int_M e(TM) = \chi(M)$$

Step 6: the low-dimensional reduction. In dimension $2m = 2$ the Euler form is $e(TM) = \frac{1}{2\pi}\,\mathrm{Pf}(\Omega) = \frac{1}{2\pi} K\,dA$, the identity $\mathrm{Pf}(\Omega) = \Omega^1_2 = K\,dA$ being the surface structure equation of example 11.3.10. The theorem then reads $\frac{1}{2\pi}\int_M K\,dA = \chi(M)$, that is $\int_M K\,dA = 2\pi\chi(M)$, which is the global Gauss–Bonnet theorem theorem 8.1.6. In every even dimension the identity $\int_M e(TM) = \chi(M)$ is the statement of theorem 8.2.1 of section 8.2, whose Euler form $e(\nabla)$ is the Pfaffian curvature form named there; the deferral of that section is thereby discharged, completing the proof.

The theorem closes the loop the chapter opened. A curvature integral, a soft geometric quantity that the pointwise construction gives no reason to be special, is forced to equal an integer read off from the topology of M with no metric in sight. The mechanism is the one visible in the proof: the Euler form represents a fixed cohomology class, the Euler class of the tangent bundle, and pairing that class against the fundamental class of M produces the Euler characteristic. The vector field and its zeros are a device for computing that pairing, converting the top-degree integral into a finite sum of local indices; the transgression form is what makes the conversion exact, exhibiting the Euler form as d of a global primitive away from the zeros. The two-dimensional case is not an analogy but the literal specialization: the Pfaffian of a 2×2 skew matrix is its single upper entry, the surface structure equation turns that entry into $K\,dA$, and the theorem becomes $\int_M K\,dA = 2\pi\chi(M)$.

Metric-independence is the substance and is worth restating in its sharpest form. On a closed oriented M^{2m}, deforming the metric arbitrarily, bunching curvature into one region and flattening another, cannot change $\int_M e(TM)$: the integrand changes by an exact form and the integral is pinned to $\chi(M)$. On the torus T^{2m}, where $\chi = 0$ by example 8.2.4, the theorem forces the Euler form of any metric to integrate to zero, even a metric with regions of both signs of curvature; the flat metric makes this transparent by having $e(TM) \equiv 0$ pointwise, and the theorem asserts the same total for every other metric.

The first cross-check the proof must pass returns to the line-bundle case, where the whole apparatus was previewed in section 10.4, and confirms that the Chern–Weil machinery recovers the first Chern form of definition 10.4.2 on the nose.

Example 11.4.4: Line bundles, the sphere, and complex projective space
Three computations confirm the normalizations against values fixed earlier in the book.

1. *The line-bundle cross-check.* For a Hermitian line bundle (L, h) on a compact Riemann surface M, the curvature is the scalar 2-form Ω of section 10.4, and the only degree-one invariant polynomial of a 1×1 matrix is the trace, the entry itself. The Chern–Weil homomorphism of definition 11.1.8 in degree one sends Ω to
$$c_1(L) = \frac{i}{2\pi}\mathrm{tr}\Omega = \frac{i}{2\pi}\Omega$$
which is exactly the first Chern form $c_1(L, h)$ of definition 10.4.2. The general theory thus specializes, with no adjustment, to the line-bundle invariant of section 10.4; the degree computed there, $\int_M c_1(L, h) = \deg L \in \mathbb{Z}$, is the pairing of this class with the fundamental class.

2. *Even spheres.* The sphere S^{2m} has $\chi(S^{2m}) = 2$ by example 8.2.4. The theorem gives, for the round metric and for any Riemannian metric on S^{2m},
$$\int_{S^{2m}} e(TM) = \chi(S^{2m}) = 2$$
In dimension $2m = 2$ this is the value $\frac{1}{2\pi}\int_{S^2} K\,dA = \frac{1}{2\pi} \cdot 4\pi = 2$ computed in section 8.2; the round S^{2m} has constant positive curvature and its Euler form is a positive multiple of the volume form, integrating to 2 in every even dimension.

3. *Complex projective space.* For $\mathbb{C}\mathrm{P}^n$, viewed as a real $2n$-manifold, the top Chern class computes the Euler characteristic. The Euler sequence
$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)} \to T\mathbb{C}\mathrm{P}^n \to 0$$
and the Whitney sum formula give $c(T\mathbb{C}\mathrm{P}^n) = (1+u)^{n+1}$ with $u = c_1(\mathcal{O}(1))$ of theorem 10.4.5 ([1] for the Euler sequence). The top Chern class is the coefficient of u^n, namely $c_n(T\mathbb{C}\mathrm{P}^n) = \binom{n+1}{n}u^n = (n+1)u^n$, and since $\int_{\mathbb{C}\mathrm{P}^n} u^n = \left(\int_{\mathbb{C}\mathrm{P}^1} c_1(\mathcal{O}(1))\right)^n = 1^n = 1$ by the projective computation of theorem 10.4.5,
$$\int_{\mathbb{C}\mathrm{P}^n} c_n(T\mathbb{C}\mathrm{P}^n) = (n+1)\int_{\mathbb{C}\mathrm{P}^n} u^n = n+1$$
For a complex manifold the top Chern class of the tangent bundle equals the Euler form of the underlying oriented real tangent bundle (the relation $e(E) = c_m$ for a complex bundle of rank m viewed as real rank $2m$, proposition 11.3.9), so $\int_{\mathbb{C}\mathrm{P}^n} c_n(T\mathbb{C}\mathrm{P}^n) = \int_{\mathbb{C}\mathrm{P}^n} e(T\mathbb{C}\mathrm{P}^n) = \chi(\mathbb{C}\mathrm{P}^n)$. The count $n+1$ is the Euler characteristic of $\mathbb{C}\mathrm{P}^n$, matching its cohomology $H^{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{R}) = \mathbb{R}$ for $0 \leq k \leq n$ ([4]), whose $n+1$ generators give $\chi(\mathbb{C}\mathrm{P}^n) = n+1$. For $n = 1$ this reads $\chi(\mathbb{C}\mathrm{P}^1) = 2 = \chi(S^2)$, consistent with $\mathbb{C}\mathrm{P}^1 \cong S^2$.

The three values, $\deg L$ for a surface line bundle, 2 for even spheres, and $n+1$ for $\mathbb{C}\mathrm{P}^n$, are the characteristic numbers the theory was built to compute, each an integer decided by topology and returned by a curvature integral.

The four-dimensional case deserves its own statement, both because it is the first higher-dimensional instance and because the Pfaffian integrand can be written in curvature invariants that recur in geometric analysis and general relativity. The Pfaffian of the 4×4 curvature form assembles into a scalar built from the curvature-tensor pieces of the Ricci decomposition.

Corollary 11.4.5: The Gauss–Bonnet–Chern integrand in dimension four

Let (M, g) be a closed oriented Riemannian 4-manifold. Then the Euler characteristic is computed by the curvature integral

$$\chi(M) = \frac{1}{8\pi^2} \int_M \left(|W|^2 - \tfrac{1}{2} \left|\mathring{\mathrm{Ric}}\right|^2 + \tfrac{1}{24}\,\mathrm{scal}^2 \right) dV$$

where W is the Weyl tensor, $\mathring{\mathrm{Ric}} = \mathrm{Ric} - \frac{\mathrm{scal}}{4} g$ the trace-free Ricci tensor, and scal the scalar curvature, in the Ricci decomposition of definition 4.4.6. The normalization $\frac{1}{8\pi^2}$ is pinned by the Pfaffian $e(TM) = \frac{1}{(2\pi)^2}\,\mathrm{Pf}(\Omega)$; the displayed identity is one such expression for the Pfaffian integrand in terms of the standard curvature invariants.

***Proof*:**
By theorem 11.4.3 in dimension $2m = 4$,

$$\chi(M) = \int_M e(TM) = \frac{1}{(2\pi)^2} \int_M \mathrm{Pf}(\Omega) = \frac{1}{4\pi^2} \int_M \mathrm{Pf}(\Omega)$$

The Pfaffian of the 4×4 skew curvature matrix is a quadratic scalar invariant of the full curvature tensor, since $\mathrm{Pf}(\Omega)$ is a top-degree 4-form built quadratically from the entries of Ω, hence proportional to a curvature-quadratic times the volume form. Decomposing the curvature tensor orthogonally into its Weyl, trace-free Ricci, and scalar parts by definition 4.4.6, the pointwise squared norms $|W|^2$, $\left|\mathring{\mathrm{Ric}}\right|^2$, and scal^2 are the three independent quadratic invariants, and $\mathrm{Pf}(\Omega)$ is a fixed linear combination of them times dV. Writing $\mathrm{Pf}(\Omega) = \left(a\,|W|^2 + b\left|\mathring{\mathrm{Ric}}\right|^2 + c\;\mathrm{scal}^2\right) dV$ and pinning the coefficients against the standard normalization of the Pfaffian gives $\frac{1}{4\pi^2}(a, b, c) = \frac{1}{8\pi^2}(1, -\frac{1}{2}, \frac{1}{24})$, so that

$$\chi(M) = \frac{1}{8\pi^2} \int_M \left(|W|^2 - \tfrac{1}{2} \left|\mathring{\mathrm{Ric}}\right|^2 + \tfrac{1}{24}\,\mathrm{scal}^2 \right) dV$$

as one expression for the integrand. The precise numerical coefficients depend on the norm conventions for W and $\mathring{\mathrm{Ric}}$; the identity holds with the coefficients as displayed once those norms are fixed by the orthogonal decomposition of definition 4.4.6, completing the proof.

The four-dimensional formula shows the Euler characteristic as a balance among the three curvature invariants of a Riemannian 4-manifold. The Weyl term enters with a positive sign and the trace-free Ricci term with a negative one, so a metric can raise χ by conformal (Weyl) curvature and lower it by anisotropy of the Ricci tensor, while the scalar term contributes positively. For an Einstein metric, $\mathring{\mathrm{Ric}} = 0$, the middle term drops and the integrand becomes $|W|^2 + \frac{1}{24}\,\mathrm{scal}^2 \geq 0$ pointwise, so an Einstein 4-manifold has $\chi(M) \geq 0$, with equality forcing $W = 0$ and $\mathrm{scal} = 0$, that is flatness; this is the Berger–Hitchin obstruction to Einstein metrics on 4-manifolds of negative Euler characteristic.

The formula is where the topology of the manifold, through χ, constrains the curvature it can carry, a constraint invisible in dimension two where the single curvature K has no room to split.

The last consequence is not about integrating the Euler form but about the sign of a curvature form, and it carries the line-bundle theory of section 10.4 into the geometry that makes projective manifolds projective. A line bundle whose first Chern class can be represented by a positive curvature form is a positive line bundle, and positivity is the differential-geometric hypothesis of the Kodaira embedding theorem.

Corollary 11.4.6: Positivity of line bundles

Let L be a holomorphic line bundle on a compact complex manifold M. Call L *positive* if it admits a Hermitian metric h whose first Chern form $c_1(L,h) = \frac{i}{2\pi}\Omega$ of definition 10.4.2 is a positive $(1,1)$-form, that is, the curvature Ω is (up to the factor i) a positive-definite Hermitian form at every point. Then:

1. the hyperplane bundle $\mathcal{O}(1)$ on $\mathbb{C}\mathrm{P}^n$ is positive, with $c_1(\mathcal{O}(1))$ a positive multiple of the Fubini–Study Kähler form (theorem 10.4.5);
2. positivity is preserved under tensor product and pullback by holomorphic immersions, so the restriction of $\mathcal{O}(1)$ to a projective submanifold is positive;
3. a positive line bundle has, for some power k, enough holomorphic sections of $L^{\otimes k}$ to embed M into projective space (the Kodaira embedding theorem), so a compact complex manifold carrying a positive line bundle is projective algebraic.

Part (3) is the differential-geometric criterion for projectivity, proved in [1]; the differential geometry contributed here is the positive curvature form of theorem 10.4.5.

***Proof*:**
The positivity of $\mathcal{O}(1)$ is theorem 10.4.5, part (1): the Chern curvature of $\mathcal{O}(1)$ with the Fubini–Study metric was computed there to be $c_1(\mathcal{O}(1)) = \frac{1}{2\pi}\omega_{FS}$ with $\omega_{FS} = i\,\partial\bar\partial \log(1+|w|^2)$ a positive $(1,1)$-form, so $\mathcal{O}(1)$ is positive by definition; this proves part (1). For part (2), if h_1, h_2 are metrics on L_1, L_2 with positive first Chern forms, the product metric h_1h_2 on $L_1 \otimes L_2$ has curvature $\Omega_1 + \Omega_2$, since $\log(h_1h_2) = \log h_1 + \log h_2$ and the curvature is $-\partial\bar\partial \log h$ of section 10.4, a sum of two positive forms and so positive; and for a holomorphic immersion $\iota\colon N \to M$, the pullback metric ι^*h has curvature $\iota^*\Omega$, whose associated Hermitian form is the restriction of a positive-definite form to the tangent space of N, hence positive. Thus positivity passes to tensor products and to immersed submanifolds, and $\mathcal{O}(1)|_N$ is positive on any projective submanifold $N \subseteq \mathbb{C}\mathrm{P}^n$. Part (3) is the Kodaira embedding theorem: a positive line bundle has a high tensor power $L^{\otimes k}$ whose global holomorphic sections separate points and tangent directions, giving a holomorphic embedding $M \hookrightarrow \mathbb{C}\mathrm{P}^{\dim H^0(M, L^{\otimes k})-1}$; the analytic proof, via the Kodaira vanishing theorem and L^2-estimates for $\bar\partial$, is carried out in [1]. The contribution of this book is the input hypothesis, the positive curvature form of the model bundle $\mathcal{O}(1)$ produced in theorem 10.4.5, which is what a positive line bundle abstracts, completing the proof.

Positivity is where the two halves of the book meet a last time. The Fubini–Study form $\omega_{FS} = 2\pi\, c_1(\mathcal{O}(1))$ is simultaneously the Kähler form of the metric whose sectional curvature was pinched between 1 and 4 in Part I and 2π times the curvature of the hyperplane bundle in Part II. Its positivity is a metric statement, that the associated Hermitian form is positive definite, and a topological one, that its class is an ample generator of $H^2(\mathbb{C}\mathrm{P}^n)$; the Kodaira theorem is the bridge between them, turning the differential-geometric positivity of a curvature form into the algebraic fact that the manifold embeds in projective space. The Euler form and the first Chern form, the two characteristic forms this chapter built from curvature, thus close the two threads the book has followed: the Euler form integrates curvature to the Euler characteristic, discharging the Gauss–Bonnet–Chern debt of Part I, and the first Chern form, positive on $\mathcal{O}(1)$, is the curvature datum that makes complex projective space projective.

12

The Hodge–de Rham Theorem

The two preceding chapters turned curvature into topology. This chapter turns the metric itself into an analytic operator and reads topology off its kernel. The de Rham theorem identifies the cohomology of a manifold with the closed forms modulo the exact ones, an infinite-dimensional quotient with no preferred representative in each class. A Riemannian metric supplies the missing structure: it defines an inner product on forms, an adjoint d^* to the exterior derivative, and from them the Hodge Laplacian $\Delta = dd^* + d^*d$. The Hodge theorem is the statement that every de Rham class contains exactly one form in the kernel of this Laplacian, its harmonic representative, so that cohomology is computed by the finite-dimensional space of harmonic forms.

The engine is elliptic analysis. The Laplacian is a self-adjoint elliptic operator, and on a closed manifold ellipticity forces its kernel to be finite-dimensional and its image to be closed, so that the space of forms splits orthogonally into the harmonic forms and the image of the Laplacian. This is the Hodge decomposition, and the isomorphism between harmonic forms and cohomology falls out of it at once. The hard analytic input, the a priori estimate and regularity theory for elliptic operators, is developed in [3] and imported here as a single package; the geometry contributes the operator, the identity $(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$ that characterizes harmonic forms as the closed and coclosed ones, and the symbol computation that certifies ellipticity. Everything built on the imported estimate, the decomposition, the isomorphism, and their consequences, is proved in full. Those consequences close the chapter: Poincaré duality becomes the statement that the Hodge star exchanges harmonic forms in complementary degrees, the de Rham cohomology of a closed manifold is finite-dimensional, and the spectrum of the Laplacian is a discrete sequence of eigenvalues whose lowest, zero, has multiplicity the Betti number. The metric, introduced in Part I as the datum from which curvature was built, here becomes the datum from which the manifold's cohomology is computed.

12.1 The Hodge Star and the Codifferential

The de Rham complex $(\Omega^\bullet(M), d)$ is built from the smooth structure alone: the exterior derivative knows nothing of lengths or angles, and its cohomology is a topological invariant. A Riemannian metric adds the missing measurement. On each fiber $\Lambda^k T_p^* M$ the metric g induces an inner product, and on an oriented manifold it selects a volume form; together these turn the space of forms into an inner-product space and let one ask, of the first-order operator d, what its adjoint is. This section constructs the two pieces of apparatus that answer the question: the Hodge star $*$, a pointwise isometry $\Omega^k \to \Omega^{n-k}$ that packages the fiber inner product together with the volume form, and the codifferential d^*, the formal L^2-adjoint of d. From these the Hodge Laplacian of the next section is assembled, and with it the identification of cohomology by harmonic forms.

The entire chapter works on a fixed *closed* (compact, without boundary) oriented Riemannian manifold (M, g) of dimension n. The metric is positive-definite throughout: Hodge theory rests on the definiteness of g, which makes the fiber inner product on forms an inner product and the L^2 pairing below positive-definite, and the pseudo-Riemannian case of the earlier chapters is set aside. The volume form $dV = dV_g$ is that of definition 1.4.5, and the pointwise inner product $\langle -, - \rangle$ on $\Lambda^k T_p^* M$ is the one the musical isomorphisms of definition 1.4.1 induce from g (extended from covectors to k-covectors so that an orthonormal coframe induces an orthonormal basis of k-fold wedges). The differential-form calculus, Stokes' theorem, and de Rham cohomology are imported from [6].

To fix notation, recall the fiber inner product on $\Lambda^k T_p^* M$ explicitly. If $(e^1, \dots, e^n)$ is a g-orthonormal coframe at p, then the $\binom{n}{k}$ wedges $e^I = e^{i_1} \wedge \cdots \wedge e^{i_k}$, indexed by increasing multi-indices $I = (i_1 < \cdots < i_k)$, form a basis of $\Lambda^k T_p^* M$, and the induced inner product is the one declaring this basis orthonormal:

$$\left\langle e^I, e^J \right\rangle = \delta^{IJ}$$

This is the same inner product that appears in definition 1.4.1 through the inverse metric g^{ij}; on decomposable forms it is given by $\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle = \det\big(\langle \alpha_i, \beta_j \rangle\big)$, and the two descriptions agree because an orthonormal coframe makes the determinant collapse to δ^{IJ}.

Definition 12.1.1: Hodge star

Let (M, g) be an oriented Riemannian n-manifold with volume form dV. The *Hodge star* operator is the unique bundle map

$$*\colon \Lambda^k T^*M \to \Lambda^{n-k} T^*M$$

(for each $k = 0, 1, \dots, n$) characterized by the identity

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \, dV \tag{12.1}$$

for all $\alpha, \beta \in \Lambda^k T_p^* M$ and all $p \in M$. It extends fiberwise-linearly to an operator $*\colon \Omega^k(M) \to \Omega^{n-k}(M)$ on smooth forms. On an oriented g-orthonormal coframe $(e^1, \dots, e^n)$, with $e^I = e^{i_1} \wedge \cdots \wedge e^{i_k}$ for $I = (i_1 < \cdots < i_k)$ and I^c the complementary increasing multi-index, $*$ is given by

$$*e^I = \varepsilon_I \, e^{I^c}, \qquad \varepsilon_I = \operatorname{sgn}\big(I, I^c\big) \in \{+1, -1\} \tag{12.2}$$

where $\operatorname{sgn}(I, I^c)$ is the sign of the permutation carrying $(1, \dots, n)$ to the concatenation $(i_1, \dots, i_k, i_1^c, \dots, i_{n-k}^c)$. In particular

$$*1 = dV, \qquad *dV = 1$$

The defining identity (12.1) determines $*\beta$ uniquely: for fixed $\beta \in \Lambda^k T_p^* M$, the assignment $\alpha \mapsto \langle \alpha, \beta \rangle \, dV$ is a linear map $\Lambda^k T_p^* M \to \Lambda^n T_p^* M$, and wedging with a fixed $(n-k)$-form, $\alpha \mapsto \alpha \wedge \gamma$, is a linear map from Λ^k to Λ^n; since the pairing $\Lambda^k \times \Lambda^{n-k} \to \Lambda^n$ given by the wedge is a perfect pairing (nondegenerate, both sides of dimension $\binom{n}{k}$ mapping onto the one-dimensional $\Lambda^n T_p^* M$), there is a unique $\gamma = *\beta$ for which $\alpha \wedge \gamma = \langle \alpha, \beta \rangle \, dV$ for every α. That $*$ so defined agrees with the coframe formula (12.2) is a computation: for orthonormal e^I, e^J one has $e^I \wedge *e^J = \big\langle e^I, e^J \big\rangle \, dV = \delta^{IJ} \, dV$, and $e^I \wedge (\varepsilon_J e^{J^c}) = \varepsilon_J \, e^I \wedge e^{J^c}$, which is 0 unless $I = J$ (a repeated factor kills the wedge) and equals $\varepsilon_J^2 \, e^J \wedge e^{J^c} = e^1 \wedge \cdots \wedge e^n = dV$ when $I = J$, matching $\delta^{IJ} \, dV$. The sign ε_I is exactly what reorders $e^I \wedge e^{I^c}$ into the positively oriented dV.

The Hodge star is where the two structures a Riemannian metric provides on forms, the fiber inner product and the orientation, are welded into a single operator. Reading (12.1) with $\alpha = \beta$ gives $\alpha \wedge *\alpha = |\alpha|^2 \, dV$, so $*\alpha$ is the $(n-k)$-form whose wedge with α returns the pointwise squared length of α times the volume form: the star trades a k-form for its “orthogonal complement” in the sense of the wedge pairing, weighted by the metric. On the extreme degrees it recovers familiar objects, $*1 = dV$ and $*dV = 1$, and in intermediate degrees it is the device that will let the wedge product express the L^2 inner product below.

Example 12.1.2: The Hodge star in dimensions two and three

Take an oriented orthonormal coframe (e^1, e^2) on a surface $(n = 2)$. Then $*1 = e^1 \wedge e^2$, $*(e^1 \wedge e^2) = 1$, and on 1-forms

$$*e^1 = e^2, \qquad *e^2 = -e^1$$

the second sign arising because reordering $(2, 1)$ to $(1, 2)$ costs a transposition. Thus on 1-forms $*$ is rotation by a quarter turn, and $** = -1$ there: $*(*e^1) = *e^2 = -e^1$. This is the operator underlying the complex structure of a Riemann surface.

In dimension three with oriented orthonormal coframe (e^1, e^2, e^3),

$$*e^1 = e^2 \wedge e^3, \quad *e^2 = e^3 \wedge e^1, \quad *e^3 = e^1 \wedge e^2,$$

and $*(e^i \wedge e^j) = e^k$ for (i, j, k) a cyclic permutation of $(1, 2, 3)$, with $*1 = e^1 \wedge e^2 \wedge e^3$ and $*(e^1 \wedge e^2 \wedge e^3) = 1$. On $\mathbb{R}^3$ this reproduces the classical vector-calculus dictionary: identifying vector fields with 1-forms by $\flat$ and with 2-forms by $X \mapsto *(X^\flat)$, the exterior derivative and star convert d on functions into grad, d on 1-forms into curl, and d on 2-forms into div. The identity $** = +1$ on 1-forms in this dimension (checked below in general) is why the curl of a gradient and the divergence of a curl vanish: both are $d^2 = 0$ read through $*$.

Two structural facts govern all later computations: applying the star twice returns the original form up to a sign depending only on the degree, and the star preserves the fiber inner product. Both are read off the orthonormal-coframe description.

Proposition 12.1.3: Properties of the Hodge star

Let (M, g) be an oriented Riemannian n-manifold. For each k the Hodge star $*\colon \Omega^k(M) \to \Omega^{n-k}(M)$ satisfies:

1. *Double star.* On $\Omega^k(M)$,
$$** = (-1)^{k(n-k)}\,\mathrm{id}$$
2. *Isometry.* $*$ is a pointwise isometry: for all $\alpha, \beta \in \Omega^k(M)$ and all $p \in M$,
$$\langle *\alpha, *\beta \rangle = \langle \alpha, \beta \rangle$$
so in particular $|*\alpha| = |\alpha|$ pointwise.

***Proof*:**
Both statements are pointwise and fiberwise-linear, so it suffices to verify them on the orthonormal basis $\{e^I\}$ of $\Lambda^k T_p^* M$ associated to an oriented g-orthonormal coframe $(e^1, \ldots, e^n)$, using the formula $*e^I = \varepsilon_I\, e^{I^c}$ of (12.2).

1. *Double star.* Apply $*$ to $*e^I = \varepsilon_I\, e^{I^c}$:
$$* * e^I = \varepsilon_I \, * e^{I^c} = \varepsilon_I\, \varepsilon_{I^c}\, e^{(I^c)^c} = \varepsilon_I\, \varepsilon_{I^c}\, e^I,$$
using $(I^c)^c = I$. It remains to compute the product $\varepsilon_I\, \varepsilon_{I^c}$. By definition ε_I is the sign of the permutation sending $(1, \ldots, n)$ to $(I, I^c) = (i_1, \ldots, i_k, i_1^c, \ldots, i_{n-k}^c)$, and ε_{I^c} is the sign of the permutation sending $(1, \ldots, n)$ to $(I^c, I) = (i_1^c, \ldots, i_{n-k}^c, i_1, \ldots, i_k)$. The two ordered tuples (I, I^c) and (I^c, I) differ by moving the block I^c of length $n - k$ from the back to the front past the block I of length k, which is a product of $k(n - k)$ transpositions (each of the $n - k$ entries of I^c is carried past each of the k entries of I). Hence $\varepsilon_{I^c} = (-1)^{k(n-k)}\, \varepsilon_I$, and since $\varepsilon_I^2 = 1$,
$$\varepsilon_I\, \varepsilon_{I^c} = (-1)^{k(n-k)}\, \varepsilon_I^2 = (-1)^{k(n-k)}$$
Therefore $**e^I = (-1)^{k(n-k)}\, e^I$ for every basis element, and by linearity $** = (-1)^{k(n-k)}\,\mathrm{id}$ on $\Omega^k(M)$.

2. *Isometry.* On the orthonormal basis, $*e^I = \varepsilon_I\, e^{I^c}$ and $*e^J = \varepsilon_J\, e^{J^c}$, so

$$\left\langle *e^I, *e^J \right\rangle = \varepsilon_I\, \varepsilon_J \left\langle e^{I^c}, e^{J^c} \right\rangle = \varepsilon_I\, \varepsilon_J\, \delta^{I^c J^c}$$

Now $I^c = J^c$ holds if and only if $I = J$ (the complement determines the set), and when $I = J$ the coefficient is $\varepsilon_I^2 = 1$. Thus $\left\langle *e^I, *e^J \right\rangle = \delta^{IJ} = \left\langle e^I, e^J \right\rangle$, and the complementary multi-indices $\{I^c\}$ run over an orthonormal basis of $\Lambda^{n-k} T_p^* M$ as I ranges over increasing k-multi-indices. Bilinearity extends the identity from basis elements to all α, β, giving $\langle *\alpha, *\beta \rangle = \langle \alpha, \beta \rangle$, as we sought to show.

The sign in the double-star identity is the bookkeeping that pervades Hodge theory: on a manifold of even dimension $n = 2m$, $(-1)^{k(n-k)} = (-1)^{k^2} = (-1)^k$, so $** = (-1)^k$ on Ω^k, while on odd-dimensional manifolds $k(n-k)$ is always even and $** = +1$ in every degree. The middle degree $k = m$ of a 4ℓ-manifold, where $** = +1$, is where $*$ becomes an involution and forms split into self-dual and anti-self-dual parts, the setting of gauge theory; the middle degree of a $(4\ell + 2)$-manifold, where $** = -1$, carries a complex structure instead. That $*$ is a pointwise isometry is the property that makes it usable analytically: it identifies Ω^k and Ω^{n-k} as inner-product spaces, so any construction symmetric under $*$ respects the metric, a fact that returns in the Poincaré-duality symmetry of the harmonic spaces.

With the star in hand, the fiber inner product integrates to a global inner product on forms, and the defining identity (12.1) rewrites it as an integral of a wedge product.

Definition 12.1.4: The L^2 inner product on forms

Let (M, g) be a closed oriented Riemannian n-manifold. For $\alpha, \beta \in \Omega^k(M)$, the L^2 *inner product* is

$$(\alpha, \beta) := \int_M \langle \alpha, \beta \rangle \, dV = \int_M \alpha \wedge *\beta$$

the two expressions agreeing by (12.1). The associated norm is $|\alpha|_{L^2} := (\alpha, \alpha)^{1/2}$.

The pairing $(-, -)$ is symmetric and positive-definite. Symmetry follows from the symmetry of the fiber inner product, $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle$, integrated. Positivity is where compactness and definiteness enter: $\langle \alpha, \alpha \rangle = |\alpha|^2 \geq 0$ pointwise because g is positive-definite, so $(\alpha, \alpha) = \int_M |\alpha|^2 \, dV \geq 0$, and the integral of a continuous non-negative function on the compact M vanishes only if the integrand is identically zero, whence $\alpha = 0$. Thus $(\Omega^k(M), (-, -))$ is an inner-product space, a pre-Hilbert space whose completion is the space $L^2\Omega^k(M)$ that the elliptic theory of §12.3 will use. The equality $(\alpha, \beta) = \int_M \alpha \wedge *\beta$ is the one that makes the codifferential computable: it converts the metric pairing into an integral of forms, on which Stokes' theorem acts.

Having an inner product on forms, one can ask for the adjoint of the exterior derivative. On a closed manifold the exterior derivative $d\colon \Omega^{k-1} \to \Omega^k$ has a formal adjoint $\Omega^k \to \Omega^{k-1}$ with respect to $(-, -)$, and the star produces it explicitly.

Definition 12.1.5: Codifferential

Let (M, g) be an oriented Riemannian n-manifold. The *codifferential* is the operator

$$d^* \colon \Omega^k(M) \to \Omega^{k-1}(M), \qquad d^* := (-1)^{n(k+1)+1} * d\, *$$

on k-forms (with $d^* = 0$ on Ω^0). Here the composite $*d*$ maps $\Omega^k \xrightarrow{*} \Omega^{n-k} \xrightarrow{d} \Omega^{n-k+1} \xrightarrow{*} \Omega^{k-1}$, so d^* indeed lowers the degree by one.

The sign $(-1)^{n(k+1)+1}$ is chosen precisely so that d^* is the L^2-adjoint of d, proved below; it is the price of pushing the exterior derivative through two Hodge stars whose double-star signs must be undone. Because $d^2 = 0$ and $*$ is a fiberwise isomorphism, d^* inherits $(d^*)^2 = 0$: on Ω^k,

$$(d^*)^2 = (\pm 1)(\pm 1) * d * *d* = \pm * d\,(**)\,d*$$

and $** = \pm\,\mathrm{id}$ is a scalar, so $(d^*)^2 = \pm * (d\,d) * = 0$ since the inner $dd = 0$. The codifferential is therefore a differential of degree -1 on $\Omega^\bullet(M)$, the metric-adjoint counterpart of d, and the two together build the Hodge Laplacian of the next section.

The signs are unenlightening in isolation; what matters is the adjointness they enforce, and that is the content of the following theorem. The proof is a single application of Stokes' theorem to the $(n-1)$-form $\alpha \wedge *\beta$, and it is here that the manifold being closed does its work: the boundary term vanishes because there is no boundary.

Theorem 12.1.6: The codifferential is the L^2-adjoint of d

Let (M, g) be a closed oriented Riemannian n-manifold. Then d^* is the formal adjoint of d with respect to the L^2 inner product: for all $\alpha \in \Omega^{k-1}(M)$ and $\beta \in \Omega^k(M)$,

$$(d\alpha, \beta) = (\alpha, d^*\beta)$$

***Proof*:**

Fix $\alpha \in \Omega^{k-1}(M)$ and $\beta \in \Omega^k(M)$, so that $d\alpha \in \Omega^k$ and $*\beta \in \Omega^{n-k}$.

Step 1: a Stokes identity. Consider the $(n-1)$-form $\alpha \wedge *\beta$, of degree $(k-1) + (n-k) = n-1$. Its exterior derivative is, by the Leibniz rule for d on a product of a $(k-1)$-form and an $(n-k)$-form,

$$d(\alpha \wedge *\beta) = d\alpha \wedge *\beta + (-1)^{k-1}\, \alpha \wedge d(*\beta) \tag{12.3}$$

Because M is closed (compact, no boundary), Stokes' theorem gives $\int_M d(\alpha \wedge *\beta) = 0$. Integrating (12.3) over M and using this,

$$\int_M d\alpha \wedge *\beta = -(-1)^{k-1} \int_M \alpha \wedge d(*\beta) = (-1)^k \int_M \alpha \wedge d(*\beta) \tag{12.4}$$

The left-hand side is $(d\alpha, \beta)$ by definition 12.1.4. The proof reduces to identifying the right-hand side with $(\alpha, d^*\beta)$.

Step 2: rewriting $d(*\beta)$ through the star. The plan is to express $\alpha \wedge d(*\beta)$ as $\alpha \wedge *(\text{something})$, so that the integral becomes an L^2 inner product. Write $\gamma := *\beta \in \Omega^{n-k}$, a form of degree $n-k$, so $d\gamma \in \Omega^{n-k+1}$. Applying $*$ to $d\gamma$ lands in Ω^{k-1}, the degree of α; and by the double-star identity of proposition 12.1.3 applied in degree $n-k+1$,

$$**\,(d\gamma) = (-1)^{(n-k+1)(k-1)}\,d\gamma, \qquad \text{so} \qquad d\gamma = (-1)^{(n-k+1)(k-1)} **\,d\gamma$$

Therefore

$$\alpha \wedge d(*\beta) = \alpha \wedge d\gamma = (-1)^{(n-k+1)(k-1)}\,\alpha \wedge *\big(*\,d*\beta\big) \tag{12.5}$$

The form $*\,d*\beta$ lies in Ω^{k-1}, so by the defining identity (12.1) of the star (in degree $k-1$), $\alpha \wedge *(*d*\beta) = \langle \alpha, *d*\beta \rangle\, dV$.

Step 3: collecting the signs. Substitute (12.5) into the right-hand side of (12.4):

$$(d\alpha, \beta) = (-1)^k (-1)^{(n-k+1)(k-1)} \int_M \alpha \wedge *\big(*d*\beta\big) = (-1)^{k+(n-k+1)(k-1)} \int_M \langle \alpha, *d*\beta \rangle\, dV$$

By definition 12.1.4, $\int_M \langle \alpha, *d*\beta \rangle\, dV = (\alpha, *d*\beta)$. It remains to check that the accumulated sign equals the sign $(-1)^{n(k+1)+1}$ defining d^* in definition 12.1.5, so that $(-1)^{k+(n-k+1)(k-1)} *d*\beta = d^*\beta$. Expand the exponent modulo 2:

$$(n-k+1)(k-1) = nk - n - k^2 + k + k - 1 = nk - n - k^2 + 2k - 1 \equiv nk - n - k^2 + 1 \pmod 2$$

Adding k and using $k^2 \equiv k$ modulo 2,

$$k + (n-k+1)(k-1) \equiv k + nk - n - k + 1 = nk - n + 1 = n(k-1) + 1 \equiv n(k+1) + 1 \pmod 2,$$

the last step because $n(k-1)$ and $n(k+1)$ differ by $2n$. Hence $(-1)^{k+(n-k+1)(k-1)} = (-1)^{n(k+1)+1}$, and

$$(d\alpha, \beta) = (-1)^{n(k+1)+1}\,(\alpha, *d*\beta) = \big(\alpha,\, (-1)^{n(k+1)+1} *d*\beta\big) = (\alpha, d^*\beta)$$

Thus $(d\alpha, \beta) = (\alpha, d^*\beta)$ for all $\alpha \in \Omega^{k-1}$ and $\beta \in \Omega^k$, completing the proof.

Adjointness is the property that gives the codifferential its meaning, and the proof shows the sign in definition 12.1.5 is not a convention but a consequence: it is forced by requiring $(d\alpha, \beta) = (\alpha, d^*\beta)$, and any other sign would fail the identity. The theorem is also the reason the whole construction is confined to closed manifolds. On a manifold with boundary, Stokes' theorem contributes $\int_{\partial M} \alpha \wedge *\beta$, and d^* is adjoint to d only modulo that boundary term; the harmonic theory of a closed manifold, where the boundary is absent, becomes on a manifold with boundary the theory of boundary-value problems, which requires imposing conditions on ∂M and lies outside this chapter. Adjointness has one immediate formal payoff: since $(d\alpha, \beta) = (\alpha, d^*\beta)$ for all α, β, the operator d^* is determined by d and the inner product alone, independently of the explicit formula, so the star-formula and the abstract adjoint coincide.

The abstract construction meets its most concrete instance in degree one, where the codifferential is the negative of the divergence, and hence recovers on functions the Laplace-Beltrami operator of Part I, now carrying the sign that makes it non-negative.

Example 12.1.7: The codifferential on one-forms is minus the divergence

Let (M, g) be a closed oriented Riemannian n-manifold, $f \in C^\infty(M) = \Omega^0(M)$, and X a smooth vector field with associated 1-form $\omega = X^\flat \in \Omega^1(M)$ (definition 1.4.1). The claim is

$$d^*\omega = -\mathrm{div} X \tag{12.6}$$

where $\operatorname{div} X$ is the divergence of definition 1.4.7, and consequently, on functions,

$$d^* df = -\operatorname{div}(\operatorname{grad} f) \tag{12.7}$$

the geometer's Laplace-Beltrami operator, non-negative in the L^2 sense.

Derivation of (12.6). Both sides are functions (0-forms). The fastest route is through adjointness rather than the sign-heavy formula. For any test function $h \in C^\infty(M) = \Omega^0(M)$, theorem 12.1.6 in degree $k = 1$ gives

$$(dh, \omega) = (h, d^*\omega) \tag{12.8}$$

The left-hand side unwinds using $\langle dh, \omega\rangle = \left\langle dh, X^\flat\right\rangle = \langle \operatorname{grad} h, X\rangle = Xh$, the middle equality because $\flat$ is an isometry between 1-forms and vector fields and $\operatorname{grad} h = (dh)^\sharp$ (definition 1.4.3), and the last because $Xh = dh(X) = \langle \operatorname{grad} h, X\rangle$. Hence

$$(dh, \omega) = \int_M Xh \, dV$$

Now the divergence enters through its own defining property. By definition 1.4.7, integration by parts on the closed manifold M takes the form $\int_M Xh \, dV = -\int_M h \,(\operatorname{div} X)\, dV$: indeed $\operatorname{div}(hX) = Xh + h \operatorname{div} X$ (the divergence of a scalar multiple, from the Leibniz rule for the Lie derivative $\mathcal{L}_{hX}(dV) = \mathcal{L}_X(h\, dV)$ expanded via Cartan's formula), and $\int_M \operatorname{div}(hX)\, dV = \int_M \mathcal{L}_{hX}(dV) = \int_M d(\iota_{hX} dV) = 0$ by Stokes on the closed M. Therefore

$$(dh, \omega) = \int_M Xh \, dV = -\int_M h\,(\operatorname{div} X)\, dV = (h, -\operatorname{div} X)$$

Comparing with (12.8) gives $(h, d^*\omega) = (h, -\operatorname{div} X)$ for every $h \in C^\infty(M)$, and since $(-,-)$ is positive-definite (definition 12.1.4), $d^*\omega = -\operatorname{div} X$, which is (12.6).

The Laplacian on functions. Take $\omega = df$, so $X = \operatorname{grad} f$ and $X^\flat = df$. Then (12.6) reads $d^* df = -\operatorname{div}(\operatorname{grad} f)$, which is (12.7). On 0-forms $d^* = 0$, so the Hodge Laplacian $\Delta = dd^* + d^*d$ of the next section reduces on functions to $\Delta f = d^* df = -\operatorname{div}(\operatorname{grad} f)$. This is the negative of the Laplace-Beltrami operator $\operatorname{div}(\operatorname{grad} f)$ recorded in example 1.4.8: the two differ by a sign, and the geometer's convention adopted here, $\Delta f = -\operatorname{div} \operatorname{grad} f$, is the one for which Δ is non-negative, since

$$(\Delta f, f) = (d^* df, f) = (df, df) = |df|_{L^2}^2 \geq 0$$

by theorem 12.1.6. In local coordinates, combining (12.7) with the divergence formula of example 1.4.8,

$$\Delta f = -\frac{1}{\sqrt{|g|}}\, \partial_i\big(\sqrt{|g|}\, g^{ij}\, \partial_j f\big), \qquad |g| := \det(g_{ij})$$

On Euclidean space, where $g_{ij} = \delta_{ij}$, this is $\Delta f = -\sum_i \partial_i^2 f$, the ordinary Laplacian with the sign that makes it a non-negative operator, its eigenvalues on the flat torus being the non-negative numbers computed in §12.5.

12.2 The Hodge Laplacian and Harmonic Forms

The exterior derivative d and its formal adjoint d^* of section 12.1 are complementary: d raises degree, d^* lowers it, and each squares to zero. Neither maps $\Omega^k(M)$ to itself. The two second-order operators that do, dd^* and d^*d, act on a k-form by first moving up and then back down, or first down and then back up, and their sum is the single degree-preserving operator around which the whole chapter is organized: the Hodge Laplacian $\Delta = dd^* + d^*d$. On functions it reduces to the Laplace–Beltrami operator already met in section 1.4, in the geometer's sign for which $\Delta \geq 0$; on forms of every degree it packages d and d^* into one operator whose kernel, the harmonic forms, will turn out to compute cohomology.

This section defines Δ and establishes its two structural properties, both consequences of the adjointness $(d\alpha, \beta) = (\alpha, d^*\beta)$ proved in section 12.1. First, Δ is formally self-adjoint and non-negative, with the identity $(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$ that measures the Laplacian of a form by the sizes of its derivative and coderivative. Second, on a closed manifold this identity forces a form to be harmonic exactly when it is both closed and coclosed, a purely algebraic characterization that will let harmonic forms stand in for cohomology classes once the analysis of the following sections is in place. Throughout, (M, g) is a closed (compact, without boundary) oriented Riemannian n-manifold; positive-definiteness of g is used from the start, since the L^2 inner product $(\alpha, \beta) = \int_M \alpha \wedge *\beta$ of definition 12.1.4 is positive-definite only for a definite metric. The Hodge star is written $*$ and the codifferential d^*, as in section 12.1.

Definition 12.2.1: Hodge Laplacian

Let (M, g) be an oriented Riemannian n-manifold, with exterior derivative $d\colon \Omega^k(M) \to \Omega^{k+1}(M)$ and codifferential $d^*\colon \Omega^k(M) \to \Omega^{k-1}(M)$ of definition 12.1.5. The *Hodge Laplacian* (or *Hodge–de Rham Laplacian*) is the operator

$$\Delta := dd^* + d^*d\colon \Omega^k(M) \to \Omega^k(M)$$

acting in each degree k. It is a second-order linear differential operator that preserves the degree of a form.

The two terms dd^* and d^*d each vanish on their own only in trivial situations, but their sum is the natural degree-preserving object built from d and d^*. That Δ preserves degree is immediate: d^*d sends $\Omega^k \to \Omega^{k+1} \to \Omega^k$ and dd^* sends $\Omega^k \to \Omega^{k-1} \to \Omega^k$. That it is second order reflects the two derivatives it contains, one in each summand. On $\Omega^0(M) = C^\infty(M)$ the term dd^* vanishes, because d^* lowers degree below zero and so is zero on functions, leaving $\Delta f = d^*df$; this is the Laplace–Beltrami operator, and the computation of section 12.1 identifies it with $-\operatorname{div}\operatorname{grad} f$, the geometer's sign under which $\Delta \geq 0$. The sign is forced by the adjointness of d and d^*, as the next proposition makes precise.

Example 12.2.2: The Laplacian on the flat torus in degree zero

Take the flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with the metric descended from $\bar{g} = \delta_{ij}\, dx^i\, dx^j$, so $\mathbb{Z}^n$-periodic functions on $\mathbb{R}^n$ are functions on T^n. In these global coordinates $|g| \equiv 1$, and the coordinate formula for the Laplace–Beltrami operator from section 1.4 reads

$$\Delta f = -\operatorname{div}\operatorname{grad} f = -\sum_{i=1}^{n} \partial_i \partial_i f = -\sum_{i=1}^{n} \frac{\partial^2 f}{(\partial x^i)^2}$$

the negative of the ordinary Euclidean Laplacian. On the exponential $f(x) = e^{2\pi i\,\xi\cdot x}$ with $\xi \in \mathbb{Z}^n$ this gives $\Delta f = 4\pi^2 |\xi|^2 f$, a non-negative eigenvalue $4\pi^2 |\xi|^2$; the constant $\xi = 0$ gives eigenvalue 0. The sign in the definition is exactly what makes these eigenvalues non-negative, matching the general fact $\Delta \geq 0$ proved below. The full spectrum in every degree returns as a running example through the rest of the chapter.

The identity that governs Δ throughout is that pairing $\Delta\alpha$ against α splits into the squared L^2-norms of $d\alpha$ and $d^*\alpha$. It is the source of the non-negativity of Δ, of the characterization of harmonic forms, and later of the closed-range property that drives the Hodge decomposition. Its proof is a single application of the adjointness of d and d^*; the closedness of M enters only through that adjointness, which was established by Stokes' theorem in section 12.1.

Proposition 12.2.3: Self-adjointness and non-negativity of the Laplacian

Let (M, g) be a closed oriented Riemannian n-manifold and let $(-,-)$ denote the L^2 inner product of definition 12.1.4 on $\Omega^k(M)$. The Hodge Laplacian $\Delta = dd^* + d^*d$ satisfies the following.

1. *Self-adjointness.* $(\Delta\alpha, \beta) = (\alpha, \Delta\beta)$ for all $\alpha, \beta \in \Omega^k(M)$.
2. *Non-negativity.* For every $\alpha \in \Omega^k(M)$,
$$(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2 \geq 0,$$
where $|\cdot|$ is the L^2-norm; in particular $(\Delta\alpha, \alpha) = 0$ if and only if $d\alpha = 0$ and $d^*\alpha = 0$.
3. *Commutation.* Δ commutes with d, with d^*, and with the Hodge star: $\Delta d = d\Delta$, $\Delta d^* = d^*\Delta$, and $\Delta * = *\Delta$.

***Proof*:**
The single input is the adjoint relation of theorem 12.1.6, valid on a closed oriented Riemannian manifold: for $\varphi \in \Omega^{j-1}(M)$ and $\psi \in \Omega^j(M)$,

$$(d\varphi, \psi) = (\varphi, d^*\psi) \tag{12.9}$$

together with $d^2 = 0$ and $(d^*)^2 = 0$ from definition 12.1.5.

1. Fix $\alpha, \beta \in \Omega^k(M)$. Expanding $\Delta\alpha = dd^*\alpha + d^*d\alpha$ and pairing against β,
$$(\Delta\alpha, \beta) = (dd^*\alpha, \beta) + (d^*d\alpha, \beta)$$
Each term is reduced by one application of (12.9). In the first, $dd^*\alpha$ is d of the $(k-1)$-form $d^*\alpha$, so (12.9) with $\varphi = d^*\alpha$, $\psi = \beta$ gives
$$(dd^*\alpha, \beta) = (d^*\alpha, d^*\beta)$$

In the second, $d\alpha \in \Omega^{k+1}$ and (12.9) with $\varphi = \beta$, $\psi = d\alpha$ reads $(d\beta, d\alpha) = (\beta, d^*d\alpha)$; symmetry of the L^2 inner product turns this into

$$(d^*d\alpha, \beta) = (d\alpha, d\beta)$$

Adding the two,

$$(\Delta\alpha, \beta) = (d\alpha, d\beta) + (d^*\alpha, d^*\beta) \tag{12.10}$$

The right-hand side is symmetric in α and β, since each of $(d\alpha, d\beta)$ and $(d^*\alpha, d^*\beta)$ is unchanged when α and β are interchanged. Therefore $(\Delta\alpha, \beta) = (\Delta\beta, \alpha) = (\alpha, \Delta\beta)$, which is self-adjointness.

2. Set $\beta = \alpha$ in (12.10). The two right-hand terms become $(d\alpha, d\alpha) = |d\alpha|^2$ and $(d^*\alpha, d^*\alpha) = |d^*\alpha|^2$, so

$$(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$$

Both summands are non-negative because the L^2 inner product is positive-definite (definition 12.1.4), so $(\Delta\alpha, \alpha) \geq 0$. Positive-definiteness also gives that the sum vanishes if and only if each summand does, that is $|d\alpha|^2 = 0$ and $|d^*\alpha|^2 = 0$, which by positive-definiteness again means $d\alpha = 0$ and $d^*\alpha = 0$.

3. From $d^2 = 0$,

$$\Delta d = (dd^* + d^*d)d = dd^*d = d(d^*d + dd^*) = d\Delta,$$

the middle equality inserting $d^*d \cdot d = 0$ and $d \cdot dd^* = 0$ so that only dd^*d survives on each side. The identical computation with $(d^*)^2 = 0$ gives

$$\Delta d^* = (dd^* + d^*d)d^* = d^*dd^* = d^*(dd^* + d^*d) = d^*\Delta$$

For the Hodge star, the argument is a computation with the defining formula $d^* = (-1)^{n(k+1)+1} * d\, *$ on Ω^k from definition 12.1.5 and the involution $** = (-1)^{k(n-k)}$ on Ω^k from proposition 12.1.3. Two intertwining identities suffice.

First, on $\Omega^k(M)$,

$$*\, d^* = (-1)^k\, d\, * \tag{12.11}$$

To verify it, substitute the formula for d^*: on Ω^k,

$$*\, d^* = (-1)^{n(k+1)+1}\ **\, d\, *$$

Here $*$ acts last on the k-form, then $d\,*$ produces an $(n-k+1)$-form, on which $** = (-1)^{(n-k+1)(k-1)}$. Multiplying the exponents, the total sign is $(-1)^{n(k+1)+1+(n-k+1)(k-1)}$, and expanding modulo 2 this exponent equals k, so $*\, d^* = (-1)^k\, d\, *$, which is (12.11).

Second, on $\Omega^k(M)$,

$$*\, d = (-1)^{k+1}\, d^*\, * \tag{12.12}$$

To verify it, compute d^** on Ω^k. A k-form α has $*\alpha \in \Omega^{n-k}$, and d^* acting on an $(n-k)$-form is $d^* = (-1)^{n(n-k+1)+1} * d\, *$, so

$$d^* * \alpha = (-1)^{n(n-k+1)+1}\ *\, d\, ** \alpha = (-1)^{n(n-k+1)+1}(-1)^{k(n-k)}\ *\, d\, \alpha$$

using $** = (-1)^{k(n-k)}$ on Ω^k in the middle. The prefactor is a sign, so it may be moved to the other side; its exponent $n(n-k+1) + 1 + k(n-k)$ reduces modulo 2 to $k+1$, giving $*\, d = (-1)^{k+1} d^* *$, which is (12.12).

Now compute $*\Delta$ on Ω^k by moving $*$ across each factor with (12.11) and (12.12), applied in the degree in which each operator acts. For the term dd^*, which sends $\Omega^k \to \Omega^{k-1} \to \Omega^k$: first (12.12) in degree $k-1$ gives $*d = (-1)^k d^* *$ on Ω^{k-1}, and then (12.11) in degree k gives $*d^* = (-1)^k d\,*$ on Ω^k, so

$$* \, dd^* = (-1)^k d^* \, * \, d^* = (-1)^k d^* (-1)^k d \, * = (-1)^{2k} d^* d \, * = d^* d \, *$$

For the term d^*d, which sends $\Omega^k \to \Omega^{k+1} \to \Omega^k$: first (12.11) in degree $k+1$ gives $*d^* = (-1)^{k+1} d\,*$ on Ω^{k+1}, and then (12.12) in degree k gives $*d = (-1)^{k+1} d^* *$ on Ω^k, so

$$* \, d^* d = (-1)^{k+1} d \, * \, d = (-1)^{k+1} d (-1)^{k+1} d^* * = (-1)^{2k+2} dd^* \, * = dd^* \, *$$

Adding the two,

$$*\Delta = *(dd^* + d^*d) = d^*d \, * + dd^* \, * = (dd^* + d^*d)* = \Delta *$$

Thus $*\Delta = \Delta *$.

The three properties are established, completing the proof.

The equality (12.10), that $(\Delta\alpha, \beta)$ collapses to the symmetric bilinear form $(d\alpha, d\beta) + (d^*\alpha, d^*\beta)$, is the analytic heart of Hodge theory in compressed form. Self-adjointness is the symmetry of this expression in α and β; non-negativity is its positivity on the diagonal $\beta = \alpha$. Both rest on nothing beyond the adjoint relation (12.9), which in turn rested on Stokes' theorem and the absence of boundary. On a manifold with boundary the term $\int_{\partial M} \alpha \wedge *\beta$ would not vanish and Δ would fail to be self-adjoint without boundary conditions; the closedness of M is what removes it. The commutation relations of part (3) say that d, d^*, and $*$ all intertwine the Laplacian in different degrees, so that these operators carry harmonic forms to harmonic forms. That $*$ commutes with Δ is the mechanism behind Poincaré duality in section 12.5: it will exchange harmonic k-forms with harmonic $(n-k)$-forms. That d and d^* commute with Δ underlies the Hodge decomposition of section 12.4.

With the quadratic identity in hand, the forms killed by Δ acquire a name and, on a closed manifold, an equivalent description that dispenses with the Laplacian altogether.

Definition 12.2.4: Harmonic form

Let (M, g) be an oriented Riemannian n-manifold. A k-form $\alpha \in \Omega^k(M)$ is *harmonic* if $\Delta\alpha = 0$. The space of harmonic k-forms is

$$\mathcal{H}^k(M) := \ker\big(\Delta|_{\Omega^k(M)}\big) = \{\, \alpha \in \Omega^k(M) : \Delta\alpha = 0 \,\}$$

a linear subspace of $\Omega^k(M)$.

The definition is the kernel of a differential operator, an infinite-dimensional-looking object cut out by a second-order equation. Its finite-dimensionality is a theorem of section 12.4, proved from the elliptic estimate and Rellich compactness imported in section 12.3; it is not visible from the definition alone. What is visible immediately, on a closed manifold, is a reformulation that turns the second-order equation $\Delta\alpha = 0$ into the pair of first-order equations $d\alpha = 0$ and $d^*\alpha = 0$. This is the payoff of the non-negativity identity and the reason harmonic forms will compute cohomology: a harmonic form is in particular closed, hence represents a de Rham class.

Theorem 12.2.5: Harmonic equals closed and coclosed

Let (M, g) be a closed oriented Riemannian n-manifold and $\alpha \in \Omega^k(M)$. Then

$$\Delta\alpha = 0 \quad \Longleftrightarrow \quad d\alpha = 0 \ \text{ and } \ d^*\alpha = 0$$

That is, a form on a closed manifold is harmonic if and only if it is both closed and coclosed.

***Proof*:**
($\Leftarrow$) Suppose $d\alpha = 0$ and $d^*\alpha = 0$. Then $d^*d\alpha = d^*0 = 0$ and $dd^*\alpha = d\,0 = 0$, so $\Delta\alpha = dd^*\alpha + d^*d\alpha = 0$. This direction uses no closedness of M and no positive-definiteness; it is formal.

($\Rightarrow$) Suppose $\Delta\alpha = 0$. Pairing against α in the L^2 inner product and applying the non-negativity identity (12.10) of proposition 12.2.3 with $\beta = \alpha$,

$$0 = (\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$$

Both terms are non-negative and sum to zero, so each vanishes: $|d\alpha|^2 = 0$ and $|d^*\alpha|^2 = 0$. Since the L^2 inner product is positive-definite (definition 12.1.4), a form of zero L^2-norm is zero, giving $d\alpha = 0$ and $d^*\alpha = 0$. This direction is where the hypotheses are spent: the identity (12.10) needed the adjointness of d and d^*, hence a closed manifold, and the passage from vanishing norm to vanishing form needed positive-definiteness of g.

Both implications hold, as we sought to show.

The theorem is the pivot on which the rest of the chapter turns. It converts the analytically defined harmonic space $\mathcal{H}^k(M) = \ker\Delta$ into an algebraically defined one, the closed and coclosed forms, and this second description is the one that connects to topology. A closed form represents a de Rham cohomology class; the extra condition $d^*\alpha = 0$ selects, within each class, a distinguished representative. That this selection is possible and unique for every class is the Hodge theorem of section 12.4. The equivalence fails without the closedness of M: on $\mathbb{R}^n$ a form can be harmonic in the sense $\Delta\alpha = 0$ (harmonic coordinate functions, for instance) without being closed, because the adjoint relation that produced the identity (12.10) used integration over a manifold without boundary. Compactness is not a convenience here; it is the hypothesis that makes "harmonic" and "closed and coclosed" the same condition.

One consequence follows at once. If α is harmonic and $\beta = d\gamma$ is exact, then $(\alpha, d\gamma) = (d^*\alpha, \gamma) = 0$ since $d^*\alpha = 0$, so harmonic forms are L^2-orthogonal to exact forms; dually, harmonic forms are orthogonal to coexact forms $d^*\eta$, because $(\alpha, d^*\eta) = (d\alpha, \eta) = 0$. These orthogonalities are the germ of the Hodge decomposition, and they are recorded here because they follow from theorem 12.2.5 with no further analysis.

The two extreme degrees can be computed by hand, without any of the elliptic machinery to come, and they furnish the first checks that harmonic forms see topology correctly.

Example 12.2.6: Harmonic forms in degrees zero and n

Let (M, g) be a closed connected oriented Riemannian n-manifold. The harmonic spaces in the top and bottom degrees are one-dimensional, and both computations use only theorem 12.2.5.

Degree zero. A 0-form is a function $f \in C^\infty(M)$, and $d^* f = 0$ automatically because d^* lowers degree below zero. By theorem 12.2.5, f is harmonic if and only if $df = 0$. A function with $df = 0$ is locally constant, and since M is connected it is globally constant. Conversely every constant is closed and coclosed, hence harmonic. Therefore

$$\mathcal{H}^0(M) = \{\,\text{constant functions}\,\} \cong \mathbb{R}$$

Without connectedness the same argument gives $\mathcal{H}^0(M) = \{\,\text{locally constant functions}\,\}$, of dimension equal to the number of connected components. The direct route confirms this: $\Delta f = d^* df$, so $(\Delta f, f) = (df, df) = |df|^2$, and $\Delta f = 0$ forces $df = 0$, recovering the same conclusion through the non-negativity identity.

Degree n. A top-degree form is $\omega = c\, dV_g$ for a function c, since $\Omega^n(M)$ is a rank-one module over $C^\infty(M)$ spanned by the volume form dV_g. The Hodge star of section 12.1 is a linear isomorphism $*\colon \Omega^0(M) \to \Omega^n(M)$ with $*1 = dV_g$, so $*$ carries Ω^0 bijectively onto Ω^n. By proposition 12.2.3(3), $*$ commutes with Δ, hence $*$ carries $\mathcal{H}^0(M) = \ker(\Delta|_{\Omega^0})$ isomorphically onto $\mathcal{H}^n(M) = \ker(\Delta|_{\Omega^n})$. Since $\mathcal{H}^0(M) = \mathbb{R}$ is spanned by the constant 1 and $*1 = dV_g$,

$$\mathcal{H}^n(M) = \mathbb{R} * 1 = \mathbb{R}\, dV_g \cong \mathbb{R},$$

the harmonic n-forms being exactly the constant multiples of the volume form. Directly: $\omega = c\, dV_g$ is automatically closed ($d\omega = 0$ as a top form), and $d^*\omega = 0$ unwinds through $d^* = (-1)^{n(k+1)+1} {*d*}$ to the condition $dc = 0$ on the function $c = *\omega$, again forcing c constant on connected M.

These match the topology. The zeroth de Rham cohomology $H^0_{dR}(M) = \mathbb{R}$ counts connected components, and $\dim \mathcal{H}^0(M) = 1$ for connected M agrees; the top cohomology $H^n_{dR}(M) = \mathbb{R}$ of a closed connected oriented n-manifold is generated by the class of dV_g (its integral is the volume, nonzero), and $\dim \mathcal{H}^n(M) = 1$ agrees. Both are the first instances of the Hodge isomorphism $\mathcal{H}^k(M) \cong H^k_{dR}(M)$ of section 12.4, verified here by hand and, through the star, seen to be exchanged by Poincaré duality in complementary degrees 0 and n.

12.3 Sobolev Spaces and the Elliptic Estimate

The Hodge Laplacian $\Delta = dd^* + d^*d$ built in section 12.2 is a second-order differential operator on smooth forms, and the harmonic space $\mathcal{H}^k(M) = \ker \Delta$ is its kernel among smooth forms. To extract the two facts the Hodge theorem will rest on, that this kernel is finite-dimensional and that the image of Δ is closed and complements the kernel, one must leave the space of smooth forms. Smoothness is not preserved under the limits the argument takes: a sequence of smooth forms with controlled derivatives converges, in general, only to a form of finite differentiability, and the natural home for such limits is a completion of $\Omega^k(M)$ in a norm that measures derivatives up to a fixed order. These completions are the Sobolev spaces of forms, and they are the setting in which Δ acquires the properties a compact operator theory needs.

This section assembles the analytic framework. It defines the Sobolev spaces $H^s\Omega^k(M)$, records that on a closed manifold the norm is independent of the auxiliary choices used to write it down, and computes the one geometric quantity that certifies Δ as an operator to which the elliptic theory applies: its principal symbol. The symbol computation, $\sigma_\xi(\Delta) = |\xi|^2 \operatorname{id}$, is carried out in full, since it is the point at which the geometry of Δ enters. The three analytic theorems the later sections consume, the a priori estimate, elliptic regularity, and Rellich compactness, are theorems of the linear analysis of elliptic operators on closed manifolds; their proofs belong to [3] and are stated here as imported inputs. Throughout, (M, g) is a closed, oriented Riemannian n-manifold with its Levi-Civita connection ∇, and the L^2 inner product on forms is the one of section 12.1. Being Riemannian, g is positive definite, so this L^2 inner product and the pointwise inner product on $\Lambda^k T^*M$ from definition 1.4.1 are genuine inner products.

The exterior derivative d and the codifferential d^* raise and lower the number of derivatives one can measure, so a norm that counts derivatives is the right yardstick. The Levi-Civita connection extends to all the tensor bundles $\Lambda^k T^*M$ (this extension is the induced connection of [6]), and iterating it produces the covariant derivatives $\nabla^j\alpha$, sections of $(T^*M)^{\otimes j} \otimes \Lambda^k T^*M$, whose pointwise norms $\left|\nabla^j\alpha\right|$ are measured by g. Integrating these against the volume form gives a family of norms indexed by the top order s.

Definition 12.3.1: Sobolev spaces of forms

Let (M, g) be a closed oriented Riemannian manifold and $s \geq 0$ an integer. For $\alpha \in \Omega^k(M)$ define the *Sobolev norm* of order s by

$$\|\alpha\|_{H^s}^2 := \sum_{j=0}^{s} \int_M \left|\nabla^j\alpha\right|^2 \, dV,$$

where $\nabla^j\alpha$ is the j-fold iterated covariant derivative and $|\cdot|$ is the pointwise norm induced by g on the relevant tensor bundle. The *Sobolev space* $H^s\Omega^k(M)$ is the completion of $\Omega^k(M)$ with respect to $\|\cdot\|_{H^s}$, a Hilbert space under the inner product

$$\langle\alpha, \beta\rangle_{H^s} := \sum_{j=0}^{s} \int_M \left\langle\nabla^j\alpha, \nabla^j\beta\right\rangle \, dV$$

For $s = 0$ this is the L^2 completion: $H^0\Omega^k(M) = L^2\Omega^k(M)$, with $\|\alpha\|_{H^0} = \|\alpha\|_{L^2}$ the norm of the L^2 inner product $(\alpha, \beta) = \int_M \langle\alpha, \beta\rangle \, dV$ of definition 12.1.4. An element of $H^s\Omega^k(M)$ is a k-form with covariant derivatives up to order s in L^2; it need not be smooth.

The definition packages a form together with a fixed number of its derivatives and asks that all of them be square-integrable. A smooth form on the closed manifold M lies in $H^s\Omega^k(M)$ for every s, because a continuous function on a compact space is bounded and its integral against dV is finite; the interesting elements of $H^s\Omega^k(M)$ are the limits that smoothness does not survive, forms whose s-th derivatives exist only in the L^2 sense. The ascending scale

$$\cdots \subseteq H^{s+1}\Omega^k(M) \subseteq H^s\Omega^k(M) \subseteq \cdots \subseteq H^0\Omega^k(M) = L^2\Omega^k(M)$$

records that demanding more derivatives is a stronger condition, and the inclusions are the ones whose compactness Rellich's theorem asserts. Two features make these spaces usable on a manifold rather than on Euclidean space. First, the definition is intrinsic: it uses only g, its Levi-Civita connection, and the volume form, with no chart or partition of unity in sight. Second, and less visible, the norm does not really depend on the connection chosen to iterate. Replacing ∇ by another connection, or building the norm instead from a finite atlas and a partition of unity as in [3], changes $\|\cdot\|_{H^s}$ to an equivalent norm, one bounded above and below by constant multiples of the present one. The equivalence uses compactness of M decisively, and it is what lets one speak of "the" Sobolev space $H^s\Omega^k(M)$ without tracking the auxiliary data.

12.3.2 Note: equivalence of Sobolev norms On a closed manifold the covariant-derivative norm of definition 12.3.1 and the partition-of-unity norm, in which one pulls each form back to a chart and sums local Euclidean Sobolev norms against a subordinate partition of unity, are equivalent norms, and both define the same Hilbert space $H^s\Omega^k(M)$ with the same underlying topology. The proof that the two are equivalent, and that different choices of atlas, partition, or connection give equivalent norms, is part of the Sobolev theory on compact manifolds and is developed in [3]; compactness of M is the hypothesis that makes it true. All statements below that name a Sobolev norm are insensitive to the choice within this equivalence class.

A concrete instance grounds the definition and previews the spectral computation of the harmonic forms.

Example 12.3.3: Sobolev norms on the flat torus

Take the flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with the metric descended from $\bar{g} = \delta_{ij}\, dx^i\, dx^j$, whose Levi-Civita connection has all Christoffel symbols zero, so ∇ is ordinary coordinate differentiation on the components. A k-form is $\alpha = \sum_{|I|=k} a_I\, dx^I$ with $\mathbb{Z}^n$-periodic coefficients $a_I \in C^\infty(T^n)$, and $\nabla^j\alpha$ has components the partial derivatives $\partial^\beta a_I$ for multi-indices β with $|\beta| = j$. Because the coframe dx^I is g-orthonormal at every point, the Sobolev norm reads off the coefficients:

$$\|\alpha\|_{H^s}^2 = \sum_{|I|=k} \sum_{|\beta|\le s} c_\beta \int_{T^n} \left|\partial^\beta a_I\right|^2 dx,$$

with combinatorial constants c_β counting the orderings of β. Expanding each coefficient in its Fourier series $a_I(x) = \sum_{\xi\in\mathbb{Z}^n} \widehat{a_I}(\xi)\, e^{2\pi i\xi\cdot x}$ turns differentiation into multiplication, $\partial^\beta a_I$ having Fourier coefficient $(2\pi i\xi)^\beta \widehat{a_I}(\xi)$, so by the Parseval identity of [9],

$$\|\alpha\|_{H^s}^2 = \sum_{|I|=k} \sum_{\xi\in\mathbb{Z}^n} \Big(\sum_{|\beta|\le s} c_\beta \left|(2\pi\xi)^\beta\right|^2 \Big) \left|\widehat{a_I}(\xi)\right|^2$$

The bracketed weight is comparable to $(1+|\xi|^2)^s$, uniformly in ξ, so up to the norm equivalence of box 12.3.2 membership in $H^s\Omega^k(T^n)$ is exactly the condition $\sum_I \sum_\xi (1+|\xi|^2)^s \left|\widehat{a_I}(\xi)\right|^2 < \infty$ on the Fourier coefficients. A form is smooth precisely when its Fourier coefficients decay faster than any power of $|\xi|$, which is the intersection $\bigcap_s H^s\Omega^k(T^n)$; this is the Sobolev embedding statement made explicit for the torus. The eigenvalue $4\pi^2\,|\xi|^2$ visible in the weight is the eigenvalue of Δ on the exponential $e^{2\pi i\xi\cdot x}$, a coincidence made structural in the symbol computation below.

With the function spaces in place, the question is which differential operators interact well with them. The exterior derivative and the codifferential each move a form up or down one degree, and each lowers the available Sobolev order by one; the Laplacian, being second order, drops it by two. What makes Δ special, and what earns it the elliptic theory, is the algebraic structure of its top-order part. That structure is captured by the principal symbol, the object obtained by keeping only the highest-order derivatives of an operator and replacing each ∂_j by a cotangent variable ξ_j. For a scalar operator the symbol is a polynomial in ξ; for an operator on forms it is, at each covector ξ, a linear endomorphism of the fiber $\Lambda^k T_p^* M$. Ellipticity is the condition that this endomorphism be invertible for every nonzero ξ, the statement that the operator's top-order part degenerates in no cotangent direction.

The symbol of Δ is assembled from the symbols of d and d^*, so those are computed first. At a point $p \in M$ and a covector $\xi \in T_p^* M$, the principal symbol of a first-order operator P on forms is the fiber endomorphism $\sigma_\xi(P)$ obtained by evaluating P on $e^{itf}\omega$ for a real function f with $df_p = \xi$ and a form ω with $\omega(p)$ the prescribed fiber value, and reading off the coefficient of the top power of t as $t \to \infty$. For the exterior derivative,

$$d\big(e^{itf}\omega\big) = e^{itf}\big(it\, df \wedge \omega + d\omega\big),$$

so the leading term is $it\,\xi \wedge \omega(p)$ and the symbol is the exterior multiplication

$$\sigma_\xi(d) = i\,\xi \wedge (\,\cdot\,) \colon \Lambda^k T_p^* M \to \Lambda^{k+1} T_p^* M$$

The codifferential $d^* = (-1)^{n(k+1)+1} * d *$ of definition 12.1.5 is the formal L^2-adjoint of d by theorem 12.1.6, and taking adjoints reverses the roles of a first-order symbol: the symbol of the adjoint is the fiber adjoint of the symbol, up to the sign that a formal adjoint introduces. The adjoint of exterior multiplication $\xi \wedge (\,\cdot\,)$ with respect to the pointwise inner product on $\Lambda^\bullet T_p^* M$ is interior multiplication (contraction) $\iota_{\xi^\sharp}$ by the metric dual vector $\xi^\sharp$, characterized by $\langle \xi \wedge \omega, \eta\rangle = \big\langle \omega, \iota_{\xi^\sharp}\eta\big\rangle$ for all ω, η. Hence

$$\sigma_\xi(d^*) = -i\,\iota_{\xi^\sharp}(\,\cdot\,) \colon \Lambda^k T_p^* M \to \Lambda^{k-1} T_p^* M,$$

the conjugate of $\sigma_\xi(d)$, as befits an L^2-adjoint. These two computations feed directly into the symbol of the Laplacian.

Proposition 12.3.4: Ellipticity of the Hodge Laplacian

Let (M, g) be a Riemannian manifold and $\Delta = dd^* + d^*d$ the Hodge Laplacian of definition 12.2.1 on $\Omega^k(M)$. For every $p \in M$ and every nonzero covector $\xi \in T_p^* M$, the principal symbol of Δ at (p, ξ) is

$$\sigma_\xi(\Delta) = |\xi|^2\ \mathrm{id} \colon \Lambda^k T_p^* M \to \Lambda^k T_p^* M,$$

scalar multiplication by $|\xi|^2 = \langle \xi, \xi\rangle_{g^{-1}}$. In particular $\sigma_\xi(\Delta)$ is invertible for every $\xi \neq 0$, so Δ is a formally self-adjoint elliptic operator of order 2.

***Proof*:**
The principal symbol is multiplicative and additive: for the composition of two first-order operators, $\sigma_\xi(PQ) = \sigma_\xi(P)\sigma_\xi(Q)$, and for a sum of operators of equal order, $\sigma_\xi(P + Q) = \sigma_\xi(P) + \sigma_\xi(Q)$. Applying this to $\Delta = dd^* + d^*d$ with the first-order symbols computed above,

$$\sigma_\xi(\Delta) = \sigma_\xi(d)\,\sigma_\xi(d^*) + \sigma_\xi(d^*)\,\sigma_\xi(d) = \big(i\,\xi\wedge\big)\big(-i\,\iota_{\xi^\sharp}\big) + \big(-i\,\iota_{\xi^\sharp}\big)\big(i\,\xi\wedge\big)$$

The two factors of $\pm i$ multiply to 1 in each term, so

$$\sigma_\xi(\Delta) = \xi \wedge \iota_{\xi^\sharp}(\,\cdot\,) + \iota_{\xi^\sharp}(\xi \wedge \,\cdot\,)$$

This is the anticommutator of exterior multiplication by ξ and interior multiplication by $\xi^\sharp$, evaluated on $\omega \in \Lambda^k T_p^* M$:

$$\sigma_\xi(\Delta)\omega = \xi \wedge \iota_{\xi^\sharp}\omega + \iota_{\xi^\sharp}(\xi \wedge \omega) \tag{12.13}$$

Interior multiplication is an antiderivation of degree -1: for a 1-form ξ and any ω,

$$\iota_{\xi^\sharp}(\xi \wedge \omega) = (\iota_{\xi^\sharp}\xi)\,\omega - \xi \wedge \iota_{\xi^\sharp}\omega = \langle \xi, \xi \rangle_{g^{-1}}\,\omega - \xi \wedge \iota_{\xi^\sharp}\omega,$$

where $\iota_{\xi^\sharp}\xi = \xi(\xi^\sharp) = \langle \xi, \xi \rangle_{g^{-1}} = |\xi|^2$ is the pairing of the 1-form ξ with its metric dual vector $\xi^\sharp$ from definition 1.4.1, equal to the squared norm of ξ in the inverse metric on $T_p^* M$. Substituting this identity into (12.13), the two terms $\xi \wedge \iota_{\xi^\sharp}\omega$ cancel:

$$\sigma_\xi(\Delta)\omega = \xi \wedge \iota_{\xi^\sharp}\omega + |\xi|^2\,\omega - \xi \wedge \iota_{\xi^\sharp}\omega = |\xi|^2\,\omega$$

Thus $\sigma_\xi(\Delta) = |\xi|^2$ id on $\Lambda^k T_p^* M$, as claimed. For $\xi \neq 0$, positive-definiteness of g gives $|\xi|^2 > 0$, so $\sigma_\xi(\Delta)$ is $|\xi|^2$ times the identity, an invertible endomorphism; ellipticity is exactly this invertibility for all nonzero ξ. Formal self-adjointness of Δ was established in proposition 12.2.3, and Δ has order 2 since its symbol is homogeneous of degree 2 in ξ, completing the proof.

The computation isolates why the Hodge Laplacian is the operator elliptic theory was built for. Its symbol is not just invertible; it is a positive scalar, the same scalar $|\xi|^2$ that appears for the scalar Laplace-Beltrami operator on functions, uniform across all degrees k and all fiber directions ω. The cross terms $\xi \wedge \iota_{\xi^\sharp}\omega$, which would encode a coupling between components of a form, cancel exactly because interior and exterior multiplication by the same covector anticommute up to the length $|\xi|^2$. In the language of Clifford algebra this is the relation $c(\xi)^2 = -|\xi|^2$ underlying the square-root structure of Δ, and it is the algebraic form of the identity $(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$ from proposition 12.2.3: the same anticommutation makes Δ nonnegative and makes its symbol a positive scalar. Positive-definiteness of g enters in one place, the step $|\xi|^2 > 0$ for $\xi \neq 0$; for an indefinite metric $|\xi|^2$ vanishes on the null cone and Δ becomes a wave operator, hyperbolic rather than elliptic, which is why the Hodge theory of this chapter requires a Riemannian metric.

Ellipticity is the input that the three theorems below convert into the finite-dimensionality and closed-range statements the Hodge theorem needs. These are theorems of linear elliptic analysis on a closed manifold, proved in [3]; they are stated here as the imported package, with the geometry entering only through the fact, established in proposition 12.3.4, that Δ is elliptic. The first is the a priori estimate, which controls a form in a higher Sobolev norm by the Laplacian applied to it in a lower one. It says that Δ, though it discards two orders of differentiability in the direction of application, loses nothing in the reverse direction: knowing $\Delta\alpha$ to order s pins down α to order $s+2$, up to a lower-order correction.

Theorem 12.3.5: Elliptic a priori estimate

Let (M, g) be a closed Riemannian manifold and Δ the Hodge Laplacian on $\Omega^k(M)$. For each integer $s \geq 0$ there is a constant $C = C(s, M, g, k) > 0$ such that

$$\|\alpha\|_{H^{s+2}} \leq C\big(\|\Delta\alpha\|_{H^s} + \|\alpha\|_{H^s}\big)$$

for all $\alpha \in \Omega^k(M)$; the estimate extends to all $\alpha \in H^{s+2}\Omega^k(M)$ by density. The estimate holds for any formally self-adjoint elliptic operator of order 2 on a closed manifold.

***Proof*:**
This is the elliptic a priori estimate for an elliptic operator on a closed manifold; a full proof, by way of the Fourier-analytic estimate for a constant-coefficient elliptic operator on Euclidean space, a freezing-coefficients argument in charts, and a partition-of-unity patching, is given in [3]. The single geometric hypothesis it consumes is that Δ is elliptic, verified in proposition 12.3.4; compactness of M supplies the uniform constant C and the equivalence of Sobolev norms of box 12.3.2 that lets the estimate be stated intrinsically. No further input from the geometry of (M, g) is used.

The estimate is the mechanism behind every finiteness statement in the chapter. Read on the harmonic forms, where $\Delta\alpha = 0$, it degenerates to $\|\alpha\|_{H^{s+2}} \leq C\,\|\alpha\|_{H^s}$: a harmonic form bounded in one Sobolev norm is bounded in every higher one, and this boundedness, combined with the compactness of the Sobolev inclusions below, is what forces the harmonic space to be finite-dimensional. Read on a general form, it says that Δ has closed range, the analytic fact from which the Hodge decomposition is built in section 12.4. The additive term $\|\alpha\|_{H^s}$ is unavoidable and harmless: it is present because Δ has a kernel, the constants and their higher-degree analogues, on which $\Delta\alpha = 0$ carries no information, and it will be absorbed by working on the orthogonal complement of the kernel.

The second input promotes weak solutions to genuine ones. A form solving $\Delta\alpha = \beta$ only in the L^2 sense of pairing against test forms is, a priori, no more than square-integrable; regularity theory says that if the right-hand side is smooth then so is the solution. The consequence the chapter needs is that weakly harmonic forms, the L^2 forms annihilated by Δ in the weak sense, are automatically smooth, so that the analytically defined kernel of Δ on L^2 coincides with the harmonic space $\mathcal{H}^k(M)$ of smooth forms.

Theorem 12.3.6: Elliptic regularity

Let (M, g) be a closed Riemannian manifold and Δ the Hodge Laplacian on forms. If $\alpha \in L^2\Omega^k(M)$ satisfies $\Delta\alpha = \beta$ weakly, meaning $(\alpha, \Delta\varphi) = (\beta, \varphi)$ for all $\varphi \in \Omega^k(M)$, and if $\beta \in H^s\Omega^k(M)$, then $\alpha \in H^{s+2}\Omega^k(M)$. In particular, if $\beta \in C^\infty$ then $\alpha \in C^\infty$; and a weakly harmonic form, one with $\Delta\alpha = 0$ weakly, is smooth.

***Proof*:**
This is elliptic regularity for the elliptic operator Δ; the proof, by difference quotients or by iterating the a priori estimate of theorem 12.3.5 up the Sobolev scale together with the Sobolev embedding $\bigcap_s H^s = C^\infty$, is carried out in [3]. As with the a priori estimate, the only geometric hypothesis is ellipticity of Δ from proposition 12.3.4. The final assertion is the case $\beta = 0$, which lies in H^s for every s, so $\alpha \in \bigcap_s H^{s+2}\Omega^k(M) = C^\infty\Omega^k(M)$; a form annihilated by Δ weakly is therefore a smooth harmonic form, completing the proof.

Elliptic regularity is what keeps the entire theory inside the smooth category even though its proofs pass through Sobolev completions. The weak formulation is the price of working in a Hilbert space, where existence of solutions is obtained by orthogonal projection or by the closed-range theorem and produces, a priori, only an L^2 object; regularity refunds the price, returning a smooth form. The last clause is the one used repeatedly: it identifies the L^2-kernel of Δ, an a priori larger space of weak solutions, with the smooth harmonic forms, so that no distinction between weak and strong harmonicity survives on a closed manifold.

The third input supplies compactness. The Sobolev inclusions $H^{s+1} \hookrightarrow H^s$ are bounded but not, on their own, compact; compactness is exactly what upgrades a bounded sequence to one with a convergent subsequence, the property that turns the a priori estimate's boundedness conclusions into finite-dimensionality and closed-range conclusions. On a closed manifold the inclusions are compact, and this is the Rellich–Kondrachov theorem.

Theorem 12.3.7: Rellich–Kondrachov compactness

Let (M, g) be a closed Riemannian manifold. For every integer $s \geq 0$ the inclusion

$$H^{s+1}\Omega^k(M) \hookrightarrow H^s\Omega^k(M)$$

is a compact operator: every sequence bounded in the H^{s+1} norm has a subsequence convergent in the H^s norm. In particular, a sequence of k-forms bounded in $H^1\Omega^k(M)$ has an L^2-convergent subsequence.

***Proof*:**
This is the Rellich–Kondrachov compactness theorem for the Sobolev spaces of a closed manifold; the proof, which localizes to charts via a partition of unity and applies the Euclidean Rellich lemma (bounded sequences in H^{s+1} of a bounded domain are precompact in H^s) on each, is given in [3]. Compactness of M is the hypothesis that makes it hold: the theorem is false on noncompact manifolds, where mass can escape to infinity and a bounded sequence can fail to have any convergent subsequence. No condition beyond compactness of (M, g) enters, completing the account of the imported inputs.

Rellich compactness is where compactness of M pays off analytically. The a priori estimate of theorem 12.3.5 bounds a family of forms in a higher Sobolev norm; Rellich converts that bound into precompactness one norm lower, producing convergent subsequences. The two theorems are used in tandem in the next section: the estimate provides the H^{s+2} bound, Rellich extracts an L^2-convergent subsequence, and the limit lands back among the smooth forms by regularity. Applied to the unit L^2-ball of the harmonic space, where the estimate gives a uniform H^{s+2} bound because Δ vanishes, Rellich yields precompactness of that ball, and a normed space whose unit ball is precompact is finite-dimensional; this is the argument that opens section 12.4. Applied to the full operator, the pair yields the closed range of Δ and hence the Hodge decomposition. The geometry has now contributed everything it will: the operator Δ, the identity making it nonnegative, and the symbol computation proposition 12.3.4 certifying it elliptic. The three imported theorems carry the rest.

12.4 The Hodge Theorem

The pieces are now in place to prove the central result of the chapter. The codifferential d^* of section 12.1 made the exterior derivative into an operator with a formal L^2-adjoint; the Hodge Laplacian $\Delta = dd^* + d^*d$ of section 12.2 is the self-adjoint, non-negative operator whose kernel, the harmonic forms $\mathcal{H}^k(M)$, consists of the closed and coclosed forms by theorem 12.2.5; and the elliptic package of section 12.3, the a priori estimate, elliptic regularity, and Rellich compactness, supplies the analytic control that the geometry alone cannot. What remains is to combine them. The metric is positive-definite throughout: (M, g) is a closed oriented Riemannian n-manifold, and definiteness is what makes the L^2 inner product $(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \, dV$ of definition 12.1.4 an inner product rather than an indefinite pairing, so that “harmonic” means $d\alpha = 0$ and $d^*\alpha = 0$ separately.

The argument has three movements. First, the harmonic space $\mathcal{H}^k(M)$ is finite-dimensional: its unit ball is bounded in a higher Sobolev norm by the elliptic estimate and therefore precompact by Rellich. Second, the Laplacian has closed range, and the space of forms splits L^2-orthogonally as $\Omega^k(M) = \mathcal{H}^k \oplus \Delta\Omega^k$; refining the second summand through $d^2 = (d^*)^2 = 0$ yields the three-term Hodge decomposition. Third, the decomposition identifies each de Rham cohomology class with its unique harmonic representative, giving the isomorphism $\mathcal{H}^k(M) \cong H^k_{dR}(M)$. This last statement is the Hodge theorem. Everything below is proved in full from the three elliptic theorems of section 12.3, which are the only imported inputs; the de Rham cohomology $H^k_{dR}(M)$ and its Betti numbers are taken from [6] and [4].

Proposition 12.4.1: Finite-dimensionality of the harmonic space

Let (M, g) be a closed oriented Riemannian manifold. For each k the space of harmonic k-forms $\mathcal{H}^k(M) = \ker(\Delta\colon \Omega^k \to \Omega^k)$ (definition 12.2.4) is finite-dimensional.

The idea is that harmonicity is a strong constraint measured by the elliptic estimate. A harmonic form satisfies $\Delta\alpha = 0$, so the a priori estimate controls its full second-order Sobolev norm by its L^2 norm alone; on the unit L^2-sphere of $\mathcal{H}^k$ this produces a set bounded in a strictly stronger norm, which Rellich compactness turns into a precompact set. An infinite-dimensional normed space has a non-precompact unit sphere, so $\mathcal{H}^k$ must be finite-dimensional.

***Proof*:**
Recall from section 12.3 that Δ is a formally self-adjoint elliptic operator of order 2 on Ω^k, with the elliptic a priori estimate (proved in [3])

$$\|\alpha\|_{H^{s+2}} \le C\big(\|\Delta\alpha\|_{H^s} + \|\alpha\|_{H^s}\big) \tag{12.14}$$

holding for every smooth k-form α and every $s \ge 0$, with a constant $C = C(s, k)$ independent of α. Here $\|-\|_{H^s}$ is the Sobolev norm on $H^s\Omega^k(M)$ of definition 12.3.1, and $\|-\|_{H^0} = \|-\|_{L^2}$ is the L^2 norm induced by definition 12.1.4.

Step 1: harmonic forms are uniformly H^2-bounded on the unit sphere. Let $\alpha \in \mathcal{H}^k(M)$, so $\Delta\alpha = 0$. Applying (12.14) with $s = 0$ and $\Delta\alpha = 0$ gives

$$\|\alpha\|_{H^2} \le C\ \|\alpha\|_{L^2}$$

Thus on the L^2-unit sphere $\Sigma = \big\{\alpha \in \mathcal{H}^k(M) \ \big|\ \|\alpha\|_{L^2} = 1\big\}$ every element satisfies $\|\alpha\|_{H^2} \le C$: the set Σ is bounded in the H^2-norm.

Step 2: precompactness in L^2. By Rellich compactness ([3]), the inclusion $H^2\Omega^k(M) \hookrightarrow L^2\Omega^k(M)$ is a compact operator: it carries H^2-bounded sets to L^2-precompact sets. Since Σ is H^2-bounded by Step 1, its image under the inclusion, namely Σ itself viewed in L^2, is precompact in the L^2-norm. Therefore the closed unit ball of the normed space $(\mathcal{H}^k(M), \|-\|_{L^2})$ is L^2-precompact, its unit sphere Σ having compact L^2-closure inside the unit ball.

Step 3: finite-dimensionality. A normed vector space whose closed unit ball is precompact is finite-dimensional. This is the Riesz criterion for local compactness: were $\mathcal{H}^k(M)$ infinite-dimensional, one could construct by the Riesz lemma a sequence $\alpha_1, \alpha_2, \ldots$ of unit vectors with $\|\alpha_i - \alpha_j\|_{L^2} \ge \frac{1}{2}$ for all $i \ne j$, a sequence in Σ with no L^2-Cauchy subsequence, contradicting the precompactness of Step 2. Hence $\mathcal{H}^k(M)$ is finite-dimensional, as we sought to show.

Finite-dimensionality is the first payoff of ellipticity and the pivot of everything to come. It says that the harmonic forms, an a priori infinite-dimensional space of solutions to a differential equation, form a space of vectors one can count, and their number will turn out to be the Betti number of M. The proof used only the estimate for the case $\Delta\alpha = 0$ together with Rellich; the same two inputs, applied to $\Delta\alpha = \beta$ with β nonzero, give the closed range that drives the decomposition below.

The analytic heart of what follows is a single fact: the range of Δ is a closed subspace of L^2, equal to the orthogonal complement of the harmonic forms. This is where the a priori estimate does its real work, and once it is in hand the decomposition is a matter of orthogonal algebra. The remaining refinement, splitting the image $\Delta\Omega^k$ into an exact part $d\Omega^{k-1}$ and a coexact part $d^*\Omega^{k+1}$, uses only $d^2 = 0$, $(d^*)^2 = 0$, and the adjunction $(d\alpha, \beta) = (\alpha, d^*\beta)$ of theorem 12.1.6. This is the three-term decomposition the chapter has been aiming at.

Theorem 12.4.2: Hodge decomposition

Let (M, g) be a closed oriented Riemannian n-manifold. For each k the space of smooth k-forms decomposes as an L^2-orthogonal direct sum

$$\Omega^k(M) = \mathcal{H}^k(M) \,\oplus\, d\,\Omega^{k-1}(M) \,\oplus\, d^*\Omega^{k+1}(M), \tag{12.15}$$

equivalently $\Omega^k(M) = \mathcal{H}^k(M) \oplus \Delta\Omega^k(M)$. The three summands are mutually L^2-orthogonal, and every $\alpha \in \Omega^k(M)$ has a unique expression

$$\alpha = h + d\eta + d^*\zeta, \qquad h \in \mathcal{H}^k(M),\ \eta \in \Omega^{k-1}(M),\ \zeta \in \Omega^{k+1}(M)$$

with h the L^2-orthogonal projection of α onto the harmonic forms.

***Proof*:**
The argument first establishes the two-term splitting $\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k$ by proving that Δ has closed range, then refines the image into its exact and coexact parts.

Step 1: harmonics and range are orthogonal, and their orthogonality is sharp. For any smooth forms α, φ with α harmonic, formal self-adjointness of Δ (proposition 12.2.3) gives $(\Delta\varphi, \alpha) = (\varphi, \Delta\alpha) = 0$, so $\Delta\Omega^k \perp \mathcal{H}^k$ in L^2. Conversely, a smooth α with $(\Delta\varphi, \alpha) = 0$ for all φ has, on taking $\varphi = \alpha$,

$$0 = (\Delta\alpha, \alpha) = |d\alpha|^2_{L^2} + |d^*\alpha|^2_{L^2}$$

by the non-negativity identity of proposition 12.2.3, forcing $d\alpha = 0$ and $d^*\alpha = 0$, hence $\Delta\alpha = 0$. So within Ω^k the L^2-orthogonal complement of $\Delta\Omega^k$ is exactly $\mathcal{H}^k$.

Step 2: an a priori estimate off the harmonics. On smooth forms orthogonal to the harmonics the L^2-norm is controlled by that of the Laplacian:

$$\|\alpha\|_{L^2} \le C_0\, \|\Delta\alpha\|_{L^2} \qquad \text{for all } \alpha \in \Omega^k(M) \text{ with } \alpha \perp \mathcal{H}^k \tag{12.16}$$

Were this false, there would be $\alpha_i \in \Omega^k$ with $\alpha_i \perp \mathcal{H}^k$, $\|\alpha_i\|_{L^2} = 1$, and $\|\Delta\alpha_i\|_{L^2} \to 0$. The elliptic estimate (12.14) with $s = 0$ gives $\|\alpha_i\|_{H^2} \le C(\|\Delta\alpha_i\|_{L^2} + 1)$, a bounded quantity, so $\{\alpha_i\}$ is H^2-bounded. By Rellich ([3]) a subsequence, still called α_i, converges in L^2 to some α_∞ with $\|\alpha_\infty\|_{L^2} = 1$. For every smooth φ,

$$(\alpha_\infty, \Delta\varphi) = \lim_i (\alpha_i, \Delta\varphi) = \lim_i (\Delta\alpha_i, \varphi) = 0,$$

using $|(\Delta\alpha_i, \varphi)| \leq \|\Delta\alpha_i\|_{L^2} \|\varphi\|_{L^2} \to 0$, so α_∞ is a weak solution of $\Delta\alpha_\infty = 0$. By elliptic regularity ([3]) α_∞ is smooth and harmonic. Orthogonality to the finite-dimensional space $\mathcal{H}^k$ passes to L^2-limits, so $\alpha_\infty \perp \mathcal{H}^k$; a form both harmonic and orthogonal to all harmonics is zero, contradicting $\|\alpha_\infty\|_{L^2} = 1$. This proves (12.16).

Step 3: $\Delta\Omega^k$ is L^2-closed and $\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k$. Let $\beta_i = \Delta\alpha_i \to \beta$ in L^2. Since Δ annihilates $\mathcal{H}^k$, projecting each α_i off $\mathcal{H}^k$ leaves $\Delta\alpha_i$ unchanged, so assume $\alpha_i \perp \mathcal{H}^k$. By (12.16), $\|\alpha_i - \alpha_j\|_{L^2} \leq C_0 \|\beta_i - \beta_j\|_{L^2} \to 0$, and then by (12.14) applied to the differences, $\|\alpha_i - \alpha_j\|_{H^2} \leq C(\|\beta_i - \beta_j\|_{L^2} + \|\alpha_i - \alpha_j\|_{L^2}) \to 0$, so $\alpha_i \to \alpha$ in H^2 for some $\alpha \in H^2\Omega^k$, and continuity of $\Delta\colon H^2\Omega^k \to L^2\Omega^k$ gives $\Delta\alpha = \beta$. When β is smooth, elliptic regularity ([3]) makes α smooth, so $\beta \in \Delta\Omega^k$: the smooth image is L^2-closed. Consequently $\Delta\Omega^k$ is a closed subspace of the Hilbert space $L^2\Omega^k$, its orthogonal complement (Step 1, extended by regularity to L^2: an L^2-form orthogonal to $\Delta\Omega^k$ is weakly harmonic, hence smooth and harmonic) is $\mathcal{H}^k$, and $L^2\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k$. Restricting to smooth forms, any $\alpha \in \Omega^k$ writes as $h + \Delta\varphi$ with $h \in \mathcal{H}^k$ smooth (proposition 12.4.1) and $\Delta\varphi = \alpha - h$ smooth (φ smooth by regularity), giving the L^2-orthogonal splitting $\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k$ of smooth forms.

Step 4: refining the image into exact and coexact parts. For $\varphi \in \Omega^k$,

$$\Delta\varphi = (dd^* + d^*d)\varphi = d(d^*\varphi) + d^*(d\varphi),$$

with $d^*\varphi \in \Omega^{k-1}$ and $d\varphi \in \Omega^{k+1}$, so every element of $\Delta\Omega^k$ is the sum of an exact form in $d\Omega^{k-1}$ and a coexact form in $d^*\Omega^{k+1}$; thus $\Delta\Omega^k \subseteq d\Omega^{k-1} + d^*\Omega^{k+1}$. For orthogonality, the adjunction of theorem 12.1.6 and $d^2 = 0$ give, for $\eta \in \Omega^{k-1}$ and $\zeta \in \Omega^{k+1}$,

$$(d\eta, d^*\zeta) = (d(d\eta), \zeta) = (0, \zeta) = 0,$$

so $d\Omega^{k-1} \perp d^*\Omega^{k+1}$; and a harmonic h has $dh = 0$, $d^*h = 0$ (theorem 12.2.5), so

$$(h, d\eta) = (d^*h, \eta) = 0, \qquad (h, d^*\zeta) = (dh, \zeta) = 0$$

Hence $\mathcal{H}^k$, $d\Omega^{k-1}$, and $d^*\Omega^{k+1}$ are pairwise L^2-orthogonal. Combining with Step 3,

$$\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k \subseteq \mathcal{H}^k + d\Omega^{k-1} + d^*\Omega^{k+1} \subseteq \Omega^k,$$

so all inclusions are equalities and the sum is an orthogonal direct sum. This is (12.15).

Step 5: uniqueness and the harmonic projection. Orthogonality forces uniqueness of the components. If $h + d\eta + d^*\zeta = h' + d\eta' + d^*\zeta'$ with both sides decomposed as in (12.15), then $(h-h')+(d\eta-d\eta')+(d^*\zeta-d^*\zeta') = 0$ is an orthogonal sum equal to zero, and taking the L^2 inner product of this identity with each summand shows each summand vanishes: $h = h'$, $d\eta = d\eta'$, and $d^*\zeta = d^*\zeta'$. The components η and ζ themselves are not unique, since $d\eta$ is unchanged by adding a closed form to η; the exact and coexact *forms* $d\eta$ and $d^*\zeta$ are the unique data. Finally, since $d\eta$ and $d^*\zeta$ are each L^2-orthogonal to $\mathcal{H}^k$, the harmonic part h is the orthogonal projection of α onto $\mathcal{H}^k$, characterized by $\alpha - h \perp \mathcal{H}^k$, as we sought to show.

The decomposition (12.15) organizes the space of forms by the two ways a form can be constructed, from below by d and from above by d^*, with the harmonic forms as the irreducible remainder that neither operator produces. It refines the elementary observation that exact forms and coexact forms are L^2-orthogonal into a complete accounting: nothing is left over except the harmonics. The next result reads cohomology off this accounting. A closed form has no coexact part, so its decomposition is $h + d\eta$, and its harmonic part is a canonical representative of its de Rham class.

Theorem 12.4.3: The Hodge theorem

Let (M, g) be a closed oriented Riemannian n-manifold. Every de Rham cohomology class in $H^k_{dR}(M)$ contains exactly one harmonic representative. The resulting map

$$\mathcal{H}^k(M) \longrightarrow H^k_{dR}(M), \qquad h \longmapsto [h],$$

sending a harmonic form to its de Rham class, is a linear isomorphism. In particular $\dim \mathcal{H}^k(M) = \dim H^k_{dR}(M) = b_k(M)$, the k-th Betti number of M.

The statement has two halves that reinforce each other: existence and uniqueness of a harmonic representative in each class, and the fact that the assignment of a representative is an isomorphism onto cohomology. The first half is a direct consequence of the decomposition; the second unpacks it into injectivity and surjectivity. That the common dimension is the Betti number then follows because $H^k_{dR}(M) \cong H^k(M; \mathbb{R})$ by the de Rham theorem ([6], [4]) and b_k is by definition its dimension.

***Proof*:**

Recall that $H^k_{dR}(M) = Z^k/B^k$, where $Z^k = \ker(d\colon \Omega^k \to \Omega^{k+1})$ is the space of closed k-forms and $B^k = d\Omega^{k-1}$ is the space of exact k-forms; the de Rham class of a closed form α is $[\alpha] = \alpha + B^k$.

Step 1: harmonic forms are closed; the map is well defined. A harmonic form h satisfies $dh = 0$ (theorem 12.2.5), so $h \in Z^k$ and its class $[h] \in H^k_{dR}(M)$ is defined. The assignment $h \mapsto [h]$ is $\mathbb{R}$-linear because d and passage to the quotient are linear.

Step 2: every class contains a harmonic representative. Let $[\alpha] \in H^k_{dR}(M)$ with $\alpha \in Z^k$ closed. Decompose α by theorem 12.4.2:

$$\alpha = h + d\eta + d^*\zeta, \qquad h \in \mathcal{H}^k,\ \eta \in \Omega^{k-1},\ \zeta \in \Omega^{k+1}$$

The coexact part vanishes. Indeed $d\alpha = 0$, and since $dh = 0$ and $d(d\eta) = 0$, applying d gives $d(d^*\zeta) = 0$; then

$$|d^*\zeta|^2_{L^2} = (d^*\zeta, d^*\zeta) = (d\,d^*\zeta, \zeta) = (0, \zeta) = 0,$$

using the adjunction of theorem 12.1.6, so $d^*\zeta = 0$. Thus $\alpha = h + d\eta$, whence $\alpha - h = d\eta$ is exact and $[\alpha] = [h]$. The harmonic form h is therefore a representative of $[\alpha]$, so the map $h \mapsto [h]$ is surjective.

Step 3: the harmonic representative is unique; the map is injective. Suppose two harmonic forms $h, h' \in \mathcal{H}^k$ lie in the same class, so $h - h' = d\beta$ is exact for some $\beta \in \Omega^{k-1}$. The difference $h - h'$ is both harmonic (a difference of harmonics) and exact. But an exact harmonic form is zero: if $\omega = d\beta \in \mathcal{H}^k$, then $d^*\omega = 0$ by theorem 12.2.5, and

$$|\omega|^2_{L^2} = (d\beta, \omega) = (\beta, d^*\omega) = (\beta, 0) = 0,$$

so $\omega = 0$. Hence $h = h'$, each class has exactly one harmonic representative, and the map $h \mapsto [h]$ is injective. Combined with Step 2, it is a linear isomorphism $\mathcal{H}^k(M) \xrightarrow{\sim} H^k_{dR}(M)$.

Step 4: dimensions. By proposition 12.4.1 the space $\mathcal{H}^k(M)$ is finite-dimensional, so the isomorphism forces $\dim H^k_{dR}(M) = \dim \mathcal{H}^k(M) < \infty$. The de Rham theorem identifies $H^k_{dR}(M)$ with the real singular cohomology $H^k(M; \mathbb{R})$ ([6], [4]), whose dimension is the Betti number $b_k(M)$; therefore $\dim \mathcal{H}^k(M) = b_k(M)$, completing the proof.

The isomorphism turns an infinite-dimensional quotient into a finite-dimensional space of solutions to a partial differential equation. Cohomology, defined purely topologically as closed forms modulo exact ones, is computed by the metric-dependent Laplacian: while the space $\mathcal{H}^k(M)$ depends on g through Δ, its dimension does not, being the topological invariant b_k. Two consequences are worth recording at once, both mandatory checks on the theory. In the extreme degrees the count is transparent. For $k = 0$, a 0-form is a function, and $\Delta f = 0$ means $df = 0$ and $d^* f = 0$; on a connected manifold $df = 0$ forces f to be constant, so $\mathcal{H}^0(M) = \mathbb{R}$ and $b_0 = 1$, the statement that a connected manifold has one component. In the top degree $k = n$, the Hodge star $*\colon \Omega^0 \to \Omega^n$ carries 1 to dV and commutes with Δ (proposition 12.2.3), so it sends $\mathcal{H}^0$ isomorphically to $\mathcal{H}^n$; hence $\mathcal{H}^n(M) = \mathbb{R}\, dV$ and $b_n = 1$ for a connected closed oriented M, recovering that the top de Rham cohomology of such a manifold is one-dimensional, generated by the class of the volume form.

A second reading of the theorem picks out the harmonic representative among all representatives of a class by a variational property: it is the shortest.

Corollary 12.4.4: The harmonic representative minimizes the L^2-norm

Let (M, g) be a closed oriented Riemannian n-manifold and let $c \in H^k_{dR}(M)$ be a de Rham class. Among all closed forms representing c, the harmonic representative $h \in \mathcal{H}^k(M) \cap c$ is the unique one of smallest L^2-norm. That is, for every $\eta \in \Omega^{k-1}(M)$,

$$\|h\|_{L^2} \leq \|h + d\eta\|_{L^2} ,$$

with equality iff $d\eta = 0$.

***Proof*:**
By theorem 12.4.3 the class c has a unique harmonic representative h, and every representative of c has the form $h + d\eta$ for some $\eta \in \Omega^{k-1}(M)$, since two closed forms in the same class differ by an exact form. The harmonic form h is coclosed, $d^* h = 0$, so by the adjunction of theorem 12.1.6,

$$(h, d\eta) = (d^* h, \eta) = (0, \eta) = 0,$$

and $h \perp d\eta$ in L^2. The Pythagorean identity then gives

$$\|h + d\eta\|^2_{L^2} = \|h\|^2_{L^2} + \|d\eta\|^2_{L^2} \geq \|h\|^2_{L^2} ,$$

with equality iff $\|d\eta\|_{L^2} = 0$, that is iff $d\eta = 0$. Hence h minimizes the L^2-norm in its class, uniquely among representatives differing by a nonzero exact form, as we sought to show.

The minimization property gives a second, geometric characterization of the harmonic representative and explains its canonicity: it is the orthogonal projection of any representative onto the affine subspace of the class, the point of the class closest to the origin in the L^2 metric. This is the exact analogue of choosing, among all solutions of an inhomogeneous linear system, the one of least norm.

To see the whole apparatus produce concrete numbers, the flat torus is the ideal test case: there the Laplacian on constant-coefficient forms can be computed by hand, and the harmonic forms turn out to be exactly the constant-coefficient ones, whose count matches the topological Betti numbers.

Example 12.4.5: Harmonic forms of the flat torus

Let $T^n = \mathbb{R}^n/\mathbb{Z}^n$ be the flat torus, the quotient of Euclidean space by the integer lattice, with the flat metric descending from $\bar{g} = \sum_i (dx^i)^2$ (the flat torus of example 1.3.6). The coordinate 1-forms $dx^1, \ldots, dx^n$ are globally defined on T^n and orthonormal at every point, and for a multi-index $I = (i_1 < \cdots < i_k)$ write $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$. A general k-form is $\alpha = \sum_I a_I\, dx^I$ with $a_I \in C^\infty(T^n)$, that is, each a_I a smooth $\mathbb{Z}^n$-periodic function on $\mathbb{R}^n$.

Because the metric is flat and the frame $\{dx^i\}$ is parallel, the Hodge Laplacian acts on such a form coefficientwise through the scalar Laplace-Beltrami operator. On functions the geometer's Laplacian is $\Delta = d^*d = -\sum_i \partial_i^2$ (example 12.1.7: on functions $\Delta f = -\operatorname{div}\operatorname{grad} f$, and on the flat torus $\operatorname{grad} f = \sum_i (\partial_i f)\partial_i$, $\operatorname{div} X = \sum_i \partial_i X^i$, so $\Delta f = -\sum_i \partial_i^2 f$). Applied to $\alpha = \sum_I a_I\, dx^I$, since $d(dx^I) = 0$ and $d^*(dx^I) = 0$ for the constant orthonormal coframe, the product rule leaves

$$\Delta\alpha = \sum_I (\Delta a_I)\, dx^I = -\sum_I \Big(\sum_{j=1}^n \partial_j^2 a_I\Big) dx^I$$

Thus α is harmonic iff every coefficient a_I satisfies $\Delta a_I = 0$, that is, iff each a_I is a harmonic periodic function on $\mathbb{R}^n$.

A harmonic function on the closed manifold T^n is constant: if $\Delta a = 0$ then $0 = (\Delta a, a) = |da|_{L^2}^2$ by proposition 12.2.3, forcing $da = 0$, hence a constant on the connected T^n. Therefore the harmonic k-forms are exactly the constant-coefficient forms

$$\mathcal{H}^k(T^n) = \Big\{ \textstyle\sum_I c_I\, dx^I : c_I \in \mathbb{R} \Big\},$$

one real parameter c_I for each increasing multi-index I of length k. The number of such multi-indices is $\binom{n}{k}$, so

$$\dim \mathcal{H}^k(T^n) = \binom{n}{k}$$

By theorem 12.4.3 this equals the Betti number $b_k(T^n)$, matching the known cohomology of the torus, $H^k_{dR}(T^n) \cong \mathbb{R}^{\binom{n}{k}}$, computed topologically from the Künneth formula for $T^n = (S^1)^n$ ([4]). The Euler characteristic assembled from these dimensions is $\sum_k (-1)^k \binom{n}{k} = (1-1)^n = 0$ for $n \geq 1$, consistent with $\chi(T^n) = 0$.

The round sphere S^n (with any Riemannian metric) provides a contrasting instance where the count is forced by the extreme-degree computations above. Since S^n is connected, closed, and oriented, $\mathcal{H}^0(S^n) = \mathbb{R}$ and $\mathcal{H}^n(S^n) = \mathbb{R}\, dV$, so $b_0 = b_n = 1$. For $0 < k < n$ the de Rham cohomology $H^k_{dR}(S^n)$ vanishes ([4]), so by theorem 12.4.3 $\mathcal{H}^k(S^n) = 0$ and the only harmonic forms on the sphere in intermediate degrees are zero. The Hodge count thus reproduces $b_k(S^n) = 1$ for $k \in \{0, n\}$ and 0 otherwise, and $\chi(S^n) = 1 + (-1)^n$, which is 2 for n even and 0 for n odd.

12.5 Poincaré Duality, Finiteness, and the Spectrum

The Hodge theorem of section 12.4 identified each de Rham cohomology class of a closed oriented Riemannian manifold with a single distinguished form, its harmonic representative, and so replaced the infinite-dimensional space of closed forms in a class by the finite-dimensional space $\mathcal{H}^k(M)$ of solutions to $\Delta\alpha = 0$. This section harvests the consequences. Three are structural. First, the Hodge star $*$, which exchanges degrees k and $n-k$ and commutes with the Laplacian, carries harmonic forms to harmonic forms and thereby produces Poincaré duality as a statement about a symmetric analytic operator rather than about triangulations or dual cell complexes. Second, the finiteness of $\mathcal{H}^k(M)$ transfers at once to de Rham cohomology, so that a closed manifold has finite Betti numbers and a well-defined Euler characteristic, closing the circle with the Gauss–Bonnet and Chern–Weil computations of section 8.2 and section 11.3. Third, the same elliptic machinery that made $\mathcal{H}^k$ finite-dimensional resolves the entire operator Δ: its spectrum is a discrete sequence of eigenvalues marching off to infinity, with smooth eigenforms furnishing an orthonormal basis of the L^2 completion.

The analytic inputs are exactly those already assembled and cited in section 12.3: the elliptic a priori estimate, elliptic regularity, and Rellich compactness, all cited to [3], together with one further imported result used only in the spectral theorem, the spectral theorem for compact self-adjoint operators on a Hilbert space, cited to [7]. Everything else is derived in full from the Hodge decomposition and the properties of $*$ and Δ. Throughout, (M,g) is a closed (compact, without boundary) oriented Riemannian n-manifold, positive-definiteness in force so that the pointwise inner product $\langle -,-\rangle$ on $\Lambda^k T_p^* M$ and the resulting L^2 inner product $(\alpha,\beta)=\int_M \alpha\wedge *\beta$ are genuine inner products; $\mathcal{H}^k(M)=\ker(\Delta|_{\Omega^k})$ is the space of harmonic k-forms, and $b_k=\dim H^k_{dR}(M)$ is the k-th Betti number.

The first result is the cleanest illustration of the principle that a geometric symmetry of Δ becomes a symmetry of cohomology. The Hodge star commutes with the Laplacian, a fact recorded in proposition 12.2.3, so it does not merely permute forms of complementary degree but restricts to an isomorphism of harmonic spaces. Transported across the Hodge isomorphism of theorem 12.4.3, this yields the duality between H^k_{dR} and H^{n-k}_{dR}.

Theorem 12.5.1: Poincaré Duality via the Hodge Star

Let (M,g) be a closed oriented Riemannian n-manifold.

1. The Hodge star restricts to a linear isomorphism
$$*\colon \mathcal{H}^k(M) \xrightarrow{\sim} \mathcal{H}^{n-k}(M)$$
on harmonic forms. Consequently $b_k=b_{n-k}$ for every k.
2. The bilinear pairing on de Rham cohomology
$$H^k_{dR}(M)\times H^{n-k}_{dR}(M)\longrightarrow \mathbb{R}, \qquad ([\alpha],[\beta])\longmapsto \int_M \alpha\wedge\beta$$
is well defined and nondegenerate.

***Proof*:**
The two parts are proved in turn.

1. *The star preserves harmonicity.* By proposition 12.2.3 the Hodge star commutes with the Laplacian, $*\Delta=\Delta *$ on every Ω^k. Hence if $\alpha\in\mathcal{H}^k(M)$, that is $\Delta\alpha=0$, then
$$\Delta(*\alpha)=*(\Delta\alpha)=*0=0$$
so $*\alpha\in\mathcal{H}^{n-k}(M)$, and $*$ maps $\mathcal{H}^k$ into $\mathcal{H}^{n-k}$. The same reasoning maps $\mathcal{H}^{n-k}$ into

$\mathcal{H}^k$. To see these two maps are mutually inverse, recall from proposition 12.1.3 that on Ω^k one has $** = (-1)^{k(n-k)}\,\mathrm{id}$. The scalar $(-1)^{k(n-k)}$ is nonzero, so $*$ has a two-sided inverse (namely $(-1)^{k(n-k)}*$) and is a linear isomorphism $\mathcal{H}^k(M) \to \mathcal{H}^{n-k}(M)$. Taking dimensions and invoking the Hodge isomorphism $\dim \mathcal{H}^k(M) = \dim H^k_{dR}(M) = b_k$ of theorem 12.4.3 gives

$$b_k = \dim \mathcal{H}^k(M) = \dim \mathcal{H}^{n-k}(M) = b_{n-k}$$

2. *The cohomological pairing is well defined and nondegenerate.* That the pairing descends to cohomology is a consequence of Stokes' theorem. If α is a closed k-form and β a closed $(n-k)$-form, then for any $(k-1)$-form η,

$$\int_M (\alpha + d\eta) \wedge \beta = \int_M \alpha \wedge \beta + \int_M d\eta \wedge \beta$$

and $d\eta \wedge \beta = d(\eta \wedge \beta) - (-1)^{k-1}\eta \wedge d\beta = d(\eta \wedge \beta)$ since $d\beta = 0$, so $\int_M d\eta \wedge \beta = \int_M d(\eta \wedge \beta) = 0$ by Stokes' theorem on the closed manifold M. The same computation in the second slot shows the value $\int_M \alpha \wedge \beta$ depends only on the classes $[\alpha]$ and $[\beta]$; the pairing is well defined.

For nondegeneracy, fix a nonzero class in $H^k_{dR}(M)$ and represent it by its harmonic representative $\alpha \in \mathcal{H}^k(M)$, which is nonzero by the uniqueness in theorem 12.4.3. Take $\beta = *\alpha \in \mathcal{H}^{n-k}(M)$, a harmonic $(n-k)$-form representing a class in $H^{n-k}_{dR}(M)$. Then, by the defining property of the star from definition 12.1.4,

$$\int_M \alpha \wedge \beta = \int_M \alpha \wedge *\alpha = \int_M \langle \alpha, \alpha \rangle \; dV = (\alpha, \alpha) = |\alpha|^2_{L^2} > 0$$

the inequality strict because $\alpha \neq 0$ and the L^2 inner product is positive-definite (positive-definiteness of g enters here). Thus the linear functional $[\beta'] \mapsto \int_M \alpha \wedge \beta'$ on $H^{n-k}_{dR}(M)$ is nonzero, so no nonzero class in $H^k_{dR}(M)$ pairs to zero with all of $H^{n-k}_{dR}(M)$. Since H^k_{dR} and H^{n-k}_{dR} have equal finite dimension by part (1), a bilinear pairing between finite-dimensional spaces of the same dimension with no left kernel has no right kernel either, and is nondegenerate, as we sought to show.

The theorem recasts a topological duality as an analytic one. On the topological side, Poincaré duality for a closed oriented n-manifold is the statement that cap product with the fundamental class identifies H^k with H_{n-k}, proved combinatorially in [4] through dual cell decompositions. The de Rham pairing above is the cohomological incarnation of that duality, and the Hodge star supplies its inverse in closed form: given a class, its Poincaré dual is represented by the star of its harmonic representative. The symmetry $b_k = b_{n-k}$ of the Betti numbers is the numerical statement of it. For a connected closed oriented M this already recovers $b_0 = b_n = 1$, since $\mathcal{H}^0(M) = \mathbb{R}$ consists of the constants and $*$ carries the constant function 1 to the volume form dV spanning $\mathcal{H}^n(M) = \mathbb{R}\, dV$, matching the cross-check that the star exchanges the bottom and top harmonic spaces. The mechanism is worth isolating: the star is an isometry of forms that intertwines Δ with itself, and any such intertwiner of a self-adjoint operator preserves its kernel. Duality is then automatic.

The finiteness of harmonic spaces, established in proposition 12.4.1 from the elliptic estimate and Rellich compactness, passes through the Hodge isomorphism to de Rham cohomology and delivers the invariant that Chapters 8 and 11 computed by curvature.

Corollary 12.5.2: Finiteness of de Rham Cohomology and the Euler Characteristic

Let (M, g) be a closed oriented Riemannian n-manifold. Then each de Rham cohomology space is finite-dimensional, with

$$\dim H^k_{dR}(M) = \dim \mathcal{H}^k(M) = b_k < \infty$$

and the Euler characteristic is computed by the alternating sum of the harmonic dimensions,

$$\chi(M) = \sum_{k=0}^{n} (-1)^k b_k = \sum_{k=0}^{n} (-1)^k \dim \mathcal{H}^k(M)$$

***Proof*:**
By proposition 12.4.1 each $\mathcal{H}^k(M)$ is finite-dimensional, and by the Hodge isomorphism theorem 12.4.3 the linear map $\mathcal{H}^k(M) \to H^k_{dR}(M)$ sending a harmonic form to its class is an isomorphism. Hence $\dim H^k_{dR}(M) = \dim \mathcal{H}^k(M) = b_k < \infty$, the first displayed equality.

The Euler characteristic of M is defined as the alternating sum of the Betti numbers, $\chi(M) = \sum_{k=0}^{n} (-1)^k b_k$ (see [4]); this sum is finite because each b_k is finite and $b_k = 0$ for $k > n$ and $k < 0$. Substituting $b_k = \dim \mathcal{H}^k(M)$ from the first part gives

$$\chi(M) = \sum_{k=0}^{n} (-1)^k b_k = \sum_{k=0}^{n} (-1)^k \dim \mathcal{H}^k(M)$$

completing the proof.

This corollary is the analytic counterpart of the topological finiteness of the cohomology of a compact manifold. The same number $\chi(M)$ has now been reached by three routes: as the alternating sum of Betti numbers from singular theory, as the integral of the Pfaffian of the curvature through the generalized Gauss–Bonnet theorem of section 8.2, and here as the alternating sum of dimensions of harmonic spaces. That the three agree is the content of the Hodge isomorphism combined with de Rham's theorem: harmonic forms compute de Rham cohomology, de Rham cohomology equals singular cohomology, and the Euler class integrates to $\chi(M)$. Poincaré duality supplies a companion constraint. For an odd-dimensional closed oriented M, pairing $b_k = b_{n-k}$ against the alternating sum forces $\chi(M) = 0$: the terms cancel in pairs $(k, n-k)$ of opposite parity, since n odd makes k and $n-k$ differ in parity. This is the manifold-analysis proof that odd-dimensional closed orientable manifolds have vanishing Euler characteristic, consistent with the vanishing of the Euler class in odd degree noted in section 11.3.

A concrete instance grounds both results at once. The flat torus makes every harmonic space explicit, so its Betti numbers, its Poincaré duality, and its Euler characteristic can all be read off by hand.

Example 12.5.3: Betti Numbers and Euler Characteristic of the Flat Torus
On the flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with its quotient flat metric, the harmonic k-forms are exactly the constant-coefficient forms $\sum_{|I|=k} a_I \, dx^I$, as computed in example 12.4.5: a form with constant coefficients has d and d^* both vanishing, hence is harmonic, and conversely averaging over the torus shows every harmonic form has constant coefficients. The space of such forms has one basis

vector $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ for each strictly increasing multi-index $I = (i_1 < \cdots < i_k)$, so

$$b_k(T^n) = \dim \mathcal{H}^k(T^n) = \binom{n}{k}$$

Poincaré duality $b_k = b_{n-k}$ is visible directly as the identity $\binom{n}{k} = \binom{n}{n-k}$, and the star $*(dx^I) = \pm\, dx^{I^c}$ sends the basis form indexed by I to the one indexed by the complementary multi-index I^c, realizing the isomorphism $\mathcal{H}^k \xrightarrow{\sim} \mathcal{H}^{n-k}$ of theorem 12.5.1. The Euler characteristic is then

$$\chi(T^n) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} = (1-1)^n = 0$$

by the binomial theorem, vanishing for every $n \geq 1$, in agreement with the Poincaré-duality argument above (and with T^n carrying a nowhere-zero vector field). For $n = 2$ the counts are $b_0 = b_2 = 1$ and $b_1 = 2$, so $\chi(T^2) = 1 - 2 + 1 = 0$, the flat structure making manifest what Gauss–Bonnet delivers as $\int_{T^2} K\, dV = 0$ for the flat metric.

The remaining consequence concerns the operator Δ in full, not only its kernel. On a closed manifold Δ acting on $\Omega^k(M)$ behaves like a self-adjoint matrix of infinite size: it is diagonalizable, with a discrete real spectrum accumulating only at infinity and finite-dimensional eigenspaces. The proof is the standard passage from an unbounded operator with compact resolvent to the spectral theorem for compact self-adjoint operators, and every geometric ingredient it needs has been prepared. The Laplacian is non-negative and formally self-adjoint by proposition 12.2.3; the shifted operator $\Delta + \mathrm{id}$ is bounded below and so admits a bounded inverse, the Green's operator, whose values the a priori estimate lands two Sobolev degrees higher; Rellich compactness turns that gain of regularity into compactness; and elliptic regularity makes the resulting eigenforms smooth. The Hodge decomposition of theorem 12.4.2, $\Omega^k = \mathcal{H}^k \oplus \Delta\Omega^k$, reappears at the end as the bottom eigenspace $\mathcal{H}^k$ sitting at $\lambda = 0$.

Theorem 12.5.4: Spectrum of the Hodge Laplacian

Let (M, g) be a closed oriented Riemannian n-manifold and fix a degree k. Let $L^2\Omega^k(M)$ denote the L^2 completion of $\Omega^k(M)$. Then there is a sequence of real numbers

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \longrightarrow +\infty$$

and an L^2-orthonormal basis $(\varphi_j)_{j \geq 0}$ of $L^2\Omega^k(M)$ consisting of *smooth* k-forms with

$$\Delta \varphi_j = \lambda_j \varphi_j$$

Each eigenvalue is non-negative and has finite multiplicity, the eigenvalues have no finite accumulation point, and the eigenspace for $\lambda = 0$ is the harmonic space $\mathcal{H}^k(M)$.

***Proof*:**
The proof produces a compact self-adjoint operator whose eigenforms are those of Δ, then invokes the spectral theorem.

Step 1: the operator $\Delta + \mathrm{id}$ *is bounded below and has closed range.* For $\alpha \in \Omega^k(M)$, the non-negativity identity of proposition 12.2.3 gives

$$((\Delta + \mathrm{id})\alpha, \alpha) = (\Delta\alpha, \alpha) + (\alpha, \alpha) = |d\alpha|_{L^2}^2 + |d^*\alpha|_{L^2}^2 + |\alpha|_{L^2}^2 \geq |\alpha|_{L^2}^2$$

so by the Cauchy–Schwarz inequality $|(\Delta + \mathrm{id})\alpha|_{L^2}\, |\alpha|_{L^2} \geq |\alpha|_{L^2}^2$, whence

$$|(\Delta + \mathrm{id})\alpha|_{L^2} \geq |\alpha|_{L^2} \tag{12.17}$$

for all $\alpha \in \Omega^k(M)$. In particular $\Delta + \mathrm{id}$ is injective on $\Omega^k(M)$. Its range is dense in $L^2\Omega^k(M)$: if $\eta \in L^2\Omega^k$ is L^2-orthogonal to the range, then $((\Delta + \mathrm{id})\alpha, \eta) = 0$ for every $\alpha \in \Omega^k(M)$, so η is a weak solution of $(\Delta + \mathrm{id})\eta = 0$; by elliptic regularity theorem 12.3.6 of section 12.3, η is smooth, hence $\eta \in \Omega^k(M)$ satisfies $(\Delta + \mathrm{id})\eta = 0$, and (12.17) forces $\eta = 0$. Thus the range of $\Delta + \mathrm{id}$ on $\Omega^k(M)$ is dense.

Step 2: the Green's operator $G = (\Delta + \mathrm{id})^{-1}$ is bounded, self-adjoint, and compact. Define G on the range of $\Delta + \mathrm{id}$ by $G((\Delta + \mathrm{id})\alpha) = \alpha$; this is single-valued by injectivity, and (12.17) reads $|G\beta|_{L^2} \leq |\beta|_{L^2}$ on the range, so G extends uniquely to a bounded operator on the closure of the range, which by Step 1 is all of $L^2\Omega^k(M)$, with operator norm at most 1. For self-adjointness, self-adjointness of Δ gives $((\Delta+\mathrm{id})\alpha, \beta) = (\alpha, (\Delta+\mathrm{id})\beta)$ for $\alpha, \beta \in \Omega^k(M)$; substituting $\alpha = G\mu$, $\beta = G\nu$ with μ, ν in the range yields $(\mu, G\nu) = (G\mu, \nu)$, so G is symmetric on a dense subspace and hence, being bounded, self-adjoint on $L^2\Omega^k(M)$. For compactness, apply the elliptic a priori estimate. If $\beta \in \Omega^k(M)$ and $\alpha = G\beta$, then $(\Delta + \mathrm{id})\alpha = \beta$, so $\Delta\alpha = \beta - \alpha$, and the estimate theorem 12.3.5 of section 12.3, in the form $\|\alpha\|_{H^2} \leq C(\|\Delta\alpha\|_{L^2} + \|\alpha\|_{L^2})$, together with $|\alpha|_{L^2} \leq |\beta|_{L^2}$, gives

$$\|G\beta\|_{H^2} = \|\alpha\|_{H^2} \leq C\big(|\beta - \alpha|_{L^2} + |\alpha|_{L^2}\big) \leq C' \, |\beta|_{L^2}$$

so $G\colon L^2\Omega^k \to H^2\Omega^k$ is bounded (by density of Ω^k in L^2 the bound holds on all of $L^2\Omega^k$). By Rellich compactness theorem 12.3.7, the inclusion $H^2\Omega^k \hookrightarrow H^0\Omega^k = L^2\Omega^k$ is compact, so the composite $G\colon L^2\Omega^k \to L^2\Omega^k$ is compact, being a bounded map into H^2 followed by a compact inclusion.

Step 3: diagonalize G by the spectral theorem. G is a compact self-adjoint operator on the Hilbert space $L^2\Omega^k(M)$. By the spectral theorem for compact self-adjoint operators ([7]), $L^2\Omega^k(M)$ admits an orthonormal basis $(\varphi_j)_{j\geq 0}$ of eigenvectors of G, with real eigenvalues $\mu_j \to 0$, each nonzero eigenvalue of finite multiplicity. Moreover G is injective. Its range contains the dense subspace $\Omega^k(M)$, since $G((\Delta + \mathrm{id})\alpha) = \alpha$ realizes every $\alpha \in \Omega^k(M)$ as a value of G; thus G has dense range. A bounded self-adjoint operator with dense range has trivial kernel, because $\ker G$ is the L^2-orthogonal complement of $\overline{\operatorname{ran} G} = L^2\Omega^k$. Hence 0 is not an eigenvalue of G, and every μ_j lies in $(0, 1]$ by the norm bound $|G| \leq 1$ of Step 2.

Step 4: transfer the eigendata back to Δ. From $G\varphi_j = \mu_j\varphi_j$ with $\mu_j \neq 0$, applying $\Delta + \mathrm{id}$ gives $\varphi_j = \mu_j(\Delta + \mathrm{id})\varphi_j$, that is

$$(\Delta + \mathrm{id})\varphi_j = \tfrac{1}{\mu_j}\varphi_j, \qquad \text{equivalently} \qquad \Delta\varphi_j = \Big(\tfrac{1}{\mu_j} - 1\Big)\varphi_j$$

Set $\lambda_j := \frac{1}{\mu_j} - 1$. A priori φ_j is only an L^2 eigenform, so the identity $\Delta\varphi_j = \lambda_j\varphi_j$ holds in the weak sense, exhibiting φ_j as a weak solution of the elliptic equation $\Delta\varphi_j - \lambda_j\varphi_j = 0$; by elliptic regularity theorem 12.3.6 of section 12.3, φ_j is smooth. Hence each $\varphi_j \in \Omega^k(M)$ and $\Delta\varphi_j = \lambda_j\varphi_j$ holds classically. Non-negativity of Δ (proposition 12.2.3) forces $\lambda_j = (\Delta\varphi_j, \varphi_j)/(\varphi_j, \varphi_j) \geq 0$, consistent with $\mu_j \in (0, 1]$; the value $\lambda_j = 0$ occurs exactly when $\mu_j = 1$, which is the eigenvalue of G on the largest eigenspace, and there $\Delta\varphi_j = 0$, so the $\lambda = 0$ eigenspace is $\mathcal{H}^k(M)$. Finally, $\mu_j \to 0$ translates to $\lambda_j = \frac{1}{\mu_j} - 1 \to +\infty$, so the eigenvalues accumulate only at $+\infty$ and each finite eigenvalue has finite multiplicity (matching the finite multiplicity of the corresponding μ_j). Reindexing the λ_j in increasing order with multiplicity gives $0 = \lambda_0 \le \lambda_1 \le \cdots \to +\infty$ and the orthonormal basis (φ_j) of smooth eigenforms, completing the proof.

The theorem is the statement that Δ, though unbounded, is as tractable as a symmetric matrix. Every k-form expands in the eigenbasis (φ_j) as a convergent L^2 series $\alpha = \sum_j (\alpha, \varphi_j)\varphi_j$, and Δ acts term by term through the multipliers λ_j. The bottom of the spectrum, $\lambda_0 = 0$, is the harmonic space, so the multiplicity of the eigenvalue 0 is the Betti number b_k; the theorem thus contains the finiteness of $\mathcal{H}^k$ as the special assertion that the zero eigenvalue has finite multiplicity. The higher eigenvalues carry the geometry not seen by cohomology. On functions ($k = 0$) the operator is the Laplace–Beltrami operator, and its spectrum is the list of frequencies at which the manifold resonates; the growth rate of λ_j with j encodes the dimension and volume of M through Weyl's asymptotic law, and the coincidence or divergence of two spectra is the question of whether one can hear the shape of a drum. The eigenforms φ_j for $\lambda_j > 0$ are the exact and coexact building blocks: since Δ commutes with d and d^*, each positive eigenspace splits into d-images and d^*-images, refining the Hodge decomposition of theorem 12.4.2 eigenvalue by eigenvalue.

The spectrum is computable in closed form for the flat torus, where the eigenforms are Fourier modes and the eigenvalues are the squared lengths of lattice vectors. This is the concrete anchor for the abstract spectral theorem and the model on which every heat-kernel and zeta-function computation in the subject is first tested.

Example 12.5.5: The Spectrum of the Laplacian on the Flat Torus

Take the flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with the flat metric descending from $\bar{g} = \sum_i (dx^i)^2$ on $\mathbb{R}^n$, and consider Δ acting on functions ($k = 0$). Here $\det(g_{ij}) = 1$ and $g^{ij} = \delta^{ij}$, so the Laplace–Beltrami operator of example 1.4.8 reduces to

$$\Delta f = -\operatorname{div}\operatorname{grad} f = -\sum_{i=1}^{n} \frac{\partial^2 f}{\partial (x^i)^2}$$

the geometer's (non-negative) Euclidean Laplacian with the leading minus sign. Smooth functions on T^n are the smooth $\mathbb{Z}^n$-periodic functions on $\mathbb{R}^n$, and the natural candidate eigenfunctions are the characters

$$\varphi_\xi(x) = e^{2\pi i\,\xi\cdot x}, \qquad \xi = (\xi_1, \dots, \xi_n) \in \mathbb{Z}^n$$

which are well defined on the quotient precisely because $\xi \in \mathbb{Z}^n$ makes φ_ξ invariant under $x \mapsto x+m$ for $m \in \mathbb{Z}^n$. Differentiating,

$$\Delta\varphi_\xi = -\sum_{i=1}^{n} \frac{\partial^2}{\partial (x^i)^2} e^{2\pi i\,\xi\cdot x} = -\sum_{i=1}^{n} (2\pi i\,\xi_i)^2\,\varphi_\xi = \sum_{i=1}^{n} 4\pi^2 \xi_i^2\,\varphi_\xi = 4\pi^2\,|\xi|^2\,\varphi_\xi$$

so φ_ξ is an eigenfunction with eigenvalue $\lambda_\xi = 4\pi^2\,|\xi|^2$, where $|\xi|^2 = \sum_i \xi_i^2$. The family $\{\varphi_\xi\}_{\xi\in\mathbb{Z}^n}$ is the standard Fourier orthonormal basis of $L^2(T^n)$ (orthonormality $\int_{T^n} \varphi_\xi \overline{\varphi_\eta}\, dV = \delta_{\xi\eta}$ is the orthogonality of characters), so by completeness these are *all* the eigenfunctions and the spectrum is exactly

$$\{\, 4\pi^2\,|\xi|^2 : \xi \in \mathbb{Z}^n \,\} = \{\, 4\pi^2 m : m \in \mathbb{Z}_{\ge 0} \text{ a sum of } n \text{ squares} \,\}$$

The smallest eigenvalue is $\lambda = 0$, attained only at $\xi = 0$ where $\varphi_0 \equiv 1$ is the constant function; its eigenspace is one-dimensional, matching $b_0(T^n) = \dim \mathcal{H}^0(T^n) = 1$ from example 12.5.3. Each positive eigenvalue $4\pi^2 \left|\xi\right|^2$ has multiplicity equal to the number of lattice points $\xi \in \mathbb{Z}^n$ on the sphere of radius $|\xi|$, a finite number, in agreement with the finite multiplicity asserted in theorem 12.5.4; for $n = 2$ this multiplicity is the number of ways to write $m = \xi_1^2 + \xi_2^2$ as a sum of two squares, an arithmetic count that fluctuates with m. The eigenvalues $4\pi^2 \left|\xi\right|^2$ are unbounded as $|\xi| \to \infty$, confirming $\lambda_j \to +\infty$, and their number below a bound R grows like the volume $\frac{\omega_n}{(2\pi)^n} R^{n/2}$ of the corresponding ball, the flat-torus instance of Weyl's law.

13

The Bochner Technique

The Hodge theorem identified the cohomology of a closed manifold with the kernel of the Laplacian, the space of harmonic forms. This final chapter of Part II asks what curvature does to that kernel, and the answer completes the arc the whole book has followed. Curvature governed geodesics and topology in Part I and characteristic classes in the preceding chapters; here it governs the harmonic forms directly, through a single identity that couples the analytic Laplacian to the geometric curvature. The identity is the Weitzenböck formula, and the method of exploiting it is Bochner's technique: a sign condition on the curvature forces the vanishing of harmonic fields, and by the Hodge theorem of vanishing Betti numbers.

The mechanism is a comparison of two Laplacians. Besides the Hodge Laplacian $\Delta = dd^* + d^*d$ built from the exterior derivative, a Riemannian manifold carries the connection Laplacian $\nabla^*\nabla$ built from the covariant derivative, and both are non-negative self-adjoint operators. They are not equal; their difference is a curvature term. For one-forms the Weitzenböck formula reads $\Delta = \nabla^*\nabla + \mathrm{Ric}$, with the Ricci curvature acting as the correction. Integrating this identity against a harmonic one-form makes the left side vanish, leaving a sum of two non-negative quantities, the L^2-norm of the covariant derivative and the integral of the Ricci curvature on the form; when the Ricci curvature is non-negative, both must vanish, so the form is parallel and the first Betti number is controlled. When the Ricci curvature is strictly positive the harmonic one-forms disappear entirely, and the first Betti number is zero. This is Bochner's theorem, and it is the analytic companion to Bonnet–Myers: where Bonnet–Myers turned positive Ricci curvature into compactness and a finite fundamental group, Bochner turns it into the vanishing of b_1. The chapter closes by placing the argument in its general frame, the Weitzenböck formulas in every degree and for every natural operator, and by pointing to where the technique goes next: the Lichnerowicz formula for the Dirac operator and the vanishing theorems of spin geometry, and the Bochner–Kodaira–Nakano identities that yield Kodaira vanishing on Kähler manifolds, developed in [2]. Curvature, the single object this book has tracked from the first chapter, is shown one last time to control the shape of a manifold, now through the harmonic forms that compute its cohomology.

13.1 The Connection Laplacian and the Weitzenböck Formula

The Hodge theorem of chapter 12 identified the harmonic k-forms $\mathcal{H}^k(M) = \ker \Delta$ with de Rham cohomology, so that the dimension of a space of solutions to a differential equation computes a topological invariant, $\dim \mathcal{H}^k(M) = b_k(M)$. That identification is silent about the geometry of M: it holds for any metric, and the Hodge Laplacian $\Delta = dd^* + d^*d$ carries no curvature in its definition. The Bochner technique breaks this silence. Its engine is a single algebraic identity that splits Δ into two pieces of opposite character: a manifestly non-negative second-order operator built from the Levi-Civita connection alone, and a zeroth-order curvature term. On 1-forms the curvature term is exactly the Ricci curvature of section 4.4. When the Ricci curvature has a sign, so does the curvature term, and the two-piece decomposition then constrains the harmonic forms, hence the Betti numbers. This section builds the two pieces and proves the identity that relates them; the next turns the identity into a vanishing theorem.

Throughout this chapter (M, g) is a closed oriented Riemannian n-manifold, so the metric is positive-definite and M is compact without boundary. This is where positive-definiteness is used: the connection Laplacian below is non-negative only for a definite metric, and the L^2 inner product $(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \, dV$ of definition 12.1.4 is an inner product rather than an indefinite pairing only then. The Levi-Civita connection ∇ of definition 2.3.2 acts on all tensor bundles by definition 2.4.6, the curvature endomorphism R is that of definition 4.1.2 in the sign convention of box 4.1.1, and the Ricci curvature Ric is that of definition 4.4.1. The Einstein summation convention is not used; frame sums are written out, and $(e_i)_{i=1}^n$ always denotes a local orthonormal frame.

The first object is the Laplacian built from the connection, without reference to d or d^*. The connection ∇ turns a tensor field T into a tensor field ∇T with one extra covariant slot, and iterating gives the second covariant derivative $\nabla^2 T$ with two extra slots. Taking the metric trace of those two slots, and negating so that the result is non-negative, produces a second-order operator on tensor fields of the original type.

Definition 13.1.1: The connection Laplacian

Let (M, g) be a Riemannian manifold with Levi-Civita connection ∇, and let T be a smooth tensor field. The *second covariant derivative* of T is the tensor field $\nabla^2 T$ whose value on vector fields X, Y is

$$\nabla^2_{X,Y} T = \nabla_X(\nabla_Y T) - \nabla_{\nabla_X Y} T$$

the correction term $-\nabla_{\nabla_X Y} T$ making $\nabla^2_{X,Y} T$ tensorial (that is, $C^\infty(M)$-linear) in both X and Y. The *connection Laplacian*, also called the *rough* or *Bochner Laplacian*, is the operator

$$\nabla^* \nabla T = -\mathrm{tr}_g \nabla^2 T = -\sum_{i=1}^n \nabla^2_{e_i, e_i} T$$

the negative metric trace of $\nabla^2 T$ over its two derivative slots, evaluated in any local orthonormal frame (e_i). It preserves the tensor type of T.

Three points fix the object. First, $\nabla^2_{X,Y}$ is not $\nabla_X \nabla_Y$: the naive composite $\nabla_X \nabla_Y T$ fails to be tensorial in Y, because replacing Y by fY produces an extra term $(Xf)\nabla_Y T$; subtracting $\nabla_{\nabla_X Y} T$ cancels it, and $\nabla^2_{X,Y} T$ is then a tensorial $(0, 2)$-slot object in X, Y, so its metric trace is well defined. Second, the trace is frame-independent: for any two orthonormal frames the sum $\sum_i \nabla^2_{e_i, e_i} T$ is the same, being the metric contraction $g^{ij}\nabla^2_{\partial_i,\partial_j}T$ of a tensor. Third, the minus sign is the geometer's convention, chosen so that $\nabla^*\nabla$ is non-negative, in parallel with the geometer's Hodge Laplacian $\Delta \geq 0$ of definition 12.2.1; the notation $\nabla^*\nabla$ anticipates the fact, proved next, that this operator is the composite of ∇ with its formal L^2-adjoint ∇^*.

The name deserves justification before the general statement, and the flat case supplies it in one line. On Euclidean space the connection is the ordinary directional derivative and the second covariant derivative is the ordinary Hessian of components, so the connection Laplacian is the componentwise negative Euclidean Laplacian.

Example 13.1.2: The connection Laplacian on Euclidean space

On $(\mathbb{R}^n, \bar{g})$ with $\bar{g} = \sum_i (dx^i)^2$ the Christoffel symbols vanish (example 4.1.6), so ∇ acting on a tensor field differentiates its components in the coordinate frame directly, with no correction terms. For a 1-form $\alpha = \sum_j a_j\, dx^j$ the frame (∂_i) is orthonormal and parallel, $\nabla_{\partial_i}\partial_j = 0$, hence $\nabla_{\nabla_{\partial_i}\partial_i}\alpha = 0$ and

$$\nabla^2_{\partial_i,\partial_i}\alpha = \nabla_{\partial_i}\nabla_{\partial_i}\alpha = \sum_j \big(\partial_i\partial_i a_j\big)\, dx^j$$

Summing over i and negating,

$$\nabla^*\nabla\alpha = -\sum_{i=1}^n \nabla^2_{\partial_i,\partial_i}\alpha = -\sum_j \Big(\sum_{i=1}^n \partial_i\partial_i a_j\Big) dx^j$$

so $\nabla^*\nabla$ acts as the negative of the ordinary Euclidean Laplacian on each coefficient a_j separately. This is the same operator, coefficientwise, that the flat torus will inherit below, and it is where the sign convention shows its purpose: the eigenvalues of $-\sum_i \partial_i^2$ on periodic functions are non-negative, matching the general non-negativity established next.

The connection Laplacian shares with the Hodge Laplacian the two structural properties that make Hodge theory run: formal self-adjointness and non-negativity. Both come from one integration by parts, and both are the analogues for ∇ of the identity $(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$ of proposition 12.2.3. The point is that pairing $\nabla^*\nabla T$ against T recovers the full L^2-norm of ∇T, so the kernel of $\nabla^*\nabla$ on a closed manifold is exactly the parallel fields. This is the property the Bochner argument will exploit: harmonic forms with vanishing curvature term are forced to be parallel, a far stronger conclusion than being closed and coclosed alone.

Proposition 13.1.3: Non-negativity of the connection Laplacian

Let (M, g) be a closed oriented Riemannian n-manifold and let T range over smooth tensor fields of a fixed type, with the L^2 inner product $(S, T) = \int_M \langle S, T\rangle \, dV$ of definition 12.1.4. Then the connection Laplacian $\nabla^*\nabla$ of definition 13.1.1 satisfies the following.

1. *Self-adjointness and the energy identity.* For all S, T,
$$(\nabla^*\nabla S, T) = (\nabla S, \nabla T) = (S, \nabla^*\nabla T)$$
in particular $(\nabla^*\nabla T, T) = |\nabla T|^2_{L^2} \geq 0$.
2. *Kernel.* $\nabla^*\nabla T = 0$ if and only if $\nabla T = 0$; that is, the kernel of $\nabla^*\nabla$ is exactly the space of parallel tensor fields.

The heart of part (1) is that ∇^*, the operator appearing in the name, really is the formal L^2-adjoint of ∇, and this is what an integration by parts on a closed manifold delivers. The divergence theorem removes the boundary term, exactly as Stokes' theorem did for the codifferential in theorem 12.1.6.

***Proof*:**
Fix a point p and work in a local orthonormal frame (e_i) that is *normal* at p, meaning $\nabla e_i(p) = 0$ (parallel transport of an orthonormal basis along radial geodesics produces such a frame from proposition 3.2.8). All frame sums below are over $i = 1, \ldots, n$.

Step 1: the pointwise divergence identity. Let S, T be tensor fields of the same type. Define the vector field W by
$$\langle W, X\rangle = \langle \nabla_X S, T\rangle \qquad \text{for all } X$$
that is, W is the metric dual of the 1-form $X \mapsto \langle \nabla_X S, T\rangle$. Its divergence, computed in the normal frame at p where $\nabla e_i(p) = 0$, is
$$\operatorname{div} W = \sum_i \langle \nabla_{e_i} W, e_i\rangle = \sum_i e_i \langle W, e_i\rangle = \sum_i e_i \langle \nabla_{e_i} S, T\rangle$$
using $\nabla_{e_i} e_i(p) = 0$ in the first equality. Expanding the derivative by metric compatibility of ∇ (definition 2.2.4), and again using $\nabla_{e_i} e_i(p) = 0$ so that $\nabla^2_{e_i, e_i} S = \nabla_{e_i} \nabla_{e_i} S$ at p,
$$\operatorname{div} W = \sum_i \Big(\langle \nabla_{e_i} \nabla_{e_i} S, T\rangle + \langle \nabla_{e_i} S, \nabla_{e_i} T\rangle \Big) = \sum_i \left\langle \nabla^2_{e_i, e_i} S, T\right\rangle + \sum_i \langle \nabla_{e_i} S, \nabla_{e_i} T\rangle$$
at p. The first sum is $-\langle \nabla^*\nabla S, T\rangle$ by definition 13.1.1, and the second is $\langle \nabla S, \nabla T\rangle$, the full contraction of the tensors ∇S and ∇T against the frame. Since p was arbitrary and both sides are tensorial expressions independent of the frame,
$$\operatorname{div} W = \langle \nabla S, \nabla T\rangle - \langle \nabla^*\nabla S, T\rangle \tag{13.1}$$
holds at every point of M.

Step 2: integrate. Integrate (13.1) over the closed manifold M. The divergence theorem gives $\int_M \operatorname{div} W \, dV = 0$, since M is compact without boundary (this is the divergence-form Stokes' theorem, with no boundary term). Hence
$$0 = \int_M \langle \nabla S, \nabla T\rangle \, dV - \int_M \langle \nabla^*\nabla S, T\rangle \, dV = (\nabla S, \nabla T) - (\nabla^*\nabla S, T)$$

so $(\nabla^*\nabla S, T) = (\nabla S, \nabla T)$. The right-hand pairing $(\nabla S, \nabla T)$ is symmetric in S and T, so running the argument with the roles of S and T exchanged gives $(\nabla^*\nabla T, S) = (\nabla T, \nabla S) = (\nabla S, \nabla T)$, whence $(\nabla^*\nabla S, T) = (S, \nabla^*\nabla T)$: the operator is formally self-adjoint. Setting $S = T$ yields the energy identity $(\nabla^*\nabla T, T) = (\nabla T, \nabla T) = |\nabla T|^2_{L^2} \geq 0$, which proves (1).

Step 3: the kernel. If $\nabla T = 0$ then $\nabla^2 T = 0$, so $\nabla^*\nabla T = -\mathrm{tr}_g \nabla^2 T = 0$; parallel fields lie in the kernel. Conversely, suppose $\nabla^*\nabla T = 0$. Pairing against T and applying the energy identity of (1),

$$0 = (\nabla^*\nabla T, T) = |\nabla T|^2_{L^2} = \int_M |\nabla T|^2 \, dV$$

The integrand $|\nabla T|^2$ is continuous and non-negative, and its integral over M vanishes, so $|\nabla T|^2 \equiv 0$, that is $\nabla T = 0$. This is where positive-definiteness of g is used, so that $|\nabla T|^2 \geq 0$ pointwise with equality only when $\nabla T = 0$. Hence T is parallel, and the kernel of $\nabla^*\nabla$ is exactly the parallel tensor fields, completing the proof.

The energy identity $(\nabla^*\nabla T, T) = |\nabla T|^2_{L^2}$ is the whole reason the connection Laplacian enters the subject. It measures the connection Laplacian of T by the size of the full covariant derivative ∇T, with no cancellation: a form can be closed and coclosed, so that $(\Delta\alpha, \alpha) = |d\alpha|^2 + |d^*\alpha|^2$ vanishes, without any of its individual covariant derivatives vanishing. The connection Laplacian sees all of ∇T, and its kernel is correspondingly smaller, consisting only of the parallel fields. A parallel 1-form on a connected manifold is determined by its value at a single point, since parallel transport is a linear isometry (proposition 2.4.4) and reconstructs the form everywhere from one value; so the space of parallel 1-forms has dimension at most n. That observation, combined with the identity of the next theorem, is the entire content of Bochner's theorem in the following section.

What connects the connection Laplacian to cohomology is that it differs from the Hodge Laplacian by a curvature term, and on 1-forms that term is the Ricci curvature. This is the Weitzenböck formula, the technical center of the chapter. Both operators are built from ∇ (the Hodge Laplacian through d and d^*, which are the antisymmetrized and traced parts of ∇), so their difference is a zeroth-order operator, a bundle endomorphism with no derivatives at all. The endomorphism is measured by how far second covariant derivatives are from commuting, and that failure is precisely the curvature. The identity that governs the commutation of covariant derivatives is the Ricci identity, which unwinds the definition of the curvature endomorphism into a statement about second derivatives of a tensor.

Before the theorem, the Ricci identity is recorded in the form the proof uses. For a vector field Z, the definition of the curvature endomorphism (definition 4.1.2) rearranges, using $\nabla^2_{X,Y} = \nabla_X\nabla_Y - \nabla_{\nabla_X Y}$ and $\nabla_X Y - \nabla_Y X = [X, Y]$ (torsion-freeness of the Levi-Civita connection, definition 2.3.2), into

$$\nabla^2_{X,Y} Z - \nabla^2_{Y,X} Z = R(X, Y)Z \tag{13.2}$$

Indeed the left-hand side is $\nabla_X\nabla_Y Z - \nabla_{\nabla_X Y} Z - \nabla_Y\nabla_X Z + \nabla_{\nabla_Y X} Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]} Z$, which is $R(X, Y)Z$ by definition 4.1.2. The same identity holds on any tensor bundle, with $R(X, Y)$ acting as the induced curvature endomorphism (definition 2.4.6); on a 1-form α the induced action is

$$\big(R(X, Y)\alpha\big)(Z) = -\alpha\big(R(X, Y)Z\big)$$

the minus sign coming from the dual pairing, exactly as ∇ on 1-forms differentiates the pairing $\alpha(Z)$ against ∇Z (definition 2.4.6). Equation (13.2) is the mechanism by which curvature enters the Weitzenböck formula, and getting its sign right is the crux of the computation.

Theorem 13.1.4: The Weitzenböck formula for one-forms

Let (M, g) be a Riemannian manifold with Hodge Laplacian $\Delta = dd^* + d^*d$ of definition 12.2.1 and connection Laplacian $\nabla^*\nabla$ of definition 13.1.1. For every smooth 1-form α,

$$\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha) \tag{13.3}$$

where $\mathrm{Ric}(\alpha)$ is the 1-form obtained by applying the Ricci curvature as an endomorphism: in a local orthonormal frame with dual coframe, if $\alpha^\sharp$ is the vector field dual to α, then $\mathrm{Ric}(\alpha) = \mathrm{Ric}(\alpha^\sharp, -)$, with components $\mathrm{Ric}(\alpha)_i = \sum_j R_{ij}\,\alpha^j$ against the Ricci tensor $R_{ij} = \mathrm{Ric}(e_i, e_j)$ of definition 4.4.1. The Ricci term enters with a plus sign.

The formula says that the two natural Laplacians on 1-forms differ by the Ricci curvature, viewed as a symmetric bundle endomorphism of T^*M. The plus sign is not a convention to be chosen freely; it is forced, in the geometer's sign for both Laplacians, by the sign convention of box 4.1.1 for curvature, and it is checked against the flat torus and the round sphere below. The proof expands $\Delta = dd^* + d^*d$ on a 1-form, matches the second-derivative terms against $\nabla^*\nabla$, and identifies the leftover as the Ricci curvature by commuting a pair of covariant derivatives through (13.2).

***Proof*:**
Fix $p \in M$ and a local orthonormal frame (e_i) that is normal at p, so $\nabla e_i(p) = 0$ (proposition 3.2.8); let (e^i) be the dual orthonormal coframe, with $e^i(e_j) = \delta_{ij}$ and $\nabla e^i(p) = 0$. All expressions are evaluated at p, where the frame is parallel, so that covariant derivatives of the frame vectors and coframe covectors drop out; the resulting identity is tensorial and therefore holds everywhere once verified at each p. Write $\alpha_i = \alpha(e_i)$ for the frame components, and abbreviate $\nabla_i = \nabla_{e_i}$ and $\nabla^2_{ij} = \nabla^2_{e_i,e_j}$. The Einstein convention is not in force; every sum is displayed.

Step 1: the two Hodge terms in the frame. On 1-forms the codifferential is minus the divergence of the dual vector field (example 12.1.7), which in the normal frame at p reads

$$d^*\alpha = -\sum_i (\nabla_{e_i}\alpha)(e_i)$$

a function. For the exterior derivative, the torsion-free formula $d\alpha(X, Y) = (\nabla_X\alpha)(Y) - (\nabla_Y\alpha)(X)$ holds because the Levi-Civita connection is torsion-free (definition 2.3.2); thus

$$d\alpha = \sum_{i<j} \big((\nabla_i\alpha)(e_j) - (\nabla_j\alpha)(e_i)\big)\, e^i \wedge e^j$$

The two second-order operators d^*d and dd^* are computed at p by differentiating these once more and using $\nabla e_i(p) = 0$, $\nabla e^i(p) = 0$.

Step 2: the term $d^*d\,\alpha$. Apply d^* to the 2-form $d\alpha$. The codifferential of a 2-form β is, at p in the normal frame, $d^*\beta = -\sum_i \iota_{e_i}(\nabla_{e_i}\beta)$, that is $(d^*\beta)(Y) = -\sum_i(\nabla_{e_i}\beta)(e_i, Y)$. With $\beta = d\alpha$ and $(d\alpha)(X, Y) = (\nabla_X\alpha)(Y) - (\nabla_Y\alpha)(X)$, differentiating and evaluating at p (where the frame is parallel, so ∇ passes onto the components) gives, on the argument e_k,

$$(d^*d\alpha)(e_k) = -\sum_i (\nabla_{e_i} d\alpha)(e_i, e_k) = -\sum_i \Big(\nabla^2_{ii}\alpha(e_k) - \nabla^2_{ik}\alpha(e_i)\Big)$$

where $\nabla^2_{ij}\alpha(e_k)$ abbreviates $(\nabla^2_{e_i,e_j}\alpha)(e_k)$. The first group of terms is

$$-\sum_i \nabla^2_{ii}\alpha(e_k) = \Big(-\mathrm{tr}_g \nabla^2\alpha\Big)(e_k) = (\nabla^*\nabla\alpha)(e_k)$$

by definition 13.1.1. So

$$(d^*d\alpha)(e_k) = (\nabla^*\nabla\alpha)(e_k) + \sum_i \nabla^2_{ik}\alpha(e_i) \tag{13.4}$$

Step 3: the term $dd^*\alpha$. The codifferential $d^*\alpha = -\sum_i(\nabla_{e_i}\alpha)(e_i)$ is a function, and $dd^*\alpha$ is its differential, so

$$(dd^*\alpha)(e_k) = \nabla_{e_k}(d^*\alpha) = -\sum_i \nabla_{e_k}\big((\nabla_{e_i}\alpha)(e_i)\big) = -\sum_i \nabla^2_{ki}\alpha(e_i)$$

at p, where the outer derivative ∇_{e_k} passes onto the components because the frame is parallel there. Thus

$$(dd^*\alpha)(e_k) = -\sum_i \nabla^2_{ki}\alpha(e_i) \tag{13.5}$$

Step 4: assemble and commute the derivatives. Add (13.4) and (13.5):

$$(\Delta\alpha)(e_k) = (\nabla^*\nabla\alpha)(e_k) + \sum_i\Big(\nabla^2_{ik}\alpha(e_i) - \nabla^2_{ki}\alpha(e_i)\Big)$$

The bracket is the commutator of the second covariant derivatives in the order (i,k) against (k,i), so the Ricci identity (13.2) for the 1-form α applies:

$$\nabla^2_{e_i,e_k}\alpha - \nabla^2_{e_k,e_i}\alpha = R(e_i,e_k)\alpha$$

as 1-forms, where $R(e_i,e_k)$ acts on 1-forms by $\big(R(e_i,e_k)\alpha\big)(Z) = -\alpha\big(R(e_i,e_k)Z\big)$. Evaluating on e_i and summing over i,

$$\sum_i\Big(\nabla^2_{ik}\alpha(e_i) - \nabla^2_{ki}\alpha(e_i)\Big) = \sum_i\big(R(e_i,e_k)\alpha\big)(e_i) = -\sum_i \alpha\big(R(e_i,e_k)e_i\big)$$

Step 5: identify the curvature sum as Ricci. The remaining sum is a contraction of the curvature endomorphism against the metric dual of α. Write $\alpha = \sum_j \alpha_j\, e^j$, so $\alpha(V) = \sum_j \alpha_j \langle V, e_j\rangle$ for any V, and

$$-\sum_i \alpha\big(R(e_i,e_k)e_i\big) = -\sum_{i,j}\alpha_j\,\langle R(e_i,e_k)e_i, e_j\rangle = \sum_{i,j}\alpha_j\,\langle R(e_i,e_k)e_j, e_i\rangle$$

the last equality by the antisymmetry $\langle R(X,Y)Z,W\rangle = -\langle R(X,Y)W,Z\rangle$ in the final pair, proposition 4.2.1(ii), applied with $Z=e_i$, $W=e_j$. The inner sum over i is now exactly the Ricci trace of definition 4.4.1, $\mathrm{Ric}(X,Y) = \sum_i \langle R(e_i,X)Y, e_i\rangle$, evaluated at $X=e_k$, $Y=e_j$:

$$\sum_i \langle R(e_i,e_k)e_j, e_i\rangle = \mathrm{Ric}(e_k,e_j) = R_{kj}$$

so the sum equals

$$\sum_j \alpha_j R_{kj} = \sum_j R_{kj}\, \alpha^j = \mathrm{Ric}(\alpha)(e_k)$$

where $\alpha^j = \alpha_j$ are the components of $\alpha^\sharp$ in the orthonormal frame, and the last equality is the definition of $\mathrm{Ric}(\alpha)$ in the statement (the Ricci tensor being symmetric, $R_{kj} = R_{jk}$, by proposition 4.4.3, either index order gives the same 1-form). Therefore

$$(\Delta\alpha)(e_k) = (\nabla^*\nabla\alpha)(e_k) + \mathrm{Ric}(\alpha)(e_k)$$

for every k at p. Both sides are components of tensors and p was arbitrary, so $\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha)$ holds on all of M, with the Ricci term entering with a plus sign, as we sought to show.

Two readings of (13.3) are worth separating. Read left to right, it computes the Hodge Laplacian, the operator that sees cohomology, in terms of the connection Laplacian, the operator that sees parallelism, plus a curvature correction. Read as a decomposition, it splits Δ into a non-negative part $\nabla^*\nabla$ and a zeroth-order part Ric; when Ric has a definite sign, the splitting pits the two parts against each other, and this is the source of every Bochner-type vanishing theorem. The sign of the Ricci term is what makes positive Ricci curvature obstruct harmonic 1-forms rather than produce them: integrating $(\Delta\alpha, \alpha)$ for a harmonic α turns (13.3) into $0 = |\nabla\alpha|^2_{L^2} + \int_M \mathrm{Ric}(\alpha, \alpha)\, dV$, a sum of two terms that are both non-negative when $\mathrm{Ric} \geq 0$, forcing each to vanish. That computation is the whole of the next section; here the formula is first checked on the one manifold where every term can be written out by hand.

The flat torus is the calibration point for the sign and for the whole method. On it the Ricci curvature vanishes identically, so the Weitzenböck formula collapses to $\Delta = \nabla^*\nabla$ on 1-forms, and the harmonic 1-forms are exactly the parallel ones, which are the constant-coefficient forms already computed cohomologically in example 12.4.5. The two counts of $\dim \mathcal{H}^1(T^n)$, one topological and one through parallelism, must agree, and they do.

Example 13.1.5: The Weitzenböck formula on the flat torus

Let $T^n = \mathbb{R}^n/\mathbb{Z}^n$ be the flat torus with the metric descended from $\bar{g} = \sum_i (dx^i)^2$ (example 1.3.6). The coordinate 1-forms $dx^1, \ldots, dx^n$ are globally defined, orthonormal at every point, and *parallel*, $\nabla(dx^j) = 0$, since the metric is flat and the coframe has constant coefficients (example 4.1.6 for the vanishing Christoffel symbols, descending to the quotient). The curvature vanishes, so by example 4.4.2 the Ricci curvature is identically zero,

$$\mathrm{Ric} \equiv 0 \qquad \text{on } T^n$$

The Weitzenböck formula (13.3) therefore reads, for every 1-form α on T^n,

$$\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha) = \nabla^*\nabla\alpha$$

the two Laplacians coincide. Writing $\alpha = \sum_j a_j\, dx^j$ with each $a_j \in C^\infty(T^n)$, both sides act coefficientwise through the scalar Laplacian, because the coframe is parallel: as in example 13.1.2, $\nabla^*\nabla\alpha = \sum_j (-\sum_i \partial_i\partial_i a_j)\, dx^j = \sum_j (\Delta a_j)\, dx^j$, matching the coefficientwise action of Δ found in example 12.4.5.

Now read off the harmonic 1-forms. By proposition 13.1.3, on the closed manifold T^n a 1-form is in the kernel of $\nabla^*\nabla$ if and only if it is parallel; since $\Delta = \nabla^*\nabla$ here, the harmonic 1-forms are exactly the parallel 1-forms. A parallel 1-form has constant coefficients in the parallel coframe

(dx^j): from $\nabla\alpha = 0$ and $\nabla(dx^j) = 0$ one gets $\sum_j (da_j) \otimes dx^j = 0$, so every $da_j = 0$ and each a_j is constant on the connected T^n. Hence

$$\mathcal{H}^1(T^n) = \left\{ \textstyle\sum_{j=1}^n c_j \, dx^j : c_j \in \mathbb{R} \right\}$$

one real parameter per coordinate direction, so $\dim \mathcal{H}^1(T^n) = n$. This is the same space and the same count that example 12.4.5 obtained from the Hodge theorem, where the harmonic 1-forms were found to be the constant-coefficient forms and $\dim \mathcal{H}^1(T^n) = \binom{n}{1} = n = b_1(T^n)$. The Bochner reading adds the geometric content the cohomological count did not display: these harmonic forms are not only closed and coclosed, they are *parallel*, the flatness of the torus being exactly what allows n independent parallel directions to exist. On a closed manifold with $\mathrm{Ric} > 0$ somewhere, the same formula will forbid any nonzero parallel harmonic 1-form, and b_1 will drop to zero; the round sphere S^n with $n \geq 2$, where $\mathrm{Ric} = \frac{n-1}{r^2} g > 0$ (example 4.4.9), is the model for that opposite extreme, and there $b_1(S^n) = 0$ in agreement with $H^1(S^n) = 0$ (example 12.4.5, closing paragraph). The torus and the sphere thus bracket the two signs of the Ricci term in (13.3), and the plus sign is what places each on the correct side.

13.2 Bochner's Theorem

The Weitzenböck formula of section 13.1 splits the Hodge Laplacian on 1-forms into two operators of opposite pedigree: the connection Laplacian $\nabla^*\nabla$, which is analytic and non-negative, and the Ricci curvature Ric, which is geometric and can carry either sign. On its own each term is transparent. Their sum is the Hodge Laplacian, whose kernel the Hodge theorem of chapter 12 identifies with cohomology. Bochner's insight is that this splitting can be read backwards: information about the sign of the curvature term constrains the kernel of the sum, and therefore constrains a topological invariant. When $\mathrm{Ric} \geq 0$ the two terms cannot cancel destructively, and a harmonic 1-form is forced to be parallel; when $\mathrm{Ric} > 0$ somewhere, no nonzero harmonic 1-form survives at all. The first Betti number, defined analytically in section 12.4 as the dimension of the space of harmonic 1-forms, is thereby bounded, and under strict positivity killed outright.

Throughout this section (M, g) is a closed oriented Riemannian n-manifold with Levi-Civita connection ∇, and the metric is positive-definite. This is where definiteness is used without reservation: the L^2 inner product $(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \, dV$ (definition 12.1.4) is an inner product, so that $(\Delta\alpha, \alpha) = 0$ with both constituent terms non-negative forces each to vanish separately. The engine is the integrated Weitzenböck identity: pairing the pointwise formula $\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha)$ of theorem 13.1.4 against a harmonic α and integrating turns a local decomposition of a differential operator into a global obstruction. The argument is short once the machinery of section 13.1 is in hand; its consequences are the payoff of the whole chapter.

The mechanism deserves to be stated before the theorem, because it is the reason positivity of a curvature term forces vanishing of a cohomology class. A harmonic form lies in the kernel of Δ, so its Weitzenböck decomposition $\nabla^*\nabla\alpha + \mathrm{Ric}(\alpha)$ vanishes as a form. Testing this against α in L^2 separates it into two integrals, one measuring the size of $\nabla\alpha$ and one measuring the Ricci curvature paired with α. Both are non-negative under $\mathrm{Ric} \geq 0$, and they sum to zero, so both are zero. A vanishing $|\nabla\alpha|_{L^2}$ says α is parallel; a vanishing Ricci integral, when Ric is strictly positive somewhere, says α vanishes there, and a parallel form vanishing at one point vanishes everywhere.

Theorem 13.2.1: Bochner's Vanishing Theorem

Let (M, g) be a closed oriented Riemannian n-manifold whose Ricci curvature (definition 4.4.1) is non-negative, $\mathrm{Ric} \geq 0$ (that is, $\mathrm{Ric}(v, v) \geq 0$ for every tangent vector v). Then:

1. every harmonic 1-form $\alpha \in \mathcal{H}^1(M)$ is parallel, $\nabla\alpha = 0$, and satisfies $\mathrm{Ric}(\alpha, \alpha) \equiv 0$;
2. the first Betti number is bounded by the dimension, $b_1(M) = \dim \mathcal{H}^1(M) \leq n$;
3. if in addition $\mathrm{Ric} > 0$ at some point of M (that is, $\mathrm{Ric}_p(v, v) > 0$ for every nonzero $v \in T_pM$ at some p), then every harmonic 1-form is zero, so $b_1(M) = 0$.

***Proof*:**
By the Hodge theorem (theorem 12.4.3) the space $\mathcal{H}^1(M)$ of harmonic 1-forms is finite-dimensional with $\dim \mathcal{H}^1(M) = b_1(M)$, and by theorem 12.2.5 a form is harmonic exactly when it is closed and coclosed, so $\Delta\alpha = 0$ for $\alpha \in \mathcal{H}^1(M)$. The proof turns entirely on integrating the Weitzenböck formula against such an α.

1. *Every harmonic 1-form is parallel.* Fix $\alpha \in \mathcal{H}^1(M)$, so $\Delta\alpha = 0$. The Weitzenböck formula of theorem 13.1.4 reads
$$\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha),$$
where $\mathrm{Ric}(\alpha)$ is the 1-form with $\mathrm{Ric}(\alpha)_i = R_{ij}\alpha^j$ in an orthonormal coframe, dual to $\mathrm{Ric}(\alpha^\sharp, -)$. Take the L^2 inner product of both sides with α and use $\Delta\alpha = 0$:
$$0 = (\Delta\alpha, \alpha) = (\nabla^*\nabla\alpha, \alpha) + (\mathrm{Ric}(\alpha), \alpha) \tag{13.6}$$
The first term on the right is computed by proposition 13.1.3: on a closed manifold $(\nabla^*\nabla\alpha, \alpha) = |\nabla\alpha|^2_{L^2}$, the L^2-norm of the full covariant derivative. The second term is the integral of the pointwise pairing $\langle \mathrm{Ric}(\alpha), \alpha\rangle$, which in the orthonormal coframe is $R_{ij}\alpha^j\alpha^i = \mathrm{Ric}(\alpha^\sharp, \alpha^\sharp)$; writing this as $\mathrm{Ric}(\alpha, \alpha)$ for the value of the Ricci quadratic form on the vector $\alpha^\sharp$ dual to α, the integrated identity (13.6) becomes
$$0 = |\nabla\alpha|^2_{L^2} + \int_M \mathrm{Ric}(\alpha, \alpha)\, dV \tag{13.7}$$
Now invoke the hypothesis. The first summand is non-negative because it is a squared L^2-norm. The second summand is non-negative because $\mathrm{Ric} \geq 0$ makes the integrand $\mathrm{Ric}(\alpha, \alpha) \geq 0$ at every point. Two non-negative quantities summing to zero are each zero:
$$|\nabla\alpha|^2_{L^2} = 0 \qquad \text{and} \qquad \int_M \mathrm{Ric}(\alpha, \alpha)\, dV = 0$$
From $|\nabla\alpha|_{L^2} = 0$ and continuity of $\nabla\alpha$ it follows that $\nabla\alpha \equiv 0$, so α is parallel. From the vanishing of the integral of the non-negative continuous function $\mathrm{Ric}(\alpha, \alpha)$ it follows that $\mathrm{Ric}(\alpha, \alpha) \equiv 0$ pointwise. This proves part (1).
2. *The dimension bound $b_1(M) \leq n$.* By part (1) every harmonic 1-form is parallel. A parallel form is determined by its value at a single point: if $\nabla\alpha = 0$ then α is invariant under parallel transport (definition 2.4.1), so for any $p, q \in M$ and any path from p to q the value

α_q is the parallel transport of α_p; since M is connected, α is determined on all of M by $\alpha_p \in T_p^*M$. Hence the evaluation map

$$\mathcal{H}^1(M) \longrightarrow T_p^*M, \qquad \alpha \longmapsto \alpha_p,$$

is linear and injective: if $\alpha_p = 0$ then parallel transport gives $\alpha \equiv 0$. Therefore

$$b_1(M) = \dim \mathcal{H}^1(M) \leq \dim T_p^*M = n,$$

which is part (2).

3. *Strict positivity kills $\mathcal{H}^1$.* Suppose $\mathrm{Ric} > 0$ at a point $p \in M$, meaning $\mathrm{Ric}_p(v, v) > 0$ for every nonzero $v \in T_pM$. Let $\alpha \in \mathcal{H}^1(M)$. By part (1), $\mathrm{Ric}(\alpha, \alpha) \equiv 0$; evaluating at p,

$$\mathrm{Ric}_p(\alpha_p^\sharp, \alpha_p^\sharp) = 0$$

Strict positivity of Ric_p forces $\alpha_p^\sharp = 0$, hence $\alpha_p = 0$. But α is parallel by part (1), so it is determined by its value at p, and $\alpha_p = 0$ gives $\alpha \equiv 0$ by the injectivity of the evaluation map of part (2). Thus $\mathcal{H}^1(M) = 0$ and $b_1(M) = 0$, completing the proof.

The proof exhibits the sign discipline the whole argument rests on. Both terms of (13.7) are non-negative precisely because the Weitzenböck formula carries the Ricci term with a *plus* sign and because $\nabla^*\nabla$ is a non-negative operator, and it is the sum of two non-negative numbers being zero that forces each to vanish. Reversing either sign would break the argument entirely. The identity (13.7) also isolates the exact role of positivity: non-negative Ricci curvature gives parallelism, a rigidity conclusion, while strict positivity at even one point gives vanishing, a topological one. The step from parallel to determined-by-one-value is where a differential-geometric fact, the invariance of parallel forms under parallel transport, converts into the linear-algebra bound $b_1 \leq n$, and the step from $\mathrm{Ric}(\alpha, \alpha) \equiv 0$ to $\alpha_p = 0$ is where strict positivity does its work at a single point and parallelism propagates the conclusion.

There is a symmetry worth marking between the two conclusions and the operator they came from. Part (1) says the kernel of Δ on 1-forms sits inside the kernel of $\nabla^*\nabla$, the parallel forms; equality of these two kernels is exactly the vanishing of the Ricci obstruction on harmonic forms. The general principle, taken up in section 13.3, is that a curvature term with a favorable sign collapses the harmonic forms onto the parallel ones, and a parallel section of a bundle is a global algebraic object governed by holonomy, far more rigid than a general harmonic form.

The strict case is significant enough to record on its own, both because it is the form in which Bochner's theorem is usually cited and because it invites a direct comparison with the diameter and fundamental-group conclusions drawn from the same hypothesis in Part I.

Corollary 13.2.2: Positive Ricci Curvature Kills the First Betti Number

Let (M, g) be a closed oriented Riemannian n-manifold with $\mathrm{Ric} > 0$ everywhere (positive Ricci curvature: $\mathrm{Ric}_p(v, v) > 0$ for every $p \in M$ and every nonzero $v \in T_pM$). Then

$$b_1(M) = 0 \qquad \text{and} \qquad H^1(M; \mathbb{R}) = 0$$

so M has no nonzero harmonic 1-forms and its first real cohomology vanishes.

***Proof*:**
Positivity everywhere is in particular positivity at some point, and it certainly implies $\mathrm{Ric} \geq 0$, so theorem 13.2.1(3) applies and gives $b_1(M) = \dim \mathcal{H}^1(M) = 0$. The de Rham theorem identifies $H^1_{dR}(M)$ with the singular cohomology $H^1(M;\mathbb{R})$ ([6], [4]), and by the Hodge isomorphism (theorem 12.4.3) its dimension is $\dim \mathcal{H}^1(M) = b_1(M) = 0$. Hence $H^1(M;\mathbb{R}) = 0$, as we sought to show.

This corollary and Bonnet–Myers (corollary 6.3.2, §6.3) extract two different topological conclusions from the same curvature hypothesis, by two different mechanisms, and it is worth setting them side by side. Bonnet–Myers assumes a uniform positive lower bound $\mathrm{Ric} \geq (n-1)k\, g$ with $k > 0$ and, through the second variation of arc length, concludes that M is compact with finite fundamental group $\pi_1(M)$. Bochner assumes only $\mathrm{Ric} > 0$ (positive but with no uniform lower bound needed, since M is already compact) and, through the integrated Weitzenböck formula, concludes $b_1(M) = 0$. The two conclusions are related but neither contains the other. Finiteness of $\pi_1(M)$ implies that its abelianization $H_1(M;\mathbb{Z})$ is finite, hence that $b_1(M) = \dim(H_1(M;\mathbb{Z}) \otimes \mathbb{R}) = 0$, so on a compact manifold the Bonnet–Myers conclusion already forces $b_1 = 0$ and Bochner's first-Betti vanishing is a formal consequence of it under the stronger hypothesis. The value of Bochner's argument is that it needs less, only pointwise $\mathrm{Ric} > 0$ rather than a uniform bound, and that it produces the vanishing directly from an analytic identity rather than through the topology of the fundamental group. The two techniques are the twin comparison partners of positive Ricci curvature: the variational one controls size and π_1, the analytic one controls harmonic forms and b_1.

Two computations make the theorem concrete and confirm the sign conventions against the cohomology already computed in chapter 12. The flat torus tests the boundary case $\mathrm{Ric} \equiv 0$, where the theorem predicts parallelism but no vanishing; the round sphere tests strict positivity, where it predicts vanishing.

Example 13.2.3: The Flat Torus and the Round Sphere
The two model cases exhibit the two regimes of theorem 13.2.1.

The flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ (example 1.3.6) is flat, so its curvature tensor vanishes identically and $\mathrm{Ric} \equiv 0$. The Ricci term in the Weitzenböck formula is therefore absent, and theorem 13.1.4 reduces on 1-forms to

$$\Delta\alpha = \nabla^*\nabla\alpha$$

the Hodge and connection Laplacians coincide. The integrated identity (13.7) becomes $0 = |\nabla\alpha|^2_{L^2}$ for a harmonic α, so every harmonic 1-form on T^n is parallel, exactly the conclusion of theorem 13.2.1(1) in the degenerate case $\mathrm{Ric} = 0$. This matches the explicit computation of example 12.4.5 from a second direction: there the harmonic 1-forms were found to be precisely the constant-coefficient forms $\sum_i c_i\, dx^i$ with $c_i \in \mathbb{R}$, and in the parallel coframe $\{dx^i\}$ of the flat metric a constant-coefficient form is exactly a parallel one, since $\nabla(dx^i) = 0$ for each i (the coordinate coframe is parallel for the flat connection) and hence $\nabla(\sum_i c_i\, dx^i) = 0$. So the two descriptions of $\mathcal{H}^1(T^n)$, "harmonic" from the Hodge theory and "parallel" from Bochner, agree. Their common dimension is n, the number of coefficients c_i:

$$b_1(T^n) = \dim \mathcal{H}^1(T^n) = n$$

This is the equality case of the bound $b_1 \leq n$ in theorem 13.2.1(2). The bound is attained precisely because T^n has a full n-dimensional space of parallel 1-forms, which happens exactly when the holonomy is trivial, as it is for the flat torus. The value n agrees with $\binom{n}{1} = n$ from the Künneth computation of example 12.4.5, closing the consistency check.

The round sphere S^n with $n \geq 2$ has positive Ricci curvature. For the sphere of radius r, example 4.4.9 gives $\mathrm{Ric} = \frac{n-1}{r^2}\, g$, so for every nonzero tangent vector v,

$$\mathrm{Ric}(v, v) = \frac{n-1}{r^2}\, |v|^2 > 0 \qquad (n \geq 2)$$

strictly positive at every point. Bochner's theorem in the form corollary 13.2.2 applies and gives $b_1(S^n) = 0$: the sphere carries no nonzero harmonic 1-form. This agrees with the topology, since $H^1(S^n; \mathbb{R}) = 0$ for $n \geq 2$ (example 12.4.5 recorded $\mathcal{H}^k(S^n) = 0$ for $0 < k < n$ from the vanishing of the intermediate de Rham cohomology). Here the Weitzenböck argument re-derives the vanishing of b_1 from curvature alone, without appealing to the known cohomology of the sphere: the strict positivity of Ric forces the Ricci integral in (13.7) to be strictly positive for any nonzero α, contradicting harmonicity, so the only harmonic 1-form is zero.

A caveat delimits what the sphere computation gives. The conclusion $b_1(S^n) = 0$ is strictly weaker than simple connectivity: it says only that the abelianized fundamental group has no free part, not that π_1 is trivial. Synge's theorem (section 8.3, §8.3) does prove that an even-dimensional orientable manifold with positive *sectional* curvature is simply connected, a stronger topological conclusion, but it demands the stronger hypothesis $K > 0$ rather than $\mathrm{Ric} > 0$. Bochner's vanishing $b_1 = 0$ needs only positive Ricci curvature, and positive Ricci curvature is strictly weaker than positive sectional curvature: a Riemannian product such as $S^2 \times S^2$ has positive Ricci curvature but has some zero sectional curvatures (the mixed planes spanned by one tangent vector from each factor are flat), so it lies outside Synge's reach while remaining inside Bochner's. On that product $b_1 = 0$ by corollary 13.2.2, and indeed $H^1(S^2 \times S^2; \mathbb{R}) = 0$ by the Künneth formula, whereas $b_2 = 2 \neq 0$, so positive Ricci curvature bounds the first Betti number without saying anything about higher ones. Controlling b_k for $k \geq 2$ requires the curvature term of the Weitzenböck formula on k-forms, the subject of section 13.3.

13.3 Weitzenböck Formulas in General

The Bochner argument of section 13.2 rested on a single identity: the Hodge Laplacian on 1-forms splits as a non-negative operator plus a curvature term, $\Delta = \nabla^*\nabla + \mathrm{Ric}$ (theorem 13.1.4). Once that identity was in place, positivity of the Ricci curvature forced every harmonic 1-form to vanish, and the first Betti number with it. Nothing in the method was special to degree 1. The exterior derivative d and the codifferential d^* are built from the same covariant derivative ∇ that defines the connection Laplacian $\nabla^*\nabla$, and the difference between the two Laplacians is always a curvature bracket, measured by commuting covariant derivatives through the Ricci identity. This section states the general form of that splitting, the Weitzenböck formula on k-forms, and then follows the method to its principal destinations: the vanishing theorem for manifolds with positive curvature operator, and the two downstream identities, Lichnerowicz's formula for the Dirac operator and the Bochner–Kodaira–Nakano identities on Kähler manifolds, that carry the same mechanism into spin geometry and complex geometry. Those two lie beyond the tools developed here and are pointed to their proper homes rather than proved.

Throughout, (M, g) is a closed oriented Riemannian n-manifold with Levi-Civita connection ∇; the metric is positive-definite, as the whole Hodge apparatus of chapter 12 requires. The Hodge Laplacian is $\Delta = dd^* + d^*d$ (definition 12.2.1), the harmonic k-forms are $\mathcal{H}^k(M) = \ker(\Delta|_{\Omega^k})$ with $\dim \mathcal{H}^k = b_k$ by the Hodge theorem (theorem 12.4.3), the connection Laplacian is $\nabla^*\nabla = -\mathrm{tr}_g \nabla^2$

(definition 13.1.1), and the L^2 inner product is $(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \, dV$ (definition 12.1.4). The curvature sign convention is that of box 4.1.1. The Einstein summation convention is in force where a local orthonormal frame is used.

The passage from 1-forms to k-forms changes only the algebra of the curvature term. On a 1-form the two Laplacians differed by the Ricci tensor acting as an endomorphism; on a k-form they differ by a curvature endomorphism $\mathcal{W}_k$ of the bundle $\Lambda^k T^*M$, assembled from the full Riemann tensor rather than from its Ricci trace alone. The mechanism producing this endomorphism is unchanged: expand $\Delta = dd^* + d^*d$ in a local frame, compare with $\nabla^*\nabla$, and collect the terms in which a pair of covariant derivatives has been commuted, each such commutation contributing a factor of the curvature through the Ricci identity $\nabla^2_{X,Y} - \nabla^2_{Y,X} = R(X, Y)$ (definition 4.1.2). What degree $k > 1$ adds is that the exterior algebra threads these curvature factors through wedge products and interior products, giving an operator that no longer collapses to a single symmetric 2-tensor.

Theorem 13.3.1: The general Weitzenböck formula

Let (M, g) be a closed oriented Riemannian n-manifold. For each k the Hodge Laplacian on k-forms decomposes as

$$\Delta = \nabla^*\nabla + \mathcal{W}_k$$

where $\nabla^*\nabla$ is the connection Laplacian of definition 13.1.1 and $\mathcal{W}_k$ is a symmetric curvature endomorphism of the bundle $\Lambda^k T^*M$, pointwise self-adjoint with respect to the metric $\langle -, - \rangle$ on $\Lambda^k T^*_p M$ and built from the Riemann curvature tensor of (M, g). In a local orthonormal frame $(e_1, \ldots, e_n)$ with dual coframe $(e^1, \ldots, e^n)$, writing $\varepsilon^i = e^i \wedge -$ for exterior multiplication and $\iota_i = \iota_{e_i}$ for interior multiplication, the operator $\mathcal{W}_k$ is

$$\mathcal{W}_k = - \sum_{i,j=1}^{n} \varepsilon^i \, \iota_j \, R(e_i, e_j) \tag{13.8}$$

where $R(e_i, e_j)$ acts on k-forms as the derivation induced by the curvature endomorphism (definition 2.4.6), and the sign is fixed by the convention of box 4.1.1. In degree $k = 1$ this reduces to $\mathcal{W}_1 = \mathrm{Ric}$ (theorem 13.1.4): the Weitzenböck endomorphism on 1-forms is the Ricci tensor viewed through the metric as an endomorphism of T^*M.

The formula is one equation carrying two operators of different character. The connection Laplacian $\nabla^*\nabla$ is universal: it depends on the metric only through ∇ and is non-negative on every closed manifold, with $(\nabla^*\nabla T, T) = |\nabla T|^2_{L^2} \geq 0$ and kernel the parallel tensor fields (proposition 13.1.3). The curvature endomorphism $\mathcal{W}_k$ carries all the geometry, and its sign at a point is what the vanishing method reads. The content of (13.8) is that the whole difference between the analyst's Laplacian Δ, defined through d and d^*, and the geometer's Laplacian $\nabla^*\nabla$, defined through parallel transport, is curvature, packaged degree by degree into $\mathcal{W}_k$.

***Proof*:**
The $k = 1$ identity is theorem 13.1.4, proved in full there by expanding $\Delta = dd^* + d^*d$ on a 1-form in a local orthonormal frame and commuting the two covariant derivatives through the Ricci identity; that proof establishes the exact statement $\mathcal{W}_1 = \mathrm{Ric}$, which anchors the sign of the general formula. The argument for general k follows the same three movements, now inside the exterior algebra.

Step 1: local frame expressions for d and d^*. Fix $p \in M$ and a local orthonormal frame $(e_1, \dots, e_n)$ that is geodesic at p, meaning $\nabla e_i(p) = 0$ (such a frame exists by parallel-transporting an orthonormal basis along the radial geodesics of a normal neighborhood, proposition 3.2.8). Let $(e^1, \dots, e^n)$ be the dual coframe; at p it too is parallel. On any k-form the exterior derivative and the codifferential are expressed through the covariant derivative by

$$d = \sum_{i=1}^{n} \varepsilon^i \, \nabla_{e_i}, \qquad d^* = -\sum_{i=1}^{n} \iota_i \, \nabla_{e_i} \tag{13.9}$$

identities that hold on all forms because d and d^* are the antisymmetrization and its metric adjoint of ∇, and because ∇ is torsion-free and metric-compatible (theorem 2.3.1). Here $\varepsilon^i = e^i \wedge -$ raises degree by one and $\iota_i = \iota_{e_i}$ lowers it by one, and the two are pointwise adjoint, $\langle \varepsilon^i \alpha, \beta \rangle = \langle \alpha, \iota_i \beta \rangle$, with the anticommutation relation $\varepsilon^i \iota_j + \iota_j \varepsilon^i = \delta^{ij}$ of the exterior algebra.

Step 2: the two Laplacians as sums of second covariant derivatives. Compose the frame expressions of (13.9). At p, where the coframe is parallel, each ∇_{e_i} commutes with the algebraic operators ε^j and ι_j, so the two halves of the Hodge Laplacian are

$$dd^* = -\sum_{i,j} \varepsilon^i \iota_j \, \nabla_{e_i} \nabla_{e_j}, \qquad d^* d = -\sum_{i,j} \iota_j \varepsilon^i \, \nabla_{e_j} \nabla_{e_i}$$

In the second sum replace $\iota_j \varepsilon^i$ using the anticommutation relation $\iota_j \varepsilon^i = \delta^{ij} - \varepsilon^i \iota_j$:

$$d^* d = -\sum_{i,j} \big(\delta^{ij} - \varepsilon^i \iota_j \big) \nabla_{e_j} \nabla_{e_i} = -\sum_{i} \nabla_{e_i} \nabla_{e_i} + \sum_{i,j} \varepsilon^i \iota_j \, \nabla_{e_j} \nabla_{e_i}$$

The pure trace $-\sum_i \nabla_{e_i} \nabla_{e_i}$ is the connection Laplacian at the geodesic frame,

$$\nabla^* \nabla = -\mathrm{tr}_g \nabla^2 = -\sum_{i=1}^{n} \big(\nabla_{e_i} \nabla_{e_i} - \nabla_{\nabla_{e_i} e_i} \big) = -\sum_{i=1}^{n} \nabla_{e_i} \nabla_{e_i} \quad \text{at } p,$$

using $\nabla_{e_i} e_i(p) = 0$. Adding dd^* and $d^* d$, the two $\varepsilon^i \iota_j$ sums combine into a single antisymmetrized second derivative, and

$$\Delta = \nabla^* \nabla - \sum_{i,j} \varepsilon^i \iota_j \big(\nabla_{e_i} \nabla_{e_j} - \nabla_{e_j} \nabla_{e_i} \big) \tag{13.10}$$

Everything to be identified as curvature now sits in the antisymmetrized second derivatives of the final sum.

Step 3: the Ricci identity converts the remainder into curvature. The antisymmetric part of the second covariant derivative is exactly the curvature action. By the Ricci identity for the induced connection on forms (definition 4.1.2, extended to tensor bundles in definition 2.4.6), for any form ω,

$$\nabla_{e_i} \nabla_{e_j} \omega - \nabla_{e_j} \nabla_{e_i} \omega = R(e_i, e_j) \omega \quad \text{at } p,$$

the bracket term $\nabla_{[e_i, e_j]} \omega$ vanishing at p because $[e_i, e_j](p) = 0$ for a frame parallel at p. Substituting into (13.10) gives, at p,

$$\Delta = \nabla^* \nabla - \sum_{i,j} \varepsilon^i \iota_j \, R(e_i, e_j)$$

The point p was arbitrary and the two Laplacians are tensorial (independent of the frame), so the identity holds everywhere, with $\mathcal{W}_k = -\sum_{i,j} \varepsilon^i \iota_j R(e_i, e_j)$; this is (13.8). The sign is the one the $k = 1$ anchor requires. On a 1-form α, the interior product ι_j picks out the e_j-component of the 1-form $R(e_i, e_j)\alpha$ and ε^i returns it to the i-slot, so $\mathcal{W}_1(\alpha)$ has i-component $-\sum_j \big(R(e_i, e_j)\alpha\big)(e_j) = \sum_{j,m} R_{ijjm}\, \alpha^m = R_{im}\alpha^m$, using the induced (dual) action of the curvature on 1-forms and the definition of Ric by tracing Riem over its first and last slots (definition 4.4.1). Thus $\mathcal{W}_1(\alpha)_i = R_{ij}\alpha^j$, that is $\mathcal{W}_1 = \mathrm{Ric}$ with the plus sign of theorem 13.1.4, and the general formula carries that sign into every degree.

Step 4: symmetry of the curvature endomorphism. That $\mathcal{W}_k$ is pointwise self-adjoint is forced by the identity itself: both Δ and $\nabla^*\nabla$ are formally self-adjoint on the closed manifold (proposition 12.2.3 and proposition 13.1.3), so their difference $\mathcal{W}_k$ is a formally self-adjoint operator; being a zeroth-order operator, a bundle endomorphism, it is pointwise self-adjoint with respect to $\langle -, - \rangle$ on each fiber $\Lambda^k T_p^* M$. This is what makes the sign of $\mathcal{W}_k$, and hence the eigenvalue condition used in the vanishing theorems, meaningful. This completes the derivation of the general Weitzenböck formula.

The formula is what the Bochner technique uses in its general form. Integrate $\Delta = \nabla^*\nabla + \mathcal{W}_k$ against a harmonic k-form α, using $(\Delta\alpha, \alpha) = 0$ and the non-negativity of the connection Laplacian, and one obtains the identity

$$0 = (\Delta\alpha, \alpha) = |\nabla\alpha|^2_{L^2} + \int_M \langle \mathcal{W}_k\alpha, \alpha \rangle \, dV$$

the exact degree-k analogue of the integrated Bochner identity for 1-forms of section 13.2. Both terms on the right are controlled by geometry: the first is non-negative always, the second is non-negative wherever $\mathcal{W}_k$ is a non-negative endomorphism. When both are non-negative and their sum is zero, each vanishes: the form is parallel and $\mathcal{W}_k\alpha \equiv 0$. Positivity of $\mathcal{W}_k$ at a single point then forces α to vanish there, and a parallel form vanishing at a point vanishes identically, so $\mathcal{H}^k = 0$ and $b_k = 0$. This is the same logic that gave $b_1 = 0$ under $\mathrm{Ric} > 0$; the only new input at each degree is a sufficient condition on $\mathcal{W}_k$.

The degree-1 case already carries the two cross-checks that calibrate the sign. On the flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ the metric is flat, so $R \equiv 0$ and $\mathcal{W}_k \equiv 0$ in every degree; the formula collapses to $\Delta = \nabla^*\nabla$, and by the non-negativity of the connection Laplacian a harmonic form is parallel, that is constant-coefficient in the parallel coframe $\{dx^i\}$. In degree 1 this recovers exactly the n parallel 1-forms $dx^1, \dots, dx^n$, giving $b_1(T^n) = n = \dim \mathcal{H}^1(T^n)$, consistent with the direct Hodge count of example 12.4.5. On the round sphere S^n with $n \geq 2$ the Ricci curvature is positive, $\mathrm{Ric} = \frac{n-1}{r^2} g > 0$ (example 4.4.2), so $\mathcal{W}_1 > 0$ everywhere and the integrated identity forces $b_1(S^n) = 0$ (theorem 13.2.1), consistent with $H^1(S^n; \mathbb{R}) = 0$. The general formula extends this dichotomy to every degree, and the condition that makes $\mathcal{W}_k$ non-negative in all degrees at once is a positivity condition not on the Ricci trace but on the full curvature operator.

The curvature operator is the symmetric endomorphism $\mathcal{R}$ of the bundle $\Lambda^2 T^* M$ of 2-forms determined by the Riemann tensor through $\langle \mathcal{R}(X \wedge Y), Z \wedge W \rangle = \mathrm{Riem}(X, Y, W, Z)$; it is positive when all its eigenvalues are positive at every point. Positive curvature operator is strictly stronger than positive sectional curvature, which is in turn stronger than positive Ricci curvature, and it is exactly the hypothesis under which every $\mathcal{W}_k$ is a positive endomorphism. Meyer's theorem draws the topological conclusion.

Corollary 13.3.2: Vanishing under a positive curvature operator

Let (M, g) be a closed oriented Riemannian n-manifold whose curvature operator $\mathcal{R}$ on $\Lambda^2 T^*M$ is positive at every point. Then the Weitzenböck endomorphism $\mathcal{W}_k$ of theorem 13.3.1 is a positive endomorphism of $\Lambda^k T^*M$ for every k with $0 < k < n$, every harmonic k-form is parallel and in fact zero, and consequently

$$b_k(M) = 0 \qquad \text{for } 0 < k < n$$

so M has the real Betti numbers of a sphere, $b_0 = b_n = 1$ and $b_k = 0$ otherwise. A closed simply connected manifold with positive curvature operator is a rational homology sphere.

This is stated, not proved here. The step from a positive curvature operator to the positivity of each $\mathcal{W}_k$ is a piece of representation theory of the orthogonal group applied to the curvature acting on $\Lambda^k T^*M$; the estimate that a positive curvature operator makes $\langle \mathcal{W}_k \alpha, \alpha \rangle$ strictly positive for $\alpha \neq 0$ is due to Meyer, refining an earlier result of Gallot and Meyer, and the argument is carried out in D. Meyer's work and in the account of the Bochner technique in Peter Petersen's *Riemannian Geometry*. Once each $\mathcal{W}_k > 0$, the integrated Weitzenböck identity above gives $\mathcal{H}^k = 0$ for $0 < k < n$ exactly as in theorem 13.2.1, and the Hodge theorem theorem 12.4.3 converts this to $b_k = 0$. The extreme degrees $b_0 = b_n = 1$ are the connectedness and orientation counts of example 12.4.5. The conclusion places the pinched-curvature sphere theorems of Part I and the vanishing theorems of Part II side by side: both derive sphere-like topology from positive curvature, the former through geodesics and the comparison method, the latter through harmonic forms and the Weitzenböck method. The vanishing route is weaker in its conclusion, homology rather than diffeomorphism type, but it needs only the pointwise algebraic positivity of a curvature endomorphism, and it generalizes to settings where geodesics give no purchase.

The same substitution, a different first-order operator in place of $d + d^*$, produces the two identities that carry the Bochner method into geometry beyond the de Rham complex. Both are Weitzenböck formulas for a square-root of a Laplacian, and both are named after the geometry in which their curvature term acquires meaning. They are recorded here as the destinations of the method, forward-pointed to the volumes that develop them.

13.3.3 Foreshadowing: Lichnerowicz and Bochner–Kodaira–Nakano Two downstream Weitzenböck identities extend the method of this chapter, each replacing the de Rham operator $d + d^*$ by a curvature-adapted square root of a Laplacian.
The *Lichnerowicz formula* governs the Dirac operator D on a spin manifold, the natural first-order operator on the spinor bundle whose square is a Laplacian. It reads

$$D^2 = \nabla^*\nabla + \tfrac{1}{4}\,\mathrm{scal}$$

with scal the scalar curvature of definition 4.4.4. The curvature term is now the scalar curvature acting by multiplication, the simplest possible Weitzenböck endomorphism. Integrating against a harmonic spinor, exactly as in the Bochner argument, shows that a closed spin manifold with positive scalar curvature carries no nonzero harmonic spinors, so the index of D vanishes; by the Atiyah–Singer index theorem that index is the $\hat{A}$-genus, a characteristic number, giving a topological obstruction to positive scalar curvature. The Dirac operator, spinor bundles, and the index-theoretic reading are developed in spin geometry and index theory; the characteristic-class side is treated in [8], and the standard reference for the analytic side is Lawson and Michelsohn's *Spin Geometry*.
The *Bochner–Kodaira–Nakano identities* govern the Dolbeault operators $\bar\partial$ and $\bar\partial^*$ on a compact Kähler manifold, the holomorphic analogue of d and d^*. There the difference between the two natural Laplacians $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ and its ∂-counterpart is a curvature term built from the Ricci form and the curvature of the coefficient bundle. Applied to a holomorphic line bundle whose curvature is positive, this yields the Kodaira vanishing theorem, the statement that the higher cohomology of a positive line bundle twisted by the canonical bundle vanishes, the analytic result underlying the projective embedding of Kähler manifolds. These identities and the vanishing theorem are proved in Hodge theory; see [2], and Griffiths and Harris's *Principles of Algebraic Geometry* for the classical account.

Both identities are the present formula with the harmonic forms replaced by harmonic sections of another bundle, and the Weitzenböck endomorphism replaced by the curvature natural to that bundle: the scalar curvature for spinors, the bundle curvature and Ricci form for holomorphic sections. The method does not change; only the operator whose square is compared with $\nabla^*\nabla$ changes, and with it the geometry that the curvature term encodes.

Example 13.3.4: The Weitzenböck method as a single mechanism
Part II has developed one idea in several forms, and the general Weitzenböck formula is its clearest statement. A geometric operator whose square is a Laplacian, the exterior derivative package $d + d^*$ here, the Dirac operator D in spin geometry, the Dolbeault operator $\bar\partial + \bar\partial^*$ on a Kähler manifold, admits a second Laplacian, the connection Laplacian $\nabla^*\nabla$, built from parallel transport rather than from the operator. The two Laplacians differ by a curvature endomorphism:

$$\text{geometric Laplacian} = \nabla^*\nabla + (\text{curvature term})$$

The connection Laplacian is non-negative on a closed manifold, with kernel the parallel fields (proposition 13.1.3). Integrating the identity against a harmonic field therefore splits zero into two contributions, $0 = |\nabla\alpha|^2_{L^2} + \int_M \langle(\text{curvature term})\,\alpha, \alpha\rangle\, dV$, both non-negative once the curvature has a sign; each must vanish, forcing the harmonic field to be parallel and to lie in the kernel of the curvature term. A strict sign at one point then eliminates the harmonic field entirely. The output is a vanishing theorem: a curvature hypothesis kills a space of harmonic objects, and, through the identification of harmonic objects with topology, a topological invariant.

The threads of Part II meet here. The curvature of a connection on a vector bundle (chapter 10) is what enters every Weitzenböck term; Chern–Weil theory (chapter 11) turns the same curvature into the characteristic numbers that the vanishing theorems constrain, as the $\hat{A}$-genus constrains positive scalar curvature; the Hodge theorem (chapter 12) is what licenses reading a dimension count of harmonic forms as a Betti number. And the curvature that drives the whole mechanism is the Ricci and sectional curvature of Part I: the Bochner vanishing $b_1 = 0$ under $\mathrm{Ric} > 0$ (theorem 13.2.1) stands beside Bonnet–Myers (corollary 6.3.2) as two conclusions drawn from the same positive Ricci hypothesis, one topological through harmonic forms, one metric through geodesics. Looking forward, the connection Laplacian and the curvature term are the objects of geometric analysis, where the heat flow $e^{-t\Delta}$ and the maximum principle refine the integrated identity into pointwise estimates; the Lichnerowicz formula is the entry to index theory; and the Bochner–Kodaira–Nakano identities open Kähler and complex-algebraic geometry. The single equation $\Delta = \nabla^*\nabla + \mathcal{W}_k$ is the seam along which curvature controls the analysis of manifolds.

14

What Next

This book has carried the reader from the definition of a metric through the Levi-Civita connection, geodesics, and curvature; through the comparison theorems, submanifold geometry, the Gauss–Bonnet and sphere theorems, and the symmetric spaces; and through the analytic half of the subject, connections and Chern–Weil theory, the Hodge theorem, and the Bochner technique. Each of these is the entrance to a subject larger than any one book can hold. This closing chapter is a map of where to go next, pairing every direction with the companion EYNTKA volume where one exists, or its codename where the volume is still to be written, and with the standard texts a working geometer keeps within reach.

Consolidating Riemannian geometry. The closest companions for a second pass over this material are do Carmo's *Riemannian Geometry*, whose treatment of the connection and the comparison theorems this book often follows; John M. Lee's *Introduction to Riemannian Manifolds*, careful and modern, whose coverage this book's Part I deliberately contains; and Petersen's *Riemannian Geometry*, which pushes the comparison theory of chapter 6 much further, into convergence theory and the structure of manifolds with curvature bounds. Cheeger and Ebin's *Comparison Theorems in Riemannian Geometry* is the classical account of the material of that chapter.

Lie groups and symmetric spaces. The homogeneous and symmetric spaces of chapter 9 were treated with only the light Lie theory this book develops, and the Cartan classification was stated rather than proved. The structure theory that completes it, root systems, the Cartan decomposition, and the classification of symmetric spaces, is the subject of the companion volume *Lie Groups* ([10]). The standard references are Helgason's *Differential Geometry, Lie Groups, and Symmetric Spaces*, the definitive account of symmetric spaces, and Knapp's *Lie Groups Beyond an Introduction* for the underlying structure theory.

Geometric analysis. The variational and elliptic methods that appeared in the second variation of chapter 5, in the minimal hypersurfaces of chapter 7, and in the Hodge theory of chapter 12 are the seed of geometric analysis, where partial differential equations are used to produce and to constrain Riemannian structures. Its three central threads are the Ricci flow, developed in

Topping's *Lectures on the Ricci Flow* and the comprehensive *Hamilton's Ricci Flow* of Chow, Lu, and Ni; minimal surfaces, in Colding and Minicozzi's *A Course in Minimal Surfaces*; and spectral geometry, the study of the Laplacian spectrum of chapter 12, in Chavel's *Eigenvalues in Riemannian Geometry*. These are gathered in the projected companion volume *Geometric Analysis* (codename `NateGeometricAnalysis`).

General relativity. The pseudo-Riemannian thread carried through this book, the signature-agnostic connection and curvature and the Lorentzian failures flagged at Hopf–Rinow and elsewhere, is the geometric foundation of general relativity, where the Einstein equations make the metric dynamical. The bridge from the present geometry to that physics is O'Neill's *Semi-Riemannian Geometry with Applications to Relativity*, which develops the Lorentzian theory in the language used here; Wald's *General Relativity* and Hawking and Ellis's *The Large Scale Structure of Space-Time* carry it into causality and the singularity theorems. The projected companion volume is *General Relativity* (codename `NateGR`).

Spin geometry, Dirac operators, and index theory. The connections and Chern–Weil theory of chapter 10 and chapter 11, and the Weitzenböck formula of chapter 13 in its Lichnerowicz form, are the entry to spin geometry and the Atiyah–Singer index theorem, where the analytic index of the Dirac operator is computed by characteristic classes. The standard reference is Lawson and Michelsohn's *Spin Geometry*, with the heat-kernel proof of the index theorem in Berline, Getzler, and Vergne's *Heat Kernels and Dirac Operators*; the topological characteristic classes on which the index formula rests are developed in the companion *K-Theory* ([8]) and in Milnor and Stasheff's *Characteristic Classes*.

Complex and Kähler geometry, Hodge theory. The real Hodge theorem of chapter 12 and the Chern–Weil theory of chapter 11 are the Riemannian input to the Hodge theory of Kähler manifolds, where the Laplacian respects the complex structure and cohomology acquires the Hodge decomposition $H^k = \bigoplus_{p+q=k} H^{p,q}$. That refinement, together with the Bochner–Kodaira–Nakano identities and the Kodaira vanishing and embedding theorems previewed in chapter 13 and chapter 10, is the subject of the companion *Hodge Theory* ([2]), which imports this book's Part II. The staple texts are Griffiths and Harris's *Principles of Algebraic Geometry*, Huybrechts's *Complex Geometry*, and Voisin's *Hodge Theory and Complex Algebraic Geometry*.

Comparison and metric geometry. The comparison theorems of chapter 6 have a synthetic continuation that dispenses with smoothness altogether, replacing sectional-curvature bounds by triangle comparison conditions on metric spaces. Alexandrov spaces and the Gromov–Hausdorff convergence of Riemannian manifolds are developed in Burago, Burago, and Ivanov's *A Course in Metric Geometry* and in the later chapters of Petersen's *Riemannian Geometry*. This is where the rigidity and finiteness theorems of comparison geometry find their natural home.

A closing suggestion. The surest way to consolidate this material is to compute with it. Re-deriving the curvature of the model spaces, running the second variation to prove Bonnet–Myers and Synge by hand, and carrying one nontrivial example through each major machine, a Jacobi field comparison, a Chern–Weil integral, a Hodge decomposition, a Weitzenböck estimate, fixes the apparatus in a way no second reading can. Curvature has been the single thread through every chapter; every direction above is a place where that thread continues.

Index

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